Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The purpose of this document is to analysis and design of **PROCESS PLANT – WAREHOUSE & WORKSHOP** foundation. In this document design & analysis of foundation have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system. Design and stability of foundation and column at each grid location has been checked, in this document design calculation of foundation and column has been shown which is most critical.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

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3. SCOPE

The analysis has been carried out in Staad pro & designed by Excel Sheet. For member strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. Load data for foundation design taken from document / drawing no. **MD-216-4000-EG-ST-GAD-2052, MD-216-4000-EG-ST-GAD-2053 & MD-216-4000-EG-ST-GAD-2054.**

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

1 DEAD LOAD (DL)
2 COLLATERAL LOAD (COLL)
3 ROOF LIVE LOAD (RLL)
4 PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)

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5 CRANE LOAD-1 (CL-1)
6 CRANE LOAD-2 (CL-2)
7 CRANE LOAD-3 (CL-3)
8 CRANE LOAD-4 (CL-4)
9 WIND LEFT-1 (WLL-1)
10 WIND LEFT-2 (WLL-2)
11 WIND RIGHT-1 (WLR-1)
12 WIND RIGHT-2 (WLR-2)
13 WIND LONG-1 (WLLG-1)
14 WIND LONG-2 (WLLG-2)
15 WIND LONG-3 (WLLG-3)
16 WIND LONG-4 (WLLG-4)
17 SEISMIC LOAD SL'+'
18 SEISMIC LOAD SL'-'
19 TEMPERATURE LOAD (TL1)
20 TEMPERATURE LOAD (TL2)
21 BRACING REACTION (BR-WLLG1)
22 BRACING REACTION (BR-WLLG2)
23 BRACING REACTION (BR-WLLG3)
24 BRACING REACTION (BR-WLLG4)
25 BRACING REACTION (BR-SL'+')
26 BRACING REACTION (BR-SL'-')

4.3 Load combination

A. The following load combination are considered for member strength calculation.

101 1.4(DL+COLL)
102 1.2(DL+COLL+CL-1)+1.6(RLL+PLL / MLL)
103 1.2(DL+COLL+CL-2)+1.6(RLL+PLL / MLL)
104 1.2(DL+COLL+CL-3)+1.6(RLL+PLL / MLL)
105 1.2(DL+COLL+CL-4)+1.6(RLL+PLL / MLL)
106 1.4(DL+COLL+CL-1)
107 1.4(DL+COLL+CL-2)
108 1.4(DL+COLL+CL-3)
109 1.4(DL+COLL+CL-4)
110 1.2(DL+COLL+CL-1+TL1)+1.6(RLL+PLL / MLL)
111 1.2(DL+COLL+CL-2+TL1)+1.6(RLL+PLL / MLL)

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112 1.2(DL+COLL+CL-3+TL1)+1.6(RLL+PLL / MLL)
113 1.2(DL+COLL+CL-4+TL1)+1.6(RLL+PLL / MLL)
114 1.2(DL+COLL+CL-1+TL2)+1.6(RLL+PLL / MLL)
115 1.2(DL+COLL+CL-2+TL2)+1.6(RLL+PLL / MLL)
116 1.2(DL+COLL+CL-3+TL2)+1.6(RLL+PLL / MLL)
117 1.2(DL+COLL+CL-4+TL2)+1.6(RLL+PLL / MLL)
118 1.2(DL+COLL+CL-1)+0.8(WLL-1)
119 1.2(DL+COLL+CL-2)+0.8(WLL-1)
120 1.2(DL+COLL+CL-3)+0.8(WLL-1)
121 1.2(DL+COLL+CL-4)+0.8(WLL-1)
122 1.2(DL+COLL+CL-1)+0.8(WLL-2)
123 1.2(DL+COLL+CL-2)+0.8(WLL-2)
124 1.2(DL+COLL+CL-3)+0.8(WLL-2)
125 1.2(DL+COLL+CL-4)+0.8(WLL-2)
126 1.2(DL+COLL+CL-1)+0.8(WLR-1)
127 1.2(DL+COLL+CL-2)+0.8(WLR-1)
128 1.2(DL+COLL+CL-3)+0.8(WLR-1)
129 1.2(DL+COLL+CL-4)+0.8(WLR-1)
130 1.2(DL+COLL+CL-1)+0.8(WLR-2)
131 1.2(DL+COLL+CL-2)+0.8(WLR-2)
132 1.2(DL+COLL+CL-3)+0.8(WLR-2)
133 1.2(DL+COLL+CL-4)+0.8(WLR-2)
134 1.2(DL+COLL+CL-1)+0.8(WLLG-1+BR-WLLG-1)
135 1.2(DL+COLL+CL-2)+0.8(WLLG-1+BR-WLLG-1)
136 1.2(DL+COLL+CL-3)+0.8(WLLG-1+BR-WLLG-1)
137 1.2(DL+COLL+CL-4)+0.8(WLLG-1+BR-WLLG-1)
138 1.2(DL+COLL+CL-1)+0.8(WLLG-2+BR-WLLG-2)
139 1.2(DL+COLL+CL-2)+0.8(WLLG-2+BR-WLLG-2)
140 1.2(DL+COLL+CL-3)+0.8(WLLG-2+BR-WLLG-2)
141 1.2(DL+COLL+CL-4)+0.8(WLLG-2+BR-WLLG-2)
142 1.2(DL+COLL+CL-1)+0.8(WLLG-3+BR-WLLG-3)
143 1.2(DL+COLL+CL-2)+0.8(WLLG-3+BR-WLLG-3)
144 1.2(DL+COLL+CL-3)+0.8(WLLG-3+BR-WLLG-3)
145 1.2(DL+COLL+CL-4)+0.8(WLLG-3+BR-WLLG-3)
146 1.2(DL+COLL+CL-1)+0.8(WLLG-4+BR-WLLG-4)
147 1.2(DL+COLL+CL-2)+0.8(WLLG-4+BR-WLLG-4)
148 1.2(DL+COLL+CL-3)+0.8(WLLG-4+BR-WLLG-4)
149 1.2(DL+COLL+CL-4)+0.8(WLLG-4+BR-WLLG-4)

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150 1.2(DL+COLL+CL-1)+1.6(WLL-1)+0.5(RLL+PLL / MLL)
151 1.2(DL+COLL+CL-2)+1.6(WLL-1)+0.5(RLL+PLL / MLL)
152 1.2(DL+COLL+CL-3)+1.6(WLL-1)+0.5(RLL+PLL / MLL)
153 1.2(DL+COLL+CL-4)+1.6(WLL-1)+0.5(RLL+PLL / MLL)
154 1.2(DL+COLL+CL-1)+1.6(WLL-2)+0.5(RLL+PLL / MLL)
155 1.2(DL+COLL+CL-2)+1.6(WLL-2)+0.5(RLL+PLL / MLL)
156 1.2(DL+COLL+CL-3)+1.6(WLL-2)+0.5(RLL+PLL / MLL)
157 1.2(DL+COLL+CL-4)+1.6(WLL-2)+0.5(RLL+PLL / MLL)
158 1.2(DL+COLL+CL-1)+1.6(WLR-1)+0.5(RLL+PLL / MLL)
159 1.2(DL+COLL+CL-2)+1.6(WLR-1)+0.5(RLL+PLL / MLL)
160 1.2(DL+COLL+CL-3)+1.6(WLR-1)+0.5(RLL+PLL / MLL)
161 1.2(DL+COLL+CL-4)+1.6(WLR-1)+0.5(RLL+PLL / MLL)
162 1.2(DL+COLL+CL-1)+1.6(WLR-2)+0.5(RLL+PLL / MLL)
163 1.2(DL+COLL+CL-2)+1.6(WLR-2)+0.5(RLL+PLL / MLL)
164 1.2(DL+COLL+CL-3)+1.6(WLR-2)+0.5(RLL+PLL / MLL)
165 1.2(DL+COLL+CL-4)+1.6(WLR-2)+0.5(RLL+PLL / MLL)
166 1.2(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+PLL / MLL)
167 1.2(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+PLL / MLL)
168 1.2(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+PLL / MLL)
169 1.2(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+PLL / MLL)
170 1.2(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+PLL / MLL)
171 1.2(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+PLL / MLL)
172 1.2(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+PLL / MLL)
173 1.2(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+PLL / MLL)
174 1.2(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+PLL / MLL)
175 1.2(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+PLL / MLL)
176 1.2(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+PLL / MLL)
177 1.2(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+PLL / MLL)
178 1.2(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PLL / MLL)
179 1.2(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PLL / MLL)
180 1.2(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PLL / MLL)
181 1.2(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PLL / MLL)
182 1.2(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / MLL)
183 1.2(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / MLL)
184 1.2(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / MLL)
185 1.2(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / MLL)
186 1.2(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL)
187 1.2(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-)+0.5(RLL+PLL / MLL)

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188 1.2(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL)
189 1.2(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL)
190 0.9(DL+COLL+CL-1)+1.6(WLL-1)
191 0.9(DL+COLL+CL-2)+1.6(WLL-1)
192 0.9(DL+COLL+CL-3)+1.6(WLL-1)
193 0.9(DL+COLL+CL-4)+1.6(WLL-1)
194 0.9(DL+COLL+CL-1)+1.6(WLL-2)
195 0.9(DL+COLL+CL-2)+1.6(WLL-2)
196 0.9(DL+COLL+CL-3)+1.6(WLL-2)
197 0.9(DL+COLL+CL-4)+1.6(WLL-2)
198 0.9(DL+COLL+CL-1)+1.6(WLR-1)
199 0.9(DL+COLL+CL-2)+1.6(WLR-1)
200 0.9(DL+COLL+CL-3)+1.6(WLR-1)
201 0.9(DL+COLL+CL-4)+1.6(WLR-1)
202 0.9(DL+COLL+CL-1)+1.6(WLR-2)
203 0.9(DL+COLL+CL-2)+1.6(WLR-2)
204 0.9(DL+COLL+CL-3)+1.6(WLR-2)
205 0.9(DL+COLL+CL-4)+1.6(WLR-2)
206 0.9(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)
207 0.9(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)
208 0.9(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)
209 0.9(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)
210 0.9(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)
211 0.9(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)
212 0.9(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)
213 0.9(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)
214 0.9(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)
215 0.9(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)
216 0.9(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)
217 0.9(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)
218 0.9(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG-4)
219 0.9(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG-4)
220 0.9(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG-4)
221 0.9(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG-4)
222 0.9(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')
223 0.9(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')
224 0.9(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')
225 0.9(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')

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226 0.9(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')
227 0.9(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-')
228 0.9(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')
229 0.9(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')

B) The following load combination are considered for base pressure, sliding & overturning check.

301 1.0(DL+COLL)
302 1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL)
303 1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL / MLL)
304 1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL / MLL)
305 1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL / MLL)
306 1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL / MLL)
307 1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL / MLL)
308 1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL / MLL)
309 1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL / MLL)
310 1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL+PLL / MLL)
311 1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL+PLL / MLL)
312 1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL+PLL / MLL)
313 1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL+PLL / MLL)
314 1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL+PLL / MLL)
315 1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL+PLL / MLL)
316 1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL+PLL / MLL)
317 1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL+PLL / MLL)
318 1.0(DL+COLL+CL-1)
319 1.0(DL+COLL+CL-2)
320 1.0(DL+COLL+CL-3)
321 1.0(DL+COLL+CL-4)
322 1.0(DL+COLL+CL-1)+1.0(WLL-1)
323 1.0(DL+COLL+CL-2)+1.0(WLL-1)
324 1.0(DL+COLL+CL-3)+1.0(WLL-1)
325 1.0(DL+COLL+CL-4)+1.0(WLL-1)
326 1.0(DL+COLL+CL-1)+1.0(WLL-2)
327 1.0(DL+COLL+CL-2)+1.0(WLL-2)
328 1.0(DL+COLL+CL-3)+1.0(WLL-2)
329 1.0(DL+COLL+CL-4)+1.0(WLL-2)
330 1.0(DL+COLL+CL-1)+1.0(WLR-1)

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331 1.0(DL+COLL+CL-2)+1.0(WLR-1)
332 1.0(DL+COLL+CL-3)+1.0(WLR-1)
333 1.0(DL+COLL+CL-4)+1.0(WLR-1)
334 1.0(DL+COLL+CL-1)+1.0(WLR-2)
335 1.0(DL+COLL+CL-2)+1.0(WLR-2)
336 1.0(DL+COLL+CL-3)+1.0(WLR-2)
337 1.0(DL+COLL+CL-4)+1.0(WLR-2)
338 1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-1)
339 1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-1)
340 1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-1)
341 1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-1)
342 1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-2)
343 1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-2)
344 1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-2)
345 1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-2)
346 1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-3)
347 1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-3)
348 1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-3)
349 1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-3)
350 1.0(DL+COLL+CL-1)+1.0(WLLG-4+BR-WLLG-4)
351 1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-4)
352 1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-4)
353 1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-4)
354 1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(RLL+PLL / MLL)
355 1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(RLL+PLL / MLL)
356 1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(RLL+PLL / MLL)
357 1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(RLL+PLL / MLL)
358 1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(RLL+PLL / MLL)
359 1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(RLL+PLL / MLL)
360 1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(RLL+PLL / MLL)
361 1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(RLL+PLL / MLL)
362 1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(RLL+PLL / MLL)
363 1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(RLL+PLL / MLL)
364 1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(RLL+PLL / MLL)
365 1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(RLL+PLL / MLL)
366 1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(RLL+PLL / MLL)
367 1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(RLL+PLL / MLL)
368 1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(RLL+PLL / MLL)

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369 1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(RLL+PLL / MLL)
370 1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-WLLG-1)+0.75(RLL+PLL / MLL)
371 1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-WLLG-1)+0.75(RLL+PLL / MLL)
372 1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-WLLG-1)+0.75(RLL+PLL / MLL)
373 1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-WLLG-1)+0.75(RLL+PLL / MLL)
374 1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-WLLG-2)+0.75(RLL+PLL / MLL)
375 1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-WLLG-2)+0.75(RLL+PLL / MLL)
376 1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-WLLG-2)+0.75(RLL+PLL / MLL)
377 1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-WLLG-2)+0.75(RLL+PLL / MLL)
378 1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-WLLG-3)+0.75(RLL+PLL / MLL)
379 1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-WLLG-3)+0.75(RLL+PLL / MLL)
380 1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-WLLG-3)+0.75(RLL+PLL / MLL)
381 1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-WLLG-3)+0.75(RLL+PLL / MLL)
382 1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-WLLG-4)+0.75(RLL+PLL / MLL)
383 1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-WLLG-4)+0.75(RLL+PLL / MLL)
384 1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-WLLG-4)+0.75(RLL+PLL / MLL)
385 1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-WLLG-4)+0.75(RLL+PLL / MLL)
386 1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+')
387 1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+")
388 1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+')
389 1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+')
390 1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')
391 1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')
392 1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')
393 1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')
394 1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'+')+0.75(RLL+PLL / MLL)
395 1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'+')+0.75(RLL+PLL / MLL)
396 1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'+')+0.75(RLL+PLL / MLL)
397 1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'+')+0.75(RLL+PLL / MLL)
398 1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-')+0.75(RLL+PLL / MLL)
399 1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-')+0.75(RLL+PLL / MLL)
400 1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-')+0.75(RLL+PLL / MLL)
401 1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-')+0.75(RLL+PLL / MLL)
402 0.6(DL+COLL+CL-1)+1.0(WLL-1)
403 0.6(DL+COLL+CL-2)+1.0(WLL-1)
404 0.6(DL+COLL+CL-3)+1.0(WLL-1)
405 0.6(DL+COLL+CL-4)+1.0(WLL-1)
406 0.6(DL+COLL+CL-1)+1.0(WLL-2)

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407 0.6(DL+COLL+CL-2)+1.0(WLL-2)
408 0.6(DL+COLL+CL-3)+1.0(WLL-2)
409 0.6(DL+COLL+CL-4)+1.0(WLL-2)
410 0.6(DL+COLL+CL-1)+1.0(WLR-1)
411 0.6(DL+COLL+CL-2)+1.0(WLR-1)
412 0.6(DL+COLL+CL-3)+1.0(WLR-1)
413 0.6(DL+COLL+CL-4)+1.0(WLR-1)
414 0.6(DL+COLL+CL-1)+1.0(WLR-2)
415 0.6(DL+COLL+CL-2)+1.0(WLR-2)
416 0.6(DL+COLL+CL-3)+1.0(WLR-2)
417 0.6(DL+COLL+CL-4)+1.0(WLR-2)
418 0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-1)
419 0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-1)
420 0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-1)
421 0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-1)
422 0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-2)
423 0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-2)
424 0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-2)
425 0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-2)
426 0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-3)
427 0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-3)
428 0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-3)
429 0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-3)
430 0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-WLLG-4)
431 0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-4)
432 0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-4)
433 0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-4)
434 0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')
435 0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')
436 0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')
437 0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')
438 0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')
439 0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')
440 0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')
441 0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')

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4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm except where welded wire fabric is used for ties and spirals the minimum rebar size shall be 10 mm.

4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 400 kN/m2 at 1.2m depth below EGL (RL 913.100) refer Table 5 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

The Vertical Modulus of Subgrade Reaction has been considered as 40500 kN/m^2/m at 1.2m depth below EGL (RL 913.100) refer Table 5 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5.1 DESIGN CALCULATION OF FOOTING F1

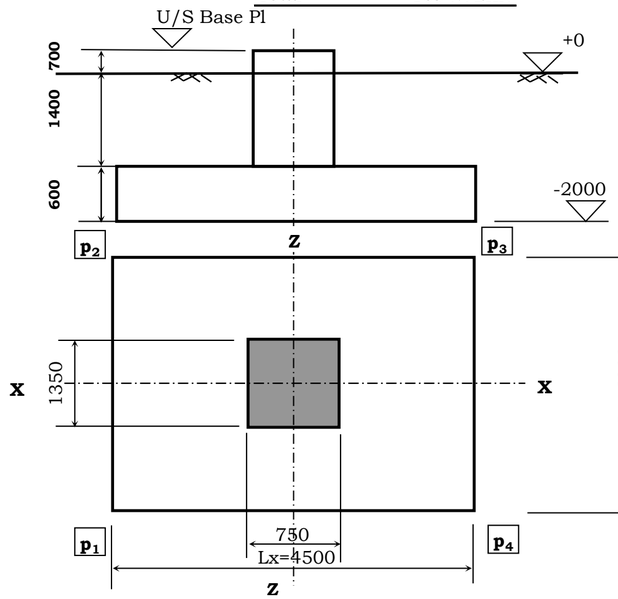
	Along-X	Along-Z
Size of column :	0.75m	1.35m
Size of fdn. raft :	4.50m	6.00m
Thickness of raft :	0.60 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-2.00 m	
U/S level Base Plate :	0.70 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

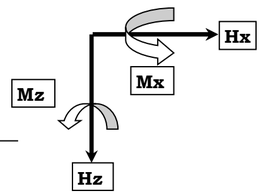
SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Crane Load	CL
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (4.5*6*0.6*24)+(0.75*1.35*2.1*24) = 439.83KN
Weight of soil = ((4.5*6)-(0.75*1.35))*1.4*18 = 654.89KN
Total = 1094.72KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 542mm = 0.542m
Effective depth of slab along Z direction = 526mm = 0.526m
Clear projection along Z direction = 2.325m
Clear projection along X direction = 1.875m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

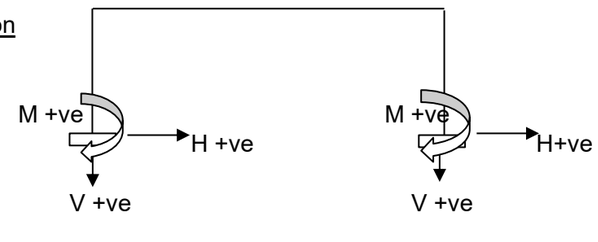
M$_X$=Moment about X-X direction

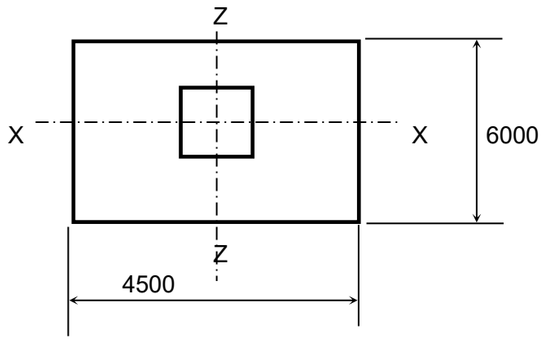
Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F1

Sign convention





Height from BOF to top of column about X (l_{ux}) = 2700mm
Height from BOF to top of column about Z (l_{uz}) = 2700mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	30.0672		21.8664		85.5336
2	COLLATERAL LOAD (COLL)	-0.0048		-0.0048		0.0048
3	ROOF LIVE LOAD (RLL)	68.052		55.8516		201.3468
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	3.5328		10.9272		64.5576
5	CRANE LOAD-1 (CL-1)	0		0		0
6	CRANE LOAD-2 (CL-2)	0		0		0
7	CRANE LOAD-3 (CL-3)	0		0		0
8	CRANE LOAD-4 (CL-4)	0		0		0
9	WIND LEFT-1 (WLL-1)	-139.517		-127.1076		-420.9948
10	WIND LEFT-2 (WLL-2)	-77.3916		-106.3572		-291.7368
11	WIND RIGHT-1 (WLR-1)	-105.104		-52.0608		-263.6508
12	WIND RIGHT-2 (WLR-2)	-43.1652		-32.3664		-141.8628
13	WIND LONG-1 (WLLG-1)	-179.395	-78.9444		-363.3672	
14	WIND LONG-2 (WLLG-2)	-117.473	-59.1216		-240.3684	
15	WIND LONG-3 (WLLG-3)	-150.955	-71.3664		-351.4164	
16	WIND LONG-4 (WLLG-4)	-89.0328	-51.5436		-228.4164	
17	SEISMIC LOAD SL'+'	-0.1668	-1.1688	-1.1688	-7.488	-7.488
18	SEISMIC LOAD SL'-'	0.1668	1.1688	1.1688	7.488	7.488
19	TEMPERATURE LOAD (TL1)	1.662		9.8352		69.972
20	TEMPERATURE LOAD (TL2)	-1.662		-9.8352		-69.972
21	BRACING REACTION (BR-WLLG1)	51.7	-41		0	
22	BRACING REACTION (BR-WLLG2)	51.7	-41		0	
23	BRACING REACTION (BR-WLLG3)	-51.7	41		0	
24	BRACING REACTION (BR-WLLG4)	-51.7	41		0	
25	BRACING REACTION (BR-SL'+')	-11	-10	-10	0	0
26	BRACING REACTION (BR-SL'-')	11	10	10	0	0

Ultimate Condition

Combined Load Cases:		Load at top of foundation			Load at bottom of foundation		
		Factored Load/Moment					
Load Case	Load Combination	Pu (kN)	Muz (kN-m)	Mux (kN-m)	Pu (kN)	Muz (kN-m)	Mux (kN-m)
101	1.4(DL+COLL)	42.09	0.0	202.4	1136.8	0.0	202.4
102	1.2(DL+COLL+CL-1)+1.6(RLL+PLL / MLL)	150.6	0.0	887.4	1245.3	0.0	887.4
103	1.2(DL+COLL+CL-2)+1.6(RLL+PLL / MLL)	150.6	0.0	887.4	1245.3	0.0	887.4
104	1.2(DL+COLL+CL-3)+1.6(RLL+PLL / MLL)	150.6	0.0	887.4	1245.3	0.0	887.4
105	1.2(DL+COLL+CL-4)+1.6(RLL+PLL / MLL)	150.6	0.0	887.4	1245.3	0.0	887.4
106	1.4(DL+COLL+CL-1)	42.1	0.0	202.4	1136.8	0.0	202.4
107	1.4(DL+COLL+CL-2)	42.1	0.0	202.4	1136.8	0.0	202.4
108	1.4(DL+COLL+CL-3)	42.1	0.0	202.4	1136.8	0.0	202.4
109	1.4(DL+COLL+CL-4)	42.1	0.0	202.4	1136.8	0.0	202.4
110	1.2(DL+COLL+CL-1+TL1)+1.6(RLL+PLL / MLL)	152.6	0.0	1003.2	1247.3	0.0	1003.2
111	1.2(DL+COLL+CL-2+TL1)+1.6(RLL+PLL / MLL)	152.6	0.0	1003.2	1247.3	0.0	1003.2
112	1.2(DL+COLL+CL-3+TL1)+1.6(RLL+PLL / MLL)	152.6	0.0	1003.2	1247.3	0.0	1003.2
113	1.2(DL+COLL+CL-4+TL1)+1.6(RLL+PLL / MLL)	152.6	0.0	1003.2	1247.3	0.0	1003.2
114	1.2(DL+COLL+CL-1+TL2)+1.6(RLL+PLL / MLL)	148.6	0.0	771.6	1243.3	0.0	771.6
115	1.2(DL+COLL+CL-2+TL2)+1.6(RLL+PLL / MLL)	148.6	0.0	771.6	1243.3	0.0	771.6
116	1.2(DL+COLL+CL-3+TL2)+1.6(RLL+PLL / MLL)	148.6	0.0	771.6	1243.3	0.0	771.6
117	1.2(DL+COLL+CL-4+TL2)+1.6(RLL+PLL / MLL)	148.6	0.0	771.6	1243.3	0.0	771.6
118	1.2(DL+COLL+CL-1)+0.8(WLL-1)	-75.5	0.0	437.9	1019.2	0.0	437.9
119	1.2(DL+COLL+CL-2)+0.8(WLL-1)	-75.5	0.0	437.9	1019.2	0.0	437.9
120	1.2(DL+COLL+CL-3)+0.8(WLL-1)	-75.5	0.0	437.9	1019.2	0.0	437.9
121	1.2(DL+COLL+CL-4)+0.8(WLL-1)	-75.5	0.0	437.9	1019.2	0.0	437.9
122	1.2(DL+COLL+CL-1)+0.8(WLL-2)	-25.8	0.0	289.6	1068.9	0.0	289.6
123	1.2(DL+COLL+CL-2)+0.8(WLL-2)	-25.8	0.0	289.6	1068.9	0.0	289.6
124	1.2(DL+COLL+CL-3)+0.8(WLL-2)	-25.8	0.0	289.6	1068.9	0.0	289.6
125	1.2(DL+COLL+CL-4)+0.8(WLL-2)	-25.8	0.0	289.6	1068.9	0.0	289.6
126	1.2(DL+COLL+CL-1)+0.8(WLR-1)	-48.0	0.0	149.9	1046.7	0.0	149.9
127	1.2(DL+COLL+CL-2)+0.8(WLR-1)	-48.0	0.0	149.9	1046.7	0.0	149.9
128	1.2(DL+COLL+CL-3)+0.8(WLR-1)	-48.0	0.0	149.9	1046.7	0.0	149.9
129	1.2(DL+COLL+CL-4)+0.8(WLR-1)	-48.0	0.0	149.9	1046.7	0.0	149.9
130	1.2(DL+COLL+CL-1)+0.8(WLR-2)	1.5	0.0	9.9	1096.3	0.0	9.9
131	1.2(DL+COLL+CL-2)+0.8(WLR-2)	1.5	0.0	9.9	1096.3	0.0	9.9
132	1.2(DL+COLL+CL-3)+0.8(WLR-2)	1.5	0.0	9.9	1096.3	0.0	9.9
133	1.2(DL+COLL+CL-4)+0.8(WLR-2)	1.5	0.0	9.9	1096.3	0.0	9.9
134	1.2(DL+COLL+CL-1)+0.8(WLLG-1+BR-WLLG-1)	-66.1	549.8	173.5	1028.6	549.8	173.5
135	1.2(DL+COLL+CL-2)+0.8(WLLG-1+BR-WLLG-1)	-66.1	549.8	173.5	1028.6	549.8	173.5
136	1.2(DL+COLL+CL-3)+0.8(WLLG-1+BR-WLLG-1)	-66.1	549.8	173.5	1028.6	549.8	173.5
137	1.2(DL+COLL+CL-4)+0.8(WLLG-1+BR-WLLG-1)	-66.1	549.8	173.5	1028.6	549.8	173.5
138	1.2(DL+COLL+CL-1)+0.8(WLLG-2+BR-WLLG-2)	-16.5	408.6	173.5	1078.2	408.6	173.5
139	1.2(DL+COLL+CL-2)+0.8(WLLG-2+BR-WLLG-2)	-16.5	408.6	173.5	1078.2	408.6	173.5
140	1.2(DL+COLL+CL-3)+0.8(WLLG-2+BR-WLLG-2)	-16.5	408.6	173.5	1078.2	408.6	173.5
141	1.2(DL+COLL+CL-4)+0.8(WLLG-2+BR-WLLG-2)	-16.5	408.6	173.5	1078.2	408.6	173.5
142	1.2(DL+COLL+CL-1)+0.8(WLLG-3+BR-WLLG-3)	-126.0	346.7	173.5	968.7	346.7	173.5
143	1.2(DL+COLL+CL-2)+0.8(WLLG-3+BR-WLLG-3)	-126.0	346.7	173.5	968.7	346.7	173.5
144	1.2(DL+COLL+CL-3)+0.8(WLLG-3+BR-WLLG-3)	-126.0	346.7	173.5	968.7	346.7	173.5
145	1.2(DL+COLL+CL-4)+0.8(WLLG-3+BR-WLLG-3)	-126.0	346.7	173.5	968.7	346.7	173.5
146	1.2(DL+COLL+CL-1)+0.8(WLLG-4+BR-WLLG-4)	-76.5	205.5	173.5	1018.2	205.5	173.5

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147	1.2(DL+COLL+CL-2)+0.8(WLLG-4+BR-WLLG-4)	-76.5	205.5	173.5	1018.2	205.5	173.5
148	1.2(DL+COLL+CL-3)+0.8(WLLG-4+BR-WLLG-4)	-76.5	205.5	173.5	1018.2	205.5	173.5
149	1.2(DL+COLL+CL-4)+0.8(WLLG-4+BR-WLLG-4)	-76.5	205.5	173.5	1018.2	205.5	173.5
150	1.2(DL+COLL+CL-1)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-151.4	0.0	826.1	943.4	0.0	826.1
151	1.2(DL+COLL+CL-2)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-151.4	0.0	826.1	943.4	0.0	826.1
152	1.2(DL+COLL+CL-3)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-151.4	0.0	826.1	943.4	0.0	826.1
153	1.2(DL+COLL+CL-4)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-151.4	0.0	826.1	943.4	0.0	826.1
154	1.2(DL+COLL+CL-1)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	-52.0	0.0	529.7	1042.8	0.0	529.7
155	1.2(DL+COLL+CL-2)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	-52.0	0.0	529.7	1042.8	0.0	529.7
156	1.2(DL+COLL+CL-3)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	-52.0	0.0	529.7	1042.8	0.0	529.7
157	1.2(DL+COLL+CL-4)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	-52.0	0.0	529.7	1042.8	0.0	529.7
158	1.2(DL+COLL+CL-1)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-96.3	0.0	250.2	998.4	0.0	250.2
159	1.2(DL+COLL+CL-2)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-96.3	0.0	250.2	998.4	0.0	250.2
160	1.2(DL+COLL+CL-3)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-96.3	0.0	250.2	998.4	0.0	250.2
161	1.2(DL+COLL+CL-4)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-96.3	0.0	250.2	998.4	0.0	250.2
162	1.2(DL+COLL+CL-1)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	2.8	0.0	29.8	1097.5	0.0	29.8
163	1.2(DL+COLL+CL-2)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	2.8	0.0	29.8	1097.5	0.0	29.8
164	1.2(DL+COLL+CL-3)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	2.8	0.0	29.8	1097.5	0.0	29.8
165	1.2(DL+COLL+CL-4)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	2.8	0.0	29.8	1097.5	0.0	29.8
166	1.2(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-132.4	1099.5	396.6	962.3	1099.5	396.6
167	1.2(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-132.4	1099.5	396.6	962.3	1099.5	396.6
168	1.2(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-132.4	1099.5	396.6	962.3	1099.5	396.6
169	1.2(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-132.4	1099.5	396.6	962.3	1099.5	396.6
170	1.2(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-33.4	817.1	396.6	1061.4	817.1	396.6
171	1.2(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-33.4	817.1	396.6	1061.4	817.1	396.6
172	1.2(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-33.4	817.1	396.6	1061.4	817.1	396.6
173	1.2(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-33.4	817.1	396.6	1061.4	817.1	396.6
174	1.2(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-252.4	693.4	396.6	842.3	693.4	396.6
175	1.2(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-252.4	693.4	396.6	842.3	693.4	396.6
176	1.2(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-252.4	693.4	396.6	842.3	693.4	396.6
177	1.2(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-252.4	693.4	396.6	842.3	693.4	396.6
178	1.2(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-153.3	411.0	396.6	941.4	411.0	396.6
179	1.2(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-153.3	411.0	396.6	941.4	411.0	396.6
180	1.2(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-153.3	411.0	396.6	941.4	411.0	396.6
181	1.2(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-153.3	411.0	396.6	941.4	411.0	396.6
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	60.7	37.6	358.9	1155.4	37.6	358.9
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	60.7	37.6	358.9	1155.4	37.6	358.9
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	60.7	37.6	358.9	1155.4	37.6	358.9
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	60.7	37.6	358.9	1155.4	37.6	358.9
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	83.0	37.6	434.2	1177.8	37.6	434.2
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-)+0.5(RLL+PLL / MLL	83.0	37.6	434.2	1177.8	37.6	434.2
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	83.0	37.6	434.2	1177.8	37.6	434.2
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	83.0	37.6	434.2	1177.8	37.6	434.2
190	0.9(DL+COLL+CL-1)+1.6(WLL-1)	-196.2	0.0	1092.6	898.5	0.0	1092.6
191	0.9(DL+COLL+CL-2)+1.6(WLL-1)	-196.2	0.0	1092.6	898.5	0.0	1092.6
192	0.9(DL+COLL+CL-3)+1.6(WLL-1)	-196.2	0.0	1092.6	898.5	0.0	1092.6
193	0.9(DL+COLL+CL-4)+1.6(WLL-1)	-196.2	0.0	1092.6	898.5	0.0	1092.6
194	0.9(DL+COLL+CL-1)+1.6(WLL-2)	-96.8	0.0	796.1	997.9	0.0	796.1
195	0.9(DL+COLL+CL-2)+1.6(WLL-2)	-96.8	0.0	796.1	997.9	0.0	796.1
196	0.9(DL+COLL+CL-3)+1.6(WLL-2)	-96.8	0.0	796.1	997.9	0.0	796.1
197	0.9(DL+COLL+CL-4)+1.6(WLL-2)	-96.8	0.0	796.1	997.9	0.0	796.1
198	0.9(DL+COLL+CL-1)+1.6(WLR-1)	-141.1	0.0	516.6	953.6	0.0	516.6
199	0.9(DL+COLL+CL-2)+1.6(WLR-1)	-141.1	0.0	516.6	953.6	0.0	516.6
200	0.9(DL+COLL+CL-3)+1.6(WLR-1)	-141.1	0.0	516.6	953.6	0.0	516.6
201	0.9(DL+COLL+CL-4)+1.6(WLR-1)	-141.1	0.0	516.6	953.6	0.0	516.6
202	0.9(DL+COLL+CL-1)+1.6(WLR-2)	-42.0	0.0	236.7	1052.7	0.0	236.7
203	0.9(DL+COLL+CL-2)+1.6(WLR-2)	-42.0	0.0	236.7	1052.7	0.0	236.7
204	0.9(DL+COLL+CL-3)+1.6(WLR-2)	-42.0	0.0	236.7	1052.7	0.0	236.7
205	0.9(DL+COLL+CL-4)+1.6(WLR-2)	-42.0	0.0	236.7	1052.7	0.0	236.7
206	0.9(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)	-177.3	1099.5	130.1	917.5	1099.5	130.1
207	0.9(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)	-177.3	1099.5	130.1	917.5	1099.5	130.1
208	0.9(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)	-177.3	1099.5	130.1	917.5	1099.5	130.1
209	0.9(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)	-177.3	1099.5	130.1	917.5	1099.5	130.1
210	0.9(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)	-78.2	817.1	130.1	1016.5	817.1	130.1
211	0.9(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)	-78.2	817.1	130.1	1016.5	817.1	130.1
212	0.9(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)	-78.2	817.1	130.1	1016.5	817.1	130.1
213	0.9(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)	-78.2	817.1	130.1	1016.5	817.1	130.1
214	0.9(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)	-297.2	693.4	130.1	797.5	693.4	130.1
215	0.9(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)	-297.2	693.4	130.1	797.5	693.4	130.1
216	0.9(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)	-297.2	693.4	130.1	797.5	693.4	130.1
217	0.9(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)	-297.2	693.4	130.1	797.5	693.4	130.1
218	0.9(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG-4)	-198.1	411.0	130.1	896.6	411.0	130.1
219	0.9(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG-4)	-198.1	411.0	130.1	896.6	411.0	130.1
220	0.9(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG-4)	-198.1	411.0	130.1	896.6	411.0	130.1
221	0.9(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG-4)	-198.1	411.0	130.1	896.6	411.0	130.1
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')	15.9	37.6	92.5	1110.6	37.6	92.5
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')	15.9	37.6	92.5	1110.6	37.6	92.5
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')	15.9	37.6	92.5	1110.6	37.6	92.5
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')	15.9	37.6	92.5	1110.6	37.6	92.5
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')	38.2	37.6	167.8	1132.9	37.6	167.8
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-')	38.2	37.6	167.8	1132.9	37.6	167.8
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')	38.2	37.6	167.8	1132.9	37.6	167.8
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')	38.2	37.6	167.8	1132.9	37.6	167.8

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Allowable stress design											
Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
301	1.0(DL+COLL)	30.0624	0	0.0	21.9	144.6	1124.8	0.0	0.0	21.9	144.6
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL	103.3092	0	0.0	98.5	687.3	1198.0	0.0	0.0	98.5	687.3
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL / MLL	103.3092	0	0.0	98.5	687.3	1198.0	0.0	0.0	98.5	687.3
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL / MLL	103.3092	0	0.0	98.5	687.3	1198.0	0.0	0.0	98.5	687.3
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL / MLL	103.3092	0	0.0	98.5	687.3	1198.0	0.0	0.0	98.5	687.3
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL / MLL	99.9852	0	0.0	78.8	494.2	1194.7	0.0	0.0	78.8	494.2
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL / MLL	99.9852	0	0.0	78.8	494.2	1194.7	0.0	0.0	78.8	494.2
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL / MLL	99.9852	0	0.0	78.8	494.2	1194.7	0.0	0.0	78.8	494.2
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL / MLL	99.9852	0	0.0	78.8	494.2	1194.7	0.0	0.0	78.8	494.2
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL+PLL	85.00	0	0.0	79.3	551.6	1179.7	0.0	0.0	79.3	551.6
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL+PLL	85.00	0	0.0	79.3	551.6	1179.7	0.0	0.0	79.3	551.6
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL+PLL	85.00	0	0.0	79.3	551.6	1179.7	0.0	0.0	79.3	551.6
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL+PLL	85.00	0	0.0	79.3	551.6	1179.7	0.0	0.0	79.3	551.6
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL+PLL	82.50	0	0.0	64.6	406.8	1177.2	0.0	0.0	64.6	406.8
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL+PLL	82.50	0	0.0	64.6	406.8	1177.2	0.0	0.0	64.6	406.8
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL+PLL	82.50	0	0.0	64.6	406.8	1177.2	0.0	0.0	64.6	406.8
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL+PLL	82.50	0	0.0	64.6	406.8	1177.2	0.0	0.0	64.6	406.8
318	1.0(DL+COLL+CL-1)	30.0624	0	0.0	21.9	144.6	1124.8	0.0	0.0	21.9	144.6
319	1.0(DL+COLL+CL-2)	30.0624	0	0.0	21.9	144.6	1124.8	0.0	0.0	21.9	144.6
320	1.0(DL+COLL+CL-3)	30.0624	0	0.0	21.9	144.6	1124.8	0.0	0.0	21.9	144.6
321	1.0(DL+COLL+CL-4)	30.0624	0	0.0	21.9	144.6	1124.8	0.0	0.0	21.9	144.6
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	-109.4544	0	0.0	-105.2	-619.6	985.3	0.0	0.0	-105.2	-619.6
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	-109.4544	0	0.0	-105.2	-619.6	985.3	0.0	0.0	-105.2	-619.6
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	-109.4544	0	0.0	-105.2	-619.6	985.3	0.0	0.0	-105.2	-619.6
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	-109.4544	0	0.0	-105.2	-619.6	985.3	0.0	0.0	-105.2	-619.6
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	-47.3292	0	0.0	-84.5	-434.3	1047.4	0.0	0.0	-84.5	-434.3
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	-47.3292	0	0.0	-84.5	-434.3	1047.4	0.0	0.0	-84.5	-434.3
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	-47.3292	0	0.0	-84.5	-434.3	1047.4	0.0	0.0	-84.5	-434.3
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	-47.3292	0	0.0	-84.5	-434.3	1047.4	0.0	0.0	-84.5	-434.3
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	-75.042	0	0.0	-30.2	-259.7	1019.7	0.0	0.0	-30.2	-259.7
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	-75.042	0	0.0	-30.2	-259.7	1019.7	0.0	0.0	-30.2	-259.7
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	-75.042	0	0.0	-30.2	-259.7	1019.7	0.0	0.0	-30.2	-259.7
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	-75.042	0	0.0	-30.2	-259.7	1019.7	0.0	0.0	-30.2	-259.7
334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	-13.1028	0	0.0	-10.5	-84.7	1081.6	0.0	0.0	-10.5	-84.7
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	-13.1028	0	0.0	-10.5	-84.7	1081.6	0.0	0.0	-10.5	-84.7
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	-13.1028	0	0.0	-10.5	-84.7	1081.6	0.0	0.0	-10.5	-84.7
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	-13.1028	0	0.0	-10.5	-84.7	1081.6	0.0	0.0	-10.5	-84.7
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	-97.6328	-119.9444	-1101.0	21.9	144.6	997.1	-119.9	-1101.0	21.9	144.6
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	-97.6328	-119.9444	-1101.0	21.9	144.6	997.1	-119.9	-1101.0	21.9	144.6
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	-97.6328	-119.9444	-1101.0	21.9	144.6	997.1	-119.9	-1101.0	21.9	144.6
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	-97.6328	-119.9444	-1101.0	21.9	144.6	997.1	-119.9	-1101.0	21.9	144.6
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	-35.7104	-100.1216	-749.1	21.9	144.6	1059.0	-100.1	-749.1	21.9	144.6
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	-35.7104	-100.1216	-749.1	21.9	144.6	1059.0	-100.1	-749.1	21.9	144.6
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	-35.7104	-100.1216	-749.1	21.9	144.6	1059.0	-100.1	-749.1	21.9	144.6
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	-35.7104	-100.1216	-749.1	21.9	144.6	1059.0	-100.1	-749.1	21.9	144.6
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-172.5928	-30.3664	-979.2	21.9	144.6	922.1	-30.4	-979.2	21.9	144.6
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	-172.5928	-30.3664	-979.2	21.9	144.6	922.1	-30.4	-979.2	21.9	144.6
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	-172.5928	-30.3664	-979.2	21.9	144.6	922.1	-30.4	-979.2	21.9	144.6
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	-172.5928	-30.3664	-979.2	21.9	144.6	922.1	-30.4	-979.2	21.9	144.6
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-110.6704	-10.5436	-627.3	21.9	144.6	984.0	-10.5	-627.3	21.9	144.6
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	-110.6704	-10.5436	-627.3	21.9	144.6	984.0	-10.5	-627.3	21.9	144.6
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	-110.6704	-10.5436	-627.3	21.9	144.6	984.0	-10.5	-627.3	21.9	144.6
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	-110.6704	-10.5436	-627.3	21.9	144.6	984.0	-10.5	-627.3	21.9	144.6
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(RLL+	-20.89	0	0.0	-23.4	-93.9	1073.8	0.0	0.0	-23.4	-93.9
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(RLL+	-20.89	0	0.0	-23.4	-93.9	1073.8	0.0	0.0	-23.4	-93.9
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(RLL+	-20.89	0	0.0	-23.4	-93.9	1073.8	0.0	0.0	-23.4	-93.9
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(RLL+	-20.89	0	0.0	-23.4	-93.9	1073.8	0.0	0.0	-23.4	-93.9
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(RLL+	25.71	0	0.0	-7.8	45.0	1120.4	0.0	0.0	-7.8	45.0
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(RLL+	25.71	0	0.0	-7.8	45.0	1120.4	0.0	0.0	-7.8	45.0
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(RLL+	25.71	0	0.0	-7.8	45.0	1120.4	0.0	0.0	-7.8	45.0
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(RLL+	25.71	0	0.0	-7.8	45.0	1120.4	0.0	0.0	-7.8	45.0
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(RLL+	4.92	0	0.0	32.9	176.1	1099.6	0.0	0.0	32.9	176.1
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(RLL+	4.92	0	0.0	32.9	176.1	1099.6	0.0	0.0	32.9	176.1
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(RLL+	4.92	0	0.0	32.9	176.1	1099.6	0.0	0.0	32.9	176.1
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(RLL+	4.92	0	0.0	32.9	176.1	1099.6	0.0	0.0	32.9	176.1
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(RLL+	51.38	0	0.0	47.7	307.3	1146.1	0.0	0.0	47.7	307.3
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(RLL+	51.38	0	0.0	47.7	307.3	1146.1	0.0	0.0	47.7	307.3
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(RLL+	51.38	0	0.0	47.7	307.3	1146.1	0.0	0.0	47.7	307.3
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(RLL+	51.38	0	0.0	47.7	307.3	1146.1	0.0	0.0	47.7	307.3
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-WLLG	-12.02	-89.9583	-825.8	71.9	479.2	1082.7	-90.0	-825.8	71.9	479.2
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-WLLG	-12.02	-89.9583	-825.8	71.9	479.2	1082.7	-90.0	-825.8	71.9	479.2
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-WLLG	-12.02	-89.9583	-825.8	71.9	479.2	1082.7	-90.0	-825.8	71.9	479.2
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-WLLG	-12.02	-89.9583	-825.8	71.9	479.2	1082.7	-90.0	-825.8	71.9	479.2
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-WLLG	34.42	-75.0912	-561.8	71.9	479.2	1129.1	-75.1	-561.8	71.9	479.2
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-WLLG	34.42	-75.0912	-561.8	71.9	479.2	1129.1	-75.1	-561.8	71.9	479.2
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-WLLG	34.42	-75.0912	-561.8	71.9	479.2	1129.1	-75.1	-561.8	71.9	479.2
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-WLLG	34.42	-75.0912	-561.8	71.9	479.2	1129.1	-75.1	-561.8	71.9	479.2
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-WLLG	-68.24	-22.7748	-734.4	71.9	479.2	1026.5	-22.8	-734.4	71.9	479.2
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-WLLG	-68.24	-22.7748	-734.4	71.9	479.2	1026.5	-22.8	-734.4	71.9	479.2
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-WLLG	-68.24	-22.7748	-734.4	71.9	479.2	1026.5	-22.8	-734.4	71.9	479.2
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-WLLG	-68.24	-22.7748	-734.4	71.9	479.2	1026.5	-22.8	-734.4	71.9	479.2
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-WLLG	-21.80	-7.9077	-470.5	71.9	479.2	1072.9	-7.9	-470.5	71.9	479.2
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-WLLG	-21.80	-7.9077	-470.5	71.9	479.2	1072.9	-7.9	-470.5	71.9	479.2
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-WLLG	-21.80	-7.9077	-470.5	71.9	479.2	1072.9	-7.9	-470.5	71.9	479.2
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-WLLG	-21.80	-7.9077	-470.5	71.9	479.2	1072.9	-7.9	-470.5	71.9	479.2
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+')	21.69	-8.3766	-23.5	13.5	121.0	1116.4	-8.4	-23.5	13.5	121.0
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+")	21.69	-8.3766	-23.5	13.5	121.0	1116.4	-8.4	-23.5	13.5	121.0
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+')	21.69	-8.3766	-23.5	13.5	121.0	1116.4	-8.4	-23.5	13.5	121.0
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+')	21.69	-8.3766	-23.5	13.5	121.0	1116.4	-8.4	-23.5	13.5	121.0

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390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	38.44	8.3766	23.5	30.2	168.1	1133.2	8.4	23.5	30.2	168.1
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	38.44	8.3766	23.5	30.2	168.1	1133.2	8.4	23.5	30.2	168.1
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	38.44	8.3766	23.5	30.2	168.1	1133.2	8.4	23.5	30.2	168.1
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	38.44	8.3766	23.5	30.2	168.1	1133.2	8.4	23.5	30.2	168.1
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'+')+0	77.89	-5.86362	-16.5	66.1	462.7	1172.6	-5.9	-16.5	66.1	462.7
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'+')+0	77.89	-5.86362	-16.5	66.1	462.7	1172.6	-5.9	-16.5	66.1	462.7
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'+')+0	77.89	-5.86362	-16.5	66.1	462.7	1172.6	-5.9	-16.5	66.1	462.7
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'+')+0	77.89	-5.86362	-16.5	66.1	462.7	1172.6	-5.9	-16.5	66.1	462.7
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-')+0.7	89.61	5.86362	16.5	77.8	495.7	1184.3	5.9	16.5	77.8	495.7
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-')+0.7	89.61	5.86362	16.5	77.8	495.7	1184.3	5.9	16.5	77.8	495.7
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-')+0.7	89.61	5.86362	16.5	77.8	495.7	1184.3	5.9	16.5	77.8	495.7
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-')+0.7	89.61	5.86362	16.5	77.8	495.7	1184.3	5.9	16.5	77.8	495.7
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	-121.48	0	0.0	-114.0	-677.4	973.2	0.0	0.0	-114.0	-677.4
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	-121.48	0	0.0	-114.0	-677.4	973.2	0.0	0.0	-114.0	-677.4
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	-121.48	0	0.0	-114.0	-677.4	973.2	0.0	0.0	-114.0	-677.4
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	-121.48	0	0.0	-114.0	-677.4	973.2	0.0	0.0	-114.0	-677.4
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	-59.35416	0	0.0	-93.2	-492.2	1035.4	0.0	0.0	-93.2	-492.2
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	-59.35416	0	0.0	-93.2	-492.2	1035.4	0.0	0.0	-93.2	-492.2
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	-59.35416	0	0.0	-93.2	-492.2	1035.4	0.0	0.0	-93.2	-492.2
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	-59.35416	0	0.0	-93.2	-492.2	1035.4	0.0	0.0	-93.2	-492.2
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	-87.06696	0	0.0	-38.9	-317.5	1007.7	0.0	0.0	-38.9	-317.5
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	-87.06696	0	0.0	-38.9	-317.5	1007.7	0.0	0.0	-38.9	-317.5
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	-87.06696	0	0.0	-38.9	-317.5	1007.7	0.0	0.0	-38.9	-317.5
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	-87.06696	0	0.0	-38.9	-317.5	1007.7	0.0	0.0	-38.9	-317.5
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	-25.12776	0	0.0	-19.2	-142.5	1069.6	0.0	0.0	-19.2	-142.5
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	-25.12776	0	0.0	-19.2	-142.5	1069.6	0.0	0.0	-19.2	-142.5
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	-25.12776	0	0.0	-19.2	-142.5	1069.6	0.0	0.0	-19.2	-142.5
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	-25.12776	0	0.0	-19.2	-142.5	1069.6	0.0	0.0	-19.2	-142.5
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	-109.6578	-119.9444	-1101.0	13.1	86.7	985.1	-119.9	-1101.0	13.1	86.7
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	-109.6578	-119.9444	-1101.0	13.1	86.7	985.1	-119.9	-1101.0	13.1	86.7
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	-109.6578	-119.9444	-1101.0	13.1	86.7	985.1	-119.9	-1101.0	13.1	86.7
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	-109.6578	-119.9444	-1101.0	13.1	86.7	985.1	-119.9	-1101.0	13.1	86.7
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	-47.73536	-100.1216	-749.1	13.1	86.7	1047.0	-100.1	-749.1	13.1	86.7
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	-47.73536	-100.1216	-749.1	13.1	86.7	1047.0	-100.1	-749.1	13.1	86.7
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	-47.73536	-100.1216	-749.1	13.1	86.7	1047.0	-100.1	-749.1	13.1	86.7
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	-47.73536	-100.1216	-749.1	13.1	86.7	1047.0	-100.1	-749.1	13.1	86.7
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-184.6178	-30.3664	-979.2	13.1	86.7	910.1	-30.4	-979.2	13.1	86.7
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	-184.6178	-30.3664	-979.2	13.1	86.7	910.1	-30.4	-979.2	13.1	86.7
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	-184.6178	-30.3664	-979.2	13.1	86.7	910.1	-30.4	-979.2	13.1	86.7
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	-184.6178	-30.3664	-979.2	13.1	86.7	910.1	-30.4	-979.2	13.1	86.7
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-WLLG-	-122.6954	-10.5436	-627.3	13.1	86.7	972.0	-10.5	-627.3	13.1	86.7
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	-122.6954	-10.5436	-627.3	13.1	86.7	972.0	-10.5	-627.3	13.1	86.7
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	-122.6954	-10.5436	-627.3	13.1	86.7	972.0	-10.5	-627.3	13.1	86.7
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	-122.6954	-10.5436	-627.3	13.1	86.7	972.0	-10.5	-627.3	13.1	86.7
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	10.22	-7.81816	-22.0	5.3	64.8	1104.9	-7.8	-22.0	5.3	64.8
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	10.22	-7.81816	-22.0	5.3	64.8	1104.9	-7.8	-22.0	5.3	64.8
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	10.22	-7.81816	-22.0	5.3	64.8	1104.9	-7.8	-22.0	5.3	64.8
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	10.22	-7.81816	-22.0	5.3	64.8	1104.9	-7.8	-22.0	5.3	64.8
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	-73.49	-7.81816	-22.0	-71.0	-393.7	1021.2	-7.8	-22.0	-71.0	-393.7
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	-36.21	-7.81816	-22.0	-58.5	-282.6	1058.5	-7.8	-22.0	-58.5	-282.6
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	-52.84	-7.81816	-22.0	-25.9	-177.8	1041.9	-7.8	-22.0	-25.9	-177.8
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	-15.68	-7.81816	-22.0	-14.1	-72.8	1079.0	-7.8	-22.0	-14.1	-72.8

Check for overturning and sliding:

Load Case	Load Combination	Load at bottom of foundation					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
301	1.0(DL+COLL)	1124.78	0.00	0.00	21.86	144.56	NA	21.01	NA	18.52
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL /	1198.03	0.00	0.00	98.48	687.30	NA	4.71	NA	4.38
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL /	1198.03	0.00	0.00	98.48	687.30	NA	4.71	NA	4.38
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL /	1198.03	0.00	0.00	98.48	687.30	NA	4.71	NA	4.38
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL /	1198.03	0.00	0.00	98.48	687.30	NA	4.71	NA	4.38
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL /	1194.71	0.00	0.00	78.81	494.24	NA	6.53	NA	5.46
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL /	1194.71	0.00	0.00	78.81	494.24	NA	6.53	NA	5.46
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL /	1194.71	0.00	0.00	78.81	494.24	NA	6.53	NA	5.46
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL /	1194.71	0.00	0.00	78.81	494.24	NA	6.53	NA	5.46
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL	1179.72	0.00	0.00	79.32	551.62	NA	5.77	NA	5.35
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL	1179.72	0.00	0.00	79.32	551.62	NA	5.77	NA	5.35
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL	1179.72	0.00	0.00	79.32	551.62	NA	5.77	NA	5.35
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL	1179.72	0.00	0.00	79.32	551.62	NA	5.77	NA	5.35
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL	1177.22	0.00	0.00	64.57	406.82	NA	7.81	NA	6.56
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL	1177.22	0.00	0.00	64.57	406.82	NA	7.81	NA	6.56
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL	1177.22	0.00	0.00	64.57	406.82	NA	7.81	NA	6.56
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL	1177.22	0.00	0.00	64.57	406.82	NA	7.81	NA	6.56
318	1.0(DL+COLL+CL-1)	1124.78	0.00	0.00	21.86	144.56	NA	21.01	NA	18.52
319	1.0(DL+COLL+CL-2)	1124.78	0.00	0.00	21.86	144.56	NA	21.01	NA	18.52
320	1.0(DL+COLL+CL-3)	1124.78	0.00	0.00	21.86	144.56	NA	21.01	NA	18.52
321	1.0(DL+COLL+CL-4)	1124.78	0.00	0.00	21.86	144.56	NA	21.01	NA	18.52
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	985.27	0.00	0.00	105.25	619.62	NA	4.29	NA	3.37
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	985.27	0.00	0.00	105.25	619.62	NA	4.29	NA	3.37
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	985.27	0.00	0.00	105.25	619.62	NA	4.29	NA	3.37
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	985.27	0.00	0.00	105.25	619.62	NA	4.29	NA	3.37
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	1047.39	0.00	0.00	84.50	434.34	NA	6.51	NA	4.46
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	1047.39	0.00	0.00	84.50	434.34	NA	6.51	NA	4.46
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	1047.39	0.00	0.00	84.50	434.34	NA	6.51	NA	4.46
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	1047.39	0.00	0.00	84.50	434.34	NA	6.51	NA	4.46
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	1019.68	0.00	0.00	30.20	259.65	NA	10.60	NA	12.16
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	1019.68	0.00	0.00	30.20	259.65	NA	10.60	NA	12.16
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	1019.68	0.00	0.00	30.20	259.65	NA	10.60	NA	12.16
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	1019.68	0.00	0.00	30.20	259.65	NA	10.60	NA	12.16

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334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	1081.62	0.00	0.00	10.50	84.69	NA	34.48	NA	37.07
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	1081.62	0.00	0.00	10.50	84.69	NA	34.48	NA	37.07
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	1081.62	0.00	0.00	10.50	84.69	NA	34.48	NA	37.07
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	1081.62	0.00	0.00	10.50	84.69	NA	34.48	NA	37.07
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	997.09	-119.94	-1101.04	21.86	144.56	1.83	18.62	3.33	16.42
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	997.09	-119.94	-1101.04	21.86	144.56	1.83	18.62	3.33	16.42
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	997.09	-119.94	-1101.04	21.86	144.56	1.83	18.62	3.33	16.42
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	997.09	-119.94	-1101.04	21.86	144.56	1.83	18.62	3.33	16.42
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	1059.01	-100.12	-749.12	21.86	144.56	2.86	19.78	4.23	17.44
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	1059.01	-100.12	-749.12	21.86	144.56	2.86	19.78	4.23	17.44
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	1059.01	-100.12	-749.12	21.86	144.56	2.86	19.78	4.23	17.44
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	1059.01	-100.12	-749.12	21.86	144.56	2.86	19.78	4.23	17.44
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	922.13	-30.37	-979.19	21.86	144.56	1.91	17.22	12.15	15.18
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	922.13	-30.37	-979.19	21.86	144.56	1.91	17.22	12.15	15.18
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	922.13	-30.37	-979.19	21.86	144.56	1.91	17.22	12.15	15.18
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	922.13	-30.37	-979.19	21.86	144.56	1.91	17.22	12.15	15.18
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	984.05	-10.54	-627.27	21.86	144.56	3.18	18.38	37.33	16.20
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	984.05	-10.54	-627.27	21.86	144.56	3.18	18.38	37.33	16.20
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	984.05	-10.54	-627.27	21.86	144.56	3.18	18.38	37.33	16.20
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	984.05	-10.54	-627.27	21.86	144.56	3.18	18.38	37.33	16.20
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(F	1073.83	0.00	0.00	23.39	93.92	NA	30.87	NA	16.53
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(F	1073.83	0.00	0.00	23.39	93.92	NA	30.87	NA	16.53
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(F	1073.83	0.00	0.00	23.39	93.92	NA	30.87	NA	16.53
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(F	1073.83	0.00	0.00	23.39	93.92	NA	30.87	NA	16.53
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(F	1120.43	0.00	0.00	7.82	45.04	NA	67.16	NA	51.57
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(F	1120.43	0.00	0.00	7.82	45.04	NA	67.16	NA	51.57
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(F	1120.43	0.00	0.00	7.82	45.04	NA	67.16	NA	51.57
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(F	1120.43	0.00	0.00	7.82	45.04	NA	67.16	NA	51.57
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(	1099.64	0.00	0.00	32.90	176.06	NA	16.86	NA	12.03
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(	1099.64	0.00	0.00	32.90	176.06	NA	16.86	NA	12.03
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(	1099.64	0.00	0.00	32.90	176.06	NA	16.86	NA	12.03
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(	1099.64	0.00	0.00	32.90	176.06	NA	16.86	NA	12.03
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(	1146.10	0.00	0.00	47.67	307.28	NA	10.07	NA	8.66
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(	1146.10	0.00	0.00	47.67	307.28	NA	10.07	NA	8.66
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(	1146.10	0.00	0.00	47.67	307.28	NA	10.07	NA	8.66
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(	1146.10	0.00	0.00	47.67	307.28	NA	10.07	NA	8.66
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-V	1082.70	-89.96	-825.78	71.95	479.22	2.66	6.10	4.81	5.42
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-V	1082.70	-89.96	-825.78	71.95	479.22	2.66	6.10	4.81	5.42
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-V	1082.70	-89.96	-825.78	71.95	479.22	2.66	6.10	4.81	5.42
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-V	1082.70	-89.96	-825.78	71.95	479.22	2.66	6.10	4.81	5.42
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-V	1129.14	-75.09	-561.84	71.95	479.22	4.07	6.36	6.01	5.65
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-V	1129.14	-75.09	-561.84	71.95	479.22	4.07	6.36	6.01	5.65
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-V	1129.14	-75.09	-561.84	71.95	479.22	4.07	6.36	6.01	5.65
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-V	1129.14	-75.09	-561.84	71.95	479.22	4.07	6.36	6.01	5.65
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-V	1026.48	-22.77	-734.39	71.95	479.22	2.83	5.78	18.03	5.14
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-V	1026.48	-22.77	-734.39	71.95	479.22	2.83	5.78	18.03	5.14
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-V	1026.48	-22.77	-734.39	71.95	479.22	2.83	5.78	18.03	5.14
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-V	1026.48	-22.77	-734.39	71.95	479.22	2.83	5.78	18.03	5.14
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-V	1072.92	-7.91	-470.45	71.95	479.22	4.62	6.05	54.27	5.37
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-V	1072.92	-7.91	-470.45	71.95	479.22	4.62	6.05	54.27	5.37
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-V	1072.92	-7.91	-470.45	71.95	479.22	4.62	6.05	54.27	5.37
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-V	1072.92	-7.91	-470.45	71.95	479.22	4.62	6.05	54.27	5.37
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+'	1116.41	-8.38	-23.54	13.49	121.02	96.04	24.91	53.31	29.80
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+'	1116.41	-8.38	-23.54	13.49	121.02	96.04	24.91	53.31	29.80
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+'	1116.41	-8.38	-23.54	13.49	121.02	96.04	24.91	53.31	29.80
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+'	1116.41	-8.38	-23.54	13.49	121.02	96.04	24.91	53.31	29.80
390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	1133.16	8.38	23.54	30.24	168.10	97.48	18.20	54.11	13.49
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	1133.16	8.38	23.54	30.24	168.10	97.48	18.20	54.11	13.49
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	1133.16	8.38	23.54	30.24	168.10	97.48	18.20	54.11	13.49
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	1133.16	8.38	23.54	30.24	168.10	97.48	18.20	54.11	13.49
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'	1172.61	-5.86	-16.48	66.08	462.74	144.10	6.84	79.99	6.39
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'	1172.61	-5.86	-16.48	66.08	462.74	144.10	6.84	79.99	6.39
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'	1172.61	-5.86	-16.48	66.08	462.74	144.10	6.84	79.99	6.39
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'	1172.61	-5.86	-16.48	66.08	462.74	144.10	6.84	79.99	6.39
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-	1184.33	5.86	16.48	77.81	495.70	145.55	6.45	80.79	5.48
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-	1184.33	5.86	16.48	77.81	495.70	145.55	6.45	80.79	5.48
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-	1184.33	5.86	16.48	77.81	495.70	145.55	6.45	80.79	5.48
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-'	1184.33	5.86	16.48	77.81	495.70	145.55	6.45	80.79	5.48
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	973.24	0.00	0.00	113.99	677.45	NA	3.88	NA	3.07
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	973.24	0.00	0.00	113.99	677.45	NA	3.88	NA	3.07
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	973.24	0.00	0.00	113.99	677.45	NA	3.88	NA	3.07
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	973.24	0.00	0.00	113.99	677.45	NA	3.88	NA	3.07
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	1035.37	0.00	0.00	93.24	492.16	NA	5.68	NA	4.00
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	1035.37	0.00	0.00	93.24	492.16	NA	5.68	NA	4.00
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	1035.37	0.00	0.00	93.24	492.16	NA	5.68	NA	4.00
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	1035.37	0.00	0.00	93.24	492.16	NA	5.68	NA	4.00
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	1007.65	0.00	0.00	38.94	317.48	NA	8.57	NA	9.31
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	1007.65	0.00	0.00	38.94	317.48	NA	8.57	NA	9.31
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	1007.65	0.00	0.00	38.94	317.48	NA	8.57	NA	9.31
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	1007.65	0.00	0.00	38.94	317.48	NA	8.57	NA	9.31
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	1069.59	0.00	0.00	19.25	142.51	NA	20.26	NA	20.00
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	1069.59	0.00	0.00	19.25	142.51	NA	20.26	NA	20.00
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	1069.59	0.00	0.00	19.25	142.51	NA	20.26	NA	20.00
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	1069.59	0.00	0.00	19.25	142.51	NA	20.26	NA	20.00
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	985.06	-119.94	-1101.04	13.12	86.74	1.81	30.66	3.29	27.04
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	985.06	-119.94	-1101.04	13.12	86.74	1.81	30.66	3.29	27.04
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	985.06	-119.94	-1101.04	13.12	86.74	1.81	30.66	3.29	27.04
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	985.06	-119.94	-1101.04	13.12	86.74	1.81	30.66	3.29	27.04
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	1046.98	-100.12	-749.12	13.12	86.74	2.83	32.59	4.18	28.73
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	1046.98	-100.12	-749.12	13.12	86.74	2.83	32.59	4.18	28.73
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	1046.98	-100.12	-749.12	13.12	86.74	2.83	32.59	4.18	28.73
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	1046.98	-100.12	-749.12	13.12	86.74	2.83	32.59	4.18	28.73
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	910.10	-30.37	-979.19	13.12	86.74	1.88	28.33	11.99	24.98
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	910.10	-30.37	-979.19	13.12	86.74	1.88	28.33	11.99	24.98
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	910.10	-30.37	-979.19	13.12	86.74	1.88	28.33	11.99	24.98
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	910.10	-30.37	-979.19	13.12	86.74	1.88	28.33	11.99	24.98
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-W	972.02	-10.54	-627.27	13.12	86.74	3.14	30.26	36.88	26.68
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	972.02	-10.54	-627.27	13.12	86.74	3.14	30.26	36.88	26.68
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	972.02	-10.54	-627.27	13.12	86.74	3.14	30.26	36.88	26.68
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	972.02	-10.54	-627.27	13.12	86.74	3.14	30.26	36.88	26.68
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	1104.94	-7.82	-21.97	5.30	64.77	101.84	46.06	56.53	75.07
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	1104.94	-7.82	-21.97	5.30	64.77	101.84	46.06	56.53	75.07
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	1104.94	-7.82	-21.97	5.30	64.77	101.84	46.06	56.53	75.07
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	1104.94	-7.82	-21.97	5.30	64.77	101.84	46.06	56.53	75.07
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	1021.23	-7.82	-21.97	70.97	393.74	94.13	7.00	52.25	5.18
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	1058.51	-7.82	-21.97	58.52	282.57	97.56	10.11	54.16	6.51
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	1041.88	-7.82	-21.97	25.94	177.76	96.03	15.83	53.31	14.46
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	1079.04	-7.82	-21.97	14.12	72.78	99.45	40.03	55.21	27.51
						Min.=	**1.81**	**3.88**	**3.29**	**3.07**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F1

Load Combination & Calculation of Base pressure

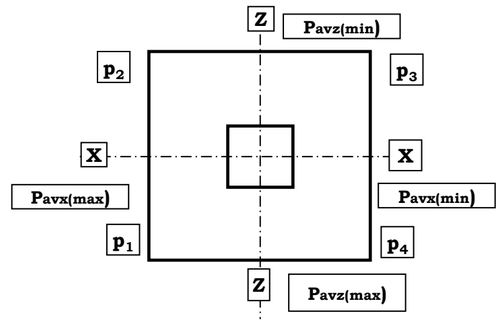
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 436 kN/m^2 For WL/SL case = 536 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
301	1.0(DL+COLL)	1124.78	0.00	144.56	41.66	0.00	5.35	47.01		36.30	47.01	36.30	11.01	41.66	41.66	47.01	36.30	O.K
302	1.0(DL+COLL+CL-1+TL1)+1.(	1198.03	0.00	687.30	44.37	0.00	25.46	69.83		18.92	69.83	18.92	33.83	44.38	44.38	69.83	18.92	O.K
303	1.0(DL+COLL+CL-2+TL1)+1.(	1198.03	0.00	687.30	44.37	0.00	25.46	69.83		18.92	69.83	18.92	33.83	44.38	44.38	69.83	18.92	O.K
304	1.0(DL+COLL+CL-3+TL1)+1.(	1198.03	0.00	687.30	44.37	0.00	25.46	69.83		18.92	69.83	18.92	33.83	44.38	44.38	69.83	18.92	O.K
305	1.0(DL+COLL+CL-4+TL1)+1.(	1198.03	0.00	687.30	44.37	0.00	25.46	69.83		18.92	69.83	18.92	33.83	44.38	44.38	69.83	18.92	O.K
306	1.0(DL+COLL+CL-1+TL2)+1.(	1194.71	0.00	494.24	44.25	0.00	18.31	62.55		25.94	62.55	25.94	26.55	44.25	44.25	62.55	25.94	O.K
307	1.0(DL+COLL+CL-2+TL2)+1.(	1194.71	0.00	494.24	44.25	0.00	18.31	62.55		25.94	62.55	25.94	26.55	44.25	44.25	62.55	25.94	O.K
308	1.0(DL+COLL+CL-3+TL2)+1.(	1194.71	0.00	494.24	44.25	0.00	18.31	62.55		25.94	62.55	25.94	26.55	44.25	44.25	62.55	25.94	O.K
309	1.0(DL+COLL+CL-4+TL2)+1.(	1194.71	0.00	494.24	44.25	0.00	18.31	62.55		25.94	62.55	25.94	26.55	44.25	44.25	62.55	25.94	O.K
310	1.0(DL+COLL+CL-1)+0.75(TL	1179.72	0.00	551.62	43.69	0.00	20.43	64.12		23.26	64.12	23.26	28.12	43.69	43.69	64.12	23.26	O.K
311	1.0(DL+COLL+CL-2)+0.75(TL	1179.72	0.00	551.62	43.69	0.00	20.43	64.12		23.26	64.12	23.26	28.12	43.69	43.69	64.12	23.26	O.K
312	1.0(DL+COLL+CL-3)+0.75(TL	1179.72	0.00	551.62	43.69	0.00	20.43	64.12		23.26	64.12	23.26	28.12	43.69	43.69	64.12	23.26	O.K
313	1.0(DL+COLL+CL-4)+0.75(TL	1179.72	0.00	551.62	43.69	0.00	20.43	64.12		23.26	64.12	23.26	28.12	43.69	43.69	64.12	23.26	O.K
314	1.0(DL+COLL+CL-1)+0.75(TL	1177.22	0.00	406.82	43.60	0.00	15.07	58.67		28.53	58.67	28.53	22.67	43.60	43.60	58.67	28.53	O.K
315	1.0(DL+COLL+CL-2)+0.75(TL	1177.22	0.00	406.82	43.60	0.00	15.07	58.67		28.53	58.67	28.53	22.67	43.60	43.60	58.67	28.53	O.K
316	1.0(DL+COLL+CL-3)+0.75(TL	1177.22	0.00	406.82	43.60	0.00	15.07	58.67		28.53	58.67	28.53	22.67	43.60	43.60	58.67	28.53	O.K

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317	1.0(DL+COLL+CL-4)+0.75(TL	1177.22	0.00	406.82	43.60	0.00	15.07	58.67		28.53	58.67	28.53	22.67	43.60	43.60	58.67	28.53	O.K
318	1.0(DL+COLL+CL-1)	1124.78	0.00	144.56	41.66	0.00	5.35	47.01		36.30	47.01	36.30	11.01	41.66	41.66	47.01	36.30	O.K
319	1.0(DL+COLL+CL-2)	1124.78	0.00	144.56	41.66	0.00	5.35	47.01		36.30	47.01	36.30	11.01	41.66	41.66	47.01	36.30	O.K
320	1.0(DL+COLL+CL-3)	1124.78	0.00	144.56	41.66	0.00	5.35	47.01		36.30	47.01	36.30	11.01	41.66	41.66	47.01	36.30	O.K
321	1.0(DL+COLL+CL-4)	1124.78	0.00	144.56	41.66	0.00	5.35	47.01		36.30	47.01	36.30	11.01	41.66	41.66	47.01	36.30	O.K
322	1.0(DL+COLL+CL-1)+1.0(WLI	985.27	0.00	-619.62	36.49	0.00	22.95	59.44		13.54	59.44	13.54	23.44	36.49	36.49	59.44	13.54	O.K
323	1.0(DL+COLL+CL-2)+1.0(WLI	985.27	0.00	-619.62	36.49	0.00	22.95	59.44		13.54	59.44	13.54	23.44	36.49	36.49	59.44	13.54	O.K
324	1.0(DL+COLL+CL-3)+1.0(WLI	985.27	0.00	-619.62	36.49	0.00	22.95	59.44		13.54	59.44	13.54	23.44	36.49	36.49	59.44	13.54	O.K
325	1.0(DL+COLL+CL-4)+1.0(WLI	985.27	0.00	-619.62	36.49	0.00	22.95	59.44		13.54	59.44	13.54	23.44	36.49	36.49	59.44	13.54	O.K
326	1.0(DL+COLL+CL-1)+1.0(WLI	1047.39	0.00	-434.34	38.79	0.00	16.09	54.88		22.71	54.88	22.71	18.88	38.80	38.80	54.88	22.71	O.K
327	1.0(DL+COLL+CL-2)+1.0(WLI	1047.39	0.00	-434.34	38.79	0.00	16.09	54.88		22.71	54.88	22.71	18.88	38.80	38.80	54.88	22.71	O.K
328	1.0(DL+COLL+CL-3)+1.0(WLI	1047.39	0.00	-434.34	38.79	0.00	16.09	54.88		22.71	54.88	22.71	18.88	38.80	38.80	54.88	22.71	O.K
329	1.0(DL+COLL+CL-4)+1.0(WLI	1047.39	0.00	-434.34	38.79	0.00	16.09	54.88		22.71	54.88	22.71	18.88	38.80	38.80	54.88	22.71	O.K
330	1.0(DL+COLL+CL-1)+1.0(WLI	1019.68	0.00	-259.65	37.77	0.00	9.62	47.38		28.15	47.38	28.15	11.38	37.77	37.77	47.38	28.15	O.K
331	1.0(DL+COLL+CL-2)+1.0(WLI	1019.68	0.00	-259.65	37.77	0.00	9.62	47.38		28.15	47.38	28.15	11.38	37.77	37.77	47.38	28.15	O.K
332	1.0(DL+COLL+CL-3)+1.0(WLI	1019.68	0.00	-259.65	37.77	0.00	9.62	47.38		28.15	47.38	28.15	11.38	37.77	37.77	47.38	28.15	O.K
333	1.0(DL+COLL+CL-4)+1.0(WLI	1019.68	0.00	-259.65	37.77	0.00	9.62	47.38		28.15	47.38	28.15	11.38	37.77	37.77	47.38	28.15	O.K
334	1.0(DL+COLL+CL-1)+1.0(WLI	1081.62	0.00	-84.69	40.06	0.00	3.14	43.20		36.92	43.20	36.92	7.20	40.06	40.06	43.20	36.92	O.K
335	1.0(DL+COLL+CL-2)+1.0(WLI	1081.62	0.00	-84.69	40.06	0.00	3.14	43.20		36.92	43.20	36.92	7.20	40.06	40.06	43.20	36.92	O.K
336	1.0(DL+COLL+CL-3)+1.0(WLI	1081.62	0.00	-84.69	40.06	0.00	3.14	43.20		36.92	43.20	36.92	7.20	40.06	40.06	43.20	36.92	O.K
337	1.0(DL+COLL+CL-4)+1.0(WLI	1081.62	0.00	-84.69	40.06	0.00	3.14	43.20		36.92	43.20	36.92	7.20	40.06	40.06	43.20	36.92	O.K
338	1.0(DL+COLL+CL-1)+1.0(WLI	997.09	-1101.04	144.56	36.93	54.37	5.35	96.66		85.95	0.00	0.00	61.66	91.81	0.00	48.83	42.98	O.K
339	1.0(DL+COLL+CL-2)+1.0(WLI	997.09	-1101.04	144.56	36.93	54.37	5.35	96.66		85.95	0.00	0.00	61.66	91.81	0.00	48.83	42.98	O.K
340	1.0(DL+COLL+CL-3)+1.0(WLI	997.09	-1101.04	144.56	36.93	54.37	5.35	96.66		85.95	0.00	0.00	61.66	91.81	0.00	48.83	42.98	O.K
341	1.0(DL+COLL+CL-4)+1.0(WLI	997.09	-1101.04	144.56	36.93	54.37	5.35	96.66		85.95	0.00	0.00	61.66	91.81	0.00	48.83	42.98	O.K
342	1.0(DL+COLL+CL-1)+1.0(WLI	1059.01	-749.12	144.56	39.22	36.99	5.35	81.57		70.86	7.58	0.00	46.57	76.72	3.79	45.08	35.43	O.K
343	1.0(DL+COLL+CL-2)+1.0(WLI	1059.01	-749.12	144.56	39.22	36.99	5.35	81.57		70.86	7.58	0.00	46.57	76.72	3.79	45.08	35.43	O.K
344	1.0(DL+COLL+CL-3)+1.0(WLI	1059.01	-749.12	144.56	39.22	36.99	5.35	81.57		70.86	7.58	0.00	46.57	76.72	3.79	45.08	35.43	O.K
345	1.0(DL+COLL+CL-4)+1.0(WLI	1059.01	-749.12	144.56	39.22	36.99	5.35	81.57		70.86	7.58	0.00	46.57	76.72	3.79	45.08	35.43	O.K
346	1.0(DL+COLL+CL-1)+1.0(WLI	922.13	-979.19	144.56	34.15	48.36	5.35	87.86		77.15	0.00	0.00	52.86	83.01	0.00	44.43	38.58	O.K
347	1.0(DL+COLL+CL-2)+1.0(WLI	922.13	-979.19	144.56	34.15	48.36	5.35	87.86		77.15	0.00	0.00	52.86	83.01	0.00	44.43	38.58	O.K

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348	1.0(DL+COLL+CL-3)+1.0(WLI	922.13	-979.19	144.56	34.15	48.36	5.35	87.86		77.15	0.00	0.00	52.86	83.01	0.00	44.43	38.58	O.K
349	1.0(DL+COLL+CL-4)+1.0(WLI	922.13	-979.19	144.56	34.15	48.36	5.35	87.86		77.15	0.00	0.00	52.86	83.01	0.00	44.43	38.58	O.K
350	1.0(DL+COLL+CL-1)+1.0(WLI	984.05	-627.27	144.56	36.45	30.98	5.35	72.78		62.07	10.82	0.12	36.78	67.43	5.47	41.80	31.10	O.K
351	1.0(DL+COLL+CL-2)+1.0(WLI	984.05	-627.27	144.56	36.45	30.98	5.35	72.78		62.07	10.82	0.12	36.78	67.43	5.47	41.80	31.10	O.K
352	1.0(DL+COLL+CL-3)+1.0(WLI	984.05	-627.27	144.56	36.45	30.98	5.35	72.78		62.07	10.82	0.12	36.78	67.43	5.47	41.80	31.10	O.K
353	1.0(DL+COLL+CL-4)+1.0(WLI	984.05	-627.27	144.56	36.45	30.98	5.35	72.78		62.07	10.82	0.12	36.78	67.43	5.47	41.80	31.10	O.K
354	1.0(DL+COLL+CL-1)+0.75(WI	1073.83	0.00	-93.92	39.77	0.00	3.48	43.25		36.29	43.25	36.29	7.25	39.77	39.77	43.25	36.29	O.K
355	1.0(DL+COLL+CL-2)+0.75(WI	1073.83	0.00	-93.92	39.77	0.00	3.48	43.25		36.29	43.25	36.29	7.25	39.77	39.77	43.25	36.29	O.K
356	1.0(DL+COLL+CL-3)+0.75(WI	1073.83	0.00	-93.92	39.77	0.00	3.48	43.25		36.29	43.25	36.29	7.25	39.77	39.77	43.25	36.29	O.K
357	1.0(DL+COLL+CL-4)+0.75(WI	1073.83	0.00	-93.92	39.77	0.00	3.48	43.25		36.29	43.25	36.29	7.25	39.77	39.77	43.25	36.29	O.K
358	1.0(DL+COLL+CL-1)+0.75(WI	1120.43	0.00	45.04	41.50	0.00	1.67	43.17		39.83	43.17	39.83	7.17	41.50	41.50	43.17	39.83	O.K
359	1.0(DL+COLL+CL-2)+0.75(WI	1120.43	0.00	45.04	41.50	0.00	1.67	43.17		39.83	43.17	39.83	7.17	41.50	41.50	43.17	39.83	O.K
360	1.0(DL+COLL+CL-3)+0.75(WI	1120.43	0.00	45.04	41.50	0.00	1.67	43.17		39.83	43.17	39.83	7.17	41.50	41.50	43.17	39.83	O.K
361	1.0(DL+COLL+CL-4)+0.75(WI	1120.43	0.00	45.04	41.50	0.00	1.67	43.17		39.83	43.17	39.83	7.17	41.50	41.50	43.17	39.83	O.K
362	1.0(DL+COLL+CL-1)+0.75(WI	1099.64	0.00	176.06	40.73	0.00	6.52	47.25		34.21	47.25	34.21	11.25	40.73	40.73	47.25	34.21	O.K
363	1.0(DL+COLL+CL-2)+0.75(WI	1099.64	0.00	176.06	40.73	0.00	6.52	47.25		34.21	47.25	34.21	11.25	40.73	40.73	47.25	34.21	O.K
364	1.0(DL+COLL+CL-3)+0.75(WI	1099.64	0.00	176.06	40.73	0.00	6.52	47.25		34.21	47.25	34.21	11.25	40.73	40.73	47.25	34.21	O.K
365	1.0(DL+COLL+CL-4)+0.75(WI	1099.64	0.00	176.06	40.73	0.00	6.52	47.25		34.21	47.25	34.21	11.25	40.73	40.73	47.25	34.21	O.K
366	1.0(DL+COLL+CL-1)+0.75(WI	1146.10	0.00	307.28	42.45	0.00	11.38	53.83		31.07	53.83	31.07	17.83	42.45	42.45	53.83	31.07	O.K
367	1.0(DL+COLL+CL-2)+0.75(WI	1146.10	0.00	307.28	42.45	0.00	11.38	53.83		31.07	53.83	31.07	17.83	42.45	42.45	53.83	31.07	O.K
368	1.0(DL+COLL+CL-3)+0.75(WI	1146.10	0.00	307.28	42.45	0.00	11.38	53.83		31.07	53.83	31.07	17.83	42.45	42.45	53.83	31.07	O.K
369	1.0(DL+COLL+CL-4)+0.75(WI	1146.10	0.00	307.28	42.45	0.00	11.38	53.83		31.07	53.83	31.07	17.83	42.45	42.45	53.83	31.07	O.K
370	1.0(DL+COLL+CL-1)+0.75(WI	1082.70	-825.78	479.22	40.10	40.78	17.75	98.63		63.13	17.07	0.00	63.63	81.38	8.54	58.35	31.57	O.K
371	1.0(DL+COLL+CL-2)+0.75(WI	1082.70	-825.78	479.22	40.10	40.78	17.75	98.63		63.13	17.07	0.00	63.63	81.38	8.54	58.35	31.57	O.K
372	1.0(DL+COLL+CL-3)+0.75(WI	1082.70	-825.78	479.22	40.10	40.78	17.75	98.63		63.13	17.07	0.00	63.63	81.38	8.54	58.35	31.57	O.K
373	1.0(DL+COLL+CL-4)+0.75(WI	1082.70	-825.78	479.22	40.10	40.78	17.75	98.63		63.13	17.07	0.00	63.63	81.38	8.54	58.35	31.57	O.K
374	1.0(DL+COLL+CL-1)+0.75(WI	1129.14	-561.84	479.22	41.82	27.75	17.75	87.31		51.82	31.82	0.00	52.31	70.07	15.91	60.07	25.91	O.K
375	1.0(DL+COLL+CL-2)+0.75(WI	1129.14	-561.84	479.22	41.82	27.75	17.75	87.31		51.82	31.82	0.00	52.31	70.07	15.91	60.07	25.91	O.K
376	1.0(DL+COLL+CL-3)+0.75(WI	1129.14	-561.84	479.22	41.82	27.75	17.75	87.31		51.82	31.82	0.00	52.31	70.07	15.91	60.07	25.91	O.K
377	1.0(DL+COLL+CL-4)+0.75(WI	1129.14	-561.84	479.22	41.82	27.75	17.75	87.31		51.82	31.82	0.00	52.31	70.07	15.91	60.07	25.91	O.K
378	1.0(DL+COLL+CL-1)+0.75(WI	1026.48	-734.39	479.22	38.02	36.27	17.75	92.03		56.54	19.50	0.00	57.03	74.79	9.75	56.27	28.27	O.K

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379	1.0(DL+COLL+CL-2)+0.75(WI	1026.48	-734.39	479.22	38.02	36.27	17.75	92.03		56.54	19.50	0.00	57.03	74.79	9.75	56.27	28.27	O.K
380	1.0(DL+COLL+CL-3)+0.75(WI	1026.48	-734.39	479.22	38.02	36.27	17.75	92.03		56.54	19.50	0.00	57.03	74.79	9.75	56.27	28.27	O.K
381	1.0(DL+COLL+CL-4)+0.75(WI	1026.48	-734.39	479.22	38.02	36.27	17.75	92.03		56.54	19.50	0.00	57.03	74.79	9.75	56.27	28.27	O.K
382	1.0(DL+COLL+CL-1)+0.75(WI	1072.92	-470.45	479.22	39.74	23.23	17.75	80.72		45.22	34.25	0.00	45.72	63.47	17.13	57.99	22.61	O.K
383	1.0(DL+COLL+CL-2)+0.75(WI	1072.92	-470.45	479.22	39.74	23.23	17.75	80.72		45.22	34.25	0.00	45.72	63.47	17.13	57.99	22.61	O.K
384	1.0(DL+COLL+CL-3)+0.75(WI	1072.92	-470.45	479.22	39.74	23.23	17.75	80.72		45.22	34.25	0.00	45.72	63.47	17.13	57.99	22.61	O.K
385	1.0(DL+COLL+CL-4)+0.75(WI	1072.92	-470.45	479.22	39.74	23.23	17.75	80.72		45.22	34.25	0.00	45.72	63.47	17.13	57.99	22.61	O.K
386	1.0(DL+COLL+CL-1)+0.75(SL	1116.41	-23.54	121.02	41.35	1.16	4.48	46.99		38.03	44.67	35.70	10.99	42.51	40.19	45.83	36.87	O.K
387	1.0(DL+COLL+CL-2)+0.75(SL	1116.41	-23.54	121.02	41.35	1.16	4.48	46.99		38.03	44.67	35.70	10.99	42.51	40.19	45.83	36.87	O.K
388	1.0(DL+COLL+CL-3)+0.75(SL	1116.41	-23.54	121.02	41.35	1.16	4.48	46.99		38.03	44.67	35.70	10.99	42.51	40.19	45.83	36.87	O.K
389	1.0(DL+COLL+CL-4)+0.75(SL	1116.41	-23.54	121.02	41.35	1.16	4.48	46.99		38.03	44.67	35.70	10.99	42.51	40.19	45.83	36.87	O.K
390	1.0(DL+COLL+CL-1)+0.75(SL	1133.16	23.54	168.10	41.97	1.16	6.23	49.36		36.91	47.03	34.58	13.36	43.14	40.81	48.20	35.75	O.K
391	1.0(DL+COLL+CL-2)+0.75(SL	1133.16	23.54	168.10	41.97	1.16	6.23	49.36		36.91	47.03	34.58	13.36	43.14	40.81	48.20	35.75	O.K
392	1.0(DL+COLL+CL-3)+0.75(SL	1133.16	23.54	168.10	41.97	1.16	6.23	49.36		36.91	47.03	34.58	13.36	43.14	40.81	48.20	35.75	O.K
393	1.0(DL+COLL+CL-4)+0.75(SL	1133.16	23.54	168.10	41.97	1.16	6.23	49.36		36.91	47.03	34.58	13.36	43.14	40.81	48.20	35.75	O.K
394	1.0(DL+COLL+CL-1)+0.525(S	1172.61	-16.48	462.74	43.43	0.81	17.14	61.38		27.11	59.75	25.48	25.38	44.25	42.62	60.57	26.30	O.K
395	1.0(DL+COLL+CL-2)+0.525(S	1172.61	-16.48	462.74	43.43	0.81	17.14	61.38		27.11	59.75	25.48	25.38	44.25	42.62	60.57	26.30	O.K
396	1.0(DL+COLL+CL-3)+0.525(S	1172.61	-16.48	462.74	43.43	0.81	17.14	61.38		27.11	59.75	25.48	25.38	44.25	42.62	60.57	26.30	O.K
397	1.0(DL+COLL+CL-4)+0.525(S	1172.61	-16.48	462.74	43.43	0.81	17.14	61.38		27.11	59.75	25.48	25.38	44.25	42.62	60.57	26.30	O.K
398	1.0(DL+COLL+CL-1)+0.525(S	1184.33	16.48	495.70	43.86	0.81	18.36	63.04		26.32	61.41	24.69	27.04	44.68	43.05	62.23	25.51	O.K
399	1.0(DL+COLL+CL-2)+0.525(S	1184.33	16.48	495.70	43.86	0.81	18.36	63.04		26.32	61.41	24.69	27.04	44.68	43.05	62.23	25.51	O.K
400	1.0(DL+COLL+CL-3)+0.525(S	1184.33	16.48	495.70	43.86	0.81	18.36	63.04		26.32	61.41	24.69	27.04	44.68	43.05	62.23	25.51	O.K
401	1.0(DL+COLL+CL-4)+0.525SI	1184.33	16.48	495.70	43.86	0.81	18.36	63.04		26.32	61.41	24.69	27.04	44.68	43.05	62.23	25.51	O.K
402	0.6(DL+COLL+CL-1)+1.0(WLI	973.24	0.00	-677.45	36.05	0.00	25.09	61.14		10.96	61.14	10.96	25.14	36.05	36.05	61.14	10.96	O.K
403	0.6(DL+COLL+CL-2)+1.0(WLI	973.24	0.00	-677.45	36.05	0.00	25.09	61.14		10.96	61.14	10.96	25.14	36.05	36.05	61.14	10.96	O.K
404	0.6(DL+COLL+CL-3)+1.0(WLI	973.24	0.00	-677.45	36.05	0.00	25.09	61.14		10.96	61.14	10.96	25.14	36.05	36.05	61.14	10.96	O.K
405	0.6(DL+COLL+CL-4)+1.0(WLI	973.24	0.00	-677.45	36.05	0.00	25.09	61.14		10.96	61.14	10.96	25.14	36.05	36.05	61.14	10.96	O.K
406	0.6(DL+COLL+CL-1)+1.0(WLI	1035.37	0.00	-492.16	38.35	0.00	18.23	56.58		20.12	56.58	20.12	20.58	38.35	38.35	56.58	20.12	O.K
407	0.6(DL+COLL+CL-2)+1.0(WLI	1035.37	0.00	-492.16	38.35	0.00	18.23	56.58		20.12	56.58	20.12	20.58	38.35	38.35	56.58	20.12	O.K
408	0.6(DL+COLL+CL-3)+1.0(WLI	1035.37	0.00	-492.16	38.35	0.00	18.23	56.58		20.12	56.58	20.12	20.58	38.35	38.35	56.58	20.12	O.K
409	0.6(DL+COLL+CL-4)+1.0(WLI	1035.37	0.00	-492.16	38.35	0.00	18.23	56.58		20.12	56.58	20.12	20.58	38.35	38.35	56.58	20.12	O.K

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410	0.6(DL+COLL+CL-1)+1.0(WLI	1007.65	0.00	-317.48	37.32	0.00	11.76	49.08		25.56	49.08	25.56	13.08	37.32	37.32	49.08	25.56	O.K
411	0.6(DL+COLL+CL-2)+1.0(WLI	1007.65	0.00	-317.48	37.32	0.00	11.76	49.08		25.56	49.08	25.56	13.08	37.32	37.32	49.08	25.56	O.K
412	0.6(DL+COLL+CL-3)+1.0(WLI	1007.65	0.00	-317.48	37.32	0.00	11.76	49.08		25.56	49.08	25.56	13.08	37.32	37.32	49.08	25.56	O.K
413	0.6(DL+COLL+CL-4)+1.0(WLI	1007.65	0.00	-317.48	37.32	0.00	11.76	49.08		25.56	49.08	25.56	13.08	37.32	37.32	49.08	25.56	O.K
414	0.6(DL+COLL+CL-1)+1.0(WLI	1069.59	0.00	-142.51	39.61	0.00	5.28	44.89		34.34	44.89	34.34	8.89	39.62	39.62	44.89	34.34	O.K
415	0.6(DL+COLL+CL-2)+1.0(WLI	1069.59	0.00	-142.51	39.61	0.00	5.28	44.89		34.34	44.89	34.34	8.89	39.62	39.62	44.89	34.34	O.K
416	0.6(DL+COLL+CL-3)+1.0(WLI	1069.59	0.00	-142.51	39.61	0.00	5.28	44.89		34.34	44.89	34.34	8.89	39.62	39.62	44.89	34.34	O.K
417	0.6(DL+COLL+CL-4)+1.0(WLI	1069.59	0.00	-142.51	39.61	0.00	5.28	44.89		34.34	44.89	34.34	8.89	39.62	39.62	44.89	34.34	O.K
418	0.6(DL+COLL+CL-1)+1.0(WLI	985.06	-1101.04	86.74	36.48	54.37	3.21	94.07		87.64	0.00	0.00	59.07	91.36	0.00	47.54	43.82	O.K
419	0.6(DL+COLL+CL-2)+1.0(WLI	985.06	-1101.04	86.74	36.48	54.37	3.21	94.07		87.64	0.00	0.00	59.07	91.36	0.00	47.54	43.82	O.K
420	0.6(DL+COLL+CL-3)+1.0(WLI	985.06	-1101.04	86.74	36.48	54.37	3.21	94.07		87.64	0.00	0.00	59.07	91.36	0.00	47.54	43.82	O.K
421	0.6(DL+COLL+CL-4)+1.0(WLI	985.06	-1101.04	86.74	36.48	54.37	3.21	94.07		87.64	0.00	0.00	59.07	91.36	0.00	47.54	43.82	O.K
422	0.6(DL+COLL+CL-1)+1.0(WLI	1046.98	-749.12	86.74	38.78	36.99	3.21	78.98		72.56	5.00	0.00	43.98	76.27	2.50	42.49	36.28	O.K
423	0.6(DL+COLL+CL-2)+1.0(WLI	1046.98	-749.12	86.74	38.78	36.99	3.21	78.98		72.56	5.00	0.00	43.98	76.27	2.50	42.49	36.28	O.K
424	0.6(DL+COLL+CL-3)+1.0(WLI	1046.98	-749.12	86.74	38.78	36.99	3.21	78.98		72.56	5.00	0.00	43.98	76.27	2.50	42.49	36.28	O.K
425	0.6(DL+COLL+CL-4)+1.0(WLI	1046.98	-749.12	86.74	38.78	36.99	3.21	78.98		72.56	5.00	0.00	43.98	76.27	2.50	42.49	36.28	O.K
426	0.6(DL+COLL+CL-1)+1.0(WLI	910.10	-979.19	86.74	33.71	48.36	3.21	85.28		78.85	0.00	0.00	50.28	82.57	0.00	43.14	39.43	O.K
427	0.6(DL+COLL+CL-2)+1.0(WLI	910.10	-979.19	86.74	33.71	48.36	3.21	85.28		78.85	0.00	0.00	50.28	82.57	0.00	43.14	39.43	O.K
428	0.6(DL+COLL+CL-3)+1.0(WLI	910.10	-979.19	86.74	33.71	48.36	3.21	85.28		78.85	0.00	0.00	50.28	82.57	0.00	43.14	39.43	O.K
429	0.6(DL+COLL+CL-4)+1.0(WLI	910.10	-979.19	86.74	33.71	48.36	3.21	85.28		78.85	0.00	0.00	50.28	82.57	0.00	43.14	39.43	O.K
430	0.6(DL+COLL+CL-1)+1.0(WLI	972.02	-627.27	86.74	36.00	30.98	3.21	70.19		63.76	8.24	1.81	34.19	66.98	5.03	39.22	32.79	O.K
431	0.6(DL+COLL+CL-2)+1.0(WLI	972.02	-627.27	86.74	36.00	30.98	3.21	70.19		63.76	8.24	1.81	34.19	66.98	5.03	39.22	32.79	O.K
432	0.6(DL+COLL+CL-3)+1.0(WLI	972.02	-627.27	86.74	36.00	30.98	3.21	70.19		63.76	8.24	1.81	34.19	66.98	5.03	39.22	32.79	O.K
433	0.6(DL+COLL+CL-4)+1.0(WLI	972.02	-627.27	86.74	36.00	30.98	3.21	70.19		63.76	8.24	1.81	34.19	66.98	5.03	39.22	32.79	O.K
434	0.6(DL+COLL+CL-1)+0.7(SL'+	1104.94	-21.97	64.77	40.92	1.08	2.40	44.41		39.61	42.24	37.44	8.41	42.01	39.84	43.33	38.53	O.K
435	0.6(DL+COLL+CL-2)+0.7(SL'+	1104.94	-21.97	64.77	40.92	1.08	2.40	44.41		39.61	42.24	37.44	8.41	42.01	39.84	43.33	38.53	O.K
436	0.6(DL+COLL+CL-3)+0.7(SL'+	1104.94	-21.97	64.77	40.92	1.08	2.40	44.41		39.61	42.24	37.44	8.41	42.01	39.84	43.33	38.53	O.K
437	0.6(DL+COLL+CL-4)+0.7(SL'+	1104.94	-21.97	64.77	40.92	1.08	2.40	44.41		39.61	42.24	37.44	8.41	42.01	39.84	43.33	38.53	O.K
438	0.6(DL+COLL+CL-1)+0.7(SL'+	1021.23	-21.97	-393.74	37.82	1.08	14.58	53.49		24.33	51.32	22.16	17.49	38.91	36.74	52.41	23.25	O.K
439	0.6(DL+COLL+CL-2)+0.7(SL'+	1058.51	-21.97	-282.57	39.20	1.08	10.47	50.75		29.82	48.58	27.65	14.75	40.29	38.12	49.67	28.74	O.K
440	0.6(DL+COLL+CL-3)+0.7(SL'+	1041.88	-21.97	-177.76	38.59	1.08	6.58	46.26		33.09	44.09	30.92	10.26	39.68	37.51	45.18	32.01	O.K
441	0.6(DL+COLL+CL-4)+0.7(SL'+	1079.04	-21.97	-72.78	39.96	1.08	2.70	43.75		38.35	41.58	36.18	7.75	41.05	38.88	42.67	37.27	O.K

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Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
	Ultimate Conditions																	
101	1.4(DL+COLL)	1136.81	0.00	202.39	42.10	0.00	7.50	49.60		34.61	49.60	34.61	13.60	42.11	42.11	49.60	34.61	
102	1.2(DL+COLL+CL-1)+1.6(RLL	1245.33	0.00	887.41	46.12	0.00	32.87	78.99		13.26	78.99	13.26	42.99	46.13	46.13	78.99	13.26	
103	1.2(DL+COLL+CL-2)+1.6(RLL	1245.33	0.00	887.41	46.12	0.00	32.87	78.99		13.26	78.99	13.26	42.99	46.13	46.13	78.99	13.26	
104	1.2(DL+COLL+CL-3)+1.6(RLL	1245.33	0.00	887.41	46.12	0.00	32.87	78.99		13.26	78.99	13.26	42.99	46.13	46.13	78.99	13.26	
105	1.2(DL+COLL+CL-4)+1.6(RLL	1245.33	0.00	887.41	46.12	0.00	32.87	78.99		13.26	78.99	13.26	42.99	46.13	46.13	78.99	13.26	
106	1.4(DL+COLL+CL-1)	1136.81	0.00	202.39	42.10	0.00	7.50	49.60		34.61	49.60	34.61	13.60	42.11	42.11	49.60	34.61	
107	1.4(DL+COLL+CL-2)	1136.81	0.00	202.39	42.10	0.00	7.50	49.60		34.61	49.60	34.61	13.60	42.11	42.11	49.60	34.61	
108	1.4(DL+COLL+CL-3)	1136.81	0.00	202.39	42.10	0.00	7.50	49.60		34.61	49.60	34.61	13.60	42.11	42.11	49.60	34.61	
109	1.4(DL+COLL+CL-4)	1136.81	0.00	202.39	42.10	0.00	7.50	49.60		34.61	49.60	34.61	13.60	42.11	42.11	49.60	34.61	
110	1.2(DL+COLL+CL-1+TL1)+1.6	1247.32	0.00	1003.24	46.20	0.00	37.16	83.35		9.04	83.35	9.04	47.35	46.20	46.20	83.35	9.04	
111	1.2(DL+COLL+CL-2+TL1)+1.6	1247.32	0.00	1003.24	46.20	0.00	37.16	83.35		9.04	83.35	9.04	47.35	46.20	46.20	83.35	9.04	
112	1.2(DL+COLL+CL-3+TL1)+1.6	1247.32	0.00	1003.24	46.20	0.00	37.16	83.35		9.04	83.35	9.04	47.35	46.20	46.20	83.35	9.04	
113	1.2(DL+COLL+CL-4+TL1)+1.6	1247.32	0.00	1003.24	46.20	0.00	37.16	83.35		9.04	83.35	9.04	47.35	46.20	46.20	83.35	9.04	
114	1.2(DL+COLL+CL-1+TL2)+1.6	1243.34	0.00	771.58	46.05	0.00	28.58	74.63		17.47	74.63	17.47	38.63	46.05	46.05	74.63	17.47	
115	1.2(DL+COLL+CL-2+TL2)+1.6	1243.34	0.00	771.58	46.05	0.00	28.58	74.63		17.47	74.63	17.47	38.63	46.05	46.05	74.63	17.47	
116	1.2(DL+COLL+CL-3+TL2)+1.6	1243.34	0.00	771.58	46.05	0.00	28.58	74.63		17.47	74.63	17.47	38.63	46.05	46.05	74.63	17.47	
117	1.2(DL+COLL+CL-4+TL2)+1.6	1243.34	0.00	771.58	46.05	0.00	28.58	74.63		17.47	74.63	17.47	38.63	46.05	46.05	74.63	17.47	
118	1.2(DL+COLL+CL-1)+0.8(WLI	1019.18	0.00	437.87	37.75	0.00	16.22	53.96		21.53	53.96	21.53	17.96	37.75	37.75	53.96	21.53	
119	1.2(DL+COLL+CL-2)+0.8(WLI	1019.18	0.00	437.87	37.75	0.00	16.22	53.96		21.53	53.96	21.53	17.96	37.75	37.75	53.96	21.53	
120	1.2(DL+COLL+CL-3)+0.8(WLI	1019.18	0.00	437.87	37.75	0.00	16.22	53.96		21.53	53.96	21.53	17.96	37.75	37.75	53.96	21.53	
121	1.2(DL+COLL+CL-4)+0.8(WLI	1019.18	0.00	437.87	37.75	0.00	16.22	53.96		21.53	53.96	21.53	17.96	37.75	37.75	53.96	21.53	
122	1.2(DL+COLL+CL-1)+0.8(WLI	1068.88	0.00	289.64	39.59	0.00	10.73	50.32		28.86	50.32	28.86	14.32	39.59	39.59	50.32	28.86	
123	1.2(DL+COLL+CL-2)+0.8(WLI	1068.88	0.00	289.64	39.59	0.00	10.73	50.32		28.86	50.32	28.86	14.32	39.59	39.59	50.32	28.86	
124	1.2(DL+COLL+CL-3)+0.8(WLI	1068.88	0.00	289.64	39.59	0.00	10.73	50.32		28.86	50.32	28.86	14.32	39.59	39.59	50.32	28.86	

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125	1.2(DL+COLL+CL-4)+0.8(WLI	1068.88	0.00	289.64	39.59	0.00	10.73	50.32		28.86	50.32	28.86	14.32	39.59	39.59	50.32	28.86	
126	1.2(DL+COLL+CL-1)+0.8(WLI	1046.71	0.00	149.89	38.77	0.00	5.55	44.32		33.22	44.32	33.22	8.32	38.77	38.77	44.32	33.22	
127	1.2(DL+COLL+CL-2)+0.8(WLI	1046.71	0.00	149.89	38.77	0.00	5.55	44.32		33.22	44.32	33.22	8.32	38.77	38.77	44.32	33.22	
128	1.2(DL+COLL+CL-3)+0.8(WLI	1046.71	0.00	149.89	38.77	0.00	5.55	44.32		33.22	44.32	33.22	8.32	38.77	38.77	44.32	33.22	
129	1.2(DL+COLL+CL-4)+0.8(WLI	1046.71	0.00	149.89	38.77	0.00	5.55	44.32		33.22	44.32	33.22	8.32	38.77	38.77	44.32	33.22	
130	1.2(DL+COLL+CL-1)+0.8(WLI	1096.26	0.00	9.92	40.60	0.00	0.37	40.97		40.23	40.97	40.23	4.97	40.60	40.60	40.97	40.23	
131	1.2(DL+COLL+CL-2)+0.8(WLI	1096.26	0.00	9.92	40.60	0.00	0.37	40.97		40.23	40.97	40.23	4.97	40.60	40.60	40.97	40.23	
132	1.2(DL+COLL+CL-3)+0.8(WLI	1096.26	0.00	9.92	40.60	0.00	0.37	40.97		40.23	40.97	40.23	4.97	40.60	40.60	40.97	40.23	
133	1.2(DL+COLL+CL-4)+0.8(WLI	1096.26	0.00	9.92	40.60	0.00	0.37	40.97		40.23	40.97	40.23	4.97	40.60	40.60	40.97	40.23	
134	1.2(DL+COLL+CL-1)+0.8(WLI	1028.64	549.77	173.48	38.10	27.15	6.43	71.67		58.82	17.37	4.52	35.67	65.25	10.95	44.52	31.67	
135	1.2(DL+COLL+CL-2)+0.8(WLI	1028.64	549.77	173.48	38.10	27.15	6.43	71.67		58.82	17.37	4.52	35.67	65.25	10.95	44.52	31.67	
136	1.2(DL+COLL+CL-3)+0.8(WLI	1028.64	549.77	173.48	38.10	27.15	6.43	71.67		58.82	17.37	4.52	35.67	65.25	10.95	44.52	31.67	
137	1.2(DL+COLL+CL-4)+0.8(WLI	1028.64	549.77	173.48	38.10	27.15	6.43	71.67		58.82	17.37	4.52	35.67	65.25	10.95	44.52	31.67	
138	1.2(DL+COLL+CL-1)+0.8(WLI	1078.18	408.56	173.48	39.93	20.18	6.43	66.53		53.68	26.18	13.33	30.53	60.11	19.76	46.36	33.51	
139	1.2(DL+COLL+CL-2)+0.8(WLI	1078.18	408.56	173.48	39.93	20.18	6.43	66.53		53.68	26.18	13.33	30.53	60.11	19.76	46.36	33.51	
140	1.2(DL+COLL+CL-3)+0.8(WLI	1078.18	408.56	173.48	39.93	20.18	6.43	66.53		53.68	26.18	13.33	30.53	60.11	19.76	46.36	33.51	
141	1.2(DL+COLL+CL-4)+0.8(WLI	1078.18	408.56	173.48	39.93	20.18	6.43	66.53		53.68	26.18	13.33	30.53	60.11	19.76	46.36	33.51	
142	1.2(DL+COLL+CL-1)+0.8(WLI	968.67	346.72	173.48	35.88	17.12	6.43	59.42		46.57	25.18	12.33	23.42	53.00	18.76	42.30	29.45	
143	1.2(DL+COLL+CL-2)+0.8(WLI	968.67	346.72	173.48	35.88	17.12	6.43	59.42		46.57	25.18	12.33	23.42	53.00	18.76	42.30	29.45	
144	1.2(DL+COLL+CL-3)+0.8(WLI	968.67	346.72	173.48	35.88	17.12	6.43	59.42		46.57	25.18	12.33	23.42	53.00	18.76	42.30	29.45	
145	1.2(DL+COLL+CL-4)+0.8(WLI	968.67	346.72	173.48	35.88	17.12	6.43	59.42		46.57	25.18	12.33	23.42	53.00	18.76	42.30	29.45	
146	1.2(DL+COLL+CL-1)+0.8(WLI	1018.21	205.51	173.48	37.71	10.15	6.43	54.29		41.43	33.99	21.14	18.29	47.86	27.57	44.14	31.29	
147	1.2(DL+COLL+CL-2)+0.8(WLI	1018.21	205.51	173.48	37.71	10.15	6.43	54.29		41.43	33.99	21.14	18.29	47.86	27.57	44.14	31.29	
148	1.2(DL+COLL+CL-3)+0.8(WLI	1018.21	205.51	173.48	37.71	10.15	6.43	54.29		41.43	33.99	21.14	18.29	47.86	27.57	44.14	31.29	
149	1.2(DL+COLL+CL-4)+0.8(WLI	1018.21	205.51	173.48	37.71	10.15	6.43	54.29		41.43	33.99	21.14	18.29	47.86	27.57	44.14	31.29	
150	1.2(DL+COLL+CL-1)+1.6(WLI	943.36	0.00	826.12	34.94	0.00	30.60	65.54		4.34	65.54	4.34	29.54	34.94	34.94	65.54	4.34	
151	1.2(DL+COLL+CL-2)+1.6(WLI	943.36	0.00	826.12	34.94	0.00	30.60	65.54		4.34	65.54	4.34	29.54	34.94	34.94	65.54	4.34	
152	1.2(DL+COLL+CL-3)+1.6(WLI	943.36	0.00	826.12	34.94	0.00	30.60	65.54		4.34	65.54	4.34	29.54	34.94	34.94	65.54	4.34	
153	1.2(DL+COLL+CL-4)+1.6(WLI	943.36	0.00	826.12	34.94	0.00	30.60	65.54		4.34	65.54	4.34	29.54	34.94	34.94	65.54	4.34	
154	1.2(DL+COLL+CL-1)+1.6(WLI	1042.76	0.00	529.66	38.62	0.00	19.62	58.24		19.00	58.24	19.00	22.24	38.62	38.62	58.24	19.00	
155	1.2(DL+COLL+CL-2)+1.6(WLI	1042.76	0.00	529.66	38.62	0.00	19.62	58.24		19.00	58.24	19.00	22.24	38.62	38.62	58.24	19.00	
156	1.2(DL+COLL+CL-3)+1.6(WLI	1042.76	0.00	529.66	38.62	0.00	19.62	58.24		19.00	58.24	19.00	22.24	38.62	38.62	58.24	19.00	

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157	1.2(DL+COLL+CL-4)+1.6(WLI	1042.76	0.00	529.66	38.62	0.00	19.62	58.24		19.00	58.24	19.00	22.24	38.62	38.62	58.24	19.00	
158	1.2(DL+COLL+CL-1)+1.6(WLI	998.42	0.00	250.16	36.98	0.00	9.27	46.24		27.71	46.24	27.71	10.24	36.98	36.98	46.24	27.71	
159	1.2(DL+COLL+CL-2)+1.6(WLI	998.42	0.00	250.16	36.98	0.00	9.27	46.24		27.71	46.24	27.71	10.24	36.98	36.98	46.24	27.71	
160	1.2(DL+COLL+CL-3)+1.6(WLI	998.42	0.00	250.16	36.98	0.00	9.27	46.24		27.71	46.24	27.71	10.24	36.98	36.98	46.24	27.71	
161	1.2(DL+COLL+CL-4)+1.6(WLI	998.42	0.00	250.16	36.98	0.00	9.27	46.24		27.71	46.24	27.71	10.24	36.98	36.98	46.24	27.71	
162	1.2(DL+COLL+CL-1)+1.6(WLI	1097.52	0.00	29.78	40.65	0.00	1.10	41.75		39.55	41.75	39.55	5.75	40.65	40.65	41.75	39.55	
163	1.2(DL+COLL+CL-2)+1.6(WLI	1097.52	0.00	29.78	40.65	0.00	1.10	41.75		39.55	41.75	39.55	5.75	40.65	40.65	41.75	39.55	
164	1.2(DL+COLL+CL-3)+1.6(WLI	1097.52	0.00	29.78	40.65	0.00	1.10	41.75		39.55	41.75	39.55	5.75	40.65	40.65	41.75	39.55	
165	1.2(DL+COLL+CL-4)+1.6(WLI	1097.52	0.00	29.78	40.65	0.00	1.10	41.75		39.55	41.75	39.55	5.75	40.65	40.65	41.75	39.55	
166	1.2(DL+COLL+CL-1)+1.6(WLI	962.27	1099.55	396.58	35.64	54.30	14.69	104.63		75.25	-3.97	-33.35	68.63	89.94	-18.66	50.33	20.95	
167	1.2(DL+COLL+CL-2)+1.6(WLI	962.27	1099.55	396.58	35.64	54.30	14.69	104.63		75.25	-3.97	-33.35	68.63	89.94	-18.66	50.33	20.95	
168	1.2(DL+COLL+CL-3)+1.6(WLI	962.27	1099.55	396.58	35.64	54.30	14.69	104.63		75.25	-3.97	-33.35	68.63	89.94	-18.66	50.33	20.95	
169	1.2(DL+COLL+CL-4)+1.6(WLI	962.27	1099.55	396.58	35.64	54.30	14.69	104.63		75.25	-3.97	-33.35	68.63	89.94	-18.66	50.33	20.95	
170	1.2(DL+COLL+CL-1)+1.6(WLI	1061.35	817.11	396.58	39.31	40.35	14.69	94.35		64.97	13.65	-15.73	58.35	79.66	-1.04	54.00	24.62	
171	1.2(DL+COLL+CL-2)+1.6(WLI	1061.35	817.11	396.58	39.31	40.35	14.69	94.35		64.97	13.65	-15.73	58.35	79.66	-1.04	54.00	24.62	
172	1.2(DL+COLL+CL-3)+1.6(WLI	1061.35	817.11	396.58	39.31	40.35	14.69	94.35		64.97	13.65	-15.73	58.35	79.66	-1.04	54.00	24.62	
173	1.2(DL+COLL+CL-4)+1.6(WLI	1061.35	817.11	396.58	39.31	40.35	14.69	94.35		64.97	13.65	-15.73	58.35	79.66	-1.04	54.00	24.62	
174	1.2(DL+COLL+CL-1)+1.6(WLI	842.34	693.45	396.58	31.20	34.24	14.69	80.13		50.75	11.64	-17.73	44.13	65.44	-3.05	45.89	16.51	
175	1.2(DL+COLL+CL-2)+1.6(WLI	842.34	693.45	396.58	31.20	34.24	14.69	80.13		50.75	11.64	-17.73	44.13	65.44	-3.05	45.89	16.51	
176	1.2(DL+COLL+CL-3)+1.6(WLI	842.34	693.45	396.58	31.20	34.24	14.69	80.13		50.75	11.64	-17.73	44.13	65.44	-3.05	45.89	16.51	
177	1.2(DL+COLL+CL-4)+1.6(WLI	842.34	693.45	396.58	31.20	34.24	14.69	80.13		50.75	11.64	-17.73	44.13	65.44	-3.05	45.89	16.51	
178	1.2(DL+COLL+CL-1)+1.6(WLI	941.41	411.01	396.58	34.87	20.30	14.69	69.85		40.48	29.26	-0.12	33.85	55.17	14.57	49.56	20.18	
179	1.2(DL+COLL+CL-2)+1.6(WLI	941.41	411.01	396.58	34.87	20.30	14.69	69.85		40.48	29.26	-0.12	33.85	55.17	14.57	49.56	20.18	
180	1.2(DL+COLL+CL-3)+1.6(WLI	941.41	411.01	396.58	34.87	20.30	14.69	69.85		40.48	29.26	-0.12	33.85	55.17	14.57	49.56	20.18	
181	1.2(DL+COLL+CL-4)+1.6(WLI	941.41	411.01	396.58	34.87	20.30	14.69	69.85		40.48	29.26	-0.12	33.85	55.17	14.57	49.56	20.18	
182	1.2(DL+COLL+CL-1)+1.0(SL'+	1155.42	37.64	358.94	42.79	1.86	13.29	57.95		31.36	54.23	27.64	21.95	44.66	40.94	56.09	29.50	
183	1.2(DL+COLL+CL-2)+1.0(SL'+	1155.42	37.64	358.94	42.79	1.86	13.29	57.95		31.36	54.23	27.64	21.95	44.66	40.94	56.09	29.50	
184	1.2(DL+COLL+CL-3)+1.0(SL'+	1155.42	37.64	358.94	42.79	1.86	13.29	57.95		31.36	54.23	27.64	21.95	44.66	40.94	56.09	29.50	
185	1.2(DL+COLL+CL-4)+1.0(SL'+	1155.42	37.64	358.94	42.79	1.86	13.29	57.95		31.36	54.23	27.64	21.95	44.66	40.94	56.09	29.50	
186	1.2(DL+COLL+CL-1)+1.0(SL'-	1177.75	37.64	434.23	43.62	1.86	16.08	61.56		29.40	57.84	25.68	25.56	45.48	41.76	59.70	27.54	
187	1.2(DL+COLL+CL-2)+1.0(SL'-	1177.75	37.64	434.23	43.62	1.86	16.08	61.56		29.40	57.84	25.68	25.56	45.48	41.76	59.70	27.54	

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188	1.2(DL+COLL+CL-3)+1.0(SL'-	1177.75	37.64	434.23	43.62	1.86	16.08	61.56		29.40	57.84	25.68	25.56	45.48	41.76	59.70	27.54	
189	1.2(DL+COLL+CL-4)+1.0(SL'-	1177.75	37.64	434.23	43.62	1.86	16.08	61.56		29.40	57.84	25.68	25.56	45.48	41.76	59.70	27.54	
190	0.9(DL+COLL+CL-1)+1.6(WLI	898.55	0.00	1092.59	33.28	0.00	40.47	73.75		-7.19	73.75	-7.19	37.75	33.28	33.28	73.75	-7.19	
191	0.9(DL+COLL+CL-2)+1.6(WLI	898.55	0.00	1092.59	33.28	0.00	40.47	73.75		-7.19	73.75	-7.19	37.75	33.28	33.28	73.75	-7.19	
192	0.9(DL+COLL+CL-3)+1.6(WLI	898.55	0.00	1092.59	33.28	0.00	40.47	73.75		-7.19	73.75	-7.19	37.75	33.28	33.28	73.75	-7.19	
193	0.9(DL+COLL+CL-4)+1.6(WLI	898.55	0.00	1092.59	33.28	0.00	40.47	73.75		-7.19	73.75	-7.19	37.75	33.28	33.28	73.75	-7.19	
194	0.9(DL+COLL+CL-1)+1.6(WLI	997.95	0.00	796.13	36.96	0.00	29.49	66.45		7.47	66.45	7.47	30.45	36.96	36.96	66.45	7.47	
195	0.9(DL+COLL+CL-2)+1.6(WLI	997.95	0.00	796.13	36.96	0.00	29.49	66.45		7.47	66.45	7.47	30.45	36.96	36.96	66.45	7.47	
196	0.9(DL+COLL+CL-3)+1.6(WLI	997.95	0.00	796.13	36.96	0.00	29.49	66.45		7.47	66.45	7.47	30.45	36.96	36.96	66.45	7.47	
197	0.9(DL+COLL+CL-4)+1.6(WLI	997.95	0.00	796.13	36.96	0.00	29.49	66.45		7.47	66.45	7.47	30.45	36.96	36.96	66.45	7.47	
198	0.9(DL+COLL+CL-1)+1.6(WLI	953.61	0.00	516.64	35.32	0.00	19.13	54.45		16.18	54.45	16.18	18.45	35.32	35.32	54.45	16.18	
199	0.9(DL+COLL+CL-2)+1.6(WLI	953.61	0.00	516.64	35.32	0.00	19.13	54.45		16.18	54.45	16.18	18.45	35.32	35.32	54.45	16.18	
200	0.9(DL+COLL+CL-3)+1.6(WLI	953.61	0.00	516.64	35.32	0.00	19.13	54.45		16.18	54.45	16.18	18.45	35.32	35.32	54.45	16.18	
201	0.9(DL+COLL+CL-4)+1.6(WLI	953.61	0.00	516.64	35.32	0.00	19.13	54.45		16.18	54.45	16.18	18.45	35.32	35.32	54.45	16.18	
202	0.9(DL+COLL+CL-1)+1.6(WLI	1052.71	0.00	236.70	38.99	0.00	8.77	47.76		30.22	47.76	30.22	11.76	38.99	38.99	47.76	30.22	
203	0.9(DL+COLL+CL-2)+1.6(WLI	1052.71	0.00	236.70	38.99	0.00	8.77	47.76		30.22	47.76	30.22	11.76	38.99	38.99	47.76	30.22	
204	0.9(DL+COLL+CL-3)+1.6(WLI	1052.71	0.00	236.70	38.99	0.00	8.77	47.76		30.22	47.76	30.22	11.76	38.99	38.99	47.76	30.22	
205	0.9(DL+COLL+CL-4)+1.6(WLI	1052.71	0.00	236.70	38.99	0.00	8.77	47.76		30.22	47.76	30.22	11.76	38.99	38.99	47.76	30.22	
206	0.9(DL+COLL+CL-1)+1.6(WLI	917.46	1099.55	130.11	33.98	54.30	4.82	93.10		83.46	-15.50	-25.14	57.10	88.28	-20.32	38.80	29.16	
207	0.9(DL+COLL+CL-2)+1.6(WLI	917.46	1099.55	130.11	33.98	54.30	4.82	93.10		83.46	-15.50	-25.14	57.10	88.28	-20.32	38.80	29.16	
208	0.9(DL+COLL+CL-3)+1.6(WLI	917.46	1099.55	130.11	33.98	54.30	4.82	93.10		83.46	-15.50	-25.14	57.10	88.28	-20.32	38.80	29.16	
209	0.9(DL+COLL+CL-4)+1.6(WLI	917.46	1099.55	130.11	33.98	54.30	4.82	93.10		83.46	-15.50	-25.14	57.10	88.28	-20.32	38.80	29.16	
210	0.9(DL+COLL+CL-1)+1.6(WLI	1016.54	817.11	130.11	37.65	40.35	4.82	82.82		73.18	2.12	-7.52	46.82	78.00	-2.70	42.47	32.83	
211	0.9(DL+COLL+CL-2)+1.6(WLI	1016.54	817.11	130.11	37.65	40.35	4.82	82.82		73.18	2.12	-7.52	46.82	78.00	-2.70	42.47	32.83	
212	0.9(DL+COLL+CL-3)+1.6(WLI	1016.54	817.11	130.11	37.65	40.35	4.82	82.82		73.18	2.12	-7.52	46.82	78.00	-2.70	42.47	32.83	
213	0.9(DL+COLL+CL-4)+1.6(WLI	1016.54	817.11	130.11	37.65	40.35	4.82	82.82		73.18	2.12	-7.52	46.82	78.00	-2.70	42.47	32.83	
214	0.9(DL+COLL+CL-1)+1.6(WLI	797.53	693.45	130.11	29.54	34.24	4.82	68.60		58.96	0.11	-9.53	32.60	63.78	-4.71	34.36	24.72	
215	0.9(DL+COLL+CL-2)+1.6(WLI	797.53	693.45	130.11	29.54	34.24	4.82	68.60		58.96	0.11	-9.53	32.60	63.78	-4.71	34.36	24.72	
216	0.9(DL+COLL+CL-3)+1.6(WLI	797.53	693.45	130.11	29.54	34.24	4.82	68.60		58.96	0.11	-9.53	32.60	63.78	-4.71	34.36	24.72	
217	0.9(DL+COLL+CL-4)+1.6(WLI	797.53	693.45	130.11	29.54	34.24	4.82	68.60		58.96	0.11	-9.53	32.60	63.78	-4.71	34.36	24.72	
218	0.9(DL+COLL+CL-1)+1.6(WLI	896.60	411.01	130.11	33.21	20.30	4.82	58.32		48.69	17.73	8.09	22.32	53.51	12.91	38.03	28.39	
219	0.9(DL+COLL+CL-2)+1.6(WLI	896.60	411.01	130.11	33.21	20.30	4.82	58.32		48.69	17.73	8.09	22.32	53.51	12.91	38.03	28.39	

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220	0.9(DL+COLL+CL-3)+1.6(WL	896.60	411.01	130.11	33.21	20.30	4.82	58.32		48.69	17.73	8.09	22.32	53.51	12.91	38.03	28.39	
221	0.9(DL+COLL+CL-4)+1.6(WL	896.60	411.01	130.11	33.21	20.30	4.82	58.32		48.69	17.73	8.09	22.32	53.51	12.91	38.03	28.39	
222	0.9(DL+COLL+CL-1)+1.0(SL'+	1110.61	37.64	92.46	41.13	1.86	3.42	46.42		39.57	42.70	35.85	10.42	43.00	39.28	44.56	37.71	
223	0.9(DL+COLL+CL-2)+1.0(SL'+	1110.61	37.64	92.46	41.13	1.86	3.42	46.42		39.57	42.70	35.85	10.42	43.00	39.28	44.56	37.71	
224	0.9(DL+COLL+CL-3)+1.0(SL'+	1110.61	37.64	92.46	41.13	1.86	3.42	46.42		39.57	42.70	35.85	10.42	43.00	39.28	44.56	37.71	
225	0.9(DL+COLL+CL-4)+1.0(SL'+	1110.61	37.64	92.46	41.13	1.86	3.42	46.42		39.57	42.70	35.85	10.42	43.00	39.28	44.56	37.71	
226	0.9(DL+COLL+CL-1)+1.0(SL'-	1132.94	37.64	167.75	41.96	1.86	6.21	50.03		37.61	46.31	33.89	14.03	43.82	40.10	48.17	35.75	
227	0.9(DL+COLL+CL-2)+1.0(SL'-	1132.94	37.64	167.75	41.96	1.86	6.21	50.03		37.61	46.31	33.89	14.03	43.82	40.10	48.17	35.75	
228	0.9(DL+COLL+CL-3)+1.0(SL'-	1132.94	37.64	167.75	41.96	1.86	6.21	50.03		37.61	46.31	33.89	14.03	43.82	40.10	48.17	35.75	
229	0.9(DL+COLL+CL-4)+1.0(SL'-	1132.94	37.64	167.75	41.96	1.86	6.21	50.03		37.61	46.31	33.89	14.03	43.82	40.10	48.17	35.75	

Max :	1198	24	687
Min :	910	-1101	-677

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 11.01 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 21.01 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 63.63 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 73.63 kN/m²

Sample calculation of values of base pressure:

Load case : 302 1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL)

Maximum pressure, p_1 (pmax) = 44.37+0.00+25.46= 69.83 kN/m²

p_2= 44.37+0.00-25.46= 18.92 kN/m²

p_3= 44.37-(0.00+25.46)= 18.92 kN/m²

p_4= 44.37-0.00+25.46= 69.83 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 69.83-2*18= 33.83 kN/m²

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DESIGN OF ISOLATED FOOTING F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

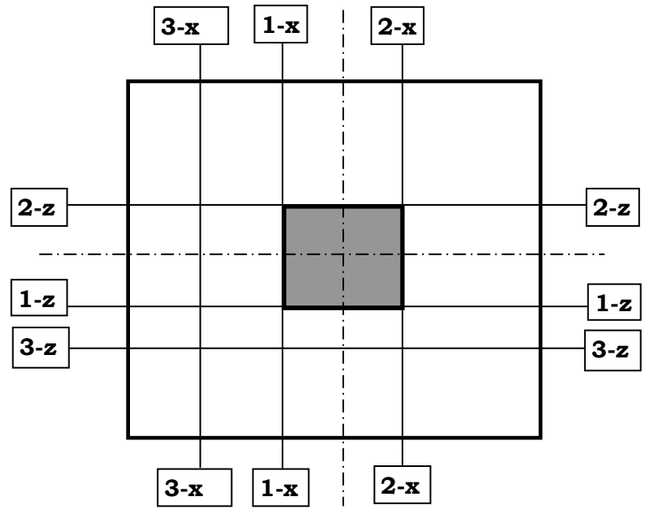
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.75+2*0.5*0.526)+(1.35+2*0.5*0.526))*2
= 6.304 m

Punching Area= ((0.75+2*0.5*0.526)*(1.35+2*0.5*0.526))
= 2.39 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.6*24+1.4*18)
= 39.60 kN/m^2

Area of footing, A_f= 27.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+COLL)	42.11	42.11	42.11	4.41	4.41	3.35	61.62
102	1.2(DL+COLL+CL-1)+1.6(R	46.13	46.13	46.13	11.48	11.48	8.70	160.54
103	1.2(DL+COLL+CL-2)+1.6(R	46.13	46.13	46.13	11.48	11.48	8.70	160.54
104	1.2(DL+COLL+CL-3)+1.6(R	46.13	46.13	46.13	11.48	11.48	8.70	160.54
105	1.2(DL+COLL+CL-4)+1.6(R	46.13	46.13	46.13	11.48	11.48	8.70	160.54
106	1.4(DL+COLL+CL-1)	42.11	42.11	42.11	4.41	4.41	3.35	61.62
107	1.4(DL+COLL+CL-2)	42.11	42.11	42.11	4.41	4.41	3.35	61.62
108	1.4(DL+COLL+CL-3)	42.11	42.11	42.11	4.41	4.41	3.35	61.62
109	1.4(DL+COLL+CL-4)	42.11	42.11	42.11	4.41	4.41	3.35	61.62
110	1.2(DL+COLL+CL-1+TL1)+	46.20	46.20	46.20	11.60	11.60	8.80	162.36
111	1.2(DL+COLL+CL-2+TL1)+	46.20	46.20	46.20	11.60	11.60	8.80	162.36
112	1.2(DL+COLL+CL-3+TL1)+	46.20	46.20	46.20	11.60	11.60	8.80	162.36

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113	1.2(DL+COLL+CL-4+TL1)+	46.20	46.20	46.20	11.60	11.60	8.80	162.36
114	1.2(DL+COLL+CL-1+TL2)+	46.05	46.05	46.05	11.34	11.34	8.60	158.72
115	1.2(DL+COLL+CL-2+TL2)+	46.05	46.05	46.05	11.34	11.34	8.60	158.72
116	1.2(DL+COLL+CL-3+TL2)+	46.05	46.05	46.05	11.34	11.34	8.60	158.72
117	1.2(DL+COLL+CL-4+TL2)+	46.05	46.05	46.05	11.34	11.34	8.60	158.72
118	1.2(DL+COLL+CL-1)+0.8(W	37.75	37.75	37.75	-3.25	-3.25	-2.47	-45.59
119	1.2(DL+COLL+CL-2)+0.8(W	37.75	37.75	37.75	-3.25	-3.25	-2.47	-45.59
120	1.2(DL+COLL+CL-3)+0.8(W	37.75	37.75	37.75	-3.25	-3.25	-2.47	-45.59
121	1.2(DL+COLL+CL-4)+0.8(W	37.75	37.75	37.75	-3.25	-3.25	-2.47	-45.59
122	1.2(DL+COLL+CL-1)+0.8(W	39.59	39.59	39.59	-0.02	-0.02	-0.01	-0.29
123	1.2(DL+COLL+CL-2)+0.8(W	39.59	39.59	39.59	-0.02	-0.02	-0.01	-0.29
124	1.2(DL+COLL+CL-3)+0.8(W	39.59	39.59	39.59	-0.02	-0.02	-0.01	-0.29
125	1.2(DL+COLL+CL-4)+0.8(W	39.59	39.59	39.59	-0.02	-0.02	-0.01	-0.29
126	1.2(DL+COLL+CL-1)+0.8(W	38.77	38.77	38.77	-1.46	-1.46	-1.11	-20.50
127	1.2(DL+COLL+CL-2)+0.8(W	38.77	38.77	38.77	-1.46	-1.46	-1.11	-20.50
128	1.2(DL+COLL+CL-3)+0.8(W	38.77	38.77	38.77	-1.46	-1.46	-1.11	-20.50
129	1.2(DL+COLL+CL-4)+0.8(W	38.77	38.77	38.77	-1.46	-1.46	-1.11	-20.50
130	1.2(DL+COLL+CL-1)+0.8(W	40.60	40.60	40.60	1.76	1.76	1.33	24.67
131	1.2(DL+COLL+CL-2)+0.8(W	40.60	40.60	40.60	1.76	1.76	1.33	24.67
132	1.2(DL+COLL+CL-3)+0.8(W	40.60	40.60	40.60	1.76	1.76	1.33	24.67
133	1.2(DL+COLL+CL-4)+0.8(W	40.60	40.60	40.60	1.76	1.76	1.33	24.67
134	1.2(DL+COLL+CL-1)+0.8(W	42.63	33.58	49.17	31.83	-37.10	23.47	-36.97
135	1.2(DL+COLL+CL-2)+0.8(W	42.63	33.58	49.17	31.83	-37.10	23.47	-36.97
136	1.2(DL+COLL+CL-3)+0.8(W	42.63	33.58	49.17	31.83	-37.10	23.47	-36.97
137	1.2(DL+COLL+CL-4)+0.8(W	42.63	33.58	49.17	31.83	-37.10	23.47	-36.97
138	1.2(DL+COLL+CL-1)+0.8(W	43.30	36.57	48.16	26.20	-25.02	19.37	8.18
139	1.2(DL+COLL+CL-2)+0.8(W	43.30	36.57	48.16	26.20	-25.02	19.37	8.18
140	1.2(DL+COLL+CL-3)+0.8(W	43.30	36.57	48.16	26.20	-25.02	19.37	8.18
141	1.2(DL+COLL+CL-4)+0.8(W	43.30	36.57	48.16	26.20	-25.02	19.37	8.18
142	1.2(DL+COLL+CL-1)+0.8(W	38.73	33.03	42.86	15.20	-28.27	11.10	-91.63
143	1.2(DL+COLL+CL-2)+0.8(W	38.73	33.03	42.86	15.20	-28.27	11.10	-91.63

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144	1.2(DL+COLL+CL-3)+0.8(W	38.73	33.03	42.86	15.20	-28.27	11.10	-91.63
145	1.2(DL+COLL+CL-4)+0.8(W	38.73	33.03	42.86	15.20	-28.27	11.10	-91.63
146	1.2(DL+COLL+CL-1)+0.8(W	39.41	36.02	41.85	9.57	-16.19	7.00	-46.48
147	1.2(DL+COLL+CL-2)+0.8(W	39.41	36.02	41.85	9.57	-16.19	7.00	-46.48
148	1.2(DL+COLL+CL-3)+0.8(W	39.41	36.02	41.85	9.57	-16.19	7.00	-46.48
149	1.2(DL+COLL+CL-4)+0.8(W	39.41	36.02	41.85	9.57	-16.19	7.00	-46.48
150	1.2(DL+COLL+CL-1)+1.6(W	34.94	34.94	34.94	-8.19	-8.19	-6.21	-114.70
151	1.2(DL+COLL+CL-2)+1.6(W	34.94	34.94	34.94	-8.19	-8.19	-6.21	-114.70
152	1.2(DL+COLL+CL-3)+1.6(W	34.94	34.94	34.94	-8.19	-8.19	-6.21	-114.70
153	1.2(DL+COLL+CL-4)+1.6(W	34.94	34.94	34.94	-8.19	-8.19	-6.21	-114.70
154	1.2(DL+COLL+CL-1)+1.6(W	38.62	38.62	38.62	-1.72	-1.72	-1.31	-24.10
155	1.2(DL+COLL+CL-2)+1.6(W	38.62	38.62	38.62	-1.72	-1.72	-1.31	-24.10
156	1.2(DL+COLL+CL-3)+1.6(W	38.62	38.62	38.62	-1.72	-1.72	-1.31	-24.10
157	1.2(DL+COLL+CL-4)+1.6(W	38.62	38.62	38.62	-1.72	-1.72	-1.31	-24.10
158	1.2(DL+COLL+CL-1)+1.6(W	36.98	36.98	36.98	-4.61	-4.61	-3.49	-64.51
159	1.2(DL+COLL+CL-2)+1.6(W	36.98	36.98	36.98	-4.61	-4.61	-3.49	-64.51
160	1.2(DL+COLL+CL-3)+1.6(W	36.98	36.98	36.98	-4.61	-4.61	-3.49	-64.51
161	1.2(DL+COLL+CL-4)+1.6(W	36.98	36.98	36.98	-4.61	-4.61	-3.49	-64.51
162	1.2(DL+COLL+CL-1)+1.6(W	40.65	40.65	40.65	1.85	1.85	1.40	25.82
163	1.2(DL+COLL+CL-2)+1.6(W	40.65	40.65	40.65	1.85	1.85	1.40	25.82
164	1.2(DL+COLL+CL-3)+1.6(W	40.65	40.65	40.65	1.85	1.85	1.40	25.82
165	1.2(DL+COLL+CL-4)+1.6(W	40.65	40.65	40.65	1.85	1.85	1.40	25.82
166	1.2(DL+COLL+CL-1)+1.6(W	44.69	26.59	57.77	61.97	-75.90	45.66	-97.46
167	1.2(DL+COLL+CL-2)+1.6(W	44.69	26.59	57.77	61.97	-75.90	45.66	-97.46
168	1.2(DL+COLL+CL-3)+1.6(W	44.69	26.59	57.77	61.97	-75.90	45.66	-97.46
169	1.2(DL+COLL+CL-4)+1.6(W	44.69	26.59	57.77	61.97	-75.90	45.66	-97.46
170	1.2(DL+COLL+CL-1)+1.6(W	46.04	32.59	55.75	50.72	-51.74	37.47	-7.15
171	1.2(DL+COLL+CL-2)+1.6(W	46.04	32.59	55.75	50.72	-51.74	37.47	-7.15
172	1.2(DL+COLL+CL-3)+1.6(W	46.04	32.59	55.75	50.72	-51.74	37.47	-7.15
173	1.2(DL+COLL+CL-4)+1.6(W	46.04	32.59	55.75	50.72	-51.74	37.47	-7.15
174	1.2(DL+COLL+CL-1)+1.6(W	36.90	25.49	45.15	28.70	-58.25	20.92	-206.78

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175	1.2(DL+COLL+CL-2)+1.6(W	36.90	25.49	45.15	28.70	-58.25	20.92	-206.78
176	1.2(DL+COLL+CL-3)+1.6(W	36.90	25.49	45.15	28.70	-58.25	20.92	-206.78
177	1.2(DL+COLL+CL-4)+1.6(W	36.90	25.49	45.15	28.70	-58.25	20.92	-206.78
178	1.2(DL+COLL+CL-1)+1.6(W	38.25	31.49	43.14	17.46	-34.09	12.74	-116.47
179	1.2(DL+COLL+CL-2)+1.6(W	38.25	31.49	43.14	17.46	-34.09	12.74	-116.47
180	1.2(DL+COLL+CL-3)+1.6(W	38.25	31.49	43.14	17.46	-34.09	12.74	-116.47
181	1.2(DL+COLL+CL-4)+1.6(W	38.25	31.49	43.14	17.46	-34.09	12.74	-116.47
182	1.2(DL+COLL+CL-1)+1.0(S	43.11	42.49	43.56	7.99	3.26	6.01	78.59
183	1.2(DL+COLL+CL-2)+1.0(S	43.11	42.49	43.56	7.99	3.26	6.01	78.59
184	1.2(DL+COLL+CL-3)+1.0(S	43.11	42.49	43.56	7.99	3.26	6.01	78.59
185	1.2(DL+COLL+CL-4)+1.0(S	43.11	42.49	43.56	7.99	3.26	6.01	78.59
186	1.2(DL+COLL+CL-1)+1.0(S	43.93	43.31	44.38	9.43	4.71	7.10	98.95
187	1.2(DL+COLL+CL-2)+1.0(S	43.93	43.31	44.38	9.43	4.71	7.10	98.95
188	1.2(DL+COLL+CL-3)+1.0(S	43.93	43.31	44.38	9.43	4.71	7.10	98.95
189	1.2(DL+COLL+CL-4)+1.0(S	43.93	43.31	44.38	9.43	4.71	7.10	98.95
190	0.9(DL+COLL+CL-1)+1.6(W	33.28	33.28	33.28	-11.11	-11.11	-8.42	-155.54
191	0.9(DL+COLL+CL-2)+1.6(W	33.28	33.28	33.28	-11.11	-11.11	-8.42	-155.54
192	0.9(DL+COLL+CL-3)+1.6(W	33.28	33.28	33.28	-11.11	-11.11	-8.42	-155.54
193	0.9(DL+COLL+CL-4)+1.6(W	33.28	33.28	33.28	-11.11	-11.11	-8.42	-155.54
194	0.9(DL+COLL+CL-1)+1.6(W	36.96	36.96	36.96	-4.64	-4.64	-3.52	-64.94
195	0.9(DL+COLL+CL-2)+1.6(W	36.96	36.96	36.96	-4.64	-4.64	-3.52	-64.94
196	0.9(DL+COLL+CL-3)+1.6(W	36.96	36.96	36.96	-4.64	-4.64	-3.52	-64.94
197	0.9(DL+COLL+CL-4)+1.6(W	36.96	36.96	36.96	-4.64	-4.64	-3.52	-64.94
198	0.9(DL+COLL+CL-1)+1.6(W	35.32	35.32	35.32	-7.52	-7.52	-5.71	-105.36
199	0.9(DL+COLL+CL-2)+1.6(W	35.32	35.32	35.32	-7.52	-7.52	-5.71	-105.36
200	0.9(DL+COLL+CL-3)+1.6(W	35.32	35.32	35.32	-7.52	-7.52	-5.71	-105.36
201	0.9(DL+COLL+CL-4)+1.6(W	35.32	35.32	35.32	-7.52	-7.52	-5.71	-105.36
202	0.9(DL+COLL+CL-1)+1.6(W	38.99	38.99	38.99	-1.07	-1.07	-0.81	-15.03
203	0.9(DL+COLL+CL-2)+1.6(W	38.99	38.99	38.99	-1.07	-1.07	-0.81	-15.03
204	0.9(DL+COLL+CL-3)+1.6(W	38.99	38.99	38.99	-1.07	-1.07	-0.81	-15.03
205	0.9(DL+COLL+CL-4)+1.6(W	38.99	38.99	38.99	-1.07	-1.07	-0.81	-15.03

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206	0.9(DL+COLL+CL-1)+1.6(W	43.03	24.93	56.11	59.06	-78.81	43.45	-138.30
207	0.9(DL+COLL+CL-2)+1.6(W	43.03	24.93	56.11	59.06	-78.81	43.45	-138.30
208	0.9(DL+COLL+CL-3)+1.6(W	43.03	24.93	56.11	59.06	-78.81	43.45	-138.30
209	0.9(DL+COLL+CL-4)+1.6(W	43.03	24.93	56.11	59.06	-78.81	43.45	-138.30
210	0.9(DL+COLL+CL-1)+1.6(W	44.38	30.93	54.09	47.80	-54.65	35.25	-48.00
211	0.9(DL+COLL+CL-2)+1.6(W	44.38	30.93	54.09	47.80	-54.65	35.25	-48.00
212	0.9(DL+COLL+CL-3)+1.6(W	44.38	30.93	54.09	47.80	-54.65	35.25	-48.00
213	0.9(DL+COLL+CL-4)+1.6(W	44.38	30.93	54.09	47.80	-54.65	35.25	-48.00
214	0.9(DL+COLL+CL-1)+1.6(W	35.24	23.83	43.49	25.78	-61.17	18.71	-247.62
215	0.9(DL+COLL+CL-2)+1.6(W	35.24	23.83	43.49	25.78	-61.17	18.71	-247.62
216	0.9(DL+COLL+CL-3)+1.6(W	35.24	23.83	43.49	25.78	-61.17	18.71	-247.62
217	0.9(DL+COLL+CL-4)+1.6(W	35.24	23.83	43.49	25.78	-61.17	18.71	-247.62
218	0.9(DL+COLL+CL-1)+1.6(W	36.59	29.83	41.48	14.54	-37.00	10.53	-157.32
219	0.9(DL+COLL+CL-2)+1.6(W	36.59	29.83	41.48	14.54	-37.00	10.53	-157.32
220	0.9(DL+COLL+CL-3)+1.6(W	36.59	29.83	41.48	14.54	-37.00	10.53	-157.32
221	0.9(DL+COLL+CL-4)+1.6(W	36.59	29.83	41.48	14.54	-37.00	10.53	-157.32
222	0.9(DL+COLL+CL-1)+1.0(S	41.45	40.83	41.90	5.07	0.35	3.80	37.74
223	0.9(DL+COLL+CL-2)+1.0(S	41.45	40.83	41.90	5.07	0.35	3.80	37.74
224	0.9(DL+COLL+CL-3)+1.0(S	41.45	40.83	41.90	5.07	0.35	3.80	37.74
225	0.9(DL+COLL+CL-4)+1.0(S	41.45	40.83	41.90	5.07	0.35	3.80	37.74
226	0.9(DL+COLL+CL-1)+1.0(S	42.27	41.65	42.72	6.51	1.79	4.89	58.10
227	0.9(DL+COLL+CL-2)+1.0(S	42.27	41.65	42.72	6.51	1.79	4.89	58.10
228	0.9(DL+COLL+CL-3)+1.0(S	42.27	41.65	42.72	6.51	1.79	4.89	58.10
229	0.9(DL+COLL+CL-4)+1.0(S	42.27	41.65	42.72	6.51	1.79	4.89	58.10

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Sample calculation of BM & SF

Load case : 101 1.4(DL+COLL)

p_{1x} = 42.11-(42.11-42.11)*1.875/4.5 = 42.11 kN/m^2 Overhang = 1.875 m

p_{2x} = 42.11+(42.11-42.11)*1.875/4.5 = 42.11 kN/m^2 Effective depth = 0.542 m

p_{3x} = 42.11-(42.11-42.11)*(1.875-0.542)/4.5 = 42.11 kN/m^2

BM_{1x} = (42.11-39.6)*1.875*1.875/2+0.5*(42.11-42.11)*1.875*1.875*0.67 = 4.41 kN-m/m

BM_{2x} = (42.11-39.6)*1.875*1.875/2-0.5*(-42.11+42.11)*1.875*1.875*0.67 = 4.41 kN-m/m

SF_{3x} = (42.11-39.6)*(1.875-0.542)+0.5*(42.11-42.11)*(1.875-0.542) = 3.35 kN/m

$S.F_{(punching)}$ = (27-2.39)*(42.10-39.6) = 61.62 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	61.97	0.21	0.28	0.001	364.73	975.60	16ϕT	206.09	300	175	1148.93
At top	-78.81	-0.27	0.36	0.001	464.44	487.80	12ϕT	231.85	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 45.66 kN/m

Shear stress , τ_v= (45.66*1000)/(1000*542)

= 0.08 N/mm^2 **O.K.**

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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 162.36 kN

Strength of Concrete in shear Vc1 = 15497.30 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 6265.53 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 6737.96 kN **Table 22.6.5.2_ (c)**

Design Vc= 6265.53 kN

Allowable shear stress, (ϕ Vc)/bd = 1.38 N/mm^2

Design punching shear stress, τ = (162.36*1000)/(6304*542)

= 0.05 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+COLL)	43.79	40.42	45.11	21.85	-8.31	13.95
102	1.2(DL+COLL+CL-1)+1.6(R	53.52	38.73	59.28	83.75	-48.47	53.14
103	1.2(DL+COLL+CL-2)+1.6(R	53.52	38.73	59.28	83.75	-48.47	53.14
104	1.2(DL+COLL+CL-3)+1.6(R	53.52	38.73	59.28	83.75	-48.47	53.14
105	1.2(DL+COLL+CL-4)+1.6(R	53.52	38.73	59.28	83.75	-48.47	53.14
106	1.4(DL+COLL+CL-1)	43.79	40.42	45.11	21.85	-8.31	13.95
107	1.4(DL+COLL+CL-2)	43.79	40.42	45.11	21.85	-8.31	13.95
108	1.4(DL+COLL+CL-3)	43.79	40.42	45.11	21.85	-8.31	13.95

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109	1.4(DL+COLL+CL-4)	43.79	40.42	45.11	21.85	-8.31	13.95
110	1.2(DL+COLL+CL-1+TL1)+	54.55	37.84	61.07	92.56	-56.91	58.66
111	1.2(DL+COLL+CL-2+TL1)+	54.55	37.84	61.07	92.56	-56.91	58.66
112	1.2(DL+COLL+CL-3+TL1)+	54.55	37.84	61.07	92.56	-56.91	58.66
113	1.2(DL+COLL+CL-4+TL1)+	54.55	37.84	61.07	92.56	-56.91	58.66
114	1.2(DL+COLL+CL-1+TL2)+	52.48	39.62	57.49	74.92	-40.06	47.60
115	1.2(DL+COLL+CL-2+TL2)+	52.48	39.62	57.49	74.92	-40.06	47.60
116	1.2(DL+COLL+CL-3+TL2)+	52.48	39.62	57.49	74.92	-40.06	47.60
117	1.2(DL+COLL+CL-4+TL2)+	52.48	39.62	57.49	74.92	-40.06	47.60
118	1.2(DL+COLL+CL-1)+0.8(W	41.39	34.10	44.24	27.60	-37.63	17.09
119	1.2(DL+COLL+CL-2)+0.8(W	41.39	34.10	44.24	27.60	-37.63	17.09
120	1.2(DL+COLL+CL-3)+0.8(W	41.39	34.10	44.24	27.60	-37.63	17.09
121	1.2(DL+COLL+CL-4)+0.8(W	41.39	34.10	44.24	27.60	-37.63	17.09
122	1.2(DL+COLL+CL-1)+0.8(W	42.00	37.18	43.89	21.56	-21.61	13.50
123	1.2(DL+COLL+CL-2)+0.8(W	42.00	37.18	43.89	21.56	-21.61	13.50
124	1.2(DL+COLL+CL-3)+0.8(W	42.00	37.18	43.89	21.56	-21.61	13.50
125	1.2(DL+COLL+CL-4)+0.8(W	42.00	37.18	43.89	21.56	-21.61	13.50
126	1.2(DL+COLL+CL-1)+0.8(W	40.02	37.52	40.99	8.92	-13.41	5.50
127	1.2(DL+COLL+CL-2)+0.8(W	40.02	37.52	40.99	8.92	-13.41	5.50
128	1.2(DL+COLL+CL-3)+0.8(W	40.02	37.52	40.99	8.92	-13.41	5.50
129	1.2(DL+COLL+CL-4)+0.8(W	40.02	37.52	40.99	8.92	-13.41	5.50
130	1.2(DL+COLL+CL-1)+0.8(W	40.68	40.52	40.75	3.45	1.96	2.27
131	1.2(DL+COLL+CL-2)+0.8(W	40.68	40.52	40.75	3.45	1.96	2.27
132	1.2(DL+COLL+CL-3)+0.8(W	40.68	40.52	40.75	3.45	1.96	2.27
133	1.2(DL+COLL+CL-4)+0.8(W	40.68	40.52	40.75	3.45	1.96	2.27
134	1.2(DL+COLL+CL-1)+0.8(W	39.54	36.65	40.67	8.86	-16.99	5.39
135	1.2(DL+COLL+CL-2)+0.8(W	39.54	36.65	40.67	8.86	-16.99	5.39
136	1.2(DL+COLL+CL-3)+0.8(W	39.54	36.65	40.67	8.86	-16.99	5.39
137	1.2(DL+COLL+CL-4)+0.8(W	39.54	36.65	40.67	8.86	-16.99	5.39
138	1.2(DL+COLL+CL-1)+0.8(W	41.38	38.49	42.51	13.83	-12.02	8.70
139	1.2(DL+COLL+CL-2)+0.8(W	41.38	38.49	42.51	13.83	-12.02	8.70
140	1.2(DL+COLL+CL-3)+0.8(W	41.38	38.49	42.51	13.83	-12.02	8.70
141	1.2(DL+COLL+CL-4)+0.8(W	41.38	38.49	42.51	13.83	-12.02	8.70

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142	1.2(DL+COLL+CL-1)+0.8(W	37.32	34.43	38.45	2.86	-22.99	1.39
143	1.2(DL+COLL+CL-2)+0.8(W	37.32	34.43	38.45	2.86	-22.99	1.39
144	1.2(DL+COLL+CL-3)+0.8(W	37.32	34.43	38.45	2.86	-22.99	1.39
145	1.2(DL+COLL+CL-4)+0.8(W	37.32	34.43	38.45	2.86	-22.99	1.39
146	1.2(DL+COLL+CL-1)+0.8(W	39.16	36.27	40.29	7.83	-18.02	4.70
147	1.2(DL+COLL+CL-2)+0.8(W	39.16	36.27	40.29	7.83	-18.02	4.70
148	1.2(DL+COLL+CL-3)+0.8(W	39.16	36.27	40.29	7.83	-18.02	4.70
149	1.2(DL+COLL+CL-4)+0.8(W	39.16	36.27	40.29	7.83	-18.02	4.70
150	1.2(DL+COLL+CL-1)+1.6(W	41.83	28.06	47.19	48.96	-74.15	30.16
151	1.2(DL+COLL+CL-2)+1.6(W	41.83	28.06	47.19	48.96	-74.15	30.16
152	1.2(DL+COLL+CL-3)+1.6(W	41.83	28.06	47.19	48.96	-74.15	30.16
153	1.2(DL+COLL+CL-4)+1.6(W	41.83	28.06	47.19	48.96	-74.15	30.16
154	1.2(DL+COLL+CL-1)+1.6(W	43.03	34.21	46.47	36.82	-42.12	22.95
155	1.2(DL+COLL+CL-2)+1.6(W	43.03	34.21	46.47	36.82	-42.12	22.95
156	1.2(DL+COLL+CL-3)+1.6(W	43.03	34.21	46.47	36.82	-42.12	22.95
157	1.2(DL+COLL+CL-4)+1.6(W	43.03	34.21	46.47	36.82	-42.12	22.95
158	1.2(DL+COLL+CL-1)+1.6(W	39.06	34.89	40.68	11.54	-25.73	6.95
159	1.2(DL+COLL+CL-2)+1.6(W	39.06	34.89	40.68	11.54	-25.73	6.95
160	1.2(DL+COLL+CL-3)+1.6(W	39.06	34.89	40.68	11.54	-25.73	6.95
161	1.2(DL+COLL+CL-4)+1.6(W	39.06	34.89	40.68	11.54	-25.73	6.95
162	1.2(DL+COLL+CL-1)+1.6(W	40.90	40.40	41.09	5.05	0.63	3.27
163	1.2(DL+COLL+CL-2)+1.6(W	40.90	40.40	41.09	5.05	0.63	3.27
164	1.2(DL+COLL+CL-3)+1.6(W	40.90	40.40	41.09	5.05	0.63	3.27
165	1.2(DL+COLL+CL-4)+1.6(W	40.90	40.40	41.09	5.05	0.63	3.27
166	1.2(DL+COLL+CL-1)+1.6(W	38.95	32.33	41.52	18.85	-40.25	11.38
167	1.2(DL+COLL+CL-2)+1.6(W	38.95	32.33	41.52	18.85	-40.25	11.38
168	1.2(DL+COLL+CL-3)+1.6(W	38.95	32.33	41.52	18.85	-40.25	11.38
169	1.2(DL+COLL+CL-4)+1.6(W	38.95	32.33	41.52	18.85	-40.25	11.38
170	1.2(DL+COLL+CL-1)+1.6(W	42.62	36.00	45.19	28.77	-30.33	17.98
171	1.2(DL+COLL+CL-2)+1.6(W	42.62	36.00	45.19	28.77	-30.33	17.98
172	1.2(DL+COLL+CL-3)+1.6(W	42.62	36.00	45.19	28.77	-30.33	17.98
173	1.2(DL+COLL+CL-4)+1.6(W	42.62	36.00	45.19	28.77	-30.33	17.98
174	1.2(DL+COLL+CL-1)+1.6(W	34.51	27.89	37.08	6.85	-52.25	3.39

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175	1.2(DL+COLL+CL-2)+1.6(W	34.51	27.89	37.08	6.85	-52.25	3.39
176	1.2(DL+COLL+CL-3)+1.6(W	34.51	27.89	37.08	6.85	-52.25	3.39
177	1.2(DL+COLL+CL-4)+1.6(W	34.51	27.89	37.08	6.85	-52.25	3.39
178	1.2(DL+COLL+CL-1)+1.6(W	38.18	31.56	40.75	16.77	-42.33	9.99
179	1.2(DL+COLL+CL-2)+1.6(W	38.18	31.56	40.75	16.77	-42.33	9.99
180	1.2(DL+COLL+CL-3)+1.6(W	38.18	31.56	40.75	16.77	-42.33	9.99
181	1.2(DL+COLL+CL-4)+1.6(W	38.18	31.56	40.75	16.77	-42.33	9.99
182	1.2(DL+COLL+CL-1)+1.0(S	45.79	39.80	48.12	35.38	-18.11	22.49
183	1.2(DL+COLL+CL-2)+1.0(S	45.79	39.80	48.12	35.38	-18.11	22.49
184	1.2(DL+COLL+CL-3)+1.0(S	45.79	39.80	48.12	35.38	-18.11	22.49
185	1.2(DL+COLL+CL-4)+1.0(S	45.79	39.80	48.12	35.38	-18.11	22.49
186	1.2(DL+COLL+CL-1)+1.0(S	47.24	40.00	50.06	43.21	-21.48	27.49
187	1.2(DL+COLL+CL-2)+1.0(S	47.24	40.00	50.06	43.21	-21.48	27.49
188	1.2(DL+COLL+CL-3)+1.0(S	47.24	40.00	50.06	43.21	-21.48	27.49
189	1.2(DL+COLL+CL-4)+1.0(S	47.24	40.00	50.06	43.21	-21.48	27.49
190	0.9(DL+COLL+CL-1)+1.6(W	42.39	24.17	49.48	64.33	-98.49	39.61
191	0.9(DL+COLL+CL-2)+1.6(W	42.39	24.17	49.48	64.33	-98.49	39.61
192	0.9(DL+COLL+CL-3)+1.6(W	42.39	24.17	49.48	64.33	-98.49	39.61
193	0.9(DL+COLL+CL-4)+1.6(W	42.39	24.17	49.48	64.33	-98.49	39.61
194	0.9(DL+COLL+CL-1)+1.6(W	43.60	30.32	48.77	52.19	-66.46	32.40
195	0.9(DL+COLL+CL-2)+1.6(W	43.60	30.32	48.77	52.19	-66.46	32.40
196	0.9(DL+COLL+CL-3)+1.6(W	43.60	30.32	48.77	52.19	-66.46	32.40
197	0.9(DL+COLL+CL-4)+1.6(W	43.60	30.32	48.77	52.19	-66.46	32.40
198	0.9(DL+COLL+CL-1)+1.6(W	39.62	31.01	42.98	26.91	-50.07	16.39
199	0.9(DL+COLL+CL-2)+1.6(W	39.62	31.01	42.98	26.91	-50.07	16.39
200	0.9(DL+COLL+CL-3)+1.6(W	39.62	31.01	42.98	26.91	-50.07	16.39
201	0.9(DL+COLL+CL-4)+1.6(W	39.62	31.01	42.98	26.91	-50.07	16.39
202	0.9(DL+COLL+CL-1)+1.6(W	40.96	37.02	42.50	15.99	-19.29	9.95
203	0.9(DL+COLL+CL-2)+1.6(W	40.96	37.02	42.50	15.99	-19.29	9.95
204	0.9(DL+COLL+CL-3)+1.6(W	40.96	37.02	42.50	15.99	-19.29	9.95
205	0.9(DL+COLL+CL-4)+1.6(W	40.96	37.02	42.50	15.99	-19.29	9.95
206	0.9(DL+COLL+CL-1)+1.6(W	35.06	32.90	35.91	-5.49	-24.89	-4.04
207	0.9(DL+COLL+CL-2)+1.6(W	35.06	32.90	35.91	-5.49	-24.89	-4.04
208	0.9(DL+COLL+CL-3)+1.6(W	35.06	32.90	35.91	-5.49	-24.89	-4.04

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209	0.9(DL+COLL+CL-4)+1.6(W	35.06	32.90	35.91	-5.49	-24.89	-4.04
210	0.9(DL+COLL+CL-1)+1.6(W	38.73	36.57	39.58	4.43	-14.97	2.56
211	0.9(DL+COLL+CL-2)+1.6(W	38.73	36.57	39.58	4.43	-14.97	2.56
212	0.9(DL+COLL+CL-3)+1.6(W	38.73	36.57	39.58	4.43	-14.97	2.56
213	0.9(DL+COLL+CL-4)+1.6(W	38.73	36.57	39.58	4.43	-14.97	2.56
214	0.9(DL+COLL+CL-1)+1.6(W	30.62	28.46	31.47	-17.49	-36.89	-12.03
215	0.9(DL+COLL+CL-2)+1.6(W	30.62	28.46	31.47	-17.49	-36.89	-12.03
216	0.9(DL+COLL+CL-3)+1.6(W	30.62	28.46	31.47	-17.49	-36.89	-12.03
217	0.9(DL+COLL+CL-4)+1.6(W	30.62	28.46	31.47	-17.49	-36.89	-12.03
218	0.9(DL+COLL+CL-1)+1.6(W	34.29	32.13	35.14	-7.58	-26.97	-5.42
219	0.9(DL+COLL+CL-2)+1.6(W	34.29	32.13	35.14	-7.58	-26.97	-5.42
220	0.9(DL+COLL+CL-3)+1.6(W	34.29	32.13	35.14	-7.58	-26.97	-5.42
221	0.9(DL+COLL+CL-4)+1.6(W	34.29	32.13	35.14	-7.58	-26.97	-5.42
222	0.9(DL+COLL+CL-1)+1.0(S	41.91	40.36	42.51	11.04	-2.74	7.08
223	0.9(DL+COLL+CL-2)+1.0(S	41.91	40.36	42.51	11.04	-2.74	7.08
224	0.9(DL+COLL+CL-3)+1.0(S	41.91	40.36	42.51	11.04	-2.74	7.08
225	0.9(DL+COLL+CL-4)+1.0(S	41.91	40.36	42.51	11.04	-2.74	7.08
226	0.9(DL+COLL+CL-1)+1.0(S	43.36	40.56	44.45	18.87	-6.11	12.07
227	0.9(DL+COLL+CL-2)+1.0(S	43.36	40.56	44.45	18.87	-6.11	12.07
228	0.9(DL+COLL+CL-3)+1.0(S	43.36	40.56	44.45	18.87	-6.11	12.07
229	0.9(DL+COLL+CL-4)+1.0(S	43.36	40.56	44.45	18.87	-6.11	12.07

Sample calculation of BM & SF

Load case : 101

p_{1z} = 49.6-(49.6-34.61)*2.325/6= 43.79 kN/m^2

p_{2z} = 34.61+(49.6-34.61)*2.325/6= 40.42 kN/m^2

p_{3z} = 49.6-(49.6-34.61)*(2.325-0.526)/6= 45.11 kN/m^2

Overhang = 2.325 m

Effective depth = 0.526 m

BM_{1z} = (43.79-39.6)*2.325*2.325/2+0.5*(49.6-43.79)*2.325*2.325*0.67 = 21.85 kN-m/m

BM_{2z} = (40.42-39.6)*2.325*2.325/2-0.5*(-34.61+40.42)*2.325*2.325*0.67 = -8.31 kN-m/m

SF_{3z} = (45.11-39.6)*(2.325-0.526)+0.5*(49.6-45.11)*(2.325-0.526) = 13.95 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	92.56	0.335	0.446	0.0011	580.04	946.80	16ϕT	212.36	300	175	1148.9
At top	-56.91	-0.206	-0.274	-0.0007	-3.42	473.40	12ϕT	238.90	250	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 58.66 kN/m

Shear stress , τ_v= (58.66*1000)/(1000*526)

= 0.11 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	**175**	c/c
	At Top	12ϕT	**200**	c/c
Along Z Direction :				
	At Bottom	16ϕT	**175**	c/c
	At Top	12ϕT	**200**	c/c

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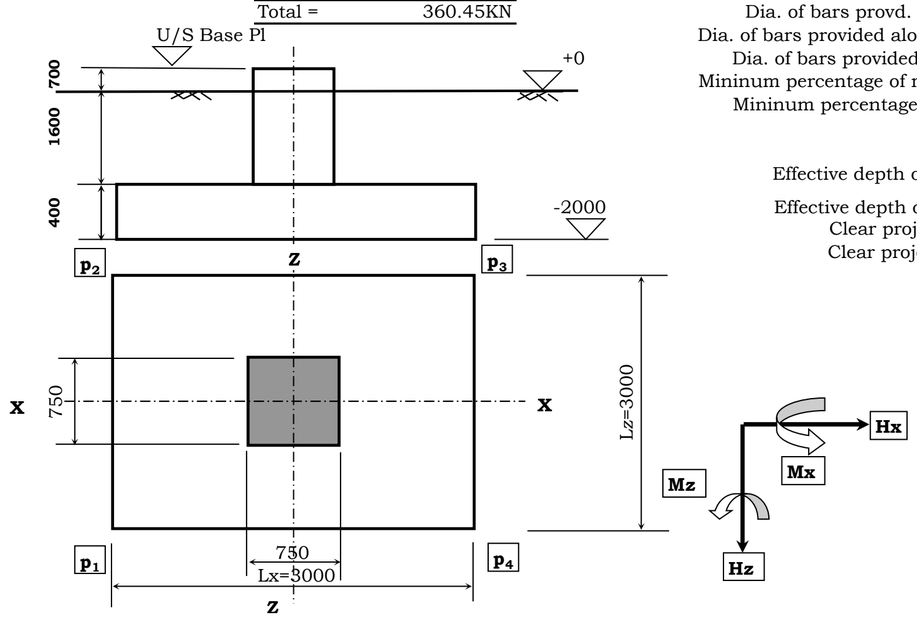
5.2 DESIGN OF CALCULATION OF FOOTING F2

	Along-X	Along-Z
Size of column :	0.75m	0.75m
Size of fdn. raft :	3.00m	3.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-2.00 m	
U/S level Base Plate :	0.70 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

Dead Load	DL
Live Load	LL
Crane Load	CL
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(3*3*0.4*24)+(0.75*0.75*2.3*24) =	117.45KN
Weight of soil =	((3*3)-(0.75*0.75))*1.6*18 =	243.00KN
	Total =	360.45KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 344mm = 0.344m

Effective depth of slab along Z direction = 332mm = 0.332m

Clear projection along Z direction = 1.125m

Clear projection along X direction = 1.125m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

M$_X$=Moment about X-X direction

Mz=Moment about Z-Z direction

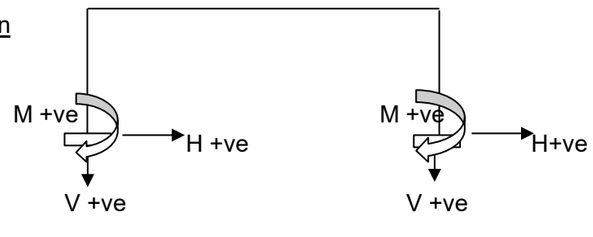
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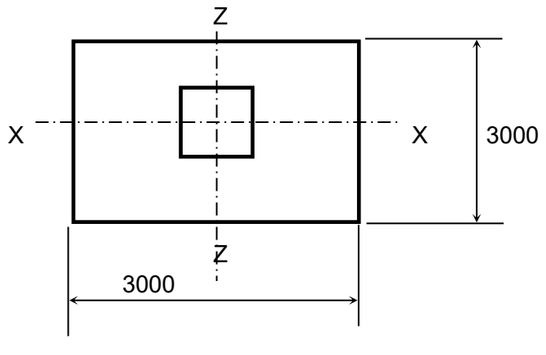
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DESIGN OF FOUNDATION MKD. F2

Sign convention





Height from BOF to top of column about X (l_{ux}) = 2700mm
Height from BOF to top of column about Z (l_{uz}) = 2700mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	13.1292		-6.702		0
2	COLLATERAL LOAD (COLL)	0.0012		-0.0024		0
3	ROOF LIVE LOAD (RLL)	36.4752		-20.7252		0
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	0.0036		-0.006		0
5	CRANE LOAD-1 (CL-1)	3.6888		-5.1264		0
6	CRANE LOAD-2 (CL-2)	-0.3096		0.4176		0
7	CRANE LOAD-3 (CL-3)	0.5304		-0.8856		0
8	CRANE LOAD-4 (CL-4)	-3.48		4.6848		0
9	WIND LEFT-1 (WLL-1)	-51.7152		18.342		0
10	WIND LEFT-2 (WLL-2)	-18.3552		7.35		0
11	WIND RIGHT-1 (WLR-1)	-82.6872		50.4228		0
12	WIND RIGHT-2 (WLR-2)	-16.1352		25.7268		0
13	WIND LONG-1 (WLLG-1)	-85.0944	29.0928		0	
14	WIND LONG-2 (WLLG-2)	-55.7412	21.4044		0	
15	WIND LONG-3 (WLLG-3)	-83.8356	27.3528		0	
16	WIND LONG-4 (WLLG-4)	-54.5208	19.8132		0	
17	SEISMIC LOAD SL'+'	0.1116	-0.306	-0.306	0	0
18	SEISMIC LOAD SL'-'	-0.1248	0.3252	0.3252	0	0
19	TEMPERATURE LOAD (TL1)	1.0284		-1.644		0
20	TEMPERATURE LOAD (TL2)	-1.0296		1.6488		0
21	BRACING REACTION (BR-WLLG1)	0	0		0	
22	BRACING REACTION (BR-WLLG2)	0	0		0	
23	BRACING REACTION (BR-WLLG3)	0	0		0	
24	BRACING REACTION (BR-WLLG4)	0	0		0	
25	BRACING REACTION (BR-SL'+')	0	0	0	0	0
26	BRACING REACTION (BR-SL'-')	0	0	0	0	0

Ultimate Condition

Combined Load Cases:		Load at top of foundation (Factored Load/Moment)			Load at bottom of foundation		
Load Case	Load Combination	Pu (kN)	Muz (kN-m)	Mux (kN-m)	Pu (kN)	Muz (kN-m)	Mux (kN-m)
101	1.4(DL+COLL)	18.38	0.0	25.3	378.8	0.0	25.3
102	1.2(DL+COLL+CL-1)+1.6(RLL+PLL / MLL)	78.5	0.0	127.9	439.0	0.0	127.9
103	1.2(DL+COLL+CL-2)+1.6(RLL+PLL / MLL)	73.8	0.0	109.9	434.2	0.0	109.9
104	1.2(DL+COLL+CL-3)+1.6(RLL+PLL / MLL)	74.8	0.0	114.2	435.2	0.0	114.2
105	1.2(DL+COLL+CL-4)+1.6(RLL+PLL / MLL)	69.9	0.0	96.1	430.4	0.0	96.1
106	1.4(DL+COLL+CL-1)	23.5	0.0	44.7	384.0	0.0	44.7
107	1.4(DL+COLL+CL-2)	17.9	0.0	23.8	378.4	0.0	23.8
108	1.4(DL+COLL+CL-3)	19.1	0.0	28.7	379.6	0.0	28.7
109	1.4(DL+COLL+CL-4)	13.5	0.0	7.6	374.0	0.0	7.6
110	1.2(DL+COLL+CL-1+TL1)+1.6(RLL+PLL / MLL)	79.8	0.0	133.2	440.2	0.0	133.2
111	1.2(DL+COLL+CL-2+TL1)+1.6(RLL+PLL / MLL)	75.0	0.0	115.3	435.4	0.0	115.3
112	1.2(DL+COLL+CL-3+TL1)+1.6(RLL+PLL / MLL)	76.0	0.0	119.5	436.4	0.0	119.5
113	1.2(DL+COLL+CL-4+TL1)+1.6(RLL+PLL / MLL)	71.2	0.0	101.4	431.6	0.0	101.4
114	1.2(DL+COLL+CL-1+TL2)+1.6(RLL+PLL / MLL)	77.3	0.0	122.5	437.8	0.0	122.5
115	1.2(DL+COLL+CL-2+TL2)+1.6(RLL+PLL / MLL)	72.5	0.0	104.6	433.0	0.0	104.6
116	1.2(DL+COLL+CL-3+TL2)+1.6(RLL+PLL / MLL)	73.5	0.0	108.8	434.0	0.0	108.8
117	1.2(DL+COLL+CL-4+TL2)+1.6(RLL+PLL / MLL)	68.7	0.0	90.8	429.2	0.0	90.8
118	1.2(DL+COLL+CL-1)+0.8(WLL-1)	-21.2	0.0	1.3	339.3	0.0	1.3
119	1.2(DL+COLL+CL-2)+0.8(WLL-1)	-26.0	0.0	19.2	334.5	0.0	19.2
120	1.2(DL+COLL+CL-3)+0.8(WLL-1)	-25.0	0.0	15.0	335.5	0.0	15.0
121	1.2(DL+COLL+CL-4)+0.8(WLL-1)	-29.8	0.0	33.1	330.7	0.0	33.1
122	1.2(DL+COLL+CL-1)+0.8(WLL-2)	5.5	0.0	22.5	365.9	0.0	22.5
123	1.2(DL+COLL+CL-2)+0.8(WLL-2)	0.7	0.0	4.5	361.2	0.0	4.5
124	1.2(DL+COLL+CL-3)+0.8(WLL-2)	1.7	0.0	8.7	362.2	0.0	8.7
125	1.2(DL+COLL+CL-4)+0.8(WLL-2)	-3.1	0.0	9.3	357.3	0.0	9.3
126	1.2(DL+COLL+CL-1)+0.8(WLR-1)	-46.0	0.0	70.6	314.5	0.0	70.6
127	1.2(DL+COLL+CL-2)+0.8(WLR-1)	-50.8	0.0	88.5	309.7	0.0	88.5
128	1.2(DL+COLL+CL-3)+0.8(WLR-1)	-49.8	0.0	84.3	310.7	0.0	84.3
129	1.2(DL+COLL+CL-4)+0.8(WLR-1)	-54.6	0.0	102.4	305.9	0.0	102.4
130	1.2(DL+COLL+CL-1)+0.8(WLR-2)	7.3	0.0	17.2	367.7	0.0	17.2
131	1.2(DL+COLL+CL-2)+0.8(WLR-2)	2.5	0.0	35.2	362.9	0.0	35.2
132	1.2(DL+COLL+CL-3)+0.8(WLR-2)	3.5	0.0	31.0	363.9	0.0	31.0
133	1.2(DL+COLL+CL-4)+0.8(WLR-2)	-1.3	0.0	49.0	359.1	0.0	49.0
134	1.2(DL+COLL+CL-1)+0.8(WLLG-1+BR-WLLG-1)	-47.9	62.8	38.3	312.6	62.8	38.3
135	1.2(DL+COLL+CL-2)+0.8(WLLG-1+BR-WLLG-1)	-52.7	62.8	20.4	307.8	62.8	20.4
136	1.2(DL+COLL+CL-3)+0.8(WLLG-1+BR-WLLG-1)	-51.7	62.8	24.6	308.8	62.8	24.6
137	1.2(DL+COLL+CL-4)+0.8(WLLG-1+BR-WLLG-1)	-56.5	62.8	6.5	304.0	62.8	6.5
138	1.2(DL+COLL+CL-1)+0.8(WLLG-2+BR-WLLG-2)	-24.4	46.2	38.3	336.0	46.2	38.3
139	1.2(DL+COLL+CL-2)+0.8(WLLG-2+BR-WLLG-2)	-29.2	46.2	20.4	331.2	46.2	20.4
140	1.2(DL+COLL+CL-3)+0.8(WLLG-2+BR-WLLG-2)	-28.2	46.2	24.6	332.3	46.2	24.6
141	1.2(DL+COLL+CL-4)+0.8(WLLG-2+BR-WLLG-2)	-33.0	46.2	6.5	327.4	46.2	6.5
142	1.2(DL+COLL+CL-1)+0.8(WLLG-3+BR-WLLG-3)	-46.9	59.1	38.3	313.6	59.1	38.3
143	1.2(DL+COLL+CL-2)+0.8(WLLG-3+BR-WLLG-3)	-51.7	59.1	20.4	308.8	59.1	20.4
144	1.2(DL+COLL+CL-3)+0.8(WLLG-3+BR-WLLG-3)	-50.7	59.1	24.6	309.8	59.1	24.6
145	1.2(DL+COLL+CL-4)+0.8(WLLG-3+BR-WLLG-3)	-55.5	59.1	6.5	305.0	59.1	6.5
146	1.2(DL+COLL+CL-1)+0.8(WLLG-4+BR-WLLG-4)	-23.4	42.8	38.3	337.0	42.8	38.3
147	1.2(DL+COLL+CL-2)+0.8(WLLG-4+BR-WLLG-4)	-28.2	42.8	20.4	332.2	42.8	20.4
148	1.2(DL+COLL+CL-3)+0.8(WLLG-4+BR-WLLG-4)	-27.2	42.8	24.6	333.2	42.8	24.6

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149	1.2(DL+COLL+CL-4)+0.8(WLLG-4+BR-WLLG-4)	-32.0	42.8	6.5	328.4	42.8	6.5
150	1.2(DL+COLL+CL-1)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-44.3	0.0	12.9	316.1	0.0	12.9
151	1.2(DL+COLL+CL-2)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-49.1	0.0	30.9	311.3	0.0	30.9
152	1.2(DL+COLL+CL-3)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-48.1	0.0	26.7	312.3	0.0	26.7
153	1.2(DL+COLL+CL-4)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-52.9	0.0	44.7	307.5	0.0	44.7
154	1.2(DL+COLL+CL-1)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	9.1	0.0	34.6	369.5	0.0	34.6
155	1.2(DL+COLL+CL-2)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	4.3	0.0	16.6	364.7	0.0	16.6
156	1.2(DL+COLL+CL-3)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	5.3	0.0	20.8	365.7	0.0	20.8
157	1.2(DL+COLL+CL-4)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	0.5	0.0	2.8	360.9	0.0	2.8
158	1.2(DL+COLL+CL-1)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-93.9	0.0	151.5	266.6	0.0	151.5
159	1.2(DL+COLL+CL-2)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-98.7	0.0	169.5	261.8	0.0	169.5
160	1.2(DL+COLL+CL-3)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-97.7	0.0	165.2	262.8	0.0	165.2
161	1.2(DL+COLL+CL-4)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-102.5	0.0	183.3	258.0	0.0	183.3
162	1.2(DL+COLL+CL-1)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	12.6	0.0	44.8	373.1	0.0	44.8
163	1.2(DL+COLL+CL-2)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	7.8	0.0	62.8	368.3	0.0	62.8
164	1.2(DL+COLL+CL-3)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	8.8	0.0	58.6	369.3	0.0	58.6
165	1.2(DL+COLL+CL-4)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	4.0	0.0	76.6	364.5	0.0	76.6
166	1.2(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-97.7	125.7	66.3	262.7	125.7	66.3
167	1.2(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-102.5	125.7	48.4	257.9	125.7	48.4
168	1.2(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-101.5	125.7	52.6	258.9	125.7	52.6
169	1.2(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-106.3	125.7	34.5	254.1	125.7	34.5
170	1.2(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-50.8	92.5	66.3	309.7	92.5	66.3
171	1.2(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-55.6	92.5	48.4	304.9	92.5	48.4
172	1.2(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-54.6	92.5	52.6	305.9	92.5	52.6
173	1.2(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-59.4	92.5	34.5	301.1	92.5	34.5
174	1.2(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-95.7	118.2	66.3	264.7	118.2	66.3
175	1.2(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-100.5	118.2	48.4	259.9	118.2	48.4
176	1.2(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-99.5	118.2	52.6	260.9	118.2	52.6
177	1.2(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-104.3	118.2	34.5	256.1	118.2	34.5
178	1.2(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-48.8	85.6	66.3	311.6	85.6	66.3
179	1.2(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-53.6	85.6	48.4	306.8	85.6	48.4
180	1.2(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-52.6	85.6	52.6	307.8	85.6	52.6
181	1.2(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-57.4	85.6	34.5	303.0	85.6	34.5
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	38.5	0.8	67.1	399.0	0.8	67.1
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	33.7	0.8	49.2	394.2	0.8	49.2
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	34.7	0.8	53.4	395.2	0.8	53.4
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	29.9	0.8	35.4	390.4	0.8	35.4
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	38.3	0.9	65.4	398.7	0.9	65.4
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-)+0.5(RLL+PLL / MLL	33.5	0.9	47.5	393.9	0.9	47.5
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	34.5	0.9	51.7	395.0	0.9	51.7
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	29.7	0.9	33.7	390.1	0.9	33.7
190	0.9(DL+COLL+CL-1)+1.6(WLL-1)	-67.6	0.0	50.5	292.8	0.0	50.5
191	0.9(DL+COLL+CL-2)+1.6(WLL-1)	-71.2	0.0	64.0	289.2	0.0	64.0
192	0.9(DL+COLL+CL-3)+1.6(WLL-1)	-70.4	0.0	60.8	290.0	0.0	60.8
193	0.9(DL+COLL+CL-4)+1.6(WLL-1)	-74.1	0.0	74.3	286.4	0.0	74.3
194	0.9(DL+COLL+CL-1)+1.6(WLL-2)	-14.2	0.0	3.0	346.2	0.0	3.0
195	0.9(DL+COLL+CL-2)+1.6(WLL-2)	-17.8	0.0	16.5	342.6	0.0	16.5
196	0.9(DL+COLL+CL-3)+1.6(WLL-2)	-17.1	0.0	13.3	343.4	0.0	13.3
197	0.9(DL+COLL+CL-4)+1.6(WLL-2)	-20.7	0.0	26.8	339.8	0.0	26.8
198	0.9(DL+COLL+CL-1)+1.6(WLR-1)	-117.2	0.0	189.1	243.3	0.0	189.1
199	0.9(DL+COLL+CL-2)+1.6(WLR-1)	-120.8	0.0	202.5	239.7	0.0	202.5
200	0.9(DL+COLL+CL-3)+1.6(WLR-1)	-120.0	0.0	199.4	240.4	0.0	199.4
201	0.9(DL+COLL+CL-4)+1.6(WLR-1)	-123.6	0.0	212.9	236.8	0.0	212.9
202	0.9(DL+COLL+CL-1)+1.6(WLR-2)	-10.7	0.0	82.4	349.8	0.0	82.4
203	0.9(DL+COLL+CL-2)+1.6(WLR-2)	-14.3	0.0	95.9	346.2	0.0	95.9
204	0.9(DL+COLL+CL-3)+1.6(WLR-2)	-13.5	0.0	92.7	346.9	0.0	92.7
205	0.9(DL+COLL+CL-4)+1.6(WLR-2)	-17.1	0.0	106.2	343.3	0.0	106.2
206	0.9(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)	-121.0	125.7	28.7	239.4	125.7	28.7
207	0.9(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)	-124.6	125.7	15.3	235.8	125.7	15.3
208	0.9(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)	-123.9	125.7	18.4	236.6	125.7	18.4
209	0.9(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)	-127.5	125.7	4.9	233.0	125.7	4.9
210	0.9(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)	-74.0	92.5	28.7	286.4	92.5	28.7
211	0.9(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)	-77.6	92.5	15.3	282.8	92.5	15.3
212	0.9(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)	-76.9	92.5	18.4	283.6	92.5	18.4
213	0.9(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)	-80.5	92.5	4.9	279.9	92.5	4.9
214	0.9(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)	-119.0	118.2	28.7	241.5	118.2	28.7
215	0.9(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)	-122.6	118.2	15.3	237.9	118.2	15.3
216	0.9(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)	-121.8	118.2	18.4	238.6	118.2	18.4
217	0.9(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)	-125.5	118.2	4.9	235.0	118.2	4.9
218	0.9(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG-4)	-72.1	85.6	28.7	288.4	85.6	28.7
219	0.9(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG-4)	-75.7	85.6	15.3	284.8	85.6	15.3
220	0.9(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG-4)	-74.9	85.6	18.4	285.5	85.6	18.4
221	0.9(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG-4)	-78.5	85.6	4.9	281.9	85.6	4.9
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')	15.2	0.8	29.6	375.7	0.8	29.6
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')	11.7	0.8	16.1	372.1	0.8	16.1
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')	12.4	0.8	19.3	372.9	0.8	19.3
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')	8.8	0.8	5.7	369.2	0.8	5.7
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')	15.0	0.9	27.9	375.5	0.9	27.9
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-')	11.4	0.9	14.4	371.9	0.9	14.4
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')	12.2	0.9	17.6	372.6	0.9	17.6
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')	8.6	0.9	4.0	369.0	0.9	4.0

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Allowable stress design											
Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
301	1.0(DL+COLL)	13.1304	0	0.0	-6.7	-18.1	373.6	0.0	0.0	-6.7	-18.1
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL	54.3264	0	0.0	-34.2	-92.4	414.8	0.0	0.0	-34.2	-92.4
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL / MLL	50.328	0	0.0	-28.7	-77.4	410.8	0.0	0.0	-28.7	-77.4
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL / MLL	51.168	0	0.0	-30.0	-80.9	411.6	0.0	0.0	-30.0	-80.9
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL / MLL	47.1576	0	0.0	-24.4	-65.9	407.6	0.0	0.0	-24.4	-65.9
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL / MLL	52.2684	0	0.0	-30.9	-83.5	412.7	0.0	0.0	-30.9	-83.5
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL / MLL	48.27	0	0.0	-25.4	-68.5	408.7	0.0	0.0	-25.4	-68.5
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL / MLL	49.11	0	0.0	-26.7	-72.0	409.6	0.0	0.0	-26.7	-72.0
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL / MLL	45.0996	0	0.0	-21.1	-57.0	405.5	0.0	0.0	-21.1	-57.0
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL+PLL	44.95	0	0.0	-28.6	-77.3	405.4	0.0	0.0	-28.6	-77.3
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL+PLL	40.95	0	0.0	-23.1	-62.3	401.4	0.0	0.0	-23.1	-62.3
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL+PLL	41.79	0	0.0	-24.4	-65.8	402.2	0.0	0.0	-24.4	-65.8
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL+PLL	37.78	0	0.0	-18.8	-50.8	398.2	0.0	0.0	-18.8	-50.8
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL+PLL	43.41	0	0.0	-26.1	-70.6	403.9	0.0	0.0	-26.1	-70.6
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL+PLL	39.41	0	0.0	-20.6	-55.6	399.9	0.0	0.0	-20.6	-55.6
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL+PLL	40.25	0	0.0	-21.9	-59.1	400.7	0.0	0.0	-21.9	-59.1
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL+PLL	36.24	0	0.0	-16.3	-44.1	396.7	0.0	0.0	-16.3	-44.1
318	1.0(DL+COLL+CL-1)	16.8192	0	0.0	-11.8	-31.9	377.3	0.0	0.0	-11.8	-31.9
319	1.0(DL+COLL+CL-2)	12.8208	0	0.0	-6.3	-17.0	373.3	0.0	0.0	-6.3	-17.0
320	1.0(DL+COLL+CL-3)	13.6608	0	0.0	-7.6	-20.5	374.1	0.0	0.0	-7.6	-20.5
321	1.0(DL+COLL+CL-4)	9.6504	0	0.0	-2.0	-5.5	370.1	0.0	0.0	-2.0	-5.5
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	-34.896	0	0.0	6.5	17.6	325.6	0.0	0.0	6.5	17.6
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	-38.8944	0	0.0	12.1	32.5	321.6	0.0	0.0	12.1	32.5
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	-38.0544	0	0.0	10.8	29.0	322.4	0.0	0.0	10.8	29.0
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	-42.0648	0	0.0	16.3	44.1	318.4	0.0	0.0	16.3	44.1
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	-1.536	0	0.0	-4.5	-12.1	358.9	0.0	0.0	-4.5	-12.1
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	-5.5344	0	0.0	1.1	2.9	354.9	0.0	0.0	1.1	2.9
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	-4.6944	0	0.0	-0.2	-0.6	355.8	0.0	0.0	-0.2	-0.6
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	-8.7048	0	0.0	5.3	14.4	351.7	0.0	0.0	5.3	14.4
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	-65.868	0	0.0	38.6	104.2	294.6	0.0	0.0	38.6	104.2
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	-69.8664	0	0.0	44.1	119.2	290.6	0.0	0.0	44.1	119.2
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	-69.0264	0	0.0	42.8	115.6	291.4	0.0	0.0	42.8	115.6
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	-73.0368	0	0.0	48.4	130.7	287.4	0.0	0.0	48.4	130.7
334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	0.684	0	0.0	13.9	37.5	361.1	0.0	0.0	13.9	37.5
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	-3.3144	0	0.0	19.4	52.5	357.1	0.0	0.0	19.4	52.5
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	-2.4744	0	0.0	18.1	49.0	358.0	0.0	0.0	18.1	49.0
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	-6.4848	0	0.0	23.7	64.0	354.0	0.0	0.0	23.7	64.0
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	-68.2752	29.0928	29.1	-6.7	-23.2	292.2	29.1	29.1	-6.7	-23.2
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	-72.2736	29.0928	29.1	-6.7	-17.7	288.2	29.1	29.1	-6.7	-17.7
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	-71.4336	29.0928	29.1	-6.7	-19.0	289.0	29.1	29.1	-6.7	-19.0
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	-75.444	29.0928	29.1	-6.7	-13.4	285.0	29.1	29.1	-6.7	-13.4
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	-38.922	21.4044	21.4	-6.7	-23.2	321.5	21.4	21.4	-6.7	-23.2
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	-42.9204	21.4044	21.4	-6.7	-17.7	317.5	21.4	21.4	-6.7	-17.7
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	-42.0804	21.4044	21.4	-6.7	-19.0	318.4	21.4	21.4	-6.7	-19.0
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	-46.0908	21.4044	21.4	-6.7	-13.4	314.4	21.4	21.4	-6.7	-13.4
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-67.0164	27.3528	27.4	-6.7	-23.2	293.4	27.4	27.4	-6.7	-23.2
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	-71.0148	27.3528	27.4	-6.7	-17.7	289.4	27.4	27.4	-6.7	-17.7
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	-70.1748	27.3528	27.4	-6.7	-19.0	290.3	27.4	27.4	-6.7	-19.0
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	-74.1852	27.3528	27.4	-6.7	-13.4	286.3	27.4	27.4	-6.7	-13.4
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-37.7016	19.8132	19.8	-6.7	-23.2	322.7	19.8	19.8	-6.7	-23.2
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	-41.7	19.8132	19.8	-6.7	-17.7	318.8	19.8	19.8	-6.7	-17.7
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	-40.86	19.8132	19.8	-6.7	-19.0	319.6	19.8	19.8	-6.7	-19.0
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	-44.8704	19.8132	19.8	-6.7	-13.4	315.6	19.8	19.8	-6.7	-13.4
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(RLL+	5.39	0	0.0	-13.6	-36.8	365.8	0.0	0.0	-13.6	-36.8
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(RLL+	1.39	0	0.0	-8.1	-21.8	361.8	0.0	0.0	-8.1	-21.8
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(RLL+	2.23	0	0.0	-9.4	-25.3	362.7	0.0	0.0	-9.4	-25.3
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(RLL+	-1.78	0	0.0	-3.8	-10.3	358.7	0.0	0.0	-3.8	-10.3
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(RLL+	30.41	0	0.0	-21.9	-59.0	390.9	0.0	0.0	-21.9	-59.0
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(RLL+	26.41	0	0.0	-16.3	-44.1	386.9	0.0	0.0	-16.3	-44.1
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(RLL+	27.25	0	0.0	-17.6	-47.6	387.7	0.0	0.0	-17.6	-47.6
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(RLL+	23.24	0	0.0	-12.1	-32.5	383.7	0.0	0.0	-12.1	-32.5
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(RLL+	-17.84	0	0.0	10.4	28.2	342.6	0.0	0.0	10.4	28.2
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(RLL+	-21.84	0	0.0	16.0	43.2	338.6	0.0	0.0	16.0	43.2
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(RLL+	-21.00	0	0.0	14.7	39.6	339.5	0.0	0.0	14.7	39.6
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(RLL+	-25.01	0	0.0	20.2	54.7	335.4	0.0	0.0	20.2	54.7
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(RLL+	32.08	0	0.0	-8.1	-21.8	392.5	0.0	0.0	-8.1	-21.8
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(RLL+	28.08	0	0.0	-2.5	-6.9	388.5	0.0	0.0	-2.5	-6.9
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(RLL+	28.92	0	0.0	-3.8	-10.4	389.4	0.0	0.0	-3.8	-10.4
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(RLL+	24.91	0	0.0	1.7	4.7	385.4	0.0	0.0	1.7	4.7
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-WLLG	-19.64	21.8196	21.8	-27.4	-73.9	340.8	21.8	21.8	-27.4	-73.9
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-WLLG	-23.64	21.8196	21.8	-21.8	-59.0	336.8	21.8	21.8	-21.8	-59.0
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-WLLG	-22.80	21.8196	21.8	-23.1	-62.5	337.6	21.8	21.8	-23.1	-62.5
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-WLLG	-26.81	21.8196	21.8	-17.6	-47.4	333.6	21.8	21.8	-17.6	-47.4
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-WLLG	2.37	16.0533	16.1	-27.4	-73.9	362.8	16.1	16.1	-27.4	-73.9
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-WLLG	-1.63	16.0533	16.1	-21.8	-59.0	358.8	16.1	16.1	-21.8	-59.0
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-WLLG	-0.79	16.0533	16.1	-23.1	-62.5	359.7	16.1	16.1	-23.1	-62.5
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-WLLG	-4.80	16.0533	16.1	-17.6	-47.4	355.7	16.1	16.1	-17.6	-47.4
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-WLLG	-18.70	20.5146	20.5	-27.4	-73.9	341.8	20.5	20.5	-27.4	-73.9
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-WLLG	-22.70	20.5146	20.5	-21.8	-59.0	337.8	20.5	20.5	-21.8	-59.0
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-WLLG	-21.86	20.5146	20.5	-23.1	-62.5	338.6	20.5	20.5	-23.1	-62.5
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-WLLG	-25.87	20.5146	20.5	-17.6	-47.4	334.6	20.5	20.5	-17.6	-47.4
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-WLLG	3.29	14.8599	14.9	-27.4	-73.9	363.7	14.9	14.9	-27.4	-73.9
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-WLLG	-0.71	14.8599	14.9	-21.8	-59.0	359.7	14.9	14.9	-21.8	-59.0
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-WLLG	0.13	14.8599	14.9	-23.1	-62.5	360.6	14.9	14.9	-23.1	-62.5
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-WLLG	-3.88	14.8599	14.9	-17.6	-47.4	356.6	14.9	14.9	-17.6	-47.4
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+')	16.90	-0.2295	-0.2	-12.1	-32.2	377.4	-0.2	-0.2	-12.1	-32.2
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+")	12.90	-0.2295	-0.2	-6.5	-17.2	373.4	-0.2	-0.2	-6.5	-17.2
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+')	13.74	-0.2295	-0.2	-7.8	-20.7	374.2	-0.2	-0.2	-7.8	-20.7

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389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+')	9.73	-0.2295	-0.2	-2.2	-5.7	370.2	-0.2	-0.2	-2.2	-5.7
390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	16.73	0.2439	0.2	-11.6	-31.7	377.2	0.2	0.2	-11.6	-31.7
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	12.73	0.2439	0.2	-6.0	-16.7	373.2	0.2	0.2	-6.0	-16.7
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	13.57	0.2439	0.2	-7.3	-20.2	374.0	0.2	0.2	-7.3	-20.2
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	9.56	0.2439	0.2	-1.8	-5.2	370.0	0.2	0.2	-1.8	-5.2
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'+')+0	44.24	-0.16065	-0.2	-27.5	-74.1	404.7	-0.2	-0.2	-27.5	-74.1
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'+')+0	40.24	-0.16065	-0.2	-22.0	-59.1	400.7	-0.2	-0.2	-22.0	-59.1
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'+')+0	41.08	-0.16065	-0.2	-23.3	-62.6	401.5	-0.2	-0.2	-23.3	-62.6
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'+')+0	37.07	-0.16065	-0.2	-17.7	-47.6	397.5	-0.2	-0.2	-17.7	-47.6
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-')+0.7	44.11	0.17073	0.2	-27.2	-73.8	404.6	0.2	0.2	-27.2	-73.8
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-')+0.7	40.11	0.17073	0.2	-21.7	-58.8	400.6	0.2	0.2	-21.7	-58.8
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-')+0.7	40.95	0.17073	0.2	-23.0	-62.3	401.4	0.2	0.2	-23.0	-62.3
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-')+0.7	36.94	0.17073	0.2	-17.4	-47.3	397.4	0.2	0.2	-17.4	-47.3
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	-41.62	0	0.0	11.2	30.4	318.8	0.0	0.0	11.2	30.4
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	-44.02	0	0.0	14.6	39.3	316.4	0.0	0.0	14.6	39.3
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	-43.52	0	0.0	13.8	37.2	316.9	0.0	0.0	13.8	37.2
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	-45.92	0	0.0	17.1	46.3	314.5	0.0	0.0	17.1	46.3
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	-8.26368	0	0.0	0.3	0.7	352.2	0.0	0.0	0.3	0.7
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	-10.66272	0	0.0	3.6	9.7	349.8	0.0	0.0	3.6	9.7
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	-10.15872	0	0.0	2.8	7.5	350.3	0.0	0.0	2.8	7.5
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	-12.56496	0	0.0	6.1	16.6	347.9	0.0	0.0	6.1	16.6
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	-72.59568	0	0.0	43.3	117.0	287.9	0.0	0.0	43.3	117.0
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	-74.99472	0	0.0	46.7	126.0	285.5	0.0	0.0	46.7	126.0
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	-74.49072	0	0.0	45.9	123.8	286.0	0.0	0.0	45.9	123.8
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	-76.89696	0	0.0	49.2	132.9	283.6	0.0	0.0	49.2	132.9
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	-6.04368	0	0.0	18.6	50.3	354.4	0.0	0.0	18.6	50.3
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	-8.44272	0	0.0	22.0	59.3	352.0	0.0	0.0	22.0	59.3
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	-7.93872	0	0.0	21.2	57.2	352.5	0.0	0.0	21.2	57.2
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	-10.34496	0	0.0	24.5	66.2	350.1	0.0	0.0	24.5	66.2
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	-75.00288	29.0928	29.1	-4.0	-13.9	285.4	29.1	29.1	-4.0	-13.9
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	-77.40192	29.0928	29.1	-4.0	-10.6	283.0	29.1	29.1	-4.0	-10.6
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	-76.89792	29.0928	29.1	-4.0	-11.4	283.6	29.1	29.1	-4.0	-11.4
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	-79.30416	29.0928	29.1	-4.0	-8.1	281.1	29.1	29.1	-4.0	-8.1
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	-45.64968	21.4044	21.4	-4.0	-13.9	314.8	21.4	21.4	-4.0	-13.9
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	-48.04872	21.4044	21.4	-4.0	-10.6	312.4	21.4	21.4	-4.0	-10.6
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	-47.54472	21.4044	21.4	-4.0	-11.4	312.9	21.4	21.4	-4.0	-11.4
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	-49.95096	21.4044	21.4	-4.0	-8.1	310.5	21.4	21.4	-4.0	-8.1
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-73.74408	27.3528	27.4	-4.0	-13.9	286.7	27.4	27.4	-4.0	-13.9
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	-76.14312	27.3528	27.4	-4.0	-10.6	284.3	27.4	27.4	-4.0	-10.6
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	-75.63912	27.3528	27.4	-4.0	-11.4	284.8	27.4	27.4	-4.0	-11.4
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	-78.04536	27.3528	27.4	-4.0	-8.1	282.4	27.4	27.4	-4.0	-8.1
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-WLLG-	-44.42928	19.8132	19.8	-4.0	-13.9	316.0	19.8	19.8	-4.0	-13.9
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	-46.82832	19.8132	19.8	-4.0	-10.6	313.6	19.8	19.8	-4.0	-10.6
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	-46.32432	19.8132	19.8	-4.0	-11.4	314.1	19.8	19.8	-4.0	-11.4
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	-48.73056	19.8132	19.8	-4.0	-8.1	311.7	19.8	19.8	-4.0	-8.1
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	10.17	-0.2142	-0.2	-7.3	-19.4	370.6	-0.2	-0.2	-7.3	-19.4
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	7.77	-0.2142	-0.2	-4.0	-10.4	368.2	-0.2	-0.2	-4.0	-10.4
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	8.27	-0.2142	-0.2	-4.8	-12.5	368.7	-0.2	-0.2	-4.8	-12.5
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	5.87	-0.2142	-0.2	-1.4	-3.5	366.3	-0.2	-0.2	-1.4	-3.5
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	-23.07	-0.2142	-0.2	6.8	18.6	337.4	-0.2	-0.2	6.8	18.6
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	-3.06	-0.2142	-0.2	0.2	0.8	357.4	-0.2	-0.2	0.2	0.8
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	-41.66	-0.2142	-0.2	26.0	70.6	318.8	-0.2	-0.2	26.0	70.6
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	-1.72	-0.2142	-0.2	11.2	30.6	358.7	-0.2	-0.2	11.2	30.6

Check for overturning and sliding:

Load Case	Load Combination	Load at bottom of foundation					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
301	1.0(DL+COLL)	373.58	0.00	0.00	6.70	18.10	NA	27.86	NA	20.06
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL /	414.78	0.00	0.00	34.21	92.36	NA	6.06	NA	4.37
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL /	410.78	0.00	0.00	28.66	77.39	NA	7.17	NA	5.16
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL /	411.62	0.00	0.00	29.97	80.91	NA	6.87	NA	4.95
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL /	407.61	0.00	0.00	24.39	65.87	NA	8.35	NA	6.02
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL /	412.72	0.00	0.00	30.91	83.47	NA	6.68	NA	4.81
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL /	408.72	0.00	0.00	25.37	68.50	NA	8.06	NA	5.80
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL /	409.56	0.00	0.00	26.67	72.02	NA	7.68	NA	5.53
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL /	405.55	0.00	0.00	21.10	56.98	NA	9.61	NA	6.92
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL	405.40	0.00	0.00	28.61	77.25	NA	7.08	NA	5.10
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL	401.40	0.00	0.00	23.07	62.28	NA	8.70	NA	6.26
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL	402.24	0.00	0.00	24.37	65.80	NA	8.25	NA	5.94
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL	398.23	0.00	0.00	18.80	50.76	NA	10.59	NA	7.63
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL	403.86	0.00	0.00	26.14	70.59	NA	7.72	NA	5.56
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL	399.86	0.00	0.00	20.60	55.62	NA	9.71	NA	6.99
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL	400.70	0.00	0.00	21.90	59.13	NA	9.15	NA	6.59
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL	396.69	0.00	0.00	16.33	44.09	NA	12.14	NA	8.74
318	1.0(DL+COLL+CL-1)	377.27	0.00	0.00	11.83	31.94	NA	15.94	NA	11.48
319	1.0(DL+COLL+CL-2)	373.27	0.00	0.00	6.29	16.97	NA	29.69	NA	21.37
320	1.0(DL+COLL+CL-3)	374.11	0.00	0.00	7.59	20.49	NA	24.64	NA	17.74
321	1.0(DL+COLL+CL-4)	370.10	0.00	0.00	2.02	5.45	NA	91.63	NA	65.97
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	325.55	0.00	0.00	6.51	17.58	NA	25.00	NA	18.00
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	321.56	0.00	0.00	12.06	32.55	NA	13.34	NA	9.60
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	322.40	0.00	0.00	10.75	29.03	NA	14.99	NA	10.79
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	318.39	0.00	0.00	16.32	44.07	NA	9.75	NA	7.02
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	358.91	0.00	0.00	4.48	12.10	NA	40.05	NA	28.84
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	354.92	0.00	0.00	1.06	2.87	NA	166.91	NA	120.17
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	355.76	0.00	0.00	0.24	0.65	NA	741.16	NA	533.63
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	351.75	0.00	0.00	5.33	14.39	NA	32.99	NA	23.76
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	294.58	0.00	0.00	38.59	104.20	NA	3.82	NA	2.75
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	290.58	0.00	0.00	44.14	119.17	NA	3.29	NA	2.37
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	291.42	0.00	0.00	42.83	115.65	NA	3.40	NA	2.45
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	287.41	0.00	0.00	48.40	130.69	NA	2.97	NA	2.14
334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	361.13	0.00	0.00	13.90	37.52	NA	12.99	NA	9.36

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335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	357.14	0.00	0.00	19.44	52.49	NA	9.19	NA	6.61
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	357.98	0.00	0.00	18.14	48.97	NA	9.87	NA	7.11
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	353.97	0.00	0.00	23.71	64.01	NA	7.47	NA	5.38
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	292.17	29.09	29.09	6.70	23.23	13.56	16.98	4.02	15.69
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	288.18	29.09	29.09	6.70	17.68	13.37	22.00	3.96	15.47
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	289.02	29.09	29.09	6.70	18.99	13.41	20.55	3.97	15.52
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	285.01	29.09	29.09	6.70	13.42	13.23	28.68	3.92	15.30
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	321.53	21.40	21.40	6.70	23.23	20.28	18.69	6.01	17.26
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	317.53	21.40	21.40	6.70	17.68	20.03	24.24	5.93	17.05
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	318.37	21.40	21.40	6.70	18.99	20.08	22.64	5.95	17.10
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	314.36	21.40	21.40	6.70	13.42	19.83	31.63	5.87	16.88
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	293.43	27.35	27.35	6.70	23.23	14.48	17.05	4.29	15.76
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	289.44	27.35	27.35	6.70	17.68	14.29	22.10	4.23	15.54
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	290.28	27.35	27.35	6.70	18.99	14.33	20.64	4.24	15.59
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	286.26	27.35	27.35	6.70	13.42	14.13	28.80	4.19	15.37
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	322.75	19.81	19.81	6.70	23.23	21.99	18.76	6.52	17.33
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	318.75	19.81	19.81	6.70	17.68	21.72	24.33	6.44	17.12
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	319.59	19.81	19.81	6.70	18.99	21.78	22.72	6.45	17.16
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	315.58	19.81	19.81	6.70	13.42	21.50	31.75	6.37	16.95
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(F	365.84	0.00	0.00	13.62	36.78	NA	13.43	NA	9.67
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(F	361.84	0.00	0.00	8.08	21.81	NA	22.39	NA	16.12
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(F	362.68	0.00	0.00	9.38	25.33	NA	19.33	NA	13.92
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(F	358.67	0.00	0.00	3.81	10.29	NA	47.05	NA	33.88
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(F	390.86	0.00	0.00	21.87	59.04	NA	8.94	NA	6.43
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(F	386.86	0.00	0.00	16.32	44.07	NA	11.85	NA	8.53
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(F	387.70	0.00	0.00	17.63	47.59	NA	11.00	NA	7.92
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(F	383.69	0.00	0.00	12.06	32.55	NA	15.91	NA	11.46
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(I	342.61	0.00	0.00	10.44	28.18	NA	16.41	NA	11.82
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(I	338.61	0.00	0.00	15.98	43.15	NA	10.59	NA	7.63
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(I	339.45	0.00	0.00	14.68	39.63	NA	11.56	NA	8.33
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(I	335.44	0.00	0.00	20.25	54.67	NA	8.28	NA	5.96
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(I	392.53	0.00	0.00	8.08	21.83	NA	24.28	NA	17.48
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(I	388.53	0.00	0.00	2.54	6.86	NA	76.48	NA	55.06
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(I	389.37	0.00	0.00	3.84	10.38	NA	50.66	NA	36.47
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(I	385.36	0.00	0.00	1.73	4.66	NA	111.56	NA	80.32
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-V	340.81	21.82	21.82	27.38	73.92	21.09	6.22	6.25	4.48
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-V	336.81	21.82	21.82	21.84	58.96	20.84	7.71	6.17	5.55
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-V	337.65	21.82	21.82	23.14	62.47	20.89	7.30	6.19	5.25
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-V	333.64	21.82	21.82	17.57	47.43	20.64	9.50	6.12	6.84
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-V	362.82	16.05	16.05	27.38	73.92	30.51	6.63	9.04	4.77
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-V	358.82	16.05	16.05	21.84	58.96	30.18	8.22	8.94	5.92
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-V	359.66	16.05	16.05	23.14	62.47	30.25	7.77	8.96	5.60
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-V	355.65	16.05	16.05	17.57	47.43	29.91	10.12	8.86	7.29
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-V	341.75	20.51	20.51	27.38	73.92	22.49	6.24	6.66	4.49
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-V	337.75	20.51	20.51	21.84	58.96	22.23	7.73	6.59	5.57
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-V	338.59	20.51	20.51	23.14	62.47	22.28	7.32	6.60	5.27
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-V	334.58	20.51	20.51	17.57	47.43	22.02	9.52	6.52	6.86
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-V	363.74	14.86	14.86	27.38	73.92	33.05	6.64	9.79	4.78
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-V	359.74	14.86	14.86	21.84	58.96	32.68	8.24	9.68	5.93
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-V	360.58	14.86	14.86	23.14	62.47	32.76	7.79	9.71	5.61
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-V	356.57	14.86	14.86	17.57	47.43	32.39	10.15	9.60	7.31
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+	377.35	-0.23	-0.23	12.06	32.17	2219.72	15.83	657.70	11.26
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+	373.35	-0.23	-0.23	6.52	17.20	2196.20	29.30	650.73	20.63
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+	374.19	-0.23	-0.23	7.82	20.72	2201.14	24.38	652.19	17.23
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+	370.18	-0.23	-0.23	2.25	5.68	2177.55	87.95	645.20	59.25
390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	377.18	0.24	0.24	11.59	31.70	2087.69	16.06	618.57	11.72
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	373.18	0.24	0.24	6.04	16.73	2065.56	30.11	612.02	22.23
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	374.02	0.24	0.24	7.35	20.25	2070.21	24.94	613.39	18.33
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	370.01	0.24	0.24	1.78	5.21	2048.01	95.89	606.82	75.01
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'	404.69	-0.16	-0.16	27.54	74.08	3400.73	7.37	1007.62	5.29
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'	400.69	-0.16	-0.16	22.00	59.12	3367.13	9.15	997.67	6.56
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'	401.53	-0.16	-0.16	23.30	62.63	3374.19	8.65	999.76	6.20
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'	397.52	-0.16	-0.16	17.73	47.59	3340.49	11.28	989.77	8.07
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-	404.56	0.17	0.17	27.21	73.75	3198.97	7.41	947.84	5.35
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-	400.56	0.17	0.17	21.66	58.78	3167.35	9.20	938.47	6.66
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-	401.40	0.17	0.17	22.97	62.30	3173.99	8.70	940.44	6.29
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-'	397.39	0.17	0.17	17.40	47.26	3142.28	11.35	931.05	8.22
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	318.83	0.00	0.00	11.24	30.36	NA	14.18	NA	10.21
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	316.43	0.00	0.00	14.57	39.34	NA	10.86	NA	7.82
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	316.93	0.00	0.00	13.79	37.23	NA	11.49	NA	8.27
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	314.53	0.00	0.00	17.13	46.25	NA	9.18	NA	6.61
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	352.19	0.00	0.00	0.25	0.68	NA	700.12	NA	504.08
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	349.79	0.00	0.00	3.58	9.66	NA	48.88	NA	35.19
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	350.29	0.00	0.00	2.80	7.55	NA	62.64	NA	45.10
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	347.89	0.00	0.00	6.14	16.57	NA	28.34	NA	20.40
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	287.85	0.00	0.00	43.32	116.98	NA	3.32	NA	2.39
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	285.46	0.00	0.00	46.65	125.96	NA	3.06	NA	2.20
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	285.96	0.00	0.00	45.87	123.85	NA	3.12	NA	2.24
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	283.55	0.00	0.00	49.21	132.87	NA	2.88	NA	2.07
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	354.41	0.00	0.00	18.63	50.30	NA	9.51	NA	6.85
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	352.01	0.00	0.00	21.95	59.28	NA	8.02	NA	5.77
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	352.51	0.00	0.00	21.17	57.17	NA	8.32	NA	5.99
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	350.11	0.00	0.00	24.52	66.19	NA	7.14	NA	5.14
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	285.45	29.09	29.09	4.02	13.94	13.25	27.65	3.92	25.55
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	283.05	29.09	29.09	4.02	10.61	13.13	36.01	3.89	25.33
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	283.55	29.09	29.09	4.02	11.39	13.16	33.60	3.90	25.38
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	281.15	29.09	29.09	4.02	8.05	13.05	47.15	3.87	25.16
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	314.80	21.40	21.40	4.02	13.94	19.85	30.49	5.88	28.17
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	312.40	21.40	21.40	4.02	10.61	19.70	39.75	5.84	27.96
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	312.91	21.40	21.40	4.02	11.39	19.74	37.08	5.85	28.00
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	310.50	21.40	21.40	4.02	8.05	19.58	52.07	5.80	27.79
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	286.71	27.35	27.35	4.02	13.94	14.15	27.77	4.19	25.66
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	284.31	27.35	27.35	4.02	10.61	14.03	36.17	4.16	25.44
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	284.81	27.35	27.35	4.02	11.39	14.06	33.75	4.16	25.49
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	282.40	27.35	27.35	4.02	8.05	13.94	47.36	4.13	25.27
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-W	316.02	19.81	19.81	4.02	13.94	21.53	30.61	6.38	28.28
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	313.62	19.81	19.81	4.02	10.61	21.37	39.90	6.33	28.07
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	314.13	19.81	19.81	4.02	11.39	21.40	37.22	6.34	28.11
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	311.72	19.81	19.81	4.02	8.05	21.24	52.27	6.29	27.90
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	370.62	-0.21	-0.21	7.31	19.38	2335.84	25.82	692.10	18.25
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	368.22	-0.21	-0.21	3.99	10.40	2320.72	47.80	687.62	33.25
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	368.72	-0.21	-0.21	4.77	12.51	2323.89	39.79	688.56	27.84
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	366.32	-0.21	-0.21	1.43	3.49	2308.73	141.86	684.07	92.48
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	337.38	-0.21	-0.21	6.77	18.64	2126.33	24.44	630.02	17.94
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	357.39	-0.21	-0.21	0.17	0.83	2252.48	580.13	667.40	743.02
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	318.79	-0.21	-0.21	26.02	70.61	2009.21	6.10	595.32	4.41
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	358.73	-0.21	-0.21	11.20	30.60	2260.87	15.83	669.89	11.53
						Min.=	**13.05**	**2.88**	**3.87**	**2.07**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F2

Load Combination & Calculation of Base pressure

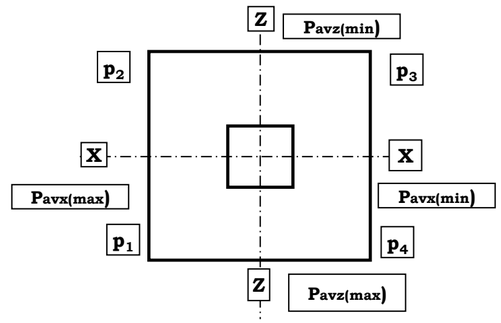
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 436 kN/m^2 For WL/SL case = 536 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
301	1.0(DL+COLL)	373.58	0.00	-18.10	41.51	0.00	4.02	45.53		37.49	45.53	37.49	9.53	41.51	41.51	45.53	37.49	**O.K**
302	1.0(DL+COLL+CL-1+TL1)+1.(	414.78	0.00	-92.36	46.09	0.00	20.52	66.61		25.56	66.61	25.56	30.61	46.09	46.09	66.61	25.56	**O.K**
303	1.0(DL+COLL+CL-2+TL1)+1.(	410.78	0.00	-77.39	45.64	0.00	17.20	62.84		28.44	62.84	28.44	26.84	45.64	45.64	62.84	28.44	**O.K**
304	1.0(DL+COLL+CL-3+TL1)+1.(	411.62	0.00	-80.91	45.74	0.00	17.98	63.71		27.76	63.71	27.76	27.71	45.74	45.74	63.71	27.76	**O.K**
305	1.0(DL+COLL+CL-4+TL1)+1.(	407.61	0.00	-65.87	45.29	0.00	14.64	59.93		30.65	59.93	30.65	23.93	45.29	45.29	59.93	30.65	**O.K**
306	1.0(DL+COLL+CL-1+TL2)+1.(	412.72	0.00	-83.47	45.86	0.00	18.55	64.41		27.31	64.41	27.31	28.41	45.86	45.86	64.41	27.31	**O.K**
307	1.0(DL+COLL+CL-2+TL2)+1.(	408.72	0.00	-68.50	45.41	0.00	15.22	60.63		30.19	60.63	30.19	24.63	45.41	45.41	60.63	30.19	**O.K**
308	1.0(DL+COLL+CL-3+TL2)+1.(	409.56	0.00	-72.02	45.51	0.00	16.00	61.51		29.50	61.51	29.50	25.51	45.51	45.51	61.51	29.50	**O.K**
309	1.0(DL+COLL+CL-4+TL2)+1.(	405.55	0.00	-56.98	45.06	0.00	12.66	57.72		32.40	57.72	32.40	21.72	45.06	45.06	57.72	32.40	**O.K**
310	1.0(DL+COLL+CL-1)+0.75(TL	405.40	0.00	-77.25	45.04	0.00	17.17	62.21		27.88	62.21	27.88	26.21	45.05	45.05	62.21	27.88	**O.K**
311	1.0(DL+COLL+CL-2)+0.75(TL	401.40	0.00	-62.28	44.60	0.00	13.84	58.44		30.76	58.44	30.76	22.44	44.60	44.60	58.44	30.76	**O.K**
312	1.0(DL+COLL+CL-3)+0.75(TL	402.24	0.00	-65.80	44.69	0.00	14.62	59.32		30.07	59.32	30.07	23.32	44.70	44.70	59.32	30.07	**O.K**
313	1.0(DL+COLL+CL-4)+0.75(TL	398.23	0.00	-50.76	44.25	0.00	11.28	55.53		32.97	55.53	32.97	19.53	44.25	44.25	55.53	32.97	**O.K**
314	1.0(DL+COLL+CL-1)+0.75(TL	403.86	0.00	-70.59	44.87	0.00	15.69	60.56		29.19	60.56	29.19	24.56	44.88	44.88	60.56	29.19	**O.K**
315	1.0(DL+COLL+CL-2)+0.75(TL	399.86	0.00	-55.62	44.43	0.00	12.36	56.79		32.07	56.79	32.07	20.79	44.43	44.43	56.79	32.07	**O.K**
316	1.0(DL+COLL+CL-3)+0.75(TL	400.70	0.00	-59.13	44.52	0.00	13.14	57.66		31.38	57.66	31.38	21.66	44.52	44.52	57.66	31.38	**O.K**

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317	1.0(DL+COLL+CL-4)+0.75(TL	396.69	0.00	-44.09	44.08	0.00	9.80	53.88		34.28	53.88	34.28	17.88	44.08	44.08	53.88	34.28	O.K
318	1.0(DL+COLL+CL-1)	377.27	0.00	-31.94	41.92	0.00	7.10	49.02		34.82	49.02	34.82	13.02	41.92	41.92	49.02	34.82	O.K
319	1.0(DL+COLL+CL-2)	373.27	0.00	-16.97	41.47	0.00	3.77	45.25		37.70	45.25	37.70	9.25	41.48	41.48	45.25	37.70	O.K
320	1.0(DL+COLL+CL-3)	374.11	0.00	-20.49	41.57	0.00	4.55	46.12		37.01	46.12	37.01	10.12	41.57	41.57	46.12	37.01	O.K
321	1.0(DL+COLL+CL-4)	370.10	0.00	-5.45	41.12	0.00	1.21	42.33		39.91	42.33	39.91	6.33	41.12	41.12	42.33	39.91	O.K
322	1.0(DL+COLL+CL-1)+1.0(WLI	325.55	0.00	17.58	36.17	0.00	3.91	40.08		32.27	40.08	32.27	4.08	36.18	36.18	40.08	32.27	O.K
323	1.0(DL+COLL+CL-2)+1.0(WLI	321.56	0.00	32.55	35.73	0.00	7.23	42.96		28.50	42.96	28.50	6.96	35.73	35.73	42.96	28.50	O.K
324	1.0(DL+COLL+CL-3)+1.0(WLI	322.40	0.00	29.03	35.82	0.00	6.45	42.27		29.37	42.27	29.37	6.27	35.82	35.82	42.27	29.37	O.K
325	1.0(DL+COLL+CL-4)+1.0(WLI	318.39	0.00	44.07	35.38	0.00	9.79	45.17		25.58	45.17	25.58	9.17	35.38	35.38	45.17	25.58	O.K
326	1.0(DL+COLL+CL-1)+1.0(WLI	358.91	0.00	-12.10	39.88	0.00	2.69	42.57		37.19	42.57	37.19	6.57	39.88	39.88	42.57	37.19	O.K
327	1.0(DL+COLL+CL-2)+1.0(WLI	354.92	0.00	2.87	39.44	0.00	0.64	40.07		38.80	40.07	38.80	4.07	39.44	39.44	40.07	38.80	O.K
328	1.0(DL+COLL+CL-3)+1.0(WLI	355.76	0.00	-0.65	39.53	0.00	0.14	39.67		39.38	39.67	39.38	3.67	39.53	39.53	39.67	39.38	O.K
329	1.0(DL+COLL+CL-4)+1.0(WLI	351.75	0.00	14.39	39.08	0.00	3.20	42.28		35.88	42.28	35.88	6.28	39.08	39.08	42.28	35.88	O.K
330	1.0(DL+COLL+CL-1)+1.0(WLI	294.58	0.00	104.20	32.73	0.00	23.16	55.89		9.58	55.89	9.58	19.89	32.74	32.74	55.89	9.58	O.K
331	1.0(DL+COLL+CL-2)+1.0(WLI	290.58	0.00	119.17	32.29	0.00	26.48	58.77		5.81	58.77	5.81	22.77	32.29	32.29	58.77	5.81	O.K
332	1.0(DL+COLL+CL-3)+1.0(WLI	291.42	0.00	115.65	32.38	0.00	25.70	58.08		6.68	58.08	6.68	22.08	32.38	32.38	58.08	6.68	O.K
333	1.0(DL+COLL+CL-4)+1.0(WLI	287.41	0.00	130.69	31.93	0.00	29.04	60.98		2.89	60.98	2.89	24.98	31.94	31.94	60.98	2.89	O.K
334	1.0(DL+COLL+CL-1)+1.0(WLI	361.13	0.00	37.52	40.13	0.00	8.34	48.46		31.79	48.46	31.79	12.46	40.13	40.13	48.46	31.79	O.K
335	1.0(DL+COLL+CL-2)+1.0(WLI	357.14	0.00	52.49	39.68	0.00	11.66	51.35		28.02	51.35	28.02	15.35	39.69	39.69	51.35	28.02	O.K
336	1.0(DL+COLL+CL-3)+1.0(WLI	357.98	0.00	48.97	39.78	0.00	10.88	50.66		28.89	50.66	28.89	14.66	39.78	39.78	50.66	28.89	O.K
337	1.0(DL+COLL+CL-4)+1.0(WLI	353.97	0.00	64.01	39.33	0.00	14.22	53.55		25.11	53.55	25.11	17.55	39.33	39.33	53.55	25.11	O.K
338	1.0(DL+COLL+CL-1)+1.0(WLI	292.17	29.09	-23.23	32.46	6.47	5.16	44.09		33.77	31.16	20.84	8.09	38.93	26.00	37.63	27.31	O.K
339	1.0(DL+COLL+CL-2)+1.0(WLI	288.18	29.09	-17.68	32.02	6.47	3.93	42.41		34.55	29.48	21.62	6.41	38.48	25.55	35.95	28.09	O.K
340	1.0(DL+COLL+CL-3)+1.0(WLI	289.02	29.09	-18.99	32.11	6.47	4.22	42.80		34.36	29.87	21.43	6.80	38.58	25.65	36.34	27.90	O.K
341	1.0(DL+COLL+CL-4)+1.0(WLI	285.01	29.09	-13.42	31.67	6.47	2.98	41.11		35.15	28.18	22.22	5.11	38.13	25.20	34.65	28.69	O.K
342	1.0(DL+COLL+CL-1)+1.0(WLI	321.53	21.40	-23.23	35.73	4.76	5.16	45.64		35.32	36.13	25.81	9.64	40.48	30.97	40.89	30.57	O.K
343	1.0(DL+COLL+CL-2)+1.0(WLI	317.53	21.40	-17.68	35.28	4.76	3.93	43.97		36.11	34.45	26.59	7.97	40.04	30.52	39.21	31.35	O.K
344	1.0(DL+COLL+CL-3)+1.0(WLI	318.37	21.40	-18.99	35.37	4.76	4.22	44.35		35.91	34.84	26.40	8.35	40.13	30.62	39.60	31.16	O.K
345	1.0(DL+COLL+CL-4)+1.0(WLI	314.36	21.40	-13.42	34.93	4.76	2.98	42.67		36.70	33.15	27.19	6.67	39.69	30.17	37.91	31.95	O.K
346	1.0(DL+COLL+CL-1)+1.0(WLI	293.43	27.35	-23.23	32.60	6.08	5.16	43.84		33.52	31.69	21.36	7.84	38.68	26.53	37.77	27.44	O.K
347	1.0(DL+COLL+CL-2)+1.0(WLI	289.44	27.35	-17.68	32.16	6.08	3.93	42.17		34.31	30.01	22.15	6.17	38.24	26.08	36.09	28.23	O.K

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348	1.0(DL+COLL+CL-3)+1.0(WLI	290.28	27.35	-18.99	32.25	6.08	4.22	42.55		34.11	30.39	21.95	6.55	38.33	26.17	36.47	28.03	O.K
349	1.0(DL+COLL+CL-4)+1.0(WLI	286.26	27.35	-13.42	31.81	6.08	2.98	40.87		34.90	28.71	22.75	4.87	37.89	25.73	34.79	28.83	O.K
350	1.0(DL+COLL+CL-1)+1.0(WLI	322.75	19.81	-23.23	35.86	4.40	5.16	45.43		35.10	36.62	26.30	9.43	40.27	31.46	41.03	30.70	O.K
351	1.0(DL+COLL+CL-2)+1.0(WLI	318.75	19.81	-17.68	35.42	4.40	3.93	43.75		35.89	34.94	27.08	7.75	39.82	31.01	39.35	31.49	O.K
352	1.0(DL+COLL+CL-3)+1.0(WLI	319.59	19.81	-18.99	35.51	4.40	4.22	44.13		35.69	35.33	26.89	8.13	39.91	31.11	39.73	31.29	O.K
353	1.0(DL+COLL+CL-4)+1.0(WLI	315.58	19.81	-13.42	35.06	4.40	2.98	42.45		36.49	33.64	27.68	6.45	39.47	30.66	38.05	32.09	O.K
354	1.0(DL+COLL+CL-1)+0.75(WI	365.84	0.00	-36.78	40.65	0.00	8.17	48.82		32.48	48.82	32.48	12.82	40.65	40.65	48.82	32.48	O.K
355	1.0(DL+COLL+CL-2)+0.75(WI	361.84	0.00	-21.81	40.20	0.00	4.85	45.05		35.36	45.05	35.36	9.05	40.21	40.21	45.05	35.36	O.K
356	1.0(DL+COLL+CL-3)+0.75(WI	362.68	0.00	-25.33	40.30	0.00	5.63	45.93		34.67	45.93	34.67	9.93	40.30	40.30	45.93	34.67	O.K
357	1.0(DL+COLL+CL-4)+0.75(WI	358.67	0.00	-10.29	39.85	0.00	2.29	42.14		37.57	42.14	37.57	6.14	39.86	39.86	42.14	37.57	O.K
358	1.0(DL+COLL+CL-1)+0.75(WI	390.86	0.00	-59.04	43.43	0.00	13.12	56.55		30.31	56.55	30.31	20.55	43.43	43.43	56.55	30.31	O.K
359	1.0(DL+COLL+CL-2)+0.75(WI	386.86	0.00	-44.07	42.98	0.00	9.79	52.78		33.19	52.78	33.19	16.78	42.99	42.99	52.78	33.19	O.K
360	1.0(DL+COLL+CL-3)+0.75(WI	387.70	0.00	-47.59	43.08	0.00	10.58	53.65		32.50	53.65	32.50	17.65	43.08	43.08	53.65	32.50	O.K
361	1.0(DL+COLL+CL-4)+0.75(WI	383.69	0.00	-32.55	42.63	0.00	7.23	49.87		35.40	49.87	35.40	13.87	42.64	42.64	49.87	35.40	O.K
362	1.0(DL+COLL+CL-1)+0.75(WI	342.61	0.00	28.18	38.07	0.00	6.26	44.33		31.81	44.33	31.81	8.33	38.07	38.07	44.33	31.81	O.K
363	1.0(DL+COLL+CL-2)+0.75(WI	338.61	0.00	43.15	37.62	0.00	9.59	47.21		28.03	47.21	28.03	11.21	37.62	37.62	47.21	28.03	O.K
364	1.0(DL+COLL+CL-3)+0.75(WI	339.45	0.00	39.63	37.72	0.00	8.81	46.52		28.91	46.52	28.91	10.52	37.72	37.72	46.52	28.91	O.K
365	1.0(DL+COLL+CL-4)+0.75(WI	335.44	0.00	54.67	37.27	0.00	12.15	49.42		25.12	49.42	25.12	13.42	37.27	37.27	49.42	25.12	O.K
366	1.0(DL+COLL+CL-1)+0.75(WI	392.53	0.00	-21.83	43.61	0.00	4.85	48.46		38.76	48.46	38.76	12.46	43.61	43.61	48.46	38.76	O.K
367	1.0(DL+COLL+CL-2)+0.75(WI	388.53	0.00	-6.86	43.17	0.00	1.52	44.69		41.65	44.69	41.65	8.69	43.17	43.17	44.69	41.65	O.K
368	1.0(DL+COLL+CL-3)+0.75(WI	389.37	0.00	-10.38	43.26	0.00	2.31	45.57		40.96	45.57	40.96	9.57	43.27	43.27	45.57	40.96	O.K
369	1.0(DL+COLL+CL-4)+0.75(WI	385.36	0.00	4.66	42.82	0.00	1.04	43.85		41.78	43.85	41.78	7.85	42.82	42.82	43.85	41.78	O.K
370	1.0(DL+COLL+CL-1)+0.75(WI	340.81	21.82	-73.92	37.87	4.85	16.43	59.14		26.29	49.45	16.59	23.14	42.72	33.02	54.30	21.44	O.K
371	1.0(DL+COLL+CL-2)+0.75(WI	336.81	21.82	-58.96	37.42	4.85	13.10	55.37		29.17	45.68	19.47	19.37	42.27	32.58	50.53	24.32	O.K
372	1.0(DL+COLL+CL-3)+0.75(WI	337.65	21.82	-62.47	37.52	4.85	13.88	56.25		28.48	46.55	18.78	20.25	42.37	32.67	51.40	23.63	O.K
373	1.0(DL+COLL+CL-4)+0.75(WI	333.64	21.82	-47.43	37.07	4.85	10.54	52.46		31.38	42.76	21.68	16.46	41.92	32.22	47.61	26.53	O.K
374	1.0(DL+COLL+CL-1)+0.75(WI	362.82	16.05	-73.92	40.31	3.57	16.43	60.31		27.45	53.17	20.32	24.31	43.88	36.75	56.74	23.89	O.K
375	1.0(DL+COLL+CL-2)+0.75(WI	358.82	16.05	-58.96	39.87	3.57	13.10	56.54		30.34	49.40	23.20	20.54	43.44	36.30	52.97	26.77	O.K
376	1.0(DL+COLL+CL-3)+0.75(WI	359.66	16.05	-62.47	39.96	3.57	13.88	57.41		29.65	50.28	22.51	21.41	43.53	36.40	53.85	26.08	O.K
377	1.0(DL+COLL+CL-4)+0.75(WI	355.65	16.05	-47.43	39.52	3.57	10.54	53.63		32.54	46.49	25.41	17.63	43.09	35.95	50.06	28.98	O.K
378	1.0(DL+COLL+CL-1)+0.75(WI	341.75	20.51	-73.92	37.97	4.56	16.43	58.96		26.10	49.84	16.99	22.96	42.53	33.42	54.40	21.55	O.K

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379	1.0(DL+COLL+CL-2)+0.75(WI	337.75	20.51	-58.96	37.53	4.56	13.10	55.19		28.99	46.07	19.87	19.19	42.09	32.97	50.63	24.43	O.K
380	1.0(DL+COLL+CL-3)+0.75(WI	338.59	20.51	-62.47	37.62	4.56	13.88	56.06		28.30	46.95	19.18	20.06	42.18	33.07	51.51	23.74	O.K
381	1.0(DL+COLL+CL-4)+0.75(WI	334.58	20.51	-47.43	37.18	4.56	10.54	52.28		31.19	43.16	22.08	16.28	41.74	32.62	47.72	26.64	O.K
382	1.0(DL+COLL+CL-1)+0.75(WI	363.74	14.86	-73.92	40.42	3.30	16.43	60.15		27.29	53.54	20.69	24.15	43.72	37.12	56.85	23.99	O.K
383	1.0(DL+COLL+CL-2)+0.75(WI	359.74	14.86	-58.96	39.97	3.30	13.10	56.37		30.17	49.77	23.57	20.37	43.27	36.67	53.07	26.87	O.K
384	1.0(DL+COLL+CL-3)+0.75(WI	360.58	14.86	-62.47	40.06	3.30	13.88	57.25		29.48	50.65	22.88	21.25	43.37	36.77	53.95	26.18	O.K
385	1.0(DL+COLL+CL-4)+0.75(WI	356.57	14.86	-47.43	39.62	3.30	10.54	53.46		32.38	46.86	25.78	17.46	42.92	36.32	50.16	29.08	O.K
386	1.0(DL+COLL+CL-1)+0.75(SL	377.35	-0.23	-32.17	41.93	0.05	7.15	49.13		34.83	49.03	34.73	13.13	41.98	41.88	49.08	34.78	O.K
387	1.0(DL+COLL+CL-2)+0.75(SL	373.35	-0.23	-17.20	41.48	0.05	3.82	45.36		37.71	45.26	37.61	9.36	41.54	41.44	45.31	37.66	O.K
388	1.0(DL+COLL+CL-3)+0.75(SL	374.19	-0.23	-20.72	41.58	0.05	4.61	46.23		37.02	46.13	36.92	10.23	41.63	41.53	46.18	36.97	O.K
389	1.0(DL+COLL+CL-4)+0.75(SL	370.18	-0.23	-5.68	41.13	0.05	1.26	42.45		39.92	42.34	39.82	6.45	41.19	41.08	42.40	39.87	O.K
390	1.0(DL+COLL+CL-1)+0.75(SL	377.18	0.24	-31.70	41.91	0.05	7.04	49.01		34.92	48.90	34.81	13.01	41.97	41.86	48.96	34.87	O.K
391	1.0(DL+COLL+CL-2)+0.75(SL	373.18	0.24	-16.73	41.46	0.05	3.72	45.24		37.80	45.13	37.69	9.24	41.52	41.41	45.19	37.75	O.K
392	1.0(DL+COLL+CL-3)+0.75(SL	374.02	0.24	-20.25	41.56	0.05	4.50	46.11		37.11	46.00	37.00	10.11	41.61	41.50	46.06	37.06	O.K
393	1.0(DL+COLL+CL-4)+0.75(SL	370.01	0.24	-5.21	41.11	0.05	1.16	42.32		40.01	42.22	39.90	6.32	41.17	41.06	42.27	39.96	O.K
394	1.0(DL+COLL+CL-1)+0.525(S	404.69	-0.16	-74.08	44.97	0.04	16.46	61.46		28.54	61.39	28.47	25.46	45.00	44.93	61.43	28.51	O.K
395	1.0(DL+COLL+CL-2)+0.525(S	400.69	-0.16	-59.12	44.52	0.04	13.14	57.69		31.42	57.62	31.35	21.69	44.56	44.49	57.66	31.39	O.K
396	1.0(DL+COLL+CL-3)+0.525(S	401.53	-0.16	-62.63	44.61	0.04	13.92	58.57		30.73	58.50	30.66	22.57	44.65	44.58	58.54	30.70	O.K
397	1.0(DL+COLL+CL-4)+0.525(S	397.52	-0.16	-47.59	44.17	0.04	10.58	54.78		33.63	54.71	33.56	18.78	44.21	44.14	54.75	33.60	O.K
398	1.0(DL+COLL+CL-1)+0.525(S	404.56	0.17	-73.75	44.95	0.04	16.39	61.38		28.60	61.30	28.52	25.38	44.99	44.91	61.34	28.56	O.K
399	1.0(DL+COLL+CL-2)+0.525(S	400.56	0.17	-58.78	44.51	0.04	13.06	57.61		31.48	57.53	31.41	21.61	44.55	44.47	57.57	31.45	O.K
400	1.0(DL+COLL+CL-3)+0.525(S	401.40	0.17	-62.30	44.60	0.04	13.85	58.48		30.79	58.41	30.72	22.48	44.64	44.57	58.45	30.76	O.K
401	1.0(DL+COLL+CL-4)+0.525SI	397.39	0.17	-47.26	44.15	0.04	10.50	54.70		33.69	54.62	33.61	18.70	44.20	44.12	54.66	33.65	O.K
402	0.6(DL+COLL+CL-1)+1.0(WLI	318.83	0.00	30.36	35.43	0.00	6.75	42.17		28.68	42.17	28.68	6.17	35.43	35.43	42.17	28.68	O.K
403	0.6(DL+COLL+CL-2)+1.0(WLI	316.43	0.00	39.34	35.16	0.00	8.74	43.90		26.42	43.90	26.42	7.90	35.16	35.16	43.90	26.42	O.K
404	0.6(DL+COLL+CL-3)+1.0(WLI	316.93	0.00	37.23	35.21	0.00	8.27	43.49		26.94	43.49	26.94	7.49	35.22	35.22	43.49	26.94	O.K
405	0.6(DL+COLL+CL-4)+1.0(WLI	314.53	0.00	46.25	34.95	0.00	10.28	45.23		24.67	45.23	24.67	9.23	34.95	34.95	45.23	24.67	O.K
406	0.6(DL+COLL+CL-1)+1.0(WLI	352.19	0.00	0.68	39.13	0.00	0.15	39.28		38.98	39.28	38.98	3.28	39.13	39.13	39.28	38.98	O.K
407	0.6(DL+COLL+CL-2)+1.0(WLI	349.79	0.00	9.66	38.87	0.00	2.15	41.01		36.72	41.01	36.72	5.01	38.87	38.87	41.01	36.72	O.K
408	0.6(DL+COLL+CL-3)+1.0(WLI	350.29	0.00	7.55	38.92	0.00	1.68	40.60		37.24	40.60	37.24	4.60	38.92	38.92	40.60	37.24	O.K
409	0.6(DL+COLL+CL-4)+1.0(WLI	347.89	0.00	16.57	38.65	0.00	3.68	42.34		34.97	42.34	34.97	6.34	38.66	38.66	42.34	34.97	O.K
410	0.6(DL+COLL+CL-1)+1.0(WLI	287.85	0.00	116.98	31.98	0.00	25.99	57.98		5.99	57.98	5.99	21.98	31.99	31.99	57.98	5.99	O.K

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411	0.6(DL+COLL+CL-2)+1.0(WLI	285.46	0.00	125.96	31.72	0.00	27.99	59.71		3.73	59.71	3.73	23.71	31.72	31.72	59.71	3.73	O.K
412	0.6(DL+COLL+CL-3)+1.0(WLI	285.96	0.00	123.85	31.77	0.00	27.52	59.29		4.25	59.29	4.25	23.29	31.77	31.77	59.29	4.25	O.K
413	0.6(DL+COLL+CL-4)+1.0(WLI	283.55	0.00	132.87	31.51	0.00	29.53	61.03		1.98	61.03	1.98	25.03	31.51	31.51	61.03	1.98	O.K
414	0.6(DL+COLL+CL-1)+1.0(WLI	354.41	0.00	50.30	39.38	0.00	11.18	50.56		28.20	50.56	28.20	14.56	39.38	39.38	50.56	28.20	O.K
415	0.6(DL+COLL+CL-2)+1.0(WLI	352.01	0.00	59.28	39.11	0.00	13.17	52.28		25.94	52.28	25.94	16.28	39.11	39.11	52.28	25.94	O.K
416	0.6(DL+COLL+CL-3)+1.0(WLI	352.51	0.00	57.17	39.17	0.00	12.70	51.87		26.46	51.87	26.46	15.87	39.17	39.17	51.87	26.46	O.K
417	0.6(DL+COLL+CL-4)+1.0(WLI	350.11	0.00	66.19	38.90	0.00	14.71	53.61		24.19	53.61	24.19	17.61	38.90	38.90	53.61	24.19	O.K
418	0.6(DL+COLL+CL-1)+1.0(WLI	285.45	29.09	-13.94	31.72	6.47	3.10	41.28		35.08	28.35	22.15	5.28	38.18	25.25	34.82	28.62	O.K
419	0.6(DL+COLL+CL-2)+1.0(WLI	283.05	29.09	-10.61	31.45	6.47	2.36	40.27		35.56	27.34	22.63	4.27	37.92	24.99	33.81	29.10	O.K
420	0.6(DL+COLL+CL-3)+1.0(WLI	283.55	29.09	-11.39	31.51	6.47	2.53	40.50		35.44	27.57	22.51	4.50	37.97	25.04	34.04	28.98	O.K
421	0.6(DL+COLL+CL-4)+1.0(WLI	281.15	29.09	-8.05	31.24	6.47	1.79	39.49		35.91	26.56	22.98	3.49	37.70	24.77	33.03	29.45	O.K
422	0.6(DL+COLL+CL-1)+1.0(WLI	314.80	21.40	-13.94	34.98	4.76	3.10	42.83		36.64	33.32	27.12	6.83	39.74	30.22	38.08	31.88	O.K
423	0.6(DL+COLL+CL-2)+1.0(WLI	312.40	21.40	-10.61	34.71	4.76	2.36	41.83		37.11	32.31	27.60	5.83	39.47	29.96	37.07	32.36	O.K
424	0.6(DL+COLL+CL-3)+1.0(WLI	312.91	21.40	-11.39	34.77	4.76	2.53	42.06		36.99	32.54	27.48	6.06	39.53	30.01	37.30	32.24	O.K
425	0.6(DL+COLL+CL-4)+1.0(WLI	310.50	21.40	-8.05	34.50	4.76	1.79	41.05		37.47	31.53	27.95	5.05	39.26	29.74	36.29	32.71	O.K
426	0.6(DL+COLL+CL-1)+1.0(WLI	286.71	27.35	-13.94	31.86	6.08	3.10	41.03		34.84	28.87	22.68	5.03	37.94	25.78	34.95	28.76	O.K
427	0.6(DL+COLL+CL-2)+1.0(WLI	284.31	27.35	-10.61	31.59	6.08	2.36	40.03		35.31	27.87	23.15	4.03	37.67	25.51	33.95	29.23	O.K
428	0.6(DL+COLL+CL-3)+1.0(WLI	284.81	27.35	-11.39	31.65	6.08	2.53	40.26		35.19	28.10	23.04	4.26	37.73	25.57	34.18	29.12	O.K
429	0.6(DL+COLL+CL-4)+1.0(WLI	282.40	27.35	-8.05	31.38	6.08	1.79	39.25		35.67	27.09	23.51	3.25	37.46	25.30	33.17	29.59	O.K
430	0.6(DL+COLL+CL-1)+1.0(WLI	316.02	19.81	-13.94	35.11	4.40	3.10	42.61		36.42	33.81	27.61	6.61	39.52	30.71	38.21	32.02	O.K
431	0.6(DL+COLL+CL-2)+1.0(WLI	313.62	19.81	-10.61	34.85	4.40	2.36	41.61		36.89	32.80	28.09	5.61	39.25	30.45	37.21	32.49	O.K
432	0.6(DL+COLL+CL-3)+1.0(WLI	314.13	19.81	-11.39	34.90	4.40	2.53	41.84		36.77	33.03	27.97	5.84	39.31	30.50	37.44	32.37	O.K
433	0.6(DL+COLL+CL-4)+1.0(WLI	311.72	19.81	-8.05	34.64	4.40	1.79	40.83		37.25	32.02	28.44	4.83	39.04	30.23	36.43	32.85	O.K
434	0.6(DL+COLL+CL-1)+0.7(SL'+	370.62	-0.21	-19.38	41.18	0.05	4.31	45.53		36.92	45.44	36.83	9.53	41.23	41.14	45.49	36.88	O.K
435	0.6(DL+COLL+CL-2)+0.7(SL'+	368.22	-0.21	-10.40	40.91	0.05	2.31	43.27		38.65	43.18	38.55	7.27	40.96	40.87	43.23	38.60	O.K
436	0.6(DL+COLL+CL-3)+0.7(SL'+	368.72	-0.21	-12.51	40.97	0.05	2.78	43.80		38.24	43.70	38.14	7.80	41.02	40.92	43.75	38.19	O.K
437	0.6(DL+COLL+CL-4)+0.7(SL'+	366.32	-0.21	-3.49	40.70	0.05	0.77	41.52		39.97	41.43	39.88	5.52	40.75	40.66	41.48	39.93	O.K
438	0.6(DL+COLL+CL-1)+0.7(SL'+	337.38	-0.21	18.64	37.49	0.05	4.14	41.68		33.39	41.58	33.30	5.68	37.54	37.44	41.63	33.35	O.K
439	0.6(DL+COLL+CL-2)+0.7(SL'+	357.39	-0.21	0.83	39.71	0.05	0.18	39.94		39.57	39.85	39.48	3.94	39.76	39.67	39.90	39.53	O.K
440	0.6(DL+COLL+CL-3)+0.7(SL'+	318.79	-0.21	70.61	35.42	0.05	15.69	51.16		19.78	51.06	19.68	15.16	35.47	35.37	51.11	19.73	O.K
441	0.6(DL+COLL+CL-4)+0.7(SL'+	358.73	-0.21	30.60	39.86	0.05	6.80	46.71		33.11	46.61	33.01	10.71	39.91	39.81	46.66	33.06	O.K

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Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
	Ultimate Conditions																	
101	1.4(DL+COLL)	378.83	0.00	25.34	42.09	0.00	5.63	47.72		36.46	47.72	36.46	11.72	42.09	42.09	47.72	36.46	
102	1.2(DL+COLL+CL-1)+1.6(RLL	439.00	0.00	127.89	48.78	0.00	28.42	77.20		20.36	77.20	20.36	41.20	48.78	48.78	77.20	20.36	
103	1.2(DL+COLL+CL-2)+1.6(RLL	434.20	0.00	109.93	48.24	0.00	24.43	72.67		23.82	72.67	23.82	36.67	48.25	48.25	72.67	23.82	
104	1.2(DL+COLL+CL-3)+1.6(RLL	435.21	0.00	114.15	48.36	0.00	25.37	73.72		22.99	73.72	22.99	37.72	48.36	48.36	73.72	22.99	
105	1.2(DL+COLL+CL-4)+1.6(RLL	430.40	0.00	96.10	47.82	0.00	21.36	69.18		26.47	69.18	26.47	33.18	47.83	47.83	69.18	26.47	
106	1.4(DL+COLL+CL-1)	384.00	0.00	44.72	42.67	0.00	9.94	52.60		32.73	52.60	32.73	16.60	42.67	42.67	52.60	32.73	
107	1.4(DL+COLL+CL-2)	378.40	0.00	23.76	42.04	0.00	5.28	47.33		36.76	47.33	36.76	11.33	42.05	42.05	47.33	36.76	
108	1.4(DL+COLL+CL-3)	379.58	0.00	28.69	42.18	0.00	6.38	48.55		35.80	48.55	35.80	12.55	42.18	42.18	48.55	35.80	
109	1.4(DL+COLL+CL-4)	373.96	0.00	7.63	41.55	0.00	1.70	43.25		39.85	43.25	39.85	7.25	41.55	41.55	43.25	39.85	
110	1.2(DL+COLL+CL-1+TL1)+1.6	440.23	0.00	133.22	48.91	0.00	29.60	78.52		19.31	78.52	19.31	42.52	48.92	48.92	78.52	19.31	
111	1.2(DL+COLL+CL-2+TL1)+1.6	435.44	0.00	115.25	48.38	0.00	25.61	73.99		22.77	73.99	22.77	37.99	48.38	48.38	73.99	22.77	
112	1.2(DL+COLL+CL-3+TL1)+1.6	436.44	0.00	119.48	48.49	0.00	26.55	75.04		21.94	75.04	21.94	39.04	48.49	48.49	75.04	21.94	
113	1.2(DL+COLL+CL-4+TL1)+1.6	431.63	0.00	101.43	47.96	0.00	22.54	70.50		25.42	70.50	25.42	34.50	47.96	47.96	70.50	25.42	
114	1.2(DL+COLL+CL-1+TL2)+1.6	437.76	0.00	122.55	48.64	0.00	27.23	75.87		21.41	75.87	21.41	39.87	48.64	48.64	75.87	21.41	
115	1.2(DL+COLL+CL-2+TL2)+1.6	432.97	0.00	104.59	48.11	0.00	23.24	71.35		24.87	71.35	24.87	35.35	48.11	48.11	71.35	24.87	
116	1.2(DL+COLL+CL-3+TL2)+1.6	433.97	0.00	108.81	48.22	0.00	24.18	72.40		24.04	72.40	24.04	36.40	48.22	48.22	72.40	24.04	
117	1.2(DL+COLL+CL-4+TL2)+1.6	429.16	0.00	90.76	47.68	0.00	20.17	67.85		27.52	67.85	27.52	31.85	47.69	47.69	67.85	27.52	
118	1.2(DL+COLL+CL-1)+0.8(WLI	339.26	0.00	1.29	37.70	0.00	0.29	37.98		37.41	37.98	37.41	1.98	37.70	37.70	37.98	37.41	
119	1.2(DL+COLL+CL-2)+0.8(WLI	334.46	0.00	19.25	37.16	0.00	4.28	41.44		32.88	41.44	32.88	5.44	37.16	37.16	41.44	32.88	
120	1.2(DL+COLL+CL-3)+0.8(WLI	335.47	0.00	15.03	37.27	0.00	3.34	40.61		33.94	40.61	33.94	4.61	37.28	37.28	40.61	33.94	
121	1.2(DL+COLL+CL-4)+0.8(WLI	330.66	0.00	33.08	36.74	0.00	7.35	44.09		29.39	44.09	29.39	8.09	36.74	36.74	44.09	29.39	
122	1.2(DL+COLL+CL-1)+0.8(WLI	365.95	0.00	22.46	40.66	0.00	4.99	45.65		35.67	45.65	35.67	9.65	40.66	40.66	45.65	35.67	
123	1.2(DL+COLL+CL-2)+0.8(WLI	361.15	0.00	4.49	40.13	0.00	1.00	41.13		39.13	41.13	39.13	5.13	40.13	40.13	41.13	39.13	
124	1.2(DL+COLL+CL-3)+0.8(WLI	362.16	0.00	8.72	40.24	0.00	1.94	42.18		38.30	42.18	38.30	6.18	40.24	40.24	42.18	38.30	

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125	1.2(DL+COLL+CL-4)+0.8(WLI	357.35	0.00	9.33	39.71	0.00	2.07	41.78		37.63	41.78	37.63	5.78	39.71	39.71	41.78	37.63	
126	1.2(DL+COLL+CL-1)+0.8(WLI	314.48	0.00	70.58	34.94	0.00	15.68	50.63		19.26	50.63	19.26	14.63	34.95	34.95	50.63	19.26	
127	1.2(DL+COLL+CL-2)+0.8(WLI	309.69	0.00	88.54	34.41	0.00	19.68	54.09		14.73	54.09	14.73	18.09	34.41	34.41	54.09	14.73	
128	1.2(DL+COLL+CL-3)+0.8(WLI	310.69	0.00	84.32	34.52	0.00	18.74	53.26		15.78	53.26	15.78	17.26	34.52	34.52	53.26	15.78	
129	1.2(DL+COLL+CL-4)+0.8(WLI	305.88	0.00	102.37	33.99	0.00	22.75	56.74		11.24	56.74	11.24	20.74	33.99	33.99	56.74	11.24	
130	1.2(DL+COLL+CL-1)+0.8(WLI	367.72	0.00	17.24	40.86	0.00	3.83	44.69		37.03	44.69	37.03	8.69	40.86	40.86	44.69	37.03	
131	1.2(DL+COLL+CL-2)+0.8(WLI	362.93	0.00	35.20	40.33	0.00	7.82	48.15		32.50	48.15	32.50	12.15	40.33	40.33	48.15	32.50	
132	1.2(DL+COLL+CL-3)+0.8(WLI	363.93	0.00	30.98	40.44	0.00	6.88	47.32		33.55	47.32	33.55	11.32	40.44	40.44	47.32	33.55	
133	1.2(DL+COLL+CL-4)+0.8(WLI	359.12	0.00	49.03	39.90	0.00	10.89	50.80		29.01	50.80	29.01	14.80	39.91	39.91	50.80	29.01	
134	1.2(DL+COLL+CL-1)+0.8(WLI	312.56	62.84	38.33	34.73	13.96	8.52	57.21		40.17	29.28	12.25	21.21	48.69	20.77	43.25	26.21	
135	1.2(DL+COLL+CL-2)+0.8(WLI	307.76	62.84	20.37	34.20	13.96	4.53	52.69		43.63	24.76	15.70	16.69	48.16	20.23	38.73	29.67	
136	1.2(DL+COLL+CL-3)+0.8(WLI	308.77	62.84	24.59	34.31	13.96	5.46	53.74		42.81	25.81	14.88	17.74	48.28	20.35	39.78	28.85	
137	1.2(DL+COLL+CL-4)+0.8(WLI	303.95	62.84	6.54	33.77	13.96	1.45	49.19		46.28	21.26	18.35	13.19	47.74	19.81	35.23	32.32	
138	1.2(DL+COLL+CL-1)+0.8(WLI	336.04	46.23	38.33	37.34	10.27	8.52	56.13		39.09	35.58	18.55	20.13	47.61	27.07	45.86	28.82	
139	1.2(DL+COLL+CL-2)+0.8(WLI	331.24	46.23	20.37	36.80	10.27	4.53	51.61		42.55	31.06	22.00	15.61	47.08	26.53	41.34	32.28	
140	1.2(DL+COLL+CL-3)+0.8(WLI	332.25	46.23	24.59	36.92	10.27	5.46	52.66		41.73	32.11	21.18	16.66	47.20	26.65	42.39	31.46	
141	1.2(DL+COLL+CL-4)+0.8(WLI	327.44	46.23	6.54	36.38	10.27	1.45	48.11		45.20	27.56	24.65	12.11	46.66	26.11	37.84	34.93	
142	1.2(DL+COLL+CL-1)+0.8(WLI	313.56	59.08	38.33	34.84	13.13	8.52	56.49		39.45	30.23	13.19	20.49	47.97	21.71	43.36	26.32	
143	1.2(DL+COLL+CL-2)+0.8(WLI	308.77	59.08	20.37	34.31	13.13	4.53	51.96		42.91	25.70	16.65	15.96	47.44	21.18	38.83	29.78	
144	1.2(DL+COLL+CL-3)+0.8(WLI	309.77	59.08	24.59	34.42	13.13	5.46	53.01		42.08	26.75	15.83	17.01	47.55	21.29	39.88	28.96	
145	1.2(DL+COLL+CL-4)+0.8(WLI	304.96	59.08	6.54	33.88	13.13	1.45	48.47		45.56	22.21	19.30	12.47	47.02	20.76	35.34	32.43	
146	1.2(DL+COLL+CL-1)+0.8(WLI	337.02	42.80	38.33	37.45	9.51	8.52	55.47		38.44	36.45	19.42	19.47	46.96	27.94	45.96	28.93	
147	1.2(DL+COLL+CL-2)+0.8(WLI	332.22	42.80	20.37	36.91	9.51	4.53	50.95		41.90	31.93	22.88	14.95	46.43	27.41	41.44	32.39	
148	1.2(DL+COLL+CL-3)+0.8(WLI	333.23	42.80	24.59	37.03	9.51	5.46	52.00		41.07	32.98	22.05	16.00	46.54	27.52	42.49	31.56	
149	1.2(DL+COLL+CL-4)+0.8(WLI	328.41	42.80	6.54	36.49	9.51	1.45	47.45		44.55	28.43	25.53	11.45	46.00	26.98	37.94	35.04	
150	1.2(DL+COLL+CL-1)+1.6(WLI	316.13	0.00	12.92	35.13	0.00	2.87	38.00		32.25	38.00	32.25	2.00	35.13	35.13	38.00	32.25	
151	1.2(DL+COLL+CL-2)+1.6(WLI	311.33	0.00	30.88	34.59	0.00	6.86	41.45		27.73	41.45	27.73	5.45	34.59	34.59	41.45	27.73	
152	1.2(DL+COLL+CL-3)+1.6(WLI	312.34	0.00	26.66	34.70	0.00	5.92	40.63		28.78	40.63	28.78	4.63	34.71	34.71	40.63	28.78	
153	1.2(DL+COLL+CL-4)+1.6(WLI	307.53	0.00	44.71	34.17	0.00	9.93	44.10		24.23	44.10	24.23	8.10	34.17	34.17	44.10	24.23	
154	1.2(DL+COLL+CL-1)+1.6(WLI	369.50	0.00	34.57	41.06	0.00	7.68	48.74		33.37	48.74	33.37	12.74	41.06	41.06	48.74	33.37	
155	1.2(DL+COLL+CL-2)+1.6(WLI	364.71	0.00	16.60	40.52	0.00	3.69	44.21		36.83	44.21	36.83	8.21	40.52	40.52	44.21	36.83	
156	1.2(DL+COLL+CL-3)+1.6(WLI	365.71	0.00	20.83	40.63	0.00	4.63	45.26		36.01	45.26	36.01	9.26	40.64	40.64	45.26	36.01	

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157	1.2(DL+COLL+CL-4)+1.6(WLI	360.90	0.00	2.78	40.10	0.00	0.62	40.72		39.48	40.72	39.48	4.72	40.10	40.10	40.72	39.48	
158	1.2(DL+COLL+CL-1)+1.6(WLI	266.57	0.00	151.51	29.62	0.00	33.67	63.29		-4.05	63.29	-4.05	27.29	29.62	29.62	63.29	-4.05	
159	1.2(DL+COLL+CL-2)+1.6(WLI	261.77	0.00	169.47	29.09	0.00	37.66	66.75		-8.57	66.75	-8.57	30.75	29.09	29.09	66.75	-8.57	
160	1.2(DL+COLL+CL-3)+1.6(WLI	262.78	0.00	165.25	29.20	0.00	36.72	65.92		-7.52	65.92	-7.52	29.92	29.20	29.20	65.92	-7.52	
161	1.2(DL+COLL+CL-4)+1.6(WLI	257.97	0.00	183.30	28.66	0.00	40.73	69.40		-12.07	69.40	-12.07	33.40	28.67	28.67	69.40	-12.07	
162	1.2(DL+COLL+CL-1)+1.6(WLI	373.06	0.00	44.82	41.45	0.00	9.96	51.41		31.49	51.41	31.49	15.41	41.45	41.45	51.41	31.49	
163	1.2(DL+COLL+CL-2)+1.6(WLI	368.26	0.00	62.78	40.92	0.00	13.95	54.87		26.97	54.87	26.97	18.87	40.92	40.92	54.87	26.97	
164	1.2(DL+COLL+CL-3)+1.6(WLI	369.27	0.00	58.56	41.03	0.00	13.01	54.04		28.02	54.04	28.02	18.04	41.03	41.03	54.04	28.02	
165	1.2(DL+COLL+CL-4)+1.6(WLI	364.45	0.00	76.61	40.49	0.00	17.02	57.52		23.47	57.52	23.47	21.52	40.50	40.50	57.52	23.47	
166	1.2(DL+COLL+CL-1)+1.6(WLI	262.72	125.68	66.32	29.19	27.93	14.74	71.86		42.38	16.00	-13.48	35.86	57.12	1.26	43.93	14.45	
167	1.2(DL+COLL+CL-2)+1.6(WLI	257.92	125.68	48.36	28.66	27.93	10.75	67.33		45.84	11.47	-10.02	31.33	56.59	0.73	39.40	17.91	
168	1.2(DL+COLL+CL-3)+1.6(WLI	258.93	125.68	52.58	28.77	27.93	11.68	68.38		45.02	12.53	-10.84	32.38	56.70	0.85	40.46	17.09	
169	1.2(DL+COLL+CL-4)+1.6(WLI	254.12	125.68	34.53	28.24	27.93	7.67	63.84		48.49	7.98	-7.37	27.84	56.17	0.31	35.91	20.56	
170	1.2(DL+COLL+CL-1)+1.6(WLI	309.69	92.47	66.32	34.41	20.55	14.74	69.70		40.22	28.60	-0.88	33.70	54.96	13.86	49.15	19.67	
171	1.2(DL+COLL+CL-2)+1.6(WLI	304.89	92.47	48.36	33.88	20.55	10.75	65.17		43.68	24.07	2.58	29.17	54.43	13.33	44.62	23.13	
172	1.2(DL+COLL+CL-3)+1.6(WLI	305.90	92.47	52.58	33.99	20.55	11.68	66.22		42.85	25.12	1.76	30.22	54.54	13.44	45.67	22.31	
173	1.2(DL+COLL+CL-4)+1.6(WLI	301.08	92.47	34.53	33.45	20.55	7.67	61.68		46.33	20.58	5.23	25.68	54.01	12.91	41.13	25.78	
174	1.2(DL+COLL+CL-1)+1.6(WLI	264.74	118.16	66.32	29.42	26.26	14.74	70.41		40.94	17.89	-11.58	34.41	55.68	3.16	44.15	14.68	
175	1.2(DL+COLL+CL-2)+1.6(WLI	259.94	118.16	48.36	28.88	26.26	10.75	65.89		44.39	13.37	-8.12	29.89	55.14	2.63	39.63	18.14	
176	1.2(DL+COLL+CL-3)+1.6(WLI	260.95	118.16	52.58	28.99	26.26	11.68	66.94		43.57	14.42	-8.95	30.94	55.26	2.74	40.68	17.31	
177	1.2(DL+COLL+CL-4)+1.6(WLI	256.13	118.16	34.53	28.46	26.26	7.67	62.39		47.04	9.87	-5.47	26.39	54.72	2.20	36.13	20.79	
178	1.2(DL+COLL+CL-1)+1.6(WLI	311.64	85.59	66.32	34.63	19.02	14.74	68.38		38.91	30.34	0.87	32.38	53.65	15.61	49.36	19.89	
179	1.2(DL+COLL+CL-2)+1.6(WLI	306.84	85.59	48.36	34.09	19.02	10.75	63.86		42.37	25.82	4.33	27.86	53.12	15.08	44.84	23.35	
180	1.2(DL+COLL+CL-3)+1.6(WLI	307.85	85.59	52.58	34.21	19.02	11.68	64.91		41.54	26.87	3.50	28.91	53.23	15.19	45.89	22.52	
181	1.2(DL+COLL+CL-4)+1.6(WLI	303.04	85.59	34.53	33.67	19.02	7.67	60.36		45.02	22.32	6.98	24.36	52.69	14.65	41.34	26.00	
182	1.2(DL+COLL+CL-1)+1.0(SL'+	398.98	0.83	67.15	44.33	0.18	14.92	59.44		29.59	59.07	29.23	23.44	44.52	44.15	59.26	29.41	
183	1.2(DL+COLL+CL-2)+1.0(SL'+	394.19	0.83	49.18	43.80	0.18	10.93	54.91		33.05	54.54	32.69	18.91	43.98	43.62	54.73	32.87	
184	1.2(DL+COLL+CL-3)+1.0(SL'+	395.19	0.83	53.40	43.91	0.18	11.87	55.96		32.23	55.59	31.86	19.96	44.10	43.73	55.78	32.05	
185	1.2(DL+COLL+CL-4)+1.0(SL'+	390.38	0.83	35.36	43.38	0.18	7.86	51.42		35.70	51.05	35.34	15.42	43.56	43.20	51.24	35.52	
186	1.2(DL+COLL+CL-1)+1.0(SL'-	398.75	0.88	65.44	44.31	0.20	14.54	59.04		29.96	58.65	29.57	23.04	44.50	44.11	58.85	29.77	
187	1.2(DL+COLL+CL-2)+1.0(SL'-	393.95	0.88	47.48	43.77	0.20	10.55	54.52		33.42	54.13	33.03	18.52	43.97	43.58	54.33	33.23	

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188	1.2(DL+COLL+CL-3)+1.0(SL'-	394.96	0.88	51.70	43.88	0.20	11.49	55.57		32.59	55.18	32.20	19.57	44.08	43.69	55.38	32.40	
189	1.2(DL+COLL+CL-4)+1.0(SL'-	390.15	0.88	33.65	43.35	0.20	7.48	51.02		36.07	50.63	35.68	15.02	43.55	43.16	50.83	35.88	
190	0.9(DL+COLL+CL-1)+1.6(WLI	292.84	0.00	50.49	32.54	0.00	11.22	43.76		21.32	43.76	21.32	7.76	32.54	32.54	43.76	21.32	
191	0.9(DL+COLL+CL-2)+1.6(WLI	289.24	0.00	63.96	32.14	0.00	14.21	46.35		17.92	46.35	17.92	10.35	32.14	32.14	46.35	17.92	
192	0.9(DL+COLL+CL-3)+1.6(WLI	290.00	0.00	60.79	32.22	0.00	13.51	45.73		18.71	45.73	18.71	9.73	32.22	32.22	45.73	18.71	
193	0.9(DL+COLL+CL-4)+1.6(WLI	286.39	0.00	74.33	31.82	0.00	16.52	48.34		15.30	48.34	15.30	12.34	31.82	31.82	48.34	15.30	
194	0.9(DL+COLL+CL-1)+1.6(WLI	346.22	0.00	3.00	38.47	0.00	0.67	39.14		37.80	39.14	37.80	3.14	38.47	38.47	39.14	37.80	
195	0.9(DL+COLL+CL-2)+1.6(WLI	342.62	0.00	16.48	38.07	0.00	3.66	41.73		34.41	41.73	34.41	5.73	38.07	38.07	41.73	34.41	
196	0.9(DL+COLL+CL-3)+1.6(WLI	343.38	0.00	13.31	38.15	0.00	2.96	41.11		35.20	41.11	35.20	5.11	38.16	38.16	41.11	35.20	
197	0.9(DL+COLL+CL-4)+1.6(WLI	339.77	0.00	26.84	37.75	0.00	5.97	43.72		31.79	43.72	31.79	7.72	37.76	37.76	43.72	31.79	
198	0.9(DL+COLL+CL-1)+1.6(WLI	243.29	0.00	189.08	27.03	0.00	42.02	69.05		-14.99	69.05	-14.99	33.05	27.03	27.03	69.05	-14.99	
199	0.9(DL+COLL+CL-2)+1.6(WLI	239.69	0.00	202.55	26.63	0.00	45.01	71.64		-18.38	71.64	-18.38	35.64	26.63	26.63	71.64	-18.38	
200	0.9(DL+COLL+CL-3)+1.6(WLI	240.45	0.00	199.38	26.72	0.00	44.31	71.02		-17.59	71.02	-17.59	35.02	26.72	26.72	71.02	-17.59	
201	0.9(DL+COLL+CL-4)+1.6(WLI	236.84	0.00	212.92	26.32	0.00	47.32	73.63		-21.00	73.63	-21.00	37.63	26.32	26.32	73.63	-21.00	
202	0.9(DL+COLL+CL-1)+1.6(WLI	349.77	0.00	82.39	38.86	0.00	18.31	57.17		20.55	57.17	20.55	21.17	38.86	38.86	57.17	20.55	
203	0.9(DL+COLL+CL-2)+1.6(WLI	346.17	0.00	95.86	38.46	0.00	21.30	59.77		17.16	59.77	17.16	23.77	38.47	38.47	59.77	17.16	
204	0.9(DL+COLL+CL-3)+1.6(WLI	346.93	0.00	92.70	38.55	0.00	20.60	59.15		17.95	59.15	17.95	23.15	38.55	38.55	59.15	17.95	
205	0.9(DL+COLL+CL-4)+1.6(WLI	343.32	0.00	106.23	38.15	0.00	23.61	61.75		14.54	61.75	14.54	25.75	38.15	38.15	61.75	14.54	
206	0.9(DL+COLL+CL-1)+1.6(WLI	239.44	125.68	28.75	26.60	27.93	6.39	60.92		48.14	5.06	-7.71	24.92	54.53	-1.33	32.99	20.22	
207	0.9(DL+COLL+CL-2)+1.6(WLI	235.84	125.68	15.28	26.20	27.93	3.39	57.53		50.74	1.67	-5.12	21.53	54.14	-1.73	29.60	22.81	
208	0.9(DL+COLL+CL-3)+1.6(WLI	236.59	125.68	18.44	26.29	27.93	4.10	58.32		50.12	2.46	-5.74	22.32	54.22	-1.64	30.39	22.19	
209	0.9(DL+COLL+CL-4)+1.6(WLI	232.98	125.68	4.91	25.89	27.93	1.09	54.91		52.73	-0.95	-3.13	18.91	53.82	-2.04	26.98	24.80	
210	0.9(DL+COLL+CL-1)+1.6(WLI	286.40	92.47	28.75	31.82	20.55	6.39	58.76		45.98	17.66	4.89	22.76	52.37	11.28	38.21	25.44	
211	0.9(DL+COLL+CL-2)+1.6(WLI	282.80	92.47	15.28	31.42	20.55	3.39	55.37		48.58	14.27	7.48	19.37	51.98	10.88	34.82	28.03	
212	0.9(DL+COLL+CL-3)+1.6(WLI	283.56	92.47	18.44	31.51	20.55	4.10	56.15		47.96	15.06	6.86	20.15	52.06	10.96	35.61	27.41	
213	0.9(DL+COLL+CL-4)+1.6(WLI	279.95	92.47	4.91	31.11	20.55	1.09	52.74		50.56	11.65	9.47	16.74	51.65	10.56	32.20	30.02	
214	0.9(DL+COLL+CL-1)+1.6(WLI	241.45	118.16	28.75	26.83	26.26	6.39	59.48		46.70	6.96	-5.82	23.48	53.09	0.57	33.22	20.44	
215	0.9(DL+COLL+CL-2)+1.6(WLI	237.85	118.16	15.28	26.43	26.26	3.39	56.08		49.29	3.56	-3.23	20.08	52.69	0.17	29.82	23.03	
216	0.9(DL+COLL+CL-3)+1.6(WLI	238.61	118.16	18.44	26.51	26.26	4.10	56.87		48.67	4.35	-3.85	20.87	52.77	0.25	30.61	22.41	
217	0.9(DL+COLL+CL-4)+1.6(WLI	235.00	118.16	4.91	26.11	26.26	1.09	53.46		51.28	0.94	-1.24	17.46	52.37	-0.15	27.20	25.02	
218	0.9(DL+COLL+CL-1)+1.6(WLI	288.35	85.59	28.75	32.04	19.02	6.39	57.45		44.67	19.41	6.63	21.45	51.06	13.02	38.43	25.65	
219	0.9(DL+COLL+CL-2)+1.6(WLI	284.76	85.59	15.28	31.64	19.02	3.39	54.06		47.27	16.01	9.22	18.06	50.67	12.62	35.04	28.25	

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220	0.9(DL+COLL+CL-3)+1.6(WL	285.51	85.59	18.44	31.72	19.02	4.10	54.84		46.65	16.80	8.60	18.84	50.75	12.70	35.82	27.63	
221	0.9(DL+COLL+CL-4)+1.6(WL	281.90	85.59	4.91	31.32	19.02	1.09	51.43		49.25	13.39	11.21	15.43	50.34	12.30	32.41	30.23	
222	0.9(DL+COLL+CL-1)+1.0(SL'+	375.70	0.83	29.58	41.74	0.18	6.57	48.50		35.36	48.13	34.99	12.50	41.93	41.56	48.32	35.18	
223	0.9(DL+COLL+CL-2)+1.0(SL'+	372.10	0.83	16.10	41.34	0.18	3.58	45.11		37.95	44.74	37.58	9.11	41.53	41.16	44.93	37.77	
224	0.9(DL+COLL+CL-3)+1.0(SL'+	372.86	0.83	19.27	41.43	0.18	4.28	45.89		37.33	45.53	36.96	9.89	41.61	41.25	45.71	37.15	
225	0.9(DL+COLL+CL-4)+1.0(SL'+	369.25	0.83	5.73	41.03	0.18	1.27	42.49		39.94	42.12	39.57	6.49	41.22	40.85	42.31	39.76	
226	0.9(DL+COLL+CL-1)+1.0(SL'-	375.46	0.88	27.87	41.72	0.20	6.19	48.11		35.72	47.72	35.33	12.11	41.92	41.53	47.92	35.53	
227	0.9(DL+COLL+CL-2)+1.0(SL'-	371.86	0.88	14.40	41.32	0.20	3.20	44.71		38.31	44.32	37.92	8.71	41.51	41.12	44.52	38.12	
228	0.9(DL+COLL+CL-3)+1.0(SL'-	372.62	0.88	17.57	41.40	0.20	3.90	45.50		37.69	45.11	37.30	9.50	41.60	41.21	45.31	37.50	
229	0.9(DL+COLL+CL-4)+1.0(SL'-	369.01	0.88	4.03	41.00	0.20	0.90	42.09		40.30	41.70	39.91	6.09	41.20	40.81	41.90	40.11	
	Max :	415	29	133														
	Min :	281	0	-92														

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 9.53 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 19.53 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 25.46 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 35.46 kN/m^2

Sample calculation of values of base pressure:

Load case : 302 1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL)

Maximum pressure, p_1 (pmax) = 46.09+0.00+20.52= 66.61 kN/m^2

p_2= 46.09+0.00-20.52= 25.56 kN/m^2

p_3= 46.09-(0.00+20.52)= 25.56 kN/m^2

p_4= 46.09-0.00+20.52= 66.61 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 66.61-2*18= 30.61 kN/m^2

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DESIGN OF ISOLATED FOOTING F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.75+2*0.5*0.332)+(0.75+2*0.5*0.332))*2

= 4.328 m

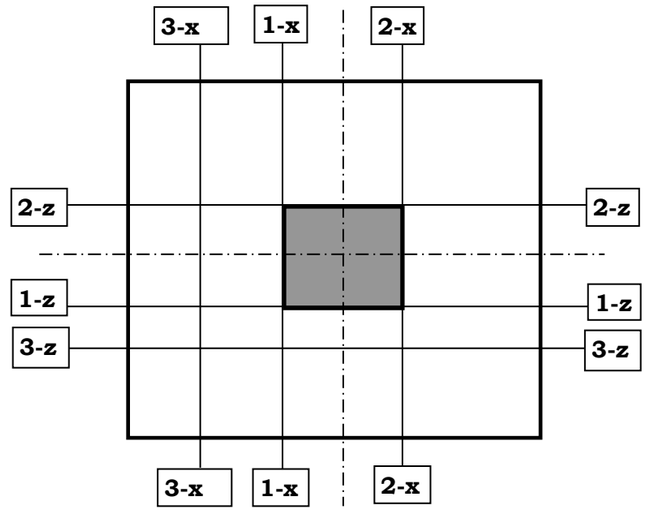
Punching Area= ((0.75+2*0.5*0.332)*(0.75+2*0.5*0.332))

= 1.17 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.6*18)

= 38.40 kN/m^2

Area of footing, A_f= 9.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+COLL)	42.09	42.09	42.09	2.34	2.34	2.88	28.91
102	1.2(DL+COLL+CL-1)+1.6(R	48.78	48.78	48.78	6.57	6.57	8.11	81.26
103	1.2(DL+COLL+CL-2)+1.6(R	48.25	48.25	48.25	6.23	6.23	7.69	77.08
104	1.2(DL+COLL+CL-3)+1.6(R	48.36	48.36	48.36	6.30	6.30	7.78	77.96
105	1.2(DL+COLL+CL-4)+1.6(R	47.83	47.83	47.83	5.97	5.97	7.36	73.77
106	1.4(DL+COLL+CL-1)	42.67	42.67	42.67	2.70	2.70	3.33	33.41
107	1.4(DL+COLL+CL-2)	42.05	42.05	42.05	2.31	2.31	2.85	28.54
108	1.4(DL+COLL+CL-3)	42.18	42.18	42.18	2.39	2.39	2.95	29.56
109	1.4(DL+COLL+CL-4)	41.55	41.55	41.55	1.99	1.99	2.46	24.67
110	1.2(DL+COLL+CL-1+TL1)+	48.92	48.92	48.92	6.66	6.66	8.22	82.33
111	1.2(DL+COLL+CL-2+TL1)+	48.38	48.38	48.38	6.32	6.32	7.79	78.16
112	1.2(DL+COLL+CL-3+TL1)+	48.49	48.49	48.49	6.39	6.39	7.88	79.03

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113	1.2(DL+COLL+CL-4+TL1)+	47.96	47.96	47.96	6.05	6.05	7.47	74.85
114	1.2(DL+COLL+CL-1+TL2)+	48.64	48.64	48.64	6.48	6.48	8.00	80.18
115	1.2(DL+COLL+CL-2+TL2)+	48.11	48.11	48.11	6.14	6.14	7.58	76.01
116	1.2(DL+COLL+CL-3+TL2)+	48.22	48.22	48.22	6.21	6.21	7.67	76.88
117	1.2(DL+COLL+CL-4+TL2)+	47.69	47.69	47.69	5.88	5.88	7.26	72.70
118	1.2(DL+COLL+CL-1)+0.8(W	37.70	37.70	37.70	-0.44	-0.44	-0.55	-5.52
119	1.2(DL+COLL+CL-2)+0.8(W	37.16	37.16	37.16	-0.78	-0.78	-0.97	-9.69
120	1.2(DL+COLL+CL-3)+0.8(W	37.28	37.28	37.28	-0.71	-0.71	-0.87	-8.81
121	1.2(DL+COLL+CL-4)+0.8(W	36.74	36.74	36.74	-1.05	-1.05	-1.30	-13.00
122	1.2(DL+COLL+CL-1)+0.8(W	40.66	40.66	40.66	1.43	1.43	1.77	17.70
123	1.2(DL+COLL+CL-2)+0.8(W	40.13	40.13	40.13	1.09	1.09	1.35	13.53
124	1.2(DL+COLL+CL-3)+0.8(W	40.24	40.24	40.24	1.16	1.16	1.44	14.41
125	1.2(DL+COLL+CL-4)+0.8(W	39.71	39.71	39.71	0.83	0.83	1.02	10.22
126	1.2(DL+COLL+CL-1)+0.8(W	34.95	34.95	34.95	-2.18	-2.18	-2.69	-27.07
127	1.2(DL+COLL+CL-2)+0.8(W	34.41	34.41	34.41	-2.52	-2.52	-3.12	-31.25
128	1.2(DL+COLL+CL-3)+0.8(W	34.52	34.52	34.52	-2.46	-2.46	-3.03	-30.37
129	1.2(DL+COLL+CL-4)+0.8(W	33.99	33.99	33.99	-2.79	-2.79	-3.44	-34.56
130	1.2(DL+COLL+CL-1)+0.8(W	40.86	40.86	40.86	1.56	1.56	1.92	19.25
131	1.2(DL+COLL+CL-2)+0.8(W	40.33	40.33	40.33	1.22	1.22	1.51	15.07
132	1.2(DL+COLL+CL-3)+0.8(W	40.44	40.44	40.44	1.29	1.29	1.59	15.95
133	1.2(DL+COLL+CL-4)+0.8(W	39.91	39.91	39.91	0.96	0.96	1.18	11.76
134	1.2(DL+COLL+CL-1)+0.8(W	38.22	31.24	41.42	4.30	-8.95	5.20	-28.75
135	1.2(DL+COLL+CL-2)+0.8(W	37.69	30.70	40.89	3.97	-9.29	4.78	-32.92
136	1.2(DL+COLL+CL-3)+0.8(W	37.81	30.82	41.01	4.04	-9.21	4.88	-32.04
137	1.2(DL+COLL+CL-4)+0.8(W	37.27	30.28	40.47	3.70	-9.55	4.46	-36.23
138	1.2(DL+COLL+CL-1)+0.8(W	39.91	34.77	42.26	4.20	-5.55	5.10	-8.32
139	1.2(DL+COLL+CL-2)+0.8(W	39.37	34.24	41.73	3.87	-5.89	4.69	-12.49
140	1.2(DL+COLL+CL-3)+0.8(W	39.49	34.36	41.85	3.94	-5.81	4.78	-11.61
141	1.2(DL+COLL+CL-4)+0.8(W	38.95	33.82	41.31	3.60	-6.15	4.36	-15.80
142	1.2(DL+COLL+CL-1)+0.8(W	38.12	31.56	41.13	3.98	-8.48	4.80	-27.87
143	1.2(DL+COLL+CL-2)+0.8(W	37.59	31.03	40.60	3.64	-8.82	4.39	-32.05

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144	1.2(DL+COLL+CL-3)+0.8(W	37.70	31.14	40.71	3.71	-8.75	4.48	-31.17
145	1.2(DL+COLL+CL-4)+0.8(W	37.17	30.61	40.18	3.38	-9.09	4.06	-35.36
146	1.2(DL+COLL+CL-1)+0.8(W	39.83	35.07	42.01	3.91	-5.11	4.75	-7.47
147	1.2(DL+COLL+CL-2)+0.8(W	39.30	34.54	41.48	3.58	-5.45	4.34	-11.64
148	1.2(DL+COLL+CL-3)+0.8(W	39.41	34.65	41.59	3.65	-5.38	4.42	-10.77
149	1.2(DL+COLL+CL-4)+0.8(W	38.87	34.11	41.05	3.30	-5.72	4.00	-14.95
150	1.2(DL+COLL+CL-1)+1.6(W	35.13	35.13	35.13	-2.07	-2.07	-2.55	-25.64
151	1.2(DL+COLL+CL-2)+1.6(W	34.59	34.59	34.59	-2.41	-2.41	-2.98	-29.81
152	1.2(DL+COLL+CL-3)+1.6(W	34.71	34.71	34.71	-2.34	-2.34	-2.88	-28.94
153	1.2(DL+COLL+CL-4)+1.6(W	34.17	34.17	34.17	-2.68	-2.68	-3.30	-33.12
154	1.2(DL+COLL+CL-1)+1.6(W	41.06	41.06	41.06	1.68	1.68	2.08	20.80
155	1.2(DL+COLL+CL-2)+1.6(W	40.52	40.52	40.52	1.34	1.34	1.66	16.62
156	1.2(DL+COLL+CL-3)+1.6(W	40.64	40.64	40.64	1.42	1.42	1.75	17.50
157	1.2(DL+COLL+CL-4)+1.6(W	40.10	40.10	40.10	1.08	1.08	1.33	13.31
158	1.2(DL+COLL+CL-1)+1.6(W	29.62	29.62	29.62	-5.56	-5.56	-6.86	-68.75
159	1.2(DL+COLL+CL-2)+1.6(W	29.09	29.09	29.09	-5.89	-5.89	-7.27	-72.93
160	1.2(DL+COLL+CL-3)+1.6(W	29.20	29.20	29.20	-5.82	-5.82	-7.19	-72.05
161	1.2(DL+COLL+CL-4)+1.6(W	28.67	28.67	28.67	-6.16	-6.16	-7.60	-76.24
162	1.2(DL+COLL+CL-1)+1.6(W	41.45	41.45	41.45	1.93	1.93	2.38	23.89
163	1.2(DL+COLL+CL-2)+1.6(W	40.92	40.92	40.92	1.59	1.59	1.97	19.71
164	1.2(DL+COLL+CL-3)+1.6(W	41.03	41.03	41.03	1.66	1.66	2.05	20.59
165	1.2(DL+COLL+CL-4)+1.6(W	40.50	40.50	40.50	1.33	1.33	1.64	16.40
166	1.2(DL+COLL+CL-1)+1.6(W	36.17	22.21	42.58	7.43	-19.08	8.94	-72.10
167	1.2(DL+COLL+CL-2)+1.6(W	35.64	21.68	42.05	7.09	-19.42	8.53	-76.28
168	1.2(DL+COLL+CL-3)+1.6(W	35.76	21.79	42.16	7.16	-19.34	8.61	-75.40
169	1.2(DL+COLL+CL-4)+1.6(W	35.22	21.26	41.63	6.83	-19.69	8.20	-79.59
170	1.2(DL+COLL+CL-1)+1.6(W	39.55	29.27	44.26	7.23	-12.28	8.76	-31.24
171	1.2(DL+COLL+CL-2)+1.6(W	39.02	28.74	43.73	6.89	-12.61	8.34	-35.42
172	1.2(DL+COLL+CL-3)+1.6(W	39.13	28.85	43.84	6.96	-12.54	8.43	-34.54
173	1.2(DL+COLL+CL-4)+1.6(W	38.60	28.32	43.31	6.63	-12.88	8.01	-38.73
174	1.2(DL+COLL+CL-1)+1.6(W	35.99	22.86	42.01	6.78	-18.15	8.16	-70.35

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175	1.2(DL+COLL+CL-2)+1.6(W	35.45	22.32	41.47	6.44	-18.48	7.74	-74.53
176	1.2(DL+COLL+CL-3)+1.6(W	35.57	22.44	41.59	6.51	-18.41	7.83	-73.65
177	1.2(DL+COLL+CL-4)+1.6(W	35.03	21.90	41.05	6.17	-18.75	7.41	-77.84
178	1.2(DL+COLL+CL-1)+1.6(W	39.39	29.88	43.75	6.64	-11.41	8.04	-29.55
179	1.2(DL+COLL+CL-2)+1.6(W	38.86	29.35	43.22	6.31	-11.75	7.63	-33.72
180	1.2(DL+COLL+CL-3)+1.6(W	38.97	29.46	43.33	6.38	-11.68	7.72	-32.84
181	1.2(DL+COLL+CL-4)+1.6(W	38.43	28.92	42.79	6.03	-12.02	7.29	-37.03
182	1.2(DL+COLL+CL-1)+1.0(S	44.38	44.29	44.42	3.84	3.67	4.74	46.44
183	1.2(DL+COLL+CL-2)+1.0(S	43.85	43.76	43.89	3.50	3.33	4.32	42.27
184	1.2(DL+COLL+CL-3)+1.0(S	43.96	43.87	44.00	3.58	3.40	4.41	43.15
185	1.2(DL+COLL+CL-4)+1.0(S	43.43	43.34	43.47	3.24	3.07	3.99	38.96
186	1.2(DL+COLL+CL-1)+1.0(S	44.35	44.26	44.40	3.83	3.64	4.72	46.24
187	1.2(DL+COLL+CL-2)+1.0(S	43.82	43.73	43.87	3.49	3.31	4.31	42.06
188	1.2(DL+COLL+CL-3)+1.0(S	43.93	43.84	43.98	3.56	3.38	4.40	42.94
189	1.2(DL+COLL+CL-4)+1.0(S	43.40	43.31	43.45	3.23	3.04	3.98	38.75
190	0.9(DL+COLL+CL-1)+1.6(W	32.54	32.54	32.54	-3.71	-3.71	-4.58	-45.90
191	0.9(DL+COLL+CL-2)+1.6(W	32.14	32.14	32.14	-3.96	-3.96	-4.89	-49.03
192	0.9(DL+COLL+CL-3)+1.6(W	32.22	32.22	32.22	-3.91	-3.91	-4.83	-48.37
193	0.9(DL+COLL+CL-4)+1.6(W	31.82	31.82	31.82	-4.16	-4.16	-5.14	-51.51
194	0.9(DL+COLL+CL-1)+1.6(W	38.47	38.47	38.47	0.04	0.04	0.05	0.54
195	0.9(DL+COLL+CL-2)+1.6(W	38.07	38.07	38.07	-0.21	-0.21	-0.26	-2.59
196	0.9(DL+COLL+CL-3)+1.6(W	38.16	38.16	38.16	-0.15	-0.15	-0.19	-1.93
197	0.9(DL+COLL+CL-4)+1.6(W	37.76	37.76	37.76	-0.41	-0.41	-0.50	-5.07
198	0.9(DL+COLL+CL-1)+1.6(W	27.03	27.03	27.03	-7.20	-7.20	-8.88	-89.01
199	0.9(DL+COLL+CL-2)+1.6(W	26.63	26.63	26.63	-7.45	-7.45	-9.19	-92.14
200	0.9(DL+COLL+CL-3)+1.6(W	26.72	26.72	26.72	-7.39	-7.39	-9.12	-91.48
201	0.9(DL+COLL+CL-4)+1.6(W	26.32	26.32	26.32	-7.64	-7.64	-9.43	-94.62
202	0.9(DL+COLL+CL-1)+1.6(W	38.86	38.86	38.86	0.29	0.29	0.36	3.63
203	0.9(DL+COLL+CL-2)+1.6(W	38.47	38.47	38.47	0.04	0.04	0.05	0.50
204	0.9(DL+COLL+CL-3)+1.6(W	38.55	38.55	38.55	0.09	0.09	0.12	1.16
205	0.9(DL+COLL+CL-4)+1.6(W	38.15	38.15	38.15	-0.16	-0.16	-0.20	-1.98

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206	0.9(DL+COLL+CL-1)+1.6(W	33.58	19.62	39.99	5.79	-20.72	6.92	-92.36
207	0.9(DL+COLL+CL-2)+1.6(W	33.19	19.22	39.60	5.54	-20.98	6.61	-95.49
208	0.9(DL+COLL+CL-3)+1.6(W	33.27	19.31	39.68	5.59	-20.92	6.68	-94.84
209	0.9(DL+COLL+CL-4)+1.6(W	32.87	18.91	39.28	5.34	-21.17	6.36	-97.98
210	0.9(DL+COLL+CL-1)+1.6(W	36.96	26.69	41.67	5.59	-13.91	6.73	-51.50
211	0.9(DL+COLL+CL-2)+1.6(W	36.57	26.29	41.28	5.34	-14.16	6.43	-54.63
212	0.9(DL+COLL+CL-3)+1.6(W	36.65	26.37	41.36	5.39	-14.11	6.49	-53.98
213	0.9(DL+COLL+CL-4)+1.6(W	36.24	25.97	40.95	5.13	-14.37	6.17	-57.12
214	0.9(DL+COLL+CL-1)+1.6(W	33.40	20.27	39.42	5.14	-19.78	6.13	-90.61
215	0.9(DL+COLL+CL-2)+1.6(W	33.00	19.87	39.02	4.89	-20.04	5.82	-93.74
216	0.9(DL+COLL+CL-3)+1.6(W	33.08	19.95	39.10	4.94	-19.99	5.88	-93.08
217	0.9(DL+COLL+CL-4)+1.6(W	32.68	19.55	38.70	4.69	-20.24	5.57	-96.22
218	0.9(DL+COLL+CL-1)+1.6(W	36.80	27.29	41.16	5.00	-13.05	6.02	-49.80
219	0.9(DL+COLL+CL-2)+1.6(W	36.40	26.89	40.76	4.75	-13.30	5.71	-52.93
220	0.9(DL+COLL+CL-3)+1.6(W	36.48	26.97	40.84	4.81	-13.25	5.78	-52.28
221	0.9(DL+COLL+CL-4)+1.6(W	36.08	26.57	40.44	4.55	-13.51	5.46	-55.42
222	0.9(DL+COLL+CL-1)+1.0(S	41.79	41.70	41.83	2.20	2.03	2.72	26.19
223	0.9(DL+COLL+CL-2)+1.0(S	41.39	41.30	41.43	1.95	1.78	2.41	23.06
224	0.9(DL+COLL+CL-3)+1.0(S	41.48	41.39	41.52	2.00	1.83	2.47	23.71
225	0.9(DL+COLL+CL-4)+1.0(S	41.08	40.99	41.12	1.76	1.58	2.16	20.57
226	0.9(DL+COLL+CL-1)+1.0(S	41.77	41.68	41.82	2.20	2.01	2.71	25.98
227	0.9(DL+COLL+CL-2)+1.0(S	41.36	41.27	41.41	1.94	1.75	2.39	22.85
228	0.9(DL+COLL+CL-3)+1.0(S	41.45	41.36	41.50	1.99	1.81	2.46	23.51
229	0.9(DL+COLL+CL-4)+1.0(S	41.05	40.96	41.10	1.74	1.56	2.15	20.37

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Sample calculation of BM & SF

Load case : 101 1.4(DL+COLL)

p_{1x} = 42.09-(42.09-42.09)*1.125/3 = 42.09 kN/m^2 Overhang = 1.125 m

p_{2x} = 42.09+(42.09-42.09)*1.125/3 = 42.09 kN/m^2 Effective depth = 0.344 m

p_{3x} = 42.09-(42.09-42.09)*(1.125-0.344)/3 = 42.09 kN/m^2

BM_{1x} = (42.09-38.4)*1.125*1.125/2+0.5*(42.09-42.09)*1.125*1.125*0.67 = 2.34 kN-m/m

BM_{2x} = (42.09-38.4)*1.125*1.125/2-0.5*(-42.09+42.09)*1.125*1.125*0.67 = 2.34 kN-m/m

SF_{3x} = (42.09-38.4)*(1.125-0.344)+0.5*(42.09-42.09)*(1.125-0.344) = 2.88 kN/m

$S.F_{(punching)}$ = (9-1.17)*(42.09-38.4) = 28.91 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	7.43	0.06	0.08	0.000	68.64	619.20	12ϕT	182.65	300	175	646.27
At top	-21.17	-0.18	0.24	0.001	196.18	309.60	10ϕT	253.68	300	250	314.16

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 8.94 kN/m

Shear stress , τ_v= (8.94*1000)/(1000*344)

= 0.03 N/mm^2 **O.K.**

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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 82.33 kN

Strength of Concrete in shear Vc1 = 4404.02 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2634.82 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2936.02 kN **Table 22.6.5.2_ (c)**

Design Vc= 2634.82 kN

Allowable shear stress, (ϕ Vc)/bd = 1.33 N/mm^2

Design punching shear stress, τ = (82.33*1000)/(4328*344)

= 0.06 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M$_{1z}$ kNm	B.M$_{2z}$ kNm	S.F$_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+COLL)	43.50	40.68	44.74	5.02	-0.35	6.21
102	1.2(DL+COLL+CL-1)+1.6(R	55.89	41.68	62.18	20.10	-6.96	24.81
103	1.2(DL+COLL+CL-2)+1.6(R	54.35	42.14	59.76	17.86	-5.40	22.06
104	1.2(DL+COLL+CL-3)+1.6(R	54.70	42.01	60.31	18.38	-5.78	22.69
105	1.2(DL+COLL+CL-4)+1.6(R	53.16	42.49	57.89	16.13	-4.20	19.93
106	1.4(DL+COLL+CL-1)	45.15	40.18	47.35	7.43	-2.03	9.18
107	1.4(DL+COLL+CL-2)	43.37	40.72	44.54	4.82	-0.21	5.97
108	1.4(DL+COLL+CL-3)	43.77	40.58	45.18	5.42	-0.65	6.71
109	1.4(DL+COLL+CL-4)	41.98	41.13	42.35	2.80	1.18	3.49
110	1.2(DL+COLL+CL-1+TL1)+	56.32	41.51	62.87	20.75	-7.44	25.61

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111	1.2(DL+COLL+CL-2+TL1)+	54.78	41.98	60.45	18.51	-5.88	22.85
112	1.2(DL+COLL+CL-3+TL1)+	55.13	41.85	61.00	19.03	-6.26	23.49
113	1.2(DL+COLL+CL-4+TL1)+	53.60	42.33	58.58	16.78	-4.68	20.73
114	1.2(DL+COLL+CL-1+TL2)+	55.45	41.83	61.47	19.45	-6.49	24.01
115	1.2(DL+COLL+CL-2+TL2)+	53.92	42.30	59.06	17.21	-4.92	21.26
116	1.2(DL+COLL+CL-3+TL2)+	54.27	42.18	59.62	17.73	-5.30	21.89
117	1.2(DL+COLL+CL-4+TL2)+	52.73	42.64	57.19	15.48	-3.73	19.13
118	1.2(DL+COLL+CL-1)+0.8(W	37.77	37.62	37.83	-0.31	-0.58	-0.39
119	1.2(DL+COLL+CL-2)+0.8(W	38.23	36.09	39.18	1.25	-2.82	1.51
120	1.2(DL+COLL+CL-3)+0.8(W	38.11	36.44	38.85	0.88	-2.30	1.05
121	1.2(DL+COLL+CL-4)+0.8(W	38.58	34.90	40.20	2.45	-4.55	2.97
122	1.2(DL+COLL+CL-1)+0.8(W	41.91	39.41	43.01	3.81	-0.95	4.70
123	1.2(DL+COLL+CL-2)+0.8(W	40.38	39.88	40.60	1.57	0.62	1.96
124	1.2(DL+COLL+CL-3)+0.8(W	40.73	39.76	41.15	2.09	0.24	2.59
125	1.2(DL+COLL+CL-4)+0.8(W	40.22	39.19	40.68	1.81	-0.16	2.25
126	1.2(DL+COLL+CL-1)+0.8(W	38.87	31.02	42.34	5.28	-9.66	6.41
127	1.2(DL+COLL+CL-2)+0.8(W	39.33	29.49	43.69	6.85	-11.90	8.32
128	1.2(DL+COLL+CL-3)+0.8(W	39.21	29.84	43.35	6.47	-11.38	7.86
129	1.2(DL+COLL+CL-4)+0.8(W	39.68	28.30	44.71	8.04	-13.62	9.77
130	1.2(DL+COLL+CL-1)+0.8(W	41.82	39.90	42.67	3.38	-0.27	4.19
131	1.2(DL+COLL+CL-2)+0.8(W	42.28	38.37	44.01	4.94	-2.51	6.09
132	1.2(DL+COLL+CL-3)+0.8(W	42.16	38.71	43.68	4.57	-1.99	5.63
133	1.2(DL+COLL+CL-4)+0.8(W	42.63	37.18	45.04	6.14	-4.24	7.55
134	1.2(DL+COLL+CL-1)+0.8(W	36.86	32.60	38.75	1.73	-6.38	2.06
135	1.2(DL+COLL+CL-2)+0.8(W	35.33	33.07	36.34	-0.50	-4.81	-0.69
136	1.2(DL+COLL+CL-3)+0.8(W	35.68	32.95	36.89	0.02	-5.19	-0.05
137	1.2(DL+COLL+CL-4)+0.8(W	34.14	33.41	34.46	-2.23	-3.62	-2.82
138	1.2(DL+COLL+CL-1)+0.8(W	39.47	35.21	41.36	3.39	-4.73	4.13
139	1.2(DL+COLL+CL-2)+0.8(W	37.94	35.68	38.95	1.15	-3.16	1.38
140	1.2(DL+COLL+CL-3)+0.8(W	38.29	35.56	39.50	1.67	-3.54	2.02
141	1.2(DL+COLL+CL-4)+0.8(W	36.75	36.02	37.07	-0.58	-1.97	-0.75
142	1.2(DL+COLL+CL-1)+0.8(W	36.97	32.71	38.86	1.80	-6.31	2.15
143	1.2(DL+COLL+CL-2)+0.8(W	35.44	33.17	36.44	-0.44	-4.75	-0.61
144	1.2(DL+COLL+CL-3)+0.8(W	35.79	33.06	36.99	0.08	-5.12	0.03

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145	1.2(DL+COLL+CL-4)+0.8(W	34.25	33.52	34.57	-2.16	-3.55	-2.73
146	1.2(DL+COLL+CL-1)+0.8(W	39.57	35.32	41.46	3.45	-4.66	4.21
147	1.2(DL+COLL+CL-2)+0.8(W	38.05	35.78	39.05	1.22	-3.09	1.46
148	1.2(DL+COLL+CL-3)+0.8(W	38.39	35.66	39.60	1.73	-3.47	2.10
149	1.2(DL+COLL+CL-4)+0.8(W	36.85	36.13	37.17	-0.52	-1.90	-0.67
150	1.2(DL+COLL+CL-1)+1.6(W	35.84	34.41	36.48	-0.70	-3.44	-0.92
151	1.2(DL+COLL+CL-2)+1.6(W	36.31	32.88	37.82	0.86	-5.68	0.98
152	1.2(DL+COLL+CL-3)+1.6(W	36.19	33.22	37.50	0.48	-5.16	0.53
153	1.2(DL+COLL+CL-4)+1.6(W	36.65	31.68	38.85	2.05	-7.41	2.44
154	1.2(DL+COLL+CL-1)+1.6(W	42.98	39.13	44.68	5.34	-1.98	6.59
155	1.2(DL+COLL+CL-2)+1.6(W	41.44	39.60	42.26	3.10	-0.42	3.83
156	1.2(DL+COLL+CL-3)+1.6(W	41.79	39.48	42.81	3.62	-0.79	4.47
157	1.2(DL+COLL+CL-4)+1.6(W	40.26	39.95	40.39	1.37	0.78	1.71
158	1.2(DL+COLL+CL-1)+1.6(W	38.04	21.20	45.49	10.48	-21.59	12.68
159	1.2(DL+COLL+CL-2)+1.6(W	38.51	19.68	46.84	12.04	-23.82	14.59
160	1.2(DL+COLL+CL-3)+1.6(W	38.38	20.02	46.51	11.66	-23.31	14.13
161	1.2(DL+COLL+CL-4)+1.6(W	38.85	18.48	47.86	13.24	-25.56	16.04
162	1.2(DL+COLL+CL-1)+1.6(W	43.94	38.96	46.14	6.67	-2.81	8.23
163	1.2(DL+COLL+CL-2)+1.6(W	44.41	37.43	47.50	8.24	-5.05	10.14
164	1.2(DL+COLL+CL-3)+1.6(W	44.28	37.78	47.16	7.86	-4.53	9.68
165	1.2(DL+COLL+CL-4)+1.6(W	44.75	36.24	48.52	9.43	-6.78	11.59
166	1.2(DL+COLL+CL-1)+1.6(W	32.88	25.51	36.14	1.19	-12.85	1.30
167	1.2(DL+COLL+CL-2)+1.6(W	31.34	25.97	33.72	-1.05	-11.28	-1.46
168	1.2(DL+COLL+CL-3)+1.6(W	31.70	25.85	34.28	-0.53	-11.66	-0.82
169	1.2(DL+COLL+CL-4)+1.6(W	30.15	26.32	31.85	-2.78	-10.09	-3.58
170	1.2(DL+COLL+CL-1)+1.6(W	38.10	30.73	41.36	4.49	-9.54	5.44
171	1.2(DL+COLL+CL-2)+1.6(W	36.56	31.19	38.94	2.25	-7.98	2.68
172	1.2(DL+COLL+CL-3)+1.6(W	36.91	31.07	39.50	2.77	-8.35	3.32
173	1.2(DL+COLL+CL-4)+1.6(W	35.37	31.54	37.07	0.53	-6.78	0.56
174	1.2(DL+COLL+CL-1)+1.6(W	33.10	25.73	36.36	1.33	-12.70	1.47
175	1.2(DL+COLL+CL-2)+1.6(W	31.57	26.20	33.95	-0.90	-11.14	-1.28
176	1.2(DL+COLL+CL-3)+1.6(W	31.92	26.07	34.50	-0.39	-11.52	-0.64
177	1.2(DL+COLL+CL-4)+1.6(W	30.38	26.54	32.08	-2.64	-9.94	-3.41
178	1.2(DL+COLL+CL-1)+1.6(W	38.31	30.94	41.57	4.63	-9.41	5.60

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179	1.2(DL+COLL+CL-2)+1.6(W	36.78	31.41	39.16	2.39	-7.84	2.85
180	1.2(DL+COLL+CL-3)+1.6(W	37.13	31.28	39.71	2.91	-8.22	3.49
181	1.2(DL+COLL+CL-4)+1.6(W	35.59	31.75	37.29	0.66	-6.65	0.72
182	1.2(DL+COLL+CL-1)+1.0(S	48.07	40.60	51.37	10.86	-3.35	13.41
183	1.2(DL+COLL+CL-2)+1.0(S	46.53	41.07	48.95	8.62	-1.79	10.66
184	1.2(DL+COLL+CL-3)+1.0(S	46.88	40.95	49.51	9.14	-2.16	11.30
185	1.2(DL+COLL+CL-4)+1.0(S	45.35	41.42	47.08	6.89	-0.59	8.53
186	1.2(DL+COLL+CL-1)+1.0(S	47.95	40.68	51.16	10.66	-3.18	13.17
187	1.2(DL+COLL+CL-2)+1.0(S	46.42	41.14	48.75	8.43	-1.62	10.42
188	1.2(DL+COLL+CL-3)+1.0(S	46.76	41.02	49.31	8.95	-2.00	11.06
189	1.2(DL+COLL+CL-4)+1.0(S	45.22	41.49	46.88	6.70	-0.42	8.29
190	0.9(DL+COLL+CL-1)+1.6(W	35.35	29.74	37.83	1.63	-9.05	1.90
191	0.9(DL+COLL+CL-2)+1.6(W	35.69	28.58	38.84	2.80	-10.73	3.32
192	0.9(DL+COLL+CL-3)+1.6(W	35.60	28.84	38.59	2.52	-10.34	2.98
193	0.9(DL+COLL+CL-4)+1.6(W	35.95	27.69	39.61	3.70	-12.03	4.42
194	0.9(DL+COLL+CL-1)+1.6(W	38.64	38.30	38.79	0.36	-0.27	0.45
195	0.9(DL+COLL+CL-2)+1.6(W	38.99	37.16	39.80	1.53	-1.95	1.87
196	0.9(DL+COLL+CL-3)+1.6(W	38.89	37.42	39.55	1.25	-1.56	1.53
197	0.9(DL+COLL+CL-4)+1.6(W	39.25	36.26	40.57	2.43	-3.25	2.97
198	0.9(DL+COLL+CL-1)+1.6(W	37.54	16.53	46.84	12.81	-27.20	15.50
199	0.9(DL+COLL+CL-2)+1.6(W	37.88	15.38	47.84	13.99	-28.88	16.92
200	0.9(DL+COLL+CL-3)+1.6(W	37.79	15.64	47.60	13.70	-28.49	16.58
201	0.9(DL+COLL+CL-4)+1.6(W	38.14	14.49	48.62	14.88	-30.18	18.02
202	0.9(DL+COLL+CL-1)+1.6(W	43.44	34.28	47.49	9.01	-8.43	11.05
203	0.9(DL+COLL+CL-2)+1.6(W	43.79	33.14	48.51	10.19	-10.10	12.48
204	0.9(DL+COLL+CL-3)+1.6(W	43.70	33.40	48.26	9.90	-9.71	12.14
205	0.9(DL+COLL+CL-4)+1.6(W	44.05	32.24	49.27	11.08	-11.40	13.57
206	0.9(DL+COLL+CL-1)+1.6(W	28.20	25.01	29.61	-4.42	-10.50	-5.63
207	0.9(DL+COLL+CL-2)+1.6(W	27.05	25.36	27.81	-6.10	-9.33	-7.69
208	0.9(DL+COLL+CL-3)+1.6(W	27.32	25.27	28.22	-5.71	-9.62	-7.21
209	0.9(DL+COLL+CL-4)+1.6(W	26.16	25.62	26.40	-7.40	-8.44	-9.28
210	0.9(DL+COLL+CL-1)+1.6(W	33.42	30.23	34.83	-1.12	-7.20	-1.49
211	0.9(DL+COLL+CL-2)+1.6(W	32.27	30.58	33.03	-2.80	-6.03	-3.55
212	0.9(DL+COLL+CL-3)+1.6(W	32.54	30.49	33.44	-2.41	-6.31	-3.07

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213	0.9(DL+COLL+CL-4)+1.6(W	31.38	30.84	31.62	-4.09	-5.13	-5.15
214	0.9(DL+COLL+CL-1)+1.6(W	28.43	25.23	29.84	-4.28	-10.36	-5.45
215	0.9(DL+COLL+CL-2)+1.6(W	27.27	25.58	28.03	-5.96	-9.19	-7.52
216	0.9(DL+COLL+CL-3)+1.6(W	27.54	25.49	28.44	-5.57	-9.48	-7.04
217	0.9(DL+COLL+CL-4)+1.6(W	26.38	25.84	26.62	-7.26	-8.30	-9.11
218	0.9(DL+COLL+CL-1)+1.6(W	33.64	30.44	35.05	-0.98	-7.07	-1.32
219	0.9(DL+COLL+CL-2)+1.6(W	32.49	30.80	33.25	-2.66	-5.89	-3.38
220	0.9(DL+COLL+CL-3)+1.6(W	32.75	30.70	33.66	-2.27	-6.17	-2.90
221	0.9(DL+COLL+CL-4)+1.6(W	31.59	31.05	31.83	-3.96	-5.00	-4.98
222	0.9(DL+COLL+CL-1)+1.0(S	43.39	40.11	44.85	5.25	-1.01	6.49
223	0.9(DL+COLL+CL-2)+1.0(S	42.25	40.46	43.04	3.57	0.16	4.43
224	0.9(DL+COLL+CL-3)+1.0(S	42.50	40.36	43.45	3.96	-0.12	4.90
225	0.9(DL+COLL+CL-4)+1.0(S	41.35	40.72	41.64	2.27	1.06	2.83
226	0.9(DL+COLL+CL-1)+1.0(S	43.27	40.18	44.64	5.05	-0.85	6.25
227	0.9(DL+COLL+CL-2)+1.0(S	42.12	40.52	42.83	3.37	0.32	4.18
228	0.9(DL+COLL+CL-3)+1.0(S	42.38	40.43	43.25	3.76	0.04	4.66
229	0.9(DL+COLL+CL-4)+1.0(S	41.23	40.78	41.43	2.07	1.22	2.59

Sample calculation of BM & SF

Load case : 101

p_{1z} = 47.72-(47.72-36.46)*1.125/3= 43.50 kN/m^2

p_{2z} = 36.46+(47.72-36.46)*1.125/3= 40.68 kN/m^2 Overhang = 1.125 m

p_{3z} = 47.72-(47.72-36.46)*(1.125-0.332)/3= 44.74 kN/m^2 Effective depth = 0.332 m

BM_{1z} = (43.50-38.4)*1.125*1.125/2+0.5*(47.72-43.50)*1.125*1.125*0.67 = 5.02 kN-m/m

BM_{2z} = (40.68-38.4)*1.125*1.125/2-0.5*(-36.46+40.68)*1.125*1.125*0.67 = -0.35 kN-m/m

SF_{3z} = (44.74-38.4)*(1.125-0.332)+0.5*(47.72-44.74)*(1.125-0.332) = 6.21 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	20.75	0.188	0.251	0.0006	206.48	597.60	12ϕT	189.25	300	175	646.3
At top	-13.62	-0.124	-0.165	-0.0004	-1.30	298.80	10ϕT	262.85	250	250	314.2

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 25.61 kN/m

Shear stress , τ_v= (25.61*1000)/(1000*332)

= 0.08 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	12ϕT	**175**	c/c
	At Top	10ϕT	**250**	c/c
Along Z Direction :				
	At Bottom	12ϕT	**175**	c/c
	At Top	10ϕT	**250**	c/c

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5.3 DESIGN OF CALCULATION OF FOOTING F3

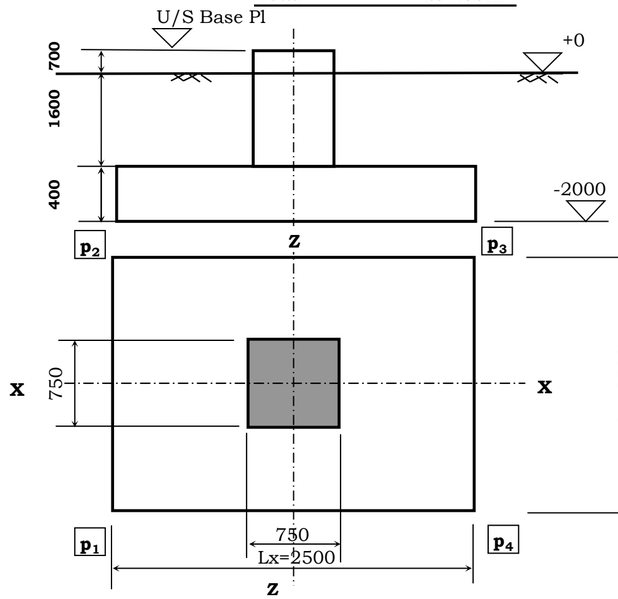
	Along-X	Along-Z
Size of column :	0.75m	0.75m
Size of fdn. raft :	2.50m	2.50m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-2.00 m	
U/S level Base Plate :	0.70 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

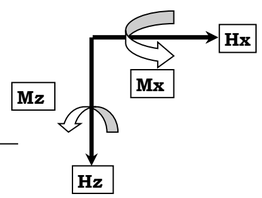
SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Crane Load	CL
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (2.5*2.5*0.4*24)+(0.75*0.75*2.3*24) = 91.05KN
Weight of soil = ((2.5*2.5)-(0.75*0.75))*1.6*18 = 163.80KN
Total = 254.85KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 344mm = 0.344m
Effective depth of slab along Z direction = 332mm = 0.332m
Clear projection along Z direction = 0.875m
Clear projection along X direction = 0.875m

H$_X$=Force along X-X direction

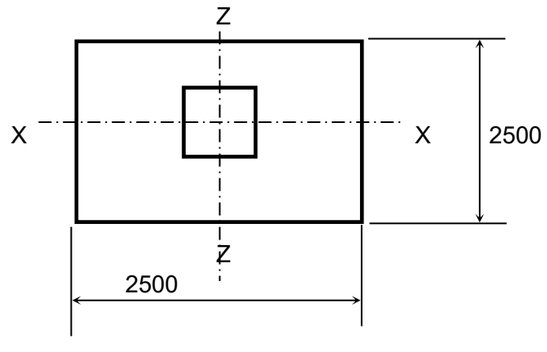
Hz=Force along Z-Z direction

M$_X$=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F3



Height from BOF to top of column about X (l_{ux}) = 2700mm
Height from BOF to top of column about Z (l_{uz}) = 2700mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	23.0856	0.0000	0		
2	COLLATERAL LOAD (COLL)	0	0.0000	0		
3	ROOF LIVE LOAD (RLL)	25.7592	0.0000	0		
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	0	0.0000	0		
5	CRANE LOAD-1 (CL-1)	0	0.0000	0		
6	CRANE LOAD-2 (CL-2)	0	0.0000	0		
7	CRANE LOAD-3 (CL-3)	0	0.0000	0		
8	CRANE LOAD-4 (CL-4)	0	0.0000	0		
9	WIND LEFT-1 (WLL-1)	-66.6984	-47.2812	-1.4136		
10	WIND LEFT-2 (WLL-2)	-44.7144	-19.0548	-0.0024		
11	WIND RIGHT-1 (WLR-1)	-61.0212	-47.4684	-12.4428		
12	WIND RIGHT-2 (WLR-2)	-36.3216	19.1244	-9.9036		
13	WIND LONG-1 (WLLG-1)	-61.3284	-47.3424	-4.6668		
14	WIND LONG-2 (WLLG-2)	-38.0136	19.0764	-2.1324		
15	WIND LONG-3 (WLLG-3)	-59.6376	-47.4072	-8.4576		
16	WIND LONG-4 (WLLG-4)	-36.1956	47.4072	-5.9148		
17	SEISMIC LOAD SL'+'	0	0.0000	0		
18	SEISMIC LOAD SL'-'	0	0.0000	0		
19	TEMPERATURE LOAD (TL1)	-27.1104	0.0000	-2.94		
20	TEMPERATURE LOAD (TL2)	27.1344	0.0000	0		
21	BRACING REACTION (BR-WLLG1)	0	0.0000	0		
22	BRACING REACTION (BR-WLLG2)	0	0.0000	0		
23	BRACING REACTION (BR-WLLG3)	0	0.0000	0		
24	BRACING REACTION (BR-WLLG4)	0	0.0000	0		
25	BRACING REACTION (BR-SL'+')	0	0.0000	0		
26	BRACING REACTION (BR-SL'-')	0	0.0000	0		

Ultimate Condition

Combined Load Cases:		Load at top of foundation			Load at bottom of foundation		
Load Case	Load Combination	Factored Load/Moment					
		Pu (kN)	Muz (kN-m)	Mux (kN-m)	Pu (kN)	Muz (kN-m)	Mux (kN-m)
101	1.4(DL+COLL)	32.32	0.0	0.0	287.2	0.0	0.0
102	1.2(DL+COLL+CL-1)+1.6(RLL+PLL / MLL)	68.9	0.0	0.0	323.8	0.0	0.0
103	1.2(DL+COLL+CL-2)+1.6(RLL+PLL / MLL)	68.9	0.0	0.0	323.8	0.0	0.0
104	1.2(DL+COLL+CL-3)+1.6(RLL+PLL / MLL)	68.9	0.0	0.0	323.8	0.0	0.0
105	1.2(DL+COLL+CL-4)+1.6(RLL+PLL / MLL)	68.9	0.0	0.0	323.8	0.0	0.0
106	1.4(DL+COLL+CL-1)	32.3	0.0	0.0	287.2	0.0	0.0
107	1.4(DL+COLL+CL-2)	32.3	0.0	0.0	287.2	0.0	0.0
108	1.4(DL+COLL+CL-3)	32.3	0.0	0.0	287.2	0.0	0.0
109	1.4(DL+COLL+CL-4)	32.3	0.0	0.0	287.2	0.0	0.0
110	1.2(DL+COLL+CL-1+TL1)+1.6(RLL+PLL / MLL)	36.4	0.0	9.5	291.2	0.0	9.5
111	1.2(DL+COLL+CL-2+TL1)+1.6(RLL+PLL / MLL)	36.4	0.0	9.5	291.2	0.0	9.5
112	1.2(DL+COLL+CL-3+TL1)+1.6(RLL+PLL / MLL)	36.4	0.0	9.5	291.2	0.0	9.5
113	1.2(DL+COLL+CL-4+TL1)+1.6(RLL+PLL / MLL)	36.4	0.0	9.5	291.2	0.0	9.5
114	1.2(DL+COLL+CL-1+TL2)+1.6(RLL+PLL / MLL)	101.5	0.0	0.0	356.3	0.0	0.0
115	1.2(DL+COLL+CL-2+TL2)+1.6(RLL+PLL / MLL)	101.5	0.0	0.0	356.3	0.0	0.0
116	1.2(DL+COLL+CL-3+TL2)+1.6(RLL+PLL / MLL)	101.5	0.0	0.0	356.3	0.0	0.0
117	1.2(DL+COLL+CL-4+TL2)+1.6(RLL+PLL / MLL)	101.5	0.0	0.0	356.3	0.0	0.0
118	1.2(DL+COLL+CL-1)+0.8(WLL-1)	-25.7	102.1	3.1	229.2	102.1	3.1
119	1.2(DL+COLL+CL-2)+0.8(WLL-1)	-25.7	102.1	3.1	229.2	102.1	3.1
120	1.2(DL+COLL+CL-3)+0.8(WLL-1)	-25.7	102.1	3.1	229.2	102.1	3.1
121	1.2(DL+COLL+CL-4)+0.8(WLL-1)	-25.7	102.1	3.1	229.2	102.1	3.1
122	1.2(DL+COLL+CL-1)+0.8(WLL-2)	-8.1	41.2	0.0	246.8	41.2	0.0
123	1.2(DL+COLL+CL-2)+0.8(WLL-2)	-8.1	41.2	0.0	246.8	41.2	0.0
124	1.2(DL+COLL+CL-3)+0.8(WLL-2)	-8.1	41.2	0.0	246.8	41.2	0.0
125	1.2(DL+COLL+CL-4)+0.8(WLL-2)	-8.1	41.2	0.0	246.8	41.2	0.0
126	1.2(DL+COLL+CL-1)+0.8(WLR-1)	-21.1	102.5	26.9	233.7	102.5	26.9
127	1.2(DL+COLL+CL-2)+0.8(WLR-1)	-21.1	102.5	26.9	233.7	102.5	26.9
128	1.2(DL+COLL+CL-3)+0.8(WLR-1)	-21.1	102.5	26.9	233.7	102.5	26.9
129	1.2(DL+COLL+CL-4)+0.8(WLR-1)	-21.1	102.5	26.9	233.7	102.5	26.9
130	1.2(DL+COLL+CL-1)+0.8(WLR-2)	-1.4	41.3	21.4	253.5	41.3	21.4
131	1.2(DL+COLL+CL-2)+0.8(WLR-2)	-1.4	41.3	21.4	253.5	41.3	21.4
132	1.2(DL+COLL+CL-3)+0.8(WLR-2)	-1.4	41.3	21.4	253.5	41.3	21.4
133	1.2(DL+COLL+CL-4)+0.8(WLR-2)	-1.4	41.3	21.4	253.5	41.3	21.4
134	1.2(DL+COLL+CL-1)+0.8(WLLG-1+BR-WLLG-1)	-21.4	102.3	10.1	233.5	102.3	10.1
135	1.2(DL+COLL+CL-2)+0.8(WLLG-1+BR-WLLG-1)	-21.4	102.3	10.1	233.5	102.3	10.1
136	1.2(DL+COLL+CL-3)+0.8(WLLG-1+BR-WLLG-1)	-21.4	102.3	10.1	233.5	102.3	10.1
137	1.2(DL+COLL+CL-4)+0.8(WLLG-1+BR-WLLG-1)	-21.4	102.3	10.1	233.5	102.3	10.1
138	1.2(DL+COLL+CL-1)+0.8(WLLG-2+BR-WLLG-2)	-2.7	41.2	4.6	252.1	41.2	4.6
139	1.2(DL+COLL+CL-2)+0.8(WLLG-2+BR-WLLG-2)	-2.7	41.2	4.6	252.1	41.2	4.6
140	1.2(DL+COLL+CL-3)+0.8(WLLG-2+BR-WLLG-2)	-2.7	41.2	4.6	252.1	41.2	4.6
141	1.2(DL+COLL+CL-4)+0.8(WLLG-2+BR-WLLG-2)	-2.7	41.2	4.6	252.1	41.2	4.6
142	1.2(DL+COLL+CL-1)+0.8(WLLG-3+BR-WLLG-3)	-20.0	102.4	18.3	234.8	102.4	18.3
143	1.2(DL+COLL+CL-2)+0.8(WLLG-3+BR-WLLG-3)	-20.0	102.4	18.3	234.8	102.4	18.3
144	1.2(DL+COLL+CL-3)+0.8(WLLG-3+BR-WLLG-3)	-20.0	102.4	18.3	234.8	102.4	18.3
145	1.2(DL+COLL+CL-4)+0.8(WLLG-3+BR-WLLG-3)	-20.0	102.4	18.3	234.8	102.4	18.3
146	1.2(DL+COLL+CL-1)+0.8(WLLG-4+BR-WLLG-4)	-1.3	102.4	12.8	253.6	102.4	12.8

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147	1.2(DL+COLL+CL-2)+0.8(WLLG-4+BR-WLLG-4)	-1.3	102.4	12.8	253.6	102.4	12.8
148	1.2(DL+COLL+CL-3)+0.8(WLLG-4+BR-WLLG-4)	-1.3	102.4	12.8	253.6	102.4	12.8
149	1.2(DL+COLL+CL-4)+0.8(WLLG-4+BR-WLLG-4)	-1.3	102.4	12.8	253.6	102.4	12.8
150	1.2(DL+COLL+CL-1)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-66.1	204.3	6.1	188.7	204.3	6.1
151	1.2(DL+COLL+CL-2)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-66.1	204.3	6.1	188.7	204.3	6.1
152	1.2(DL+COLL+CL-3)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-66.1	204.3	6.1	188.7	204.3	6.1
153	1.2(DL+COLL+CL-4)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	-66.1	204.3	6.1	188.7	204.3	6.1
154	1.2(DL+COLL+CL-1)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	-31.0	82.3	0.0	223.9	82.3	0.0
155	1.2(DL+COLL+CL-2)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	-31.0	82.3	0.0	223.9	82.3	0.0
156	1.2(DL+COLL+CL-3)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	-31.0	82.3	0.0	223.9	82.3	0.0
157	1.2(DL+COLL+CL-4)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	-31.0	82.3	0.0	223.9	82.3	0.0
158	1.2(DL+COLL+CL-1)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-57.1	205.1	53.8	197.8	205.1	53.8
159	1.2(DL+COLL+CL-2)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-57.1	205.1	53.8	197.8	205.1	53.8
160	1.2(DL+COLL+CL-3)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-57.1	205.1	53.8	197.8	205.1	53.8
161	1.2(DL+COLL+CL-4)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	-57.1	205.1	53.8	197.8	205.1	53.8
162	1.2(DL+COLL+CL-1)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	-17.5	205.1	42.8	237.3	205.1	42.8
163	1.2(DL+COLL+CL-2)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	-17.5	205.1	42.8	237.3	205.1	42.8
164	1.2(DL+COLL+CL-3)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	-17.5	205.1	42.8	237.3	205.1	42.8
165	1.2(DL+COLL+CL-4)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	-17.5	82.6	42.8	237.3	82.6	42.8
166	1.2(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-57.5	204.5	20.2	197.3	204.5	20.2
167	1.2(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-57.5	204.5	20.2	197.3	204.5	20.2
168	1.2(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-57.5	204.5	20.2	197.3	204.5	20.2
169	1.2(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-57.5	204.5	20.2	197.3	204.5	20.2
170	1.2(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-20.2	82.4	9.2	234.6	82.4	9.2
171	1.2(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-20.2	82.4	9.2	234.6	82.4	9.2
172	1.2(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-20.2	82.4	9.2	234.6	82.4	9.2
173	1.2(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-20.2	82.4	9.2	234.6	82.4	9.2
174	1.2(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-54.8	204.8	36.5	200.0	204.8	36.5
175	1.2(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-54.8	204.8	36.5	200.0	204.8	36.5
176	1.2(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-54.8	204.8	36.5	200.0	204.8	36.5
177	1.2(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	-54.8	204.8	36.5	200.0	204.8	36.5
178	1.2(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-17.3	204.8	25.6	237.5	204.8	25.6
179	1.2(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-17.3	204.8	25.6	237.5	204.8	25.6
180	1.2(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-17.3	204.8	25.6	237.5	204.8	25.6
181	1.2(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-17.3	204.8	25.6	237.5	204.8	25.6
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	40.6	0.0	0.0	295.4	0.0	0.0
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	40.6	0.0	0.0	295.4	0.0	0.0
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	40.6	0.0	0.0	295.4	0.0	0.0
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	40.6	0.0	0.0	295.4	0.0	0.0
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	40.6	0.0	0.0	295.4	0.0	0.0
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-)+0.5(RLL+PLL / MLL	40.6	0.0	0.0	295.4	0.0	0.0
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	40.6	0.0	0.0	295.4	0.0	0.0
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	40.6	0.0	0.0	295.4	0.0	0.0
190	0.9(DL+COLL+CL-1)+1.6(WLL-1)	-85.9	204.3	6.1	168.9	204.3	6.1
191	0.9(DL+COLL+CL-2)+1.6(WLL-1)	-85.9	204.3	6.1	168.9	204.3	6.1
192	0.9(DL+COLL+CL-3)+1.6(WLL-1)	-85.9	204.3	6.1	168.9	204.3	6.1
193	0.9(DL+COLL+CL-4)+1.6(WLL-1)	-85.9	204.3	6.1	168.9	204.3	6.1
194	0.9(DL+COLL+CL-1)+1.6(WLL-2)	-50.8	82.3	0.0	204.1	82.3	0.0
195	0.9(DL+COLL+CL-2)+1.6(WLL-2)	-50.8	82.3	0.0	204.1	82.3	0.0
196	0.9(DL+COLL+CL-3)+1.6(WLL-2)	-50.8	82.3	0.0	204.1	82.3	0.0
197	0.9(DL+COLL+CL-4)+1.6(WLL-2)	-50.8	82.3	0.0	204.1	82.3	0.0
198	0.9(DL+COLL+CL-1)+1.6(WLR-1)	-76.9	205.1	53.8	178.0	205.1	53.8
199	0.9(DL+COLL+CL-2)+1.6(WLR-1)	-76.9	205.1	53.8	178.0	205.1	53.8
200	0.9(DL+COLL+CL-3)+1.6(WLR-1)	-76.9	205.1	53.8	178.0	205.1	53.8
201	0.9(DL+COLL+CL-4)+1.6(WLR-1)	-76.9	205.1	53.8	178.0	205.1	53.8
202	0.9(DL+COLL+CL-1)+1.6(WLR-2)	-37.3	82.6	42.8	217.5	82.6	42.8
203	0.9(DL+COLL+CL-2)+1.6(WLR-2)	-37.3	82.6	42.8	217.5	82.6	42.8
204	0.9(DL+COLL+CL-3)+1.6(WLR-2)	-37.3	82.6	42.8	217.5	82.6	42.8
205	0.9(DL+COLL+CL-4)+1.6(WLR-2)	-37.3	82.6	42.8	217.5	82.6	42.8
206	0.9(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)	-77.3	204.5	20.2	177.5	204.5	20.2
207	0.9(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)	-77.3	204.5	20.2	177.5	204.5	20.2
208	0.9(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)	-77.3	204.5	20.2	177.5	204.5	20.2
209	0.9(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)	-77.3	204.5	20.2	177.5	204.5	20.2
210	0.9(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)	-40.0	82.4	9.2	214.8	82.4	9.2
211	0.9(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)	-40.0	82.4	9.2	214.8	82.4	9.2
212	0.9(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)	-40.0	82.4	9.2	214.8	82.4	9.2
213	0.9(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)	-40.0	82.4	9.2	214.8	82.4	9.2
214	0.9(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)	-74.6	204.8	36.5	180.2	204.8	36.5
215	0.9(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)	-74.6	204.8	36.5	180.2	204.8	36.5
216	0.9(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)	-74.6	204.8	36.5	180.2	204.8	36.5
217	0.9(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)	-74.6	204.8	36.5	180.2	204.8	36.5
218	0.9(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG-4)	-37.1	204.8	25.6	217.7	204.8	25.6
219	0.9(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG-4)	-37.1	204.8	25.6	217.7	204.8	25.6
220	0.9(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG-4)	-37.1	204.8	25.6	217.7	204.8	25.6
221	0.9(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG-4)	-37.1	204.8	25.6	217.7	204.8	25.6
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')	20.8	0.0	0.0	275.6	0.0	0.0
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')	20.8	0.0	0.0	275.6	0.0	0.0
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')	20.8	0.0	0.0	275.6	0.0	0.0
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')	20.8	0.0	0.0	275.6	0.0	0.0
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')	20.8	0.0	0.0	275.6	0.0	0.0
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-')	20.8	0.0	0.0	275.6	0.0	0.0
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')	20.8	0.0	0.0	275.6	0.0	0.0
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')	20.8	0.0	0.0	275.6	0.0	0.0

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Allowable stress design

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
301	1.0(DL+COLL)	23.0856	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL	21.7344	0	0.0	-2.9	-7.9	276.6	0.0	0.0	-2.9	-7.9
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL / MLL	21.7344	0	0.0	-2.9	-7.9	276.6	0.0	0.0	-2.9	-7.9
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL / MLL	21.7344	0	0.0	-2.9	-7.9	276.6	0.0	0.0	-2.9	-7.9
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL / MLL	21.7344	0	0.0	-2.9	-7.9	276.6	0.0	0.0	-2.9	-7.9
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL / MLL	75.9792	0	0.0	0.0	0.0	330.8	0.0	0.0	0.0	0.0
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL / MLL	75.9792	0	0.0	0.0	0.0	330.8	0.0	0.0	0.0	0.0
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL / MLL	75.9792	0	0.0	0.0	0.0	330.8	0.0	0.0	0.0	0.0
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL / MLL	75.9792	0	0.0	0.0	0.0	330.8	0.0	0.0	0.0	0.0
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL+PLL	22.07	0	0.0	-2.2	-6.0	276.9	0.0	0.0	-2.2	-6.0
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL+PLL	22.07	0	0.0	-2.2	-6.0	276.9	0.0	0.0	-2.2	-6.0
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL+PLL	22.07	0	0.0	-2.2	-6.0	276.9	0.0	0.0	-2.2	-6.0
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL+PLL	22.07	0	0.0	-2.2	-6.0	276.9	0.0	0.0	-2.2	-6.0
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL+PLL	62.76	0	0.0	0.0	0.0	317.6	0.0	0.0	0.0	0.0
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL+PLL	62.76	0	0.0	0.0	0.0	317.6	0.0	0.0	0.0	0.0
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL+PLL	62.76	0	0.0	0.0	0.0	317.6	0.0	0.0	0.0	0.0
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL+PLL	62.76	0	0.0	0.0	0.0	317.6	0.0	0.0	0.0	0.0
318	1.0(DL+COLL+CL-1)	23.0856	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
319	1.0(DL+COLL+CL-2)	23.0856	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
320	1.0(DL+COLL+CL-3)	23.0856	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
321	1.0(DL+COLL+CL-4)	23.0856	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	-43.6128	-47.2812	-47.3	-1.4	-3.8	211.2	-47.3	-47.3	-1.4	-3.8
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	-43.6128	-47.2812	-47.3	-1.4	-3.8	211.2	-47.3	-47.3	-1.4	-3.8
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	-43.6128	-47.2812	-47.3	-1.4	-3.8	211.2	-47.3	-47.3	-1.4	-3.8
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	-43.6128	-47.2812	-47.3	-1.4	-3.8	211.2	-47.3	-47.3	-1.4	-3.8
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	-21.6288	-19.0548	-19.1	0.0	0.0	233.2	-19.1	-19.1	0.0	0.0
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	-21.6288	-19.0548	-19.1	0.0	0.0	233.2	-19.1	-19.1	0.0	0.0
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	-21.6288	-19.0548	-19.1	0.0	0.0	233.2	-19.1	-19.1	0.0	0.0
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	-21.6288	-19.0548	-19.1	0.0	0.0	233.2	-19.1	-19.1	0.0	0.0
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	-37.9356	-47.4684	-47.5	-12.4	-33.6	216.9	-47.5	-47.5	-12.4	-33.6
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	-37.9356	-47.4684	-47.5	-12.4	-33.6	216.9	-47.5	-47.5	-12.4	-33.6
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	-37.9356	-47.4684	-47.5	-12.4	-33.6	216.9	-47.5	-47.5	-12.4	-33.6
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	-37.9356	-47.4684	-47.5	-12.4	-33.6	216.9	-47.5	-47.5	-12.4	-33.6
334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	-13.236	19.1244	19.1	-9.9	-26.7	241.6	19.1	19.1	-9.9	-26.7
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	-13.236	19.1244	19.1	-9.9	-26.7	241.6	19.1	19.1	-9.9	-26.7
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	-13.236	19.1244	19.1	-9.9	-26.7	241.6	19.1	19.1	-9.9	-26.7
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	-13.236	19.1244	19.1	-9.9	-26.7	241.6	19.1	19.1	-9.9	-26.7
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	-38.2428	-47.3424	-47.3	-4.7	-4.7	216.6	-47.3	-47.3	-4.7	-4.7
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	-38.2428	-47.3424	-47.3	-4.7	-4.7	216.6	-47.3	-47.3	-4.7	-4.7
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	-38.2428	-47.3424	-47.3	-4.7	-4.7	216.6	-47.3	-47.3	-4.7	-4.7
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	-38.2428	-47.3424	-47.3	-4.7	-4.7	216.6	-47.3	-47.3	-4.7	-4.7
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	-14.928	19.0764	19.1	-2.1	-2.1	239.9	19.1	19.1	-2.1	-2.1
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	-14.928	19.0764	19.1	-2.1	-2.1	239.9	19.1	19.1	-2.1	-2.1
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	-14.928	19.0764	19.1	-2.1	-2.1	239.9	19.1	19.1	-2.1	-2.1
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	-14.928	19.0764	19.1	-2.1	-2.1	239.9	19.1	19.1	-2.1	-2.1
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-36.552	-47.4072	-47.4	-8.5	-8.5	218.3	-47.4	-47.4	-8.5	-8.5
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	-36.552	-47.4072	-47.4	-8.5	-8.5	218.3	-47.4	-47.4	-8.5	-8.5
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	-36.552	-47.4072	-47.4	-8.5	-8.5	218.3	-47.4	-47.4	-8.5	-8.5
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	-36.552	-47.4072	-47.4	-8.5	-8.5	218.3	-47.4	-47.4	-8.5	-8.5
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-13.11	47.4072	47.4	-5.9	-5.9	241.7	47.4	47.4	-5.9	-5.9
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	-13.11	47.4072	47.4	-5.9	-5.9	241.7	47.4	47.4	-5.9	-5.9
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	-13.11	47.4072	47.4	-5.9	-5.9	241.7	47.4	47.4	-5.9	-5.9
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	-13.11	47.4072	47.4	-5.9	-5.9	241.7	47.4	47.4	-5.9	-5.9
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(RLL+	-7.62	-35.4609	-95.7	-1.1	-2.9	247.2	-35.5	-95.7	-1.1	-2.9
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(RLL+	-7.62	-35.4609	-95.7	-1.1	-2.9	247.2	-35.5	-95.7	-1.1	-2.9
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(RLL+	-7.62	-35.4609	-95.7	-1.1	-2.9	247.2	-35.5	-95.7	-1.1	-2.9
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(RLL+	-7.62	-35.4609	-95.7	-1.1	-2.9	247.2	-35.5	-95.7	-1.1	-2.9
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(RLL+	8.87	-14.2911	-38.6	0.0	0.0	263.7	-14.3	-38.6	0.0	0.0
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(RLL+	8.87	-14.2911	-38.6	0.0	0.0	263.7	-14.3	-38.6	0.0	0.0
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(RLL+	8.87	-14.2911	-38.6	0.0	0.0	263.7	-14.3	-38.6	0.0	0.0
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(RLL+	8.87	-14.2911	-38.6	0.0	0.0	263.7	-14.3	-38.6	0.0	0.0
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(RLL+	-3.36	-35.6013	-96.1	-9.3	-25.2	251.5	-35.6	-96.1	-9.3	-25.2
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(RLL+	-3.36	-35.6013	-96.1	-9.3	-25.2	251.5	-35.6	-96.1	-9.3	-25.2
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(RLL+	-3.36	-35.6013	-96.1	-9.3	-25.2	251.5	-35.6	-96.1	-9.3	-25.2
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(RLL+	-3.36	-35.6013	-96.1	-9.3	-25.2	251.5	-35.6	-96.1	-9.3	-25.2
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(RLL+	15.16	14.3433	38.7	-7.4	-20.1	270.0	14.3	38.7	-7.4	-20.1
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(RLL+	15.16	14.3433	38.7	-7.4	-20.1	270.0	14.3	38.7	-7.4	-20.1
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(RLL+	15.16	14.3433	38.7	-7.4	-20.1	270.0	14.3	38.7	-7.4	-20.1
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(RLL+	15.16	14.3433	38.7	-7.4	-20.1	270.0	14.3	38.7	-7.4	-20.1
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-WLLG	-3.59	-35.5068	-35.5	-3.5	-3.5	251.3	-35.5	-35.5	-3.5	-3.5
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-WLLG	-3.59	-35.5068	-35.5	-3.5	-3.5	251.3	-35.5	-35.5	-3.5	-3.5
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-WLLG	-3.59	-35.5068	-35.5	-3.5	-3.5	251.3	-35.5	-35.5	-3.5	-3.5
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-WLLG	-3.59	-35.5068	-35.5	-3.5	-3.5	251.3	-35.5	-35.5	-3.5	-3.5
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-WLLG	13.89	14.3073	14.3	-1.6	-1.6	268.7	14.3	14.3	-1.6	-1.6
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-WLLG	13.89	14.3073	14.3	-1.6	-1.6	268.7	14.3	14.3	-1.6	-1.6
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-WLLG	13.89	14.3073	14.3	-1.6	-1.6	268.7	14.3	14.3	-1.6	-1.6
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-WLLG	13.89	14.3073	14.3	-1.6	-1.6	268.7	14.3	14.3	-1.6	-1.6
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-WLLG	-2.32	-35.5554	-35.6	-6.3	-6.3	252.5	-35.6	-35.6	-6.3	-6.3
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-WLLG	-2.32	-35.5554	-35.6	-6.3	-6.3	252.5	-35.6	-35.6	-6.3	-6.3
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-WLLG	-2.32	-35.5554	-35.6	-6.3	-6.3	252.5	-35.6	-35.6	-6.3	-6.3
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-WLLG	-2.32	-35.5554	-35.6	-6.3	-6.3	252.5	-35.6	-35.6	-6.3	-6.3
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-WLLG	15.26	35.5554	35.6	-4.4	-4.4	270.1	35.6	35.6	-4.4	-4.4
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-WLLG	15.26	35.5554	35.6	-4.4	-4.4	270.1	35.6	35.6	-4.4	-4.4
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-WLLG	15.26	35.5554	35.6	-4.4	-4.4	270.1	35.6	35.6	-4.4	-4.4
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-WLLG	15.26	35.5554	35.6	-4.4	-4.4	270.1	35.6	35.6	-4.4	-4.4
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+')	23.09	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+")	23.09	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+')	23.09	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+')	23.09	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0

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390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	23.09	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	23.09	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	23.09	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	23.09	0	0.0	0.0	0.0	277.9	0.0	0.0	0.0	0.0
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'+')+0	42.41	0	0.0	0.0	0.0	297.3	0.0	0.0	0.0	0.0
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'+')+0	42.41	0	0.0	0.0	0.0	297.3	0.0	0.0	0.0	0.0
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'+')+0	42.41	0	0.0	0.0	0.0	297.3	0.0	0.0	0.0	0.0
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'+')+0	42.41	0	0.0	0.0	0.0	297.3	0.0	0.0	0.0	0.0
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-')+0.7	42.41	0	0.0	0.0	0.0	297.3	0.0	0.0	0.0	0.0
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-')+0.7	42.41	0	0.0	0.0	0.0	297.3	0.0	0.0	0.0	0.0
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-')+0.7	42.41	0	0.0	0.0	0.0	297.3	0.0	0.0	0.0	0.0
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-')+0.7	42.41	0	0.0	0.0	0.0	297.3	0.0	0.0	0.0	0.0
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	-52.85	-47.2812	-47.3	-1.4	-3.8	202.0	-47.3	-47.3	-1.4	-3.8
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	-52.85	-47.2812	-47.3	-1.4	-3.8	202.0	-47.3	-47.3	-1.4	-3.8
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	-52.85	-47.2812	-47.3	-1.4	-3.8	202.0	-47.3	-47.3	-1.4	-3.8
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	-52.85	-47.2812	-47.3	-1.4	-3.8	202.0	-47.3	-47.3	-1.4	-3.8
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	-30.86304	-19.0548	-19.1	0.0	0.0	224.0	-19.1	-19.1	0.0	0.0
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	-30.86304	-19.0548	-19.1	0.0	0.0	224.0	-19.1	-19.1	0.0	0.0
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	-30.86304	-19.0548	-19.1	0.0	0.0	224.0	-19.1	-19.1	0.0	0.0
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	-30.86304	-19.0548	-19.1	0.0	0.0	224.0	-19.1	-19.1	0.0	0.0
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	-47.16984	-47.4684	-47.5	-12.4	-33.6	207.7	-47.5	-47.5	-12.4	-33.6
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	-47.16984	-47.4684	-47.5	-12.4	-33.6	207.7	-47.5	-47.5	-12.4	-33.6
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	-47.16984	-47.4684	-47.5	-12.4	-33.6	207.7	-47.5	-47.5	-12.4	-33.6
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	-47.16984	-47.4684	-47.5	-12.4	-33.6	207.7	-47.5	-47.5	-12.4	-33.6
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	-22.47024	19.1244	19.1	-9.9	-26.7	232.4	19.1	19.1	-9.9	-26.7
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	-22.47024	19.1244	19.1	-9.9	-26.7	232.4	19.1	19.1	-9.9	-26.7
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	-22.47024	19.1244	19.1	-9.9	-26.7	232.4	19.1	19.1	-9.9	-26.7
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	-22.47024	19.1244	19.1	-9.9	-26.7	232.4	19.1	19.1	-9.9	-26.7
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	-47.47704	-47.3424	-47.3	-4.7	-4.7	207.4	-47.3	-47.3	-4.7	-4.7
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	-47.47704	-47.3424	-47.3	-4.7	-4.7	207.4	-47.3	-47.3	-4.7	-4.7
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	-47.47704	-47.3424	-47.3	-4.7	-4.7	207.4	-47.3	-47.3	-4.7	-4.7
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	-47.47704	-47.3424	-47.3	-4.7	-4.7	207.4	-47.3	-47.3	-4.7	-4.7
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	-24.16224	19.0764	19.1	-2.1	-2.1	230.7	19.1	19.1	-2.1	-2.1
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	-24.16224	19.0764	19.1	-2.1	-2.1	230.7	19.1	19.1	-2.1	-2.1
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	-24.16224	19.0764	19.1	-2.1	-2.1	230.7	19.1	19.1	-2.1	-2.1
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	-24.16224	19.0764	19.1	-2.1	-2.1	230.7	19.1	19.1	-2.1	-2.1
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-45.78624	-47.4072	-47.4	-8.5	-8.5	209.1	-47.4	-47.4	-8.5	-8.5
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	-45.78624	-47.4072	-47.4	-8.5	-8.5	209.1	-47.4	-47.4	-8.5	-8.5
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	-45.78624	-47.4072	-47.4	-8.5	-8.5	209.1	-47.4	-47.4	-8.5	-8.5
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	-45.78624	-47.4072	-47.4	-8.5	-8.5	209.1	-47.4	-47.4	-8.5	-8.5
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-WLLG-	-22.34424	47.4072	47.4	-5.9	-5.9	232.5	47.4	47.4	-5.9	-5.9
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	-22.34424	47.4072	47.4	-5.9	-5.9	232.5	47.4	47.4	-5.9	-5.9
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	-22.34424	47.4072	47.4	-5.9	-5.9	232.5	47.4	47.4	-5.9	-5.9
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	-22.34424	47.4072	47.4	-5.9	-5.9	232.5	47.4	47.4	-5.9	-5.9
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	13.85	0	0.0	0.0	0.0	268.7	0.0	0.0	0.0	0.0
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	13.85	0	0.0	0.0	0.0	268.7	0.0	0.0	0.0	0.0
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	13.85	0	0.0	0.0	0.0	268.7	0.0	0.0	0.0	0.0
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	13.85	0	0.0	0.0	0.0	268.7	0.0	0.0	0.0	0.0
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	-26.17	-28.36872	-76.6	-0.8	-2.3	228.7	-28.4	-76.6	-0.8	-2.3
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	-12.98	-11.43288	-30.9	0.0	0.0	241.9	-11.4	-30.9	0.0	0.0
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	-22.76	-28.48104	-76.9	-7.5	-20.2	232.1	-28.5	-76.9	-7.5	-20.2
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	-7.94	11.47464	31.0	-5.9	-16.0	246.9	11.5	31.0	-5.9	-16.0

Check for overturning and sliding:

Load Case	Load Combination	Load at bottom of foundation					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
301	1.0(DL+COLL)	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL /	276.58	0.00	0.00	2.94	7.94	NA	39.20	NA	33.87
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL /	276.58	0.00	0.00	2.94	7.94	NA	39.20	NA	33.87
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL /	276.58	0.00	0.00	2.94	7.94	NA	39.20	NA	33.87
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL /	276.58	0.00	0.00	2.94	7.94	NA	39.20	NA	33.87
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL /	330.83	0.00	0.00	0.00	0.00	NA	NA	NA	NA
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL /	330.83	0.00	0.00	0.00	0.00	NA	NA	NA	NA
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL /	330.83	0.00	0.00	0.00	0.00	NA	NA	NA	NA
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL /	330.83	0.00	0.00	0.00	0.00	NA	NA	NA	NA
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL	276.92	0.00	0.00	2.21	5.95	NA	52.33	NA	45.21
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL	276.92	0.00	0.00	2.21	5.95	NA	52.33	NA	45.21
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL	276.92	0.00	0.00	2.21	5.95	NA	52.33	NA	45.21
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL	276.92	0.00	0.00	2.21	5.95	NA	52.33	NA	45.21
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL	317.61	0.00	0.00	0.00	0.00	NA	NA	NA	NA
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL	317.61	0.00	0.00	0.00	0.00	NA	NA	NA	NA
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL	317.61	0.00	0.00	0.00	0.00	NA	NA	NA	NA
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL	317.61	0.00	0.00	0.00	0.00	NA	NA	NA	NA
318	1.0(DL+COLL+CL-1)	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
319	1.0(DL+COLL+CL-2)	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
320	1.0(DL+COLL+CL-3)	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
321	1.0(DL+COLL+CL-4)	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	211.24	-47.28	-47.28	1.41	3.82	5.03	62.26	1.79	53.80
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	211.24	-47.28	-47.28	1.41	3.82	5.03	62.26	1.79	53.80
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	211.24	-47.28	-47.28	1.41	3.82	5.03	62.26	1.79	53.80
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	211.24	-47.28	-47.28	1.41	3.82	5.03	62.26	1.79	53.80
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	233.22	-19.05	-19.05	0.00	0.01	13.77	40489.79	4.90	34983.18
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	233.22	-19.05	-19.05	0.00	0.01	13.77	40489.79	4.90	34983.18
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	233.22	-19.05	-19.05	0.00	0.01	13.77	40489.79	4.90	34983.18
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	233.22	-19.05	-19.05	0.00	0.01	13.77	40489.79	4.90	34983.18
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	216.91	-47.47	-47.47	12.44	33.60	5.14	7.26	1.83	6.28
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	216.91	-47.47	-47.47	12.44	33.60	5.14	7.26	1.83	6.28
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	216.91	-47.47	-47.47	12.44	33.60	5.14	7.26	1.83	6.28
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	216.91	-47.47	-47.47	12.44	33.60	5.14	7.26	1.83	6.28

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334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	241.61	19.12	19.12	9.90	26.74	14.21	10.17	5.05	8.78
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	241.61	19.12	19.12	9.90	26.74	14.21	10.17	5.05	8.78
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	241.61	19.12	19.12	9.90	26.74	14.21	10.17	5.05	8.78
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	241.61	19.12	19.12	9.90	26.74	14.21	10.17	5.05	8.78
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	216.61	-47.34	-47.34	4.67	4.67	5.15	52.22	1.83	16.71
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	216.61	-47.34	-47.34	4.67	4.67	5.15	52.22	1.83	16.71
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	216.61	-47.34	-47.34	4.67	4.67	5.15	52.22	1.83	16.71
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	216.61	-47.34	-47.34	4.67	4.67	5.15	52.22	1.83	16.71
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	239.92	19.08	19.08	2.13	2.13	14.15	126.58	5.03	40.50
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	239.92	19.08	19.08	2.13	2.13	14.15	126.58	5.03	40.50
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	239.92	19.08	19.08	2.13	2.13	14.15	126.58	5.03	40.50
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	239.92	19.08	19.08	2.13	2.13	14.15	126.58	5.03	40.50
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	218.30	-47.41	-47.41	8.46	8.46	5.18	29.04	1.84	9.29
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	218.30	-47.41	-47.41	8.46	8.46	5.18	29.04	1.84	9.29
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	218.30	-47.41	-47.41	8.46	8.46	5.18	29.04	1.84	9.29
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	218.30	-47.41	-47.41	8.46	8.46	5.18	29.04	1.84	9.29
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	241.74	47.41	47.41	5.91	5.91	5.74	45.98	2.04	14.71
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	241.74	47.41	47.41	5.91	5.91	5.74	45.98	2.04	14.71
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	241.74	47.41	47.41	5.91	5.91	5.74	45.98	2.04	14.71
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	241.74	47.41	47.41	5.91	5.91	5.74	45.98	2.04	14.71
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(F	247.23	-35.46	-95.74	1.06	2.86	2.90	97.16	2.79	83.95
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(F	247.23	-35.46	-95.74	1.06	2.86	2.90	97.16	2.79	83.95
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(F	247.23	-35.46	-95.74	1.06	2.86	2.90	97.16	2.79	83.95
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(F	247.23	-35.46	-95.74	1.06	2.86	2.90	97.16	2.79	83.95
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(F	263.72	-14.29	-38.59	0.00	0.00	7.69	61046.11	7.38	52743.84
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(F	263.72	-14.29	-38.59	0.00	0.00	7.69	61046.11	7.38	52743.84
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(F	263.72	-14.29	-38.59	0.00	0.00	7.69	61046.11	7.38	52743.84
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(F	263.72	-14.29	-38.59	0.00	0.00	7.69	61046.11	7.38	52743.84
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(	251.49	-35.60	-96.12	9.33	25.20	2.94	11.23	2.83	9.70
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(	251.49	-35.60	-96.12	9.33	25.20	2.94	11.23	2.83	9.70
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(	251.49	-35.60	-96.12	9.33	25.20	2.94	11.23	2.83	9.70
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(	251.49	-35.60	-96.12	9.33	25.20	2.94	11.23	2.83	9.70
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(	270.01	14.34	38.73	7.43	20.05	7.84	15.15	7.53	13.09
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(	270.01	14.34	38.73	7.43	20.05	7.84	15.15	7.53	13.09
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(	270.01	14.34	38.73	7.43	20.05	7.84	15.15	7.53	13.09
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(	270.01	14.34	38.73	7.43	20.05	7.84	15.15	7.53	13.09
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-V	251.26	-35.51	-35.51	3.50	3.50	7.96	80.76	2.83	25.84
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-V	251.26	-35.51	-35.51	3.50	3.50	7.96	80.76	2.83	25.84
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-V	251.26	-35.51	-35.51	3.50	3.50	7.96	80.76	2.83	25.84
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-V	251.26	-35.51	-35.51	3.50	3.50	7.96	80.76	2.83	25.84
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-V	268.74	14.31	14.31	1.60	1.60	21.13	189.04	7.51	60.49
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-V	268.74	14.31	14.31	1.60	1.60	21.13	189.04	7.51	60.49
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-V	268.74	14.31	14.31	1.60	1.60	21.13	189.04	7.51	60.49
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-V	268.74	14.31	14.31	1.60	1.60	21.13	189.04	7.51	60.49
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-V	252.53	-35.56	-35.56	6.34	6.34	7.99	44.79	2.84	14.33
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-V	252.53	-35.56	-35.56	6.34	6.34	7.99	44.79	2.84	14.33
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-V	252.53	-35.56	-35.56	6.34	6.34	7.99	44.79	2.84	14.33
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-V	252.53	-35.56	-35.56	6.34	6.34	7.99	44.79	2.84	14.33
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-V	270.11	35.56	35.56	4.44	4.44	8.55	68.50	3.04	21.92
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-V	270.11	35.56	35.56	4.44	4.44	8.55	68.50	3.04	21.92
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-V	270.11	35.56	35.56	4.44	4.44	8.55	68.50	3.04	21.92
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-V	270.11	35.56	35.56	4.44	4.44	8.55	68.50	3.04	21.92
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+'	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+'	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+'	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+'	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	277.94	0.00	0.00	0.00	0.00	NA	NA	NA	NA
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'	297.26	0.00	0.00	0.00	0.00	NA	NA	NA	NA
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'	297.26	0.00	0.00	0.00	0.00	NA	NA	NA	NA
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'	297.26	0.00	0.00	0.00	0.00	NA	NA	NA	NA
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'	297.26	0.00	0.00	0.00	0.00	NA	NA	NA	NA
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-	297.26	0.00	0.00	0.00	0.00	NA	NA	NA	NA
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-	297.26	0.00	0.00	0.00	0.00	NA	NA	NA	NA
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-	297.26	0.00	0.00	0.00	0.00	NA	NA	NA	NA
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-'	297.26	0.00	0.00	0.00	0.00	NA	NA	NA	NA
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	202.00	-47.28	-47.28	1.41	3.82	4.81	59.54	1.71	51.44
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	202.00	-47.28	-47.28	1.41	3.82	4.81	59.54	1.71	51.44
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	202.00	-47.28	-47.28	1.41	3.82	4.81	59.54	1.71	51.44
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	202.00	-47.28	-47.28	1.41	3.82	4.81	59.54	1.71	51.44
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	223.99	-19.05	-19.05	0.00	0.01	13.22	38886.63	4.70	33598.04
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	223.99	-19.05	-19.05	0.00	0.01	13.22	38886.63	4.70	33598.04
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	223.99	-19.05	-19.05	0.00	0.01	13.22	38886.63	4.70	33598.04
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	223.99	-19.05	-19.05	0.00	0.01	13.22	38886.63	4.70	33598.04
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	207.68	-47.47	-47.47	12.44	33.60	4.92	6.95	1.75	6.01
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	207.68	-47.47	-47.47	12.44	33.60	4.92	6.95	1.75	6.01
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	207.68	-47.47	-47.47	12.44	33.60	4.92	6.95	1.75	6.01
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	207.68	-47.47	-47.47	12.44	33.60	4.92	6.95	1.75	6.01
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	232.38	19.12	19.12	9.90	26.74	13.67	9.78	4.86	8.45
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	232.38	19.12	19.12	9.90	26.74	13.67	9.78	4.86	8.45
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	232.38	19.12	19.12	9.90	26.74	13.67	9.78	4.86	8.45
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	232.38	19.12	19.12	9.90	26.74	13.67	9.78	4.86	8.45
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	207.37	-47.34	-47.34	4.67	4.67	4.93	49.99	1.75	16.00
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	207.37	-47.34	-47.34	4.67	4.67	4.93	49.99	1.75	16.00
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	207.37	-47.34	-47.34	4.67	4.67	4.93	49.99	1.75	16.00
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	207.37	-47.34	-47.34	4.67	4.67	4.93	49.99	1.75	16.00
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	230.69	19.08	19.08	2.13	2.13	13.60	121.70	4.84	38.95
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	230.69	19.08	19.08	2.13	2.13	13.60	121.70	4.84	38.95
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	230.69	19.08	19.08	2.13	2.13	13.60	121.70	4.84	38.95
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	230.69	19.08	19.08	2.13	2.13	13.60	121.70	4.84	38.95
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	209.06	-47.41	-47.41	8.46	8.46	4.96	27.81	1.76	8.90
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	209.06	-47.41	-47.41	8.46	8.46	4.96	27.81	1.76	8.90
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	209.06	-47.41	-47.41	8.46	8.46	4.96	27.81	1.76	8.90
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	209.06	-47.41	-47.41	8.46	8.46	4.96	27.81	1.76	8.90
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-W	232.51	47.41	47.41	5.91	5.91	5.52	44.22	1.96	14.15
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	232.51	47.41	47.41	5.91	5.91	5.52	44.22	1.96	14.15
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	232.51	47.41	47.41	5.91	5.91	5.52	44.22	1.96	14.15
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	232.51	47.41	47.41	5.91	5.91	5.52	44.22	1.96	14.15
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	268.70	0.00	0.00	0.00	0.00	NA	NA	NA	NA
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	268.70	0.00	0.00	0.00	0.00	NA	NA	NA	NA
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	268.70	0.00	0.00	0.00	0.00	NA	NA	NA	NA
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	268.70	0.00	0.00	0.00	0.00	NA	NA	NA	NA
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	228.68	-28.37	-76.60	0.85	2.29	3.36	112.34	3.22	97.06
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	241.87	-11.43	-30.87	0.00	0.00	8.81	69986.32	8.46	60468.18
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	232.09	-28.48	-76.90	7.47	20.16	3.40	12.95	3.26	11.19
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	246.91	11.47	30.98	5.94	16.04	8.97	17.31	8.61	14.96
						Min.=	**2.90**	**6.95**	**1.71**	**6.01**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F3

Load Combination & Calculation of Base pressure

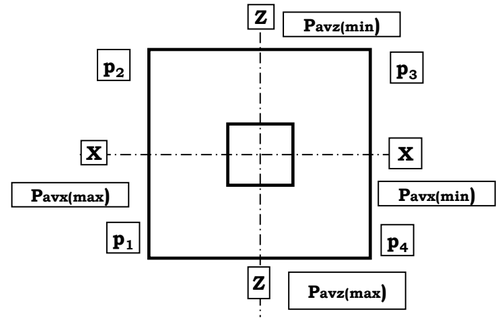
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 436 kN/m^2 For WL/SL case = 536 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
301	1.0(DL+COLL)	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
302	1.0(DL+COLL+CL-1+TL1)+1.(	276.58	0.00	-7.94	44.25	0.00	3.05	47.30		41.21	47.30	41.21	11.30	44.26	44.26	47.30	41.21	O.K
303	1.0(DL+COLL+CL-2+TL1)+1.(	276.58	0.00	-7.94	44.25	0.00	3.05	47.30		41.21	47.30	41.21	11.30	44.26	44.26	47.30	41.21	O.K
304	1.0(DL+COLL+CL-3+TL1)+1.(	276.58	0.00	-7.94	44.25	0.00	3.05	47.30		41.21	47.30	41.21	11.30	44.26	44.26	47.30	41.21	O.K
305	1.0(DL+COLL+CL-4+TL1)+1.(	276.58	0.00	-7.94	44.25	0.00	3.05	47.30		41.21	47.30	41.21	11.30	44.26	44.26	47.30	41.21	O.K
306	1.0(DL+COLL+CL-1+TL2)+1.(	330.83	0.00	0.00	52.93	0.00	0.00	52.93		52.93	52.93	52.93	16.93	52.93	52.93	52.93	52.93	O.K
307	1.0(DL+COLL+CL-2+TL2)+1.(	330.83	0.00	0.00	52.93	0.00	0.00	52.93		52.93	52.93	52.93	16.93	52.93	52.93	52.93	52.93	O.K
308	1.0(DL+COLL+CL-3+TL2)+1.(	330.83	0.00	0.00	52.93	0.00	0.00	52.93		52.93	52.93	52.93	16.93	52.93	52.93	52.93	52.93	O.K
309	1.0(DL+COLL+CL-4+TL2)+1.(	330.83	0.00	0.00	52.93	0.00	0.00	52.93		52.93	52.93	52.93	16.93	52.93	52.93	52.93	52.93	O.K
310	1.0(DL+COLL+CL-1)+0.75(TL	276.92	0.00	-5.95	44.31	0.00	2.29	46.59		42.02	46.59	42.02	10.59	44.31	44.31	46.59	42.02	O.K
311	1.0(DL+COLL+CL-2)+0.75(TL	276.92	0.00	-5.95	44.31	0.00	2.29	46.59		42.02	46.59	42.02	10.59	44.31	44.31	46.59	42.02	O.K
312	1.0(DL+COLL+CL-3)+0.75(TL	276.92	0.00	-5.95	44.31	0.00	2.29	46.59		42.02	46.59	42.02	10.59	44.31	44.31	46.59	42.02	O.K
313	1.0(DL+COLL+CL-4)+0.75(TL	276.92	0.00	-5.95	44.31	0.00	2.29	46.59		42.02	46.59	42.02	10.59	44.31	44.31	46.59	42.02	O.K
314	1.0(DL+COLL+CL-1)+0.75(TL	317.61	0.00	0.00	50.82	0.00	0.00	50.82		50.82	50.82	50.82	14.82	50.82	50.82	50.82	50.82	O.K
315	1.0(DL+COLL+CL-2)+0.75(TL	317.61	0.00	0.00	50.82	0.00	0.00	50.82		50.82	50.82	50.82	14.82	50.82	50.82	50.82	50.82	O.K
316	1.0(DL+COLL+CL-3)+0.75(TL	317.61	0.00	0.00	50.82	0.00	0.00	50.82		50.82	50.82	50.82	14.82	50.82	50.82	50.82	50.82	O.K

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317	1.0(DL+COLL+CL-4)+0.75(TL	317.61	0.00	0.00	50.82	0.00	0.00	50.82		50.82	50.82	50.82	14.82	50.82	50.82	50.82	50.82	O.K
318	1.0(DL+COLL+CL-1)	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
319	1.0(DL+COLL+CL-2)	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
320	1.0(DL+COLL+CL-3)	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
321	1.0(DL+COLL+CL-4)	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
322	1.0(DL+COLL+CL-1)+1.0(WLI	211.24	-47.28	-3.82	33.80	18.16	1.47	53.42		50.49	17.11	14.18	17.42	51.96	15.65	35.27	32.34	O.K
323	1.0(DL+COLL+CL-2)+1.0(WLI	211.24	-47.28	-3.82	33.80	18.16	1.47	53.42		50.49	17.11	14.18	17.42	51.96	15.65	35.27	32.34	O.K
324	1.0(DL+COLL+CL-3)+1.0(WLI	211.24	-47.28	-3.82	33.80	18.16	1.47	53.42		50.49	17.11	14.18	17.42	51.96	15.65	35.27	32.34	O.K
325	1.0(DL+COLL+CL-4)+1.0(WLI	211.24	-47.28	-3.82	33.80	18.16	1.47	53.42		50.49	17.11	14.18	17.42	51.96	15.65	35.27	32.34	O.K
326	1.0(DL+COLL+CL-1)+1.0(WLI	233.22	-19.05	-0.01	37.32	7.32	0.00	44.63		44.63	30.00	30.00	8.63	44.63	30.00	37.32	37.32	O.K
327	1.0(DL+COLL+CL-2)+1.0(WLI	233.22	-19.05	-0.01	37.32	7.32	0.00	44.63		44.63	30.00	30.00	8.63	44.63	30.00	37.32	37.32	O.K
328	1.0(DL+COLL+CL-3)+1.0(WLI	233.22	-19.05	-0.01	37.32	7.32	0.00	44.63		44.63	30.00	30.00	8.63	44.63	30.00	37.32	37.32	O.K
329	1.0(DL+COLL+CL-4)+1.0(WLI	233.22	-19.05	-0.01	37.32	7.32	0.00	44.63		44.63	30.00	30.00	8.63	44.63	30.00	37.32	37.32	O.K
330	1.0(DL+COLL+CL-1)+1.0(WLI	216.91	-47.47	-33.60	34.71	18.23	12.90	65.83		40.03	29.38	3.58	29.83	52.93	16.48	47.61	21.81	O.K
331	1.0(DL+COLL+CL-2)+1.0(WLI	216.91	-47.47	-33.60	34.71	18.23	12.90	65.83		40.03	29.38	3.58	29.83	52.93	16.48	47.61	21.81	O.K
332	1.0(DL+COLL+CL-3)+1.0(WLI	216.91	-47.47	-33.60	34.71	18.23	12.90	65.83		40.03	29.38	3.58	29.83	52.93	16.48	47.61	21.81	O.K
333	1.0(DL+COLL+CL-4)+1.0(WLI	216.91	-47.47	-33.60	34.71	18.23	12.90	65.83		40.03	29.38	3.58	29.83	52.93	16.48	47.61	21.81	O.K
334	1.0(DL+COLL+CL-1)+1.0(WLI	241.61	19.12	-26.74	38.66	7.34	10.27	56.27		35.73	41.58	21.05	20.27	46.00	31.32	48.93	28.39	O.K
335	1.0(DL+COLL+CL-2)+1.0(WLI	241.61	19.12	-26.74	38.66	7.34	10.27	56.27		35.73	41.58	21.05	20.27	46.00	31.32	48.93	28.39	O.K
336	1.0(DL+COLL+CL-3)+1.0(WLI	241.61	19.12	-26.74	38.66	7.34	10.27	56.27		35.73	41.58	21.05	20.27	46.00	31.32	48.93	28.39	O.K
337	1.0(DL+COLL+CL-4)+1.0(WLI	241.61	19.12	-26.74	38.66	7.34	10.27	56.27		35.73	41.58	21.05	20.27	46.00	31.32	48.93	28.39	O.K
338	1.0(DL+COLL+CL-1)+1.0(WLI	216.61	-47.34	-4.67	34.66	18.18	1.79	54.63		51.04	18.27	14.69	18.63	52.84	16.48	36.45	32.87	O.K
339	1.0(DL+COLL+CL-2)+1.0(WLI	216.61	-47.34	-4.67	34.66	18.18	1.79	54.63		51.04	18.27	14.69	18.63	52.84	16.48	36.45	32.87	O.K
340	1.0(DL+COLL+CL-3)+1.0(WLI	216.61	-47.34	-4.67	34.66	18.18	1.79	54.63		51.04	18.27	14.69	18.63	52.84	16.48	36.45	32.87	O.K
341	1.0(DL+COLL+CL-4)+1.0(WLI	216.61	-47.34	-4.67	34.66	18.18	1.79	54.63		51.04	18.27	14.69	18.63	52.84	16.48	36.45	32.87	O.K
342	1.0(DL+COLL+CL-1)+1.0(WLI	239.92	19.08	-2.13	38.39	7.33	0.82	46.53		44.89	31.88	30.24	10.53	45.71	31.06	39.21	37.57	O.K
343	1.0(DL+COLL+CL-2)+1.0(WLI	239.92	19.08	-2.13	38.39	7.33	0.82	46.53		44.89	31.88	30.24	10.53	45.71	31.06	39.21	37.57	O.K
344	1.0(DL+COLL+CL-3)+1.0(WLI	239.92	19.08	-2.13	38.39	7.33	0.82	46.53		44.89	31.88	30.24	10.53	45.71	31.06	39.21	37.57	O.K
345	1.0(DL+COLL+CL-4)+1.0(WLI	239.92	19.08	-2.13	38.39	7.33	0.82	46.53		44.89	31.88	30.24	10.53	45.71	31.06	39.21	37.57	O.K
346	1.0(DL+COLL+CL-1)+1.0(WLI	218.30	-47.41	-8.46	34.93	18.20	3.25	56.38		49.88	19.97	13.48	20.38	53.13	16.73	38.18	31.68	O.K
347	1.0(DL+COLL+CL-2)+1.0(WLI	218.30	-47.41	-8.46	34.93	18.20	3.25	56.38		49.88	19.97	13.48	20.38	53.13	16.73	38.18	31.68	O.K

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348	1.0(DL+COLL+CL-3)+1.0(WL	218.30	-47.41	-8.46	34.93	18.20	3.25	56.38		49.88	19.97	13.48	20.38	53.13	16.73	38.18	31.68	O.K
349	1.0(DL+COLL+CL-4)+1.0(WL	218.30	-47.41	-8.46	34.93	18.20	3.25	56.38		49.88	19.97	13.48	20.38	53.13	16.73	38.18	31.68	O.K
350	1.0(DL+COLL+CL-1)+1.0(WL	241.74	47.41	-5.91	38.68	18.20	2.27	59.15		54.61	22.75	18.20	23.15	56.88	20.48	40.95	36.41	O.K
351	1.0(DL+COLL+CL-2)+1.0(WL	241.74	47.41	-5.91	38.68	18.20	2.27	59.15		54.61	22.75	18.20	23.15	56.88	20.48	40.95	36.41	O.K
352	1.0(DL+COLL+CL-3)+1.0(WL	241.74	47.41	-5.91	38.68	18.20	2.27	59.15		54.61	22.75	18.20	23.15	56.88	20.48	40.95	36.41	O.K
353	1.0(DL+COLL+CL-4)+1.0(WL	241.74	47.41	-5.91	38.68	18.20	2.27	59.15		54.61	22.75	18.20	23.15	56.88	20.48	40.95	36.41	O.K
354	1.0(DL+COLL+CL-1)+0.75(WI	247.23	-95.74	-2.86	39.56	36.77	1.10	77.42		75.22	3.89	1.69	41.42	76.32	2.79	40.66	38.46	O.K
355	1.0(DL+COLL+CL-2)+0.75(WI	247.23	-95.74	-2.86	39.56	36.77	1.10	77.42		75.22	3.89	1.69	41.42	76.32	2.79	40.66	38.46	O.K
356	1.0(DL+COLL+CL-3)+0.75(WI	247.23	-95.74	-2.86	39.56	36.77	1.10	77.42		75.22	3.89	1.69	41.42	76.32	2.79	40.66	38.46	O.K
357	1.0(DL+COLL+CL-4)+0.75(WI	247.23	-95.74	-2.86	39.56	36.77	1.10	77.42		75.22	3.89	1.69	41.42	76.32	2.79	40.66	38.46	O.K
358	1.0(DL+COLL+CL-1)+0.75(WI	263.72	-38.59	0.00	42.20	14.82	0.00	57.01		57.01	27.38	27.38	21.01	57.01	27.38	42.20	42.20	O.K
359	1.0(DL+COLL+CL-2)+0.75(WI	263.72	-38.59	0.00	42.20	14.82	0.00	57.01		57.01	27.38	27.38	21.01	57.01	27.38	42.20	42.20	O.K
360	1.0(DL+COLL+CL-3)+0.75(WI	263.72	-38.59	0.00	42.20	14.82	0.00	57.01		57.01	27.38	27.38	21.01	57.01	27.38	42.20	42.20	O.K
361	1.0(DL+COLL+CL-4)+0.75(WI	263.72	-38.59	0.00	42.20	14.82	0.00	57.01		57.01	27.38	27.38	21.01	57.01	27.38	42.20	42.20	O.K
362	1.0(DL+COLL+CL-1)+0.75(WI	251.49	-96.12	-25.20	40.24	36.91	9.68	86.83		67.47	13.00	0.00	51.83	77.65	6.50	50.42	33.74	O.K
363	1.0(DL+COLL+CL-2)+0.75(WI	251.49	-96.12	-25.20	40.24	36.91	9.68	86.83		67.47	13.00	0.00	51.83	77.65	6.50	50.42	33.74	O.K
364	1.0(DL+COLL+CL-3)+0.75(WI	251.49	-96.12	-25.20	40.24	36.91	9.68	86.83		67.47	13.00	0.00	51.83	77.65	6.50	50.42	33.74	O.K
365	1.0(DL+COLL+CL-4)+0.75(WI	251.49	-96.12	-25.20	40.24	36.91	9.68	86.83		67.47	13.00	0.00	51.83	77.65	6.50	50.42	33.74	O.K
366	1.0(DL+COLL+CL-1)+0.75(WI	270.01	38.73	-20.05	43.20	14.87	7.70	65.77		50.37	36.03	20.63	29.77	58.07	28.33	50.90	35.50	O.K
367	1.0(DL+COLL+CL-2)+0.75(WI	270.01	38.73	-20.05	43.20	14.87	7.70	65.77		50.37	36.03	20.63	29.77	58.07	28.33	50.90	35.50	O.K
368	1.0(DL+COLL+CL-3)+0.75(WI	270.01	38.73	-20.05	43.20	14.87	7.70	65.77		50.37	36.03	20.63	29.77	58.07	28.33	50.90	35.50	O.K
369	1.0(DL+COLL+CL-4)+0.75(WI	270.01	38.73	-20.05	43.20	14.87	7.70	65.77		50.37	36.03	20.63	29.77	58.07	28.33	50.90	35.50	O.K
370	1.0(DL+COLL+CL-1)+0.75(WI	251.26	-35.51	-3.50	40.20	13.63	1.34	55.18		52.49	27.91	25.22	19.18	53.84	26.57	41.55	38.86	O.K
371	1.0(DL+COLL+CL-2)+0.75(WI	251.26	-35.51	-3.50	40.20	13.63	1.34	55.18		52.49	27.91	25.22	19.18	53.84	26.57	41.55	38.86	O.K
372	1.0(DL+COLL+CL-3)+0.75(WI	251.26	-35.51	-3.50	40.20	13.63	1.34	55.18		52.49	27.91	25.22	19.18	53.84	26.57	41.55	38.86	O.K
373	1.0(DL+COLL+CL-4)+0.75(WI	251.26	-35.51	-3.50	40.20	13.63	1.34	55.18		52.49	27.91	25.22	19.18	53.84	26.57	41.55	38.86	O.K
374	1.0(DL+COLL+CL-1)+0.75(WI	268.74	14.31	-1.60	43.00	5.49	0.61	49.11		47.88	38.12	36.89	13.11	48.50	37.51	43.62	42.39	O.K
375	1.0(DL+COLL+CL-2)+0.75(WI	268.74	14.31	-1.60	43.00	5.49	0.61	49.11		47.88	38.12	36.89	13.11	48.50	37.51	43.62	42.39	O.K
376	1.0(DL+COLL+CL-3)+0.75(WI	268.74	14.31	-1.60	43.00	5.49	0.61	49.11		47.88	38.12	36.89	13.11	48.50	37.51	43.62	42.39	O.K
377	1.0(DL+COLL+CL-4)+0.75(WI	268.74	14.31	-1.60	43.00	5.49	0.61	49.11		47.88	38.12	36.89	13.11	48.50	37.51	43.62	42.39	O.K
378	1.0(DL+COLL+CL-1)+0.75(WI	252.53	-35.56	-6.34	40.40	13.65	2.44	56.49		51.62	29.19	24.32	20.49	54.06	26.76	42.84	37.97	O.K

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379	1.0(DL+COLL+CL-2)+0.75(WI	252.53	-35.56	-6.34	40.40	13.65	2.44	56.49		51.62	29.19	24.32	20.49	54.06	26.76	42.84	37.97	O.K
380	1.0(DL+COLL+CL-3)+0.75(WI	252.53	-35.56	-6.34	40.40	13.65	2.44	56.49		51.62	29.19	24.32	20.49	54.06	26.76	42.84	37.97	O.K
381	1.0(DL+COLL+CL-4)+0.75(WI	252.53	-35.56	-6.34	40.40	13.65	2.44	56.49		51.62	29.19	24.32	20.49	54.06	26.76	42.84	37.97	O.K
382	1.0(DL+COLL+CL-1)+0.75(WI	270.11	35.56	-4.44	43.22	13.65	1.70	58.57		55.17	31.27	27.86	22.57	56.87	29.57	44.92	41.52	O.K
383	1.0(DL+COLL+CL-2)+0.75(WI	270.11	35.56	-4.44	43.22	13.65	1.70	58.57		55.17	31.27	27.86	22.57	56.87	29.57	44.92	41.52	O.K
384	1.0(DL+COLL+CL-3)+0.75(WI	270.11	35.56	-4.44	43.22	13.65	1.70	58.57		55.17	31.27	27.86	22.57	56.87	29.57	44.92	41.52	O.K
385	1.0(DL+COLL+CL-4)+0.75(WI	270.11	35.56	-4.44	43.22	13.65	1.70	58.57		55.17	31.27	27.86	22.57	56.87	29.57	44.92	41.52	O.K
386	1.0(DL+COLL+CL-1)+0.75(SL	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
387	1.0(DL+COLL+CL-2)+0.75(SL	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
388	1.0(DL+COLL+CL-3)+0.75(SL	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
389	1.0(DL+COLL+CL-4)+0.75(SL	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
390	1.0(DL+COLL+CL-1)+0.75(SL	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
391	1.0(DL+COLL+CL-2)+0.75(SL	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
392	1.0(DL+COLL+CL-3)+0.75(SL	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
393	1.0(DL+COLL+CL-4)+0.75(SL	277.94	0.00	0.00	44.47	0.00	0.00	44.47		44.47	44.47	44.47	8.47	44.47	44.47	44.47	44.47	O.K
394	1.0(DL+COLL+CL-1)+0.525(S	297.26	0.00	0.00	47.56	0.00	0.00	47.56		47.56	47.56	47.56	11.56	47.56	47.56	47.56	47.56	O.K
395	1.0(DL+COLL+CL-2)+0.525(S	297.26	0.00	0.00	47.56	0.00	0.00	47.56		47.56	47.56	47.56	11.56	47.56	47.56	47.56	47.56	O.K
396	1.0(DL+COLL+CL-3)+0.525(S	297.26	0.00	0.00	47.56	0.00	0.00	47.56		47.56	47.56	47.56	11.56	47.56	47.56	47.56	47.56	O.K
397	1.0(DL+COLL+CL-4)+0.525(S	297.26	0.00	0.00	47.56	0.00	0.00	47.56		47.56	47.56	47.56	11.56	47.56	47.56	47.56	47.56	O.K
398	1.0(DL+COLL+CL-1)+0.525(S	297.26	0.00	0.00	47.56	0.00	0.00	47.56		47.56	47.56	47.56	11.56	47.56	47.56	47.56	47.56	O.K
399	1.0(DL+COLL+CL-2)+0.525(S	297.26	0.00	0.00	47.56	0.00	0.00	47.56		47.56	47.56	47.56	11.56	47.56	47.56	47.56	47.56	O.K
400	1.0(DL+COLL+CL-3)+0.525(S	297.26	0.00	0.00	47.56	0.00	0.00	47.56		47.56	47.56	47.56	11.56	47.56	47.56	47.56	47.56	O.K
401	1.0(DL+COLL+CL-4)+0.525SI	297.26	0.00	0.00	47.56	0.00	0.00	47.56		47.56	47.56	47.56	11.56	47.56	47.56	47.56	47.56	O.K
402	0.6(DL+COLL+CL-1)+1.0(WLI	202.00	-47.28	-3.82	32.32	18.16	1.47	51.94		49.01	15.63	12.70	15.94	50.48	14.17	33.79	30.86	O.K
403	0.6(DL+COLL+CL-2)+1.0(WLI	202.00	-47.28	-3.82	32.32	18.16	1.47	51.94		49.01	15.63	12.70	15.94	50.48	14.17	33.79	30.86	O.K
404	0.6(DL+COLL+CL-3)+1.0(WLI	202.00	-47.28	-3.82	32.32	18.16	1.47	51.94		49.01	15.63	12.70	15.94	50.48	14.17	33.79	30.86	O.K
405	0.6(DL+COLL+CL-4)+1.0(WLI	202.00	-47.28	-3.82	32.32	18.16	1.47	51.94		49.01	15.63	12.70	15.94	50.48	14.17	33.79	30.86	O.K
406	0.6(DL+COLL+CL-1)+1.0(WLI	223.99	-19.05	-0.01	35.84	7.32	0.00	43.16		43.15	28.52	28.52	7.16	43.16	28.52	35.84	35.84	O.K
407	0.6(DL+COLL+CL-2)+1.0(WLI	223.99	-19.05	-0.01	35.84	7.32	0.00	43.16		43.15	28.52	28.52	7.16	43.16	28.52	35.84	35.84	O.K
408	0.6(DL+COLL+CL-3)+1.0(WLI	223.99	-19.05	-0.01	35.84	7.32	0.00	43.16		43.15	28.52	28.52	7.16	43.16	28.52	35.84	35.84	O.K
409	0.6(DL+COLL+CL-4)+1.0(WLI	223.99	-19.05	-0.01	35.84	7.32	0.00	43.16		43.15	28.52	28.52	7.16	43.16	28.52	35.84	35.84	O.K

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410	0.6(DL+COLL+CL-1)+1.0(WLI	207.68	-47.47	-33.60	33.23	18.23	12.90	64.36		38.56	27.90	2.10	28.36	51.46	15.00	46.13	20.33	O.K
411	0.6(DL+COLL+CL-2)+1.0(WLI	207.68	-47.47	-33.60	33.23	18.23	12.90	64.36		38.56	27.90	2.10	28.36	51.46	15.00	46.13	20.33	O.K
412	0.6(DL+COLL+CL-3)+1.0(WLI	207.68	-47.47	-33.60	33.23	18.23	12.90	64.36		38.56	27.90	2.10	28.36	51.46	15.00	46.13	20.33	O.K
413	0.6(DL+COLL+CL-4)+1.0(WLI	207.68	-47.47	-33.60	33.23	18.23	12.90	64.36		38.56	27.90	2.10	28.36	51.46	15.00	46.13	20.33	O.K
414	0.6(DL+COLL+CL-1)+1.0(WLI	232.38	19.12	-26.74	37.18	7.34	10.27	54.79		34.26	40.11	19.57	18.79	44.53	29.84	47.45	26.92	O.K
415	0.6(DL+COLL+CL-2)+1.0(WLI	232.38	19.12	-26.74	37.18	7.34	10.27	54.79		34.26	40.11	19.57	18.79	44.53	29.84	47.45	26.92	O.K
416	0.6(DL+COLL+CL-3)+1.0(WLI	232.38	19.12	-26.74	37.18	7.34	10.27	54.79		34.26	40.11	19.57	18.79	44.53	29.84	47.45	26.92	O.K
417	0.6(DL+COLL+CL-4)+1.0(WLI	232.38	19.12	-26.74	37.18	7.34	10.27	54.79		34.26	40.11	19.57	18.79	44.53	29.84	47.45	26.92	O.K
418	0.6(DL+COLL+CL-1)+1.0(WLI	207.37	-47.34	-4.67	33.18	18.18	1.79	53.15		49.57	16.79	13.21	17.15	51.36	15.00	34.97	31.39	O.K
419	0.6(DL+COLL+CL-2)+1.0(WLI	207.37	-47.34	-4.67	33.18	18.18	1.79	53.15		49.57	16.79	13.21	17.15	51.36	15.00	34.97	31.39	O.K
420	0.6(DL+COLL+CL-3)+1.0(WLI	207.37	-47.34	-4.67	33.18	18.18	1.79	53.15		49.57	16.79	13.21	17.15	51.36	15.00	34.97	31.39	O.K
421	0.6(DL+COLL+CL-4)+1.0(WLI	207.37	-47.34	-4.67	33.18	18.18	1.79	53.15		49.57	16.79	13.21	17.15	51.36	15.00	34.97	31.39	O.K
422	0.6(DL+COLL+CL-1)+1.0(WLI	230.69	19.08	-2.13	36.91	7.33	0.82	45.05		43.42	30.40	28.77	9.05	44.24	29.59	37.73	36.10	O.K
423	0.6(DL+COLL+CL-2)+1.0(WLI	230.69	19.08	-2.13	36.91	7.33	0.82	45.05		43.42	30.40	28.77	9.05	44.24	29.59	37.73	36.10	O.K
424	0.6(DL+COLL+CL-3)+1.0(WLI	230.69	19.08	-2.13	36.91	7.33	0.82	45.05		43.42	30.40	28.77	9.05	44.24	29.59	37.73	36.10	O.K
425	0.6(DL+COLL+CL-4)+1.0(WLI	230.69	19.08	-2.13	36.91	7.33	0.82	45.05		43.42	30.40	28.77	9.05	44.24	29.59	37.73	36.10	O.K
426	0.6(DL+COLL+CL-1)+1.0(WLI	209.06	-47.41	-8.46	33.45	18.20	3.25	54.90		48.41	18.49	12.00	18.90	51.66	15.25	36.70	30.21	O.K
427	0.6(DL+COLL+CL-2)+1.0(WLI	209.06	-47.41	-8.46	33.45	18.20	3.25	54.90		48.41	18.49	12.00	18.90	51.66	15.25	36.70	30.21	O.K
428	0.6(DL+COLL+CL-3)+1.0(WLI	209.06	-47.41	-8.46	33.45	18.20	3.25	54.90		48.41	18.49	12.00	18.90	51.66	15.25	36.70	30.21	O.K
429	0.6(DL+COLL+CL-4)+1.0(WLI	209.06	-47.41	-8.46	33.45	18.20	3.25	54.90		48.41	18.49	12.00	18.90	51.66	15.25	36.70	30.21	O.K
430	0.6(DL+COLL+CL-1)+1.0(WLI	232.51	47.41	-5.91	37.20	18.20	2.27	57.68		53.13	21.27	16.73	21.68	55.41	19.00	39.48	34.93	O.K
431	0.6(DL+COLL+CL-2)+1.0(WLI	232.51	47.41	-5.91	37.20	18.20	2.27	57.68		53.13	21.27	16.73	21.68	55.41	19.00	39.48	34.93	O.K
432	0.6(DL+COLL+CL-3)+1.0(WLI	232.51	47.41	-5.91	37.20	18.20	2.27	57.68		53.13	21.27	16.73	21.68	55.41	19.00	39.48	34.93	O.K
433	0.6(DL+COLL+CL-4)+1.0(WLI	232.51	47.41	-5.91	37.20	18.20	2.27	57.68		53.13	21.27	16.73	21.68	55.41	19.00	39.48	34.93	O.K
434	0.6(DL+COLL+CL-1)+0.7(SL'+	268.70	0.00	0.00	42.99	0.00	0.00	42.99		42.99	42.99	42.99	6.99	42.99	42.99	42.99	42.99	O.K
435	0.6(DL+COLL+CL-2)+0.7(SL'+	268.70	0.00	0.00	42.99	0.00	0.00	42.99		42.99	42.99	42.99	6.99	42.99	42.99	42.99	42.99	O.K
436	0.6(DL+COLL+CL-3)+0.7(SL'+	268.70	0.00	0.00	42.99	0.00	0.00	42.99		42.99	42.99	42.99	6.99	42.99	42.99	42.99	42.99	O.K
437	0.6(DL+COLL+CL-4)+0.7(SL'+	268.70	0.00	0.00	42.99	0.00	0.00	42.99		42.99	42.99	42.99	6.99	42.99	42.99	42.99	42.99	O.K
438	0.6(DL+COLL+CL-1)+0.7(SL'+	228.68	-76.60	-2.29	36.59	29.41	0.88	66.88		65.12	8.06	6.30	30.88	66.00	7.18	37.47	35.71	O.K
439	0.6(DL+COLL+CL-2)+0.7(SL'+	241.87	-30.87	0.00	38.70	11.85	0.00	50.55		50.55	26.85	26.84	14.55	50.55	26.85	38.70	38.70	O.K
440	0.6(DL+COLL+CL-3)+0.7(SL'+	232.09	-76.90	-20.16	37.13	29.53	7.74	74.40		58.92	15.35	-0.14	38.40	66.66	7.61	44.88	29.39	-
441	0.6(DL+COLL+CL-4)+0.7(SL'+	246.91	30.98	-16.04	39.51	11.90	6.16	57.56		45.24	33.77	21.45	21.56	51.40	27.61	45.67	33.35	O.K

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Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
	Ultimate Conditions																	
101	1.4(DL+COLL)	287.17	0.00	0.00	45.95	0.00	0.00	45.95		45.95	45.95	45.95	9.95	45.95	45.95	45.95	45.95	
102	1.2(DL+COLL+CL-1)+1.6(RLL	323.77	0.00	0.00	51.80	0.00	0.00	51.80		51.80	51.80	51.80	15.80	51.80	51.80	51.80	51.80	
103	1.2(DL+COLL+CL-2)+1.6(RLL	323.77	0.00	0.00	51.80	0.00	0.00	51.80		51.80	51.80	51.80	15.80	51.80	51.80	51.80	51.80	
104	1.2(DL+COLL+CL-3)+1.6(RLL	323.77	0.00	0.00	51.80	0.00	0.00	51.80		51.80	51.80	51.80	15.80	51.80	51.80	51.80	51.80	
105	1.2(DL+COLL+CL-4)+1.6(RLL	323.77	0.00	0.00	51.80	0.00	0.00	51.80		51.80	51.80	51.80	15.80	51.80	51.80	51.80	51.80	
106	1.4(DL+COLL+CL-1)	287.17	0.00	0.00	45.95	0.00	0.00	45.95		45.95	45.95	45.95	9.95	45.95	45.95	45.95	45.95	
107	1.4(DL+COLL+CL-2)	287.17	0.00	0.00	45.95	0.00	0.00	45.95		45.95	45.95	45.95	9.95	45.95	45.95	45.95	45.95	
108	1.4(DL+COLL+CL-3)	287.17	0.00	0.00	45.95	0.00	0.00	45.95		45.95	45.95	45.95	9.95	45.95	45.95	45.95	45.95	
109	1.4(DL+COLL+CL-4)	287.17	0.00	0.00	45.95	0.00	0.00	45.95		45.95	45.95	45.95	9.95	45.95	45.95	45.95	45.95	
110	1.2(DL+COLL+CL-1+TL1)+1.6	291.23	0.00	9.53	46.60	0.00	3.66	50.26		42.94	50.26	42.94	14.26	46.60	46.60	50.26	42.94	
111	1.2(DL+COLL+CL-2+TL1)+1.6	291.23	0.00	9.53	46.60	0.00	3.66	50.26		42.94	50.26	42.94	14.26	46.60	46.60	50.26	42.94	
112	1.2(DL+COLL+CL-3+TL1)+1.6	291.23	0.00	9.53	46.60	0.00	3.66	50.26		42.94	50.26	42.94	14.26	46.60	46.60	50.26	42.94	
113	1.2(DL+COLL+CL-4+TL1)+1.6	291.23	0.00	9.53	46.60	0.00	3.66	50.26		42.94	50.26	42.94	14.26	46.60	46.60	50.26	42.94	
114	1.2(DL+COLL+CL-1+TL2)+1.6	356.33	0.00	0.00	57.01	0.00	0.00	57.01		57.01	57.01	57.01	21.01	57.01	57.01	57.01	57.01	
115	1.2(DL+COLL+CL-2+TL2)+1.6	356.33	0.00	0.00	57.01	0.00	0.00	57.01		57.01	57.01	57.01	21.01	57.01	57.01	57.01	57.01	
116	1.2(DL+COLL+CL-3+TL2)+1.6	356.33	0.00	0.00	57.01	0.00	0.00	57.01		57.01	57.01	57.01	21.01	57.01	57.01	57.01	57.01	
117	1.2(DL+COLL+CL-4+TL2)+1.6	356.33	0.00	0.00	57.01	0.00	0.00	57.01		57.01	57.01	57.01	21.01	57.01	57.01	57.01	57.01	
118	1.2(DL+COLL+CL-1)+0.8(WLI	229.19	102.13	3.05	36.67	39.22	1.17	77.06		74.72	-1.37	-3.72	41.06	75.89	-2.55	37.85	35.50	
119	1.2(DL+COLL+CL-2)+0.8(WLI	229.19	102.13	3.05	36.67	39.22	1.17	77.06		74.72	-1.37	-3.72	41.06	75.89	-2.55	37.85	35.50	
120	1.2(DL+COLL+CL-3)+0.8(WLI	229.19	102.13	3.05	36.67	39.22	1.17	77.06		74.72	-1.37	-3.72	41.06	75.89	-2.55	37.85	35.50	
121	1.2(DL+COLL+CL-4)+0.8(WLI	229.19	102.13	3.05	36.67	39.22	1.17	77.06		74.72	-1.37	-3.72	41.06	75.89	-2.55	37.85	35.50	
122	1.2(DL+COLL+CL-1)+0.8(WLI	246.78	41.16	0.01	39.48	15.80	0.00	55.29		55.29	23.68	23.68	19.29	55.29	23.68	39.49	39.49	
123	1.2(DL+COLL+CL-2)+0.8(WLI	246.78	41.16	0.01	39.48	15.80	0.00	55.29		55.29	23.68	23.68	19.29	55.29	23.68	39.49	39.49	
124	1.2(DL+COLL+CL-3)+0.8(WLI	246.78	41.16	0.01	39.48	15.80	0.00	55.29		55.29	23.68	23.68	19.29	55.29	23.68	39.49	39.49	

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125	1.2(DL+COLL+CL-4)+0.8(WLI	246.78	41.16	0.01	39.48	15.80	0.00	55.29		55.29	23.68	23.68	19.29	55.29	23.68	39.49	39.49	
126	1.2(DL+COLL+CL-1)+0.8(WLI	233.74	102.53	26.88	37.40	39.37	10.32	87.09		66.45	8.35	-12.30	51.09	76.77	-1.98	47.72	27.08	
127	1.2(DL+COLL+CL-2)+0.8(WLI	233.74	102.53	26.88	37.40	39.37	10.32	87.09		66.45	8.35	-12.30	51.09	76.77	-1.98	47.72	27.08	
128	1.2(DL+COLL+CL-3)+0.8(WLI	233.74	102.53	26.88	37.40	39.37	10.32	87.09		66.45	8.35	-12.30	51.09	76.77	-1.98	47.72	27.08	
129	1.2(DL+COLL+CL-4)+0.8(WLI	233.74	102.53	26.88	37.40	39.37	10.32	87.09		66.45	8.35	-12.30	51.09	76.77	-1.98	47.72	27.08	
130	1.2(DL+COLL+CL-1)+0.8(WLI	253.50	41.31	21.39	40.56	15.86	8.21	64.64		48.21	32.91	16.48	28.64	56.43	24.70	48.78	32.35	
131	1.2(DL+COLL+CL-2)+0.8(WLI	253.50	41.31	21.39	40.56	15.86	8.21	64.64		48.21	32.91	16.48	28.64	56.43	24.70	48.78	32.35	
132	1.2(DL+COLL+CL-3)+0.8(WLI	253.50	41.31	21.39	40.56	15.86	8.21	64.64		48.21	32.91	16.48	28.64	56.43	24.70	48.78	32.35	
133	1.2(DL+COLL+CL-4)+0.8(WLI	253.50	41.31	21.39	40.56	15.86	8.21	64.64		48.21	32.91	16.48	28.64	56.43	24.70	48.78	32.35	
134	1.2(DL+COLL+CL-1)+0.8(WLI	233.49	102.26	10.08	37.36	39.27	3.87	80.50		72.76	1.96	-5.78	44.50	76.63	-1.91	41.23	33.49	
135	1.2(DL+COLL+CL-2)+0.8(WLI	233.49	102.26	10.08	37.36	39.27	3.87	80.50		72.76	1.96	-5.78	44.50	76.63	-1.91	41.23	33.49	
136	1.2(DL+COLL+CL-3)+0.8(WLI	233.49	102.26	10.08	37.36	39.27	3.87	80.50		72.76	1.96	-5.78	44.50	76.63	-1.91	41.23	33.49	
137	1.2(DL+COLL+CL-4)+0.8(WLI	233.49	102.26	10.08	37.36	39.27	3.87	80.50		72.76	1.96	-5.78	44.50	76.63	-1.91	41.23	33.49	
138	1.2(DL+COLL+CL-1)+0.8(WLI	252.14	41.21	4.61	40.34	15.82	1.77	57.93		54.40	26.29	22.75	21.93	56.17	24.52	42.11	38.58	
139	1.2(DL+COLL+CL-2)+0.8(WLI	252.14	41.21	4.61	40.34	15.82	1.77	57.93		54.40	26.29	22.75	21.93	56.17	24.52	42.11	38.58	
140	1.2(DL+COLL+CL-3)+0.8(WLI	252.14	41.21	4.61	40.34	15.82	1.77	57.93		54.40	26.29	22.75	21.93	56.17	24.52	42.11	38.58	
141	1.2(DL+COLL+CL-4)+0.8(WLI	252.14	41.21	4.61	40.34	15.82	1.77	57.93		54.40	26.29	22.75	21.93	56.17	24.52	42.11	38.58	
142	1.2(DL+COLL+CL-1)+0.8(WLI	234.84	102.40	18.27	37.57	39.32	7.02	83.91		69.88	5.27	-8.76	47.91	76.90	-1.75	44.59	30.56	
143	1.2(DL+COLL+CL-2)+0.8(WLI	234.84	102.40	18.27	37.57	39.32	7.02	83.91		69.88	5.27	-8.76	47.91	76.90	-1.75	44.59	30.56	
144	1.2(DL+COLL+CL-3)+0.8(WLI	234.84	102.40	18.27	37.57	39.32	7.02	83.91		69.88	5.27	-8.76	47.91	76.90	-1.75	44.59	30.56	
145	1.2(DL+COLL+CL-4)+0.8(WLI	234.84	102.40	18.27	37.57	39.32	7.02	83.91		69.88	5.27	-8.76	47.91	76.90	-1.75	44.59	30.56	
146	1.2(DL+COLL+CL-1)+0.8(WLI	253.60	102.40	12.78	40.58	39.32	4.91	84.80		74.99	6.16	-3.65	48.80	79.90	1.26	45.48	35.67	
147	1.2(DL+COLL+CL-2)+0.8(WLI	253.60	102.40	12.78	40.58	39.32	4.91	84.80		74.99	6.16	-3.65	48.80	79.90	1.26	45.48	35.67	
148	1.2(DL+COLL+CL-3)+0.8(WLI	253.60	102.40	12.78	40.58	39.32	4.91	84.80		74.99	6.16	-3.65	48.80	79.90	1.26	45.48	35.67	
149	1.2(DL+COLL+CL-4)+0.8(WLI	253.60	102.40	12.78	40.58	39.32	4.91	84.80		74.99	6.16	-3.65	48.80	79.90	1.26	45.48	35.67	
150	1.2(DL+COLL+CL-1)+1.6(WLI	188.71	204.25	6.11	30.19	78.43	2.34	110.97		106.28	-45.89	-50.58	74.97	108.63	-48.24	32.54	27.85	
151	1.2(DL+COLL+CL-2)+1.6(WLI	188.71	204.25	6.11	30.19	78.43	2.34	110.97		106.28	-45.89	-50.58	74.97	108.63	-48.24	32.54	27.85	
152	1.2(DL+COLL+CL-3)+1.6(WLI	188.71	204.25	6.11	30.19	78.43	2.34	110.97		106.28	-45.89	-50.58	74.97	108.63	-48.24	32.54	27.85	
153	1.2(DL+COLL+CL-4)+1.6(WLI	188.71	204.25	6.11	30.19	78.43	2.34	110.97		106.28	-45.89	-50.58	74.97	108.63	-48.24	32.54	27.85	
154	1.2(DL+COLL+CL-1)+1.6(WLI	223.89	82.32	0.01	35.82	31.61	0.00	67.44		67.43	4.22	4.21	31.44	67.44	4.22	35.83	35.82	
155	1.2(DL+COLL+CL-2)+1.6(WLI	223.89	82.32	0.01	35.82	31.61	0.00	67.44		67.43	4.22	4.21	31.44	67.44	4.22	35.83	35.82	
156	1.2(DL+COLL+CL-3)+1.6(WLI	223.89	82.32	0.01	35.82	31.61	0.00	67.44		67.43	4.22	4.21	31.44	67.44	4.22	35.83	35.82	

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157	1.2(DL+COLL+CL-4)+1.6(WLI	223.89	82.32	0.01	35.82	31.61	0.00	67.44		67.43	4.22	4.21	31.44	67.44	4.22	35.83	35.82	
158	1.2(DL+COLL+CL-1)+1.6(WLI	197.80	205.06	53.75	31.65	78.74	20.64	131.03		89.75	-26.46	-67.74	95.03	110.39	-47.10	52.29	11.01	
159	1.2(DL+COLL+CL-2)+1.6(WLI	197.80	205.06	53.75	31.65	78.74	20.64	131.03		89.75	-26.46	-67.74	95.03	110.39	-47.10	52.29	11.01	
160	1.2(DL+COLL+CL-3)+1.6(WLI	197.80	205.06	53.75	31.65	78.74	20.64	131.03		89.75	-26.46	-67.74	95.03	110.39	-47.10	52.29	11.01	
161	1.2(DL+COLL+CL-4)+1.6(WLI	197.80	205.06	53.75	31.65	78.74	20.64	131.03		89.75	-26.46	-67.74	95.03	110.39	-47.10	52.29	11.01	
162	1.2(DL+COLL+CL-1)+1.6(WLI	237.32	205.06	42.78	37.97	78.74	16.43	133.14		100.29	-24.34	-57.20	97.14	116.72	-40.77	54.40	21.55	
163	1.2(DL+COLL+CL-2)+1.6(WLI	237.32	205.06	42.78	37.97	78.74	16.43	133.14		100.29	-24.34	-57.20	97.14	116.72	-40.77	54.40	21.55	
164	1.2(DL+COLL+CL-3)+1.6(WLI	237.32	205.06	42.78	37.97	78.74	16.43	133.14		100.29	-24.34	-57.20	97.14	116.72	-40.77	54.40	21.55	
165	1.2(DL+COLL+CL-4)+1.6(WLI	237.32	82.62	42.78	37.97	31.73	16.43	86.12		53.27	22.67	-10.18	50.12	69.70	6.25	54.40	21.55	
166	1.2(DL+COLL+CL-1)+1.6(WLI	197.31	204.52	20.16	31.57	78.54	7.74	117.85		102.36	-39.22	-54.71	81.85	110.11	-46.97	39.32	23.83	
167	1.2(DL+COLL+CL-2)+1.6(WLI	197.31	204.52	20.16	31.57	78.54	7.74	117.85		102.36	-39.22	-54.71	81.85	110.11	-46.97	39.32	23.83	
168	1.2(DL+COLL+CL-3)+1.6(WLI	197.31	204.52	20.16	31.57	78.54	7.74	117.85		102.36	-39.22	-54.71	81.85	110.11	-46.97	39.32	23.83	
169	1.2(DL+COLL+CL-4)+1.6(WLI	197.31	204.52	20.16	31.57	78.54	7.74	117.85		102.36	-39.22	-54.71	81.85	110.11	-46.97	39.32	23.83	
170	1.2(DL+COLL+CL-1)+1.6(WLI	234.61	82.41	9.21	37.54	31.65	3.54	72.72		65.65	9.43	2.35	36.72	69.19	5.89	41.08	34.00	
171	1.2(DL+COLL+CL-2)+1.6(WLI	234.61	82.41	9.21	37.54	31.65	3.54	72.72		65.65	9.43	2.35	36.72	69.19	5.89	41.08	34.00	
172	1.2(DL+COLL+CL-3)+1.6(WLI	234.61	82.41	9.21	37.54	31.65	3.54	72.72		65.65	9.43	2.35	36.72	69.19	5.89	41.08	34.00	
173	1.2(DL+COLL+CL-4)+1.6(WLI	234.61	82.41	9.21	37.54	31.65	3.54	72.72		65.65	9.43	2.35	36.72	69.19	5.89	41.08	34.00	
174	1.2(DL+COLL+CL-1)+1.6(WLI	200.01	204.80	36.54	32.00	78.64	14.03	124.67		96.61	-32.61	-60.67	88.67	110.64	-46.64	46.03	17.97	
175	1.2(DL+COLL+CL-2)+1.6(WLI	200.01	204.80	36.54	32.00	78.64	14.03	124.67		96.61	-32.61	-60.67	88.67	110.64	-46.64	46.03	17.97	
176	1.2(DL+COLL+CL-3)+1.6(WLI	200.01	204.80	36.54	32.00	78.64	14.03	124.67		96.61	-32.61	-60.67	88.67	110.64	-46.64	46.03	17.97	
177	1.2(DL+COLL+CL-4)+1.6(WLI	200.01	204.80	36.54	32.00	78.64	14.03	124.67		96.61	-32.61	-60.67	88.67	110.64	-46.64	46.03	17.97	
178	1.2(DL+COLL+CL-1)+1.6(WLI	237.52	204.80	25.55	38.00	78.64	9.81	126.46		106.83	-30.83	-50.45	90.46	116.65	-40.64	47.82	28.19	
179	1.2(DL+COLL+CL-2)+1.6(WLI	237.52	204.80	25.55	38.00	78.64	9.81	126.46		106.83	-30.83	-50.45	90.46	116.65	-40.64	47.82	28.19	
180	1.2(DL+COLL+CL-3)+1.6(WLI	237.52	204.80	25.55	38.00	78.64	9.81	126.46		106.83	-30.83	-50.45	90.46	116.65	-40.64	47.82	28.19	
181	1.2(DL+COLL+CL-4)+1.6(WLI	237.52	204.80	25.55	38.00	78.64	9.81	126.46		106.83	-30.83	-50.45	90.46	116.65	-40.64	47.82	28.19	
182	1.2(DL+COLL+CL-1)+1.0(SL'+	295.43	0.00	0.00	47.27	0.00	0.00	47.27		47.27	47.27	47.27	11.27	47.27	47.27	47.27	47.27	
183	1.2(DL+COLL+CL-2)+1.0(SL'+	295.43	0.00	0.00	47.27	0.00	0.00	47.27		47.27	47.27	47.27	11.27	47.27	47.27	47.27	47.27	
184	1.2(DL+COLL+CL-3)+1.0(SL'+	295.43	0.00	0.00	47.27	0.00	0.00	47.27		47.27	47.27	47.27	11.27	47.27	47.27	47.27	47.27	
185	1.2(DL+COLL+CL-4)+1.0(SL'+	295.43	0.00	0.00	47.27	0.00	0.00	47.27		47.27	47.27	47.27	11.27	47.27	47.27	47.27	47.27	
186	1.2(DL+COLL+CL-1)+1.0(SL'-	295.43	0.00	0.00	47.27	0.00	0.00	47.27		47.27	47.27	47.27	11.27	47.27	47.27	47.27	47.27	
187	1.2(DL+COLL+CL-2)+1.0(SL'-	295.43	0.00	0.00	47.27	0.00	0.00	47.27		47.27	47.27	47.27	11.27	47.27	47.27	47.27	47.27	

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188	1.2(DL+COLL+CL-3)+1.0(SL'-	295.43	0.00	0.00	47.27	0.00	0.00	47.27		47.27	47.27	47.27	11.27	47.27	47.27	47.27	47.27	
189	1.2(DL+COLL+CL-4)+1.0(SL'-	295.43	0.00	0.00	47.27	0.00	0.00	47.27		47.27	47.27	47.27	11.27	47.27	47.27	47.27	47.27	
190	0.9(DL+COLL+CL-1)+1.6(WLI	168.91	204.25	6.11	27.03	78.43	2.34	107.80		103.11	-49.06	-53.75	71.80	105.46	-51.41	29.37	24.68	
191	0.9(DL+COLL+CL-2)+1.6(WLI	168.91	204.25	6.11	27.03	78.43	2.34	107.80		103.11	-49.06	-53.75	71.80	105.46	-51.41	29.37	24.68	
192	0.9(DL+COLL+CL-3)+1.6(WLI	168.91	204.25	6.11	27.03	78.43	2.34	107.80		103.11	-49.06	-53.75	71.80	105.46	-51.41	29.37	24.68	
193	0.9(DL+COLL+CL-4)+1.6(WLI	168.91	204.25	6.11	27.03	78.43	2.34	107.80		103.11	-49.06	-53.75	71.80	105.46	-51.41	29.37	24.68	
194	0.9(DL+COLL+CL-1)+1.6(WLI	204.08	82.32	0.01	32.65	31.61	0.00	64.27		64.26	1.05	1.04	28.27	64.27	1.05	32.66	32.65	
195	0.9(DL+COLL+CL-2)+1.6(WLI	204.08	82.32	0.01	32.65	31.61	0.00	64.27		64.26	1.05	1.04	28.27	64.27	1.05	32.66	32.65	
196	0.9(DL+COLL+CL-3)+1.6(WLI	204.08	82.32	0.01	32.65	31.61	0.00	64.27		64.26	1.05	1.04	28.27	64.27	1.05	32.66	32.65	
197	0.9(DL+COLL+CL-4)+1.6(WLI	204.08	82.32	0.01	32.65	31.61	0.00	64.27		64.26	1.05	1.04	28.27	64.27	1.05	32.66	32.65	
198	0.9(DL+COLL+CL-1)+1.6(WLI	177.99	205.06	53.75	28.48	78.74	20.64	127.86		86.58	-29.62	-70.91	91.86	107.22	-50.27	49.12	7.84	
199	0.9(DL+COLL+CL-2)+1.6(WLI	177.99	205.06	53.75	28.48	78.74	20.64	127.86		86.58	-29.62	-70.91	91.86	107.22	-50.27	49.12	7.84	
200	0.9(DL+COLL+CL-3)+1.6(WLI	177.99	205.06	53.75	28.48	78.74	20.64	127.86		86.58	-29.62	-70.91	91.86	107.22	-50.27	49.12	7.84	
201	0.9(DL+COLL+CL-4)+1.6(WLI	177.99	205.06	53.75	28.48	78.74	20.64	127.86		86.58	-29.62	-70.91	91.86	107.22	-50.27	49.12	7.84	
202	0.9(DL+COLL+CL-1)+1.6(WLI	217.51	82.62	42.78	34.80	31.73	16.43	82.96		50.10	19.51	-13.35	46.96	66.53	3.08	51.24	18.38	
203	0.9(DL+COLL+CL-2)+1.6(WLI	217.51	82.62	42.78	34.80	31.73	16.43	82.96		50.10	19.51	-13.35	46.96	66.53	3.08	51.24	18.38	
204	0.9(DL+COLL+CL-3)+1.6(WLI	217.51	82.62	42.78	34.80	31.73	16.43	82.96		50.10	19.51	-13.35	46.96	66.53	3.08	51.24	18.38	
205	0.9(DL+COLL+CL-4)+1.6(WLI	217.51	82.62	42.78	34.80	31.73	16.43	82.96		50.10	19.51	-13.35	46.96	66.53	3.08	51.24	18.38	
206	0.9(DL+COLL+CL-1)+1.6(WLI	177.50	204.52	20.16	28.40	78.54	7.74	114.68		99.19	-42.39	-57.88	78.68	106.94	-50.14	36.15	20.66	
207	0.9(DL+COLL+CL-2)+1.6(WLI	177.50	204.52	20.16	28.40	78.54	7.74	114.68		99.19	-42.39	-57.88	78.68	106.94	-50.14	36.15	20.66	
208	0.9(DL+COLL+CL-3)+1.6(WLI	177.50	204.52	20.16	28.40	78.54	7.74	114.68		99.19	-42.39	-57.88	78.68	106.94	-50.14	36.15	20.66	
209	0.9(DL+COLL+CL-4)+1.6(WLI	177.50	204.52	20.16	28.40	78.54	7.74	114.68		99.19	-42.39	-57.88	78.68	106.94	-50.14	36.15	20.66	
210	0.9(DL+COLL+CL-1)+1.6(WLI	214.81	82.41	9.21	34.37	31.65	3.54	69.55		62.48	6.26	-0.81	33.55	66.02	2.73	37.91	30.84	
211	0.9(DL+COLL+CL-2)+1.6(WLI	214.81	82.41	9.21	34.37	31.65	3.54	69.55		62.48	6.26	-0.81	33.55	66.02	2.73	37.91	30.84	
212	0.9(DL+COLL+CL-3)+1.6(WLI	214.81	82.41	9.21	34.37	31.65	3.54	69.55		62.48	6.26	-0.81	33.55	66.02	2.73	37.91	30.84	
213	0.9(DL+COLL+CL-4)+1.6(WLI	214.81	82.41	9.21	34.37	31.65	3.54	69.55		62.48	6.26	-0.81	33.55	66.02	2.73	37.91	30.84	
214	0.9(DL+COLL+CL-1)+1.6(WLI	180.21	204.80	36.54	28.83	78.64	14.03	121.51		93.45	-35.78	-63.84	85.51	107.48	-49.81	42.87	14.81	
215	0.9(DL+COLL+CL-2)+1.6(WLI	180.21	204.80	36.54	28.83	78.64	14.03	121.51		93.45	-35.78	-63.84	85.51	107.48	-49.81	42.87	14.81	
216	0.9(DL+COLL+CL-3)+1.6(WLI	180.21	204.80	36.54	28.83	78.64	14.03	121.51		93.45	-35.78	-63.84	85.51	107.48	-49.81	42.87	14.81	
217	0.9(DL+COLL+CL-4)+1.6(WLI	180.21	204.80	36.54	28.83	78.64	14.03	121.51		93.45	-35.78	-63.84	85.51	107.48	-49.81	42.87	14.81	
218	0.9(DL+COLL+CL-1)+1.6(WLI	217.71	204.80	25.55	34.83	78.64	9.81	123.29		103.67	-34.00	-53.62	87.29	113.48	-43.81	44.65	25.03	
219	0.9(DL+COLL+CL-2)+1.6(WLI	217.71	204.80	25.55	34.83	78.64	9.81	123.29		103.67	-34.00	-53.62	87.29	113.48	-43.81	44.65	25.03	

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220	0.9(DL+COLL+CL-3)+1.6(WL	217.71	204.80	25.55	34.83	78.64	9.81	123.29		103.67	-34.00	-53.62	87.29	113.48	-43.81	44.65	25.03	
221	0.9(DL+COLL+CL-4)+1.6(WL	217.71	204.80	25.55	34.83	78.64	9.81	123.29		103.67	-34.00	-53.62	87.29	113.48	-43.81	44.65	25.03	
222	0.9(DL+COLL+CL-1)+1.0(SL'+	275.63	0.00	0.00	44.10	0.00	0.00	44.10		44.10	44.10	44.10	8.10	44.10	44.10	44.10	44.10	
223	0.9(DL+COLL+CL-2)+1.0(SL'+	275.63	0.00	0.00	44.10	0.00	0.00	44.10		44.10	44.10	44.10	8.10	44.10	44.10	44.10	44.10	
224	0.9(DL+COLL+CL-3)+1.0(SL'+	275.63	0.00	0.00	44.10	0.00	0.00	44.10		44.10	44.10	44.10	8.10	44.10	44.10	44.10	44.10	
225	0.9(DL+COLL+CL-4)+1.0(SL'+	275.63	0.00	0.00	44.10	0.00	0.00	44.10		44.10	44.10	44.10	8.10	44.10	44.10	44.10	44.10	
226	0.9(DL+COLL+CL-1)+1.0(SL'-	275.63	0.00	0.00	44.10	0.00	0.00	44.10		44.10	44.10	44.10	8.10	44.10	44.10	44.10	44.10	
227	0.9(DL+COLL+CL-2)+1.0(SL'-	275.63	0.00	0.00	44.10	0.00	0.00	44.10		44.10	44.10	44.10	8.10	44.10	44.10	44.10	44.10	
228	0.9(DL+COLL+CL-3)+1.0(SL'-	275.63	0.00	0.00	44.10	0.00	0.00	44.10		44.10	44.10	44.10	8.10	44.10	44.10	44.10	44.10	
229	0.9(DL+COLL+CL-4)+1.0(SL'-	275.63	0.00	0.00	44.10	0.00	0.00	44.10		44.10	44.10	44.10	8.10	44.10	44.10	44.10	44.10	

Max :	331	47	0
Min :	202	-96	-34

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 8.47 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 18.47 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 51.83 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 61.83 kN/m²

Sample calculation of values of base pressure:

Load case : 302 1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL)

Maximum pressure, p_1 (pmax) = 44.25+0.00+3.05= 47.30 kN/m²

p_2= 44.25+0.00-3.05= 41.21 kN/m²

p_3= 44.25-(0.00+3.05)= 41.21 kN/m²

p_4= 44.25-0.00+3.05= 47.30 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 47.30-2*18= 11.30 kN/m²

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DESIGN OF ISOLATED FOOTING F3

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

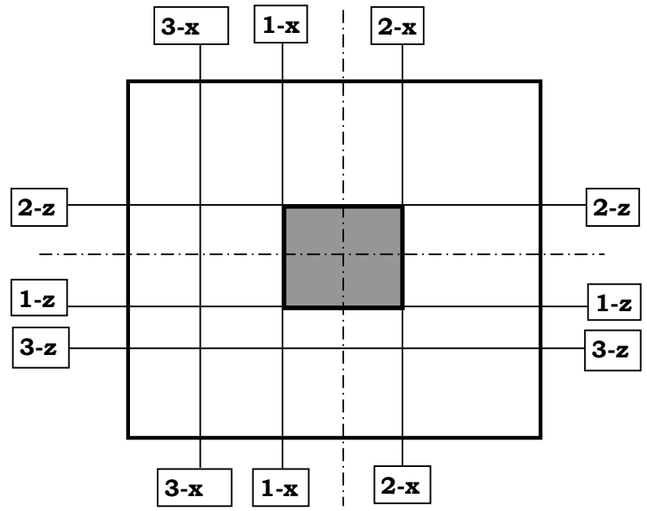
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.75+2*0.5*0.332)+(0.75+2*0.5*0.332))*2
= 4.328 m

Punching Area= ((0.75+2*0.5*0.332)*(0.75+2*0.5*0.332))
= 1.17 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.6*18)
= 38.40 kN/m^2

Area of footing, A_f= 6.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+COLL)	45.95	45.95	45.95	2.89	2.89	4.01	38.34
102	1.2(DL+COLL+CL-1)+1.6(R	51.80	51.80	51.80	5.13	5.13	7.12	68.09
103	1.2(DL+COLL+CL-2)+1.6(R	51.80	51.80	51.80	5.13	5.13	7.12	68.09
104	1.2(DL+COLL+CL-3)+1.6(R	51.80	51.80	51.80	5.13	5.13	7.12	68.09
105	1.2(DL+COLL+CL-4)+1.6(R	51.80	51.80	51.80	5.13	5.13	7.12	68.09
106	1.4(DL+COLL+CL-1)	45.95	45.95	45.95	2.89	2.89	4.01	38.34
107	1.4(DL+COLL+CL-2)	45.95	45.95	45.95	2.89	2.89	4.01	38.34
108	1.4(DL+COLL+CL-3)	45.95	45.95	45.95	2.89	2.89	4.01	38.34
109	1.4(DL+COLL+CL-4)	45.95	45.95	45.95	2.89	2.89	4.01	38.34
110	1.2(DL+COLL+CL-1+TL1)+	46.60	46.60	46.60	3.14	3.14	4.35	41.64
111	1.2(DL+COLL+CL-2+TL1)+	46.60	46.60	46.60	3.14	3.14	4.35	41.64
112	1.2(DL+COLL+CL-3+TL1)+	46.60	46.60	46.60	3.14	3.14	4.35	41.64

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113	1.2(DL+COLL+CL-4+TL1)+	46.60	46.60	46.60	3.14	3.14	4.35	41.64
114	1.2(DL+COLL+CL-1+TL2)+	57.01	57.01	57.01	7.12	7.12	9.88	94.55
115	1.2(DL+COLL+CL-2+TL2)+	57.01	57.01	57.01	7.12	7.12	9.88	94.55
116	1.2(DL+COLL+CL-3+TL2)+	57.01	57.01	57.01	7.12	7.12	9.88	94.55
117	1.2(DL+COLL+CL-4+TL2)+	57.01	57.01	57.01	7.12	7.12	9.88	94.55
118	1.2(DL+COLL+CL-1)+0.8(W	48.44	24.90	59.23	10.85	-12.17	15.48	-8.78
119	1.2(DL+COLL+CL-2)+0.8(W	48.44	24.90	59.23	10.85	-12.17	15.48	-8.78
120	1.2(DL+COLL+CL-3)+0.8(W	48.44	24.90	59.23	10.85	-12.17	15.48	-8.78
121	1.2(DL+COLL+CL-4)+0.8(W	48.44	24.90	59.23	10.85	-12.17	15.48	-8.78
122	1.2(DL+COLL+CL-1)+0.8(W	44.23	34.74	48.58	5.05	-4.22	7.19	5.51
123	1.2(DL+COLL+CL-2)+0.8(W	44.23	34.74	48.58	5.05	-4.22	7.19	5.51
124	1.2(DL+COLL+CL-3)+0.8(W	44.23	34.74	48.58	5.05	-4.22	7.19	5.51
125	1.2(DL+COLL+CL-4)+0.8(W	44.23	34.74	48.58	5.05	-4.22	7.19	5.51
126	1.2(DL+COLL+CL-1)+0.8(W	49.21	25.58	60.04	11.17	-11.94	15.93	-5.09
127	1.2(DL+COLL+CL-2)+0.8(W	49.21	25.58	60.04	11.17	-11.94	15.93	-5.09
128	1.2(DL+COLL+CL-3)+0.8(W	49.21	25.58	60.04	11.17	-11.94	15.93	-5.09
129	1.2(DL+COLL+CL-4)+0.8(W	49.21	25.58	60.04	11.17	-11.94	15.93	-5.09
130	1.2(DL+COLL+CL-1)+0.8(W	45.32	35.81	49.69	5.49	-3.83	7.78	10.97
131	1.2(DL+COLL+CL-2)+0.8(W	45.32	35.81	49.69	5.49	-3.83	7.78	10.97
132	1.2(DL+COLL+CL-3)+0.8(W	45.32	35.81	49.69	5.49	-3.83	7.78	10.97
133	1.2(DL+COLL+CL-4)+0.8(W	45.32	35.81	49.69	5.49	-3.83	7.78	10.97
134	1.2(DL+COLL+CL-1)+0.8(W	49.14	25.58	59.95	11.13	-11.92	15.87	-5.29
135	1.2(DL+COLL+CL-2)+0.8(W	49.14	25.58	59.95	11.13	-11.92	15.87	-5.29
136	1.2(DL+COLL+CL-3)+0.8(W	49.14	25.58	59.95	11.13	-11.92	15.87	-5.29
137	1.2(DL+COLL+CL-4)+0.8(W	49.14	25.58	59.95	11.13	-11.92	15.87	-5.29
138	1.2(DL+COLL+CL-1)+0.8(W	45.09	35.60	49.45	5.39	-3.90	7.65	9.87
139	1.2(DL+COLL+CL-2)+0.8(W	45.09	35.60	49.45	5.39	-3.90	7.65	9.87
140	1.2(DL+COLL+CL-3)+0.8(W	45.09	35.60	49.45	5.39	-3.90	7.65	9.87
141	1.2(DL+COLL+CL-4)+0.8(W	45.09	35.60	49.45	5.39	-3.90	7.65	9.87
142	1.2(DL+COLL+CL-1)+0.8(W	49.37	25.78	60.19	11.23	-11.86	16.01	-4.19
143	1.2(DL+COLL+CL-2)+0.8(W	49.37	25.78	60.19	11.23	-11.86	16.01	-4.19

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144	1.2(DL+COLL+CL-3)+0.8(W	49.37	25.78	60.19	11.23	-11.86	16.01	-4.19
145	1.2(DL+COLL+CL-4)+0.8(W	49.37	25.78	60.19	11.23	-11.86	16.01	-4.19
146	1.2(DL+COLL+CL-1)+0.8(W	52.38	28.78	63.20	12.37	-10.71	17.60	11.05
147	1.2(DL+COLL+CL-2)+0.8(W	52.38	28.78	63.20	12.37	-10.71	17.60	11.05
148	1.2(DL+COLL+CL-3)+0.8(W	52.38	28.78	63.20	12.37	-10.71	17.60	11.05
149	1.2(DL+COLL+CL-4)+0.8(W	52.38	28.78	63.20	12.37	-10.71	17.60	11.05
150	1.2(DL+COLL+CL-1)+1.6(W	53.73	6.66	75.31	19.88	-26.16	28.45	-41.68
151	1.2(DL+COLL+CL-2)+1.6(W	53.73	6.66	75.31	19.88	-26.16	28.45	-41.68
152	1.2(DL+COLL+CL-3)+1.6(W	53.73	6.66	75.31	19.88	-26.16	28.45	-41.68
153	1.2(DL+COLL+CL-4)+1.6(W	53.73	6.66	75.31	19.88	-26.16	28.45	-41.68
154	1.2(DL+COLL+CL-1)+1.6(W	45.31	26.35	54.01	8.29	-10.26	11.86	-13.09
155	1.2(DL+COLL+CL-2)+1.6(W	45.31	26.35	54.01	8.29	-10.26	11.86	-13.09
156	1.2(DL+COLL+CL-3)+1.6(W	45.31	26.35	54.01	8.29	-10.26	11.86	-13.09
157	1.2(DL+COLL+CL-4)+1.6(W	45.31	26.35	54.01	8.29	-10.26	11.86	-13.09
158	1.2(DL+COLL+CL-1)+1.6(W	55.27	8.02	76.94	20.52	-25.70	29.35	-34.30
159	1.2(DL+COLL+CL-2)+1.6(W	55.27	8.02	76.94	20.52	-25.70	29.35	-34.30
160	1.2(DL+COLL+CL-3)+1.6(W	55.27	8.02	76.94	20.52	-25.70	29.35	-34.30
161	1.2(DL+COLL+CL-4)+1.6(W	55.27	8.02	76.94	20.52	-25.70	29.35	-34.30
162	1.2(DL+COLL+CL-1)+1.6(W	61.60	14.35	83.27	22.95	-23.27	32.71	-2.18
163	1.2(DL+COLL+CL-2)+1.6(W	61.60	14.35	83.27	22.95	-23.27	32.71	-2.18
164	1.2(DL+COLL+CL-3)+1.6(W	61.60	14.35	83.27	22.95	-23.27	32.71	-2.18
165	1.2(DL+COLL+CL-4)+1.6(W	47.49	28.46	56.22	9.15	-9.47	13.04	-2.18
166	1.2(DL+COLL+CL-1)+1.6(W	55.13	8.01	76.75	20.44	-25.67	29.22	-34.70
167	1.2(DL+COLL+CL-2)+1.6(W	55.13	8.01	76.75	20.44	-25.67	29.22	-34.70
168	1.2(DL+COLL+CL-3)+1.6(W	55.13	8.01	76.75	20.44	-25.67	29.22	-34.70
169	1.2(DL+COLL+CL-4)+1.6(W	55.13	8.01	76.75	20.44	-25.67	29.22	-34.70
170	1.2(DL+COLL+CL-1)+1.6(W	47.04	28.05	55.75	8.96	-9.62	12.78	-4.38
171	1.2(DL+COLL+CL-2)+1.6(W	47.04	28.05	55.75	8.96	-9.62	12.78	-4.38
172	1.2(DL+COLL+CL-3)+1.6(W	47.04	28.05	55.75	8.96	-9.62	12.78	-4.38
173	1.2(DL+COLL+CL-4)+1.6(W	47.04	28.05	55.75	8.96	-9.62	12.78	-4.38
174	1.2(DL+COLL+CL-1)+1.6(W	55.59	8.41	77.23	20.63	-25.53	29.49	-32.50

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175	1.2(DL+COLL+CL-2)+1.6(W	55.59	8.41	77.23	20.63	-25.53	29.49	-32.50
176	1.2(DL+COLL+CL-3)+1.6(W	55.59	8.41	77.23	20.63	-25.53	29.49	-32.50
177	1.2(DL+COLL+CL-4)+1.6(W	55.59	8.41	77.23	20.63	-25.53	29.49	-32.50
178	1.2(DL+COLL+CL-1)+1.6(W	61.60	14.41	83.24	22.93	-23.23	32.68	-2.02
179	1.2(DL+COLL+CL-2)+1.6(W	61.60	14.41	83.24	22.93	-23.23	32.68	-2.02
180	1.2(DL+COLL+CL-3)+1.6(W	61.60	14.41	83.24	22.93	-23.23	32.68	-2.02
181	1.2(DL+COLL+CL-4)+1.6(W	61.60	14.41	83.24	22.93	-23.23	32.68	-2.02
182	1.2(DL+COLL+CL-1)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.71	45.06
183	1.2(DL+COLL+CL-2)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.71	45.06
184	1.2(DL+COLL+CL-3)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.71	45.06
185	1.2(DL+COLL+CL-4)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.71	45.06
186	1.2(DL+COLL+CL-1)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.71	45.06
187	1.2(DL+COLL+CL-2)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.71	45.06
188	1.2(DL+COLL+CL-3)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.71	45.06
189	1.2(DL+COLL+CL-4)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.71	45.06
190	0.9(DL+COLL+CL-1)+1.6(W	50.56	3.49	72.14	18.67	-27.37	26.76	-57.78
191	0.9(DL+COLL+CL-2)+1.6(W	50.56	3.49	72.14	18.67	-27.37	26.76	-57.78
192	0.9(DL+COLL+CL-3)+1.6(W	50.56	3.49	72.14	18.67	-27.37	26.76	-57.78
193	0.9(DL+COLL+CL-4)+1.6(W	50.56	3.49	72.14	18.67	-27.37	26.76	-57.78
194	0.9(DL+COLL+CL-1)+1.6(W	42.14	23.18	50.84	7.08	-11.47	10.17	-29.19
195	0.9(DL+COLL+CL-2)+1.6(W	42.14	23.18	50.84	7.08	-11.47	10.17	-29.19
196	0.9(DL+COLL+CL-3)+1.6(W	42.14	23.18	50.84	7.08	-11.47	10.17	-29.19
197	0.9(DL+COLL+CL-4)+1.6(W	42.14	23.18	50.84	7.08	-11.47	10.17	-29.19
198	0.9(DL+COLL+CL-1)+1.6(W	52.10	4.85	73.77	19.31	-26.91	27.66	-50.40
199	0.9(DL+COLL+CL-2)+1.6(W	52.10	4.85	73.77	19.31	-26.91	27.66	-50.40
200	0.9(DL+COLL+CL-3)+1.6(W	52.10	4.85	73.77	19.31	-26.91	27.66	-50.40
201	0.9(DL+COLL+CL-4)+1.6(W	52.10	4.85	73.77	19.31	-26.91	27.66	-50.40
202	0.9(DL+COLL+CL-1)+1.6(W	44.32	25.29	53.05	7.93	-10.69	11.36	-18.28
203	0.9(DL+COLL+CL-2)+1.6(W	44.32	25.29	53.05	7.93	-10.69	11.36	-18.28
204	0.9(DL+COLL+CL-3)+1.6(W	44.32	25.29	53.05	7.93	-10.69	11.36	-18.28
205	0.9(DL+COLL+CL-4)+1.6(W	44.32	25.29	53.05	7.93	-10.69	11.36	-18.28

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206	0.9(DL+COLL+CL-1)+1.6(W	51.96	4.84	73.58	19.22	-26.88	27.54	-50.80
207	0.9(DL+COLL+CL-2)+1.6(W	51.96	4.84	73.58	19.22	-26.88	27.54	-50.80
208	0.9(DL+COLL+CL-3)+1.6(W	51.96	4.84	73.58	19.22	-26.88	27.54	-50.80
209	0.9(DL+COLL+CL-4)+1.6(W	51.96	4.84	73.58	19.22	-26.88	27.54	-50.80
210	0.9(DL+COLL+CL-1)+1.6(W	43.87	24.88	52.58	7.75	-10.83	11.10	-20.48
211	0.9(DL+COLL+CL-2)+1.6(W	43.87	24.88	52.58	7.75	-10.83	11.10	-20.48
212	0.9(DL+COLL+CL-3)+1.6(W	43.87	24.88	52.58	7.75	-10.83	11.10	-20.48
213	0.9(DL+COLL+CL-4)+1.6(W	43.87	24.88	52.58	7.75	-10.83	11.10	-20.48
214	0.9(DL+COLL+CL-1)+1.6(W	52.43	5.24	74.07	19.42	-26.74	27.81	-48.60
215	0.9(DL+COLL+CL-2)+1.6(W	52.43	5.24	74.07	19.42	-26.74	27.81	-48.60
216	0.9(DL+COLL+CL-3)+1.6(W	52.43	5.24	74.07	19.42	-26.74	27.81	-48.60
217	0.9(DL+COLL+CL-4)+1.6(W	52.43	5.24	74.07	19.42	-26.74	27.81	-48.60
218	0.9(DL+COLL+CL-1)+1.6(W	58.43	11.24	80.07	21.72	-24.45	31.00	-18.11
219	0.9(DL+COLL+CL-2)+1.6(W	58.43	11.24	80.07	21.72	-24.45	31.00	-18.11
220	0.9(DL+COLL+CL-3)+1.6(W	58.43	11.24	80.07	21.72	-24.45	31.00	-18.11
221	0.9(DL+COLL+CL-4)+1.6(W	58.43	11.24	80.07	21.72	-24.45	31.00	-18.11
222	0.9(DL+COLL+CL-1)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.03	28.96
223	0.9(DL+COLL+CL-2)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.03	28.96
224	0.9(DL+COLL+CL-3)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.03	28.96
225	0.9(DL+COLL+CL-4)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.03	28.96
226	0.9(DL+COLL+CL-1)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.03	28.96
227	0.9(DL+COLL+CL-2)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.03	28.96
228	0.9(DL+COLL+CL-3)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.03	28.96
229	0.9(DL+COLL+CL-4)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.03	28.96

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Sample calculation of BM & SF

Load case : 101 1.4(DL+COLL)

p_{1x} = 45.95-(45.95-45.95)*0.875/2.5 = 45.95 kN/m^2 Overhang = 0.875 m

p_{2x} = 45.95+(45.95-45.95)*0.875/2.5 = 45.95 kN/m^2 Effective depth = 0.344 m

p_{3x} = 45.95-(45.95-45.95)*(0.875-0.344)/2.5 = 45.95 kN/m^2

BM_{1x} = (45.95-38.4)*0.875*0.875/2+0.5*(45.95-45.95)*0.875*0.875*0.67 = 2.89 kN-m/m

BM_{2x} = (45.95-38.4)*0.875*0.875/2-0.5*(-45.95+45.95)*0.875*0.875*0.67 = 2.89 kN-m/m

SF_{3x} = (45.95-38.4)*(0.875-0.344)+0.5*(45.95-45.95)*(0.875-0.344) = 4.01 kN/m

$S.F_{(punching)}$ = (6.25-1.17)*(45.95-38.4) = 38.34 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	22.95	0.19	0.26	0.001	212.71	619.20	12ϕT	182.65	300	175	646.27
At top	-27.37	-0.23	0.31	0.001	253.95	309.60	10ϕT	253.68	300	250	314.16

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 32.71 kN/m

Shear stress , τ_v= (32.71*1000)/(1000*344)

= 0.10 N/mm^2 **O.K.**

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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 94.55 kN

Strength of Concrete in shear Vc1 = 4404.02 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2634.82 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2936.02 kN **Table 22.6.5.2_ (c)**

Design Vc= 2634.82 kN

Allowable shear stress, (ϕ Vc)/bd = 1.33 N/mm^2

Design punching shear stress, τ = (94.55*1000)/(4328*344)

= 0.06 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+COLL)	45.95	45.95	45.95	2.89	2.89	4.10
102	1.2(DL+COLL+CL-1)+1.6(R	51.80	51.80	51.80	5.13	5.13	7.28
103	1.2(DL+COLL+CL-2)+1.6(R	51.80	51.80	51.80	5.13	5.13	7.28
104	1.2(DL+COLL+CL-3)+1.6(R	51.80	51.80	51.80	5.13	5.13	7.28
105	1.2(DL+COLL+CL-4)+1.6(R	51.80	51.80	51.80	5.13	5.13	7.28
106	1.4(DL+COLL+CL-1)	45.95	45.95	45.95	2.89	2.89	4.10
107	1.4(DL+COLL+CL-2)	45.95	45.95	45.95	2.89	2.89	4.10
108	1.4(DL+COLL+CL-3)	45.95	45.95	45.95	2.89	2.89	4.10

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109	1.4(DL+COLL+CL-4)	45.95	45.95	45.95	2.89	2.89	4.10
110	1.2(DL+COLL+CL-1+TL1)+	47.70	45.50	48.67	4.22	2.06	6.01
111	1.2(DL+COLL+CL-2+TL1)+	47.70	45.50	48.67	4.22	2.06	6.01
112	1.2(DL+COLL+CL-3+TL1)+	47.70	45.50	48.67	4.22	2.06	6.01
113	1.2(DL+COLL+CL-4+TL1)+	47.70	45.50	48.67	4.22	2.06	6.01
114	1.2(DL+COLL+CL-1+TL2)+	57.01	57.01	57.01	7.12	7.12	10.11
115	1.2(DL+COLL+CL-2+TL2)+	57.01	57.01	57.01	7.12	7.12	10.11
116	1.2(DL+COLL+CL-3+TL2)+	57.01	57.01	57.01	7.12	7.12	10.11
117	1.2(DL+COLL+CL-4+TL2)+	57.01	57.01	57.01	7.12	7.12	10.11
118	1.2(DL+COLL+CL-1)+0.8(W	37.03	36.32	37.34	-0.31	-1.01	-0.44
119	1.2(DL+COLL+CL-2)+0.8(W	37.03	36.32	37.34	-0.31	-1.01	-0.44
120	1.2(DL+COLL+CL-3)+0.8(W	37.03	36.32	37.34	-0.31	-1.01	-0.44
121	1.2(DL+COLL+CL-4)+0.8(W	37.03	36.32	37.34	-0.31	-1.01	-0.44
122	1.2(DL+COLL+CL-1)+0.8(W	39.49	39.49	39.49	0.42	0.42	0.59
123	1.2(DL+COLL+CL-2)+0.8(W	39.49	39.49	39.49	0.42	0.42	0.59
124	1.2(DL+COLL+CL-3)+0.8(W	39.49	39.49	39.49	0.42	0.42	0.59
125	1.2(DL+COLL+CL-4)+0.8(W	39.49	39.49	39.49	0.42	0.42	0.59
126	1.2(DL+COLL+CL-1)+0.8(W	40.50	34.30	43.24	2.66	-3.42	3.84
127	1.2(DL+COLL+CL-2)+0.8(W	40.50	34.30	43.24	2.66	-3.42	3.84
128	1.2(DL+COLL+CL-3)+0.8(W	40.50	34.30	43.24	2.66	-3.42	3.84
129	1.2(DL+COLL+CL-4)+0.8(W	40.50	34.30	43.24	2.66	-3.42	3.84
130	1.2(DL+COLL+CL-1)+0.8(W	43.03	38.10	45.21	3.25	-1.59	4.67
131	1.2(DL+COLL+CL-2)+0.8(W	43.03	38.10	45.21	3.25	-1.59	4.67
132	1.2(DL+COLL+CL-3)+0.8(W	43.03	38.10	45.21	3.25	-1.59	4.67
133	1.2(DL+COLL+CL-4)+0.8(W	43.03	38.10	45.21	3.25	-1.59	4.67
134	1.2(DL+COLL+CL-1)+0.8(W	38.52	36.20	39.55	0.74	-1.54	1.08
135	1.2(DL+COLL+CL-2)+0.8(W	38.52	36.20	39.55	0.74	-1.54	1.08
136	1.2(DL+COLL+CL-3)+0.8(W	38.52	36.20	39.55	0.74	-1.54	1.08
137	1.2(DL+COLL+CL-4)+0.8(W	38.52	36.20	39.55	0.74	-1.54	1.08
138	1.2(DL+COLL+CL-1)+0.8(W	40.87	39.82	41.34	1.26	0.22	1.81
139	1.2(DL+COLL+CL-2)+0.8(W	40.87	39.82	41.34	1.26	0.22	1.81
140	1.2(DL+COLL+CL-3)+0.8(W	40.87	39.82	41.34	1.26	0.22	1.81
141	1.2(DL+COLL+CL-4)+0.8(W	40.87	39.82	41.34	1.26	0.22	1.81

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142	1.2(DL+COLL+CL-1)+0.8(W	39.68	35.47	41.54	1.75	-2.38	2.53
143	1.2(DL+COLL+CL-2)+0.8(W	39.68	35.47	41.54	1.75	-2.38	2.53
144	1.2(DL+COLL+CL-3)+0.8(W	39.68	35.47	41.54	1.75	-2.38	2.53
145	1.2(DL+COLL+CL-4)+0.8(W	39.68	35.47	41.54	1.75	-2.38	2.53
146	1.2(DL+COLL+CL-1)+0.8(W	42.05	39.10	43.35	2.28	-0.61	3.27
147	1.2(DL+COLL+CL-2)+0.8(W	42.05	39.10	43.35	2.28	-0.61	3.27
148	1.2(DL+COLL+CL-3)+0.8(W	42.05	39.10	43.35	2.28	-0.61	3.27
149	1.2(DL+COLL+CL-4)+0.8(W	42.05	39.10	43.35	2.28	-0.61	3.27
150	1.2(DL+COLL+CL-1)+1.6(W	30.90	29.49	31.52	-2.45	-3.83	-3.46
151	1.2(DL+COLL+CL-2)+1.6(W	30.90	29.49	31.52	-2.45	-3.83	-3.46
152	1.2(DL+COLL+CL-3)+1.6(W	30.90	29.49	31.52	-2.45	-3.83	-3.46
153	1.2(DL+COLL+CL-4)+1.6(W	30.90	29.49	31.52	-2.45	-3.83	-3.46
154	1.2(DL+COLL+CL-1)+1.6(W	35.83	35.82	35.83	-0.98	-0.99	-1.40
155	1.2(DL+COLL+CL-2)+1.6(W	35.83	35.82	35.83	-0.98	-0.99	-1.40
156	1.2(DL+COLL+CL-3)+1.6(W	35.83	35.82	35.83	-0.98	-0.99	-1.40
157	1.2(DL+COLL+CL-4)+1.6(W	35.83	35.82	35.83	-0.98	-0.99	-1.40
158	1.2(DL+COLL+CL-1)+1.6(W	37.84	25.46	43.32	3.49	-8.66	5.11
159	1.2(DL+COLL+CL-2)+1.6(W	37.84	25.46	43.32	3.49	-8.66	5.11
160	1.2(DL+COLL+CL-3)+1.6(W	37.84	25.46	43.32	3.49	-8.66	5.11
161	1.2(DL+COLL+CL-4)+1.6(W	37.84	25.46	43.32	3.49	-8.66	5.11
162	1.2(DL+COLL+CL-1)+1.6(W	42.90	33.05	47.26	4.67	-5.00	6.75
163	1.2(DL+COLL+CL-2)+1.6(W	42.90	33.05	47.26	4.67	-5.00	6.75
164	1.2(DL+COLL+CL-3)+1.6(W	42.90	33.05	47.26	4.67	-5.00	6.75
165	1.2(DL+COLL+CL-4)+1.6(W	42.90	33.05	47.26	4.67	-5.00	6.75
166	1.2(DL+COLL+CL-1)+1.6(W	33.90	29.25	35.96	-0.33	-4.89	-0.41
167	1.2(DL+COLL+CL-2)+1.6(W	33.90	29.25	35.96	-0.33	-4.89	-0.41
168	1.2(DL+COLL+CL-3)+1.6(W	33.90	29.25	35.96	-0.33	-4.89	-0.41
169	1.2(DL+COLL+CL-4)+1.6(W	33.90	29.25	35.96	-0.33	-4.89	-0.41
170	1.2(DL+COLL+CL-1)+1.6(W	38.60	36.48	39.54	0.71	-1.37	1.04
171	1.2(DL+COLL+CL-2)+1.6(W	38.60	36.48	39.54	0.71	-1.37	1.04
172	1.2(DL+COLL+CL-3)+1.6(W	38.60	36.48	39.54	0.71	-1.37	1.04
173	1.2(DL+COLL+CL-4)+1.6(W	38.60	36.48	39.54	0.71	-1.37	1.04
174	1.2(DL+COLL+CL-1)+1.6(W	36.21	27.79	39.94	1.68	-6.58	2.49

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175	1.2(DL+COLL+CL-2)+1.6(W	36.21	27.79	39.94	1.68	-6.58	2.49
176	1.2(DL+COLL+CL-3)+1.6(W	36.21	27.79	39.94	1.68	-6.58	2.49
177	1.2(DL+COLL+CL-4)+1.6(W	36.21	27.79	39.94	1.68	-6.58	2.49
178	1.2(DL+COLL+CL-1)+1.6(W	40.95	35.06	43.56	2.74	-3.04	3.96
179	1.2(DL+COLL+CL-2)+1.6(W	40.95	35.06	43.56	2.74	-3.04	3.96
180	1.2(DL+COLL+CL-3)+1.6(W	40.95	35.06	43.56	2.74	-3.04	3.96
181	1.2(DL+COLL+CL-4)+1.6(W	40.95	35.06	43.56	2.74	-3.04	3.96
182	1.2(DL+COLL+CL-1)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.82
183	1.2(DL+COLL+CL-2)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.82
184	1.2(DL+COLL+CL-3)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.82
185	1.2(DL+COLL+CL-4)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.82
186	1.2(DL+COLL+CL-1)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.82
187	1.2(DL+COLL+CL-2)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.82
188	1.2(DL+COLL+CL-3)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.82
189	1.2(DL+COLL+CL-4)+1.0(S	47.27	47.27	47.27	3.40	3.40	4.82
190	0.9(DL+COLL+CL-1)+1.6(W	27.73	26.32	28.35	-3.66	-5.04	-5.18
191	0.9(DL+COLL+CL-2)+1.6(W	27.73	26.32	28.35	-3.66	-5.04	-5.18
192	0.9(DL+COLL+CL-3)+1.6(W	27.73	26.32	28.35	-3.66	-5.04	-5.18
193	0.9(DL+COLL+CL-4)+1.6(W	27.73	26.32	28.35	-3.66	-5.04	-5.18
194	0.9(DL+COLL+CL-1)+1.6(W	32.66	32.65	32.66	-2.20	-2.20	-3.12
195	0.9(DL+COLL+CL-2)+1.6(W	32.66	32.65	32.66	-2.20	-2.20	-3.12
196	0.9(DL+COLL+CL-3)+1.6(W	32.66	32.65	32.66	-2.20	-2.20	-3.12
197	0.9(DL+COLL+CL-4)+1.6(W	32.66	32.65	32.66	-2.20	-2.20	-3.12
198	0.9(DL+COLL+CL-1)+1.6(W	34.67	22.29	40.15	2.28	-9.87	3.39
199	0.9(DL+COLL+CL-2)+1.6(W	34.67	22.29	40.15	2.28	-9.87	3.39
200	0.9(DL+COLL+CL-3)+1.6(W	34.67	22.29	40.15	2.28	-9.87	3.39
201	0.9(DL+COLL+CL-4)+1.6(W	34.67	22.29	40.15	2.28	-9.87	3.39
202	0.9(DL+COLL+CL-1)+1.6(W	39.74	29.88	44.10	3.46	-6.21	5.03
203	0.9(DL+COLL+CL-2)+1.6(W	39.74	29.88	44.10	3.46	-6.21	5.03
204	0.9(DL+COLL+CL-3)+1.6(W	39.74	29.88	44.10	3.46	-6.21	5.03
205	0.9(DL+COLL+CL-4)+1.6(W	39.74	29.88	44.10	3.46	-6.21	5.03
206	0.9(DL+COLL+CL-1)+1.6(W	30.73	26.08	32.79	-1.55	-6.11	-2.14
207	0.9(DL+COLL+CL-2)+1.6(W	30.73	26.08	32.79	-1.55	-6.11	-2.14
208	0.9(DL+COLL+CL-3)+1.6(W	30.73	26.08	32.79	-1.55	-6.11	-2.14

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209	0.9(DL+COLL+CL-4)+1.6(W	30.73	26.08	32.79	-1.55	-6.11	-2.14
210	0.9(DL+COLL+CL-1)+1.6(W	35.44	33.31	36.37	-0.50	-2.58	-0.68
211	0.9(DL+COLL+CL-2)+1.6(W	35.44	33.31	36.37	-0.50	-2.58	-0.68
212	0.9(DL+COLL+CL-3)+1.6(W	35.44	33.31	36.37	-0.50	-2.58	-0.68
213	0.9(DL+COLL+CL-4)+1.6(W	35.44	33.31	36.37	-0.50	-2.58	-0.68
214	0.9(DL+COLL+CL-1)+1.6(W	33.05	24.63	36.78	0.47	-7.79	0.77
215	0.9(DL+COLL+CL-2)+1.6(W	33.05	24.63	36.78	0.47	-7.79	0.77
216	0.9(DL+COLL+CL-3)+1.6(W	33.05	24.63	36.78	0.47	-7.79	0.77
217	0.9(DL+COLL+CL-4)+1.6(W	33.05	24.63	36.78	0.47	-7.79	0.77
218	0.9(DL+COLL+CL-1)+1.6(W	37.78	31.90	40.39	1.53	-4.25	2.24
219	0.9(DL+COLL+CL-2)+1.6(W	37.78	31.90	40.39	1.53	-4.25	2.24
220	0.9(DL+COLL+CL-3)+1.6(W	37.78	31.90	40.39	1.53	-4.25	2.24
221	0.9(DL+COLL+CL-4)+1.6(W	37.78	31.90	40.39	1.53	-4.25	2.24
222	0.9(DL+COLL+CL-1)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.10
223	0.9(DL+COLL+CL-2)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.10
224	0.9(DL+COLL+CL-3)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.10
225	0.9(DL+COLL+CL-4)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.10
226	0.9(DL+COLL+CL-1)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.10
227	0.9(DL+COLL+CL-2)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.10
228	0.9(DL+COLL+CL-3)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.10
229	0.9(DL+COLL+CL-4)+1.0(S	44.10	44.10	44.10	2.18	2.18	3.10

Sample calculation of BM & SF

Load case : 101

p_{1z} = 45.95-(45.95-45.95)*0.875/2.5= 45.95 kN/m^2

p_{2z} = 45.95+(45.95-45.95)*0.875/2.5= 45.95 kN/m^2 Overhang = 0.875 m

p_{3z} = 45.95-(45.95-45.95)*(0.875-0.332)/2.5= 45.95 kN/m^2 Effective depth = 0.332 m

BM_{1z} = (45.95-38.4)*0.875*0.875/2+0.5*(45.95-45.95)*0.875*0.875*0.67 = 2.89 kN-m/m

BM_{2z} = (45.95-38.4)*0.875*0.875/2-0.5*(-45.95+45.95)*0.875*0.875*0.67 = 2.89 kN-m/m

SF_{3z} = (45.95-38.4)*(0.875-0.332)+0.5*(45.95-45.95)*(0.875-0.332) = 4.10 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	7.12	0.065	0.086	0.0002	70.69	597.60	12ϕT	189.25	300	175	646.3
At top	-3.42	-0.031	-0.041	-0.0001	-0.33	298.80	10ϕT	262.85	250	250	314.2

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 10.11 kN/m

Shear stress , τ_v= (10.11*1000)/(1000*332)

= 0.03 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	12ϕT	**175**	c/c
	At Top	10ϕT	**250**	c/c
Along Z Direction :				
	At Bottom	12ϕT	**175**	c/c
	At Top	10ϕT	**250**	c/c

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5.4 DESIGN OF CALCULATION OF FOOTING F4

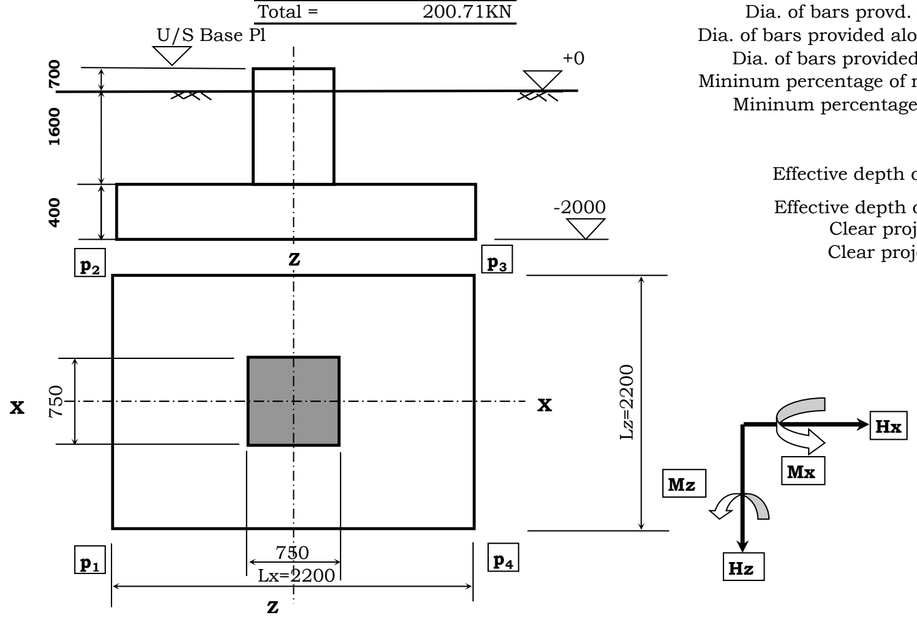
	Along-X	Along-Z
Size of column :	0.75m	0.75m
Size of fdn. raft :	2.20m	2.20m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-2.00 m	
U/S level Base Plate :	0.70 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Crane Load	CL
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (2.2*2.2*0.4*24)+(0.75*0.75*2.3*24) = 77.51KN
Weight of soil = ((2.2*2.2)-(0.75*0.75))*1.6*18 = 123.19KN
Total = 200.71KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

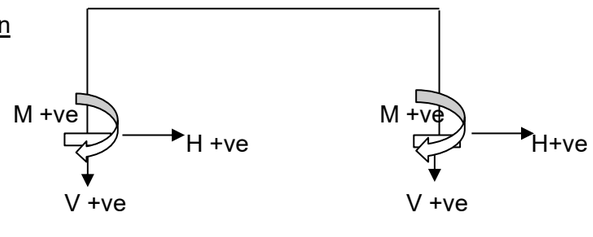
Effective depth of slab along X direction = 344mm = 0.344m
Effective depth of slab along Z direction = 332mm = 0.332m
Clear projection along Z direction = 0.725m
Clear projection along X direction = 0.725m

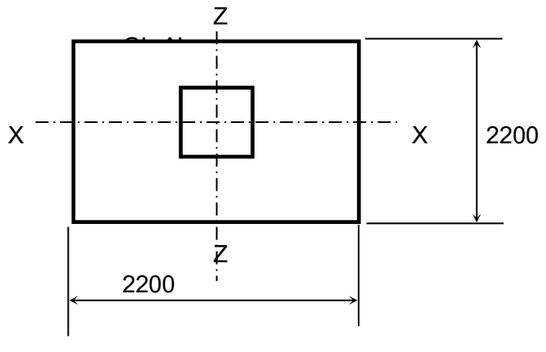
H$_X$=Force along X-X direction
Hz=Force along Z-Z direction
M$_X$=Moment about X-X direction
Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F4

Sign convention





Height from BOF to top of column about X (l_{ux}) = 2700mm
Height from BOF to top of column about Z (l_{uz}) = 2700mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	10.8312		-3.7404		0
2	COLLATERAL LOAD (COLL)	0		0		0
3	ROOF LIVE LOAD (RLL)	31.7064		-13.9656		0
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	-2.1492		2.2884		0
5	CRANE LOAD-1 (CL-1)	0		0		0
6	CRANE LOAD-2 (CL-2)	0		0		0
7	CRANE LOAD-3 (CL-3)	0		0		0
8	CRANE LOAD-4 (CL-4)	0		0		0
9	WIND LEFT-1 (WLL-1)	24.3684		-12.8784		0
10	WIND LEFT-2 (WLL-2)	-14.3292		9.09		0
11	WIND RIGHT-1 (WLR-1)	59.184		-22.7892		0
12	WIND RIGHT-2 (WLR-2)	-53.9268		20.9052		0
13	WIND LONG-1 (WLLG-1)	-17.4384	8.0988		0	
14	WIND LONG-2 (WLLG-2)	-34.0584	15.0924		0	
15	WIND LONG-3 (WLLG-3)	-17.1708	7.8		0	
16	WIND LONG-4 (WLLG-4)	-33.8844	14.9016		0	
17	SEISMIC LOAD SL'+'	0.078	-0.0984	-0.0984	0	0
18	SEISMIC LOAD SL'-'	-0.078	0.0984	0.0984	0	0
19	TEMPERATURE LOAD (TL1)	1.3188		-1.8888		0
20	TEMPERATURE LOAD (TL2)	-1.3188		1.8888		0
21	BRACING REACTION (BR-WLLG1)	0	0		0	
22	BRACING REACTION (BR-WLLG2)	0	0		0	
23	BRACING REACTION (BR-WLLG3)	0	0		0	
24	BRACING REACTION (BR-WLLG4)	0	0		0	
25	BRACING REACTION (BR-SL'+')	0	0	0	0	0
26	BRACING REACTION (BR-SL'-')	0	0	0	0	0

Ultimate Condition

Combined Load Cases:		Load at top of foundation (Factored Load/Moment)			Load at bottom of foundation		
Load Case	Load Combination	Pu (kN)	Muz (kN-m)	Mux (kN-m)	Pu (kN)	Muz (kN-m)	Mux (kN-m)
101	1.4(DL+COLL)	15.16	0.0	14.1	215.9	0.0	14.1
102	1.2(DL+COLL+CL-1)+1.6(RLL+PLL / MLL)	60.3	0.0	62.6	261.0	0.0	62.6
103	1.2(DL+COLL+CL-2)+1.6(RLL+PLL / MLL)	60.3	0.0	62.6	261.0	0.0	62.6
104	1.2(DL+COLL+CL-3)+1.6(RLL+PLL / MLL)	60.3	0.0	62.6	261.0	0.0	62.6
105	1.2(DL+COLL+CL-4)+1.6(RLL+PLL / MLL)	60.3	0.0	62.6	261.0	0.0	62.6
106	1.4(DL+COLL+CL-1)	15.2	0.0	14.1	215.9	0.0	14.1
107	1.4(DL+COLL+CL-2)	15.2	0.0	14.1	215.9	0.0	14.1
108	1.4(DL+COLL+CL-3)	15.2	0.0	14.1	215.9	0.0	14.1
109	1.4(DL+COLL+CL-4)	15.2	0.0	14.1	215.9	0.0	14.1
110	1.2(DL+COLL+CL-1+TL1)+1.6(RLL+PLL / MLL)	61.9	0.0	68.7	262.6	0.0	68.7
111	1.2(DL+COLL+CL-2+TL1)+1.6(RLL+PLL / MLL)	61.9	0.0	68.7	262.6	0.0	68.7
112	1.2(DL+COLL+CL-3+TL1)+1.6(RLL+PLL / MLL)	61.9	0.0	68.7	262.6	0.0	68.7
113	1.2(DL+COLL+CL-4+TL1)+1.6(RLL+PLL / MLL)	61.9	0.0	68.7	262.6	0.0	68.7
114	1.2(DL+COLL+CL-1+TL2)+1.6(RLL+PLL / MLL)	58.7	0.0	56.4	259.4	0.0	56.4
115	1.2(DL+COLL+CL-2+TL2)+1.6(RLL+PLL / MLL)	58.7	0.0	56.4	259.4	0.0	56.4
116	1.2(DL+COLL+CL-3+TL2)+1.6(RLL+PLL / MLL)	58.7	0.0	56.4	259.4	0.0	56.4
117	1.2(DL+COLL+CL-4+TL2)+1.6(RLL+PLL / MLL)	58.7	0.0	56.4	259.4	0.0	56.4
118	1.2(DL+COLL+CL-1)+0.8(WLL-1)	32.5	0.0	39.9	233.2	0.0	39.9
119	1.2(DL+COLL+CL-2)+0.8(WLL-1)	32.5	0.0	39.9	233.2	0.0	39.9
120	1.2(DL+COLL+CL-3)+0.8(WLL-1)	32.5	0.0	39.9	233.2	0.0	39.9
121	1.2(DL+COLL+CL-4)+0.8(WLL-1)	32.5	0.0	39.9	233.2	0.0	39.9
122	1.2(DL+COLL+CL-1)+0.8(WLL-2)	1.5	0.0	7.5	202.2	0.0	7.5
123	1.2(DL+COLL+CL-2)+0.8(WLL-2)	1.5	0.0	7.5	202.2	0.0	7.5
124	1.2(DL+COLL+CL-3)+0.8(WLL-2)	1.5	0.0	7.5	202.2	0.0	7.5
125	1.2(DL+COLL+CL-4)+0.8(WLL-2)	1.5	0.0	7.5	202.2	0.0	7.5
126	1.2(DL+COLL+CL-1)+0.8(WLR-1)	60.3	0.0	61.3	261.1	0.0	61.3
127	1.2(DL+COLL+CL-2)+0.8(WLR-1)	60.3	0.0	61.3	261.1	0.0	61.3
128	1.2(DL+COLL+CL-3)+0.8(WLR-1)	60.3	0.0	61.3	261.1	0.0	61.3
129	1.2(DL+COLL+CL-4)+0.8(WLR-1)	60.3	0.0	61.3	261.1	0.0	61.3
130	1.2(DL+COLL+CL-1)+0.8(WLR-2)	-30.1	0.0	33.0	170.6	0.0	33.0
131	1.2(DL+COLL+CL-2)+0.8(WLR-2)	-30.1	0.0	33.0	170.6	0.0	33.0
132	1.2(DL+COLL+CL-3)+0.8(WLR-2)	-30.1	0.0	33.0	170.6	0.0	33.0
133	1.2(DL+COLL+CL-4)+0.8(WLR-2)	-30.1	0.0	33.0	170.6	0.0	33.0
134	1.2(DL+COLL+CL-1)+0.8(WLLG-1+BR-WLLG-1)	-1.0	17.5	12.1	199.8	17.5	12.1
135	1.2(DL+COLL+CL-2)+0.8(WLLG-1+BR-WLLG-1)	-1.0	17.5	12.1	199.8	17.5	12.1
136	1.2(DL+COLL+CL-3)+0.8(WLLG-1+BR-WLLG-1)	-1.0	17.5	12.1	199.8	17.5	12.1
137	1.2(DL+COLL+CL-4)+0.8(WLLG-1+BR-WLLG-1)	-1.0	17.5	12.1	199.8	17.5	12.1
138	1.2(DL+COLL+CL-1)+0.8(WLLG-2+BR-WLLG-2)	-14.2	32.6	12.1	186.5	32.6	12.1
139	1.2(DL+COLL+CL-2)+0.8(WLLG-2+BR-WLLG-2)	-14.2	32.6	12.1	186.5	32.6	12.1
140	1.2(DL+COLL+CL-3)+0.8(WLLG-2+BR-WLLG-2)	-14.2	32.6	12.1	186.5	32.6	12.1
141	1.2(DL+COLL+CL-4)+0.8(WLLG-2+BR-WLLG-2)	-14.2	32.6	12.1	186.5	32.6	12.1
142	1.2(DL+COLL+CL-1)+0.8(WLLG-3+BR-WLLG-3)	-0.7	16.8	12.1	200.0	16.8	12.1
143	1.2(DL+COLL+CL-2)+0.8(WLLG-3+BR-WLLG-3)	-0.7	16.8	12.1	200.0	16.8	12.1
144	1.2(DL+COLL+CL-3)+0.8(WLLG-3+BR-WLLG-3)	-0.7	16.8	12.1	200.0	16.8	12.1
145	1.2(DL+COLL+CL-4)+0.8(WLLG-3+BR-WLLG-3)	-0.7	16.8	12.1	200.0	16.8	12.1
146	1.2(DL+COLL+CL-1)+0.8(WLLG-4+BR-WLLG-4)	-14.1	32.2	12.1	186.6	32.2	12.1

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147	1.2(DL+COLL+CL-2)+0.8(WLLG-4+BR-WLLG-4)	-14.1	32.2	12.1	186.6	32.2	12.1
148	1.2(DL+COLL+CL-3)+0.8(WLLG-4+BR-WLLG-4)	-14.1	32.2	12.1	186.6	32.2	12.1
149	1.2(DL+COLL+CL-4)+0.8(WLLG-4+BR-WLLG-4)	-14.1	32.2	12.1	186.6	32.2	12.1
150	1.2(DL+COLL+CL-1)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	66.8	0.0	83.5	267.5	0.0	83.5
151	1.2(DL+COLL+CL-2)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	66.8	0.0	83.5	267.5	0.0	83.5
152	1.2(DL+COLL+CL-3)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	66.8	0.0	83.5	267.5	0.0	83.5
153	1.2(DL+COLL+CL-4)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	66.8	0.0	83.5	267.5	0.0	83.5
154	1.2(DL+COLL+CL-1)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	4.8	0.0	11.4	205.6	0.0	11.4
155	1.2(DL+COLL+CL-2)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	4.8	0.0	11.4	205.6	0.0	11.4
156	1.2(DL+COLL+CL-3)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	4.8	0.0	11.4	205.6	0.0	11.4
157	1.2(DL+COLL+CL-4)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	4.8	0.0	11.4	205.6	0.0	11.4
158	1.2(DL+COLL+CL-1)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	122.5	0.0	126.3	323.2	0.0	126.3
159	1.2(DL+COLL+CL-2)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	122.5	0.0	126.3	323.2	0.0	126.3
160	1.2(DL+COLL+CL-3)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	122.5	0.0	126.3	323.2	0.0	126.3
161	1.2(DL+COLL+CL-4)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	122.5	0.0	126.3	323.2	0.0	126.3
162	1.2(DL+COLL+CL-1)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	-58.5	0.0	62.4	142.2	0.0	62.4
163	1.2(DL+COLL+CL-2)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	-58.5	0.0	62.4	142.2	0.0	62.4
164	1.2(DL+COLL+CL-3)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	-58.5	0.0	62.4	142.2	0.0	62.4
165	1.2(DL+COLL+CL-4)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	-58.5	0.0	62.4	142.2	0.0	62.4
166	1.2(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-0.1	35.0	27.9	200.6	35.0	27.9
167	1.2(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-0.1	35.0	27.9	200.6	35.0	27.9
168	1.2(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-0.1	35.0	27.9	200.6	35.0	27.9
169	1.2(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	-0.1	35.0	27.9	200.6	35.0	27.9
170	1.2(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-26.7	65.2	27.9	174.0	65.2	27.9
171	1.2(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-26.7	65.2	27.9	174.0	65.2	27.9
172	1.2(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-26.7	65.2	27.9	174.0	65.2	27.9
173	1.2(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	-26.7	65.2	27.9	174.0	65.2	27.9
174	1.2(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	0.3	33.7	27.9	201.0	33.7	27.9
175	1.2(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	0.3	33.7	27.9	201.0	33.7	27.9
176	1.2(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	0.3	33.7	27.9	201.0	33.7	27.9
177	1.2(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	0.3	33.7	27.9	201.0	33.7	27.9
178	1.2(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-26.4	64.4	27.9	174.3	64.4	27.9
179	1.2(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-26.4	64.4	27.9	174.3	64.4	27.9
180	1.2(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-26.4	64.4	27.9	174.3	64.4	27.9
181	1.2(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	-26.4	64.4	27.9	174.3	64.4	27.9
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	27.9	0.3	28.1	228.6	0.3	28.1
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	27.9	0.3	28.1	228.6	0.3	28.1
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	27.9	0.3	28.1	228.6	0.3	28.1
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	27.9	0.3	28.1	228.6	0.3	28.1
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	27.7	0.3	27.6	228.4	0.3	27.6
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-)+0.5(RLL+PLL / MLL	27.7	0.3	27.6	228.4	0.3	27.6
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	27.7	0.3	27.6	228.4	0.3	27.6
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	27.7	0.3	27.6	228.4	0.3	27.6
190	0.9(DL+COLL+CL-1)+1.6(WLL-1)	48.7	0.0	64.7	249.4	0.0	64.7
191	0.9(DL+COLL+CL-2)+1.6(WLL-1)	48.7	0.0	64.7	249.4	0.0	64.7
192	0.9(DL+COLL+CL-3)+1.6(WLL-1)	48.7	0.0	64.7	249.4	0.0	64.7
193	0.9(DL+COLL+CL-4)+1.6(WLL-1)	48.7	0.0	64.7	249.4	0.0	64.7
194	0.9(DL+COLL+CL-1)+1.6(WLL-2)	-13.2	0.0	30.2	187.5	0.0	30.2
195	0.9(DL+COLL+CL-2)+1.6(WLL-2)	-13.2	0.0	30.2	187.5	0.0	30.2
196	0.9(DL+COLL+CL-3)+1.6(WLL-2)	-13.2	0.0	30.2	187.5	0.0	30.2
197	0.9(DL+COLL+CL-4)+1.6(WLL-2)	-13.2	0.0	30.2	187.5	0.0	30.2
198	0.9(DL+COLL+CL-1)+1.6(WLR-1)	104.4	0.0	107.5	305.2	0.0	107.5
199	0.9(DL+COLL+CL-2)+1.6(WLR-1)	104.4	0.0	107.5	305.2	0.0	107.5
200	0.9(DL+COLL+CL-3)+1.6(WLR-1)	104.4	0.0	107.5	305.2	0.0	107.5
201	0.9(DL+COLL+CL-4)+1.6(WLR-1)	104.4	0.0	107.5	305.2	0.0	107.5
202	0.9(DL+COLL+CL-1)+1.6(WLR-2)	-76.5	0.0	81.2	124.2	0.0	81.2
203	0.9(DL+COLL+CL-2)+1.6(WLR-2)	-76.5	0.0	81.2	124.2	0.0	81.2
204	0.9(DL+COLL+CL-3)+1.6(WLR-2)	-76.5	0.0	81.2	124.2	0.0	81.2
205	0.9(DL+COLL+CL-4)+1.6(WLR-2)	-76.5	0.0	81.2	124.2	0.0	81.2
206	0.9(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)	-18.2	35.0	9.1	182.6	35.0	9.1
207	0.9(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)	-18.2	35.0	9.1	182.6	35.0	9.1
208	0.9(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)	-18.2	35.0	9.1	182.6	35.0	9.1
209	0.9(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)	-18.2	35.0	9.1	182.6	35.0	9.1
210	0.9(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)	-44.7	65.2	9.1	156.0	65.2	9.1
211	0.9(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)	-44.7	65.2	9.1	156.0	65.2	9.1
212	0.9(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)	-44.7	65.2	9.1	156.0	65.2	9.1
213	0.9(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)	-44.7	65.2	9.1	156.0	65.2	9.1
214	0.9(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)	-17.7	33.7	9.1	183.0	33.7	9.1
215	0.9(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)	-17.7	33.7	9.1	183.0	33.7	9.1
216	0.9(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)	-17.7	33.7	9.1	183.0	33.7	9.1
217	0.9(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)	-17.7	33.7	9.1	183.0	33.7	9.1
218	0.9(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG-4)	-44.5	64.4	9.1	156.2	64.4	9.1
219	0.9(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG-4)	-44.5	64.4	9.1	156.2	64.4	9.1
220	0.9(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG-4)	-44.5	64.4	9.1	156.2	64.4	9.1
221	0.9(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG-4)	-44.5	64.4	9.1	156.2	64.4	9.1
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')	9.8	0.3	9.4	210.5	0.3	9.4
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')	9.8	0.3	9.4	210.5	0.3	9.4
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')	9.8	0.3	9.4	210.5	0.3	9.4
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')	9.8	0.3	9.4	210.5	0.3	9.4
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')	9.7	0.3	8.8	210.4	0.3	8.8
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-')	9.7	0.3	8.8	210.4	0.3	8.8
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')	9.7	0.3	8.8	210.4	0.3	8.8
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')	9.7	0.3	8.8	210.4	0.3	8.8

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Allowable stress design

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
301	1.0(DL+COLL)	10.8312	0	0.0	-3.7	-10.1	211.5	0.0	0.0	-3.7	-10.1
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL	41.7072	0	0.0	-17.3	-46.7	242.4	0.0	0.0	-17.3	-46.7
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL / MLL	41.7072	0	0.0	-17.3	-46.7	242.4	0.0	0.0	-17.3	-46.7
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL / MLL	41.7072	0	0.0	-17.3	-46.7	242.4	0.0	0.0	-17.3	-46.7
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL / MLL	41.7072	0	0.0	-17.3	-46.7	242.4	0.0	0.0	-17.3	-46.7
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL / MLL	39.0696	0	0.0	-13.5	-36.5	239.8	0.0	0.0	-13.5	-36.5
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL / MLL	39.0696	0	0.0	-13.5	-36.5	239.8	0.0	0.0	-13.5	-36.5
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL / MLL	39.0696	0	0.0	-13.5	-36.5	239.8	0.0	0.0	-13.5	-36.5
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL / MLL	39.0696	0	0.0	-13.5	-36.5	239.8	0.0	0.0	-13.5	-36.5
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL+PLL	33.99	0	0.0	-13.9	-37.6	234.7	0.0	0.0	-13.9	-37.6
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL+PLL	33.99	0	0.0	-13.9	-37.6	234.7	0.0	0.0	-13.9	-37.6
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL+PLL	33.99	0	0.0	-13.9	-37.6	234.7	0.0	0.0	-13.9	-37.6
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL+PLL	33.99	0	0.0	-13.9	-37.6	234.7	0.0	0.0	-13.9	-37.6
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL+PLL	32.01	0	0.0	-11.1	-29.9	232.7	0.0	0.0	-11.1	-29.9
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL+PLL	32.01	0	0.0	-11.1	-29.9	232.7	0.0	0.0	-11.1	-29.9
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL+PLL	32.01	0	0.0	-11.1	-29.9	232.7	0.0	0.0	-11.1	-29.9
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL+PLL	32.01	0	0.0	-11.1	-29.9	232.7	0.0	0.0	-11.1	-29.9
318	1.0(DL+COLL+CL-1)	10.8312	0	0.0	-3.7	-10.1	211.5	0.0	0.0	-3.7	-10.1
319	1.0(DL+COLL+CL-2)	10.8312	0	0.0	-3.7	-10.1	211.5	0.0	0.0	-3.7	-10.1
320	1.0(DL+COLL+CL-3)	10.8312	0	0.0	-3.7	-10.1	211.5	0.0	0.0	-3.7	-10.1
321	1.0(DL+COLL+CL-4)	10.8312	0	0.0	-3.7	-10.1	211.5	0.0	0.0	-3.7	-10.1
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	35.1996	0	0.0	-16.6	-44.9	235.9	0.0	0.0	-16.6	-44.9
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	35.1996	0	0.0	-16.6	-44.9	235.9	0.0	0.0	-16.6	-44.9
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	35.1996	0	0.0	-16.6	-44.9	235.9	0.0	0.0	-16.6	-44.9
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	35.1996	0	0.0	-16.6	-44.9	235.9	0.0	0.0	-16.6	-44.9
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	-3.498	0	0.0	5.3	14.4	197.2	0.0	0.0	5.3	14.4
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	-3.498	0	0.0	5.3	14.4	197.2	0.0	0.0	5.3	14.4
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	-3.498	0	0.0	5.3	14.4	197.2	0.0	0.0	5.3	14.4
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	-3.498	0	0.0	5.3	14.4	197.2	0.0	0.0	5.3	14.4
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	70.0152	0	0.0	-26.5	-71.6	270.7	0.0	0.0	-26.5	-71.6
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	70.0152	0	0.0	-26.5	-71.6	270.7	0.0	0.0	-26.5	-71.6
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	70.0152	0	0.0	-26.5	-71.6	270.7	0.0	0.0	-26.5	-71.6
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	70.0152	0	0.0	-26.5	-71.6	270.7	0.0	0.0	-26.5	-71.6
334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	-43.0956	0	0.0	17.2	46.3	157.6	0.0	0.0	17.2	46.3
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	-43.0956	0	0.0	17.2	46.3	157.6	0.0	0.0	17.2	46.3
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	-43.0956	0	0.0	17.2	46.3	157.6	0.0	0.0	17.2	46.3
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	-43.0956	0	0.0	17.2	46.3	157.6	0.0	0.0	17.2	46.3
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	-6.6072	8.0988	8.1	-3.7	-10.1	194.1	8.1	8.1	-3.7	-10.1
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	-6.6072	8.0988	8.1	-3.7	-10.1	194.1	8.1	8.1	-3.7	-10.1
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	-6.6072	8.0988	8.1	-3.7	-10.1	194.1	8.1	8.1	-3.7	-10.1
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	-6.6072	8.0988	8.1	-3.7	-10.1	194.1	8.1	8.1	-3.7	-10.1
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	-23.2272	15.0924	15.1	-3.7	-10.1	177.5	15.1	15.1	-3.7	-10.1
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	-23.2272	15.0924	15.1	-3.7	-10.1	177.5	15.1	15.1	-3.7	-10.1
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	-23.2272	15.0924	15.1	-3.7	-10.1	177.5	15.1	15.1	-3.7	-10.1
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	-23.2272	15.0924	15.1	-3.7	-10.1	177.5	15.1	15.1	-3.7	-10.1
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-6.3396	7.8	7.8	-3.7	-10.1	194.4	7.8	7.8	-3.7	-10.1
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	-6.3396	7.8	7.8	-3.7	-10.1	194.4	7.8	7.8	-3.7	-10.1
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	-6.3396	7.8	7.8	-3.7	-10.1	194.4	7.8	7.8	-3.7	-10.1
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	-6.3396	7.8	7.8	-3.7	-10.1	194.4	7.8	7.8	-3.7	-10.1
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-23.0532	14.9016	14.9	-3.7	-10.1	177.7	14.9	14.9	-3.7	-10.1
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	-23.0532	14.9016	14.9	-3.7	-10.1	177.7	14.9	14.9	-3.7	-10.1
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	-23.0532	14.9016	14.9	-3.7	-10.1	177.7	14.9	14.9	-3.7	-10.1
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	-23.0532	14.9016	14.9	-3.7	-10.1	177.7	14.9	14.9	-3.7	-10.1
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(RLL+	51.28	0	0.0	-22.2	-59.8	252.0	0.0	0.0	-22.2	-59.8
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(RLL+	51.28	0	0.0	-22.2	-59.8	252.0	0.0	0.0	-22.2	-59.8
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(RLL+	51.28	0	0.0	-22.2	-59.8	252.0	0.0	0.0	-22.2	-59.8
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(RLL+	51.28	0	0.0	-22.2	-59.8	252.0	0.0	0.0	-22.2	-59.8
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(RLL+	22.25	0	0.0	-5.7	-15.3	223.0	0.0	0.0	-5.7	-15.3
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(RLL+	22.25	0	0.0	-5.7	-15.3	223.0	0.0	0.0	-5.7	-15.3
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(RLL+	22.25	0	0.0	-5.7	-15.3	223.0	0.0	0.0	-5.7	-15.3
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(RLL+	22.25	0	0.0	-5.7	-15.3	223.0	0.0	0.0	-5.7	-15.3
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(RLL+	77.39	0	0.0	-29.6	-79.9	278.1	0.0	0.0	-29.6	-79.9
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(RLL+	77.39	0	0.0	-29.6	-79.9	278.1	0.0	0.0	-29.6	-79.9
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(RLL+	77.39	0	0.0	-29.6	-79.9	278.1	0.0	0.0	-29.6	-79.9
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(RLL+	77.39	0	0.0	-29.6	-79.9	278.1	0.0	0.0	-29.6	-79.9
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(RLL+	-7.45	0	0.0	3.2	8.6	193.3	0.0	0.0	3.2	8.6
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(RLL+	-7.45	0	0.0	3.2	8.6	193.3	0.0	0.0	3.2	8.6
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(RLL+	-7.45	0	0.0	3.2	8.6	193.3	0.0	0.0	3.2	8.6
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(RLL+	-7.45	0	0.0	3.2	8.6	193.3	0.0	0.0	3.2	8.6
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-WLLG	19.92	6.0741	6.1	-12.5	-33.7	220.6	6.1	6.1	-12.5	-33.7
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-WLLG	19.92	6.0741	6.1	-12.5	-33.7	220.6	6.1	6.1	-12.5	-33.7
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-WLLG	19.92	6.0741	6.1	-12.5	-33.7	220.6	6.1	6.1	-12.5	-33.7
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-WLLG	19.92	6.0741	6.1	-12.5	-33.7	220.6	6.1	6.1	-12.5	-33.7
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-WLLG	7.46	11.3193	11.3	-12.5	-33.7	208.2	11.3	11.3	-12.5	-33.7
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-WLLG	7.46	11.3193	11.3	-12.5	-33.7	208.2	11.3	11.3	-12.5	-33.7
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-WLLG	7.46	11.3193	11.3	-12.5	-33.7	208.2	11.3	11.3	-12.5	-33.7
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-WLLG	7.46	11.3193	11.3	-12.5	-33.7	208.2	11.3	11.3	-12.5	-33.7
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-WLLG	20.12	5.85	5.9	-12.5	-33.7	220.8	5.9	5.9	-12.5	-33.7
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-WLLG	20.12	5.85	5.9	-12.5	-33.7	220.8	5.9	5.9	-12.5	-33.7
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-WLLG	20.12	5.85	5.9	-12.5	-33.7	220.8	5.9	5.9	-12.5	-33.7
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-WLLG	20.12	5.85	5.9	-12.5	-33.7	220.8	5.9	5.9	-12.5	-33.7
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-WLLG	7.59	11.1762	11.2	-12.5	-33.7	208.3	11.2	11.2	-12.5	-33.7
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-WLLG	7.59	11.1762	11.2	-12.5	-33.7	208.3	11.2	11.2	-12.5	-33.7
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-WLLG	7.59	11.1762	11.2	-12.5	-33.7	208.3	11.2	11.2	-12.5	-33.7
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-WLLG	7.59	11.1762	11.2	-12.5	-33.7	208.3	11.2	11.2	-12.5	-33.7
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+')	10.89	-0.0738	-0.1	-3.8	-10.2	211.6	-0.1	-0.1	-3.8	-10.2
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+")	10.89	-0.0738	-0.1	-3.8	-10.2	211.6	-0.1	-0.1	-3.8	-10.2
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+')	10.89	-0.0738	-0.1	-3.8	-10.2	211.6	-0.1	-0.1	-3.8	-10.2
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+')	10.89	-0.0738	-0.1	-3.8	-10.2	211.6	-0.1	-0.1	-3.8	-10.2

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390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	10.77	0.0738	0.1	-3.7	-10.0	211.5	0.1	0.1	-3.7	-10.0
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	10.77	0.0738	0.1	-3.7	-10.0	211.5	0.1	0.1	-3.7	-10.0
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	10.77	0.0738	0.1	-3.7	-10.0	211.5	0.1	0.1	-3.7	-10.0
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	10.77	0.0738	0.1	-3.7	-10.0	211.5	0.1	0.1	-3.7	-10.0
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'+')+0	33.04	-0.05166	-0.1	-12.5	-33.8	233.8	-0.1	-0.1	-12.5	-33.8
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'+')+0	33.04	-0.05166	-0.1	-12.5	-33.8	233.8	-0.1	-0.1	-12.5	-33.8
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'+')+0	33.04	-0.05166	-0.1	-12.5	-33.8	233.8	-0.1	-0.1	-12.5	-33.8
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'+')+0	33.04	-0.05166	-0.1	-12.5	-33.8	233.8	-0.1	-0.1	-12.5	-33.8
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-')+0.7	32.96	0.05166	0.1	-12.4	-33.7	233.7	0.1	0.1	-12.4	-33.7
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-')+0.7	32.96	0.05166	0.1	-12.4	-33.7	233.7	0.1	0.1	-12.4	-33.7
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-')+0.7	32.96	0.05166	0.1	-12.4	-33.7	233.7	0.1	0.1	-12.4	-33.7
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-')+0.7	32.96	0.05166	0.1	-12.4	-33.7	233.7	0.1	0.1	-12.4	-33.7
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	30.87	0	0.0	-15.1	-40.8	231.6	0.0	0.0	-15.1	-40.8
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	30.87	0	0.0	-15.1	-40.8	231.6	0.0	0.0	-15.1	-40.8
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	30.87	0	0.0	-15.1	-40.8	231.6	0.0	0.0	-15.1	-40.8
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	30.87	0	0.0	-15.1	-40.8	231.6	0.0	0.0	-15.1	-40.8
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	-7.83048	0	0.0	6.8	18.5	192.9	0.0	0.0	6.8	18.5
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	-7.83048	0	0.0	6.8	18.5	192.9	0.0	0.0	6.8	18.5
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	-7.83048	0	0.0	6.8	18.5	192.9	0.0	0.0	6.8	18.5
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	-7.83048	0	0.0	6.8	18.5	192.9	0.0	0.0	6.8	18.5
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	65.68272	0	0.0	-25.0	-67.6	266.4	0.0	0.0	-25.0	-67.6
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	65.68272	0	0.0	-25.0	-67.6	266.4	0.0	0.0	-25.0	-67.6
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	65.68272	0	0.0	-25.0	-67.6	266.4	0.0	0.0	-25.0	-67.6
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	65.68272	0	0.0	-25.0	-67.6	266.4	0.0	0.0	-25.0	-67.6
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	-47.42808	0	0.0	18.7	50.4	153.3	0.0	0.0	18.7	50.4
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	-47.42808	0	0.0	18.7	50.4	153.3	0.0	0.0	18.7	50.4
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	-47.42808	0	0.0	18.7	50.4	153.3	0.0	0.0	18.7	50.4
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	-47.42808	0	0.0	18.7	50.4	153.3	0.0	0.0	18.7	50.4
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	-10.93968	8.0988	8.1	-2.2	-6.1	189.8	8.1	8.1	-2.2	-6.1
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	-10.93968	8.0988	8.1	-2.2	-6.1	189.8	8.1	8.1	-2.2	-6.1
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	-10.93968	8.0988	8.1	-2.2	-6.1	189.8	8.1	8.1	-2.2	-6.1
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	-10.93968	8.0988	8.1	-2.2	-6.1	189.8	8.1	8.1	-2.2	-6.1
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	-27.55968	15.0924	15.1	-2.2	-6.1	173.2	15.1	15.1	-2.2	-6.1
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	-27.55968	15.0924	15.1	-2.2	-6.1	173.2	15.1	15.1	-2.2	-6.1
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	-27.55968	15.0924	15.1	-2.2	-6.1	173.2	15.1	15.1	-2.2	-6.1
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	-27.55968	15.0924	15.1	-2.2	-6.1	173.2	15.1	15.1	-2.2	-6.1
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-10.67208	7.8	7.8	-2.2	-6.1	190.0	7.8	7.8	-2.2	-6.1
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	-10.67208	7.8	7.8	-2.2	-6.1	190.0	7.8	7.8	-2.2	-6.1
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	-10.67208	7.8	7.8	-2.2	-6.1	190.0	7.8	7.8	-2.2	-6.1
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	-10.67208	7.8	7.8	-2.2	-6.1	190.0	7.8	7.8	-2.2	-6.1
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-WLLG-	-27.38568	14.9016	14.9	-2.2	-6.1	173.3	14.9	14.9	-2.2	-6.1
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	-27.38568	14.9016	14.9	-2.2	-6.1	173.3	14.9	14.9	-2.2	-6.1
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	-27.38568	14.9016	14.9	-2.2	-6.1	173.3	14.9	14.9	-2.2	-6.1
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	-27.38568	14.9016	14.9	-2.2	-6.1	173.3	14.9	14.9	-2.2	-6.1
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	6.55	-0.06888	-0.1	-2.3	-6.1	207.3	-0.1	-0.1	-2.3	-6.1
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	6.55	-0.06888	-0.1	-2.3	-6.1	207.3	-0.1	-0.1	-2.3	-6.1
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	6.55	-0.06888	-0.1	-2.3	-6.1	207.3	-0.1	-0.1	-2.3	-6.1
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	6.55	-0.06888	-0.1	-2.3	-6.1	207.3	-0.1	-0.1	-2.3	-6.1
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	21.17	-0.06888	-0.1	-10.0	-27.0	221.9	-0.1	-0.1	-10.0	-27.0
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	-2.04	-0.06888	-0.1	3.1	8.6	198.7	-0.1	-0.1	3.1	8.6
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	42.06	-0.06888	-0.1	-16.0	-43.0	242.8	-0.1	-0.1	-16.0	-43.0
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	-25.80	-0.06888	-0.1	10.2	27.7	174.9	-0.1	-0.1	10.2	27.7

Check for overturning and sliding:

Load Case	Load Combination	Load at bottom of foundation					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
301	1.0(DL+COLL)	211.54	0.00	0.00	3.74	10.10	NA	20.74	NA	20.36
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL /	242.42	0.00	0.00	17.31	46.73	NA	5.14	NA	5.04
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL /	242.42	0.00	0.00	17.31	46.73	NA	5.14	NA	5.04
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL /	242.42	0.00	0.00	17.31	46.73	NA	5.14	NA	5.04
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL /	242.42	0.00	0.00	17.31	46.73	NA	5.14	NA	5.04
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL /	239.78	0.00	0.00	13.53	36.53	NA	6.50	NA	6.38
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL /	239.78	0.00	0.00	13.53	36.53	NA	6.50	NA	6.38
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL /	239.78	0.00	0.00	13.53	36.53	NA	6.50	NA	6.38
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL /	239.78	0.00	0.00	13.53	36.53	NA	6.50	NA	6.38
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL	234.70	0.00	0.00	13.91	37.57	NA	6.18	NA	6.07
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL	234.70	0.00	0.00	13.91	37.57	NA	6.18	NA	6.07
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL	234.70	0.00	0.00	13.91	37.57	NA	6.18	NA	6.07
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL	234.70	0.00	0.00	13.91	37.57	NA	6.18	NA	6.07
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL	232.72	0.00	0.00	11.08	29.92	NA	7.70	NA	7.56
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL	232.72	0.00	0.00	11.08	29.92	NA	7.70	NA	7.56
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL	232.72	0.00	0.00	11.08	29.92	NA	7.70	NA	7.56
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL	232.72	0.00	0.00	11.08	29.92	NA	7.70	NA	7.56
318	1.0(DL+COLL+CL-1)	211.54	0.00	0.00	3.74	10.10	NA	20.74	NA	20.36
319	1.0(DL+COLL+CL-2)	211.54	0.00	0.00	3.74	10.10	NA	20.74	NA	20.36
320	1.0(DL+COLL+CL-3)	211.54	0.00	0.00	3.74	10.10	NA	20.74	NA	20.36
321	1.0(DL+COLL+CL-4)	211.54	0.00	0.00	3.74	10.10	NA	20.74	NA	20.36
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	235.91	0.00	0.00	16.62	44.87	NA	5.20	NA	5.11
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	235.91	0.00	0.00	16.62	44.87	NA	5.20	NA	5.11
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	235.91	0.00	0.00	16.62	44.87	NA	5.20	NA	5.11
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	235.91	0.00	0.00	16.62	44.87	NA	5.20	NA	5.11
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	197.21	0.00	0.00	5.35	14.44	NA	13.52	NA	13.27
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	197.21	0.00	0.00	5.35	14.44	NA	13.52	NA	13.27
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	197.21	0.00	0.00	5.35	14.44	NA	13.52	NA	13.27
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	197.21	0.00	0.00	5.35	14.44	NA	13.52	NA	13.27
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	270.73	0.00	0.00	26.53	71.63	NA	3.74	NA	3.67
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	270.73	0.00	0.00	26.53	71.63	NA	3.74	NA	3.67
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	270.73	0.00	0.00	26.53	71.63	NA	3.74	NA	3.67
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	270.73	0.00	0.00	26.53	71.63	NA	3.74	NA	3.67

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334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	157.61	0.00	0.00	17.16	46.34	NA	3.37	NA	3.31
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	157.61	0.00	0.00	17.16	46.34	NA	3.37	NA	3.31
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	157.61	0.00	0.00	17.16	46.34	NA	3.37	NA	3.31
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	157.61	0.00	0.00	17.16	46.34	NA	3.37	NA	3.31
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	194.10	8.10	8.10	3.74	10.10	23.73	19.03	9.59	18.68
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	194.10	8.10	8.10	3.74	10.10	23.73	19.03	9.59	18.68
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	194.10	8.10	8.10	3.74	10.10	23.73	19.03	9.59	18.68
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	194.10	8.10	8.10	3.74	10.10	23.73	19.03	9.59	18.68
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	177.48	15.09	15.09	3.74	10.10	11.64	17.40	4.70	17.08
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	177.48	15.09	15.09	3.74	10.10	11.64	17.40	4.70	17.08
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	177.48	15.09	15.09	3.74	10.10	11.64	17.40	4.70	17.08
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	177.48	15.09	15.09	3.74	10.10	11.64	17.40	4.70	17.08
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	194.37	7.80	7.80	3.74	10.10	24.67	19.05	9.97	18.71
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	194.37	7.80	7.80	3.74	10.10	24.67	19.05	9.97	18.71
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	194.37	7.80	7.80	3.74	10.10	24.67	19.05	9.97	18.71
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	194.37	7.80	7.80	3.74	10.10	24.67	19.05	9.97	18.71
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	177.66	14.90	14.90	3.74	10.10	11.80	17.42	4.77	17.10
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	177.66	14.90	14.90	3.74	10.10	11.80	17.42	4.77	17.10
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	177.66	14.90	14.90	3.74	10.10	11.80	17.42	4.77	17.10
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	177.66	14.90	14.90	3.74	10.10	11.80	17.42	4.77	17.10
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(F	251.99	0.00	0.00	22.16	59.82	NA	4.17	NA	4.09
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(F	251.99	0.00	0.00	22.16	59.82	NA	4.17	NA	4.09
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(F	251.99	0.00	0.00	22.16	59.82	NA	4.17	NA	4.09
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(F	251.99	0.00	0.00	22.16	59.82	NA	4.17	NA	4.09
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(F	222.96	0.00	0.00	5.68	15.34	NA	14.39	NA	14.13
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(F	222.96	0.00	0.00	5.68	15.34	NA	14.39	NA	14.13
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(F	222.96	0.00	0.00	5.68	15.34	NA	14.39	NA	14.13
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(F	222.96	0.00	0.00	5.68	15.34	NA	14.39	NA	14.13
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(	278.10	0.00	0.00	29.59	79.89	NA	3.45	NA	3.38
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(	278.10	0.00	0.00	29.59	79.89	NA	3.45	NA	3.38
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(	278.10	0.00	0.00	29.59	79.89	NA	3.45	NA	3.38
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(	278.10	0.00	0.00	29.59	79.89	NA	3.45	NA	3.38
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(	193.26	0.00	0.00	3.18	8.59	NA	22.28	NA	21.87
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(	193.26	0.00	0.00	3.18	8.59	NA	22.28	NA	21.87
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(	193.26	0.00	0.00	3.18	8.59	NA	22.28	NA	21.87
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(	193.26	0.00	0.00	3.18	8.59	NA	22.28	NA	21.87
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-V	220.63	6.07	6.07	12.50	33.75	35.96	6.47	14.53	6.36
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-V	220.63	6.07	6.07	12.50	33.75	35.96	6.47	14.53	6.36
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-V	220.63	6.07	6.07	12.50	33.75	35.96	6.47	14.53	6.36
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-V	220.63	6.07	6.07	12.50	33.75	35.96	6.47	14.53	6.36
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-V	208.17	11.32	11.32	12.50	33.75	18.21	6.11	7.36	6.00
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-V	208.17	11.32	11.32	12.50	33.75	18.21	6.11	7.36	6.00
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-V	208.17	11.32	11.32	12.50	33.75	18.21	6.11	7.36	6.00
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-V	208.17	11.32	11.32	12.50	33.75	18.21	6.11	7.36	6.00
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-V	220.83	5.85	5.85	12.50	33.75	37.37	6.48	15.10	6.36
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-V	220.83	5.85	5.85	12.50	33.75	37.37	6.48	15.10	6.36
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-V	220.83	5.85	5.85	12.50	33.75	37.37	6.48	15.10	6.36
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-V	220.83	5.85	5.85	12.50	33.75	37.37	6.48	15.10	6.36
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-V	208.30	11.18	11.18	12.50	33.75	18.45	6.11	7.45	6.00
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-V	208.30	11.18	11.18	12.50	33.75	18.45	6.11	7.45	6.00
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-V	208.30	11.18	11.18	12.50	33.75	18.45	6.11	7.45	6.00
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-V	208.30	11.18	11.18	12.50	33.75	18.45	6.11	7.45	6.00
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+'	211.60	-0.07	-0.07	3.81	10.17	2838.53	20.59	1146.88	19.97
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+'	211.60	-0.07	-0.07	3.81	10.17	2838.53	20.59	1146.88	19.97
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+'	211.60	-0.07	-0.07	3.81	10.17	2838.53	20.59	1146.88	19.97
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+'	211.60	-0.07	-0.07	3.81	10.17	2838.53	20.59	1146.88	19.97
390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	211.48	0.07	0.07	3.67	10.03	2836.96	20.88	1146.25	20.76
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	211.48	0.07	0.07	3.67	10.03	2836.96	20.88	1146.25	20.76
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	211.48	0.07	0.07	3.67	10.03	2836.96	20.88	1146.25	20.76
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	211.48	0.07	0.07	3.67	10.03	2836.96	20.88	1146.25	20.76
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'	233.75	-0.05	-0.05	12.55	33.80	4479.53	6.85	1809.91	6.71
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'	233.75	-0.05	-0.05	12.55	33.80	4479.53	6.85	1809.91	6.71
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'	233.75	-0.05	-0.05	12.55	33.80	4479.53	6.85	1809.91	6.71
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'	233.75	-0.05	-0.05	12.55	33.80	4479.53	6.85	1809.91	6.71
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-	233.67	0.05	0.05	12.45	33.69	4477.96	6.87	1809.28	6.76
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-	233.67	0.05	0.05	12.45	33.69	4477.96	6.87	1809.28	6.76
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-	233.67	0.05	0.05	12.45	33.69	4477.96	6.87	1809.28	6.76
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-'	233.67	0.05	0.05	12.45	33.69	4477.96	6.87	1809.28	6.76
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	231.58	0.00	0.00	15.12	40.83	NA	5.61	NA	5.51
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	231.58	0.00	0.00	15.12	40.83	NA	5.61	NA	5.51
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	231.58	0.00	0.00	15.12	40.83	NA	5.61	NA	5.51
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	231.58	0.00	0.00	15.12	40.83	NA	5.61	NA	5.51
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	192.88	0.00	0.00	6.85	18.48	NA	10.33	NA	10.14
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	192.88	0.00	0.00	6.85	18.48	NA	10.33	NA	10.14
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	192.88	0.00	0.00	6.85	18.48	NA	10.33	NA	10.14
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	192.88	0.00	0.00	6.85	18.48	NA	10.33	NA	10.14
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	266.39	0.00	0.00	25.03	67.59	NA	3.90	NA	3.83
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	266.39	0.00	0.00	25.03	67.59	NA	3.90	NA	3.83
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	266.39	0.00	0.00	25.03	67.59	NA	3.90	NA	3.83
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	266.39	0.00	0.00	25.03	67.59	NA	3.90	NA	3.83
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	153.28	0.00	0.00	18.66	50.38	NA	3.01	NA	2.96
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	153.28	0.00	0.00	18.66	50.38	NA	3.01	NA	2.96
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	153.28	0.00	0.00	18.66	50.38	NA	3.01	NA	2.96
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	153.28	0.00	0.00	18.66	50.38	NA	3.01	NA	2.96
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	189.77	8.10	8.10	2.24	6.06	23.20	31.00	9.37	30.44
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	189.77	8.10	8.10	2.24	6.06	23.20	31.00	9.37	30.44
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	189.77	8.10	8.10	2.24	6.06	23.20	31.00	9.37	30.44
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	189.77	8.10	8.10	2.24	6.06	23.20	31.00	9.37	30.44
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	173.15	15.09	15.09	2.24	6.06	11.36	28.29	4.59	27.78
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	173.15	15.09	15.09	2.24	6.06	11.36	28.29	4.59	27.78
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	173.15	15.09	15.09	2.24	6.06	11.36	28.29	4.59	27.78
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	173.15	15.09	15.09	2.24	6.06	11.36	28.29	4.59	27.78
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	190.04	7.80	7.80	2.24	6.06	24.12	31.05	9.75	30.48
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	190.04	7.80	7.80	2.24	6.06	24.12	31.05	9.75	30.48
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	190.04	7.80	7.80	2.24	6.06	24.12	31.05	9.75	30.48
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	190.04	7.80	7.80	2.24	6.06	24.12	31.05	9.75	30.48
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-W	173.32	14.90	14.90	2.24	6.06	11.51	28.32	4.65	27.80
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	173.32	14.90	14.90	2.24	6.06	11.51	28.32	4.65	27.80
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	173.32	14.90	14.90	2.24	6.06	11.51	28.32	4.65	27.80
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	173.32	14.90	14.90	2.24	6.06	11.51	28.32	4.65	27.80
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	207.26	-0.07	-0.07	2.31	6.13	2978.96	33.48	1203.62	32.26
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	207.26	-0.07	-0.07	2.31	6.13	2978.96	33.48	1203.62	32.26
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	207.26	-0.07	-0.07	2.31	6.13	2978.96	33.48	1203.62	32.26
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	207.26	-0.07	-0.07	2.31	6.13	2978.96	33.48	1203.62	32.26
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	221.88	-0.07	-0.07	10.04	26.99	3189.10	8.14	1288.53	7.96
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	198.67	-0.07	-0.07	3.14	8.60	2855.39	22.88	1153.69	22.77
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	242.77	-0.07	-0.07	15.99	43.05	3489.34	5.58	1409.84	5.47
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	174.91	-0.07	-0.07	10.23	27.74	2513.91	6.24	1015.72	6.16
						Min.=	**11.36**	**3.01**	**4.59**	**2.96**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F4

Load Combination & Calculation of Base pressure

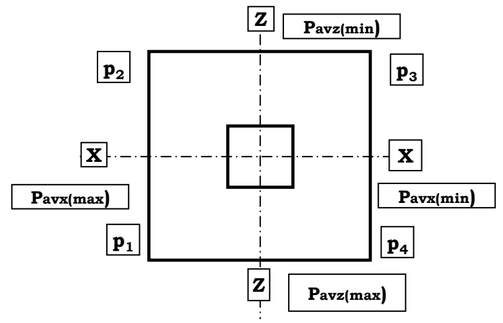
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 436 kN/m^2 For WL/SL case = 536 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
301	1.0(DL+COLL)	211.54	0.00	-10.10	43.71	0.00	5.69	49.40		38.02	49.40	38.02	13.40	43.71	43.71	49.40	38.02	O.K
302	1.0(DL+COLL+CL-1+TL1)+1.(	242.42	0.00	-46.73	50.09	0.00	26.33	76.42		23.76	76.42	23.76	40.42	50.09	50.09	76.42	23.76	O.K
303	1.0(DL+COLL+CL-2+TL1)+1.(	242.42	0.00	-46.73	50.09	0.00	26.33	76.42		23.76	76.42	23.76	40.42	50.09	50.09	76.42	23.76	O.K
304	1.0(DL+COLL+CL-3+TL1)+1.(	242.42	0.00	-46.73	50.09	0.00	26.33	76.42		23.76	76.42	23.76	40.42	50.09	50.09	76.42	23.76	O.K
305	1.0(DL+COLL+CL-4+TL1)+1.(	242.42	0.00	-46.73	50.09	0.00	26.33	76.42		23.76	76.42	23.76	40.42	50.09	50.09	76.42	23.76	O.K
306	1.0(DL+COLL+CL-1+TL2)+1.(	239.78	0.00	-36.53	49.54	0.00	20.58	70.12		28.96	70.12	28.96	34.12	49.54	49.54	70.12	28.96	O.K
307	1.0(DL+COLL+CL-2+TL2)+1.(	239.78	0.00	-36.53	49.54	0.00	20.58	70.12		28.96	70.12	28.96	34.12	49.54	49.54	70.12	28.96	O.K
308	1.0(DL+COLL+CL-3+TL2)+1.(	239.78	0.00	-36.53	49.54	0.00	20.58	70.12		28.96	70.12	28.96	34.12	49.54	49.54	70.12	28.96	O.K
309	1.0(DL+COLL+CL-4+TL2)+1.(	239.78	0.00	-36.53	49.54	0.00	20.58	70.12		28.96	70.12	28.96	34.12	49.54	49.54	70.12	28.96	O.K
310	1.0(DL+COLL+CL-1)+0.75(TL	234.70	0.00	-37.57	48.49	0.00	21.17	69.66		27.32	69.66	27.32	33.66	48.49	48.49	69.66	27.32	O.K
311	1.0(DL+COLL+CL-2)+0.75(TL	234.70	0.00	-37.57	48.49	0.00	21.17	69.66		27.32	69.66	27.32	33.66	48.49	48.49	69.66	27.32	O.K
312	1.0(DL+COLL+CL-3)+0.75(TL	234.70	0.00	-37.57	48.49	0.00	21.17	69.66		27.32	69.66	27.32	33.66	48.49	48.49	69.66	27.32	O.K
313	1.0(DL+COLL+CL-4)+0.75(TL	234.70	0.00	-37.57	48.49	0.00	21.17	69.66		27.32	69.66	27.32	33.66	48.49	48.49	69.66	27.32	O.K
314	1.0(DL+COLL+CL-1)+0.75(TL	232.72	0.00	-29.92	48.08	0.00	16.86	64.94		31.22	64.94	31.22	28.94	48.08	48.08	64.94	31.22	O.K
315	1.0(DL+COLL+CL-2)+0.75(TL	232.72	0.00	-29.92	48.08	0.00	16.86	64.94		31.22	64.94	31.22	28.94	48.08	48.08	64.94	31.22	O.K
316	1.0(DL+COLL+CL-3)+0.75(TL	232.72	0.00	-29.92	48.08	0.00	16.86	64.94		31.22	64.94	31.22	28.94	48.08	48.08	64.94	31.22	O.K

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317	1.0(DL+COLL+CL-4)+0.75(TL	232.72	0.00	-29.92	48.08	0.00	16.86	64.94		31.22	64.94	31.22	28.94	48.08	48.08	64.94	31.22	O.K
318	1.0(DL+COLL+CL-1)	211.54	0.00	-10.10	43.71	0.00	5.69	49.40		38.02	49.40	38.02	13.40	43.71	43.71	49.40	38.02	O.K
319	1.0(DL+COLL+CL-2)	211.54	0.00	-10.10	43.71	0.00	5.69	49.40		38.02	49.40	38.02	13.40	43.71	43.71	49.40	38.02	O.K
320	1.0(DL+COLL+CL-3)	211.54	0.00	-10.10	43.71	0.00	5.69	49.40		38.02	49.40	38.02	13.40	43.71	43.71	49.40	38.02	O.K
321	1.0(DL+COLL+CL-4)	211.54	0.00	-10.10	43.71	0.00	5.69	49.40		38.02	49.40	38.02	13.40	43.71	43.71	49.40	38.02	O.K
322	1.0(DL+COLL+CL-1)+1.0(WLI	235.91	0.00	-44.87	48.74	0.00	25.28	74.03		23.46	74.03	23.46	38.03	48.75	48.75	74.03	23.46	O.K
323	1.0(DL+COLL+CL-2)+1.0(WLI	235.91	0.00	-44.87	48.74	0.00	25.28	74.03		23.46	74.03	23.46	38.03	48.75	48.75	74.03	23.46	O.K
324	1.0(DL+COLL+CL-3)+1.0(WLI	235.91	0.00	-44.87	48.74	0.00	25.28	74.03		23.46	74.03	23.46	38.03	48.75	48.75	74.03	23.46	O.K
325	1.0(DL+COLL+CL-4)+1.0(WLI	235.91	0.00	-44.87	48.74	0.00	25.28	74.03		23.46	74.03	23.46	38.03	48.75	48.75	74.03	23.46	O.K
326	1.0(DL+COLL+CL-1)+1.0(WLI	197.21	0.00	14.44	40.75	0.00	8.14	48.89		32.61	48.89	32.61	12.89	40.75	40.75	48.89	32.61	O.K
327	1.0(DL+COLL+CL-2)+1.0(WLI	197.21	0.00	14.44	40.75	0.00	8.14	48.89		32.61	48.89	32.61	12.89	40.75	40.75	48.89	32.61	O.K
328	1.0(DL+COLL+CL-3)+1.0(WLI	197.21	0.00	14.44	40.75	0.00	8.14	48.89		32.61	48.89	32.61	12.89	40.75	40.75	48.89	32.61	O.K
329	1.0(DL+COLL+CL-4)+1.0(WLI	197.21	0.00	14.44	40.75	0.00	8.14	48.89		32.61	48.89	32.61	12.89	40.75	40.75	48.89	32.61	O.K
330	1.0(DL+COLL+CL-1)+1.0(WLI	270.73	0.00	-71.63	55.93	0.00	40.36	96.30		15.57	96.30	15.57	60.30	55.94	55.94	96.30	15.57	O.K
331	1.0(DL+COLL+CL-2)+1.0(WLI	270.73	0.00	-71.63	55.93	0.00	40.36	96.30		15.57	96.30	15.57	60.30	55.94	55.94	96.30	15.57	O.K
332	1.0(DL+COLL+CL-3)+1.0(WLI	270.73	0.00	-71.63	55.93	0.00	40.36	96.30		15.57	96.30	15.57	60.30	55.94	55.94	96.30	15.57	O.K
333	1.0(DL+COLL+CL-4)+1.0(WLI	270.73	0.00	-71.63	55.93	0.00	40.36	96.30		15.57	96.30	15.57	60.30	55.94	55.94	96.30	15.57	O.K
334	1.0(DL+COLL+CL-1)+1.0(WLI	157.61	0.00	46.34	32.56	0.00	26.11	58.68		6.45	58.68	6.45	22.68	32.57	32.57	58.68	6.45	O.K
335	1.0(DL+COLL+CL-2)+1.0(WLI	157.61	0.00	46.34	32.56	0.00	26.11	58.68		6.45	58.68	6.45	22.68	32.57	32.57	58.68	6.45	O.K
336	1.0(DL+COLL+CL-3)+1.0(WLI	157.61	0.00	46.34	32.56	0.00	26.11	58.68		6.45	58.68	6.45	22.68	32.57	32.57	58.68	6.45	O.K
337	1.0(DL+COLL+CL-4)+1.0(WLI	157.61	0.00	46.34	32.56	0.00	26.11	58.68		6.45	58.68	6.45	22.68	32.57	32.57	58.68	6.45	O.K
338	1.0(DL+COLL+CL-1)+1.0(WLI	194.10	8.10	-10.10	40.10	4.56	5.69	50.36		38.98	41.23	29.85	14.36	44.67	35.54	45.80	34.42	O.K
339	1.0(DL+COLL+CL-2)+1.0(WLI	194.10	8.10	-10.10	40.10	4.56	5.69	50.36		38.98	41.23	29.85	14.36	44.67	35.54	45.80	34.42	O.K
340	1.0(DL+COLL+CL-3)+1.0(WLI	194.10	8.10	-10.10	40.10	4.56	5.69	50.36		38.98	41.23	29.85	14.36	44.67	35.54	45.80	34.42	O.K
341	1.0(DL+COLL+CL-4)+1.0(WLI	194.10	8.10	-10.10	40.10	4.56	5.69	50.36		38.98	41.23	29.85	14.36	44.67	35.54	45.80	34.42	O.K
342	1.0(DL+COLL+CL-1)+1.0(WLI	177.48	15.09	-10.10	36.67	8.50	5.69	50.87		39.48	33.86	22.47	14.87	45.18	28.17	42.37	30.98	O.K
343	1.0(DL+COLL+CL-2)+1.0(WLI	177.48	15.09	-10.10	36.67	8.50	5.69	50.87		39.48	33.86	22.47	14.87	45.18	28.17	42.37	30.98	O.K
344	1.0(DL+COLL+CL-3)+1.0(WLI	177.48	15.09	-10.10	36.67	8.50	5.69	50.87		39.48	33.86	22.47	14.87	45.18	28.17	42.37	30.98	O.K
345	1.0(DL+COLL+CL-4)+1.0(WLI	177.48	15.09	-10.10	36.67	8.50	5.69	50.87		39.48	33.86	22.47	14.87	45.18	28.17	42.37	30.98	O.K
346	1.0(DL+COLL+CL-1)+1.0(WLI	194.37	7.80	-10.10	40.16	4.40	5.69	50.25		38.86	41.45	30.07	14.25	44.56	35.76	45.85	34.47	O.K
347	1.0(DL+COLL+CL-2)+1.0(WLI	194.37	7.80	-10.10	40.16	4.40	5.69	50.25		38.86	41.45	30.07	14.25	44.56	35.76	45.85	34.47	O.K

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348	1.0(DL+COLL+CL-3)+1.0(WL	194.37	7.80	-10.10	40.16	4.40	5.69	50.25		38.86	41.45	30.07	14.25	44.56	35.76	45.85	34.47	O.K
349	1.0(DL+COLL+CL-4)+1.0(WL	194.37	7.80	-10.10	40.16	4.40	5.69	50.25		38.86	41.45	30.07	14.25	44.56	35.76	45.85	34.47	O.K
350	1.0(DL+COLL+CL-1)+1.0(WL	177.66	14.90	-10.10	36.71	8.40	5.69	50.79		39.41	34.00	22.62	14.79	45.10	28.31	42.40	31.02	O.K
351	1.0(DL+COLL+CL-2)+1.0(WL	177.66	14.90	-10.10	36.71	8.40	5.69	50.79		39.41	34.00	22.62	14.79	45.10	28.31	42.40	31.02	O.K
352	1.0(DL+COLL+CL-3)+1.0(WL	177.66	14.90	-10.10	36.71	8.40	5.69	50.79		39.41	34.00	22.62	14.79	45.10	28.31	42.40	31.02	O.K
353	1.0(DL+COLL+CL-4)+1.0(WL	177.66	14.90	-10.10	36.71	8.40	5.69	50.79		39.41	34.00	22.62	14.79	45.10	28.31	42.40	31.02	O.K
354	1.0(DL+COLL+CL-1)+0.75(WI	251.99	0.00	-59.82	52.06	0.00	33.71	85.77		18.35	85.77	18.35	49.77	52.06	52.06	85.77	18.35	O.K
355	1.0(DL+COLL+CL-2)+0.75(WI	251.99	0.00	-59.82	52.06	0.00	33.71	85.77		18.35	85.77	18.35	49.77	52.06	52.06	85.77	18.35	O.K
356	1.0(DL+COLL+CL-3)+0.75(WI	251.99	0.00	-59.82	52.06	0.00	33.71	85.77		18.35	85.77	18.35	49.77	52.06	52.06	85.77	18.35	O.K
357	1.0(DL+COLL+CL-4)+0.75(WI	251.99	0.00	-59.82	52.06	0.00	33.71	85.77		18.35	85.77	18.35	49.77	52.06	52.06	85.77	18.35	O.K
358	1.0(DL+COLL+CL-1)+0.75(WI	222.96	0.00	-15.34	46.07	0.00	8.64	54.71		37.42	54.71	37.42	18.71	46.07	46.07	54.71	37.42	O.K
359	1.0(DL+COLL+CL-2)+0.75(WI	222.96	0.00	-15.34	46.07	0.00	8.64	54.71		37.42	54.71	37.42	18.71	46.07	46.07	54.71	37.42	O.K
360	1.0(DL+COLL+CL-3)+0.75(WI	222.96	0.00	-15.34	46.07	0.00	8.64	54.71		37.42	54.71	37.42	18.71	46.07	46.07	54.71	37.42	O.K
361	1.0(DL+COLL+CL-4)+0.75(WI	222.96	0.00	-15.34	46.07	0.00	8.64	54.71		37.42	54.71	37.42	18.71	46.07	46.07	54.71	37.42	O.K
362	1.0(DL+COLL+CL-1)+0.75(WI	278.10	0.00	-79.89	57.46	0.00	45.02	102.48		12.44	102.48	12.44	66.48	57.46	57.46	102.48	12.44	O.K
363	1.0(DL+COLL+CL-2)+0.75(WI	278.10	0.00	-79.89	57.46	0.00	45.02	102.48		12.44	102.48	12.44	66.48	57.46	57.46	102.48	12.44	O.K
364	1.0(DL+COLL+CL-3)+0.75(WI	278.10	0.00	-79.89	57.46	0.00	45.02	102.48		12.44	102.48	12.44	66.48	57.46	57.46	102.48	12.44	O.K
365	1.0(DL+COLL+CL-4)+0.75(WI	278.10	0.00	-79.89	57.46	0.00	45.02	102.48		12.44	102.48	12.44	66.48	57.46	57.46	102.48	12.44	O.K
366	1.0(DL+COLL+CL-1)+0.75(WI	193.26	0.00	8.59	39.93	0.00	4.84	44.77		35.09	44.77	35.09	8.77	39.93	39.93	44.77	35.09	O.K
367	1.0(DL+COLL+CL-2)+0.75(WI	193.26	0.00	8.59	39.93	0.00	4.84	44.77		35.09	44.77	35.09	8.77	39.93	39.93	44.77	35.09	O.K
368	1.0(DL+COLL+CL-3)+0.75(WI	193.26	0.00	8.59	39.93	0.00	4.84	44.77		35.09	44.77	35.09	8.77	39.93	39.93	44.77	35.09	O.K
369	1.0(DL+COLL+CL-4)+0.75(WI	193.26	0.00	8.59	39.93	0.00	4.84	44.77		35.09	44.77	35.09	8.77	39.93	39.93	44.77	35.09	O.K
370	1.0(DL+COLL+CL-1)+0.75(WI	220.63	6.07	-33.75	45.58	3.42	19.02	68.02		29.99	61.18	23.15	32.02	49.01	42.17	64.60	26.57	O.K
371	1.0(DL+COLL+CL-2)+0.75(WI	220.63	6.07	-33.75	45.58	3.42	19.02	68.02		29.99	61.18	23.15	32.02	49.01	42.17	64.60	26.57	O.K
372	1.0(DL+COLL+CL-3)+0.75(WI	220.63	6.07	-33.75	45.58	3.42	19.02	68.02		29.99	61.18	23.15	32.02	49.01	42.17	64.60	26.57	O.K
373	1.0(DL+COLL+CL-4)+0.75(WI	220.63	6.07	-33.75	45.58	3.42	19.02	68.02		29.99	61.18	23.15	32.02	49.01	42.17	64.60	26.57	O.K
374	1.0(DL+COLL+CL-1)+0.75(WI	208.17	11.32	-33.75	43.01	6.38	19.02	68.40		30.37	55.65	17.62	32.40	49.39	36.64	62.03	24.00	O.K
375	1.0(DL+COLL+CL-2)+0.75(WI	208.17	11.32	-33.75	43.01	6.38	19.02	68.40		30.37	55.65	17.62	32.40	49.39	36.64	62.03	24.00	O.K
376	1.0(DL+COLL+CL-3)+0.75(WI	208.17	11.32	-33.75	43.01	6.38	19.02	68.40		30.37	55.65	17.62	32.40	49.39	36.64	62.03	24.00	O.K
377	1.0(DL+COLL+CL-4)+0.75(WI	208.17	11.32	-33.75	43.01	6.38	19.02	68.40		30.37	55.65	17.62	32.40	49.39	36.64	62.03	24.00	O.K
378	1.0(DL+COLL+CL-1)+0.75(WI	220.83	5.85	-33.75	45.63	3.30	19.02	67.94		29.91	61.34	23.31	31.94	48.93	42.33	64.64	26.61	O.K

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379	1.0(DL+COLL+CL-2)+0.75(WI	220.83	5.85	-33.75	45.63	3.30	19.02	67.94		29.91	61.34	23.31	31.94	48.93	42.33	64.64	26.61	O.K
380	1.0(DL+COLL+CL-3)+0.75(WI	220.83	5.85	-33.75	45.63	3.30	19.02	67.94		29.91	61.34	23.31	31.94	48.93	42.33	64.64	26.61	O.K
381	1.0(DL+COLL+CL-4)+0.75(WI	220.83	5.85	-33.75	45.63	3.30	19.02	67.94		29.91	61.34	23.31	31.94	48.93	42.33	64.64	26.61	O.K
382	1.0(DL+COLL+CL-1)+0.75(WI	208.30	11.18	-33.75	43.04	6.30	19.02	68.35		30.32	55.75	17.72	32.35	49.34	36.74	62.05	24.02	O.K
383	1.0(DL+COLL+CL-2)+0.75(WI	208.30	11.18	-33.75	43.04	6.30	19.02	68.35		30.32	55.75	17.72	32.35	49.34	36.74	62.05	24.02	O.K
384	1.0(DL+COLL+CL-3)+0.75(WI	208.30	11.18	-33.75	43.04	6.30	19.02	68.35		30.32	55.75	17.72	32.35	49.34	36.74	62.05	24.02	O.K
385	1.0(DL+COLL+CL-4)+0.75(WI	208.30	11.18	-33.75	43.04	6.30	19.02	68.35		30.32	55.75	17.72	32.35	49.34	36.74	62.05	24.02	O.K
386	1.0(DL+COLL+CL-1)+0.75(SL	211.60	-0.07	-10.17	43.72	0.04	5.73	49.49		38.03	49.41	37.95	13.49	43.76	43.68	49.45	37.99	O.K
387	1.0(DL+COLL+CL-2)+0.75(SL	211.60	-0.07	-10.17	43.72	0.04	5.73	49.49		38.03	49.41	37.95	13.49	43.76	43.68	49.45	37.99	O.K
388	1.0(DL+COLL+CL-3)+0.75(SL	211.60	-0.07	-10.17	43.72	0.04	5.73	49.49		38.03	49.41	37.95	13.49	43.76	43.68	49.45	37.99	O.K
389	1.0(DL+COLL+CL-4)+0.75(SL	211.60	-0.07	-10.17	43.72	0.04	5.73	49.49		38.03	49.41	37.95	13.49	43.76	43.68	49.45	37.99	O.K
390	1.0(DL+COLL+CL-1)+0.75(SL	211.48	0.07	-10.03	43.69	0.04	5.65	49.39		38.09	49.30	38.00	13.39	43.74	43.65	49.35	38.05	O.K
391	1.0(DL+COLL+CL-2)+0.75(SL	211.48	0.07	-10.03	43.69	0.04	5.65	49.39		38.09	49.30	38.00	13.39	43.74	43.65	49.35	38.05	O.K
392	1.0(DL+COLL+CL-3)+0.75(SL	211.48	0.07	-10.03	43.69	0.04	5.65	49.39		38.09	49.30	38.00	13.39	43.74	43.65	49.35	38.05	O.K
393	1.0(DL+COLL+CL-4)+0.75(SL	211.48	0.07	-10.03	43.69	0.04	5.65	49.39		38.09	49.30	38.00	13.39	43.74	43.65	49.35	38.05	O.K
394	1.0(DL+COLL+CL-1)+0.525(S	233.75	-0.05	-33.80	48.30	0.03	19.04	67.37		29.28	67.31	29.22	31.37	48.33	48.27	67.34	29.25	O.K
395	1.0(DL+COLL+CL-2)+0.525(S	233.75	-0.05	-33.80	48.30	0.03	19.04	67.37		29.28	67.31	29.22	31.37	48.33	48.27	67.34	29.25	O.K
396	1.0(DL+COLL+CL-3)+0.525(S	233.75	-0.05	-33.80	48.30	0.03	19.04	67.37		29.28	67.31	29.22	31.37	48.33	48.27	67.34	29.25	O.K
397	1.0(DL+COLL+CL-4)+0.525(S	233.75	-0.05	-33.80	48.30	0.03	19.04	67.37		29.28	67.31	29.22	31.37	48.33	48.27	67.34	29.25	O.K
398	1.0(DL+COLL+CL-1)+0.525(S	233.67	0.05	-33.69	48.28	0.03	18.99	67.29		29.32	67.24	29.26	31.29	48.31	48.25	67.27	29.29	O.K
399	1.0(DL+COLL+CL-2)+0.525(S	233.67	0.05	-33.69	48.28	0.03	18.99	67.29		29.32	67.24	29.26	31.29	48.31	48.25	67.27	29.29	O.K
400	1.0(DL+COLL+CL-3)+0.525(S	233.67	0.05	-33.69	48.28	0.03	18.99	67.29		29.32	67.24	29.26	31.29	48.31	48.25	67.27	29.29	O.K
401	1.0(DL+COLL+CL-4)+0.525SI	233.67	0.05	-33.69	48.28	0.03	18.99	67.29		29.32	67.24	29.26	31.29	48.31	48.25	67.27	29.29	O.K
402	0.6(DL+COLL+CL-1)+1.0(WLI	231.58	0.00	-40.83	47.85	0.00	23.01	70.85		24.84	70.85	24.84	34.85	47.85	47.85	70.85	24.84	O.K
403	0.6(DL+COLL+CL-2)+1.0(WLI	231.58	0.00	-40.83	47.85	0.00	23.01	70.85		24.84	70.85	24.84	34.85	47.85	47.85	70.85	24.84	O.K
404	0.6(DL+COLL+CL-3)+1.0(WLI	231.58	0.00	-40.83	47.85	0.00	23.01	70.85		24.84	70.85	24.84	34.85	47.85	47.85	70.85	24.84	O.K
405	0.6(DL+COLL+CL-4)+1.0(WLI	231.58	0.00	-40.83	47.85	0.00	23.01	70.85		24.84	70.85	24.84	34.85	47.85	47.85	70.85	24.84	O.K
406	0.6(DL+COLL+CL-1)+1.0(WLI	192.88	0.00	18.48	39.85	0.00	10.42	50.27		29.44	50.27	29.44	14.27	39.86	39.86	50.27	29.44	O.K
407	0.6(DL+COLL+CL-2)+1.0(WLI	192.88	0.00	18.48	39.85	0.00	10.42	50.27		29.44	50.27	29.44	14.27	39.86	39.86	50.27	29.44	O.K
408	0.6(DL+COLL+CL-3)+1.0(WLI	192.88	0.00	18.48	39.85	0.00	10.42	50.27		29.44	50.27	29.44	14.27	39.86	39.86	50.27	29.44	O.K
409	0.6(DL+COLL+CL-4)+1.0(WLI	192.88	0.00	18.48	39.85	0.00	10.42	50.27		29.44	50.27	29.44	14.27	39.86	39.86	50.27	29.44	O.K

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410	0.6(DL+COLL+CL-1)+1.0(WLI	266.39	0.00	-67.59	55.04	0.00	38.09	93.13		16.95	93.13	16.95	57.13	55.04	55.04	93.13	16.95	O.K
411	0.6(DL+COLL+CL-2)+1.0(WLI	266.39	0.00	-67.59	55.04	0.00	38.09	93.13		16.95	93.13	16.95	57.13	55.04	55.04	93.13	16.95	O.K
412	0.6(DL+COLL+CL-3)+1.0(WLI	266.39	0.00	-67.59	55.04	0.00	38.09	93.13		16.95	93.13	16.95	57.13	55.04	55.04	93.13	16.95	O.K
413	0.6(DL+COLL+CL-4)+1.0(WLI	266.39	0.00	-67.59	55.04	0.00	38.09	93.13		16.95	93.13	16.95	57.13	55.04	55.04	93.13	16.95	O.K
414	0.6(DL+COLL+CL-1)+1.0(WLI	153.28	0.00	50.38	31.67	0.00	28.39	60.06		3.28	60.06	3.28	24.06	31.67	31.67	60.06	3.28	O.K
415	0.6(DL+COLL+CL-2)+1.0(WLI	153.28	0.00	50.38	31.67	0.00	28.39	60.06		3.28	60.06	3.28	24.06	31.67	31.67	60.06	3.28	O.K
416	0.6(DL+COLL+CL-3)+1.0(WLI	153.28	0.00	50.38	31.67	0.00	28.39	60.06		3.28	60.06	3.28	24.06	31.67	31.67	60.06	3.28	O.K
417	0.6(DL+COLL+CL-4)+1.0(WLI	153.28	0.00	50.38	31.67	0.00	28.39	60.06		3.28	60.06	3.28	24.06	31.67	31.67	60.06	3.28	O.K
418	0.6(DL+COLL+CL-1)+1.0(WLI	189.77	8.10	-6.06	39.21	4.56	3.41	47.19		40.36	38.06	31.23	11.19	43.78	34.65	42.63	35.80	O.K
419	0.6(DL+COLL+CL-2)+1.0(WLI	189.77	8.10	-6.06	39.21	4.56	3.41	47.19		40.36	38.06	31.23	11.19	43.78	34.65	42.63	35.80	O.K
420	0.6(DL+COLL+CL-3)+1.0(WLI	189.77	8.10	-6.06	39.21	4.56	3.41	47.19		40.36	38.06	31.23	11.19	43.78	34.65	42.63	35.80	O.K
421	0.6(DL+COLL+CL-4)+1.0(WLI	189.77	8.10	-6.06	39.21	4.56	3.41	47.19		40.36	38.06	31.23	11.19	43.78	34.65	42.63	35.80	O.K
422	0.6(DL+COLL+CL-1)+1.0(WLI	173.15	15.09	-6.06	35.77	8.50	3.41	47.69		40.86	30.68	23.86	11.69	44.28	27.27	39.19	32.36	O.K
423	0.6(DL+COLL+CL-2)+1.0(WLI	173.15	15.09	-6.06	35.77	8.50	3.41	47.69		40.86	30.68	23.86	11.69	44.28	27.27	39.19	32.36	O.K
424	0.6(DL+COLL+CL-3)+1.0(WLI	173.15	15.09	-6.06	35.77	8.50	3.41	47.69		40.86	30.68	23.86	11.69	44.28	27.27	39.19	32.36	O.K
425	0.6(DL+COLL+CL-4)+1.0(WLI	173.15	15.09	-6.06	35.77	8.50	3.41	47.69		40.86	30.68	23.86	11.69	44.28	27.27	39.19	32.36	O.K
426	0.6(DL+COLL+CL-1)+1.0(WLI	190.04	7.80	-6.06	39.26	4.40	3.41	47.07		40.24	38.28	31.45	11.07	43.66	34.87	42.68	35.85	O.K
427	0.6(DL+COLL+CL-2)+1.0(WLI	190.04	7.80	-6.06	39.26	4.40	3.41	47.07		40.24	38.28	31.45	11.07	43.66	34.87	42.68	35.85	O.K
428	0.6(DL+COLL+CL-3)+1.0(WLI	190.04	7.80	-6.06	39.26	4.40	3.41	47.07		40.24	38.28	31.45	11.07	43.66	34.87	42.68	35.85	O.K
429	0.6(DL+COLL+CL-4)+1.0(WLI	190.04	7.80	-6.06	39.26	4.40	3.41	47.07		40.24	38.28	31.45	11.07	43.66	34.87	42.68	35.85	O.K
430	0.6(DL+COLL+CL-1)+1.0(WLI	173.32	14.90	-6.06	35.81	8.40	3.41	47.62		40.79	30.83	24.00	11.62	44.21	27.42	39.23	32.40	O.K
431	0.6(DL+COLL+CL-2)+1.0(WLI	173.32	14.90	-6.06	35.81	8.40	3.41	47.62		40.79	30.83	24.00	11.62	44.21	27.42	39.23	32.40	O.K
432	0.6(DL+COLL+CL-3)+1.0(WLI	173.32	14.90	-6.06	35.81	8.40	3.41	47.62		40.79	30.83	24.00	11.62	44.21	27.42	39.23	32.40	O.K
433	0.6(DL+COLL+CL-4)+1.0(WLI	173.32	14.90	-6.06	35.81	8.40	3.41	47.62		40.79	30.83	24.00	11.62	44.21	27.42	39.23	32.40	O.K
434	0.6(DL+COLL+CL-1)+0.7(SL'+	207.26	-0.07	-6.13	42.82	0.04	3.45	46.32		39.41	46.24	39.33	10.32	42.87	42.79	46.28	39.37	O.K
435	0.6(DL+COLL+CL-2)+0.7(SL'+	207.26	-0.07	-6.13	42.82	0.04	3.45	46.32		39.41	46.24	39.33	10.32	42.87	42.79	46.28	39.37	O.K
436	0.6(DL+COLL+CL-3)+0.7(SL'+	207.26	-0.07	-6.13	42.82	0.04	3.45	46.32		39.41	46.24	39.33	10.32	42.87	42.79	46.28	39.37	O.K
437	0.6(DL+COLL+CL-4)+0.7(SL'+	207.26	-0.07	-6.13	42.82	0.04	3.45	46.32		39.41	46.24	39.33	10.32	42.87	42.79	46.28	39.37	O.K
438	0.6(DL+COLL+CL-1)+0.7(SL'+	221.88	-0.07	-26.99	45.84	0.04	15.21	61.09		30.67	61.01	30.60	25.09	45.88	45.81	61.05	30.64	O.K
439	0.6(DL+COLL+CL-2)+0.7(SL'+	198.67	-0.07	8.60	41.05	0.04	4.84	45.93		36.24	45.85	36.16	9.93	41.09	41.01	45.89	36.20	O.K
440	0.6(DL+COLL+CL-3)+0.7(SL'+	242.77	-0.07	-43.05	50.16	0.04	24.26	74.45		25.94	74.38	25.86	38.45	50.20	50.12	74.42	25.90	O.K
441	0.6(DL+COLL+CL-4)+0.7(SL'+	174.91	-0.07	27.74	36.14	0.04	15.63	51.81		20.55	51.73	20.47	15.81	36.18	36.10	51.77	20.51	O.K

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Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
	Ultimate Conditions																	
101	1.4(DL+COLL)	215.87	0.00	14.14	44.60	0.00	7.97	52.57		36.64	52.57	36.64	16.57	44.61	44.61	52.57	36.64	
102	1.2(DL+COLL+CL-1)+1.6(RLL	261.00	0.00	62.56	53.93	0.00	35.25	89.18		18.67	89.18	18.67	53.18	53.93	53.93	89.18	18.67	
103	1.2(DL+COLL+CL-2)+1.6(RLL	261.00	0.00	62.56	53.93	0.00	35.25	89.18		18.67	89.18	18.67	53.18	53.93	53.93	89.18	18.67	
104	1.2(DL+COLL+CL-3)+1.6(RLL	261.00	0.00	62.56	53.93	0.00	35.25	89.18		18.67	89.18	18.67	53.18	53.93	53.93	89.18	18.67	
105	1.2(DL+COLL+CL-4)+1.6(RLL	261.00	0.00	62.56	53.93	0.00	35.25	89.18		18.67	89.18	18.67	53.18	53.93	53.93	89.18	18.67	
106	1.4(DL+COLL+CL-1)	215.87	0.00	14.14	44.60	0.00	7.97	52.57		36.64	52.57	36.64	16.57	44.61	44.61	52.57	36.64	
107	1.4(DL+COLL+CL-2)	215.87	0.00	14.14	44.60	0.00	7.97	52.57		36.64	52.57	36.64	16.57	44.61	44.61	52.57	36.64	
108	1.4(DL+COLL+CL-3)	215.87	0.00	14.14	44.60	0.00	7.97	52.57		36.64	52.57	36.64	16.57	44.61	44.61	52.57	36.64	
109	1.4(DL+COLL+CL-4)	215.87	0.00	14.14	44.60	0.00	7.97	52.57		36.64	52.57	36.64	16.57	44.61	44.61	52.57	36.64	
110	1.2(DL+COLL+CL-1+TL1)+1.6	262.58	0.00	68.68	54.25	0.00	38.70	92.95		15.55	92.95	15.55	56.95	54.25	54.25	92.95	15.55	
111	1.2(DL+COLL+CL-2+TL1)+1.6	262.58	0.00	68.68	54.25	0.00	38.70	92.95		15.55	92.95	15.55	56.95	54.25	54.25	92.95	15.55	
112	1.2(DL+COLL+CL-3+TL1)+1.6	262.58	0.00	68.68	54.25	0.00	38.70	92.95		15.55	92.95	15.55	56.95	54.25	54.25	92.95	15.55	
113	1.2(DL+COLL+CL-4+TL1)+1.6	262.58	0.00	68.68	54.25	0.00	38.70	92.95		15.55	92.95	15.55	56.95	54.25	54.25	92.95	15.55	
114	1.2(DL+COLL+CL-1+TL2)+1.6	259.42	0.00	56.44	53.60	0.00	31.81	85.40		21.79	85.40	21.79	49.40	53.60	53.60	85.40	21.79	
115	1.2(DL+COLL+CL-2+TL2)+1.6	259.42	0.00	56.44	53.60	0.00	31.81	85.40		21.79	85.40	21.79	49.40	53.60	53.60	85.40	21.79	
116	1.2(DL+COLL+CL-3+TL2)+1.6	259.42	0.00	56.44	53.60	0.00	31.81	85.40		21.79	85.40	21.79	49.40	53.60	53.60	85.40	21.79	
117	1.2(DL+COLL+CL-4+TL2)+1.6	259.42	0.00	56.44	53.60	0.00	31.81	85.40		21.79	85.40	21.79	49.40	53.60	53.60	85.40	21.79	
118	1.2(DL+COLL+CL-1)+0.8(WLI	233.20	0.00	39.94	48.18	0.00	22.50	70.69		25.68	70.69	25.68	34.69	48.19	48.19	70.69	25.68	
119	1.2(DL+COLL+CL-2)+0.8(WLI	233.20	0.00	39.94	48.18	0.00	22.50	70.69		25.68	70.69	25.68	34.69	48.19	48.19	70.69	25.68	
120	1.2(DL+COLL+CL-3)+0.8(WLI	233.20	0.00	39.94	48.18	0.00	22.50	70.69		25.68	70.69	25.68	34.69	48.19	48.19	70.69	25.68	
121	1.2(DL+COLL+CL-4)+0.8(WLI	233.20	0.00	39.94	48.18	0.00	22.50	70.69		25.68	70.69	25.68	34.69	48.19	48.19	70.69	25.68	
122	1.2(DL+COLL+CL-1)+0.8(WLI	202.24	0.00	7.52	41.79	0.00	4.23	46.02		37.55	46.02	37.55	10.02	41.79	41.79	46.02	37.55	
123	1.2(DL+COLL+CL-2)+0.8(WLI	202.24	0.00	7.52	41.79	0.00	4.23	46.02		37.55	46.02	37.55	10.02	41.79	41.79	46.02	37.55	
124	1.2(DL+COLL+CL-3)+0.8(WLI	202.24	0.00	7.52	41.79	0.00	4.23	46.02		37.55	46.02	37.55	10.02	41.79	41.79	46.02	37.55	

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125	1.2(DL+COLL+CL-4)+0.8(WLI	202.24	0.00	7.52	41.79	0.00	4.23	46.02		37.55	46.02	37.55	10.02	41.79	41.79	46.02	37.55	
126	1.2(DL+COLL+CL-1)+0.8(WLI	261.05	0.00	61.34	53.94	0.00	34.57	88.50		19.37	88.50	19.37	52.50	53.94	53.94	88.50	19.37	
127	1.2(DL+COLL+CL-2)+0.8(WLI	261.05	0.00	61.34	53.94	0.00	34.57	88.50		19.37	88.50	19.37	52.50	53.94	53.94	88.50	19.37	
128	1.2(DL+COLL+CL-3)+0.8(WLI	261.05	0.00	61.34	53.94	0.00	34.57	88.50		19.37	88.50	19.37	52.50	53.94	53.94	88.50	19.37	
129	1.2(DL+COLL+CL-4)+0.8(WLI	261.05	0.00	61.34	53.94	0.00	34.57	88.50		19.37	88.50	19.37	52.50	53.94	53.94	88.50	19.37	
130	1.2(DL+COLL+CL-1)+0.8(WLI	170.57	0.00	33.04	35.24	0.00	18.62	53.86		16.63	53.86	16.63	17.86	35.25	35.25	53.86	16.63	
131	1.2(DL+COLL+CL-2)+0.8(WLI	170.57	0.00	33.04	35.24	0.00	18.62	53.86		16.63	53.86	16.63	17.86	35.25	35.25	53.86	16.63	
132	1.2(DL+COLL+CL-3)+0.8(WLI	170.57	0.00	33.04	35.24	0.00	18.62	53.86		16.63	53.86	16.63	17.86	35.25	35.25	53.86	16.63	
133	1.2(DL+COLL+CL-4)+0.8(WLI	170.57	0.00	33.04	35.24	0.00	18.62	53.86		16.63	53.86	16.63	17.86	35.25	35.25	53.86	16.63	
134	1.2(DL+COLL+CL-1)+0.8(WLI	199.76	17.49	12.12	41.27	9.86	6.83	57.96		44.30	38.24	24.59	21.96	51.13	31.42	48.10	34.45	
135	1.2(DL+COLL+CL-2)+0.8(WLI	199.76	17.49	12.12	41.27	9.86	6.83	57.96		44.30	38.24	24.59	21.96	51.13	31.42	48.10	34.45	
136	1.2(DL+COLL+CL-3)+0.8(WLI	199.76	17.49	12.12	41.27	9.86	6.83	57.96		44.30	38.24	24.59	21.96	51.13	31.42	48.10	34.45	
137	1.2(DL+COLL+CL-4)+0.8(WLI	199.76	17.49	12.12	41.27	9.86	6.83	57.96		44.30	38.24	24.59	21.96	51.13	31.42	48.10	34.45	
138	1.2(DL+COLL+CL-1)+0.8(WLI	186.46	32.60	12.12	38.52	18.37	6.83	63.72		50.07	26.98	13.33	27.72	56.90	20.16	45.35	31.70	
139	1.2(DL+COLL+CL-2)+0.8(WLI	186.46	32.60	12.12	38.52	18.37	6.83	63.72		50.07	26.98	13.33	27.72	56.90	20.16	45.35	31.70	
140	1.2(DL+COLL+CL-3)+0.8(WLI	186.46	32.60	12.12	38.52	18.37	6.83	63.72		50.07	26.98	13.33	27.72	56.90	20.16	45.35	31.70	
141	1.2(DL+COLL+CL-4)+0.8(WLI	186.46	32.60	12.12	38.52	18.37	6.83	63.72		50.07	26.98	13.33	27.72	56.90	20.16	45.35	31.70	
142	1.2(DL+COLL+CL-1)+0.8(WLI	199.97	16.85	12.12	41.32	9.49	6.83	57.64		43.98	38.65	24.99	21.64	50.81	31.82	48.15	34.49	
143	1.2(DL+COLL+CL-2)+0.8(WLI	199.97	16.85	12.12	41.32	9.49	6.83	57.64		43.98	38.65	24.99	21.64	50.81	31.82	48.15	34.49	
144	1.2(DL+COLL+CL-3)+0.8(WLI	199.97	16.85	12.12	41.32	9.49	6.83	57.64		43.98	38.65	24.99	21.64	50.81	31.82	48.15	34.49	
145	1.2(DL+COLL+CL-4)+0.8(WLI	199.97	16.85	12.12	41.32	9.49	6.83	57.64		43.98	38.65	24.99	21.64	50.81	31.82	48.15	34.49	
146	1.2(DL+COLL+CL-1)+0.8(WLI	186.60	32.19	12.12	38.55	18.14	6.83	63.52		49.86	27.25	13.59	27.52	56.69	20.42	45.39	31.73	
147	1.2(DL+COLL+CL-2)+0.8(WLI	186.60	32.19	12.12	38.55	18.14	6.83	63.52		49.86	27.25	13.59	27.52	56.69	20.42	45.39	31.73	
148	1.2(DL+COLL+CL-3)+0.8(WLI	186.60	32.19	12.12	38.55	18.14	6.83	63.52		49.86	27.25	13.59	27.52	56.69	20.42	45.39	31.73	
149	1.2(DL+COLL+CL-4)+0.8(WLI	186.60	32.19	12.12	38.55	18.14	6.83	63.52		49.86	27.25	13.59	27.52	56.69	20.42	45.39	31.73	
150	1.2(DL+COLL+CL-1)+1.6(WLI	267.48	0.00	83.52	55.26	0.00	47.06	102.32		8.20	102.32	8.20	66.32	55.26	55.26	102.32	8.20	
151	1.2(DL+COLL+CL-2)+1.6(WLI	267.48	0.00	83.52	55.26	0.00	47.06	102.32		8.20	102.32	8.20	66.32	55.26	55.26	102.32	8.20	
152	1.2(DL+COLL+CL-3)+1.6(WLI	267.48	0.00	83.52	55.26	0.00	47.06	102.32		8.20	102.32	8.20	66.32	55.26	55.26	102.32	8.20	
153	1.2(DL+COLL+CL-4)+1.6(WLI	267.48	0.00	83.52	55.26	0.00	47.06	102.32		8.20	102.32	8.20	66.32	55.26	55.26	102.32	8.20	
154	1.2(DL+COLL+CL-1)+1.6(WLI	205.56	0.00	11.39	42.47	0.00	6.42	48.89		36.06	48.89	36.06	12.89	42.48	42.48	48.89	36.06	
155	1.2(DL+COLL+CL-2)+1.6(WLI	205.56	0.00	11.39	42.47	0.00	6.42	48.89		36.06	48.89	36.06	12.89	42.48	42.48	48.89	36.06	
156	1.2(DL+COLL+CL-3)+1.6(WLI	205.56	0.00	11.39	42.47	0.00	6.42	48.89		36.06	48.89	36.06	12.89	42.48	42.48	48.89	36.06	

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157	1.2(DL+COLL+CL-4)+1.6(WLI	205.56	0.00	11.39	42.47	0.00	6.42	48.89		36.06	48.89	36.06	12.89	42.48	42.48	48.89	36.06	
158	1.2(DL+COLL+CL-1)+1.6(WLI	323.18	0.00	126.33	66.77	0.00	71.19	137.96		-4.41	137.96	-4.41	101.96	66.78	66.78	137.96	-4.41	
159	1.2(DL+COLL+CL-2)+1.6(WLI	323.18	0.00	126.33	66.77	0.00	71.19	137.96		-4.41	137.96	-4.41	101.96	66.78	66.78	137.96	-4.41	
160	1.2(DL+COLL+CL-3)+1.6(WLI	323.18	0.00	126.33	66.77	0.00	71.19	137.96		-4.41	137.96	-4.41	101.96	66.78	66.78	137.96	-4.41	
161	1.2(DL+COLL+CL-4)+1.6(WLI	323.18	0.00	126.33	66.77	0.00	71.19	137.96		-4.41	137.96	-4.41	101.96	66.78	66.78	137.96	-4.41	
162	1.2(DL+COLL+CL-1)+1.6(WLI	142.20	0.00	62.43	29.38	0.00	35.18	64.56		-5.80	64.56	-5.80	28.56	29.38	29.38	64.56	-5.80	
163	1.2(DL+COLL+CL-2)+1.6(WLI	142.20	0.00	62.43	29.38	0.00	35.18	64.56		-5.80	64.56	-5.80	28.56	29.38	29.38	64.56	-5.80	
164	1.2(DL+COLL+CL-3)+1.6(WLI	142.20	0.00	62.43	29.38	0.00	35.18	64.56		-5.80	64.56	-5.80	28.56	29.38	29.38	64.56	-5.80	
165	1.2(DL+COLL+CL-4)+1.6(WLI	142.20	0.00	62.43	29.38	0.00	35.18	64.56		-5.80	64.56	-5.80	28.56	29.38	29.38	64.56	-5.80	
166	1.2(DL+COLL+CL-1)+1.6(WLI	200.58	34.99	27.88	41.44	19.71	15.71	76.87		45.45	37.44	6.02	40.87	61.16	21.73	57.16	25.74	
167	1.2(DL+COLL+CL-2)+1.6(WLI	200.58	34.99	27.88	41.44	19.71	15.71	76.87		45.45	37.44	6.02	40.87	61.16	21.73	57.16	25.74	
168	1.2(DL+COLL+CL-3)+1.6(WLI	200.58	34.99	27.88	41.44	19.71	15.71	76.87		45.45	37.44	6.02	40.87	61.16	21.73	57.16	25.74	
169	1.2(DL+COLL+CL-4)+1.6(WLI	200.58	34.99	27.88	41.44	19.71	15.71	76.87		45.45	37.44	6.02	40.87	61.16	21.73	57.16	25.74	
170	1.2(DL+COLL+CL-1)+1.6(WLI	173.99	65.20	27.88	35.95	36.74	15.71	88.40		56.98	14.92	-16.50	52.40	72.69	-0.79	51.66	20.24	
171	1.2(DL+COLL+CL-2)+1.6(WLI	173.99	65.20	27.88	35.95	36.74	15.71	88.40		56.98	14.92	-16.50	52.40	72.69	-0.79	51.66	20.24	
172	1.2(DL+COLL+CL-3)+1.6(WLI	173.99	65.20	27.88	35.95	36.74	15.71	88.40		56.98	14.92	-16.50	52.40	72.69	-0.79	51.66	20.24	
173	1.2(DL+COLL+CL-4)+1.6(WLI	173.99	65.20	27.88	35.95	36.74	15.71	88.40		56.98	14.92	-16.50	52.40	72.69	-0.79	51.66	20.24	
174	1.2(DL+COLL+CL-1)+1.6(WLI	201.01	33.70	27.88	41.53	18.99	15.71	76.23		44.81	38.26	6.83	40.23	60.52	22.55	57.25	25.82	
175	1.2(DL+COLL+CL-2)+1.6(WLI	201.01	33.70	27.88	41.53	18.99	15.71	76.23		44.81	38.26	6.83	40.23	60.52	22.55	57.25	25.82	
176	1.2(DL+COLL+CL-3)+1.6(WLI	201.01	33.70	27.88	41.53	18.99	15.71	76.23		44.81	38.26	6.83	40.23	60.52	22.55	57.25	25.82	
177	1.2(DL+COLL+CL-4)+1.6(WLI	201.01	33.70	27.88	41.53	18.99	15.71	76.23		44.81	38.26	6.83	40.23	60.52	22.55	57.25	25.82	
178	1.2(DL+COLL+CL-1)+1.6(WLI	174.27	64.37	27.88	36.01	36.27	15.71	87.99		56.57	15.44	-15.98	51.99	72.28	-0.27	51.72	20.30	
179	1.2(DL+COLL+CL-2)+1.6(WLI	174.27	64.37	27.88	36.01	36.27	15.71	87.99		56.57	15.44	-15.98	51.99	72.28	-0.27	51.72	20.30	
180	1.2(DL+COLL+CL-3)+1.6(WLI	174.27	64.37	27.88	36.01	36.27	15.71	87.99		56.57	15.44	-15.98	51.99	72.28	-0.27	51.72	20.30	
181	1.2(DL+COLL+CL-4)+1.6(WLI	174.27	64.37	27.88	36.01	36.27	15.71	87.99		56.57	15.44	-15.98	51.99	72.28	-0.27	51.72	20.30	
182	1.2(DL+COLL+CL-1)+1.0(SL'+	228.56	0.27	28.15	47.22	0.15	15.86	63.24		31.51	62.94	31.21	27.24	47.38	47.08	63.09	31.36	
183	1.2(DL+COLL+CL-2)+1.0(SL'+	228.56	0.27	28.15	47.22	0.15	15.86	63.24		31.51	62.94	31.21	27.24	47.38	47.08	63.09	31.36	
184	1.2(DL+COLL+CL-3)+1.0(SL'+	228.56	0.27	28.15	47.22	0.15	15.86	63.24		31.51	62.94	31.21	27.24	47.38	47.08	63.09	31.36	
185	1.2(DL+COLL+CL-4)+1.0(SL'+	228.56	0.27	28.15	47.22	0.15	15.86	63.24		31.51	62.94	31.21	27.24	47.38	47.08	63.09	31.36	
186	1.2(DL+COLL+CL-1)+1.0(SL'-	228.41	0.27	27.62	47.19	0.15	15.56	62.90		31.78	62.60	31.48	26.90	47.34	47.04	62.75	31.63	
187	1.2(DL+COLL+CL-2)+1.0(SL'-	228.41	0.27	27.62	47.19	0.15	15.56	62.90		31.78	62.60	31.48	26.90	47.34	47.04	62.75	31.63	

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188	1.2(DL+COLL+CL-3)+1.0(SL'-	228.41	0.27	27.62	47.19	0.15	15.56	62.90		31.78	62.60	31.48	26.90	47.34	47.04	62.75	31.63	
189	1.2(DL+COLL+CL-4)+1.0(SL'-	228.41	0.27	27.62	47.19	0.15	15.56	62.90		31.78	62.60	31.48	26.90	47.34	47.04	62.75	31.63	
190	0.9(DL+COLL+CL-1)+1.6(WLI	249.45	0.00	64.72	51.54	0.00	36.47	88.01		15.07	88.01	15.07	52.01	51.54	51.54	88.01	15.07	
191	0.9(DL+COLL+CL-2)+1.6(WLI	249.45	0.00	64.72	51.54	0.00	36.47	88.01		15.07	88.01	15.07	52.01	51.54	51.54	88.01	15.07	
192	0.9(DL+COLL+CL-3)+1.6(WLI	249.45	0.00	64.72	51.54	0.00	36.47	88.01		15.07	88.01	15.07	52.01	51.54	51.54	88.01	15.07	
193	0.9(DL+COLL+CL-4)+1.6(WLI	249.45	0.00	64.72	51.54	0.00	36.47	88.01		15.07	88.01	15.07	52.01	51.54	51.54	88.01	15.07	
194	0.9(DL+COLL+CL-1)+1.6(WLI	187.53	0.00	30.18	38.75	0.00	17.01	55.75		21.74	55.75	21.74	19.75	38.75	38.75	55.75	21.74	
195	0.9(DL+COLL+CL-2)+1.6(WLI	187.53	0.00	30.18	38.75	0.00	17.01	55.75		21.74	55.75	21.74	19.75	38.75	38.75	55.75	21.74	
196	0.9(DL+COLL+CL-3)+1.6(WLI	187.53	0.00	30.18	38.75	0.00	17.01	55.75		21.74	55.75	21.74	19.75	38.75	38.75	55.75	21.74	
197	0.9(DL+COLL+CL-4)+1.6(WLI	187.53	0.00	30.18	38.75	0.00	17.01	55.75		21.74	55.75	21.74	19.75	38.75	38.75	55.75	21.74	
198	0.9(DL+COLL+CL-1)+1.6(WLI	305.15	0.00	107.54	63.05	0.00	60.60	123.64		2.45	123.64	2.45	87.64	63.05	63.05	123.64	2.45	
199	0.9(DL+COLL+CL-2)+1.6(WLI	305.15	0.00	107.54	63.05	0.00	60.60	123.64		2.45	123.64	2.45	87.64	63.05	63.05	123.64	2.45	
200	0.9(DL+COLL+CL-3)+1.6(WLI	305.15	0.00	107.54	63.05	0.00	60.60	123.64		2.45	123.64	2.45	87.64	63.05	63.05	123.64	2.45	
201	0.9(DL+COLL+CL-4)+1.6(WLI	305.15	0.00	107.54	63.05	0.00	60.60	123.64		2.45	123.64	2.45	87.64	63.05	63.05	123.64	2.45	
202	0.9(DL+COLL+CL-1)+1.6(WLI	124.18	0.00	81.22	25.66	0.00	45.77	71.42		-20.11	71.42	-20.11	35.42	25.66	25.66	71.42	-20.11	
203	0.9(DL+COLL+CL-2)+1.6(WLI	124.18	0.00	81.22	25.66	0.00	45.77	71.42		-20.11	71.42	-20.11	35.42	25.66	25.66	71.42	-20.11	
204	0.9(DL+COLL+CL-3)+1.6(WLI	124.18	0.00	81.22	25.66	0.00	45.77	71.42		-20.11	71.42	-20.11	35.42	25.66	25.66	71.42	-20.11	
205	0.9(DL+COLL+CL-4)+1.6(WLI	124.18	0.00	81.22	25.66	0.00	45.77	71.42		-20.11	71.42	-20.11	35.42	25.66	25.66	71.42	-20.11	
206	0.9(DL+COLL+CL-1)+1.6(WLI	182.56	34.99	9.09	37.72	19.71	5.12	62.55		52.31	23.13	12.88	26.55	57.43	18.01	42.84	32.60	
207	0.9(DL+COLL+CL-2)+1.6(WLI	182.56	34.99	9.09	37.72	19.71	5.12	62.55		52.31	23.13	12.88	26.55	57.43	18.01	42.84	32.60	
208	0.9(DL+COLL+CL-3)+1.6(WLI	182.56	34.99	9.09	37.72	19.71	5.12	62.55		52.31	23.13	12.88	26.55	57.43	18.01	42.84	32.60	
209	0.9(DL+COLL+CL-4)+1.6(WLI	182.56	34.99	9.09	37.72	19.71	5.12	62.55		52.31	23.13	12.88	26.55	57.43	18.01	42.84	32.60	
210	0.9(DL+COLL+CL-1)+1.6(WLI	155.96	65.20	9.09	32.22	36.74	5.12	74.08		63.84	0.61	-9.64	38.08	68.96	-4.52	37.35	27.10	
211	0.9(DL+COLL+CL-2)+1.6(WLI	155.96	65.20	9.09	32.22	36.74	5.12	74.08		63.84	0.61	-9.64	38.08	68.96	-4.52	37.35	27.10	
212	0.9(DL+COLL+CL-3)+1.6(WLI	155.96	65.20	9.09	32.22	36.74	5.12	74.08		63.84	0.61	-9.64	38.08	68.96	-4.52	37.35	27.10	
213	0.9(DL+COLL+CL-4)+1.6(WLI	155.96	65.20	9.09	32.22	36.74	5.12	74.08		63.84	0.61	-9.64	38.08	68.96	-4.52	37.35	27.10	
214	0.9(DL+COLL+CL-1)+1.6(WLI	182.98	33.70	9.09	37.81	18.99	5.12	61.92		51.67	23.94	13.70	25.92	56.80	18.82	42.93	32.69	
215	0.9(DL+COLL+CL-2)+1.6(WLI	182.98	33.70	9.09	37.81	18.99	5.12	61.92		51.67	23.94	13.70	25.92	56.80	18.82	42.93	32.69	
216	0.9(DL+COLL+CL-3)+1.6(WLI	182.98	33.70	9.09	37.81	18.99	5.12	61.92		51.67	23.94	13.70	25.92	56.80	18.82	42.93	32.69	
217	0.9(DL+COLL+CL-4)+1.6(WLI	182.98	33.70	9.09	37.81	18.99	5.12	61.92		51.67	23.94	13.70	25.92	56.80	18.82	42.93	32.69	
218	0.9(DL+COLL+CL-1)+1.6(WLI	156.24	64.37	9.09	32.28	36.27	5.12	73.68		63.43	1.13	-9.11	37.68	68.56	-3.99	37.41	27.16	
219	0.9(DL+COLL+CL-2)+1.6(WLI	156.24	64.37	9.09	32.28	36.27	5.12	73.68		63.43	1.13	-9.11	37.68	68.56	-3.99	37.41	27.16	

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220	0.9(DL+COLL+CL-3)+1.6(WL	156.24	64.37	9.09	32.28	36.27	5.12	73.68		63.43	1.13	-9.11	37.68	68.56	-3.99	37.41	27.16	
221	0.9(DL+COLL+CL-4)+1.6(WL	156.24	64.37	9.09	32.28	36.27	5.12	73.68		63.43	1.13	-9.11	37.68	68.56	-3.99	37.41	27.16	
222	0.9(DL+COLL+CL-1)+1.0(SL'+	210.54	0.27	9.35	43.50	0.15	5.27	48.92		38.38	48.62	38.08	12.92	43.65	43.35	48.77	38.23	
223	0.9(DL+COLL+CL-2)+1.0(SL'+	210.54	0.27	9.35	43.50	0.15	5.27	48.92		38.38	48.62	38.08	12.92	43.65	43.35	48.77	38.23	
224	0.9(DL+COLL+CL-3)+1.0(SL'+	210.54	0.27	9.35	43.50	0.15	5.27	48.92		38.38	48.62	38.08	12.92	43.65	43.35	48.77	38.23	
225	0.9(DL+COLL+CL-4)+1.0(SL'+	210.54	0.27	9.35	43.50	0.15	5.27	48.92		38.38	48.62	38.08	12.92	43.65	43.35	48.77	38.23	
226	0.9(DL+COLL+CL-1)+1.0(SL'-	210.38	0.27	8.82	43.47	0.15	4.97	48.59		38.64	48.29	38.35	12.59	43.62	43.32	48.44	38.50	
227	0.9(DL+COLL+CL-2)+1.0(SL'-	210.38	0.27	8.82	43.47	0.15	4.97	48.59		38.64	48.29	38.35	12.59	43.62	43.32	48.44	38.50	
228	0.9(DL+COLL+CL-3)+1.0(SL'-	210.38	0.27	8.82	43.47	0.15	4.97	48.59		38.64	48.29	38.35	12.59	43.62	43.32	48.44	38.50	
229	0.9(DL+COLL+CL-4)+1.0(SL'-	210.38	0.27	8.82	43.47	0.15	4.97	48.59		38.64	48.29	38.35	12.59	43.62	43.32	48.44	38.50	

Max :	278	15	50
Min :	153	0	-80

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 13.40 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 23.40 kN/m²
Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 66.48 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 76.48 kN/m²

Sample calculation of values of base pressure:

Load case : 302 1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL)

Maximum pressure, p_1 (pmax) = 50.09+0.00+26.33= 76.42 kN/m²

p_2= 50.09+0.00-26.33= 23.76 kN/m²

p_3= 50.09-(0.00+26.33)= 23.76 kN/m²

p_4= 50.09-0.00+26.33= 76.42 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 76.42-2*18= 40.42 kN/m²

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DESIGN OF ISOLATED FOOTING F4

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

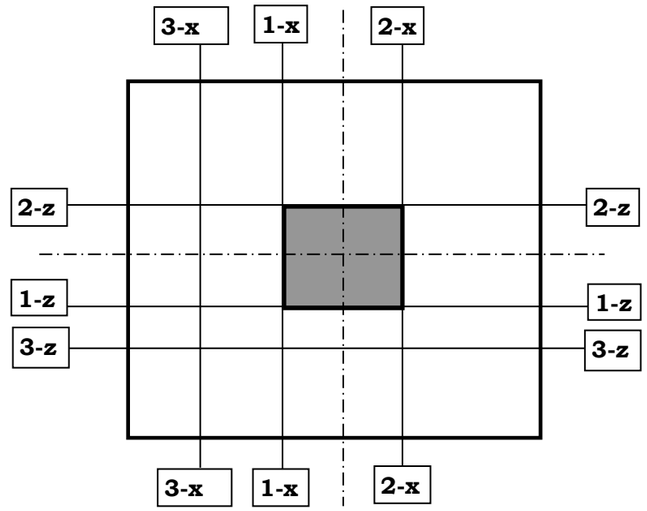
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.75+2*0.5*0.332)+(0.75+2*0.5*0.332))*2
= 4.328 m

Punching Area= ((0.75+2*0.5*0.332)*(0.75+2*0.5*0.332))
= 1.17 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.6*18)
= 38.40 kN/m^2

Area of footing, A_f= 4.84 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+COLL)	44.61	44.61	44.61	1.63	1.63	2.37	22.76
102	1.2(DL+COLL+CL-1)+1.6(R	53.93	53.93	53.93	4.08	4.08	5.92	56.98
103	1.2(DL+COLL+CL-2)+1.6(R	53.93	53.93	53.93	4.08	4.08	5.92	56.98
104	1.2(DL+COLL+CL-3)+1.6(R	53.93	53.93	53.93	4.08	4.08	5.92	56.98
105	1.2(DL+COLL+CL-4)+1.6(R	53.93	53.93	53.93	4.08	4.08	5.92	56.98
106	1.4(DL+COLL+CL-1)	44.61	44.61	44.61	1.63	1.63	2.37	22.76
107	1.4(DL+COLL+CL-2)	44.61	44.61	44.61	1.63	1.63	2.37	22.76
108	1.4(DL+COLL+CL-3)	44.61	44.61	44.61	1.63	1.63	2.37	22.76
109	1.4(DL+COLL+CL-4)	44.61	44.61	44.61	1.63	1.63	2.37	22.76
110	1.2(DL+COLL+CL-1+TL1)+	54.25	54.25	54.25	4.17	4.17	6.04	58.18
111	1.2(DL+COLL+CL-2+TL1)+	54.25	54.25	54.25	4.17	4.17	6.04	58.18
112	1.2(DL+COLL+CL-3+TL1)+	54.25	54.25	54.25	4.17	4.17	6.04	58.18

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113	1.2(DL+COLL+CL-4+TL1)+	54.25	54.25	54.25	4.17	4.17	6.04	58.18
114	1.2(DL+COLL+CL-1+TL2)+	53.60	53.60	53.60	3.99	3.99	5.79	55.78
115	1.2(DL+COLL+CL-2+TL2)+	53.60	53.60	53.60	3.99	3.99	5.79	55.78
116	1.2(DL+COLL+CL-3+TL2)+	53.60	53.60	53.60	3.99	3.99	5.79	55.78
117	1.2(DL+COLL+CL-4+TL2)+	53.60	53.60	53.60	3.99	3.99	5.79	55.78
118	1.2(DL+COLL+CL-1)+0.8(W	48.19	48.19	48.19	2.57	2.57	3.73	35.90
119	1.2(DL+COLL+CL-2)+0.8(W	48.19	48.19	48.19	2.57	2.57	3.73	35.90
120	1.2(DL+COLL+CL-3)+0.8(W	48.19	48.19	48.19	2.57	2.57	3.73	35.90
121	1.2(DL+COLL+CL-4)+0.8(W	48.19	48.19	48.19	2.57	2.57	3.73	35.90
122	1.2(DL+COLL+CL-1)+0.8(W	41.79	41.79	41.79	0.89	0.89	1.29	12.43
123	1.2(DL+COLL+CL-2)+0.8(W	41.79	41.79	41.79	0.89	0.89	1.29	12.43
124	1.2(DL+COLL+CL-3)+0.8(W	41.79	41.79	41.79	0.89	0.89	1.29	12.43
125	1.2(DL+COLL+CL-4)+0.8(W	41.79	41.79	41.79	0.89	0.89	1.29	12.43
126	1.2(DL+COLL+CL-1)+0.8(W	53.94	53.94	53.94	4.08	4.08	5.92	57.02
127	1.2(DL+COLL+CL-2)+0.8(W	53.94	53.94	53.94	4.08	4.08	5.92	57.02
128	1.2(DL+COLL+CL-3)+0.8(W	53.94	53.94	53.94	4.08	4.08	5.92	57.02
129	1.2(DL+COLL+CL-4)+0.8(W	53.94	53.94	53.94	4.08	4.08	5.92	57.02
130	1.2(DL+COLL+CL-1)+0.8(W	35.25	35.25	35.25	-0.83	-0.83	-1.20	-11.59
131	1.2(DL+COLL+CL-2)+0.8(W	35.25	35.25	35.25	-0.83	-0.83	-1.20	-11.59
132	1.2(DL+COLL+CL-3)+0.8(W	35.25	35.25	35.25	-0.83	-0.83	-1.20	-11.59
133	1.2(DL+COLL+CL-4)+0.8(W	35.25	35.25	35.25	-0.83	-0.83	-1.20	-11.59
134	1.2(DL+COLL+CL-1)+0.8(W	44.63	37.92	47.72	2.78	-1.27	4.20	10.54
135	1.2(DL+COLL+CL-2)+0.8(W	44.63	37.92	47.72	2.78	-1.27	4.20	10.54
136	1.2(DL+COLL+CL-3)+0.8(W	44.63	37.92	47.72	2.78	-1.27	4.20	10.54
137	1.2(DL+COLL+CL-4)+0.8(W	44.63	37.92	47.72	2.78	-1.27	4.20	10.54
138	1.2(DL+COLL+CL-1)+0.8(W	44.79	32.27	50.54	3.80	-3.73	5.84	0.46
139	1.2(DL+COLL+CL-2)+0.8(W	44.79	32.27	50.54	3.80	-3.73	5.84	0.46
140	1.2(DL+COLL+CL-3)+0.8(W	44.79	32.27	50.54	3.80	-3.73	5.84	0.46
141	1.2(DL+COLL+CL-4)+0.8(W	44.79	32.27	50.54	3.80	-3.73	5.84	0.46
142	1.2(DL+COLL+CL-1)+0.8(W	44.55	38.08	47.52	2.71	-1.18	4.10	10.70
143	1.2(DL+COLL+CL-2)+0.8(W	44.55	38.08	47.52	2.71	-1.18	4.10	10.70

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144	1.2(DL+COLL+CL-3)+0.8(W	44.55	38.08	47.52	2.71	-1.18	4.10	10.70
145	1.2(DL+COLL+CL-4)+0.8(W	44.55	38.08	47.52	2.71	-1.18	4.10	10.70
146	1.2(DL+COLL+CL-1)+0.8(W	44.74	32.37	50.41	3.76	-3.68	5.77	0.56
147	1.2(DL+COLL+CL-2)+0.8(W	44.74	32.37	50.41	3.76	-3.68	5.77	0.56
148	1.2(DL+COLL+CL-3)+0.8(W	44.74	32.37	50.41	3.76	-3.68	5.77	0.56
149	1.2(DL+COLL+CL-4)+0.8(W	44.74	32.37	50.41	3.76	-3.68	5.77	0.56
150	1.2(DL+COLL+CL-1)+1.6(W	55.26	55.26	55.26	4.43	4.43	6.42	61.89
151	1.2(DL+COLL+CL-2)+1.6(W	55.26	55.26	55.26	4.43	4.43	6.42	61.89
152	1.2(DL+COLL+CL-3)+1.6(W	55.26	55.26	55.26	4.43	4.43	6.42	61.89
153	1.2(DL+COLL+CL-4)+1.6(W	55.26	55.26	55.26	4.43	4.43	6.42	61.89
154	1.2(DL+COLL+CL-1)+1.6(W	42.48	42.48	42.48	1.07	1.07	1.55	14.94
155	1.2(DL+COLL+CL-2)+1.6(W	42.48	42.48	42.48	1.07	1.07	1.55	14.94
156	1.2(DL+COLL+CL-3)+1.6(W	42.48	42.48	42.48	1.07	1.07	1.55	14.94
157	1.2(DL+COLL+CL-4)+1.6(W	42.48	42.48	42.48	1.07	1.07	1.55	14.94
158	1.2(DL+COLL+CL-1)+1.6(W	66.78	66.78	66.78	7.46	7.46	10.81	104.13
159	1.2(DL+COLL+CL-2)+1.6(W	66.78	66.78	66.78	7.46	7.46	10.81	104.13
160	1.2(DL+COLL+CL-3)+1.6(W	66.78	66.78	66.78	7.46	7.46	10.81	104.13
161	1.2(DL+COLL+CL-4)+1.6(W	66.78	66.78	66.78	7.46	7.46	10.81	104.13
162	1.2(DL+COLL+CL-1)+1.6(W	29.38	29.38	29.38	-2.37	-2.37	-3.44	-33.10
163	1.2(DL+COLL+CL-2)+1.6(W	29.38	29.38	29.38	-2.37	-2.37	-3.44	-33.10
164	1.2(DL+COLL+CL-3)+1.6(W	29.38	29.38	29.38	-2.37	-2.37	-3.44	-33.10
165	1.2(DL+COLL+CL-4)+1.6(W	29.38	29.38	29.38	-2.37	-2.37	-3.44	-33.10
166	1.2(DL+COLL+CL-1)+1.6(W	48.17	34.72	54.33	4.84	-3.24	7.37	11.17
167	1.2(DL+COLL+CL-2)+1.6(W	48.17	34.72	54.33	4.84	-3.24	7.37	11.17
168	1.2(DL+COLL+CL-3)+1.6(W	48.17	34.72	54.33	4.84	-3.24	7.37	11.17
169	1.2(DL+COLL+CL-4)+1.6(W	48.17	34.72	54.33	4.84	-3.24	7.37	11.17
170	1.2(DL+COLL+CL-1)+1.6(W	48.48	23.43	59.96	6.89	-8.18	10.64	-9.00
171	1.2(DL+COLL+CL-2)+1.6(W	48.48	23.43	59.96	6.89	-8.18	10.64	-9.00
172	1.2(DL+COLL+CL-3)+1.6(W	48.48	23.43	59.96	6.89	-8.18	10.64	-9.00
173	1.2(DL+COLL+CL-4)+1.6(W	48.48	23.43	59.96	6.89	-8.18	10.64	-9.00
174	1.2(DL+COLL+CL-1)+1.6(W	48.01	35.06	53.94	4.72	-3.07	7.18	11.49

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175	1.2(DL+COLL+CL-2)+1.6(W	48.01	35.06	53.94	4.72	-3.07	7.18	11.49
176	1.2(DL+COLL+CL-3)+1.6(W	48.01	35.06	53.94	4.72	-3.07	7.18	11.49
177	1.2(DL+COLL+CL-4)+1.6(W	48.01	35.06	53.94	4.72	-3.07	7.18	11.49
178	1.2(DL+COLL+CL-1)+1.6(W	48.37	23.64	59.72	6.81	-8.07	10.51	-8.78
179	1.2(DL+COLL+CL-2)+1.6(W	48.37	23.64	59.72	6.81	-8.07	10.51	-8.78
180	1.2(DL+COLL+CL-3)+1.6(W	48.37	23.64	59.72	6.81	-8.07	10.51	-8.78
181	1.2(DL+COLL+CL-4)+1.6(W	48.37	23.64	59.72	6.81	-8.07	10.51	-8.78
182	1.2(DL+COLL+CL-1)+1.0(S	47.28	47.18	47.33	2.35	2.29	3.41	32.38
183	1.2(DL+COLL+CL-2)+1.0(S	47.28	47.18	47.33	2.35	2.29	3.41	32.38
184	1.2(DL+COLL+CL-3)+1.0(S	47.28	47.18	47.33	2.35	2.29	3.41	32.38
185	1.2(DL+COLL+CL-4)+1.0(S	47.28	47.18	47.33	2.35	2.29	3.41	32.38
186	1.2(DL+COLL+CL-1)+1.0(S	47.24	47.14	47.29	2.34	2.28	3.40	32.27
187	1.2(DL+COLL+CL-2)+1.0(S	47.24	47.14	47.29	2.34	2.28	3.40	32.27
188	1.2(DL+COLL+CL-3)+1.0(S	47.24	47.14	47.29	2.34	2.28	3.40	32.27
189	1.2(DL+COLL+CL-4)+1.0(S	47.24	47.14	47.29	2.34	2.28	3.40	32.27
190	0.9(DL+COLL+CL-1)+1.6(W	51.54	51.54	51.54	3.45	3.45	5.01	48.22
191	0.9(DL+COLL+CL-2)+1.6(W	51.54	51.54	51.54	3.45	3.45	5.01	48.22
192	0.9(DL+COLL+CL-3)+1.6(W	51.54	51.54	51.54	3.45	3.45	5.01	48.22
193	0.9(DL+COLL+CL-4)+1.6(W	51.54	51.54	51.54	3.45	3.45	5.01	48.22
194	0.9(DL+COLL+CL-1)+1.6(W	38.75	38.75	38.75	0.09	0.09	0.13	1.27
195	0.9(DL+COLL+CL-2)+1.6(W	38.75	38.75	38.75	0.09	0.09	0.13	1.27
196	0.9(DL+COLL+CL-3)+1.6(W	38.75	38.75	38.75	0.09	0.09	0.13	1.27
197	0.9(DL+COLL+CL-4)+1.6(W	38.75	38.75	38.75	0.09	0.09	0.13	1.27
198	0.9(DL+COLL+CL-1)+1.6(W	63.05	63.05	63.05	6.48	6.48	9.39	90.46
199	0.9(DL+COLL+CL-2)+1.6(W	63.05	63.05	63.05	6.48	6.48	9.39	90.46
200	0.9(DL+COLL+CL-3)+1.6(W	63.05	63.05	63.05	6.48	6.48	9.39	90.46
201	0.9(DL+COLL+CL-4)+1.6(W	63.05	63.05	63.05	6.48	6.48	9.39	90.46
202	0.9(DL+COLL+CL-1)+1.6(W	25.66	25.66	25.66	-3.35	-3.35	-4.85	-46.77
203	0.9(DL+COLL+CL-2)+1.6(W	25.66	25.66	25.66	-3.35	-3.35	-4.85	-46.77
204	0.9(DL+COLL+CL-3)+1.6(W	25.66	25.66	25.66	-3.35	-3.35	-4.85	-46.77
205	0.9(DL+COLL+CL-4)+1.6(W	25.66	25.66	25.66	-3.35	-3.35	-4.85	-46.77

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206	0.9(DL+COLL+CL-1)+1.6(W	44.44	31.00	50.60	3.86	-4.22	5.95	-2.50
207	0.9(DL+COLL+CL-2)+1.6(W	44.44	31.00	50.60	3.86	-4.22	5.95	-2.50
208	0.9(DL+COLL+CL-3)+1.6(W	44.44	31.00	50.60	3.86	-4.22	5.95	-2.50
209	0.9(DL+COLL+CL-4)+1.6(W	44.44	31.00	50.60	3.86	-4.22	5.95	-2.50
210	0.9(DL+COLL+CL-1)+1.6(W	44.75	19.70	56.23	5.91	-9.16	9.22	-22.67
211	0.9(DL+COLL+CL-2)+1.6(W	44.75	19.70	56.23	5.91	-9.16	9.22	-22.67
212	0.9(DL+COLL+CL-3)+1.6(W	44.75	19.70	56.23	5.91	-9.16	9.22	-22.67
213	0.9(DL+COLL+CL-4)+1.6(W	44.75	19.70	56.23	5.91	-9.16	9.22	-22.67
214	0.9(DL+COLL+CL-1)+1.6(W	44.28	31.34	50.22	3.74	-4.05	5.76	-2.18
215	0.9(DL+COLL+CL-2)+1.6(W	44.28	31.34	50.22	3.74	-4.05	5.76	-2.18
216	0.9(DL+COLL+CL-3)+1.6(W	44.28	31.34	50.22	3.74	-4.05	5.76	-2.18
217	0.9(DL+COLL+CL-4)+1.6(W	44.28	31.34	50.22	3.74	-4.05	5.76	-2.18
218	0.9(DL+COLL+CL-1)+1.6(W	44.65	19.92	56.00	5.83	-9.05	9.10	-22.45
219	0.9(DL+COLL+CL-2)+1.6(W	44.65	19.92	56.00	5.83	-9.05	9.10	-22.45
220	0.9(DL+COLL+CL-3)+1.6(W	44.65	19.92	56.00	5.83	-9.05	9.10	-22.45
221	0.9(DL+COLL+CL-4)+1.6(W	44.65	19.92	56.00	5.83	-9.05	9.10	-22.45
222	0.9(DL+COLL+CL-1)+1.0(S	43.55	43.45	43.60	1.37	1.31	1.99	18.71
223	0.9(DL+COLL+CL-2)+1.0(S	43.55	43.45	43.60	1.37	1.31	1.99	18.71
224	0.9(DL+COLL+CL-3)+1.0(S	43.55	43.45	43.60	1.37	1.31	1.99	18.71
225	0.9(DL+COLL+CL-4)+1.0(S	43.55	43.45	43.60	1.37	1.31	1.99	18.71
226	0.9(DL+COLL+CL-1)+1.0(S	43.52	43.42	43.57	1.36	1.30	1.98	18.60
227	0.9(DL+COLL+CL-2)+1.0(S	43.52	43.42	43.57	1.36	1.30	1.98	18.60
228	0.9(DL+COLL+CL-3)+1.0(S	43.52	43.42	43.57	1.36	1.30	1.98	18.60
229	0.9(DL+COLL+CL-4)+1.0(S	43.52	43.42	43.57	1.36	1.30	1.98	18.60

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Sample calculation of BM & SF

Load case : 101 1.4(DL+COLL)

p_{1x} = 44.61-(44.61-44.61)*0.725/2.2 = 44.61 kN/m^2 Overhang = 0.725 m

p_{2x} = 44.61+(44.61-44.61)*0.725/2.2 = 44.61 kN/m^2 Effective depth = 0.344 m

p_{3x} = 44.61-(44.61-44.61)*(0.725-0.344)/2.2 = 44.61 kN/m^2

BM_{1x} = (44.61-38.4)*0.725*0.725/2+0.5*(44.61-44.61)*0.725*0.725*0.67 = 1.63 kN-m/m

BM_{2x} = (44.61-38.4)*0.725*0.725/2-0.5*(-44.61+44.61)*0.725*0.725*0.67 = 1.63 kN-m/m

SF_{3x} = (44.61-38.4)*(0.725-0.344)+0.5*(44.61-44.61)*(0.725-0.344) = 2.37 kN/m

$S.F_{(punching)}$ = (4.84-1.17)*(44.60-38.4) = 22.76 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	7.46	0.06	0.08	0.000	68.93	619.20	12ϕT	182.65	300	175	646.27
At top	-9.16	-0.08	0.10	0.000	84.67	309.60	10ϕT	253.68	300	250	314.16

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 10.81 kN/m

Shear stress , τ_v= (10.81*1000)/(1000*344)

= 0.03 N/mm^2 **O.K.**

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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 58.18 kN

Strength of Concrete in shear Vc1 = 4404.02 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2634.82 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2936.02 kN **Table 22.6.5.2_ (c)**

Design Vc= 2634.82 kN

Allowable shear stress, (ϕ Vc)/bd = 1.33 N/mm^2

Design punching shear stress, τ = (58.18*1000)/(4328*344)

= 0.04 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+COLL)	47.32	41.89	49.72	3.27	-0.01	5.01
102	1.2(DL+COLL+CL-1)+1.6(R	65.94	41.91	76.58	11.33	-3.17	17.48
103	1.2(DL+COLL+CL-2)+1.6(R	65.94	41.91	76.58	11.33	-3.17	17.48
104	1.2(DL+COLL+CL-3)+1.6(R	65.94	41.91	76.58	11.33	-3.17	17.48
105	1.2(DL+COLL+CL-4)+1.6(R	65.94	41.91	76.58	11.33	-3.17	17.48
106	1.4(DL+COLL+CL-1)	47.32	41.89	49.72	3.27	-0.01	5.01
107	1.4(DL+COLL+CL-2)	47.32	41.89	49.72	3.27	-0.01	5.01
108	1.4(DL+COLL+CL-3)	47.32	41.89	49.72	3.27	-0.01	5.01

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109	1.4(DL+COLL+CL-4)	47.32	41.89	49.72	3.27	-0.01	5.01
110	1.2(DL+COLL+CL-1+TL1)+	67.44	41.06	79.12	12.12	-3.79	18.72
111	1.2(DL+COLL+CL-2+TL1)+	67.44	41.06	79.12	12.12	-3.79	18.72
112	1.2(DL+COLL+CL-3+TL1)+	67.44	41.06	79.12	12.12	-3.79	18.72
113	1.2(DL+COLL+CL-4+TL1)+	67.44	41.06	79.12	12.12	-3.79	18.72
114	1.2(DL+COLL+CL-1+TL2)+	64.44	42.75	74.04	10.53	-2.55	16.24
115	1.2(DL+COLL+CL-2+TL2)+	64.44	42.75	74.04	10.53	-2.55	16.24
116	1.2(DL+COLL+CL-3+TL2)+	64.44	42.75	74.04	10.53	-2.55	16.24
117	1.2(DL+COLL+CL-4+TL2)+	64.44	42.75	74.04	10.53	-2.55	16.24
118	1.2(DL+COLL+CL-1)+0.8(W	55.86	40.51	62.65	7.20	-2.06	11.11
119	1.2(DL+COLL+CL-2)+0.8(W	55.86	40.51	62.65	7.20	-2.06	11.11
120	1.2(DL+COLL+CL-3)+0.8(W	55.86	40.51	62.65	7.20	-2.06	11.11
121	1.2(DL+COLL+CL-4)+0.8(W	55.86	40.51	62.65	7.20	-2.06	11.11
122	1.2(DL+COLL+CL-1)+0.8(W	43.23	40.34	44.51	1.76	0.02	2.70
123	1.2(DL+COLL+CL-2)+0.8(W	43.23	40.34	44.51	1.76	0.02	2.70
124	1.2(DL+COLL+CL-3)+0.8(W	43.23	40.34	44.51	1.76	0.02	2.70
125	1.2(DL+COLL+CL-4)+0.8(W	43.23	40.34	44.51	1.76	0.02	2.70
126	1.2(DL+COLL+CL-1)+0.8(W	65.72	42.15	76.15	11.19	-3.03	17.26
127	1.2(DL+COLL+CL-2)+0.8(W	65.72	42.15	76.15	11.19	-3.03	17.26
128	1.2(DL+COLL+CL-3)+0.8(W	65.72	42.15	76.15	11.19	-3.03	17.26
129	1.2(DL+COLL+CL-4)+0.8(W	65.72	42.15	76.15	11.19	-3.03	17.26
130	1.2(DL+COLL+CL-1)+0.8(W	41.59	28.90	47.21	3.00	-4.66	4.77
131	1.2(DL+COLL+CL-2)+0.8(W	41.59	28.90	47.21	3.00	-4.66	4.77
132	1.2(DL+COLL+CL-3)+0.8(W	41.59	28.90	47.21	3.00	-4.66	4.77
133	1.2(DL+COLL+CL-4)+0.8(W	41.59	28.90	47.21	3.00	-4.66	4.77
134	1.2(DL+COLL+CL-1)+0.8(W	43.60	38.95	45.66	2.16	-0.65	3.33
135	1.2(DL+COLL+CL-2)+0.8(W	43.60	38.95	45.66	2.16	-0.65	3.33
136	1.2(DL+COLL+CL-3)+0.8(W	43.60	38.95	45.66	2.16	-0.65	3.33
137	1.2(DL+COLL+CL-4)+0.8(W	43.60	38.95	45.66	2.16	-0.65	3.33
138	1.2(DL+COLL+CL-1)+0.8(W	40.85	36.20	42.91	1.44	-1.37	2.25
139	1.2(DL+COLL+CL-2)+0.8(W	40.85	36.20	42.91	1.44	-1.37	2.25
140	1.2(DL+COLL+CL-3)+0.8(W	40.85	36.20	42.91	1.44	-1.37	2.25
141	1.2(DL+COLL+CL-4)+0.8(W	40.85	36.20	42.91	1.44	-1.37	2.25

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142	1.2(DL+COLL+CL-1)+0.8(W	43.65	38.99	45.71	2.17	-0.64	3.35
143	1.2(DL+COLL+CL-2)+0.8(W	43.65	38.99	45.71	2.17	-0.64	3.35
144	1.2(DL+COLL+CL-3)+0.8(W	43.65	38.99	45.71	2.17	-0.64	3.35
145	1.2(DL+COLL+CL-4)+0.8(W	43.65	38.99	45.71	2.17	-0.64	3.35
146	1.2(DL+COLL+CL-1)+0.8(W	40.89	36.23	42.95	1.45	-1.36	2.27
147	1.2(DL+COLL+CL-2)+0.8(W	40.89	36.23	42.95	1.45	-1.36	2.27
148	1.2(DL+COLL+CL-3)+0.8(W	40.89	36.23	42.95	1.45	-1.36	2.27
149	1.2(DL+COLL+CL-4)+0.8(W	40.89	36.23	42.95	1.45	-1.36	2.27
150	1.2(DL+COLL+CL-1)+1.6(W	71.30	39.22	85.51	14.11	-5.25	21.82
151	1.2(DL+COLL+CL-2)+1.6(W	71.30	39.22	85.51	14.11	-5.25	21.82
152	1.2(DL+COLL+CL-3)+1.6(W	71.30	39.22	85.51	14.11	-5.25	21.82
153	1.2(DL+COLL+CL-4)+1.6(W	71.30	39.22	85.51	14.11	-5.25	21.82
154	1.2(DL+COLL+CL-1)+1.6(W	44.66	40.29	46.60	2.39	-0.25	3.67
155	1.2(DL+COLL+CL-2)+1.6(W	44.66	40.29	46.60	2.39	-0.25	3.67
156	1.2(DL+COLL+CL-3)+1.6(W	44.66	40.29	46.60	2.39	-0.25	3.67
157	1.2(DL+COLL+CL-4)+1.6(W	44.66	40.29	46.60	2.39	-0.25	3.67
158	1.2(DL+COLL+CL-1)+1.6(W	91.04	42.51	112.53	22.10	-7.18	34.13
159	1.2(DL+COLL+CL-2)+1.6(W	91.04	42.51	112.53	22.10	-7.18	34.13
160	1.2(DL+COLL+CL-3)+1.6(W	91.04	42.51	112.53	22.10	-7.18	34.13
161	1.2(DL+COLL+CL-4)+1.6(W	91.04	42.51	112.53	22.10	-7.18	34.13
162	1.2(DL+COLL+CL-1)+1.6(W	41.37	17.39	51.99	4.86	-9.61	7.81
163	1.2(DL+COLL+CL-2)+1.6(W	41.37	17.39	51.99	4.86	-9.61	7.81
164	1.2(DL+COLL+CL-3)+1.6(W	41.37	17.39	51.99	4.86	-9.61	7.81
165	1.2(DL+COLL+CL-4)+1.6(W	41.37	17.39	51.99	4.86	-9.61	7.81
166	1.2(DL+COLL+CL-1)+1.6(W	46.81	36.09	51.55	4.03	-2.43	6.27
167	1.2(DL+COLL+CL-2)+1.6(W	46.81	36.09	51.55	4.03	-2.43	6.27
168	1.2(DL+COLL+CL-3)+1.6(W	46.81	36.09	51.55	4.03	-2.43	6.27
169	1.2(DL+COLL+CL-4)+1.6(W	46.81	36.09	51.55	4.03	-2.43	6.27
170	1.2(DL+COLL+CL-1)+1.6(W	41.31	30.59	46.05	2.59	-3.87	4.11
171	1.2(DL+COLL+CL-2)+1.6(W	41.31	30.59	46.05	2.59	-3.87	4.11
172	1.2(DL+COLL+CL-3)+1.6(W	41.31	30.59	46.05	2.59	-3.87	4.11
173	1.2(DL+COLL+CL-4)+1.6(W	41.31	30.59	46.05	2.59	-3.87	4.11
174	1.2(DL+COLL+CL-1)+1.6(W	46.89	36.18	51.64	4.06	-2.41	6.30

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175	1.2(DL+COLL+CL-2)+1.6(W	46.89	36.18	51.64	4.06	-2.41	6.30
176	1.2(DL+COLL+CL-3)+1.6(W	46.89	36.18	51.64	4.06	-2.41	6.30
177	1.2(DL+COLL+CL-4)+1.6(W	46.89	36.18	51.64	4.06	-2.41	6.30
178	1.2(DL+COLL+CL-1)+1.6(W	41.37	30.65	46.11	2.60	-3.86	4.13
179	1.2(DL+COLL+CL-2)+1.6(W	41.37	30.65	46.11	2.60	-3.86	4.13
180	1.2(DL+COLL+CL-3)+1.6(W	41.37	30.65	46.11	2.60	-3.86	4.13
181	1.2(DL+COLL+CL-4)+1.6(W	41.37	30.65	46.11	2.60	-3.86	4.13
182	1.2(DL+COLL+CL-1)+1.0(S	52.63	41.82	57.42	5.58	-0.94	8.59
183	1.2(DL+COLL+CL-2)+1.0(S	52.63	41.82	57.42	5.58	-0.94	8.59
184	1.2(DL+COLL+CL-3)+1.0(S	52.63	41.82	57.42	5.58	-0.94	8.59
185	1.2(DL+COLL+CL-4)+1.0(S	52.63	41.82	57.42	5.58	-0.94	8.59
186	1.2(DL+COLL+CL-1)+1.0(S	52.49	41.89	57.19	5.51	-0.89	8.48
187	1.2(DL+COLL+CL-2)+1.0(S	52.49	41.89	57.19	5.51	-0.89	8.48
188	1.2(DL+COLL+CL-3)+1.0(S	52.49	41.89	57.19	5.51	-0.89	8.48
189	1.2(DL+COLL+CL-4)+1.0(S	52.49	41.89	57.19	5.51	-0.89	8.48
190	0.9(DL+COLL+CL-1)+1.6(W	63.97	39.11	74.98	10.95	-4.05	16.94
191	0.9(DL+COLL+CL-2)+1.6(W	63.97	39.11	74.98	10.95	-4.05	16.94
192	0.9(DL+COLL+CL-3)+1.6(W	63.97	39.11	74.98	10.95	-4.05	16.94
193	0.9(DL+COLL+CL-4)+1.6(W	63.97	39.11	74.98	10.95	-4.05	16.94
194	0.9(DL+COLL+CL-1)+1.6(W	44.54	32.95	49.67	3.59	-3.41	5.62
195	0.9(DL+COLL+CL-2)+1.6(W	44.54	32.95	49.67	3.59	-3.41	5.62
196	0.9(DL+COLL+CL-3)+1.6(W	44.54	32.95	49.67	3.59	-3.41	5.62
197	0.9(DL+COLL+CL-4)+1.6(W	44.54	32.95	49.67	3.59	-3.41	5.62
198	0.9(DL+COLL+CL-1)+1.6(W	83.70	42.39	101.99	18.94	-5.98	29.25
199	0.9(DL+COLL+CL-2)+1.6(W	83.70	42.39	101.99	18.94	-5.98	29.25
200	0.9(DL+COLL+CL-3)+1.6(W	83.70	42.39	101.99	18.94	-5.98	29.25
201	0.9(DL+COLL+CL-4)+1.6(W	83.70	42.39	101.99	18.94	-5.98	29.25
202	0.9(DL+COLL+CL-1)+1.6(W	41.26	10.05	55.07	6.06	-12.76	9.76
203	0.9(DL+COLL+CL-2)+1.6(W	41.26	10.05	55.07	6.06	-12.76	9.76
204	0.9(DL+COLL+CL-3)+1.6(W	41.26	10.05	55.07	6.06	-12.76	9.76
205	0.9(DL+COLL+CL-4)+1.6(W	41.26	10.05	55.07	6.06	-12.76	9.76
206	0.9(DL+COLL+CL-1)+1.6(W	39.47	35.97	41.01	0.87	-1.23	1.39
207	0.9(DL+COLL+CL-2)+1.6(W	39.47	35.97	41.01	0.87	-1.23	1.39
208	0.9(DL+COLL+CL-3)+1.6(W	39.47	35.97	41.01	0.87	-1.23	1.39

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209	0.9(DL+COLL+CL-4)+1.6(W	39.47	35.97	41.01	0.87	-1.23	1.39
210	0.9(DL+COLL+CL-1)+1.6(W	33.97	30.48	35.52	-0.57	-2.68	-0.77
211	0.9(DL+COLL+CL-2)+1.6(W	33.97	30.48	35.52	-0.57	-2.68	-0.77
212	0.9(DL+COLL+CL-3)+1.6(W	33.97	30.48	35.52	-0.57	-2.68	-0.77
213	0.9(DL+COLL+CL-4)+1.6(W	33.97	30.48	35.52	-0.57	-2.68	-0.77
214	0.9(DL+COLL+CL-1)+1.6(W	39.56	36.06	41.10	0.90	-1.21	1.42
215	0.9(DL+COLL+CL-2)+1.6(W	39.56	36.06	41.10	0.90	-1.21	1.42
216	0.9(DL+COLL+CL-3)+1.6(W	39.56	36.06	41.10	0.90	-1.21	1.42
217	0.9(DL+COLL+CL-4)+1.6(W	39.56	36.06	41.10	0.90	-1.21	1.42
218	0.9(DL+COLL+CL-1)+1.6(W	34.03	30.54	35.58	-0.55	-2.66	-0.75
219	0.9(DL+COLL+CL-2)+1.6(W	34.03	30.54	35.58	-0.55	-2.66	-0.75
220	0.9(DL+COLL+CL-3)+1.6(W	34.03	30.54	35.58	-0.55	-2.66	-0.75
221	0.9(DL+COLL+CL-4)+1.6(W	34.03	30.54	35.58	-0.55	-2.66	-0.75
222	0.9(DL+COLL+CL-1)+1.0(S	45.30	41.70	46.89	2.42	0.26	3.71
223	0.9(DL+COLL+CL-2)+1.0(S	45.30	41.70	46.89	2.42	0.26	3.71
224	0.9(DL+COLL+CL-3)+1.0(S	45.30	41.70	46.89	2.42	0.26	3.71
225	0.9(DL+COLL+CL-4)+1.0(S	45.30	41.70	46.89	2.42	0.26	3.71
226	0.9(DL+COLL+CL-1)+1.0(S	45.16	41.78	46.66	2.35	0.31	3.60
227	0.9(DL+COLL+CL-2)+1.0(S	45.16	41.78	46.66	2.35	0.31	3.60
228	0.9(DL+COLL+CL-3)+1.0(S	45.16	41.78	46.66	2.35	0.31	3.60
229	0.9(DL+COLL+CL-4)+1.0(S	45.16	41.78	46.66	2.35	0.31	3.60

Sample calculation of BM & SF

Load case : 101

p_{1z} = 52.57-(52.57-36.64)*0.725/2.2= 47.32 kN/m^2

p_{2z} = 36.64+(52.57-36.64)*0.725/2.2= 41.89 kN/m^2

p_{3z} = 52.57-(52.57-36.64)*(0.725-0.332)/2.2= 49.72 kN/m^2

Overhang = 0.725 m

Effective depth = 0.332 m

BM_{1z} = (47.32-38.4)*0.725*0.725/2+0.5*(52.57-47.32)*0.725*0.725*0.67 = 3.27 kN-m/m

BM_{2z} = (41.89-38.4)*0.725*0.725/2-0.5*(-36.64+41.89)*0.725*0.725*0.67 = -0.01 kN-m/m

SF_{3z} = (49.72-38.4)*(0.725-0.332)+0.5*(52.57-49.72)*(0.725-0.332) = 5.01 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	22.10	0.200	0.267	0.0006	219.92	597.60	12ϕT	189.25	300	175	646.3
At top	-3.79	-0.034	-0.046	-0.0001	-0.36	298.80	10ϕT	262.85	250	250	314.2

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 34.13 kN/m

Shear stress , τ_v= (34.13*1000)/(1000*332)

= 0.10 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	12ϕT	**175**	c/c
	At Top	10ϕT	**250**	c/c
Along Z Direction :				
	At Bottom	12ϕT	**175**	c/c
	At Top	10ϕT	**250**	c/c

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5.5 DESIGN OF CALCULATION OF FOOTING F5

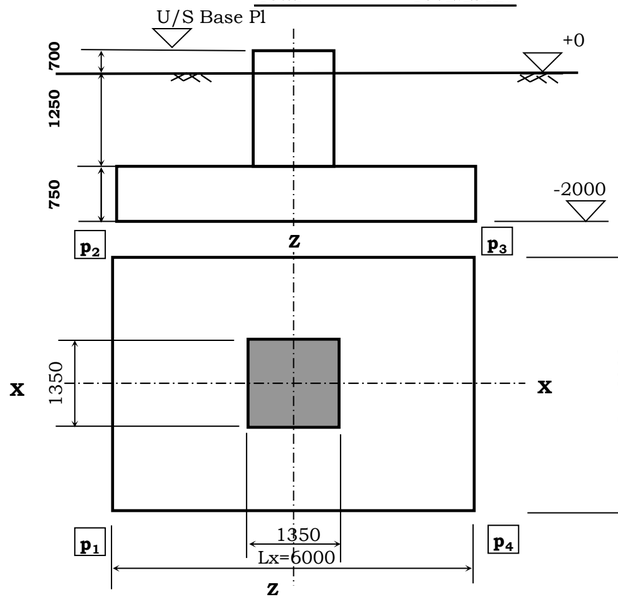
	Along-X	Along-Z
Size of column :	1.35m	1.35m
Size of fdn. raft :	6.00m	6.00m
Thickness of raft :	0.75 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-2.00 m	
U/S level Base Plate :	0.70 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Crane Load	CL
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (6*6*0.75*24)+(1.35*1.35*1.95*24) = 733.29KN
Weight of soil = ((6*6)-(1.35*1.35))*1.25*18 = 768.99KN
Total = 1502.29KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	25mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	25mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 687.5mm = 0.6875m
Effective depth of slab along Z direction = 662.5mm = 0.6625m
Clear projection along Z direction = 2.325m
Clear projection along X direction = 2.325m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

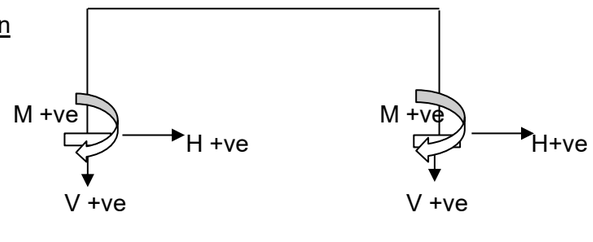
M_X=Moment about X-X direction

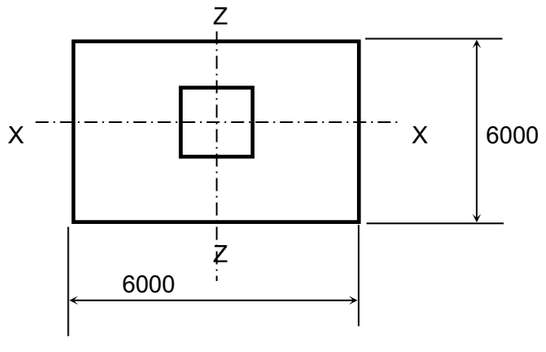
Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F5

Sign convention





Height from BOF to top of column about X (l_{ux}) = 2700mm
Height from BOF to top of column about Z (l_{uz}) = 2700mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	533.598		-154.95		-207.0156
2	COLLATERAL LOAD (COLL)	-0.0048		-0.0048		0.0048
3	ROOF LIVE LOAD (RLL)	114.132		-31.7076		-114.1872
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	1539.994		-445.7676		-479.6868
5	CRANE LOAD-1 (CL-1)	0		0		0
6	CRANE LOAD-2 (CL-2)	0		0		0
7	CRANE LOAD-3 (CL-3)	0		0		0
8	CRANE LOAD-4 (CL-4)	0		0		0
9	WIND LEFT-1 (WLL-1)	-40.7604		76.8		190.8324
10	WIND LEFT-2 (WLL-2)	-109.501		12.198		63.2544
11	WIND RIGHT-1 (WLR-1)	-117.401		115.3872		311.694
12	WIND RIGHT-2 (WLR-2)	-104.03		34.7412		116.862
13	WIND LONG-1 (WLLG-1)	-189.367	86.5608		272.4492	
14	WIND LONG-2 (WLLG-2)	-143.381	44.7336		159.9156	
15	WIND LONG-3 (WLLG-3)	-208.013	73.3788		224.2368	
16	WIND LONG-4 (WLLG-4)	-162.026	31.5516		111.702	
17	SEISMIC LOAD SL'+'	6.606	-23.5272	-23.5272	-69.4092	-69.4092
18	SEISMIC LOAD SL'-'	-6.606	23.5272	23.5272	69.4092	69.4092
19	TEMPERATURE LOAD (TL1)	-10.194		-125.8584		-222.756
20	TEMPERATURE LOAD (TL2)	10.194		125.8584		222.756
21	BRACING REACTION (BR-WLLG1)	51.7	-41		0	
22	BRACING REACTION (BR-WLLG2)	51.7	-41		0	
23	BRACING REACTION (BR-WLLG3)	-51.7	41		0	
24	BRACING REACTION (BR-WLLG4)	-51.7	41		0	
25	BRACING REACTION (BR-SL'+')	91.34	-96.33	-96.33	0	0
26	BRACING REACTION (BR-SL'-')	-91.34	96.33	96.33	0	0

Ultimate Condition

Combined Load Cases:		Load at top of foundation			Load at bottom of foundation		
Load Case	Load Combination	Factored Load/Moment					
		Pu (kN)	Muz (kN-m)	Mux (kN-m)	Pu (kN)	Muz (kN-m)	Mux (kN-m)
101	1.4(DL+COLL)	747.03	0.0	875.5	2249.3	0.0	875.5
102	1.2(DL+COLL+CL-1)+1.6(RLL+PLL / MLL)	3286.9	0.0	3763.4	4789.2	0.0	3763.4
103	1.2(DL+COLL+CL-2)+1.6(RLL+PLL / MLL)	3286.9	0.0	3763.4	4789.2	0.0	3763.4
104	1.2(DL+COLL+CL-3)+1.6(RLL+PLL / MLL)	3286.9	0.0	3763.4	4789.2	0.0	3763.4
105	1.2(DL+COLL+CL-4)+1.6(RLL+PLL / MLL)	3286.9	0.0	3763.4	4789.2	0.0	3763.4
106	1.4(DL+COLL+CL-1)	747.0	0.0	875.5	2249.3	0.0	875.5
107	1.4(DL+COLL+CL-2)	747.0	0.0	875.5	2249.3	0.0	875.5
108	1.4(DL+COLL+CL-3)	747.0	0.0	875.5	2249.3	0.0	875.5
109	1.4(DL+COLL+CL-4)	747.0	0.0	875.5	2249.3	0.0	875.5
110	1.2(DL+COLL+CL-1+TL1)+1.6(RLL+PLL / MLL)	3274.7	0.0	4438.4	4777.0	0.0	4438.4
111	1.2(DL+COLL+CL-2+TL1)+1.6(RLL+PLL / MLL)	3274.7	0.0	4438.4	4777.0	0.0	4438.4
112	1.2(DL+COLL+CL-3+TL1)+1.6(RLL+PLL / MLL)	3274.7	0.0	4438.4	4777.0	0.0	4438.4
113	1.2(DL+COLL+CL-4+TL1)+1.6(RLL+PLL / MLL)	3274.7	0.0	4438.4	4777.0	0.0	4438.4
114	1.2(DL+COLL+CL-1+TL2)+1.6(RLL+PLL / MLL)	3299.1	0.0	3088.3	4801.4	0.0	3088.3
115	1.2(DL+COLL+CL-2+TL2)+1.6(RLL+PLL / MLL)	3299.1	0.0	3088.3	4801.4	0.0	3088.3
116	1.2(DL+COLL+CL-3+TL2)+1.6(RLL+PLL / MLL)	3299.1	0.0	3088.3	4801.4	0.0	3088.3
117	1.2(DL+COLL+CL-4+TL2)+1.6(RLL+PLL / MLL)	3299.1	0.0	3088.3	4801.4	0.0	3088.3
118	1.2(DL+COLL+CL-1)+0.8(WLL-1)	607.7	0.0	431.9	2110.0	0.0	431.9
119	1.2(DL+COLL+CL-2)+0.8(WLL-1)	607.7	0.0	431.9	2110.0	0.0	431.9
120	1.2(DL+COLL+CL-3)+0.8(WLL-1)	607.7	0.0	431.9	2110.0	0.0	431.9
121	1.2(DL+COLL+CL-4)+0.8(WLL-1)	607.7	0.0	431.9	2110.0	0.0	431.9
122	1.2(DL+COLL+CL-1)+0.8(WLL-2)	552.7	0.0	673.5	2055.0	0.0	673.5
123	1.2(DL+COLL+CL-2)+0.8(WLL-2)	552.7	0.0	673.5	2055.0	0.0	673.5
124	1.2(DL+COLL+CL-3)+0.8(WLL-2)	552.7	0.0	673.5	2055.0	0.0	673.5
125	1.2(DL+COLL+CL-4)+0.8(WLL-2)	552.7	0.0	673.5	2055.0	0.0	673.5
126	1.2(DL+COLL+CL-1)+0.8(WLR-1)	546.4	0.0	251.9	2048.7	0.0	251.9
127	1.2(DL+COLL+CL-2)+0.8(WLR-1)	546.4	0.0	251.9	2048.7	0.0	251.9
128	1.2(DL+COLL+CL-3)+0.8(WLR-1)	546.4	0.0	251.9	2048.7	0.0	251.9
129	1.2(DL+COLL+CL-4)+0.8(WLR-1)	546.4	0.0	251.9	2048.7	0.0	251.9
130	1.2(DL+COLL+CL-1)+0.8(WLR-2)	557.1	0.0	581.9	2059.4	0.0	581.9
131	1.2(DL+COLL+CL-2)+0.8(WLR-2)	557.1	0.0	581.9	2059.4	0.0	581.9
132	1.2(DL+COLL+CL-3)+0.8(WLR-2)	557.1	0.0	581.9	2059.4	0.0	581.9
133	1.2(DL+COLL+CL-4)+0.8(WLR-2)	557.1	0.0	581.9	2059.4	0.0	581.9
134	1.2(DL+COLL+CL-1)+0.8(WLLG-1+BR-WLLG-1)	530.2	316.4	750.5	2032.5	316.4	750.5
135	1.2(DL+COLL+CL-2)+0.8(WLLG-1+BR-WLLG-1)	530.2	316.4	750.5	2032.5	316.4	750.5
136	1.2(DL+COLL+CL-3)+0.8(WLLG-1+BR-WLLG-1)	530.2	316.4	750.5	2032.5	316.4	750.5
137	1.2(DL+COLL+CL-4)+0.8(WLLG-1+BR-WLLG-1)	530.2	316.4	750.5	2032.5	316.4	750.5
138	1.2(DL+COLL+CL-1)+0.8(WLLG-2+BR-WLLG-2)	567.0	136.0	750.5	2069.3	136.0	750.5
139	1.2(DL+COLL+CL-2)+0.8(WLLG-2+BR-WLLG-2)	567.0	136.0	750.5	2069.3	136.0	750.5
140	1.2(DL+COLL+CL-3)+0.8(WLLG-2+BR-WLLG-2)	567.0	136.0	750.5	2069.3	136.0	750.5
141	1.2(DL+COLL+CL-4)+0.8(WLLG-2+BR-WLLG-2)	567.0	136.0	750.5	2069.3	136.0	750.5
142	1.2(DL+COLL+CL-1)+0.8(WLLG-3+BR-WLLG-3)	432.5	426.4	750.5	1934.8	426.4	750.5
143	1.2(DL+COLL+CL-2)+0.8(WLLG-3+BR-WLLG-3)	432.5	426.4	750.5	1934.8	426.4	750.5
144	1.2(DL+COLL+CL-3)+0.8(WLLG-3+BR-WLLG-3)	432.5	426.4	750.5	1934.8	426.4	750.5
145	1.2(DL+COLL+CL-4)+0.8(WLLG-3+BR-WLLG-3)	432.5	426.4	750.5	1934.8	426.4	750.5
146	1.2(DL+COLL+CL-1)+0.8(WLLG-4+BR-WLLG-4)	469.3	246.1	750.5	1971.6	246.1	750.5

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147	1.2(DL+COLL+CL-2)+0.8(WLLG-4+BR-WLLG-4)	469.3	246.1	750.5	1971.6	246.1	750.5
148	1.2(DL+COLL+CL-3)+0.8(WLLG-4+BR-WLLG-4)	469.3	246.1	750.5	1971.6	246.1	750.5
149	1.2(DL+COLL+CL-4)+0.8(WLLG-4+BR-WLLG-4)	469.3	246.1	750.5	1971.6	246.1	750.5
150	1.2(DL+COLL+CL-1)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	1402.2	0.0	1054.9	2904.4	0.0	1054.9
151	1.2(DL+COLL+CL-2)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	1402.2	0.0	1054.9	2904.4	0.0	1054.9
152	1.2(DL+COLL+CL-3)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	1402.2	0.0	1054.9	2904.4	0.0	1054.9
153	1.2(DL+COLL+CL-4)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	1402.2	0.0	1054.9	2904.4	0.0	1054.9
154	1.2(DL+COLL+CL-1)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	1292.2	0.0	1538.1	2794.5	0.0	1538.1
155	1.2(DL+COLL+CL-2)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	1292.2	0.0	1538.1	2794.5	0.0	1538.1
156	1.2(DL+COLL+CL-3)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	1292.2	0.0	1538.1	2794.5	0.0	1538.1
157	1.2(DL+COLL+CL-4)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	1292.2	0.0	1538.1	2794.5	0.0	1538.1
158	1.2(DL+COLL+CL-1)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	1279.5	0.0	694.8	2781.8	0.0	694.8
159	1.2(DL+COLL+CL-2)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	1279.5	0.0	694.8	2781.8	0.0	694.8
160	1.2(DL+COLL+CL-3)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	1279.5	0.0	694.8	2781.8	0.0	694.8
161	1.2(DL+COLL+CL-4)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	1279.5	0.0	694.8	2781.8	0.0	694.8
162	1.2(DL+COLL+CL-1)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	1300.9	0.0	1354.9	2803.2	0.0	1354.9
163	1.2(DL+COLL+CL-2)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	1300.9	0.0	1354.9	2803.2	0.0	1354.9
164	1.2(DL+COLL+CL-3)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	1300.9	0.0	1354.9	2803.2	0.0	1354.9
165	1.2(DL+COLL+CL-4)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	1300.9	0.0	1354.9	2803.2	0.0	1354.9
166	1.2(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	1247.1	632.7	1692.0	2749.4	632.7	1692.0
167	1.2(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	1247.1	632.7	1692.0	2749.4	632.7	1692.0
168	1.2(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	1247.1	632.7	1692.0	2749.4	632.7	1692.0
169	1.2(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	1247.1	632.7	1692.0	2749.4	632.7	1692.0
170	1.2(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	1320.7	272.0	1692.0	2823.0	272.0	1692.0
171	1.2(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	1320.7	272.0	1692.0	2823.0	272.0	1692.0
172	1.2(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	1320.7	272.0	1692.0	2823.0	272.0	1692.0
173	1.2(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	1320.7	272.0	1692.0	2823.0	272.0	1692.0
174	1.2(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	1051.8	852.9	1692.0	2554.1	852.9	1692.0
175	1.2(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	1051.8	852.9	1692.0	2554.1	852.9	1692.0
176	1.2(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	1051.8	852.9	1692.0	2554.1	852.9	1692.0
177	1.2(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	1051.8	852.9	1692.0	2554.1	852.9	1692.0
178	1.2(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	1125.4	492.1	1692.0	2627.7	492.1	1692.0
179	1.2(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	1125.4	492.1	1692.0	2627.7	492.1	1692.0
180	1.2(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	1125.4	492.1	1692.0	2627.7	492.1	1692.0
181	1.2(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	1125.4	492.1	1692.0	2627.7	492.1	1692.0
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	1565.3	393.0	2085.0	3067.6	393.0	2085.0
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	1565.3	393.0	2085.0	3067.6	393.0	2085.0
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	1565.3	393.0	2085.0	3067.6	393.0	2085.0
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	1565.3	393.0	2085.0	3067.6	393.0	2085.0
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	1369.4	393.0	1299.0	2871.7	393.0	1299.0
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-)+0.5(RLL+PLL / MLL	1369.4	393.0	1299.0	2871.7	393.0	1299.0
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	1369.4	393.0	1299.0	2871.7	393.0	1299.0
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	1369.4	393.0	1299.0	2871.7	393.0	1299.0
190	0.9(DL+COLL+CL-1)+1.6(WLL-1)	415.0	0.0	74.3	1917.3	0.0	74.3
191	0.9(DL+COLL+CL-2)+1.6(WLL-1)	415.0	0.0	74.3	1917.3	0.0	74.3
192	0.9(DL+COLL+CL-3)+1.6(WLL-1)	415.0	0.0	74.3	1917.3	0.0	74.3
193	0.9(DL+COLL+CL-4)+1.6(WLL-1)	415.0	0.0	74.3	1917.3	0.0	74.3
194	0.9(DL+COLL+CL-1)+1.6(WLL-2)	305.0	0.0	408.9	1807.3	0.0	408.9
195	0.9(DL+COLL+CL-2)+1.6(WLL-2)	305.0	0.0	408.9	1807.3	0.0	408.9
196	0.9(DL+COLL+CL-3)+1.6(WLL-2)	305.0	0.0	408.9	1807.3	0.0	408.9
197	0.9(DL+COLL+CL-4)+1.6(WLL-2)	305.0	0.0	408.9	1807.3	0.0	408.9
198	0.9(DL+COLL+CL-1)+1.6(WLR-1)	292.4	0.0	434.3	1794.7	0.0	434.3
199	0.9(DL+COLL+CL-2)+1.6(WLR-1)	292.4	0.0	434.3	1794.7	0.0	434.3
200	0.9(DL+COLL+CL-3)+1.6(WLR-1)	292.4	0.0	434.3	1794.7	0.0	434.3
201	0.9(DL+COLL+CL-4)+1.6(WLR-1)	292.4	0.0	434.3	1794.7	0.0	434.3
202	0.9(DL+COLL+CL-1)+1.6(WLR-2)	313.8	0.0	225.8	1816.1	0.0	225.8
203	0.9(DL+COLL+CL-2)+1.6(WLR-2)	313.8	0.0	225.8	1816.1	0.0	225.8
204	0.9(DL+COLL+CL-3)+1.6(WLR-2)	313.8	0.0	225.8	1816.1	0.0	225.8
205	0.9(DL+COLL+CL-4)+1.6(WLR-2)	313.8	0.0	225.8	1816.1	0.0	225.8
206	0.9(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)	260.0	632.7	562.8	1762.3	632.7	562.8
207	0.9(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)	260.0	632.7	562.8	1762.3	632.7	562.8
208	0.9(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)	260.0	632.7	562.8	1762.3	632.7	562.8
209	0.9(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)	260.0	632.7	562.8	1762.3	632.7	562.8
210	0.9(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)	333.5	272.0	562.8	1835.8	272.0	562.8
211	0.9(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)	333.5	272.0	562.8	1835.8	272.0	562.8
212	0.9(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)	333.5	272.0	562.8	1835.8	272.0	562.8
213	0.9(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)	333.5	272.0	562.8	1835.8	272.0	562.8
214	0.9(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)	64.7	852.9	562.8	1567.0	852.9	562.8
215	0.9(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)	64.7	852.9	562.8	1567.0	852.9	562.8
216	0.9(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)	64.7	852.9	562.8	1567.0	852.9	562.8
217	0.9(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)	64.7	852.9	562.8	1567.0	852.9	562.8
218	0.9(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG-4)	138.3	492.1	562.8	1640.6	492.1	562.8
219	0.9(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG-4)	138.3	492.1	562.8	1640.6	492.1	562.8
220	0.9(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG-4)	138.3	492.1	562.8	1640.6	492.1	562.8
221	0.9(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG-4)	138.3	492.1	562.8	1640.6	492.1	562.8
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')	578.2	393.0	955.9	2080.5	393.0	955.9
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')	578.2	393.0	955.9	2080.5	393.0	955.9
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')	578.2	393.0	955.9	2080.5	393.0	955.9
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')	578.2	393.0	955.9	2080.5	393.0	955.9
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')	382.3	393.0	169.8	1884.6	393.0	169.8
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-')	382.3	393.0	169.8	1884.6	393.0	169.8
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')	382.3	393.0	169.8	1884.6	393.0	169.8
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')	382.3	393.0	169.8	1884.6	393.0	169.8

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Allowable stress design

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
301	1.0(DL+COLL)	533.5932	0	0.0	-155.0	-625.4	2035.9	0.0	0.0	-155.0	-625.4
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL	2177.525	0	0.0	-758.3	-3071.0	3679.8	0.0	0.0	-758.3	-3071.0
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL / MLL	2177.525	0	0.0	-758.3	-3071.0	3679.8	0.0	0.0	-758.3	-3071.0
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL / MLL	2177.525	0	0.0	-758.3	-3071.0	3679.8	0.0	0.0	-758.3	-3071.0
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL / MLL	2177.525	0	0.0	-758.3	-3071.0	3679.8	0.0	0.0	-758.3	-3071.0
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL / MLL	2197.913	0	0.0	-506.6	-1945.9	3700.2	0.0	0.0	-506.6	-1945.9
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL / MLL	2197.913	0	0.0	-506.6	-1945.9	3700.2	0.0	0.0	-506.6	-1945.9
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL / MLL	2197.913	0	0.0	-506.6	-1945.9	3700.2	0.0	0.0	-506.6	-1945.9
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL / MLL	2197.913	0	0.0	-506.6	-1945.9	3700.2	0.0	0.0	-506.6	-1945.9
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL+PLL	1766.54	0	0.0	-607.5	-2459.6	3268.8	0.0	0.0	-607.5	-2459.6
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL+PLL	1766.54	0	0.0	-607.5	-2459.6	3268.8	0.0	0.0	-607.5	-2459.6
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL+PLL	1766.54	0	0.0	-607.5	-2459.6	3268.8	0.0	0.0	-607.5	-2459.6
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL+PLL	1766.54	0	0.0	-607.5	-2459.6	3268.8	0.0	0.0	-607.5	-2459.6
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL+PLL	1781.83	0	0.0	-418.7	-1615.8	3284.1	0.0	0.0	-418.7	-1615.8
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL+PLL	1781.83	0	0.0	-418.7	-1615.8	3284.1	0.0	0.0	-418.7	-1615.8
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL+PLL	1781.83	0	0.0	-418.7	-1615.8	3284.1	0.0	0.0	-418.7	-1615.8
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL+PLL	1781.83	0	0.0	-418.7	-1615.8	3284.1	0.0	0.0	-418.7	-1615.8
318	1.0(DL+COLL+CL-1)	533.5932	0	0.0	-155.0	-625.4	2035.9	0.0	0.0	-155.0	-625.4
319	1.0(DL+COLL+CL-2)	533.5932	0	0.0	-155.0	-625.4	2035.9	0.0	0.0	-155.0	-625.4
320	1.0(DL+COLL+CL-3)	533.5932	0	0.0	-155.0	-625.4	2035.9	0.0	0.0	-155.0	-625.4
321	1.0(DL+COLL+CL-4)	533.5932	0	0.0	-155.0	-625.4	2035.9	0.0	0.0	-155.0	-625.4
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	492.8328	0	0.0	-78.2	-227.2	1995.1	0.0	0.0	-78.2	-227.2
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	492.8328	0	0.0	-78.2	-227.2	1995.1	0.0	0.0	-78.2	-227.2
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	492.8328	0	0.0	-78.2	-227.2	1995.1	0.0	0.0	-78.2	-227.2
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	492.8328	0	0.0	-78.2	-227.2	1995.1	0.0	0.0	-78.2	-227.2
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	424.092	0	0.0	-142.8	-529.2	1926.4	0.0	0.0	-142.8	-529.2
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	424.092	0	0.0	-142.8	-529.2	1926.4	0.0	0.0	-142.8	-529.2
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	424.092	0	0.0	-142.8	-529.2	1926.4	0.0	0.0	-142.8	-529.2
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	424.092	0	0.0	-142.8	-529.2	1926.4	0.0	0.0	-142.8	-529.2
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	416.1924	0	0.0	-39.6	-2.1	1918.5	0.0	0.0	-39.6	-2.1
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	416.1924	0	0.0	-39.6	-2.1	1918.5	0.0	0.0	-39.6	-2.1
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	416.1924	0	0.0	-39.6	-2.1	1918.5	0.0	0.0	-39.6	-2.1
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	416.1924	0	0.0	-39.6	-2.1	1918.5	0.0	0.0	-39.6	-2.1
334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	429.5628	0	0.0	-120.2	-414.7	1931.9	0.0	0.0	-120.2	-414.7
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	429.5628	0	0.0	-120.2	-414.7	1931.9	0.0	0.0	-120.2	-414.7
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	429.5628	0	0.0	-120.2	-414.7	1931.9	0.0	0.0	-120.2	-414.7
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	429.5628	0	0.0	-120.2	-414.7	1931.9	0.0	0.0	-120.2	-414.7
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	395.926	45.5608	781.2	-155.0	-625.4	1898.2	45.6	781.2	-155.0	-625.4
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	395.926	45.5608	781.2	-155.0	-625.4	1898.2	45.6	781.2	-155.0	-625.4
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	395.926	45.5608	781.2	-155.0	-625.4	1898.2	45.6	781.2	-155.0	-625.4
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	395.926	45.5608	781.2	-155.0	-625.4	1898.2	45.6	781.2	-155.0	-625.4
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	441.9124	3.7336	435.5	-155.0	-625.4	1944.2	3.7	435.5	-155.0	-625.4
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	441.9124	3.7336	435.5	-155.0	-625.4	1944.2	3.7	435.5	-155.0	-625.4
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	441.9124	3.7336	435.5	-155.0	-625.4	1944.2	3.7	435.5	-155.0	-625.4
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	441.9124	3.7336	435.5	-155.0	-625.4	1944.2	3.7	435.5	-155.0	-625.4
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	273.8804	114.3788	719.8	-155.0	-625.4	1776.2	114.4	719.8	-155.0	-625.4
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	273.8804	114.3788	719.8	-155.0	-625.4	1776.2	114.4	719.8	-155.0	-625.4
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	273.8804	114.3788	719.8	-155.0	-625.4	1776.2	114.4	719.8	-155.0	-625.4
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	273.8804	114.3788	719.8	-155.0	-625.4	1776.2	114.4	719.8	-155.0	-625.4
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	319.8668	72.5516	374.1	-155.0	-625.4	1822.2	72.6	374.1	-155.0	-625.4
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	319.8668	72.5516	374.1	-155.0	-625.4	1822.2	72.6	374.1	-155.0	-625.4
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	319.8668	72.5516	374.1	-155.0	-625.4	1822.2	72.6	374.1	-155.0	-625.4
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	319.8668	72.5516	374.1	-155.0	-625.4	1822.2	72.6	374.1	-155.0	-625.4
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(RLL+	1743.62	0	0.0	-455.5	-1739.0	3245.9	0.0	0.0	-455.5	-1739.0
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(RLL+	1743.62	0	0.0	-455.5	-1739.0	3245.9	0.0	0.0	-455.5	-1739.0
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(RLL+	1743.62	0	0.0	-455.5	-1739.0	3245.9	0.0	0.0	-455.5	-1739.0
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(RLL+	1743.62	0	0.0	-455.5	-1739.0	3245.9	0.0	0.0	-455.5	-1739.0
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(RLL+	1692.06	0	0.0	-503.9	-1965.5	3194.4	0.0	0.0	-503.9	-1965.5
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(RLL+	1692.06	0	0.0	-503.9	-1965.5	3194.4	0.0	0.0	-503.9	-1965.5
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(RLL+	1692.06	0	0.0	-503.9	-1965.5	3194.4	0.0	0.0	-503.9	-1965.5
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(RLL+	1692.06	0	0.0	-503.9	-1965.5	3194.4	0.0	0.0	-503.9	-1965.5
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(RLL+	1686.14	0	0.0	-426.5	-1570.3	3188.4	0.0	0.0	-426.5	-1570.3
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(RLL+	1686.14	0	0.0	-426.5	-1570.3	3188.4	0.0	0.0	-426.5	-1570.3
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(RLL+	1686.14	0	0.0	-426.5	-1570.3	3188.4	0.0	0.0	-426.5	-1570.3
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(RLL+	1686.14	0	0.0	-426.5	-1570.3	3188.4	0.0	0.0	-426.5	-1570.3
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(RLL+	1696.16	0	0.0	-487.0	-1879.7	3198.5	0.0	0.0	-487.0	-1879.7
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(RLL+	1696.16	0	0.0	-487.0	-1879.7	3198.5	0.0	0.0	-487.0	-1879.7
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(RLL+	1696.16	0	0.0	-487.0	-1879.7	3198.5	0.0	0.0	-487.0	-1879.7
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(RLL+	1696.16	0	0.0	-487.0	-1879.7	3198.5	0.0	0.0	-487.0	-1879.7
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-WLLG	1670.94	34.1706	585.9	-513.1	-2037.7	3173.2	34.2	585.9	-513.1	-2037.7
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-WLLG	1670.94	34.1706	585.9	-513.1	-2037.7	3173.2	34.2	585.9	-513.1	-2037.7
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-WLLG	1670.94	34.1706	585.9	-513.1	-2037.7	3173.2	34.2	585.9	-513.1	-2037.7
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-WLLG	1670.94	34.1706	585.9	-513.1	-2037.7	3173.2	34.2	585.9	-513.1	-2037.7
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-WLLG	1705.43	2.8002	326.6	-513.1	-2037.7	3207.7	2.8	326.6	-513.1	-2037.7
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-WLLG	1705.43	2.8002	326.6	-513.1	-2037.7	3207.7	2.8	326.6	-513.1	-2037.7
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-WLLG	1705.43	2.8002	326.6	-513.1	-2037.7	3207.7	2.8	326.6	-513.1	-2037.7
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-WLLG	1705.43	2.8002	326.6	-513.1	-2037.7	3207.7	2.8	326.6	-513.1	-2037.7
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-WLLG	1579.40	85.7841	539.9	-513.1	-2037.7	3081.7	85.8	539.9	-513.1	-2037.7
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-WLLG	1579.40	85.7841	539.9	-513.1	-2037.7	3081.7	85.8	539.9	-513.1	-2037.7
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-WLLG	1579.40	85.7841	539.9	-513.1	-2037.7	3081.7	85.8	539.9	-513.1	-2037.7
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-WLLG	1579.40	85.7841	539.9	-513.1	-2037.7	3081.7	85.8	539.9	-513.1	-2037.7
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-WLLG	1613.89	54.4137	280.6	-513.1	-2037.7	3116.2	54.4	280.6	-513.1	-2037.7
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-WLLG	1613.89	54.4137	280.6	-513.1	-2037.7	3116.2	54.4	280.6	-513.1	-2037.7
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-WLLG	1613.89	54.4137	280.6	-513.1	-2037.7	3116.2	54.4	280.6	-513.1	-2037.7
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-WLLG	1613.89	54.4137	280.6	-513.1	-2037.7	3116.2	54.4	280.6	-513.1	-2037.7
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+')	607.05	-89.8929	-230.4	-244.8	-855.8	2109.3	-89.9	-230.4	-244.8	-855.8
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+")	607.05	-89.8929	-230.4	-244.8	-855.8	2109.3	-89.9	-230.4	-244.8	-855.8
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+')	607.05	-89.8929	-230.4	-244.8	-855.8	2109.3	-89.9	-230.4	-244.8	-855.8
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+')	607.05	-89.8929	-230.4	-244.8	-855.8	2109.3	-89.9	-230.4	-244.8	-855.8

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390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	460.13	89.8929	230.4	-65.1	-394.9	1962.4	89.9	230.4	-65.1	-394.9
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	460.13	89.8929	230.4	-65.1	-394.9	1962.4	89.9	230.4	-65.1	-394.9
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	460.13	89.8929	230.4	-65.1	-394.9	1962.4	89.9	230.4	-65.1	-394.9
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	460.13	89.8929	230.4	-65.1	-394.9	1962.4	89.9	230.4	-65.1	-394.9
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'+')+0	1825.61	-62.92503	-161.3	-576.0	-2199.0	3327.9	-62.9	-161.3	-576.0	-2199.0
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'+')+0	1825.61	-62.92503	-161.3	-576.0	-2199.0	3327.9	-62.9	-161.3	-576.0	-2199.0
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'+')+0	1825.61	-62.92503	-161.3	-576.0	-2199.0	3327.9	-62.9	-161.3	-576.0	-2199.0
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'+')+0	1825.61	-62.92503	-161.3	-576.0	-2199.0	3327.9	-62.9	-161.3	-576.0	-2199.0
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-')+0.7	1722.77	62.92503	161.3	-450.1	-1876.4	3225.1	62.9	161.3	-450.1	-1876.4
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-')+0.7	1722.77	62.92503	161.3	-450.1	-1876.4	3225.1	62.9	161.3	-450.1	-1876.4
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-')+0.7	1722.77	62.92503	161.3	-450.1	-1876.4	3225.1	62.9	161.3	-450.1	-1876.4
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-')+0.7	1722.77	62.92503	161.3	-450.1	-1876.4	3225.1	62.9	161.3	-450.1	-1876.4
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	279.40	0	0.0	-16.2	23.0	1781.7	0.0	0.0	-16.2	23.0
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	279.40	0	0.0	-16.2	23.0	1781.7	0.0	0.0	-16.2	23.0
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	279.40	0	0.0	-16.2	23.0	1781.7	0.0	0.0	-16.2	23.0
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	279.40	0	0.0	-16.2	23.0	1781.7	0.0	0.0	-16.2	23.0
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	210.6547	0	0.0	-80.8	-279.0	1712.9	0.0	0.0	-80.8	-279.0
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	210.6547	0	0.0	-80.8	-279.0	1712.9	0.0	0.0	-80.8	-279.0
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	210.6547	0	0.0	-80.8	-279.0	1712.9	0.0	0.0	-80.8	-279.0
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	210.6547	0	0.0	-80.8	-279.0	1712.9	0.0	0.0	-80.8	-279.0
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	202.7551	0	0.0	22.4	248.0	1705.0	0.0	0.0	22.4	248.0
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	202.7551	0	0.0	22.4	248.0	1705.0	0.0	0.0	22.4	248.0
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	202.7551	0	0.0	22.4	248.0	1705.0	0.0	0.0	22.4	248.0
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	202.7551	0	0.0	22.4	248.0	1705.0	0.0	0.0	22.4	248.0
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	216.1255	0	0.0	-58.2	-164.6	1718.4	0.0	0.0	-58.2	-164.6
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	216.1255	0	0.0	-58.2	-164.6	1718.4	0.0	0.0	-58.2	-164.6
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	216.1255	0	0.0	-58.2	-164.6	1718.4	0.0	0.0	-58.2	-164.6
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	216.1255	0	0.0	-58.2	-164.6	1718.4	0.0	0.0	-58.2	-164.6
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	182.4887	45.5608	781.2	-93.0	-375.2	1684.8	45.6	781.2	-93.0	-375.2
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	182.4887	45.5608	781.2	-93.0	-375.2	1684.8	45.6	781.2	-93.0	-375.2
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	182.4887	45.5608	781.2	-93.0	-375.2	1684.8	45.6	781.2	-93.0	-375.2
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	182.4887	45.5608	781.2	-93.0	-375.2	1684.8	45.6	781.2	-93.0	-375.2
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	228.4751	3.7336	435.5	-93.0	-375.2	1730.8	3.7	435.5	-93.0	-375.2
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	228.4751	3.7336	435.5	-93.0	-375.2	1730.8	3.7	435.5	-93.0	-375.2
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	228.4751	3.7336	435.5	-93.0	-375.2	1730.8	3.7	435.5	-93.0	-375.2
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	228.4751	3.7336	435.5	-93.0	-375.2	1730.8	3.7	435.5	-93.0	-375.2
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	60.44312	114.3788	719.8	-93.0	-375.2	1562.7	114.4	719.8	-93.0	-375.2
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	60.44312	114.3788	719.8	-93.0	-375.2	1562.7	114.4	719.8	-93.0	-375.2
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	60.44312	114.3788	719.8	-93.0	-375.2	1562.7	114.4	719.8	-93.0	-375.2
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	60.44312	114.3788	719.8	-93.0	-375.2	1562.7	114.4	719.8	-93.0	-375.2
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-WLLG-	106.4295	72.5516	374.1	-93.0	-375.2	1608.7	72.6	374.1	-93.0	-375.2
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	106.4295	72.5516	374.1	-93.0	-375.2	1608.7	72.6	374.1	-93.0	-375.2
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	106.4295	72.5516	374.1	-93.0	-375.2	1608.7	72.6	374.1	-93.0	-375.2
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	106.4295	72.5516	374.1	-93.0	-375.2	1608.7	72.6	374.1	-93.0	-375.2
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	388.72	-83.90004	-215.1	-176.9	-590.3	1891.0	-83.9	-215.1	-176.9	-590.3
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	388.72	-83.90004	-215.1	-176.9	-590.3	1891.0	-83.9	-215.1	-176.9	-590.3
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	388.72	-83.90004	-215.1	-176.9	-590.3	1891.0	-83.9	-215.1	-176.9	-590.3
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	388.72	-83.90004	-215.1	-176.9	-590.3	1891.0	-83.9	-215.1	-176.9	-590.3
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	364.26	-83.90004	-215.1	-130.8	-351.4	1866.6	-83.9	-215.1	-130.8	-351.4
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	323.02	-83.90004	-215.1	-169.6	-532.6	1825.3	-83.9	-215.1	-169.6	-532.6
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	318.28	-83.90004	-215.1	-107.6	-216.4	1820.6	-83.9	-215.1	-107.6	-216.4
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	326.30	-83.90004	-215.1	-156.0	-463.9	1828.6	-83.9	-215.1	-156.0	-463.9

Check for overturning and sliding:

Load Case	Load Combination	Load at bottom of foundation					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
301	1.0(DL+COLL)	2035.88	0.00	0.00	154.95	625.39	NA	8.79	NA	4.73
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL /	3679.81	0.00	0.00	758.29	3071.02	NA	3.24	NA	1.75
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL /	3679.81	0.00	0.00	758.29	3071.02	NA	3.24	NA	1.75
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL /	3679.81	0.00	0.00	758.29	3071.02	NA	3.24	NA	1.75
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL /	3679.81	0.00	0.00	758.29	3071.02	NA	3.24	NA	1.75
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL /	3700.20	0.00	0.00	506.57	1945.87	NA	5.13	NA	2.63
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL /	3700.20	0.00	0.00	506.57	1945.87	NA	5.13	NA	2.63
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL /	3700.20	0.00	0.00	506.57	1945.87	NA	5.13	NA	2.63
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL /	3700.20	0.00	0.00	506.57	1945.87	NA	5.13	NA	2.63
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL	3268.83	0.00	0.00	607.46	2459.61	NA	3.59	NA	1.94
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL	3268.83	0.00	0.00	607.46	2459.61	NA	3.59	NA	1.94
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL	3268.83	0.00	0.00	607.46	2459.61	NA	3.59	NA	1.94
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL	3268.83	0.00	0.00	607.46	2459.61	NA	3.59	NA	1.94
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL	3284.12	0.00	0.00	418.67	1615.75	NA	5.49	NA	2.82
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL	3284.12	0.00	0.00	418.67	1615.75	NA	5.49	NA	2.82
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL	3284.12	0.00	0.00	418.67	1615.75	NA	5.49	NA	2.82
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL	3284.12	0.00	0.00	418.67	1615.75	NA	5.49	NA	2.82
318	1.0(DL+COLL+CL-1)	2035.88	0.00	0.00	154.95	625.39	NA	8.79	NA	4.73
319	1.0(DL+COLL+CL-2)	2035.88	0.00	0.00	154.95	625.39	NA	8.79	NA	4.73
320	1.0(DL+COLL+CL-3)	2035.88	0.00	0.00	154.95	625.39	NA	8.79	NA	4.73
321	1.0(DL+COLL+CL-4)	2035.88	0.00	0.00	154.95	625.39	NA	8.79	NA	4.73
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	1995.12	0.00	0.00	78.15	227.20	NA	23.71	NA	9.19
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	1995.12	0.00	0.00	78.15	227.20	NA	23.71	NA	9.19
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	1995.12	0.00	0.00	78.15	227.20	NA	23.71	NA	9.19
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	1995.12	0.00	0.00	78.15	227.20	NA	23.71	NA	9.19
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	1926.38	0.00	0.00	142.76	529.20	NA	9.83	NA	4.86
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	1926.38	0.00	0.00	142.76	529.20	NA	9.83	NA	4.86
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	1926.38	0.00	0.00	142.76	529.20	NA	9.83	NA	4.86
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	1926.38	0.00	0.00	142.76	529.20	NA	9.83	NA	4.86
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	1918.48	0.00	0.00	39.57	2.15	NA	2410.02	NA	17.46
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	1918.48	0.00	0.00	39.57	2.15	NA	2410.02	NA	17.46
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	1918.48	0.00	0.00	39.57	2.15	NA	2410.02	NA	17.46
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	1918.48	0.00	0.00	39.57	2.15	NA	2410.02	NA	17.46

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334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	1931.85	0.00	0.00	120.21	414.73	NA	12.58	NA	5.79
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	1931.85	0.00	0.00	120.21	414.73	NA	12.58	NA	5.79
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	1931.85	0.00	0.00	120.21	414.73	NA	12.58	NA	5.79
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	1931.85	0.00	0.00	120.21	414.73	NA	12.58	NA	5.79
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	1898.22	45.56	781.17	154.95	625.39	6.56	8.20	16.67	4.41
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	1898.22	45.56	781.17	154.95	625.39	6.56	8.20	16.67	4.41
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	1898.22	45.56	781.17	154.95	625.39	6.56	8.20	16.67	4.41
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	1898.22	45.56	781.17	154.95	625.39	6.56	8.20	16.67	4.41
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	1944.20	3.73	435.51	154.95	625.39	12.05	8.39	208.29	4.52
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	1944.20	3.73	435.51	154.95	625.39	12.05	8.39	208.29	4.52
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	1944.20	3.73	435.51	154.95	625.39	12.05	8.39	208.29	4.52
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	1944.20	3.73	435.51	154.95	625.39	12.05	8.39	208.29	4.52
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	1776.17	114.38	719.82	154.95	625.39	6.66	7.67	6.21	4.13
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	1776.17	114.38	719.82	154.95	625.39	6.66	7.67	6.21	4.13
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	1776.17	114.38	719.82	154.95	625.39	6.66	7.67	6.21	4.13
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	1776.17	114.38	719.82	154.95	625.39	6.66	7.67	6.21	4.13
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	1822.16	72.55	374.15	154.95	625.39	13.15	7.87	10.05	4.23
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	1822.16	72.55	374.15	154.95	625.39	13.15	7.87	10.05	4.23
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	1822.16	72.55	374.15	154.95	625.39	13.15	7.87	10.05	4.23
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	1822.16	72.55	374.15	154.95	625.39	13.15	7.87	10.05	4.23
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(F	3245.91	0.00	0.00	455.46	1739.04	NA	5.04	NA	2.57
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(F	3245.91	0.00	0.00	455.46	1739.04	NA	5.04	NA	2.57
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(F	3245.91	0.00	0.00	455.46	1739.04	NA	5.04	NA	2.57
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(F	3245.91	0.00	0.00	455.46	1739.04	NA	5.04	NA	2.57
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(F	3194.35	0.00	0.00	503.91	1965.54	NA	4.39	NA	2.28
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(F	3194.35	0.00	0.00	503.91	1965.54	NA	4.39	NA	2.28
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(F	3194.35	0.00	0.00	503.91	1965.54	NA	4.39	NA	2.28
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(F	3194.35	0.00	0.00	503.91	1965.54	NA	4.39	NA	2.28
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(	3188.43	0.00	0.00	426.52	1570.25	NA	5.48	NA	2.69
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(	3188.43	0.00	0.00	426.52	1570.25	NA	5.48	NA	2.69
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(	3188.43	0.00	0.00	426.52	1570.25	NA	5.48	NA	2.69
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(	3188.43	0.00	0.00	426.52	1570.25	NA	5.48	NA	2.69
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(	3198.45	0.00	0.00	487.01	1879.68	NA	4.59	NA	2.36
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(	3198.45	0.00	0.00	487.01	1879.68	NA	4.59	NA	2.36
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(	3198.45	0.00	0.00	487.01	1879.68	NA	4.59	NA	2.36
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(	3198.45	0.00	0.00	487.01	1879.68	NA	4.59	NA	2.36
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-V	3173.23	34.17	585.88	513.06	2037.68	14.62	4.20	37.15	2.23
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-V	3173.23	34.17	585.88	513.06	2037.68	14.62	4.20	37.15	2.23
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-V	3173.23	34.17	585.88	513.06	2037.68	14.62	4.20	37.15	2.23
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-V	3173.23	34.17	585.88	513.06	2037.68	14.62	4.20	37.15	2.23
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-V	3207.72	2.80	326.63	513.06	2037.68	26.52	4.25	458.21	2.25
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-V	3207.72	2.80	326.63	513.06	2037.68	26.52	4.25	458.21	2.25
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-V	3207.72	2.80	326.63	513.06	2037.68	26.52	4.25	458.21	2.25
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-V	3207.72	2.80	326.63	513.06	2037.68	26.52	4.25	458.21	2.25
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-V	3081.69	85.78	539.86	513.06	2037.68	15.41	4.08	14.37	2.16
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-V	3081.69	85.78	539.86	513.06	2037.68	15.41	4.08	14.37	2.16
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-V	3081.69	85.78	539.86	513.06	2037.68	15.41	4.08	14.37	2.16
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-V	3081.69	85.78	539.86	513.06	2037.68	15.41	4.08	14.37	2.16
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-V	3116.18	54.41	280.61	513.06	2037.68	29.98	4.13	22.91	2.19
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-V	3116.18	54.41	280.61	513.06	2037.68	29.98	4.13	22.91	2.19
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-V	3116.18	54.41	280.61	513.06	2037.68	29.98	4.13	22.91	2.19
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-V	3116.18	54.41	280.61	513.06	2037.68	29.98	4.13	22.91	2.19
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+	2109.34	-89.89	-230.45	244.85	855.84	24.71	6.65	9.39	3.10
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+	2109.34	-89.89	-230.45	244.85	855.84	24.71	6.65	9.39	3.10
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+	2109.34	-89.89	-230.45	244.85	855.84	24.71	6.65	9.39	3.10
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+	2109.34	-89.89	-230.45	244.85	855.84	24.71	6.65	9.39	3.10
390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	1962.42	89.89	230.45	65.06	394.94	22.99	13.42	8.73	10.86
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	1962.42	89.89	230.45	65.06	394.94	22.99	13.42	8.73	10.86
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	1962.42	89.89	230.45	65.06	394.94	22.99	13.42	8.73	10.86
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	1962.42	89.89	230.45	65.06	394.94	22.99	13.42	8.73	10.86
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'	3327.90	-62.93	-161.31	575.99	2198.99	55.70	4.09	21.15	2.08
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'	3327.90	-62.93	-161.31	575.99	2198.99	55.70	4.09	21.15	2.08
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'	3327.90	-62.93	-161.31	575.99	2198.99	55.70	4.09	21.15	2.08
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'	3327.90	-62.93	-161.31	575.99	2198.99	55.70	4.09	21.15	2.08
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-	3225.06	62.93	161.31	450.14	1876.37	53.98	4.64	20.50	2.58
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-	3225.06	62.93	161.31	450.14	1876.37	53.98	4.64	20.50	2.58
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-	3225.06	62.93	161.31	450.14	1876.37	53.98	4.64	20.50	2.58
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-'	3225.06	62.93	161.31	450.14	1876.37	53.98	4.64	20.50	2.58
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	1781.69	0.00	0.00	16.17	22.96	NA	209.53	NA	39.66
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	1781.69	0.00	0.00	16.17	22.96	NA	209.53	NA	39.66
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	1781.69	0.00	0.00	16.17	22.96	NA	209.53	NA	39.66
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	1781.69	0.00	0.00	16.17	22.96	NA	209.53	NA	39.66
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	1712.94	0.00	0.00	80.77	279.04	NA	16.57	NA	7.63
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	1712.94	0.00	0.00	80.77	279.04	NA	16.57	NA	7.63
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	1712.94	0.00	0.00	80.77	279.04	NA	16.57	NA	7.63
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	1712.94	0.00	0.00	80.77	279.04	NA	16.57	NA	7.63
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	1705.05	0.00	0.00	22.41	248.01	NA	18.56	NA	27.39
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	1705.05	0.00	0.00	22.41	248.01	NA	18.56	NA	27.39
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	1705.05	0.00	0.00	22.41	248.01	NA	18.56	NA	27.39
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	1705.05	0.00	0.00	22.41	248.01	NA	18.56	NA	27.39
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	1718.42	0.00	0.00	58.23	164.57	NA	28.19	NA	10.62
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	1718.42	0.00	0.00	58.23	164.57	NA	28.19	NA	10.62
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	1718.42	0.00	0.00	58.23	164.57	NA	28.19	NA	10.62
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	1718.42	0.00	0.00	58.23	164.57	NA	28.19	NA	10.62
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	1684.78	45.56	781.17	92.97	375.23	5.82	12.12	14.79	6.52
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	1684.78	45.56	781.17	92.97	375.23	5.82	12.12	14.79	6.52
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	1684.78	45.56	781.17	92.97	375.23	5.82	12.12	14.79	6.52
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	1684.78	45.56	781.17	92.97	375.23	5.82	12.12	14.79	6.52
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	1730.77	3.73	435.51	92.97	375.23	10.73	12.45	185.43	6.70
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	1730.77	3.73	435.51	92.97	375.23	10.73	12.45	185.43	6.70
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	1730.77	3.73	435.51	92.97	375.23	10.73	12.45	185.43	6.70
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	1730.77	3.73	435.51	92.97	375.23	10.73	12.45	185.43	6.70
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	1562.73	114.38	719.82	92.97	375.23	5.86	11.24	5.47	6.05
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	1562.73	114.38	719.82	92.97	375.23	5.86	11.24	5.47	6.05
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	1562.73	114.38	719.82	92.97	375.23	5.86	11.24	5.47	6.05
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	1562.73	114.38	719.82	92.97	375.23	5.86	11.24	5.47	6.05
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-W	1608.72	72.55	374.15	92.97	375.23	11.61	11.58	8.87	6.23
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	1608.72	72.55	374.15	92.97	375.23	11.61	11.58	8.87	6.23
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	1608.72	72.55	374.15	92.97	375.23	11.61	11.58	8.87	6.23
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	1608.72	72.55	374.15	92.97	375.23	11.61	11.58	8.87	6.23
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	1891.01	-83.90	-215.08	176.87	590.32	23.74	8.65	9.02	3.85
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	1891.01	-83.90	-215.08	176.87	590.32	23.74	8.65	9.02	3.85
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	1891.01	-83.90	-215.08	176.87	590.32	23.74	8.65	9.02	3.85
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	1891.01	-83.90	-215.08	176.87	590.32	23.74	8.65	9.02	3.85
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	1866.55	-83.90	-215.08	130.79	351.40	23.43	14.34	8.90	5.14
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	1825.31	-83.90	-215.08	169.55	532.60	22.91	9.25	8.70	3.88
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	1820.57	-83.90	-215.08	107.64	216.37	22.85	22.72	8.68	6.09
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	1828.59	-83.90	-215.08	156.03	463.92	22.95	10.64	8.72	4.22
						Min.=	**5.82**	**3.24**	**5.47**	**1.75**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F5

Load Combination & Calculation of Base pressure

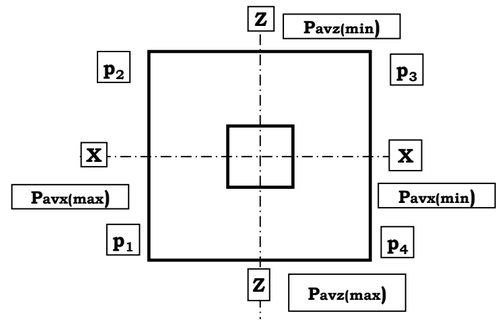
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 436 kN/m^2 For WL/SL case = 536 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
301	1.0(DL+COLL)	2035.88	0.00	-625.39	56.55	0.00	17.37	73.92		39.18	73.92	39.18	37.92	56.55	56.55	73.92	39.18	O.K
302	1.0(DL+COLL+CL-1+TL1)+1.(	3679.81	0.00	-3071.02	102.22	0.00	85.31	187.52		16.91	187.52	16.91	151.52	102.22	102.22	187.52	16.91	O.K
303	1.0(DL+COLL+CL-2+TL1)+1.(	3679.81	0.00	-3071.02	102.22	0.00	85.31	187.52		16.91	187.52	16.91	151.52	102.22	102.22	187.52	16.91	O.K
304	1.0(DL+COLL+CL-3+TL1)+1.(	3679.81	0.00	-3071.02	102.22	0.00	85.31	187.52		16.91	187.52	16.91	151.52	102.22	102.22	187.52	16.91	O.K
305	1.0(DL+COLL+CL-4+TL1)+1.(	3679.81	0.00	-3071.02	102.22	0.00	85.31	187.52		16.91	187.52	16.91	151.52	102.22	102.22	187.52	16.91	O.K
306	1.0(DL+COLL+CL-1+TL2)+1.(	3700.20	0.00	-1945.87	102.78	0.00	54.05	156.84		48.73	156.84	48.73	120.84	102.79	102.79	156.84	48.73	O.K
307	1.0(DL+COLL+CL-2+TL2)+1.(	3700.20	0.00	-1945.87	102.78	0.00	54.05	156.84		48.73	156.84	48.73	120.84	102.79	102.79	156.84	48.73	O.K
308	1.0(DL+COLL+CL-3+TL2)+1.(	3700.20	0.00	-1945.87	102.78	0.00	54.05	156.84		48.73	156.84	48.73	120.84	102.79	102.79	156.84	48.73	O.K
309	1.0(DL+COLL+CL-4+TL2)+1.(	3700.20	0.00	-1945.87	102.78	0.00	54.05	156.84		48.73	156.84	48.73	120.84	102.79	102.79	156.84	48.73	O.K
310	1.0(DL+COLL+CL-1)+0.75(TL	3268.83	0.00	-2459.61	90.80	0.00	68.32	159.12		22.48	159.12	22.48	123.12	90.80	90.80	159.12	22.48	O.K
311	1.0(DL+COLL+CL-2)+0.75(TL	3268.83	0.00	-2459.61	90.80	0.00	68.32	159.12		22.48	159.12	22.48	123.12	90.80	90.80	159.12	22.48	O.K
312	1.0(DL+COLL+CL-3)+0.75(TL	3268.83	0.00	-2459.61	90.80	0.00	68.32	159.12		22.48	159.12	22.48	123.12	90.80	90.80	159.12	22.48	O.K
313	1.0(DL+COLL+CL-4)+0.75(TL	3268.83	0.00	-2459.61	90.80	0.00	68.32	159.12		22.48	159.12	22.48	123.12	90.80	90.80	159.12	22.48	O.K
314	1.0(DL+COLL+CL-1)+0.75(TL	3284.12	0.00	-1615.75	91.23	0.00	44.88	136.11		46.34	136.11	46.34	100.11	91.23	91.23	136.11	46.34	O.K
315	1.0(DL+COLL+CL-2)+0.75(TL	3284.12	0.00	-1615.75	91.23	0.00	44.88	136.11		46.34	136.11	46.34	100.11	91.23	91.23	136.11	46.34	O.K
316	1.0(DL+COLL+CL-3)+0.75(TL	3284.12	0.00	-1615.75	91.23	0.00	44.88	136.11		46.34	136.11	46.34	100.11	91.23	91.23	136.11	46.34	O.K

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317	1.0(DL+COLL+CL-4)+0.75(TL	3284.12	0.00	-1615.75	91.23	0.00	44.88	136.11		46.34	136.11	46.34	100.11	91.23	91.23	136.11	46.34	O.K
318	1.0(DL+COLL+CL-1)	2035.88	0.00	-625.39	56.55	0.00	17.37	73.92		39.18	73.92	39.18	37.92	56.55	56.55	73.92	39.18	O.K
319	1.0(DL+COLL+CL-2)	2035.88	0.00	-625.39	56.55	0.00	17.37	73.92		39.18	73.92	39.18	37.92	56.55	56.55	73.92	39.18	O.K
320	1.0(DL+COLL+CL-3)	2035.88	0.00	-625.39	56.55	0.00	17.37	73.92		39.18	73.92	39.18	37.92	56.55	56.55	73.92	39.18	O.K
321	1.0(DL+COLL+CL-4)	2035.88	0.00	-625.39	56.55	0.00	17.37	73.92		39.18	73.92	39.18	37.92	56.55	56.55	73.92	39.18	O.K
322	1.0(DL+COLL+CL-1)+1.0(WLI	1995.12	0.00	-227.20	55.42	0.00	6.31	61.73		49.11	61.73	49.11	25.73	55.42	55.42	61.73	49.11	O.K
323	1.0(DL+COLL+CL-2)+1.0(WLI	1995.12	0.00	-227.20	55.42	0.00	6.31	61.73		49.11	61.73	49.11	25.73	55.42	55.42	61.73	49.11	O.K
324	1.0(DL+COLL+CL-3)+1.0(WLI	1995.12	0.00	-227.20	55.42	0.00	6.31	61.73		49.11	61.73	49.11	25.73	55.42	55.42	61.73	49.11	O.K
325	1.0(DL+COLL+CL-4)+1.0(WLI	1995.12	0.00	-227.20	55.42	0.00	6.31	61.73		49.11	61.73	49.11	25.73	55.42	55.42	61.73	49.11	O.K
326	1.0(DL+COLL+CL-1)+1.0(WLI	1926.38	0.00	-529.20	53.51	0.00	14.70	68.21		38.81	68.21	38.81	32.21	53.51	53.51	68.21	38.81	O.K
327	1.0(DL+COLL+CL-2)+1.0(WLI	1926.38	0.00	-529.20	53.51	0.00	14.70	68.21		38.81	68.21	38.81	32.21	53.51	53.51	68.21	38.81	O.K
328	1.0(DL+COLL+CL-3)+1.0(WLI	1926.38	0.00	-529.20	53.51	0.00	14.70	68.21		38.81	68.21	38.81	32.21	53.51	53.51	68.21	38.81	O.K
329	1.0(DL+COLL+CL-4)+1.0(WLI	1926.38	0.00	-529.20	53.51	0.00	14.70	68.21		38.81	68.21	38.81	32.21	53.51	53.51	68.21	38.81	O.K
330	1.0(DL+COLL+CL-1)+1.0(WLI	1918.48	0.00	-2.15	53.29	0.00	0.06	53.35		53.23	53.35	53.23	17.35	53.29	53.29	53.35	53.23	O.K
331	1.0(DL+COLL+CL-2)+1.0(WLI	1918.48	0.00	-2.15	53.29	0.00	0.06	53.35		53.23	53.35	53.23	17.35	53.29	53.29	53.35	53.23	O.K
332	1.0(DL+COLL+CL-3)+1.0(WLI	1918.48	0.00	-2.15	53.29	0.00	0.06	53.35		53.23	53.35	53.23	17.35	53.29	53.29	53.35	53.23	O.K
333	1.0(DL+COLL+CL-4)+1.0(WLI	1918.48	0.00	-2.15	53.29	0.00	0.06	53.35		53.23	53.35	53.23	17.35	53.29	53.29	53.35	53.23	O.K
334	1.0(DL+COLL+CL-1)+1.0(WLI	1931.85	0.00	-414.73	53.66	0.00	11.52	65.18		42.14	65.18	42.14	29.18	53.66	53.66	65.18	42.14	O.K
335	1.0(DL+COLL+CL-2)+1.0(WLI	1931.85	0.00	-414.73	53.66	0.00	11.52	65.18		42.14	65.18	42.14	29.18	53.66	53.66	65.18	42.14	O.K
336	1.0(DL+COLL+CL-3)+1.0(WLI	1931.85	0.00	-414.73	53.66	0.00	11.52	65.18		42.14	65.18	42.14	29.18	53.66	53.66	65.18	42.14	O.K
337	1.0(DL+COLL+CL-4)+1.0(WLI	1931.85	0.00	-414.73	53.66	0.00	11.52	65.18		42.14	65.18	42.14	29.18	53.66	53.66	65.18	42.14	O.K
338	1.0(DL+COLL+CL-1)+1.0(WLI	1898.22	781.17	-625.39	52.73	21.70	17.37	91.80		57.06	48.40	13.66	55.80	74.43	31.03	70.10	35.36	O.K
339	1.0(DL+COLL+CL-2)+1.0(WLI	1898.22	781.17	-625.39	52.73	21.70	17.37	91.80		57.06	48.40	13.66	55.80	74.43	31.03	70.10	35.36	O.K
340	1.0(DL+COLL+CL-3)+1.0(WLI	1898.22	781.17	-625.39	52.73	21.70	17.37	91.80		57.06	48.40	13.66	55.80	74.43	31.03	70.10	35.36	O.K
341	1.0(DL+COLL+CL-4)+1.0(WLI	1898.22	781.17	-625.39	52.73	21.70	17.37	91.80		57.06	48.40	13.66	55.80	74.43	31.03	70.10	35.36	O.K
342	1.0(DL+COLL+CL-1)+1.0(WLI	1944.20	435.51	-625.39	54.01	12.10	17.37	83.47		48.73	59.28	24.54	47.47	66.10	41.91	71.38	36.64	O.K
343	1.0(DL+COLL+CL-2)+1.0(WLI	1944.20	435.51	-625.39	54.01	12.10	17.37	83.47		48.73	59.28	24.54	47.47	66.10	41.91	71.38	36.64	O.K
344	1.0(DL+COLL+CL-3)+1.0(WLI	1944.20	435.51	-625.39	54.01	12.10	17.37	83.47		48.73	59.28	24.54	47.47	66.10	41.91	71.38	36.64	O.K
345	1.0(DL+COLL+CL-4)+1.0(WLI	1944.20	435.51	-625.39	54.01	12.10	17.37	83.47		48.73	59.28	24.54	47.47	66.10	41.91	71.38	36.64	O.K
346	1.0(DL+COLL+CL-1)+1.0(WLI	1776.17	719.82	-625.39	49.34	19.99	17.37	86.70		51.96	46.72	11.97	50.70	69.33	29.35	66.71	31.97	O.K
347	1.0(DL+COLL+CL-2)+1.0(WLI	1776.17	719.82	-625.39	49.34	19.99	17.37	86.70		51.96	46.72	11.97	50.70	69.33	29.35	66.71	31.97	O.K

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348	1.0(DL+COLL+CL-3)+1.0(WL	1776.17	719.82	-625.39	49.34	19.99	17.37	86.70		51.96	46.72	11.97	50.70	69.33	29.35	66.71	31.97	O.K
349	1.0(DL+COLL+CL-4)+1.0(WL	1776.17	719.82	-625.39	49.34	19.99	17.37	86.70		51.96	46.72	11.97	50.70	69.33	29.35	66.71	31.97	O.K
350	1.0(DL+COLL+CL-1)+1.0(WL	1822.16	374.15	-625.39	50.62	10.39	17.37	78.38		43.64	57.59	22.85	42.38	61.01	40.22	67.99	33.25	O.K
351	1.0(DL+COLL+CL-2)+1.0(WL	1822.16	374.15	-625.39	50.62	10.39	17.37	78.38		43.64	57.59	22.85	42.38	61.01	40.22	67.99	33.25	O.K
352	1.0(DL+COLL+CL-3)+1.0(WL	1822.16	374.15	-625.39	50.62	10.39	17.37	78.38		43.64	57.59	22.85	42.38	61.01	40.22	67.99	33.25	O.K
353	1.0(DL+COLL+CL-4)+1.0(WL	1822.16	374.15	-625.39	50.62	10.39	17.37	78.38		43.64	57.59	22.85	42.38	61.01	40.22	67.99	33.25	O.K
354	1.0(DL+COLL+CL-1)+0.75(WI	3245.91	0.00	-1739.04	90.16	0.00	48.31	138.47		41.86	138.47	41.86	102.47	90.17	90.17	138.47	41.86	O.K
355	1.0(DL+COLL+CL-2)+0.75(WI	3245.91	0.00	-1739.04	90.16	0.00	48.31	138.47		41.86	138.47	41.86	102.47	90.17	90.17	138.47	41.86	O.K
356	1.0(DL+COLL+CL-3)+0.75(WI	3245.91	0.00	-1739.04	90.16	0.00	48.31	138.47		41.86	138.47	41.86	102.47	90.17	90.17	138.47	41.86	O.K
357	1.0(DL+COLL+CL-4)+0.75(WI	3245.91	0.00	-1739.04	90.16	0.00	48.31	138.47		41.86	138.47	41.86	102.47	90.17	90.17	138.47	41.86	O.K
358	1.0(DL+COLL+CL-1)+0.75(WI	3194.35	0.00	-1965.54	88.73	0.00	54.60	143.33		34.13	143.33	34.13	107.33	88.73	88.73	143.33	34.13	O.K
359	1.0(DL+COLL+CL-2)+0.75(WI	3194.35	0.00	-1965.54	88.73	0.00	54.60	143.33		34.13	143.33	34.13	107.33	88.73	88.73	143.33	34.13	O.K
360	1.0(DL+COLL+CL-3)+0.75(WI	3194.35	0.00	-1965.54	88.73	0.00	54.60	143.33		34.13	143.33	34.13	107.33	88.73	88.73	143.33	34.13	O.K
361	1.0(DL+COLL+CL-4)+0.75(WI	3194.35	0.00	-1965.54	88.73	0.00	54.60	143.33		34.13	143.33	34.13	107.33	88.73	88.73	143.33	34.13	O.K
362	1.0(DL+COLL+CL-1)+0.75(WI	3188.43	0.00	-1570.25	88.57	0.00	43.62	132.19		44.95	132.19	44.95	96.19	88.57	88.57	132.19	44.95	O.K
363	1.0(DL+COLL+CL-2)+0.75(WI	3188.43	0.00	-1570.25	88.57	0.00	43.62	132.19		44.95	132.19	44.95	96.19	88.57	88.57	132.19	44.95	O.K
364	1.0(DL+COLL+CL-3)+0.75(WI	3188.43	0.00	-1570.25	88.57	0.00	43.62	132.19		44.95	132.19	44.95	96.19	88.57	88.57	132.19	44.95	O.K
365	1.0(DL+COLL+CL-4)+0.75(WI	3188.43	0.00	-1570.25	88.57	0.00	43.62	132.19		44.95	132.19	44.95	96.19	88.57	88.57	132.19	44.95	O.K
366	1.0(DL+COLL+CL-1)+0.75(WI	3198.45	0.00	-1879.68	88.85	0.00	52.21	141.06		36.63	141.06	36.63	105.06	88.85	88.85	141.06	36.63	O.K
367	1.0(DL+COLL+CL-2)+0.75(WI	3198.45	0.00	-1879.68	88.85	0.00	52.21	141.06		36.63	141.06	36.63	105.06	88.85	88.85	141.06	36.63	O.K
368	1.0(DL+COLL+CL-3)+0.75(WI	3198.45	0.00	-1879.68	88.85	0.00	52.21	141.06		36.63	141.06	36.63	105.06	88.85	88.85	141.06	36.63	O.K
369	1.0(DL+COLL+CL-4)+0.75(WI	3198.45	0.00	-1879.68	88.85	0.00	52.21	141.06		36.63	141.06	36.63	105.06	88.85	88.85	141.06	36.63	O.K
370	1.0(DL+COLL+CL-1)+0.75(WI	3173.23	585.88	-2037.68	88.15	16.27	56.60	161.02		47.82	128.47	15.27	125.02	104.42	71.87	144.75	31.55	O.K
371	1.0(DL+COLL+CL-2)+0.75(WI	3173.23	585.88	-2037.68	88.15	16.27	56.60	161.02		47.82	128.47	15.27	125.02	104.42	71.87	144.75	31.55	O.K
372	1.0(DL+COLL+CL-3)+0.75(WI	3173.23	585.88	-2037.68	88.15	16.27	56.60	161.02		47.82	128.47	15.27	125.02	104.42	71.87	144.75	31.55	O.K
373	1.0(DL+COLL+CL-4)+0.75(WI	3173.23	585.88	-2037.68	88.15	16.27	56.60	161.02		47.82	128.47	15.27	125.02	104.42	71.87	144.75	31.55	O.K
374	1.0(DL+COLL+CL-1)+0.75(WI	3207.72	326.63	-2037.68	89.10	9.07	56.60	154.78		41.57	136.63	23.43	118.78	98.18	80.03	145.71	32.50	O.K
375	1.0(DL+COLL+CL-2)+0.75(WI	3207.72	326.63	-2037.68	89.10	9.07	56.60	154.78		41.57	136.63	23.43	118.78	98.18	80.03	145.71	32.50	O.K
376	1.0(DL+COLL+CL-3)+0.75(WI	3207.72	326.63	-2037.68	89.10	9.07	56.60	154.78		41.57	136.63	23.43	118.78	98.18	80.03	145.71	32.50	O.K
377	1.0(DL+COLL+CL-4)+0.75(WI	3207.72	326.63	-2037.68	89.10	9.07	56.60	154.78		41.57	136.63	23.43	118.78	98.18	80.03	145.71	32.50	O.K
378	1.0(DL+COLL+CL-1)+0.75(WI	3081.69	539.86	-2037.68	85.60	15.00	56.60	157.20		44.00	127.21	14.00	121.20	100.60	70.61	142.21	29.00	O.K

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379	1.0(DL+COLL+CL-2)+0.75(WI	3081.69	539.86	-2037.68	85.60	15.00	56.60	157.20		44.00	127.21	14.00	121.20	100.60	70.61	142.21	29.00	O.K
380	1.0(DL+COLL+CL-3)+0.75(WI	3081.69	539.86	-2037.68	85.60	15.00	56.60	157.20		44.00	127.21	14.00	121.20	100.60	70.61	142.21	29.00	O.K
381	1.0(DL+COLL+CL-4)+0.75(WI	3081.69	539.86	-2037.68	85.60	15.00	56.60	157.20		44.00	127.21	14.00	121.20	100.60	70.61	142.21	29.00	O.K
382	1.0(DL+COLL+CL-1)+0.75(WI	3116.18	280.61	-2037.68	86.56	7.79	56.60	150.96		37.75	135.37	22.16	114.96	94.36	78.77	143.17	29.96	O.K
383	1.0(DL+COLL+CL-2)+0.75(WI	3116.18	280.61	-2037.68	86.56	7.79	56.60	150.96		37.75	135.37	22.16	114.96	94.36	78.77	143.17	29.96	O.K
384	1.0(DL+COLL+CL-3)+0.75(WI	3116.18	280.61	-2037.68	86.56	7.79	56.60	150.96		37.75	135.37	22.16	114.96	94.36	78.77	143.17	29.96	O.K
385	1.0(DL+COLL+CL-4)+0.75(WI	3116.18	280.61	-2037.68	86.56	7.79	56.60	150.96		37.75	135.37	22.16	114.96	94.36	78.77	143.17	29.96	O.K
386	1.0(DL+COLL+CL-1)+0.75(SL	2109.34	-230.45	-855.84	58.59	6.40	23.77	88.77		41.22	75.96	28.42	52.77	65.00	52.19	82.37	34.82	O.K
387	1.0(DL+COLL+CL-2)+0.75(SL	2109.34	-230.45	-855.84	58.59	6.40	23.77	88.77		41.22	75.96	28.42	52.77	65.00	52.19	82.37	34.82	O.K
388	1.0(DL+COLL+CL-3)+0.75(SL	2109.34	-230.45	-855.84	58.59	6.40	23.77	88.77		41.22	75.96	28.42	52.77	65.00	52.19	82.37	34.82	O.K
389	1.0(DL+COLL+CL-4)+0.75(SL	2109.34	-230.45	-855.84	58.59	6.40	23.77	88.77		41.22	75.96	28.42	52.77	65.00	52.19	82.37	34.82	O.K
390	1.0(DL+COLL+CL-1)+0.75(SL	1962.42	230.45	-394.94	54.51	6.40	10.97	71.88		49.94	59.08	37.14	35.88	60.91	48.11	65.48	43.54	O.K
391	1.0(DL+COLL+CL-2)+0.75(SL	1962.42	230.45	-394.94	54.51	6.40	10.97	71.88		49.94	59.08	37.14	35.88	60.91	48.11	65.48	43.54	O.K
392	1.0(DL+COLL+CL-3)+0.75(SL	1962.42	230.45	-394.94	54.51	6.40	10.97	71.88		49.94	59.08	37.14	35.88	60.91	48.11	65.48	43.54	O.K
393	1.0(DL+COLL+CL-4)+0.75(SL	1962.42	230.45	-394.94	54.51	6.40	10.97	71.88		49.94	59.08	37.14	35.88	60.91	48.11	65.48	43.54	O.K
394	1.0(DL+COLL+CL-1)+0.525(S	3327.90	-161.31	-2198.99	92.44	4.48	61.08	158.01		35.84	149.04	26.88	122.01	96.93	87.96	153.53	31.36	O.K
395	1.0(DL+COLL+CL-2)+0.525(S	3327.90	-161.31	-2198.99	92.44	4.48	61.08	158.01		35.84	149.04	26.88	122.01	96.93	87.96	153.53	31.36	O.K
396	1.0(DL+COLL+CL-3)+0.525(S	3327.90	-161.31	-2198.99	92.44	4.48	61.08	158.01		35.84	149.04	26.88	122.01	96.93	87.96	153.53	31.36	O.K
397	1.0(DL+COLL+CL-4)+0.525(S	3327.90	-161.31	-2198.99	92.44	4.48	61.08	158.01		35.84	149.04	26.88	122.01	96.93	87.96	153.53	31.36	O.K
398	1.0(DL+COLL+CL-1)+0.525(S	3225.06	161.31	-1876.37	89.58	4.48	52.12	146.19		41.94	137.23	32.98	110.19	94.07	85.11	141.71	37.46	O.K
399	1.0(DL+COLL+CL-2)+0.525(S	3225.06	161.31	-1876.37	89.58	4.48	52.12	146.19		41.94	137.23	32.98	110.19	94.07	85.11	141.71	37.46	O.K
400	1.0(DL+COLL+CL-3)+0.525(S	3225.06	161.31	-1876.37	89.58	4.48	52.12	146.19		41.94	137.23	32.98	110.19	94.07	85.11	141.71	37.46	O.K
401	1.0(DL+COLL+CL-4)+0.525SI	3225.06	161.31	-1876.37	89.58	4.48	52.12	146.19		41.94	137.23	32.98	110.19	94.07	85.11	141.71	37.46	O.K
402	0.6(DL+COLL+CL-1)+1.0(WLI	1781.69	0.00	22.96	49.49	0.00	0.64	50.13		48.85	50.13	48.85	14.13	49.49	49.49	50.13	48.85	O.K
403	0.6(DL+COLL+CL-2)+1.0(WLI	1781.69	0.00	22.96	49.49	0.00	0.64	50.13		48.85	50.13	48.85	14.13	49.49	49.49	50.13	48.85	O.K
404	0.6(DL+COLL+CL-3)+1.0(WLI	1781.69	0.00	22.96	49.49	0.00	0.64	50.13		48.85	50.13	48.85	14.13	49.49	49.49	50.13	48.85	O.K
405	0.6(DL+COLL+CL-4)+1.0(WLI	1781.69	0.00	22.96	49.49	0.00	0.64	50.13		48.85	50.13	48.85	14.13	49.49	49.49	50.13	48.85	O.K
406	0.6(DL+COLL+CL-1)+1.0(WLI	1712.94	0.00	-279.04	47.58	0.00	7.75	55.33		39.83	55.33	39.83	19.33	47.58	47.58	55.33	39.83	O.K
407	0.6(DL+COLL+CL-2)+1.0(WLI	1712.94	0.00	-279.04	47.58	0.00	7.75	55.33		39.83	55.33	39.83	19.33	47.58	47.58	55.33	39.83	O.K
408	0.6(DL+COLL+CL-3)+1.0(WLI	1712.94	0.00	-279.04	47.58	0.00	7.75	55.33		39.83	55.33	39.83	19.33	47.58	47.58	55.33	39.83	O.K
409	0.6(DL+COLL+CL-4)+1.0(WLI	1712.94	0.00	-279.04	47.58	0.00	7.75	55.33		39.83	55.33	39.83	19.33	47.58	47.58	55.33	39.83	O.K

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410	0.6(DL+COLL+CL-1)+1.0(WLI	1705.05	0.00	248.01	47.36	0.00	6.89	54.25		40.47	54.25	40.47	18.25	47.36	47.36	54.25	40.47	O.K
411	0.6(DL+COLL+CL-2)+1.0(WLI	1705.05	0.00	248.01	47.36	0.00	6.89	54.25		40.47	54.25	40.47	18.25	47.36	47.36	54.25	40.47	O.K
412	0.6(DL+COLL+CL-3)+1.0(WLI	1705.05	0.00	248.01	47.36	0.00	6.89	54.25		40.47	54.25	40.47	18.25	47.36	47.36	54.25	40.47	O.K
413	0.6(DL+COLL+CL-4)+1.0(WLI	1705.05	0.00	248.01	47.36	0.00	6.89	54.25		40.47	54.25	40.47	18.25	47.36	47.36	54.25	40.47	O.K
414	0.6(DL+COLL+CL-1)+1.0(WLI	1718.42	0.00	-164.57	47.73	0.00	4.57	52.31		43.16	52.31	43.16	16.31	47.74	47.74	52.31	43.16	O.K
415	0.6(DL+COLL+CL-2)+1.0(WLI	1718.42	0.00	-164.57	47.73	0.00	4.57	52.31		43.16	52.31	43.16	16.31	47.74	47.74	52.31	43.16	O.K
416	0.6(DL+COLL+CL-3)+1.0(WLI	1718.42	0.00	-164.57	47.73	0.00	4.57	52.31		43.16	52.31	43.16	16.31	47.74	47.74	52.31	43.16	O.K
417	0.6(DL+COLL+CL-4)+1.0(WLI	1718.42	0.00	-164.57	47.73	0.00	4.57	52.31		43.16	52.31	43.16	16.31	47.74	47.74	52.31	43.16	O.K
418	0.6(DL+COLL+CL-1)+1.0(WLI	1684.78	781.17	-375.23	46.80	21.70	10.42	78.92		58.08	35.52	14.68	42.92	68.50	25.10	57.22	36.38	O.K
419	0.6(DL+COLL+CL-2)+1.0(WLI	1684.78	781.17	-375.23	46.80	21.70	10.42	78.92		58.08	35.52	14.68	42.92	68.50	25.10	57.22	36.38	O.K
420	0.6(DL+COLL+CL-3)+1.0(WLI	1684.78	781.17	-375.23	46.80	21.70	10.42	78.92		58.08	35.52	14.68	42.92	68.50	25.10	57.22	36.38	O.K
421	0.6(DL+COLL+CL-4)+1.0(WLI	1684.78	781.17	-375.23	46.80	21.70	10.42	78.92		58.08	35.52	14.68	42.92	68.50	25.10	57.22	36.38	O.K
422	0.6(DL+COLL+CL-1)+1.0(WLI	1730.77	435.51	-375.23	48.08	12.10	10.42	70.60		49.75	46.40	25.56	34.60	60.18	35.98	58.50	37.66	O.K
423	0.6(DL+COLL+CL-2)+1.0(WLI	1730.77	435.51	-375.23	48.08	12.10	10.42	70.60		49.75	46.40	25.56	34.60	60.18	35.98	58.50	37.66	O.K
424	0.6(DL+COLL+CL-3)+1.0(WLI	1730.77	435.51	-375.23	48.08	12.10	10.42	70.60		49.75	46.40	25.56	34.60	60.18	35.98	58.50	37.66	O.K
425	0.6(DL+COLL+CL-4)+1.0(WLI	1730.77	435.51	-375.23	48.08	12.10	10.42	70.60		49.75	46.40	25.56	34.60	60.18	35.98	58.50	37.66	O.K
426	0.6(DL+COLL+CL-1)+1.0(WLI	1562.73	719.82	-375.23	43.41	19.99	10.42	73.83		52.98	33.84	12.99	37.83	63.41	23.42	53.84	32.99	O.K
427	0.6(DL+COLL+CL-2)+1.0(WLI	1562.73	719.82	-375.23	43.41	19.99	10.42	73.83		52.98	33.84	12.99	37.83	63.41	23.42	53.84	32.99	O.K
428	0.6(DL+COLL+CL-3)+1.0(WLI	1562.73	719.82	-375.23	43.41	19.99	10.42	73.83		52.98	33.84	12.99	37.83	63.41	23.42	53.84	32.99	O.K
429	0.6(DL+COLL+CL-4)+1.0(WLI	1562.73	719.82	-375.23	43.41	19.99	10.42	73.83		52.98	33.84	12.99	37.83	63.41	23.42	53.84	32.99	O.K
430	0.6(DL+COLL+CL-1)+1.0(WLI	1608.72	374.15	-375.23	44.69	10.39	10.42	65.50		44.66	44.72	23.87	29.50	55.08	34.30	55.11	34.27	O.K
431	0.6(DL+COLL+CL-2)+1.0(WLI	1608.72	374.15	-375.23	44.69	10.39	10.42	65.50		44.66	44.72	23.87	29.50	55.08	34.30	55.11	34.27	O.K
432	0.6(DL+COLL+CL-3)+1.0(WLI	1608.72	374.15	-375.23	44.69	10.39	10.42	65.50		44.66	44.72	23.87	29.50	55.08	34.30	55.11	34.27	O.K
433	0.6(DL+COLL+CL-4)+1.0(WLI	1608.72	374.15	-375.23	44.69	10.39	10.42	65.50		44.66	44.72	23.87	29.50	55.08	34.30	55.11	34.27	O.K
434	0.6(DL+COLL+CL-1)+0.7(SL'+	1891.01	-215.08	-590.32	52.53	5.97	16.40	74.90		42.10	62.95	30.16	38.90	58.50	46.56	68.93	36.13	O.K
435	0.6(DL+COLL+CL-2)+0.7(SL'+	1891.01	-215.08	-590.32	52.53	5.97	16.40	74.90		42.10	62.95	30.16	38.90	58.50	46.56	68.93	36.13	O.K
436	0.6(DL+COLL+CL-3)+0.7(SL'+	1891.01	-215.08	-590.32	52.53	5.97	16.40	74.90		42.10	62.95	30.16	38.90	58.50	46.56	68.93	36.13	O.K
437	0.6(DL+COLL+CL-4)+0.7(SL'+	1891.01	-215.08	-590.32	52.53	5.97	16.40	74.90		42.10	62.95	30.16	38.90	58.50	46.56	68.93	36.13	O.K
438	0.6(DL+COLL+CL-1)+0.7(SL'+	1866.55	-215.08	-351.40	51.85	5.97	9.76	67.58		48.06	55.64	36.11	31.58	57.82	45.88	61.61	42.09	O.K
439	0.6(DL+COLL+CL-2)+0.7(SL'+	1825.31	-215.08	-532.60	50.70	5.97	14.79	71.47		41.88	59.52	29.93	35.47	56.68	44.73	65.50	35.91	O.K
440	0.6(DL+COLL+CL-3)+0.7(SL'+	1820.57	-215.08	-216.37	50.57	5.97	6.01	62.56		50.54	50.61	38.59	26.56	56.55	44.60	56.59	44.57	O.K
441	0.6(DL+COLL+CL-4)+0.7(SL'+	1828.59	-215.08	-463.92	50.79	5.97	12.89	69.66		43.88	57.71	31.93	33.66	56.77	44.82	63.69	37.91	O.K

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Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
	Ultimate Conditions																	
101	1.4(DL+COLL)	2249.32	0.00	875.54	62.48	0.00	24.32	86.80		38.16	86.80	38.16	50.80	62.48	62.48	86.80	38.16	
102	1.2(DL+COLL+CL-1)+1.6(RLL	4789.20	0.00	3763.36	133.03	0.00	104.54	237.57		28.50	237.57	28.50	201.57	133.04	133.04	237.57	28.50	
103	1.2(DL+COLL+CL-2)+1.6(RLL	4789.20	0.00	3763.36	133.03	0.00	104.54	237.57		28.50	237.57	28.50	201.57	133.04	133.04	237.57	28.50	
104	1.2(DL+COLL+CL-3)+1.6(RLL	4789.20	0.00	3763.36	133.03	0.00	104.54	237.57		28.50	237.57	28.50	201.57	133.04	133.04	237.57	28.50	
105	1.2(DL+COLL+CL-4)+1.6(RLL	4789.20	0.00	3763.36	133.03	0.00	104.54	237.57		28.50	237.57	28.50	201.57	133.04	133.04	237.57	28.50	
106	1.4(DL+COLL+CL-1)	2249.32	0.00	875.54	62.48	0.00	24.32	86.80		38.16	86.80	38.16	50.80	62.48	62.48	86.80	38.16	
107	1.4(DL+COLL+CL-2)	2249.32	0.00	875.54	62.48	0.00	24.32	86.80		38.16	86.80	38.16	50.80	62.48	62.48	86.80	38.16	
108	1.4(DL+COLL+CL-3)	2249.32	0.00	875.54	62.48	0.00	24.32	86.80		38.16	86.80	38.16	50.80	62.48	62.48	86.80	38.16	
109	1.4(DL+COLL+CL-4)	2249.32	0.00	875.54	62.48	0.00	24.32	86.80		38.16	86.80	38.16	50.80	62.48	62.48	86.80	38.16	
110	1.2(DL+COLL+CL-1+TL1)+1.6	4776.97	0.00	4438.45	132.69	0.00	123.29	255.98		9.40	255.98	9.40	219.98	132.69	132.69	255.98	9.40	
111	1.2(DL+COLL+CL-2+TL1)+1.6	4776.97	0.00	4438.45	132.69	0.00	123.29	255.98		9.40	255.98	9.40	219.98	132.69	132.69	255.98	9.40	
112	1.2(DL+COLL+CL-3+TL1)+1.6	4776.97	0.00	4438.45	132.69	0.00	123.29	255.98		9.40	255.98	9.40	219.98	132.69	132.69	255.98	9.40	
113	1.2(DL+COLL+CL-4+TL1)+1.6	4776.97	0.00	4438.45	132.69	0.00	123.29	255.98		9.40	255.98	9.40	219.98	132.69	132.69	255.98	9.40	
114	1.2(DL+COLL+CL-1+TL2)+1.6	4801.44	0.00	3088.27	133.37	0.00	85.79	219.16		47.59	219.16	47.59	183.16	133.38	133.38	219.16	47.59	
115	1.2(DL+COLL+CL-2+TL2)+1.6	4801.44	0.00	3088.27	133.37	0.00	85.79	219.16		47.59	219.16	47.59	183.16	133.38	133.38	219.16	47.59	
116	1.2(DL+COLL+CL-3+TL2)+1.6	4801.44	0.00	3088.27	133.37	0.00	85.79	219.16		47.59	219.16	47.59	183.16	133.38	133.38	219.16	47.59	
117	1.2(DL+COLL+CL-4+TL2)+1.6	4801.44	0.00	3088.27	133.37	0.00	85.79	219.16		47.59	219.16	47.59	183.16	133.38	133.38	219.16	47.59	
118	1.2(DL+COLL+CL-1)+0.8(WLI	2109.99	0.00	431.91	58.61	0.00	12.00	70.61		46.61	70.61	46.61	34.61	58.61	58.61	70.61	46.61	
119	1.2(DL+COLL+CL-2)+0.8(WLI	2109.99	0.00	431.91	58.61	0.00	12.00	70.61		46.61	70.61	46.61	34.61	58.61	58.61	70.61	46.61	
120	1.2(DL+COLL+CL-3)+0.8(WLI	2109.99	0.00	431.91	58.61	0.00	12.00	70.61		46.61	70.61	46.61	34.61	58.61	58.61	70.61	46.61	
121	1.2(DL+COLL+CL-4)+0.8(WLI	2109.99	0.00	431.91	58.61	0.00	12.00	70.61		46.61	70.61	46.61	34.61	58.61	58.61	70.61	46.61	
122	1.2(DL+COLL+CL-1)+0.8(WLI	2055.00	0.00	673.52	57.08	0.00	18.71	75.79		38.37	75.79	38.37	39.79	57.08	57.08	75.79	38.37	
123	1.2(DL+COLL+CL-2)+0.8(WLI	2055.00	0.00	673.52	57.08	0.00	18.71	75.79		38.37	75.79	38.37	39.79	57.08	57.08	75.79	38.37	
124	1.2(DL+COLL+CL-3)+0.8(WLI	2055.00	0.00	673.52	57.08	0.00	18.71	75.79		38.37	75.79	38.37	39.79	57.08	57.08	75.79	38.37	

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125	1.2(DL+COLL+CL-4)+0.8(WLI	2055.00	0.00	673.52	57.08	0.00	18.71	75.79		38.37	75.79	38.37	39.79	57.08	57.08	75.79	38.37	
126	1.2(DL+COLL+CL-1)+0.8(WLI	2048.68	0.00	251.87	56.91	0.00	7.00	63.90		49.91	63.90	49.91	27.90	56.91	56.91	63.90	49.91	
127	1.2(DL+COLL+CL-2)+0.8(WLI	2048.68	0.00	251.87	56.91	0.00	7.00	63.90		49.91	63.90	49.91	27.90	56.91	56.91	63.90	49.91	
128	1.2(DL+COLL+CL-3)+0.8(WLI	2048.68	0.00	251.87	56.91	0.00	7.00	63.90		49.91	63.90	49.91	27.90	56.91	56.91	63.90	49.91	
129	1.2(DL+COLL+CL-4)+0.8(WLI	2048.68	0.00	251.87	56.91	0.00	7.00	63.90		49.91	63.90	49.91	27.90	56.91	56.91	63.90	49.91	
130	1.2(DL+COLL+CL-1)+0.8(WLI	2059.38	0.00	581.94	57.20	0.00	16.16	73.37		41.04	73.37	41.04	37.37	57.21	57.21	73.37	41.04	
131	1.2(DL+COLL+CL-2)+0.8(WLI	2059.38	0.00	581.94	57.20	0.00	16.16	73.37		41.04	73.37	41.04	37.37	57.21	57.21	73.37	41.04	
132	1.2(DL+COLL+CL-3)+0.8(WLI	2059.38	0.00	581.94	57.20	0.00	16.16	73.37		41.04	73.37	41.04	37.37	57.21	57.21	73.37	41.04	
133	1.2(DL+COLL+CL-4)+0.8(WLI	2059.38	0.00	581.94	57.20	0.00	16.16	73.37		41.04	73.37	41.04	37.37	57.21	57.21	73.37	41.04	
134	1.2(DL+COLL+CL-1)+0.8(WLI	2032.47	316.37	750.47	56.46	8.79	20.85	86.09		44.40	68.52	26.82	50.09	65.25	47.67	77.31	35.61	
135	1.2(DL+COLL+CL-2)+0.8(WLI	2032.47	316.37	750.47	56.46	8.79	20.85	86.09		44.40	68.52	26.82	50.09	65.25	47.67	77.31	35.61	
136	1.2(DL+COLL+CL-3)+0.8(WLI	2032.47	316.37	750.47	56.46	8.79	20.85	86.09		44.40	68.52	26.82	50.09	65.25	47.67	77.31	35.61	
137	1.2(DL+COLL+CL-4)+0.8(WLI	2032.47	316.37	750.47	56.46	8.79	20.85	86.09		44.40	68.52	26.82	50.09	65.25	47.67	77.31	35.61	
138	1.2(DL+COLL+CL-1)+0.8(WLI	2069.26	136.00	750.47	57.48	3.78	20.85	82.10		40.41	74.55	32.86	46.10	61.26	53.71	78.33	36.64	
139	1.2(DL+COLL+CL-2)+0.8(WLI	2069.26	136.00	750.47	57.48	3.78	20.85	82.10		40.41	74.55	32.86	46.10	61.26	53.71	78.33	36.64	
140	1.2(DL+COLL+CL-3)+0.8(WLI	2069.26	136.00	750.47	57.48	3.78	20.85	82.10		40.41	74.55	32.86	46.10	61.26	53.71	78.33	36.64	
141	1.2(DL+COLL+CL-4)+0.8(WLI	2069.26	136.00	750.47	57.48	3.78	20.85	82.10		40.41	74.55	32.86	46.10	61.26	53.71	78.33	36.64	
142	1.2(DL+COLL+CL-1)+0.8(WLI	1934.83	426.45	750.47	53.75	11.85	20.85	86.44		44.74	62.75	21.05	50.44	65.59	41.90	74.60	32.90	
143	1.2(DL+COLL+CL-2)+0.8(WLI	1934.83	426.45	750.47	53.75	11.85	20.85	86.44		44.74	62.75	21.05	50.44	65.59	41.90	74.60	32.90	
144	1.2(DL+COLL+CL-3)+0.8(WLI	1934.83	426.45	750.47	53.75	11.85	20.85	86.44		44.74	62.75	21.05	50.44	65.59	41.90	74.60	32.90	
145	1.2(DL+COLL+CL-4)+0.8(WLI	1934.83	426.45	750.47	53.75	11.85	20.85	86.44		44.74	62.75	21.05	50.44	65.59	41.90	74.60	32.90	
146	1.2(DL+COLL+CL-1)+0.8(WLI	1971.62	246.07	750.47	54.77	6.84	20.85	82.45		40.76	68.78	27.09	46.45	61.61	47.94	75.62	33.93	
147	1.2(DL+COLL+CL-2)+0.8(WLI	1971.62	246.07	750.47	54.77	6.84	20.85	82.45		40.76	68.78	27.09	46.45	61.61	47.94	75.62	33.93	
148	1.2(DL+COLL+CL-3)+0.8(WLI	1971.62	246.07	750.47	54.77	6.84	20.85	82.45		40.76	68.78	27.09	46.45	61.61	47.94	75.62	33.93	
149	1.2(DL+COLL+CL-4)+0.8(WLI	1971.62	246.07	750.47	54.77	6.84	20.85	82.45		40.76	68.78	27.09	46.45	61.61	47.94	75.62	33.93	
150	1.2(DL+COLL+CL-1)+1.6(WLI	2904.45	0.00	1054.89	80.68	0.00	29.30	109.98		51.38	109.98	51.38	73.98	80.68	80.68	109.98	51.38	
151	1.2(DL+COLL+CL-2)+1.6(WLI	2904.45	0.00	1054.89	80.68	0.00	29.30	109.98		51.38	109.98	51.38	73.98	80.68	80.68	109.98	51.38	
152	1.2(DL+COLL+CL-3)+1.6(WLI	2904.45	0.00	1054.89	80.68	0.00	29.30	109.98		51.38	109.98	51.38	73.98	80.68	80.68	109.98	51.38	
153	1.2(DL+COLL+CL-4)+1.6(WLI	2904.45	0.00	1054.89	80.68	0.00	29.30	109.98		51.38	109.98	51.38	73.98	80.68	80.68	109.98	51.38	
154	1.2(DL+COLL+CL-1)+1.6(WLI	2794.46	0.00	1538.09	77.62	0.00	42.72	120.35		34.90	120.35	34.90	84.35	77.63	77.63	120.35	34.90	
155	1.2(DL+COLL+CL-2)+1.6(WLI	2794.46	0.00	1538.09	77.62	0.00	42.72	120.35		34.90	120.35	34.90	84.35	77.63	77.63	120.35	34.90	
156	1.2(DL+COLL+CL-3)+1.6(WLI	2794.46	0.00	1538.09	77.62	0.00	42.72	120.35		34.90	120.35	34.90	84.35	77.63	77.63	120.35	34.90	

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157	1.2(DL+COLL+CL-4)+1.6(WLI	2794.46	0.00	1538.09	77.62	0.00	42.72	120.35		34.90	120.35	34.90	84.35	77.63	77.63	120.35	34.90	
158	1.2(DL+COLL+CL-1)+1.6(WLI	2781.82	0.00	694.81	77.27	0.00	19.30	96.57		57.97	96.57	57.97	60.57	77.27	77.27	96.57	57.97	
159	1.2(DL+COLL+CL-2)+1.6(WLI	2781.82	0.00	694.81	77.27	0.00	19.30	96.57		57.97	96.57	57.97	60.57	77.27	77.27	96.57	57.97	
160	1.2(DL+COLL+CL-3)+1.6(WLI	2781.82	0.00	694.81	77.27	0.00	19.30	96.57		57.97	96.57	57.97	60.57	77.27	77.27	96.57	57.97	
161	1.2(DL+COLL+CL-4)+1.6(WLI	2781.82	0.00	694.81	77.27	0.00	19.30	96.57		57.97	96.57	57.97	60.57	77.27	77.27	96.57	57.97	
162	1.2(DL+COLL+CL-1)+1.6(WLI	2803.22	0.00	1354.93	77.87	0.00	37.64	115.50		40.23	115.50	40.23	79.50	77.87	77.87	115.50	40.23	
163	1.2(DL+COLL+CL-2)+1.6(WLI	2803.22	0.00	1354.93	77.87	0.00	37.64	115.50		40.23	115.50	40.23	79.50	77.87	77.87	115.50	40.23	
164	1.2(DL+COLL+CL-3)+1.6(WLI	2803.22	0.00	1354.93	77.87	0.00	37.64	115.50		40.23	115.50	40.23	79.50	77.87	77.87	115.50	40.23	
165	1.2(DL+COLL+CL-4)+1.6(WLI	2803.22	0.00	1354.93	77.87	0.00	37.64	115.50		40.23	115.50	40.23	79.50	77.87	77.87	115.50	40.23	
166	1.2(DL+COLL+CL-1)+1.6(WLI	2749.40	632.74	1692.00	76.37	17.58	47.00	140.95		46.95	105.80	11.80	104.95	93.95	58.80	123.38	29.38	
167	1.2(DL+COLL+CL-2)+1.6(WLI	2749.40	632.74	1692.00	76.37	17.58	47.00	140.95		46.95	105.80	11.80	104.95	93.95	58.80	123.38	29.38	
168	1.2(DL+COLL+CL-3)+1.6(WLI	2749.40	632.74	1692.00	76.37	17.58	47.00	140.95		46.95	105.80	11.80	104.95	93.95	58.80	123.38	29.38	
169	1.2(DL+COLL+CL-4)+1.6(WLI	2749.40	632.74	1692.00	76.37	17.58	47.00	140.95		46.95	105.80	11.80	104.95	93.95	58.80	123.38	29.38	
170	1.2(DL+COLL+CL-1)+1.6(WLI	2822.98	271.99	1692.00	78.42	7.56	47.00	132.97		38.97	117.86	23.86	96.97	85.97	70.86	125.42	31.42	
171	1.2(DL+COLL+CL-2)+1.6(WLI	2822.98	271.99	1692.00	78.42	7.56	47.00	132.97		38.97	117.86	23.86	96.97	85.97	70.86	125.42	31.42	
172	1.2(DL+COLL+CL-3)+1.6(WLI	2822.98	271.99	1692.00	78.42	7.56	47.00	132.97		38.97	117.86	23.86	96.97	85.97	70.86	125.42	31.42	
173	1.2(DL+COLL+CL-4)+1.6(WLI	2822.98	271.99	1692.00	78.42	7.56	47.00	132.97		38.97	117.86	23.86	96.97	85.97	70.86	125.42	31.42	
174	1.2(DL+COLL+CL-1)+1.6(WLI	2554.12	852.90	1692.00	70.95	23.69	47.00	141.64		47.64	94.26	0.26	105.64	94.64	47.26	117.95	23.95	
175	1.2(DL+COLL+CL-2)+1.6(WLI	2554.12	852.90	1692.00	70.95	23.69	47.00	141.64		47.64	94.26	0.26	105.64	94.64	47.26	117.95	23.95	
176	1.2(DL+COLL+CL-3)+1.6(WLI	2554.12	852.90	1692.00	70.95	23.69	47.00	141.64		47.64	94.26	0.26	105.64	94.64	47.26	117.95	23.95	
177	1.2(DL+COLL+CL-4)+1.6(WLI	2554.12	852.90	1692.00	70.95	23.69	47.00	141.64		47.64	94.26	0.26	105.64	94.64	47.26	117.95	23.95	
178	1.2(DL+COLL+CL-1)+1.6(WLI	2627.70	492.15	1692.00	72.99	13.67	47.00	133.66		39.66	106.32	12.32	97.66	86.66	59.32	119.99	25.99	
179	1.2(DL+COLL+CL-2)+1.6(WLI	2627.70	492.15	1692.00	72.99	13.67	47.00	133.66		39.66	106.32	12.32	97.66	86.66	59.32	119.99	25.99	
180	1.2(DL+COLL+CL-3)+1.6(WLI	2627.70	492.15	1692.00	72.99	13.67	47.00	133.66		39.66	106.32	12.32	97.66	86.66	59.32	119.99	25.99	
181	1.2(DL+COLL+CL-4)+1.6(WLI	2627.70	492.15	1692.00	72.99	13.67	47.00	133.66		39.66	106.32	12.32	97.66	86.66	59.32	119.99	25.99	
182	1.2(DL+COLL+CL-1)+1.0(SL'+	3067.61	393.02	2085.02	85.21	10.92	57.92	154.05		38.21	132.21	16.38	118.05	96.13	74.30	143.13	27.30	
183	1.2(DL+COLL+CL-2)+1.0(SL'+	3067.61	393.02	2085.02	85.21	10.92	57.92	154.05		38.21	132.21	16.38	118.05	96.13	74.30	143.13	27.30	
184	1.2(DL+COLL+CL-3)+1.0(SL'+	3067.61	393.02	2085.02	85.21	10.92	57.92	154.05		38.21	132.21	16.38	118.05	96.13	74.30	143.13	27.30	
185	1.2(DL+COLL+CL-4)+1.0(SL'+	3067.61	393.02	2085.02	85.21	10.92	57.92	154.05		38.21	132.21	16.38	118.05	96.13	74.30	143.13	27.30	
186	1.2(DL+COLL+CL-1)+1.0(SL'-	2871.72	393.02	1298.97	79.77	10.92	36.08	126.77		54.60	104.94	32.77	90.77	90.69	68.86	115.86	43.69	
187	1.2(DL+COLL+CL-2)+1.0(SL'-	2871.72	393.02	1298.97	79.77	10.92	36.08	126.77		54.60	104.94	32.77	90.77	90.69	68.86	115.86	43.69	

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188	1.2(DL+COLL+CL-3)+1.0(SL'-	2871.72	393.02	1298.97	79.77	10.92	36.08	126.77		54.60	104.94	32.77	90.77	90.69	68.86	115.86	43.69	
189	1.2(DL+COLL+CL-4)+1.0(SL'-	2871.72	393.02	1298.97	79.77	10.92	36.08	126.77		54.60	104.94	32.77	90.77	90.69	68.86	115.86	43.69	
190	0.9(DL+COLL+CL-1)+1.6(WLI	1917.31	0.00	74.26	53.26	0.00	2.06	55.32		51.20	55.32	51.20	19.32	53.26	53.26	55.32	51.20	
191	0.9(DL+COLL+CL-2)+1.6(WLI	1917.31	0.00	74.26	53.26	0.00	2.06	55.32		51.20	55.32	51.20	19.32	53.26	53.26	55.32	51.20	
192	0.9(DL+COLL+CL-3)+1.6(WLI	1917.31	0.00	74.26	53.26	0.00	2.06	55.32		51.20	55.32	51.20	19.32	53.26	53.26	55.32	51.20	
193	0.9(DL+COLL+CL-4)+1.6(WLI	1917.31	0.00	74.26	53.26	0.00	2.06	55.32		51.20	55.32	51.20	19.32	53.26	53.26	55.32	51.20	
194	0.9(DL+COLL+CL-1)+1.6(WLI	1807.32	0.00	408.95	50.20	0.00	11.36	61.56		38.84	61.56	38.84	25.56	50.20	50.20	61.56	38.84	
195	0.9(DL+COLL+CL-2)+1.6(WLI	1807.32	0.00	408.95	50.20	0.00	11.36	61.56		38.84	61.56	38.84	25.56	50.20	50.20	61.56	38.84	
196	0.9(DL+COLL+CL-3)+1.6(WLI	1807.32	0.00	408.95	50.20	0.00	11.36	61.56		38.84	61.56	38.84	25.56	50.20	50.20	61.56	38.84	
197	0.9(DL+COLL+CL-4)+1.6(WLI	1807.32	0.00	408.95	50.20	0.00	11.36	61.56		38.84	61.56	38.84	25.56	50.20	50.20	61.56	38.84	
198	0.9(DL+COLL+CL-1)+1.6(WLI	1794.68	0.00	434.33	49.85	0.00	12.06	61.92		37.79	61.92	37.79	25.92	49.86	49.86	61.92	37.79	
199	0.9(DL+COLL+CL-2)+1.6(WLI	1794.68	0.00	434.33	49.85	0.00	12.06	61.92		37.79	61.92	37.79	25.92	49.86	49.86	61.92	37.79	
200	0.9(DL+COLL+CL-3)+1.6(WLI	1794.68	0.00	434.33	49.85	0.00	12.06	61.92		37.79	61.92	37.79	25.92	49.86	49.86	61.92	37.79	
201	0.9(DL+COLL+CL-4)+1.6(WLI	1794.68	0.00	434.33	49.85	0.00	12.06	61.92		37.79	61.92	37.79	25.92	49.86	49.86	61.92	37.79	
202	0.9(DL+COLL+CL-1)+1.6(WLI	1816.08	0.00	225.79	50.45	0.00	6.27	56.72		44.17	56.72	44.17	20.72	50.45	50.45	56.72	44.17	
203	0.9(DL+COLL+CL-2)+1.6(WLI	1816.08	0.00	225.79	50.45	0.00	6.27	56.72		44.17	56.72	44.17	20.72	50.45	50.45	56.72	44.17	
204	0.9(DL+COLL+CL-3)+1.6(WLI	1816.08	0.00	225.79	50.45	0.00	6.27	56.72		44.17	56.72	44.17	20.72	50.45	50.45	56.72	44.17	
205	0.9(DL+COLL+CL-4)+1.6(WLI	1816.08	0.00	225.79	50.45	0.00	6.27	56.72		44.17	56.72	44.17	20.72	50.45	50.45	56.72	44.17	
206	0.9(DL+COLL+CL-1)+1.6(WLI	1762.26	632.74	562.85	48.95	17.58	15.63	82.16		50.89	47.01	15.74	46.16	66.53	31.38	64.59	33.32	
207	0.9(DL+COLL+CL-2)+1.6(WLI	1762.26	632.74	562.85	48.95	17.58	15.63	82.16		50.89	47.01	15.74	46.16	66.53	31.38	64.59	33.32	
208	0.9(DL+COLL+CL-3)+1.6(WLI	1762.26	632.74	562.85	48.95	17.58	15.63	82.16		50.89	47.01	15.74	46.16	66.53	31.38	64.59	33.32	
209	0.9(DL+COLL+CL-4)+1.6(WLI	1762.26	632.74	562.85	48.95	17.58	15.63	82.16		50.89	47.01	15.74	46.16	66.53	31.38	64.59	33.32	
210	0.9(DL+COLL+CL-1)+1.6(WLI	1835.83	271.99	562.85	51.00	7.56	15.63	74.19		42.92	59.07	27.81	38.19	58.56	43.44	66.63	35.37	
211	0.9(DL+COLL+CL-2)+1.6(WLI	1835.83	271.99	562.85	51.00	7.56	15.63	74.19		42.92	59.07	27.81	38.19	58.56	43.44	66.63	35.37	
212	0.9(DL+COLL+CL-3)+1.6(WLI	1835.83	271.99	562.85	51.00	7.56	15.63	74.19		42.92	59.07	27.81	38.19	58.56	43.44	66.63	35.37	
213	0.9(DL+COLL+CL-4)+1.6(WLI	1835.83	271.99	562.85	51.00	7.56	15.63	74.19		42.92	59.07	27.81	38.19	58.56	43.44	66.63	35.37	
214	0.9(DL+COLL+CL-1)+1.6(WLI	1566.98	852.90	562.85	43.53	23.69	15.63	82.85		51.58	35.47	4.20	46.85	67.22	19.84	59.16	27.89	
215	0.9(DL+COLL+CL-2)+1.6(WLI	1566.98	852.90	562.85	43.53	23.69	15.63	82.85		51.58	35.47	4.20	46.85	67.22	19.84	59.16	27.89	
216	0.9(DL+COLL+CL-3)+1.6(WLI	1566.98	852.90	562.85	43.53	23.69	15.63	82.85		51.58	35.47	4.20	46.85	67.22	19.84	59.16	27.89	
217	0.9(DL+COLL+CL-4)+1.6(WLI	1566.98	852.90	562.85	43.53	23.69	15.63	82.85		51.58	35.47	4.20	46.85	67.22	19.84	59.16	27.89	
218	0.9(DL+COLL+CL-1)+1.6(WLI	1640.56	492.15	562.85	45.57	13.67	15.63	74.88		43.61	47.54	16.27	38.88	59.25	31.91	61.21	29.94	
219	0.9(DL+COLL+CL-2)+1.6(WLI	1640.56	492.15	562.85	45.57	13.67	15.63	74.88		43.61	47.54	16.27	38.88	59.25	31.91	61.21	29.94	

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220	0.9(DL+COLL+CL-3)+1.6(WL	1640.56	492.15	562.85	45.57	13.67	15.63	74.88		43.61	47.54	16.27	38.88	59.25	31.91	61.21	29.94	
221	0.9(DL+COLL+CL-4)+1.6(WL	1640.56	492.15	562.85	45.57	13.67	15.63	74.88		43.61	47.54	16.27	38.88	59.25	31.91	61.21	29.94	
222	0.9(DL+COLL+CL-1)+1.0(SL'+	2080.47	393.02	955.87	57.79	10.92	26.55	95.26		42.16	73.43	20.32	59.26	68.71	46.88	84.35	31.24	
223	0.9(DL+COLL+CL-2)+1.0(SL'+	2080.47	393.02	955.87	57.79	10.92	26.55	95.26		42.16	73.43	20.32	59.26	68.71	46.88	84.35	31.24	
224	0.9(DL+COLL+CL-3)+1.0(SL'+	2080.47	393.02	955.87	57.79	10.92	26.55	95.26		42.16	73.43	20.32	59.26	68.71	46.88	84.35	31.24	
225	0.9(DL+COLL+CL-4)+1.0(SL'+	2080.47	393.02	955.87	57.79	10.92	26.55	95.26		42.16	73.43	20.32	59.26	68.71	46.88	84.35	31.24	
226	0.9(DL+COLL+CL-1)+1.0(SL'-	1884.58	393.02	169.83	52.35	10.92	4.72	67.98		58.55	46.15	36.71	31.98	63.27	41.43	57.07	47.63	
227	0.9(DL+COLL+CL-2)+1.0(SL'-	1884.58	393.02	169.83	52.35	10.92	4.72	67.98		58.55	46.15	36.71	31.98	63.27	41.43	57.07	47.63	
228	0.9(DL+COLL+CL-3)+1.0(SL'-	1884.58	393.02	169.83	52.35	10.92	4.72	67.98		58.55	46.15	36.71	31.98	63.27	41.43	57.07	47.63	
229	0.9(DL+COLL+CL-4)+1.0(SL'-	1884.58	393.02	169.83	52.35	10.92	4.72	67.98		58.55	46.15	36.71	31.98	63.27	41.43	57.07	47.63	

Max :	3700	781	248
Min :	1563	-230	-3071

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 37.92 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 47.92 kN/m²
Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 125.02 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 135.02 kN/m²

Sample calculation of values of base pressure:

Load case : 302 1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL)

Maximum pressure, p_1 (pmax) = 102.22+0.00+85.31= 187.52 kN/m²
p_2= 102.22+0.00-85.31= 16.91 kN/m²
p_3= 102.22-(0.00+85.31)= 16.91 kN/m²
p_4= 102.22-0.00+85.31= 187.52 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 187.52-2*18= 151.52 kN/m²

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DESIGN OF ISOLATED FOOTING F5

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

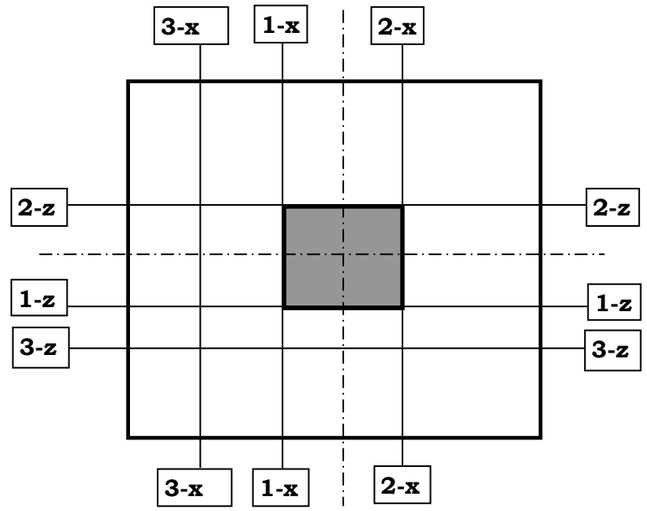
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((1.35+2*0.5*0.6625)+(1.35+2*0.5*0.6625))*2
= 8.05 m

Punching Area= ((1.35+2*0.5*0.6625)*(1.35+2*0.5*0.6625))
= 4.05 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.75*24+1.25*18)
= 40.50 kN/m^2

Area of footing, A_f= 36.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+COLL)	62.48	62.48	62.48	59.41	59.41	35.99	702.30
102	1.2(DL+COLL+CL-1)+1.6(R	133.04	133.04	133.04	250.12	250.12	151.53	2956.44
103	1.2(DL+COLL+CL-2)+1.6(R	133.04	133.04	133.04	250.12	250.12	151.53	2956.44
104	1.2(DL+COLL+CL-3)+1.6(R	133.04	133.04	133.04	250.12	250.12	151.53	2956.44
105	1.2(DL+COLL+CL-4)+1.6(R	133.04	133.04	133.04	250.12	250.12	151.53	2956.44
106	1.4(DL+COLL+CL-1)	62.48	62.48	62.48	59.41	59.41	35.99	702.30
107	1.4(DL+COLL+CL-2)	62.48	62.48	62.48	59.41	59.41	35.99	702.30
108	1.4(DL+COLL+CL-3)	62.48	62.48	62.48	59.41	59.41	35.99	702.30
109	1.4(DL+COLL+CL-4)	62.48	62.48	62.48	59.41	59.41	35.99	702.30
110	1.2(DL+COLL+CL-1+TL1)+	132.69	132.69	132.69	249.17	249.17	150.96	2945.59
111	1.2(DL+COLL+CL-2+TL1)+	132.69	132.69	132.69	249.17	249.17	150.96	2945.59
112	1.2(DL+COLL+CL-3+TL1)+	132.69	132.69	132.69	249.17	249.17	150.96	2945.59

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113	1.2(DL+COLL+CL-4+TL1)+	132.69	132.69	132.69	249.17	249.17	150.96	2945.59
114	1.2(DL+COLL+CL-1+TL2)+	133.38	133.38	133.38	251.04	251.04	152.09	2967.30
115	1.2(DL+COLL+CL-2+TL2)+	133.38	133.38	133.38	251.04	251.04	152.09	2967.30
116	1.2(DL+COLL+CL-3+TL2)+	133.38	133.38	133.38	251.04	251.04	152.09	2967.30
117	1.2(DL+COLL+CL-4+TL2)+	133.38	133.38	133.38	251.04	251.04	152.09	2967.30
118	1.2(DL+COLL+CL-1)+0.8(W	58.61	58.61	58.61	48.95	48.95	29.66	578.64
119	1.2(DL+COLL+CL-2)+0.8(W	58.61	58.61	58.61	48.95	48.95	29.66	578.64
120	1.2(DL+COLL+CL-3)+0.8(W	58.61	58.61	58.61	48.95	48.95	29.66	578.64
121	1.2(DL+COLL+CL-4)+0.8(W	58.61	58.61	58.61	48.95	48.95	29.66	578.64
122	1.2(DL+COLL+CL-1)+0.8(W	57.08	57.08	57.08	44.81	44.81	27.15	529.84
123	1.2(DL+COLL+CL-2)+0.8(W	57.08	57.08	57.08	44.81	44.81	27.15	529.84
124	1.2(DL+COLL+CL-3)+0.8(W	57.08	57.08	57.08	44.81	44.81	27.15	529.84
125	1.2(DL+COLL+CL-4)+0.8(W	57.08	57.08	57.08	44.81	44.81	27.15	529.84
126	1.2(DL+COLL+CL-1)+0.8(W	56.91	56.91	56.91	44.35	44.35	26.87	524.23
127	1.2(DL+COLL+CL-2)+0.8(W	56.91	56.91	56.91	44.35	44.35	26.87	524.23
128	1.2(DL+COLL+CL-3)+0.8(W	56.91	56.91	56.91	44.35	44.35	26.87	524.23
129	1.2(DL+COLL+CL-4)+0.8(W	56.91	56.91	56.91	44.35	44.35	26.87	524.23
130	1.2(DL+COLL+CL-1)+0.8(W	57.21	57.21	57.21	45.16	45.16	27.36	533.72
131	1.2(DL+COLL+CL-2)+0.8(W	57.21	57.21	57.21	45.16	45.16	27.36	533.72
132	1.2(DL+COLL+CL-3)+0.8(W	57.21	57.21	57.21	45.16	45.16	27.36	533.72
133	1.2(DL+COLL+CL-4)+0.8(W	57.21	57.21	57.21	45.16	45.16	27.36	533.72
134	1.2(DL+COLL+CL-1)+0.8(W	58.44	54.48	60.45	60.76	25.52	36.60	509.84
135	1.2(DL+COLL+CL-2)+0.8(W	58.44	54.48	60.45	60.76	25.52	36.60	509.84
136	1.2(DL+COLL+CL-3)+0.8(W	58.44	54.48	60.45	60.76	25.52	36.60	509.84
137	1.2(DL+COLL+CL-4)+0.8(W	58.44	54.48	60.45	60.76	25.52	36.60	509.84
138	1.2(DL+COLL+CL-1)+0.8(W	58.33	56.64	59.20	53.47	38.34	32.31	542.49
139	1.2(DL+COLL+CL-2)+0.8(W	58.33	56.64	59.20	53.47	38.34	32.31	542.49
140	1.2(DL+COLL+CL-3)+0.8(W	58.33	56.64	59.20	53.47	38.34	32.31	542.49
141	1.2(DL+COLL+CL-4)+0.8(W	58.33	56.64	59.20	53.47	38.34	32.31	542.49
142	1.2(DL+COLL+CL-1)+0.8(W	56.41	51.08	59.12	59.54	12.05	35.79	423.19
143	1.2(DL+COLL+CL-2)+0.8(W	56.41	51.08	59.12	59.54	12.05	35.79	423.19

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144	1.2(DL+COLL+CL-3)+0.8(W	56.41	51.08	59.12	59.54	12.05	35.79	423.19
145	1.2(DL+COLL+CL-4)+0.8(W	56.41	51.08	59.12	59.54	12.05	35.79	423.19
146	1.2(DL+COLL+CL-1)+0.8(W	56.31	53.24	57.88	52.28	24.88	31.51	455.84
147	1.2(DL+COLL+CL-2)+0.8(W	56.31	53.24	57.88	52.28	24.88	31.51	455.84
148	1.2(DL+COLL+CL-3)+0.8(W	56.31	53.24	57.88	52.28	24.88	31.51	455.84
149	1.2(DL+COLL+CL-4)+0.8(W	56.31	53.24	57.88	52.28	24.88	31.51	455.84
150	1.2(DL+COLL+CL-1)+1.6(W	80.68	80.68	80.68	108.60	108.60	65.79	1283.72
151	1.2(DL+COLL+CL-2)+1.6(W	80.68	80.68	80.68	108.60	108.60	65.79	1283.72
152	1.2(DL+COLL+CL-3)+1.6(W	80.68	80.68	80.68	108.60	108.60	65.79	1283.72
153	1.2(DL+COLL+CL-4)+1.6(W	80.68	80.68	80.68	108.60	108.60	65.79	1283.72
154	1.2(DL+COLL+CL-1)+1.6(W	77.63	77.63	77.63	100.36	100.36	60.80	1186.11
155	1.2(DL+COLL+CL-2)+1.6(W	77.63	77.63	77.63	100.36	100.36	60.80	1186.11
156	1.2(DL+COLL+CL-3)+1.6(W	77.63	77.63	77.63	100.36	100.36	60.80	1186.11
157	1.2(DL+COLL+CL-4)+1.6(W	77.63	77.63	77.63	100.36	100.36	60.80	1186.11
158	1.2(DL+COLL+CL-1)+1.6(W	77.27	77.27	77.27	99.38	99.38	60.21	1174.89
159	1.2(DL+COLL+CL-2)+1.6(W	77.27	77.27	77.27	99.38	99.38	60.21	1174.89
160	1.2(DL+COLL+CL-3)+1.6(W	77.27	77.27	77.27	99.38	99.38	60.21	1174.89
161	1.2(DL+COLL+CL-4)+1.6(W	77.27	77.27	77.27	99.38	99.38	60.21	1174.89
162	1.2(DL+COLL+CL-1)+1.6(W	77.87	77.87	77.87	101.00	101.00	61.19	1193.88
163	1.2(DL+COLL+CL-2)+1.6(W	77.87	77.87	77.87	101.00	101.00	61.19	1193.88
164	1.2(DL+COLL+CL-3)+1.6(W	77.87	77.87	77.87	101.00	101.00	61.19	1193.88
165	1.2(DL+COLL+CL-4)+1.6(W	77.87	77.87	77.87	101.00	101.00	61.19	1193.88
166	1.2(DL+COLL+CL-1)+1.6(W	80.33	72.42	84.36	132.19	61.73	79.67	1146.11
167	1.2(DL+COLL+CL-2)+1.6(W	80.33	72.42	84.36	132.19	61.73	79.67	1146.11
168	1.2(DL+COLL+CL-3)+1.6(W	80.33	72.42	84.36	132.19	61.73	79.67	1146.11
169	1.2(DL+COLL+CL-4)+1.6(W	80.33	72.42	84.36	132.19	61.73	79.67	1146.11
170	1.2(DL+COLL+CL-1)+1.6(W	80.11	76.72	81.85	117.62	87.33	71.08	1211.42
171	1.2(DL+COLL+CL-2)+1.6(W	80.11	76.72	81.85	117.62	87.33	71.08	1211.42
172	1.2(DL+COLL+CL-3)+1.6(W	80.11	76.72	81.85	117.62	87.33	71.08	1211.42
173	1.2(DL+COLL+CL-4)+1.6(W	80.11	76.72	81.85	117.62	87.33	71.08	1211.42
174	1.2(DL+COLL+CL-1)+1.6(W	76.28	65.62	81.71	129.79	34.81	78.07	972.81

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175	1.2(DL+COLL+CL-2)+1.6(W	76.28	65.62	81.71	129.79	34.81	78.07	972.81
176	1.2(DL+COLL+CL-3)+1.6(W	76.28	65.62	81.71	129.79	34.81	78.07	972.81
177	1.2(DL+COLL+CL-4)+1.6(W	76.28	65.62	81.71	129.79	34.81	78.07	972.81
178	1.2(DL+COLL+CL-1)+1.6(W	76.07	69.91	79.20	115.22	60.41	69.48	1038.11
179	1.2(DL+COLL+CL-2)+1.6(W	76.07	69.91	79.20	115.22	60.41	69.48	1038.11
180	1.2(DL+COLL+CL-3)+1.6(W	76.07	69.91	79.20	115.22	60.41	69.48	1038.11
181	1.2(DL+COLL+CL-4)+1.6(W	76.07	69.91	79.20	115.22	60.41	69.48	1038.11
182	1.2(DL+COLL+CL-1)+1.0(S	87.67	82.76	90.17	142.74	98.98	86.22	1428.53
183	1.2(DL+COLL+CL-2)+1.0(S	87.67	82.76	90.17	142.74	98.98	86.22	1428.53
184	1.2(DL+COLL+CL-3)+1.0(S	87.67	82.76	90.17	142.74	98.98	86.22	1428.53
185	1.2(DL+COLL+CL-4)+1.0(S	87.67	82.76	90.17	142.74	98.98	86.22	1428.53
186	1.2(DL+COLL+CL-1)+1.0(S	82.23	77.32	84.73	128.03	84.27	77.31	1254.68
187	1.2(DL+COLL+CL-2)+1.0(S	82.23	77.32	84.73	128.03	84.27	77.31	1254.68
188	1.2(DL+COLL+CL-3)+1.0(S	82.23	77.32	84.73	128.03	84.27	77.31	1254.68
189	1.2(DL+COLL+CL-4)+1.0(S	82.23	77.32	84.73	128.03	84.27	77.31	1254.68
190	0.9(DL+COLL+CL-1)+1.6(W	53.26	53.26	53.26	34.49	34.49	20.89	407.64
191	0.9(DL+COLL+CL-2)+1.6(W	53.26	53.26	53.26	34.49	34.49	20.89	407.64
192	0.9(DL+COLL+CL-3)+1.6(W	53.26	53.26	53.26	34.49	34.49	20.89	407.64
193	0.9(DL+COLL+CL-4)+1.6(W	53.26	53.26	53.26	34.49	34.49	20.89	407.64
194	0.9(DL+COLL+CL-1)+1.6(W	50.20	50.20	50.20	26.22	26.22	15.88	310.02
195	0.9(DL+COLL+CL-2)+1.6(W	50.20	50.20	50.20	26.22	26.22	15.88	310.02
196	0.9(DL+COLL+CL-3)+1.6(W	50.20	50.20	50.20	26.22	26.22	15.88	310.02
197	0.9(DL+COLL+CL-4)+1.6(W	50.20	50.20	50.20	26.22	26.22	15.88	310.02
198	0.9(DL+COLL+CL-1)+1.6(W	49.86	49.86	49.86	25.30	25.30	15.33	298.81
199	0.9(DL+COLL+CL-2)+1.6(W	49.86	49.86	49.86	25.30	25.30	15.33	298.81
200	0.9(DL+COLL+CL-3)+1.6(W	49.86	49.86	49.86	25.30	25.30	15.33	298.81
201	0.9(DL+COLL+CL-4)+1.6(W	49.86	49.86	49.86	25.30	25.30	15.33	298.81
202	0.9(DL+COLL+CL-1)+1.6(W	50.45	50.45	50.45	26.89	26.89	16.29	317.79
203	0.9(DL+COLL+CL-2)+1.6(W	50.45	50.45	50.45	26.89	26.89	16.29	317.79
204	0.9(DL+COLL+CL-3)+1.6(W	50.45	50.45	50.45	26.89	26.89	16.29	317.79
205	0.9(DL+COLL+CL-4)+1.6(W	50.45	50.45	50.45	26.89	26.89	16.29	317.79

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206	0.9(DL+COLL+CL-1)+1.6(W	52.91	45.00	56.94	58.08	-12.38	34.77	270.03
207	0.9(DL+COLL+CL-2)+1.6(W	52.91	45.00	56.94	58.08	-12.38	34.77	270.03
208	0.9(DL+COLL+CL-3)+1.6(W	52.91	45.00	56.94	58.08	-12.38	34.77	270.03
209	0.9(DL+COLL+CL-4)+1.6(W	52.91	45.00	56.94	58.08	-12.38	34.77	270.03
210	0.9(DL+COLL+CL-1)+1.6(W	52.70	49.30	54.43	43.53	13.22	26.19	335.33
211	0.9(DL+COLL+CL-2)+1.6(W	52.70	49.30	54.43	43.53	13.22	26.19	335.33
212	0.9(DL+COLL+CL-3)+1.6(W	52.70	49.30	54.43	43.53	13.22	26.19	335.33
213	0.9(DL+COLL+CL-4)+1.6(W	52.70	49.30	54.43	43.53	13.22	26.19	335.33
214	0.9(DL+COLL+CL-1)+1.6(W	48.86	38.20	54.29	55.68	-39.30	33.17	96.72
215	0.9(DL+COLL+CL-2)+1.6(W	48.86	38.20	54.29	55.68	-39.30	33.17	96.72
216	0.9(DL+COLL+CL-3)+1.6(W	48.86	38.20	54.29	55.68	-39.30	33.17	96.72
217	0.9(DL+COLL+CL-4)+1.6(W	48.86	38.20	54.29	55.68	-39.30	33.17	96.72
218	0.9(DL+COLL+CL-1)+1.6(W	48.66	42.50	51.79	41.13	-13.67	24.59	162.02
219	0.9(DL+COLL+CL-2)+1.6(W	48.66	42.50	51.79	41.13	-13.67	24.59	162.02
220	0.9(DL+COLL+CL-3)+1.6(W	48.66	42.50	51.79	41.13	-13.67	24.59	162.02
221	0.9(DL+COLL+CL-4)+1.6(W	48.66	42.50	51.79	41.13	-13.67	24.59	162.02
222	0.9(DL+COLL+CL-1)+1.0(S	60.25	55.34	62.75	68.63	24.87	41.32	552.44
223	0.9(DL+COLL+CL-2)+1.0(S	60.25	55.34	62.75	68.63	24.87	41.32	552.44
224	0.9(DL+COLL+CL-3)+1.0(S	60.25	55.34	62.75	68.63	24.87	41.32	552.44
225	0.9(DL+COLL+CL-4)+1.0(S	60.25	55.34	62.75	68.63	24.87	41.32	552.44
226	0.9(DL+COLL+CL-1)+1.0(S	54.81	49.89	57.31	53.92	10.14	32.41	378.59
227	0.9(DL+COLL+CL-2)+1.0(S	54.81	49.89	57.31	53.92	10.14	32.41	378.59
228	0.9(DL+COLL+CL-3)+1.0(S	54.81	49.89	57.31	53.92	10.14	32.41	378.59
229	0.9(DL+COLL+CL-4)+1.0(S	54.81	49.89	57.31	53.92	10.14	32.41	378.59

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Sample calculation of BM & SF

Load case : 101 1.4(DL+COLL)

p_{1x} = 62.48-(62.48-62.48)*2.325/6 = 62.48 kN/m^2 Overhang = 2.325 m

p_{2x} = 62.48+(62.48-62.48)*2.325/6 = 62.48 kN/m^2 Effective depth = 0.6875 m

p_{3x} = 62.48-(62.48-62.48)*(2.325-0.6875)/6 = 62.48 kN/m^2

BM_{1x} = (62.48-40.5)*2.325*2.325/2+0.5*(62.48-62.48)*2.325*2.325*0.67 = 59.41 kN-m/m

BM_{2x} = (62.48-40.5)*2.325*2.325/2-0.5*(-62.48+62.48)*2.325*2.325*0.67 = 59.41 kN-m/m

SF_{3x} = (62.48-40.5)*(2.325-0.6875)+0.5*(62.48-62.48)*(2.325-0.6875) = 35.99 kN/m

$S.F_{(punching)}$ = (36-4.05)*(62.48-40.5) = 702.30 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	251.04	0.53	0.71	0.002	1173.33	1237.50	25ϕT	396.67	300	150	3272.49
At top	-39.30	-0.08	0.11	0.000	181.81	618.75	12ϕT	182.78	300	150	753.98

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 152.09 kN/m

Shear stress , τ_v= (152.09*1000)/(1000*687.5)

= 0.22 N/mm^2 **O.K.**

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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 2967.30 kN

Strength of Concrete in shear Vc1 = 16370.90 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 10117.42 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 10913.93 kN **Table 22.6.5.2_ (c)**

Design Vc= 10117.42 kN

Allowable shear stress, (ϕ Vc)/bd = 1.37 N/mm^2

Design punching shear stress, τ = (2967.3*1000)/(8050*687.5)

= 0.54 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M$_{1z}$ kNm	B.M$_{2z}$ kNm	S.F$_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+COLL)	67.95	57.01	73.32	108.33	10.49	65.77
102	1.2(DL+COLL+CL-1)+1.6(R	156.56	109.51	179.64	460.38	39.83	279.47
103	1.2(DL+COLL+CL-2)+1.6(R	156.56	109.51	179.64	460.38	39.83	279.47
104	1.2(DL+COLL+CL-3)+1.6(R	156.56	109.51	179.64	460.38	39.83	279.47
105	1.2(DL+COLL+CL-4)+1.6(R	156.56	109.51	179.64	460.38	39.83	279.47
106	1.4(DL+COLL+CL-1)	67.95	57.01	73.32	108.33	10.49	65.77
107	1.4(DL+COLL+CL-2)	67.95	57.01	73.32	108.33	10.49	65.77
108	1.4(DL+COLL+CL-3)	67.95	57.01	73.32	108.33	10.49	65.77

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109	1.4(DL+COLL+CL-4)	67.95	57.01	73.32	108.33	10.49	65.77
110	1.2(DL+COLL+CL-1+TL1)+	160.43	104.95	187.66	497.18	1.17	301.44
111	1.2(DL+COLL+CL-2+TL1)+	160.43	104.95	187.66	497.18	1.17	301.44
112	1.2(DL+COLL+CL-3+TL1)+	160.43	104.95	187.66	497.18	1.17	301.44
113	1.2(DL+COLL+CL-4+TL1)+	160.43	104.95	187.66	497.18	1.17	301.44
114	1.2(DL+COLL+CL-1+TL2)+	152.68	114.07	171.62	423.59	78.46	257.51
115	1.2(DL+COLL+CL-2+TL2)+	152.68	114.07	171.62	423.59	78.46	257.51
116	1.2(DL+COLL+CL-3+TL2)+	152.68	114.07	171.62	423.59	78.46	257.51
117	1.2(DL+COLL+CL-4+TL2)+	152.68	114.07	171.62	423.59	78.46	257.51
118	1.2(DL+COLL+CL-1)+0.8(W	61.31	55.91	63.96	73.09	24.81	44.53
119	1.2(DL+COLL+CL-2)+0.8(W	61.31	55.91	63.96	73.09	24.81	44.53
120	1.2(DL+COLL+CL-3)+0.8(W	61.31	55.91	63.96	73.09	24.81	44.53
121	1.2(DL+COLL+CL-4)+0.8(W	61.31	55.91	63.96	73.09	24.81	44.53
122	1.2(DL+COLL+CL-1)+0.8(W	61.29	52.87	65.42	82.45	7.18	50.05
123	1.2(DL+COLL+CL-2)+0.8(W	61.29	52.87	65.42	82.45	7.18	50.05
124	1.2(DL+COLL+CL-3)+0.8(W	61.29	52.87	65.42	82.45	7.18	50.05
125	1.2(DL+COLL+CL-4)+0.8(W	61.29	52.87	65.42	82.45	7.18	50.05
126	1.2(DL+COLL+CL-1)+0.8(W	58.48	55.33	60.02	58.41	30.27	35.68
127	1.2(DL+COLL+CL-2)+0.8(W	58.48	55.33	60.02	58.41	30.27	35.68
128	1.2(DL+COLL+CL-3)+0.8(W	58.48	55.33	60.02	58.41	30.27	35.68
129	1.2(DL+COLL+CL-4)+0.8(W	58.48	55.33	60.02	58.41	30.27	35.68
130	1.2(DL+COLL+CL-1)+0.8(W	60.84	53.57	64.41	77.67	12.63	47.20
131	1.2(DL+COLL+CL-2)+0.8(W	60.84	53.57	64.41	77.67	12.63	47.20
132	1.2(DL+COLL+CL-3)+0.8(W	60.84	53.57	64.41	77.67	12.63	47.20
133	1.2(DL+COLL+CL-4)+0.8(W	60.84	53.57	64.41	77.67	12.63	47.20
134	1.2(DL+COLL+CL-1)+0.8(W	61.15	51.77	65.76	85.08	1.20	51.59
135	1.2(DL+COLL+CL-2)+0.8(W	61.15	51.77	65.76	85.08	1.20	51.59
136	1.2(DL+COLL+CL-3)+0.8(W	61.15	51.77	65.76	85.08	1.20	51.59
137	1.2(DL+COLL+CL-4)+0.8(W	61.15	51.77	65.76	85.08	1.20	51.59
138	1.2(DL+COLL+CL-1)+0.8(W	62.18	52.79	66.78	87.84	3.98	53.29
139	1.2(DL+COLL+CL-2)+0.8(W	62.18	52.79	66.78	87.84	3.98	53.29
140	1.2(DL+COLL+CL-3)+0.8(W	62.18	52.79	66.78	87.84	3.98	53.29
141	1.2(DL+COLL+CL-4)+0.8(W	62.18	52.79	66.78	87.84	3.98	53.29

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142	1.2(DL+COLL+CL-1)+0.8(W	58.44	49.06	63.05	77.75	-6.13	47.09
143	1.2(DL+COLL+CL-2)+0.8(W	58.44	49.06	63.05	77.75	-6.13	47.09
144	1.2(DL+COLL+CL-3)+0.8(W	58.44	49.06	63.05	77.75	-6.13	47.09
145	1.2(DL+COLL+CL-4)+0.8(W	58.44	49.06	63.05	77.75	-6.13	47.09
146	1.2(DL+COLL+CL-1)+0.8(W	59.47	50.08	64.07	80.51	-3.35	48.78
147	1.2(DL+COLL+CL-2)+0.8(W	59.47	50.08	64.07	80.51	-3.35	48.78
148	1.2(DL+COLL+CL-3)+0.8(W	59.47	50.08	64.07	80.51	-3.35	48.78
149	1.2(DL+COLL+CL-4)+0.8(W	59.47	50.08	64.07	80.51	-3.35	48.78
150	1.2(DL+COLL+CL-1)+1.6(W	87.27	74.09	93.74	167.54	49.66	102.01
151	1.2(DL+COLL+CL-2)+1.6(W	87.27	74.09	93.74	167.54	49.66	102.01
152	1.2(DL+COLL+CL-3)+1.6(W	87.27	74.09	93.74	167.54	49.66	102.01
153	1.2(DL+COLL+CL-4)+1.6(W	87.27	74.09	93.74	167.54	49.66	102.01
154	1.2(DL+COLL+CL-1)+1.6(W	87.24	68.01	96.67	186.29	14.40	113.07
155	1.2(DL+COLL+CL-2)+1.6(W	87.24	68.01	96.67	186.29	14.40	113.07
156	1.2(DL+COLL+CL-3)+1.6(W	87.24	68.01	96.67	186.29	14.40	113.07
157	1.2(DL+COLL+CL-4)+1.6(W	87.24	68.01	96.67	186.29	14.40	113.07
158	1.2(DL+COLL+CL-1)+1.6(W	81.61	72.93	85.87	138.21	60.56	84.33
159	1.2(DL+COLL+CL-2)+1.6(W	81.61	72.93	85.87	138.21	60.56	84.33
160	1.2(DL+COLL+CL-3)+1.6(W	81.61	72.93	85.87	138.21	60.56	84.33
161	1.2(DL+COLL+CL-4)+1.6(W	81.61	72.93	85.87	138.21	60.56	84.33
162	1.2(DL+COLL+CL-1)+1.6(W	86.33	69.40	94.64	176.70	25.29	107.35
163	1.2(DL+COLL+CL-2)+1.6(W	86.33	69.40	94.64	176.70	25.29	107.35
164	1.2(DL+COLL+CL-3)+1.6(W	86.33	69.40	94.64	176.70	25.29	107.35
165	1.2(DL+COLL+CL-4)+1.6(W	86.33	69.40	94.64	176.70	25.29	107.35
166	1.2(DL+COLL+CL-1)+1.6(W	86.96	65.81	97.33	191.52	2.43	116.14
167	1.2(DL+COLL+CL-2)+1.6(W	86.96	65.81	97.33	191.52	2.43	116.14
168	1.2(DL+COLL+CL-3)+1.6(W	86.96	65.81	97.33	191.52	2.43	116.14
169	1.2(DL+COLL+CL-4)+1.6(W	86.96	65.81	97.33	191.52	2.43	116.14
170	1.2(DL+COLL+CL-1)+1.6(W	89.00	67.85	99.37	197.03	7.95	119.53
171	1.2(DL+COLL+CL-2)+1.6(W	89.00	67.85	99.37	197.03	7.95	119.53
172	1.2(DL+COLL+CL-3)+1.6(W	89.00	67.85	99.37	197.03	7.95	119.53
173	1.2(DL+COLL+CL-4)+1.6(W	89.00	67.85	99.37	197.03	7.95	119.53
174	1.2(DL+COLL+CL-1)+1.6(W	81.53	60.38	91.90	176.84	-12.24	107.11

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175	1.2(DL+COLL+CL-2)+1.6(W	81.53	60.38	91.90	176.84	-12.24	107.11
176	1.2(DL+COLL+CL-3)+1.6(W	81.53	60.38	91.90	176.84	-12.24	107.11
177	1.2(DL+COLL+CL-4)+1.6(W	81.53	60.38	91.90	176.84	-12.24	107.11
178	1.2(DL+COLL+CL-1)+1.6(W	83.57	62.42	93.94	182.36	-6.73	110.50
179	1.2(DL+COLL+CL-2)+1.6(W	83.57	62.42	93.94	182.36	-6.73	110.50
180	1.2(DL+COLL+CL-3)+1.6(W	83.57	62.42	93.94	182.36	-6.73	110.50
181	1.2(DL+COLL+CL-4)+1.6(W	83.57	62.42	93.94	182.36	-6.73	110.50
182	1.2(DL+COLL+CL-1)+1.0(S	98.25	72.18	111.04	237.36	4.36	143.94
183	1.2(DL+COLL+CL-2)+1.0(S	98.25	72.18	111.04	237.36	4.36	143.94
184	1.2(DL+COLL+CL-3)+1.0(S	98.25	72.18	111.04	237.36	4.36	143.94
185	1.2(DL+COLL+CL-4)+1.0(S	98.25	72.18	111.04	237.36	4.36	143.94
186	1.2(DL+COLL+CL-1)+1.0(S	87.89	71.66	95.86	178.74	33.57	108.66
187	1.2(DL+COLL+CL-2)+1.0(S	87.89	71.66	95.86	178.74	33.57	108.66
188	1.2(DL+COLL+CL-3)+1.0(S	87.89	71.66	95.86	178.74	33.57	108.66
189	1.2(DL+COLL+CL-4)+1.0(S	87.89	71.66	95.86	178.74	33.57	108.66
190	0.9(DL+COLL+CL-1)+1.6(W	53.72	52.80	54.18	38.63	30.34	23.69
191	0.9(DL+COLL+CL-2)+1.6(W	53.72	52.80	54.18	38.63	30.34	23.69
192	0.9(DL+COLL+CL-3)+1.6(W	53.72	52.80	54.18	38.63	30.34	23.69
193	0.9(DL+COLL+CL-4)+1.6(W	53.72	52.80	54.18	38.63	30.34	23.69
194	0.9(DL+COLL+CL-1)+1.6(W	52.76	47.64	55.26	49.07	3.37	29.78
195	0.9(DL+COLL+CL-2)+1.6(W	52.76	47.64	55.26	49.07	3.37	29.78
196	0.9(DL+COLL+CL-3)+1.6(W	52.76	47.64	55.26	49.07	3.37	29.78
197	0.9(DL+COLL+CL-4)+1.6(W	52.76	47.64	55.26	49.07	3.37	29.78
198	0.9(DL+COLL+CL-1)+1.6(W	52.57	47.14	55.23	49.55	1.02	30.05
199	0.9(DL+COLL+CL-2)+1.6(W	52.57	47.14	55.23	49.55	1.02	30.05
200	0.9(DL+COLL+CL-3)+1.6(W	52.57	47.14	55.23	49.55	1.02	30.05
201	0.9(DL+COLL+CL-4)+1.6(W	52.57	47.14	55.23	49.55	1.02	30.05
202	0.9(DL+COLL+CL-1)+1.6(W	51.86	49.03	53.24	39.50	14.26	24.08
203	0.9(DL+COLL+CL-2)+1.6(W	51.86	49.03	53.24	39.50	14.26	24.08
204	0.9(DL+COLL+CL-3)+1.6(W	51.86	49.03	53.24	39.50	14.26	24.08
205	0.9(DL+COLL+CL-4)+1.6(W	51.86	49.03	53.24	39.50	14.26	24.08
206	0.9(DL+COLL+CL-1)+1.6(W	52.47	45.44	55.93	54.30	-8.60	32.85
207	0.9(DL+COLL+CL-2)+1.6(W	52.47	45.44	55.93	54.30	-8.60	32.85
208	0.9(DL+COLL+CL-3)+1.6(W	52.47	45.44	55.93	54.30	-8.60	32.85

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209	0.9(DL+COLL+CL-4)+1.6(W	52.47	45.44	55.93	54.30	-8.60	32.85
210	0.9(DL+COLL+CL-1)+1.6(W	54.52	47.48	57.97	59.82	-3.06	36.24
211	0.9(DL+COLL+CL-2)+1.6(W	54.52	47.48	57.97	59.82	-3.06	36.24
212	0.9(DL+COLL+CL-3)+1.6(W	54.52	47.48	57.97	59.82	-3.06	36.24
213	0.9(DL+COLL+CL-4)+1.6(W	54.52	47.48	57.97	59.82	-3.06	36.24
214	0.9(DL+COLL+CL-1)+1.6(W	47.04	40.01	50.50	39.63	-23.27	23.82
215	0.9(DL+COLL+CL-2)+1.6(W	47.04	40.01	50.50	39.63	-23.27	23.82
216	0.9(DL+COLL+CL-3)+1.6(W	47.04	40.01	50.50	39.63	-23.27	23.82
217	0.9(DL+COLL+CL-4)+1.6(W	47.04	40.01	50.50	39.63	-23.27	23.82
218	0.9(DL+COLL+CL-1)+1.6(W	49.09	42.06	52.55	45.17	-17.73	27.23
219	0.9(DL+COLL+CL-2)+1.6(W	49.09	42.06	52.55	45.17	-17.73	27.23
220	0.9(DL+COLL+CL-3)+1.6(W	49.09	42.06	52.55	45.17	-17.73	27.23
221	0.9(DL+COLL+CL-4)+1.6(W	49.09	42.06	52.55	45.17	-17.73	27.23
222	0.9(DL+COLL+CL-1)+1.0(S	63.77	51.82	69.63	100.16	-6.67	60.67
223	0.9(DL+COLL+CL-2)+1.0(S	63.77	51.82	69.63	100.16	-6.67	60.67
224	0.9(DL+COLL+CL-3)+1.0(S	63.77	51.82	69.63	100.16	-6.67	60.67
225	0.9(DL+COLL+CL-4)+1.0(S	63.77	51.82	69.63	100.16	-6.67	60.67
226	0.9(DL+COLL+CL-1)+1.0(S	53.41	51.29	54.45	41.52	22.53	25.37
227	0.9(DL+COLL+CL-2)+1.0(S	53.41	51.29	54.45	41.52	22.53	25.37
228	0.9(DL+COLL+CL-3)+1.0(S	53.41	51.29	54.45	41.52	22.53	25.37
229	0.9(DL+COLL+CL-4)+1.0(S	53.41	51.29	54.45	41.52	22.53	25.37

Sample calculation of BM & SF

Load case : 101

p_{1z} = 86.8-(86.8-38.16)*2.325/6= 67.95 kN/m^2

p_{2z} = 38.16+(86.8-38.16)*2.325/6= 57.01 kN/m^2

p_{3z} = 86.8-(86.8-38.16)*(2.325-0.6625)/6= 73.32 kN/m^2

Overhang = 2.325 m

Effective depth = 0.6625 m

BM_{1z} = (67.95-40.5)*2.325*2.325/2+0.5*(86.8-67.95)*2.325*2.325*0.67 = 108.33 kN-m/m

BM_{2z} = (57.01-40.5)*2.325*2.325/2-0.5*(-38.16+57.01)*2.325*2.325*0.67 = 10.49 kN-m/m

SF_{3z} = (73.32-40.5)*(2.325-0.6625)+0.5*(86.8-73.32)*(2.325-0.6625) = 65.77 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	497.18	1.133	1.510	0.0037	2538.47	1192.50	25ϕT	193.37	300	150	3272.5
At top	0.00	0.000	0.000	0.0000	0.00	596.25	12ϕT	189.68	250	150	754.0

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 301.44 kN/m

Shear stress , τ_v= (301.44*1000)/(1000*662.5)

= 0.46 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	25ϕT	**150**	c/c
	At Top	12ϕT	**150**	c/c
Along Z Direction :				
	At Bottom	25ϕT	**150**	c/c
	At Top	12ϕT	**150**	c/c

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5.6 DESIGN OF CALCULATION OF FOOTING F6

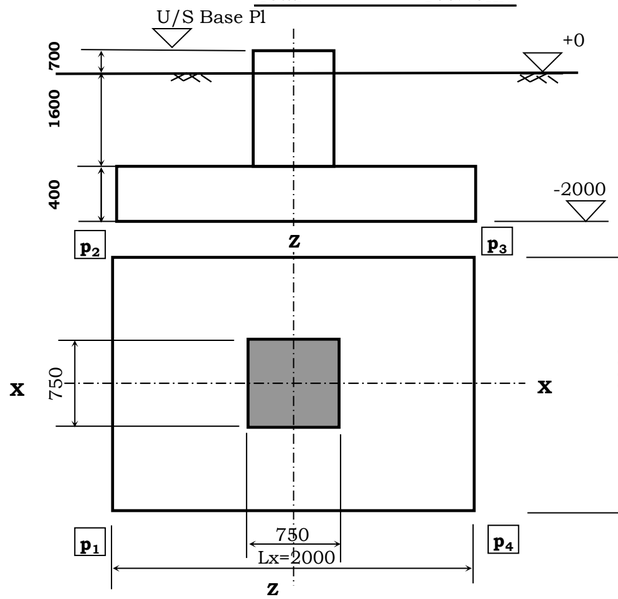
	Along-X	Along-Z
Size of column :	0.75m	0.75m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-2.00 m	
U/S level Base Plate :	0.70 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

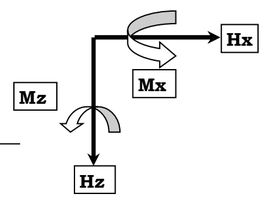
SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Crane Load	CL
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (2*2*0.4*24)+(0.75*0.75*2.3*24) = 69.45KN
Weight of soil = ((2*2)-(0.75*0.75))*1.6*18 = 99.00KN
Total = 168.45KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 342mm = 0.342m
Effective depth of slab along Z direction = 326mm = 0.326m
Clear projection along Z direction = 0.625m
Clear projection along X direction = 0.625m

H$_X$=Force along X-X direction

Hz=Force along Z-Z direction

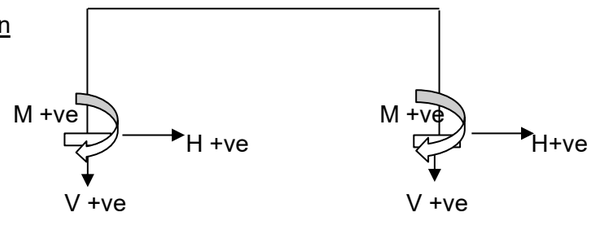
M$_X$=Moment about X-X direction

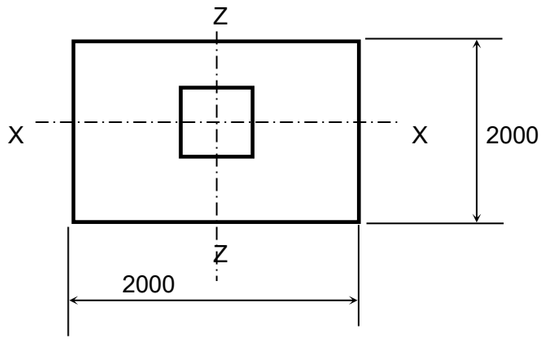
Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F6

Sign convention





Height from BOF to top of column about X (l_{ux}) = 2700mm
Height from BOF to top of column about Z (l_{uz}) = 2700mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	145.698		-16.4172		0
2	COLLATERAL LOAD (COLL)	0		0		0
3	ROOF LIVE LOAD (RLL)	0		0		0
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	438.948		-51.6228		0
5	CRANE LOAD-1 (CL-1)	0		0		0
6	CRANE LOAD-2 (CL-2)	0		0		0
7	CRANE LOAD-3 (CL-3)	0		0		0
8	CRANE LOAD-4 (CL-4)	0		0		0
9	WIND LEFT-1 (WLL-1)	0		0		0
10	WIND LEFT-2 (WLL-2)	0		0		0
11	WIND RIGHT-1 (WLR-1)	0		0		0
12	WIND RIGHT-2 (WLR-2)	0		0		0
13	WIND LONG-1 (WLLG-1)	0	0		0	
14	WIND LONG-2 (WLLG-2)	0	0		0	
15	WIND LONG-3 (WLLG-3)	0	0		0	
16	WIND LONG-4 (WLLG-4)	0	0		0	
17	SEISMIC LOAD SL'+'	-19.001	-23.494	-23.494	0	0
18	SEISMIC LOAD SL'-'	19.001	23.494	23.494	0	0
19	TEMPERATURE LOAD (TL1)	0		-15.1896		0
20	TEMPERATURE LOAD (TL2)	0		15.1896		0
21	BRACING REACTION (BR-WLLG1)	0	0		0	
22	BRACING REACTION (BR-WLLG2)	0	0		0	
23	BRACING REACTION (BR-WLLG3)	0	0		0	
24	BRACING REACTION (BR-WLLG4)	0	0		0	
25	BRACING REACTION (BR-SL'+')	0	0	0	0	0
26	BRACING REACTION (BR-SL'-')	0	0	0	0	0

Ultimate Condition

Combined Load Cases:		Load at top of foundation			Load at bottom of foundation		
		Factored Load/Moment					
Load Case	Load Combination	Pu (kN)	Muz (kN-m)	Mux (kN-m)	Pu (kN)	Muz (kN-m)	Mux (kN-m)
101	1.4(DL+COLL)	203.98	0.0	62.1	372.4	0.0	62.1
102	1.2(DL+COLL+CL-1)+1.6(RLL+PLL / MLL)	877.2	0.0	276.2	1045.6	0.0	276.2
103	1.2(DL+COLL+CL-2)+1.6(RLL+PLL / MLL)	877.2	0.0	276.2	1045.6	0.0	276.2
104	1.2(DL+COLL+CL-3)+1.6(RLL+PLL / MLL)	877.2	0.0	276.2	1045.6	0.0	276.2
105	1.2(DL+COLL+CL-4)+1.6(RLL+PLL / MLL)	877.2	0.0	276.2	1045.6	0.0	276.2
106	1.4(DL+COLL+CL-1)	204.0	0.0	62.1	372.4	0.0	62.1
107	1.4(DL+COLL+CL-2)	204.0	0.0	62.1	372.4	0.0	62.1
108	1.4(DL+COLL+CL-3)	204.0	0.0	62.1	372.4	0.0	62.1
109	1.4(DL+COLL+CL-4)	204.0	0.0	62.1	372.4	0.0	62.1
110	1.2(DL+COLL+CL-1+TL1)+1.6(RLL+PLL / MLL)	877.2	0.0	325.4	1045.6	0.0	325.4
111	1.2(DL+COLL+CL-2+TL1)+1.6(RLL+PLL / MLL)	877.2	0.0	325.4	1045.6	0.0	325.4
112	1.2(DL+COLL+CL-3+TL1)+1.6(RLL+PLL / MLL)	877.2	0.0	325.4	1045.6	0.0	325.4
113	1.2(DL+COLL+CL-4+TL1)+1.6(RLL+PLL / MLL)	877.2	0.0	325.4	1045.6	0.0	325.4
114	1.2(DL+COLL+CL-1+TL2)+1.6(RLL+PLL / MLL)	877.2	0.0	227.0	1045.6	0.0	227.0
115	1.2(DL+COLL+CL-2+TL2)+1.6(RLL+PLL / MLL)	877.2	0.0	227.0	1045.6	0.0	227.0
116	1.2(DL+COLL+CL-3+TL2)+1.6(RLL+PLL / MLL)	877.2	0.0	227.0	1045.6	0.0	227.0
117	1.2(DL+COLL+CL-4+TL2)+1.6(RLL+PLL / MLL)	877.2	0.0	227.0	1045.6	0.0	227.0
118	1.2(DL+COLL+CL-1)+0.8(WLL-1)	174.8	0.0	53.2	343.3	0.0	53.2
119	1.2(DL+COLL+CL-2)+0.8(WLL-1)	174.8	0.0	53.2	343.3	0.0	53.2
120	1.2(DL+COLL+CL-3)+0.8(WLL-1)	174.8	0.0	53.2	343.3	0.0	53.2
121	1.2(DL+COLL+CL-4)+0.8(WLL-1)	174.8	0.0	53.2	343.3	0.0	53.2
122	1.2(DL+COLL+CL-1)+0.8(WLL-2)	174.8	0.0	53.2	343.3	0.0	53.2
123	1.2(DL+COLL+CL-2)+0.8(WLL-2)	174.8	0.0	53.2	343.3	0.0	53.2
124	1.2(DL+COLL+CL-3)+0.8(WLL-2)	174.8	0.0	53.2	343.3	0.0	53.2
125	1.2(DL+COLL+CL-4)+0.8(WLL-2)	174.8	0.0	53.2	343.3	0.0	53.2
126	1.2(DL+COLL+CL-1)+0.8(WLR-1)	174.8	0.0	53.2	343.3	0.0	53.2
127	1.2(DL+COLL+CL-2)+0.8(WLR-1)	174.8	0.0	53.2	343.3	0.0	53.2
128	1.2(DL+COLL+CL-3)+0.8(WLR-1)	174.8	0.0	53.2	343.3	0.0	53.2
129	1.2(DL+COLL+CL-4)+0.8(WLR-1)	174.8	0.0	53.2	343.3	0.0	53.2
130	1.2(DL+COLL+CL-1)+0.8(WLR-2)	174.8	0.0	53.2	343.3	0.0	53.2
131	1.2(DL+COLL+CL-2)+0.8(WLR-2)	174.8	0.0	53.2	343.3	0.0	53.2
132	1.2(DL+COLL+CL-3)+0.8(WLR-2)	174.8	0.0	53.2	343.3	0.0	53.2
133	1.2(DL+COLL+CL-4)+0.8(WLR-2)	174.8	0.0	53.2	343.3	0.0	53.2
134	1.2(DL+COLL+CL-1)+0.8(WLLG-1+BR-WLLG-1)	174.8	0.0	53.2	343.3	0.0	53.2
135	1.2(DL+COLL+CL-2)+0.8(WLLG-1+BR-WLLG-1)	174.8	0.0	53.2	343.3	0.0	53.2
136	1.2(DL+COLL+CL-3)+0.8(WLLG-1+BR-WLLG-1)	174.8	0.0	53.2	343.3	0.0	53.2
137	1.2(DL+COLL+CL-4)+0.8(WLLG-1+BR-WLLG-1)	174.8	0.0	53.2	343.3	0.0	53.2
138	1.2(DL+COLL+CL-1)+0.8(WLLG-2+BR-WLLG-2)	174.8	0.0	53.2	343.3	0.0	53.2
139	1.2(DL+COLL+CL-2)+0.8(WLLG-2+BR-WLLG-2)	174.8	0.0	53.2	343.3	0.0	53.2
140	1.2(DL+COLL+CL-3)+0.8(WLLG-2+BR-WLLG-2)	174.8	0.0	53.2	343.3	0.0	53.2
141	1.2(DL+COLL+CL-4)+0.8(WLLG-2+BR-WLLG-2)	174.8	0.0	53.2	343.3	0.0	53.2
142	1.2(DL+COLL+CL-1)+0.8(WLLG-3+BR-WLLG-3)	174.8	0.0	53.2	343.3	0.0	53.2
143	1.2(DL+COLL+CL-2)+0.8(WLLG-3+BR-WLLG-3)	174.8	0.0	53.2	343.3	0.0	53.2
144	1.2(DL+COLL+CL-3)+0.8(WLLG-3+BR-WLLG-3)	174.8	0.0	53.2	343.3	0.0	53.2
145	1.2(DL+COLL+CL-4)+0.8(WLLG-3+BR-WLLG-3)	174.8	0.0	53.2	343.3	0.0	53.2
146	1.2(DL+COLL+CL-1)+0.8(WLLG-4+BR-WLLG-4)	174.8	0.0	53.2	343.3	0.0	53.2
147	1.2(DL+COLL+CL-2)+0.8(WLLG-4+BR-WLLG-4)	174.8	0.0	53.2	343.3	0.0	53.2
148	1.2(DL+COLL+CL-3)+0.8(WLLG-4+BR-WLLG-4)	174.8	0.0	53.2	343.3	0.0	53.2

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149	1.2(DL+COLL+CL-4)+0.8(WLLG-4+BR-WLLG-4)	174.8	0.0	53.2	343.3	0.0	53.2
150	1.2(DL+COLL+CL-1)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
151	1.2(DL+COLL+CL-2)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
152	1.2(DL+COLL+CL-3)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
153	1.2(DL+COLL+CL-4)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
154	1.2(DL+COLL+CL-1)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
155	1.2(DL+COLL+CL-2)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
156	1.2(DL+COLL+CL-3)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
157	1.2(DL+COLL+CL-4)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
158	1.2(DL+COLL+CL-1)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
159	1.2(DL+COLL+CL-2)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
160	1.2(DL+COLL+CL-3)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
161	1.2(DL+COLL+CL-4)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
162	1.2(DL+COLL+CL-1)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
163	1.2(DL+COLL+CL-2)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
164	1.2(DL+COLL+CL-3)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
165	1.2(DL+COLL+CL-4)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	394.3	0.0	122.9	562.8	0.0	122.9
166	1.2(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
167	1.2(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
168	1.2(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
169	1.2(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
170	1.2(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
171	1.2(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
172	1.2(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
173	1.2(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
174	1.2(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
175	1.2(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
176	1.2(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
177	1.2(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	394.3	0.0	122.9	562.8	0.0	122.9
178	1.2(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	394.3	0.0	122.9	562.8	0.0	122.9
179	1.2(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	394.3	0.0	122.9	562.8	0.0	122.9
180	1.2(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	394.3	0.0	122.9	562.8	0.0	122.9
181	1.2(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	394.3	0.0	122.9	562.8	0.0	122.9
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	375.3	63.4	186.3	543.8	63.4	186.3
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	375.3	63.4	186.3	543.8	63.4	186.3
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	375.3	63.4	186.3	543.8	63.4	186.3
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	375.3	63.4	186.3	543.8	63.4	186.3
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	413.3	63.4	59.4	581.8	63.4	59.4
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-)+0.5(RLL+PLL / MLL	413.3	63.4	59.4	581.8	63.4	59.4
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	413.3	63.4	59.4	581.8	63.4	59.4
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	413.3	63.4	59.4	581.8	63.4	59.4
190	0.9(DL+COLL+CL-1)+1.6(WLL-1)	131.1	0.0	39.9	299.6	0.0	39.9
191	0.9(DL+COLL+CL-2)+1.6(WLL-1)	131.1	0.0	39.9	299.6	0.0	39.9
192	0.9(DL+COLL+CL-3)+1.6(WLL-1)	131.1	0.0	39.9	299.6	0.0	39.9
193	0.9(DL+COLL+CL-4)+1.6(WLL-1)	131.1	0.0	39.9	299.6	0.0	39.9
194	0.9(DL+COLL+CL-1)+1.6(WLL-2)	131.1	0.0	39.9	299.6	0.0	39.9
195	0.9(DL+COLL+CL-2)+1.6(WLL-2)	131.1	0.0	39.9	299.6	0.0	39.9
196	0.9(DL+COLL+CL-3)+1.6(WLL-2)	131.1	0.0	39.9	299.6	0.0	39.9
197	0.9(DL+COLL+CL-4)+1.6(WLL-2)	131.1	0.0	39.9	299.6	0.0	39.9
198	0.9(DL+COLL+CL-1)+1.6(WLR-1)	131.1	0.0	39.9	299.6	0.0	39.9
199	0.9(DL+COLL+CL-2)+1.6(WLR-1)	131.1	0.0	39.9	299.6	0.0	39.9
200	0.9(DL+COLL+CL-3)+1.6(WLR-1)	131.1	0.0	39.9	299.6	0.0	39.9
201	0.9(DL+COLL+CL-4)+1.6(WLR-1)	131.1	0.0	39.9	299.6	0.0	39.9
202	0.9(DL+COLL+CL-1)+1.6(WLR-2)	131.1	0.0	39.9	299.6	0.0	39.9
203	0.9(DL+COLL+CL-2)+1.6(WLR-2)	131.1	0.0	39.9	299.6	0.0	39.9
204	0.9(DL+COLL+CL-3)+1.6(WLR-2)	131.1	0.0	39.9	299.6	0.0	39.9
205	0.9(DL+COLL+CL-4)+1.6(WLR-2)	131.1	0.0	39.9	299.6	0.0	39.9
206	0.9(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)	131.1	0.0	39.9	299.6	0.0	39.9
207	0.9(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)	131.1	0.0	39.9	299.6	0.0	39.9
208	0.9(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)	131.1	0.0	39.9	299.6	0.0	39.9
209	0.9(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)	131.1	0.0	39.9	299.6	0.0	39.9
210	0.9(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)	131.1	0.0	39.9	299.6	0.0	39.9
211	0.9(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)	131.1	0.0	39.9	299.6	0.0	39.9
212	0.9(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)	131.1	0.0	39.9	299.6	0.0	39.9
213	0.9(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)	131.1	0.0	39.9	299.6	0.0	39.9
214	0.9(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)	131.1	0.0	39.9	299.6	0.0	39.9
215	0.9(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)	131.1	0.0	39.9	299.6	0.0	39.9
216	0.9(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)	131.1	0.0	39.9	299.6	0.0	39.9
217	0.9(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)	131.1	0.0	39.9	299.6	0.0	39.9
218	0.9(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG-4)	131.1	0.0	39.9	299.6	0.0	39.9
219	0.9(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG-4)	131.1	0.0	39.9	299.6	0.0	39.9
220	0.9(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG-4)	131.1	0.0	39.9	299.6	0.0	39.9
221	0.9(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG-4)	131.1	0.0	39.9	299.6	0.0	39.9
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')	112.1	63.4	103.3	280.6	63.4	103.3
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')	112.1	63.4	103.3	280.6	63.4	103.3
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')	112.1	63.4	103.3	280.6	63.4	103.3
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')	112.1	63.4	103.3	280.6	63.4	103.3
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')	150.1	63.4	23.5	318.6	63.4	23.5
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-')	150.1	63.4	23.5	318.6	63.4	23.5
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')	150.1	63.4	23.5	318.6	63.4	23.5
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')	150.1	63.4	23.5	318.6	63.4	23.5

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Allowable stress design											
Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
301	1.0(DL+COLL)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL	584.646	0	0.0	-83.2	-224.7	753.1	0.0	0.0	-83.2	-224.7
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL / MLL	584.646	0	0.0	-83.2	-224.7	753.1	0.0	0.0	-83.2	-224.7
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL / MLL	584.646	0	0.0	-83.2	-224.7	753.1	0.0	0.0	-83.2	-224.7
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL / MLL	584.646	0	0.0	-83.2	-224.7	753.1	0.0	0.0	-83.2	-224.7
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL / MLL	584.646	0	0.0	-52.9	-142.7	753.1	0.0	0.0	-52.9	-142.7
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL / MLL	584.646	0	0.0	-52.9	-142.7	753.1	0.0	0.0	-52.9	-142.7
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL / MLL	584.646	0	0.0	-52.9	-142.7	753.1	0.0	0.0	-52.9	-142.7
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL / MLL	584.646	0	0.0	-52.9	-142.7	753.1	0.0	0.0	-52.9	-142.7
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL+PLL	474.91	0	0.0	-66.5	-179.6	643.4	0.0	0.0	-66.5	-179.6
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL+PLL	474.91	0	0.0	-66.5	-179.6	643.4	0.0	0.0	-66.5	-179.6
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL+PLL	474.91	0	0.0	-66.5	-179.6	643.4	0.0	0.0	-66.5	-179.6
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL+PLL	474.91	0	0.0	-66.5	-179.6	643.4	0.0	0.0	-66.5	-179.6
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL+PLL	474.91	0	0.0	-43.7	-118.1	643.4	0.0	0.0	-43.7	-118.1
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL+PLL	474.91	0	0.0	-43.7	-118.1	643.4	0.0	0.0	-43.7	-118.1
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL+PLL	474.91	0	0.0	-43.7	-118.1	643.4	0.0	0.0	-43.7	-118.1
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL+PLL	474.91	0	0.0	-43.7	-118.1	643.4	0.0	0.0	-43.7	-118.1
318	1.0(DL+COLL+CL-1)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
319	1.0(DL+COLL+CL-2)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
320	1.0(DL+COLL+CL-3)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
321	1.0(DL+COLL+CL-4)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	145.698	0	0.0	-16.4	-44.3	314.1	0.0	0.0	-16.4	-44.3
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(RLL+	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-WLLG	474.91	0	0.0	-55.1	-148.9	643.4	0.0	0.0	-55.1	-148.9
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+')	131.45	-17.6205	-17.6	-34.0	-61.9	299.9	-17.6	-17.6	-34.0	-61.9
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+")	131.45	-17.6205	-17.6	-34.0	-61.9	299.9	-17.6	-17.6	-34.0	-61.9
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+')	131.45	-17.6205	-17.6	-34.0	-61.9	299.9	-17.6	-17.6	-34.0	-61.9

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389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+')	131.45	-17.6205	-17.6	-34.0	-61.9	299.9	-17.6	-17.6	-34.0	-61.9
390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	159.95	17.6205	17.6	1.2	-26.7	328.4	17.6	17.6	1.2	-26.7
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	159.95	17.6205	17.6	1.2	-26.7	328.4	17.6	17.6	1.2	-26.7
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	159.95	17.6205	17.6	1.2	-26.7	328.4	17.6	17.6	1.2	-26.7
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	159.95	17.6205	17.6	1.2	-26.7	328.4	17.6	17.6	1.2	-26.7
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'+')+0	464.93	-12.33435	-12.3	-67.5	-161.2	633.4	-12.3	-12.3	-67.5	-161.2
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'+')+0	464.93	-12.33435	-12.3	-67.5	-161.2	633.4	-12.3	-12.3	-67.5	-161.2
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'+')+0	464.93	-12.33435	-12.3	-67.5	-161.2	633.4	-12.3	-12.3	-67.5	-161.2
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'+')+0	464.93	-12.33435	-12.3	-67.5	-161.2	633.4	-12.3	-12.3	-67.5	-161.2
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-')+0.7	484.88	12.33435	12.3	-42.8	-136.5	653.3	12.3	12.3	-42.8	-136.5
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-')+0.7	484.88	12.33435	12.3	-42.8	-136.5	653.3	12.3	12.3	-42.8	-136.5
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-')+0.7	484.88	12.33435	12.3	-42.8	-136.5	653.3	12.3	12.3	-42.8	-136.5
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-')+0.7	484.88	12.33435	12.3	-42.8	-136.5	653.3	12.3	12.3	-42.8	-136.5
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	87.42	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	87.42	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	87.42	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	87.42	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	87.4188	0	0.0	-9.9	-26.6	255.9	0.0	0.0	-9.9	-26.6
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	74.12	-16.4458	-16.4	-26.3	-43.0	242.6	-16.4	-16.4	-26.3	-43.0
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	74.12	-16.4458	-16.4	-26.3	-43.0	242.6	-16.4	-16.4	-26.3	-43.0
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	74.12	-16.4458	-16.4	-26.3	-43.0	242.6	-16.4	-16.4	-26.3	-43.0
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	74.12	-16.4458	-16.4	-26.3	-43.0	242.6	-16.4	-16.4	-26.3	-43.0
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	74.12	-16.4458	-16.4	-26.3	-43.0	242.6	-16.4	-16.4	-26.3	-43.0
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	74.12	-16.4458	-16.4	-26.3	-43.0	242.6	-16.4	-16.4	-26.3	-43.0
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	74.12	-16.4458	-16.4	-26.3	-43.0	242.6	-16.4	-16.4	-26.3	-43.0
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	74.12	-16.4458	-16.4	-26.3	-43.0	242.6	-16.4	-16.4	-26.3	-43.0

Check for overturning and sliding:

Load Case	Load Combination	Load at bottom of foundation					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
301	1.0(DL+COLL)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL /	753.10	0.00	0.00	83.23	224.72	NA	3.02	NA	3.26
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL /	753.10	0.00	0.00	83.23	224.72	NA	3.02	NA	3.26
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL /	753.10	0.00	0.00	83.23	224.72	NA	3.02	NA	3.26
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL /	753.10	0.00	0.00	83.23	224.72	NA	3.02	NA	3.26
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL /	753.10	0.00	0.00	52.85	142.70	NA	4.75	NA	5.13
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL /	753.10	0.00	0.00	52.85	142.70	NA	4.75	NA	5.13
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL /	753.10	0.00	0.00	52.85	142.70	NA	4.75	NA	5.13
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL /	753.10	0.00	0.00	52.85	142.70	NA	4.75	NA	5.13
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL	643.36	0.00	0.00	66.53	179.62	NA	3.22	NA	3.48
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL	643.36	0.00	0.00	66.53	179.62	NA	3.22	NA	3.48
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL	643.36	0.00	0.00	66.53	179.62	NA	3.22	NA	3.48
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL	643.36	0.00	0.00	66.53	179.62	NA	3.22	NA	3.48
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL	643.36	0.00	0.00	43.74	118.10	NA	4.90	NA	5.29
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL	643.36	0.00	0.00	43.74	118.10	NA	4.90	NA	5.29
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL	643.36	0.00	0.00	43.74	118.10	NA	4.90	NA	5.29
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL	643.36	0.00	0.00	43.74	118.10	NA	4.90	NA	5.29
318	1.0(DL+COLL+CL-1)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
319	1.0(DL+COLL+CL-2)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
320	1.0(DL+COLL+CL-3)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
321	1.0(DL+COLL+CL-4)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89

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335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	314.15	0.00	0.00	16.42	44.33	NA	6.38	NA	6.89
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(F	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(F	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(F	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(F	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(F	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(F	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(F	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(F	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(I	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(I	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(I	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(I	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(I	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(I	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(I	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(I	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-V	643.36	0.00	0.00	55.13	148.86	NA	3.89	NA	4.20
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+	299.90	-17.62	-17.62	34.04	61.95	15.32	4.36	6.81	3.17
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+	299.90	-17.62	-17.62	34.04	61.95	15.32	4.36	6.81	3.17
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+	299.90	-17.62	-17.62	34.04	61.95	15.32	4.36	6.81	3.17
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+	299.90	-17.62	-17.62	34.04	61.95	15.32	4.36	6.81	3.17
390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	328.40	17.62	17.62	1.20	26.71	16.77	11.07	7.45	98.25
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	328.40	17.62	17.62	1.20	26.71	16.77	11.07	7.45	98.25
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	328.40	17.62	17.62	1.20	26.71	16.77	11.07	7.45	98.25
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	328.40	17.62	17.62	1.20	26.71	16.77	11.07	7.45	98.25
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'	633.38	-12.33	-12.33	67.47	161.20	46.22	3.54	20.54	3.38
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'	633.38	-12.33	-12.33	67.47	161.20	46.22	3.54	20.54	3.38
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'	633.38	-12.33	-12.33	67.47	161.20	46.22	3.54	20.54	3.38
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'	633.38	-12.33	-12.33	67.47	161.20	46.22	3.54	20.54	3.38
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-	653.33	12.33	12.33	42.80	136.53	47.67	4.31	21.19	5.50
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-	653.33	12.33	12.33	42.80	136.53	47.67	4.31	21.19	5.50
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-	653.33	12.33	12.33	42.80	136.53	47.67	4.31	21.19	5.50
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-'	653.33	12.33	12.33	42.80	136.53	47.67	4.31	21.19	5.50
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	255.87	0.00	0.00	9.85	26.60	NA	8.66	NA	9.35
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	242.57	-16.45	-16.45	26.30	43.04	13.27	5.07	5.90	3.32
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	242.57	-16.45	-16.45	26.30	43.04	13.27	5.07	5.90	3.32
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	242.57	-16.45	-16.45	26.30	43.04	13.27	5.07	5.90	3.32
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	242.57	-16.45	-16.45	26.30	43.04	13.27	5.07	5.90	3.32
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	242.57	-16.45	-16.45	26.30	43.04	13.27	5.07	5.90	3.32
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	242.57	-16.45	-16.45	26.30	43.04	13.27	5.07	5.90	3.32
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	242.57	-16.45	-16.45	26.30	43.04	13.27	5.07	5.90	3.32
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	242.57	-16.45	-16.45	26.30	43.04	13.27	5.07	5.90	3.32
						Min.=	**13.27**	**3.02**	**5.90**	**3.17**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F6

Load Combination & Calculation of Base pressure

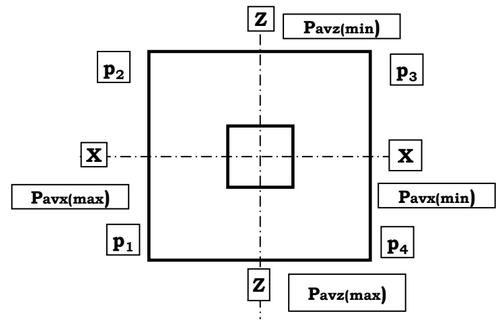
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 400 kN/m² For WL/SL case = 500 kN/m²

Gross SBC of soil for DL, LL= 436 kN/m² For WL/SL case = 536 kN/m²



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/(Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/Lx.(Lz)²	Base Pressure (kN/m²) p1 (pmax)		p2	p4	p3 (pmin)	(pmax-γh)	pavx(max)	pavx(min)	pavz(max)	pavz(min)	Remarks
301	1.0(DL+COLL)	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
302	1.0(DL+COLL+CL-1+TL1)+1.(	753.10	0.00	-224.72	188.27	0.00	168.54	356.81		19.73	356.81	19.73	320.81	188.27	188.27	356.81	19.73	O.K
303	1.0(DL+COLL+CL-2+TL1)+1.(	753.10	0.00	-224.72	188.27	0.00	168.54	356.81		19.73	356.81	19.73	320.81	188.27	188.27	356.81	19.73	O.K
304	1.0(DL+COLL+CL-3+TL1)+1.(	753.10	0.00	-224.72	188.27	0.00	168.54	356.81		19.73	356.81	19.73	320.81	188.27	188.27	356.81	19.73	O.K
305	1.0(DL+COLL+CL-4+TL1)+1.(	753.10	0.00	-224.72	188.27	0.00	168.54	356.81		19.73	356.81	19.73	320.81	188.27	188.27	356.81	19.73	O.K
306	1.0(DL+COLL+CL-1+TL2)+1.(	753.10	0.00	-142.70	188.27	0.00	107.02	295.30		81.25	295.30	81.25	259.30	188.28	188.28	295.30	81.25	O.K
307	1.0(DL+COLL+CL-2+TL2)+1.(	753.10	0.00	-142.70	188.27	0.00	107.02	295.30		81.25	295.30	81.25	259.30	188.28	188.28	295.30	81.25	O.K
308	1.0(DL+COLL+CL-3+TL2)+1.(	753.10	0.00	-142.70	188.27	0.00	107.02	295.30		81.25	295.30	81.25	259.30	188.28	188.28	295.30	81.25	O.K
309	1.0(DL+COLL+CL-4+TL2)+1.(	753.10	0.00	-142.70	188.27	0.00	107.02	295.30		81.25	295.30	81.25	259.30	188.28	188.28	295.30	81.25	O.K
310	1.0(DL+COLL+CL-1)+0.75(TL	643.36	0.00	-179.62	160.84	0.00	134.72	295.56		26.12	295.56	26.12	259.56	160.84	160.84	295.56	26.12	O.K
311	1.0(DL+COLL+CL-2)+0.75(TL	643.36	0.00	-179.62	160.84	0.00	134.72	295.56		26.12	295.56	26.12	259.56	160.84	160.84	295.56	26.12	O.K
312	1.0(DL+COLL+CL-3)+0.75(TL	643.36	0.00	-179.62	160.84	0.00	134.72	295.56		26.12	295.56	26.12	259.56	160.84	160.84	295.56	26.12	O.K
313	1.0(DL+COLL+CL-4)+0.75(TL	643.36	0.00	-179.62	160.84	0.00	134.72	295.56		26.12	295.56	26.12	259.56	160.84	160.84	295.56	26.12	O.K
314	1.0(DL+COLL+CL-1)+0.75(TL	643.36	0.00	-118.10	160.84	0.00	88.58	249.42		72.26	249.42	72.26	213.42	160.84	160.84	249.42	72.26	O.K
315	1.0(DL+COLL+CL-2)+0.75(TL	643.36	0.00	-118.10	160.84	0.00	88.58	249.42		72.26	249.42	72.26	213.42	160.84	160.84	249.42	72.26	O.K
316	1.0(DL+COLL+CL-3)+0.75(TL	643.36	0.00	-118.10	160.84	0.00	88.58	249.42		72.26	249.42	72.26	213.42	160.84	160.84	249.42	72.26	O.K

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317	1.0(DL+COLL+CL-4)+0.75(TL	643.36	0.00	-118.10	160.84	0.00	88.58	249.42		72.26	249.42	72.26	213.42	160.84	160.84	249.42	72.26	O.K
318	1.0(DL+COLL+CL-1)	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
319	1.0(DL+COLL+CL-2)	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
320	1.0(DL+COLL+CL-3)	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
321	1.0(DL+COLL+CL-4)	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
322	1.0(DL+COLL+CL-1)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
323	1.0(DL+COLL+CL-2)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
324	1.0(DL+COLL+CL-3)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
325	1.0(DL+COLL+CL-4)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
326	1.0(DL+COLL+CL-1)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
327	1.0(DL+COLL+CL-2)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
328	1.0(DL+COLL+CL-3)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
329	1.0(DL+COLL+CL-4)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
330	1.0(DL+COLL+CL-1)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
331	1.0(DL+COLL+CL-2)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
332	1.0(DL+COLL+CL-3)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
333	1.0(DL+COLL+CL-4)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
334	1.0(DL+COLL+CL-1)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
335	1.0(DL+COLL+CL-2)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
336	1.0(DL+COLL+CL-3)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
337	1.0(DL+COLL+CL-4)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
338	1.0(DL+COLL+CL-1)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
339	1.0(DL+COLL+CL-2)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
340	1.0(DL+COLL+CL-3)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
341	1.0(DL+COLL+CL-4)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
342	1.0(DL+COLL+CL-1)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
343	1.0(DL+COLL+CL-2)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
344	1.0(DL+COLL+CL-3)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
345	1.0(DL+COLL+CL-4)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
346	1.0(DL+COLL+CL-1)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
347	1.0(DL+COLL+CL-2)+1.0(WLI	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K

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348	1.0(DL+COLL+CL-3)+1.0(WL	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
349	1.0(DL+COLL+CL-4)+1.0(WL	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
350	1.0(DL+COLL+CL-1)+1.0(WL	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
351	1.0(DL+COLL+CL-2)+1.0(WL	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
352	1.0(DL+COLL+CL-3)+1.0(WL	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
353	1.0(DL+COLL+CL-4)+1.0(WL	314.15	0.00	-44.33	78.54	0.00	33.24	111.78		45.29	111.78	45.29	75.78	78.54	78.54	111.78	45.29	O.K
354	1.0(DL+COLL+CL-1)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
355	1.0(DL+COLL+CL-2)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
356	1.0(DL+COLL+CL-3)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
357	1.0(DL+COLL+CL-4)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
358	1.0(DL+COLL+CL-1)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
359	1.0(DL+COLL+CL-2)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
360	1.0(DL+COLL+CL-3)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
361	1.0(DL+COLL+CL-4)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
362	1.0(DL+COLL+CL-1)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
363	1.0(DL+COLL+CL-2)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
364	1.0(DL+COLL+CL-3)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
365	1.0(DL+COLL+CL-4)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
366	1.0(DL+COLL+CL-1)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
367	1.0(DL+COLL+CL-2)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
368	1.0(DL+COLL+CL-3)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
369	1.0(DL+COLL+CL-4)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
370	1.0(DL+COLL+CL-1)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
371	1.0(DL+COLL+CL-2)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
372	1.0(DL+COLL+CL-3)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
373	1.0(DL+COLL+CL-4)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
374	1.0(DL+COLL+CL-1)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
375	1.0(DL+COLL+CL-2)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
376	1.0(DL+COLL+CL-3)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
377	1.0(DL+COLL+CL-4)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
378	1.0(DL+COLL+CL-1)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K

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379	1.0(DL+COLL+CL-2)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
380	1.0(DL+COLL+CL-3)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
381	1.0(DL+COLL+CL-4)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
382	1.0(DL+COLL+CL-1)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
383	1.0(DL+COLL+CL-2)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
384	1.0(DL+COLL+CL-3)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
385	1.0(DL+COLL+CL-4)+0.75(WI	643.36	0.00	-148.86	160.84	0.00	111.65	272.49		49.19	272.49	49.19	236.49	160.84	160.84	272.49	49.19	O.K
386	1.0(DL+COLL+CL-1)+0.75(SL	299.90	-17.62	-61.95	74.97	13.22	46.46	134.65		41.73	108.22	15.30	98.65	88.19	61.76	121.44	28.52	O.K
387	1.0(DL+COLL+CL-2)+0.75(SL	299.90	-17.62	-61.95	74.97	13.22	46.46	134.65		41.73	108.22	15.30	98.65	88.19	61.76	121.44	28.52	O.K
388	1.0(DL+COLL+CL-3)+0.75(SL	299.90	-17.62	-61.95	74.97	13.22	46.46	134.65		41.73	108.22	15.30	98.65	88.19	61.76	121.44	28.52	O.K
389	1.0(DL+COLL+CL-4)+0.75(SL	299.90	-17.62	-61.95	74.97	13.22	46.46	134.65		41.73	108.22	15.30	98.65	88.19	61.76	121.44	28.52	O.K
390	1.0(DL+COLL+CL-1)+0.75(SL	328.40	17.62	-26.71	82.10	13.22	20.03	115.34		75.29	88.91	48.85	79.34	95.32	68.88	102.13	62.07	O.K
391	1.0(DL+COLL+CL-2)+0.75(SL	328.40	17.62	-26.71	82.10	13.22	20.03	115.34		75.29	88.91	48.85	79.34	95.32	68.88	102.13	62.07	O.K
392	1.0(DL+COLL+CL-3)+0.75(SL	328.40	17.62	-26.71	82.10	13.22	20.03	115.34		75.29	88.91	48.85	79.34	95.32	68.88	102.13	62.07	O.K
393	1.0(DL+COLL+CL-4)+0.75(SL	328.40	17.62	-26.71	82.10	13.22	20.03	115.34		75.29	88.91	48.85	79.34	95.32	68.88	102.13	62.07	O.K
394	1.0(DL+COLL+CL-1)+0.525(S	633.38	-12.33	-161.20	158.35	9.25	120.90	288.49		46.70	269.99	28.20	252.49	167.60	149.10	279.24	37.45	O.K
395	1.0(DL+COLL+CL-2)+0.525(S	633.38	-12.33	-161.20	158.35	9.25	120.90	288.49		46.70	269.99	28.20	252.49	167.60	149.10	279.24	37.45	O.K
396	1.0(DL+COLL+CL-3)+0.525(S	633.38	-12.33	-161.20	158.35	9.25	120.90	288.49		46.70	269.99	28.20	252.49	167.60	149.10	279.24	37.45	O.K
397	1.0(DL+COLL+CL-4)+0.525(S	633.38	-12.33	-161.20	158.35	9.25	120.90	288.49		46.70	269.99	28.20	252.49	167.60	149.10	279.24	37.45	O.K
398	1.0(DL+COLL+CL-1)+0.525(S	653.33	12.33	-136.53	163.33	9.25	102.40	274.98		70.19	256.48	51.69	238.98	172.59	154.09	265.73	60.94	O.K
399	1.0(DL+COLL+CL-2)+0.525(S	653.33	12.33	-136.53	163.33	9.25	102.40	274.98		70.19	256.48	51.69	238.98	172.59	154.09	265.73	60.94	O.K
400	1.0(DL+COLL+CL-3)+0.525(S	653.33	12.33	-136.53	163.33	9.25	102.40	274.98		70.19	256.48	51.69	238.98	172.59	154.09	265.73	60.94	O.K
401	1.0(DL+COLL+CL-4)+0.525SI	653.33	12.33	-136.53	163.33	9.25	102.40	274.98		70.19	256.48	51.69	238.98	172.59	154.09	265.73	60.94	O.K
402	0.6(DL+COLL+CL-1)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
403	0.6(DL+COLL+CL-2)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
404	0.6(DL+COLL+CL-3)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
405	0.6(DL+COLL+CL-4)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
406	0.6(DL+COLL+CL-1)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
407	0.6(DL+COLL+CL-2)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
408	0.6(DL+COLL+CL-3)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
409	0.6(DL+COLL+CL-4)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
410	0.6(DL+COLL+CL-1)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K

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411	0.6(DL+COLL+CL-2)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
412	0.6(DL+COLL+CL-3)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
413	0.6(DL+COLL+CL-4)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
414	0.6(DL+COLL+CL-1)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
415	0.6(DL+COLL+CL-2)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
416	0.6(DL+COLL+CL-3)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
417	0.6(DL+COLL+CL-4)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
418	0.6(DL+COLL+CL-1)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
419	0.6(DL+COLL+CL-2)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
420	0.6(DL+COLL+CL-3)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
421	0.6(DL+COLL+CL-4)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
422	0.6(DL+COLL+CL-1)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
423	0.6(DL+COLL+CL-2)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
424	0.6(DL+COLL+CL-3)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
425	0.6(DL+COLL+CL-4)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
426	0.6(DL+COLL+CL-1)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
427	0.6(DL+COLL+CL-2)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
428	0.6(DL+COLL+CL-3)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
429	0.6(DL+COLL+CL-4)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
430	0.6(DL+COLL+CL-1)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
431	0.6(DL+COLL+CL-2)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
432	0.6(DL+COLL+CL-3)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
433	0.6(DL+COLL+CL-4)+1.0(WLI	255.87	0.00	-26.60	63.97	0.00	19.95	83.91		44.02	83.91	44.02	47.91	63.97	63.97	83.91	44.02	O.K
434	0.6(DL+COLL+CL-1)+0.7(SL'+	242.57	-16.45	-43.04	60.64	12.33	32.28	105.26		40.70	80.59	16.03	69.26	72.98	48.31	92.93	28.37	O.K
435	0.6(DL+COLL+CL-2)+0.7(SL'+	242.57	-16.45	-43.04	60.64	12.33	32.28	105.26		40.70	80.59	16.03	69.26	72.98	48.31	92.93	28.37	O.K
436	0.6(DL+COLL+CL-3)+0.7(SL'+	242.57	-16.45	-43.04	60.64	12.33	32.28	105.26		40.70	80.59	16.03	69.26	72.98	48.31	92.93	28.37	O.K
437	0.6(DL+COLL+CL-4)+0.7(SL'+	242.57	-16.45	-43.04	60.64	12.33	32.28	105.26		40.70	80.59	16.03	69.26	72.98	48.31	92.93	28.37	O.K
438	0.6(DL+COLL+CL-1)+0.7(SL'+	242.57	-16.45	-43.04	60.64	12.33	32.28	105.26		40.70	80.59	16.03	69.26	72.98	48.31	92.93	28.37	O.K
439	0.6(DL+COLL+CL-2)+0.7(SL'+	242.57	-16.45	-43.04	60.64	12.33	32.28	105.26		40.70	80.59	16.03	69.26	72.98	48.31	92.93	28.37	O.K
440	0.6(DL+COLL+CL-3)+0.7(SL'+	242.57	-16.45	-43.04	60.64	12.33	32.28	105.26		40.70	80.59	16.03	69.26	72.98	48.31	92.93	28.37	O.K
441	0.6(DL+COLL+CL-4)+0.7(SL'+	242.57	-16.45	-43.04	60.64	12.33	32.28	105.26		40.70	80.59	16.03	69.26	72.98	48.31	92.93	28.37	O.K

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Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
	Ultimate Conditions																	
101	1.4(DL+COLL)	372.43	0.00	62.06	93.11	0.00	46.54	139.65		46.56	139.65	46.56	103.65	93.11	93.11	139.65	46.56	
102	1.2(DL+COLL+CL-1)+1.6(RLL	1045.60	0.00	276.20	261.40	0.00	207.15	468.55		54.25	468.55	54.25	432.55	261.40	261.40	468.55	54.25	
103	1.2(DL+COLL+CL-2)+1.6(RLL	1045.60	0.00	276.20	261.40	0.00	207.15	468.55		54.25	468.55	54.25	432.55	261.40	261.40	468.55	54.25	
104	1.2(DL+COLL+CL-3)+1.6(RLL	1045.60	0.00	276.20	261.40	0.00	207.15	468.55		54.25	468.55	54.25	432.55	261.40	261.40	468.55	54.25	
105	1.2(DL+COLL+CL-4)+1.6(RLL	1045.60	0.00	276.20	261.40	0.00	207.15	468.55		54.25	468.55	54.25	432.55	261.40	261.40	468.55	54.25	
106	1.4(DL+COLL+CL-1)	372.43	0.00	62.06	93.11	0.00	46.54	139.65		46.56	139.65	46.56	103.65	93.11	93.11	139.65	46.56	
107	1.4(DL+COLL+CL-2)	372.43	0.00	62.06	93.11	0.00	46.54	139.65		46.56	139.65	46.56	103.65	93.11	93.11	139.65	46.56	
108	1.4(DL+COLL+CL-3)	372.43	0.00	62.06	93.11	0.00	46.54	139.65		46.56	139.65	46.56	103.65	93.11	93.11	139.65	46.56	
109	1.4(DL+COLL+CL-4)	372.43	0.00	62.06	93.11	0.00	46.54	139.65		46.56	139.65	46.56	103.65	93.11	93.11	139.65	46.56	
110	1.2(DL+COLL+CL-1+TL1)+1.6	1045.60	0.00	325.42	261.40	0.00	244.06	505.46		17.34	505.46	17.34	469.46	261.40	261.40	505.46	17.34	
111	1.2(DL+COLL+CL-2+TL1)+1.6	1045.60	0.00	325.42	261.40	0.00	244.06	505.46		17.34	505.46	17.34	469.46	261.40	261.40	505.46	17.34	
112	1.2(DL+COLL+CL-3+TL1)+1.6	1045.60	0.00	325.42	261.40	0.00	244.06	505.46		17.34	505.46	17.34	469.46	261.40	261.40	505.46	17.34	
113	1.2(DL+COLL+CL-4+TL1)+1.6	1045.60	0.00	325.42	261.40	0.00	244.06	505.46		17.34	505.46	17.34	469.46	261.40	261.40	505.46	17.34	
114	1.2(DL+COLL+CL-1+TL2)+1.6	1045.60	0.00	226.99	261.40	0.00	170.24	431.64		91.16	431.64	91.16	395.64	261.40	261.40	431.64	91.16	
115	1.2(DL+COLL+CL-2+TL2)+1.6	1045.60	0.00	226.99	261.40	0.00	170.24	431.64		91.16	431.64	91.16	395.64	261.40	261.40	431.64	91.16	
116	1.2(DL+COLL+CL-3+TL2)+1.6	1045.60	0.00	226.99	261.40	0.00	170.24	431.64		91.16	431.64	91.16	395.64	261.40	261.40	431.64	91.16	
117	1.2(DL+COLL+CL-4+TL2)+1.6	1045.60	0.00	226.99	261.40	0.00	170.24	431.64		91.16	431.64	91.16	395.64	261.40	261.40	431.64	91.16	
118	1.2(DL+COLL+CL-1)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
119	1.2(DL+COLL+CL-2)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
120	1.2(DL+COLL+CL-3)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
121	1.2(DL+COLL+CL-4)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
122	1.2(DL+COLL+CL-1)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
123	1.2(DL+COLL+CL-2)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
124	1.2(DL+COLL+CL-3)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	

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125	1.2(DL+COLL+CL-4)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
126	1.2(DL+COLL+CL-1)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
127	1.2(DL+COLL+CL-2)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
128	1.2(DL+COLL+CL-3)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
129	1.2(DL+COLL+CL-4)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
130	1.2(DL+COLL+CL-1)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
131	1.2(DL+COLL+CL-2)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
132	1.2(DL+COLL+CL-3)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
133	1.2(DL+COLL+CL-4)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
134	1.2(DL+COLL+CL-1)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
135	1.2(DL+COLL+CL-2)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
136	1.2(DL+COLL+CL-3)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
137	1.2(DL+COLL+CL-4)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
138	1.2(DL+COLL+CL-1)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
139	1.2(DL+COLL+CL-2)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
140	1.2(DL+COLL+CL-3)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
141	1.2(DL+COLL+CL-4)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
142	1.2(DL+COLL+CL-1)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
143	1.2(DL+COLL+CL-2)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
144	1.2(DL+COLL+CL-3)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
145	1.2(DL+COLL+CL-4)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
146	1.2(DL+COLL+CL-1)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
147	1.2(DL+COLL+CL-2)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
148	1.2(DL+COLL+CL-3)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
149	1.2(DL+COLL+CL-4)+0.8(WLI	343.29	0.00	53.19	85.82	0.00	39.89	125.72		45.93	125.72	45.93	89.72	85.83	85.83	125.72	45.93	
150	1.2(DL+COLL+CL-1)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
151	1.2(DL+COLL+CL-2)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
152	1.2(DL+COLL+CL-3)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
153	1.2(DL+COLL+CL-4)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
154	1.2(DL+COLL+CL-1)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
155	1.2(DL+COLL+CL-2)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
156	1.2(DL+COLL+CL-3)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	

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157	1.2(DL+COLL+CL-4)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
158	1.2(DL+COLL+CL-1)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
159	1.2(DL+COLL+CL-2)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
160	1.2(DL+COLL+CL-3)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
161	1.2(DL+COLL+CL-4)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
162	1.2(DL+COLL+CL-1)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
163	1.2(DL+COLL+CL-2)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
164	1.2(DL+COLL+CL-3)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
165	1.2(DL+COLL+CL-4)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
166	1.2(DL+COLL+CL-1)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
167	1.2(DL+COLL+CL-2)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
168	1.2(DL+COLL+CL-3)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
169	1.2(DL+COLL+CL-4)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
170	1.2(DL+COLL+CL-1)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
171	1.2(DL+COLL+CL-2)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
172	1.2(DL+COLL+CL-3)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
173	1.2(DL+COLL+CL-4)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
174	1.2(DL+COLL+CL-1)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
175	1.2(DL+COLL+CL-2)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
176	1.2(DL+COLL+CL-3)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
177	1.2(DL+COLL+CL-4)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
178	1.2(DL+COLL+CL-1)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
179	1.2(DL+COLL+CL-2)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
180	1.2(DL+COLL+CL-3)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
181	1.2(DL+COLL+CL-4)+1.6(WLI	562.76	0.00	122.88	140.69	0.00	92.16	232.85		48.53	232.85	48.53	196.85	140.69	140.69	232.85	48.53	
182	1.2(DL+COLL+CL-1)+1.0(SL'+	543.76	63.43	186.32	135.94	47.58	139.74	323.25		43.78	228.10	-51.37	287.25	183.52	88.37	275.68	-3.80	
183	1.2(DL+COLL+CL-2)+1.0(SL'+	543.76	63.43	186.32	135.94	47.58	139.74	323.25		43.78	228.10	-51.37	287.25	183.52	88.37	275.68	-3.80	
184	1.2(DL+COLL+CL-3)+1.0(SL'+	543.76	63.43	186.32	135.94	47.58	139.74	323.25		43.78	228.10	-51.37	287.25	183.52	88.37	275.68	-3.80	
185	1.2(DL+COLL+CL-4)+1.0(SL'+	543.76	63.43	186.32	135.94	47.58	139.74	323.25		43.78	228.10	-51.37	287.25	183.52	88.37	275.68	-3.80	
186	1.2(DL+COLL+CL-1)+1.0(SL'-	581.76	63.43	59.45	145.44	47.58	44.59	237.60		148.43	142.45	53.28	201.60	193.02	97.87	190.03	100.86	
187	1.2(DL+COLL+CL-2)+1.0(SL'-	581.76	63.43	59.45	145.44	47.58	44.59	237.60		148.43	142.45	53.28	201.60	193.02	97.87	190.03	100.86	

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188	1.2(DL+COLL+CL-3)+1.0(SL'-	581.76	63.43	59.45	145.44	47.58	44.59	237.60		148.43	142.45	53.28	201.60	193.02	97.87	190.03	100.86	
189	1.2(DL+COLL+CL-4)+1.0(SL'-	581.76	63.43	59.45	145.44	47.58	44.59	237.60		148.43	142.45	53.28	201.60	193.02	97.87	190.03	100.86	
190	0.9(DL+COLL+CL-1)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
191	0.9(DL+COLL+CL-2)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
192	0.9(DL+COLL+CL-3)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
193	0.9(DL+COLL+CL-4)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
194	0.9(DL+COLL+CL-1)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
195	0.9(DL+COLL+CL-2)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
196	0.9(DL+COLL+CL-3)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
197	0.9(DL+COLL+CL-4)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
198	0.9(DL+COLL+CL-1)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
199	0.9(DL+COLL+CL-2)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
200	0.9(DL+COLL+CL-3)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
201	0.9(DL+COLL+CL-4)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
202	0.9(DL+COLL+CL-1)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
203	0.9(DL+COLL+CL-2)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
204	0.9(DL+COLL+CL-3)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
205	0.9(DL+COLL+CL-4)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
206	0.9(DL+COLL+CL-1)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
207	0.9(DL+COLL+CL-2)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
208	0.9(DL+COLL+CL-3)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
209	0.9(DL+COLL+CL-4)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
210	0.9(DL+COLL+CL-1)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
211	0.9(DL+COLL+CL-2)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
212	0.9(DL+COLL+CL-3)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
213	0.9(DL+COLL+CL-4)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
214	0.9(DL+COLL+CL-1)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
215	0.9(DL+COLL+CL-2)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
216	0.9(DL+COLL+CL-3)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
217	0.9(DL+COLL+CL-4)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
218	0.9(DL+COLL+CL-1)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
219	0.9(DL+COLL+CL-2)+1.6(WLI	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	

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220	0.9(DL+COLL+CL-3)+1.6(WL	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
221	0.9(DL+COLL+CL-4)+1.6(WL	299.58	0.00	39.89	74.89	0.00	29.92	104.81		44.97	104.81	44.97	68.81	74.89	74.89	104.81	44.97	
222	0.9(DL+COLL+CL-1)+1.0(SL'+	280.58	63.43	103.33	70.14	47.58	77.50	195.22		40.22	100.06	-54.93	159.22	117.72	22.57	147.64	-7.36	
223	0.9(DL+COLL+CL-2)+1.0(SL'+	280.58	63.43	103.33	70.14	47.58	77.50	195.22		40.22	100.06	-54.93	159.22	117.72	22.57	147.64	-7.36	
224	0.9(DL+COLL+CL-3)+1.0(SL'+	280.58	63.43	103.33	70.14	47.58	77.50	195.22		40.22	100.06	-54.93	159.22	117.72	22.57	147.64	-7.36	
225	0.9(DL+COLL+CL-4)+1.0(SL'+	280.58	63.43	103.33	70.14	47.58	77.50	195.22		40.22	100.06	-54.93	159.22	117.72	22.57	147.64	-7.36	
226	0.9(DL+COLL+CL-1)+1.0(SL'-	318.58	63.43	23.54	79.64	47.58	17.66	144.88		109.57	49.72	14.41	108.88	127.23	32.07	97.30	61.99	
227	0.9(DL+COLL+CL-2)+1.0(SL'-	318.58	63.43	23.54	79.64	47.58	17.66	144.88		109.57	49.72	14.41	108.88	127.23	32.07	97.30	61.99	
228	0.9(DL+COLL+CL-3)+1.0(SL'-	318.58	63.43	23.54	79.64	47.58	17.66	144.88		109.57	49.72	14.41	108.88	127.23	32.07	97.30	61.99	
229	0.9(DL+COLL+CL-4)+1.0(SL'-	318.58	63.43	23.54	79.64	47.58	17.66	144.88		109.57	49.72	14.41	108.88	127.23	32.07	97.30	61.99	
	Max :	753	18	-27														
	Min :	243	-18	-225														

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 75.78 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 85.78 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 252.49 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 262.49 kN/m^2

Sample calculation of values of base pressure:

Load case : 302 1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL)

Maximum pressure, p_1 (pmax) = 188.27+0.00+168.54= 356.81 kN/m^2

p_2= 188.27+0.00-168.54= 19.73 kN/m^2

p_3= 188.27-(0.00+168.54)= 19.73 kN/m^2

p_4= 188.27-0.00+168.54= 356.81 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 356.81-2*18= 320.81 kN/m^2

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DESIGN OF ISOLATED FOOTING F6

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

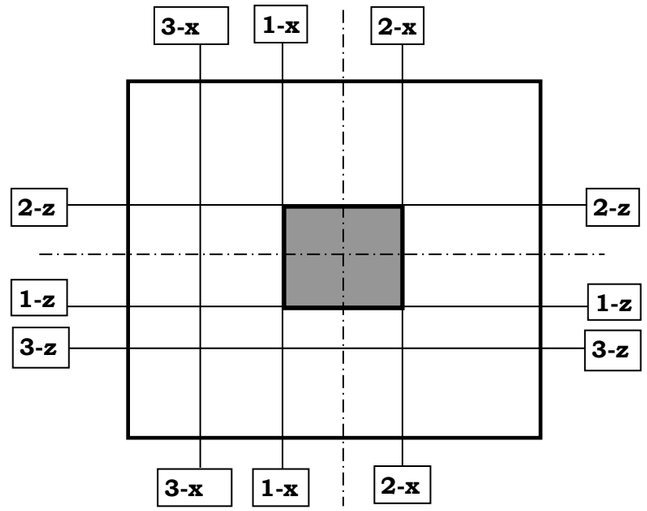
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.75+2*0.5*0.326)+(0.75+2*0.5*0.326))*2
= 4.304 m

Punching Area= ((0.75+2*0.5*0.326)*(0.75+2*0.5*0.326))
= 1.16 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.6*18)
= 38.40 kN/m^2

Area of footing, A_f= 4.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+COLL)	93.11	93.11	93.11	10.69	10.69	15.48	155.37
102	1.2(DL+COLL+CL-1)+1.6(R	261.40	261.40	261.40	43.55	43.55	63.11	633.32
103	1.2(DL+COLL+CL-2)+1.6(R	261.40	261.40	261.40	43.55	43.55	63.11	633.32
104	1.2(DL+COLL+CL-3)+1.6(R	261.40	261.40	261.40	43.55	43.55	63.11	633.32
105	1.2(DL+COLL+CL-4)+1.6(R	261.40	261.40	261.40	43.55	43.55	63.11	633.32
106	1.4(DL+COLL+CL-1)	93.11	93.11	93.11	10.69	10.69	15.48	155.37
107	1.4(DL+COLL+CL-2)	93.11	93.11	93.11	10.69	10.69	15.48	155.37
108	1.4(DL+COLL+CL-3)	93.11	93.11	93.11	10.69	10.69	15.48	155.37
109	1.4(DL+COLL+CL-4)	93.11	93.11	93.11	10.69	10.69	15.48	155.37
110	1.2(DL+COLL+CL-1+TL1)+	261.40	261.40	261.40	43.55	43.55	63.11	633.32
111	1.2(DL+COLL+CL-2+TL1)+	261.40	261.40	261.40	43.55	43.55	63.11	633.32
112	1.2(DL+COLL+CL-3+TL1)+	261.40	261.40	261.40	43.55	43.55	63.11	633.32

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113	1.2(DL+COLL+CL-4+TL1)+	261.40	261.40	261.40	43.55	43.55	63.11	633.32
114	1.2(DL+COLL+CL-1+TL2)+	261.40	261.40	261.40	43.55	43.55	63.11	633.32
115	1.2(DL+COLL+CL-2+TL2)+	261.40	261.40	261.40	43.55	43.55	63.11	633.32
116	1.2(DL+COLL+CL-3+TL2)+	261.40	261.40	261.40	43.55	43.55	63.11	633.32
117	1.2(DL+COLL+CL-4+TL2)+	261.40	261.40	261.40	43.55	43.55	63.11	633.32
118	1.2(DL+COLL+CL-1)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
119	1.2(DL+COLL+CL-2)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
120	1.2(DL+COLL+CL-3)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
121	1.2(DL+COLL+CL-4)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
122	1.2(DL+COLL+CL-1)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
123	1.2(DL+COLL+CL-2)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
124	1.2(DL+COLL+CL-3)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
125	1.2(DL+COLL+CL-4)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
126	1.2(DL+COLL+CL-1)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
127	1.2(DL+COLL+CL-2)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
128	1.2(DL+COLL+CL-3)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
129	1.2(DL+COLL+CL-4)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
130	1.2(DL+COLL+CL-1)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
131	1.2(DL+COLL+CL-2)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
132	1.2(DL+COLL+CL-3)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
133	1.2(DL+COLL+CL-4)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
134	1.2(DL+COLL+CL-1)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
135	1.2(DL+COLL+CL-2)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
136	1.2(DL+COLL+CL-3)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
137	1.2(DL+COLL+CL-4)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
138	1.2(DL+COLL+CL-1)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
139	1.2(DL+COLL+CL-2)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
140	1.2(DL+COLL+CL-3)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
141	1.2(DL+COLL+CL-4)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
142	1.2(DL+COLL+CL-1)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
143	1.2(DL+COLL+CL-2)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68

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144	1.2(DL+COLL+CL-3)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
145	1.2(DL+COLL+CL-4)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
146	1.2(DL+COLL+CL-1)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
147	1.2(DL+COLL+CL-2)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
148	1.2(DL+COLL+CL-3)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
149	1.2(DL+COLL+CL-4)+0.8(W	85.83	85.83	85.83	9.26	9.26	13.42	134.68
150	1.2(DL+COLL+CL-1)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
151	1.2(DL+COLL+CL-2)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
152	1.2(DL+COLL+CL-3)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
153	1.2(DL+COLL+CL-4)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
154	1.2(DL+COLL+CL-1)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
155	1.2(DL+COLL+CL-2)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
156	1.2(DL+COLL+CL-3)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
157	1.2(DL+COLL+CL-4)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
158	1.2(DL+COLL+CL-1)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
159	1.2(DL+COLL+CL-2)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
160	1.2(DL+COLL+CL-3)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
161	1.2(DL+COLL+CL-4)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
162	1.2(DL+COLL+CL-1)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
163	1.2(DL+COLL+CL-2)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
164	1.2(DL+COLL+CL-3)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
165	1.2(DL+COLL+CL-4)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
166	1.2(DL+COLL+CL-1)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
167	1.2(DL+COLL+CL-2)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
168	1.2(DL+COLL+CL-3)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
169	1.2(DL+COLL+CL-4)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
170	1.2(DL+COLL+CL-1)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
171	1.2(DL+COLL+CL-2)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
172	1.2(DL+COLL+CL-3)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
173	1.2(DL+COLL+CL-4)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
174	1.2(DL+COLL+CL-1)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50

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175	1.2(DL+COLL+CL-2)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
176	1.2(DL+COLL+CL-3)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
177	1.2(DL+COLL+CL-4)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
178	1.2(DL+COLL+CL-1)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
179	1.2(DL+COLL+CL-2)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
180	1.2(DL+COLL+CL-3)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
181	1.2(DL+COLL+CL-4)+1.6(W	140.69	140.69	140.69	19.98	19.98	28.95	290.50
182	1.2(DL+COLL+CL-1)+1.0(S	153.79	118.10	170.06	26.41	11.70	39.16	277.01
183	1.2(DL+COLL+CL-2)+1.0(S	153.79	118.10	170.06	26.41	11.70	39.16	277.01
184	1.2(DL+COLL+CL-3)+1.0(S	153.79	118.10	170.06	26.41	11.70	39.16	277.01
185	1.2(DL+COLL+CL-4)+1.0(S	153.79	118.10	170.06	26.41	11.70	39.16	277.01
186	1.2(DL+COLL+CL-1)+1.0(S	163.29	127.60	179.56	28.26	13.55	41.85	304.00
187	1.2(DL+COLL+CL-2)+1.0(S	163.29	127.60	179.56	28.26	13.55	41.85	304.00
188	1.2(DL+COLL+CL-3)+1.0(S	163.29	127.60	179.56	28.26	13.55	41.85	304.00
189	1.2(DL+COLL+CL-4)+1.0(S	163.29	127.60	179.56	28.26	13.55	41.85	304.00
190	0.9(DL+COLL+CL-1)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
191	0.9(DL+COLL+CL-2)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
192	0.9(DL+COLL+CL-3)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
193	0.9(DL+COLL+CL-4)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
194	0.9(DL+COLL+CL-1)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
195	0.9(DL+COLL+CL-2)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
196	0.9(DL+COLL+CL-3)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
197	0.9(DL+COLL+CL-4)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
198	0.9(DL+COLL+CL-1)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
199	0.9(DL+COLL+CL-2)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
200	0.9(DL+COLL+CL-3)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
201	0.9(DL+COLL+CL-4)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
202	0.9(DL+COLL+CL-1)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
203	0.9(DL+COLL+CL-2)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
204	0.9(DL+COLL+CL-3)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
205	0.9(DL+COLL+CL-4)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64

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206	0.9(DL+COLL+CL-1)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
207	0.9(DL+COLL+CL-2)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
208	0.9(DL+COLL+CL-3)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
209	0.9(DL+COLL+CL-4)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
210	0.9(DL+COLL+CL-1)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
211	0.9(DL+COLL+CL-2)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
212	0.9(DL+COLL+CL-3)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
213	0.9(DL+COLL+CL-4)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
214	0.9(DL+COLL+CL-1)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
215	0.9(DL+COLL+CL-2)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
216	0.9(DL+COLL+CL-3)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
217	0.9(DL+COLL+CL-4)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
218	0.9(DL+COLL+CL-1)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
219	0.9(DL+COLL+CL-2)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
220	0.9(DL+COLL+CL-3)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
221	0.9(DL+COLL+CL-4)+1.6(W	74.89	74.89	74.89	7.13	7.13	10.33	103.64
222	0.9(DL+COLL+CL-1)+1.0(S	87.99	52.30	104.26	13.56	-1.16	20.54	90.15
223	0.9(DL+COLL+CL-2)+1.0(S	87.99	52.30	104.26	13.56	-1.16	20.54	90.15
224	0.9(DL+COLL+CL-3)+1.0(S	87.99	52.30	104.26	13.56	-1.16	20.54	90.15
225	0.9(DL+COLL+CL-4)+1.0(S	87.99	52.30	104.26	13.56	-1.16	20.54	90.15
226	0.9(DL+COLL+CL-1)+1.0(S	97.49	61.81	113.76	15.41	0.70	23.23	117.14
227	0.9(DL+COLL+CL-2)+1.0(S	97.49	61.81	113.76	15.41	0.70	23.23	117.14
228	0.9(DL+COLL+CL-3)+1.0(S	97.49	61.81	113.76	15.41	0.70	23.23	117.14
229	0.9(DL+COLL+CL-4)+1.0(S	97.49	61.81	113.76	15.41	0.70	23.23	117.14

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Sample calculation of BM & SF

Load case : 101 1.4(DL+COLL)

p_{1x} = 93.11-(93.11-93.11)*0.625/2 = 93.11 kN/m^2 Overhang = 0.625 m

p_{2x} = 93.11+(93.11-93.11)*0.625/2 = 93.11 kN/m^2 Effective depth = 0.342 m

p_{3x} = 93.11-(93.11-93.11)*(0.625-0.342)/2 = 93.11 kN/m^2

BM_{1x} = (93.11-38.4)*0.625*0.625/2+0.5*(93.11-93.11)*0.625*0.625*0.67 = 10.69 kN-m/m

BM_{2x} = (93.11-38.4)*0.625*0.625/2-0.5*(-93.11+93.11)*0.625*0.625*0.67 = 10.69 kN-m/m

SF_{3x} = (93.11-38.4)*(0.625-0.342)+0.5*(93.11-93.11)*(0.625-0.342) = 15.48 kN/m

$S.F_{(punching)}$ = (4-1.16)*(93.11-38.4) = 155.37 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	43.55	0.37	0.50	0.001	407.73	615.60	16ϕT	326.61	300	200	1005.31
At top	-1.16	-0.01	0.01	0.000	10.73	307.80	10ϕT	255.17	300	250	314.16

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 63.11 kN/m

Shear stress , τ_v= (63.11*1000)/(1000*342)

= 0.18 N/mm^2 **O.K.**

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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 633.32 kN

Strength of Concrete in shear Vc1 = 4354.14 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2604.66 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2902.76 kN **Table 22.6.5.2_ (c)**

Design Vc= 2604.66 kN

Allowable shear stress, (ϕ Vc)/bd = 1.33 N/mm^2

Design punching shear stress, τ = (633.32*1000)/(4304*342)

= 0.43 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M$_{1z}$ kNm	B.M$_{2z}$ kNm	S.F$_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+COLL)	110.56	75.65	125.73	17.90	3.47	28.19
102	1.2(DL+COLL+CL-1)+1.6(R	339.08	183.72	406.61	75.67	11.44	119.36
103	1.2(DL+COLL+CL-2)+1.6(R	339.08	183.72	406.61	75.67	11.44	119.36
104	1.2(DL+COLL+CL-3)+1.6(R	339.08	183.72	406.61	75.67	11.44	119.36
105	1.2(DL+COLL+CL-4)+1.6(R	339.08	183.72	406.61	75.67	11.44	119.36
106	1.4(DL+COLL+CL-1)	110.56	75.65	125.73	17.90	3.47	28.19
107	1.4(DL+COLL+CL-2)	110.56	75.65	125.73	17.90	3.47	28.19
108	1.4(DL+COLL+CL-3)	110.56	75.65	125.73	17.90	3.47	28.19
109	1.4(DL+COLL+CL-4)	110.56	75.65	125.73	17.90	3.47	28.19
110	1.2(DL+COLL+CL-1+TL1)+	352.92	169.88	432.49	81.39	5.72	128.74

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111	1.2(DL+COLL+CL-2+TL1)+	352.92	169.88	432.49	81.39	5.72	128.74
112	1.2(DL+COLL+CL-3+TL1)+	352.92	169.88	432.49	81.39	5.72	128.74
113	1.2(DL+COLL+CL-4+TL1)+	352.92	169.88	432.49	81.39	5.72	128.74
114	1.2(DL+COLL+CL-1+TL2)+	325.24	197.56	380.74	69.95	17.16	109.97
115	1.2(DL+COLL+CL-2+TL2)+	325.24	197.56	380.74	69.95	17.16	109.97
116	1.2(DL+COLL+CL-3+TL2)+	325.24	197.56	380.74	69.95	17.16	109.97
117	1.2(DL+COLL+CL-4+TL2)+	325.24	197.56	380.74	69.95	17.16	109.97
118	1.2(DL+COLL+CL-1)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
119	1.2(DL+COLL+CL-2)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
120	1.2(DL+COLL+CL-3)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
121	1.2(DL+COLL+CL-4)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
122	1.2(DL+COLL+CL-1)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
123	1.2(DL+COLL+CL-2)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
124	1.2(DL+COLL+CL-3)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
125	1.2(DL+COLL+CL-4)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
126	1.2(DL+COLL+CL-1)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
127	1.2(DL+COLL+CL-2)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
128	1.2(DL+COLL+CL-3)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
129	1.2(DL+COLL+CL-4)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
130	1.2(DL+COLL+CL-1)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
131	1.2(DL+COLL+CL-2)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
132	1.2(DL+COLL+CL-3)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
133	1.2(DL+COLL+CL-4)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
134	1.2(DL+COLL+CL-1)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
135	1.2(DL+COLL+CL-2)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
136	1.2(DL+COLL+CL-3)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
137	1.2(DL+COLL+CL-4)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
138	1.2(DL+COLL+CL-1)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
139	1.2(DL+COLL+CL-2)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
140	1.2(DL+COLL+CL-3)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
141	1.2(DL+COLL+CL-4)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
142	1.2(DL+COLL+CL-1)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
143	1.2(DL+COLL+CL-2)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
144	1.2(DL+COLL+CL-3)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33

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145	1.2(DL+COLL+CL-4)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
146	1.2(DL+COLL+CL-1)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
147	1.2(DL+COLL+CL-2)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
148	1.2(DL+COLL+CL-3)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
149	1.2(DL+COLL+CL-4)+0.8(W	100.79	70.86	113.79	15.45	3.08	24.33
150	1.2(DL+COLL+CL-1)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
151	1.2(DL+COLL+CL-2)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
152	1.2(DL+COLL+CL-3)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
153	1.2(DL+COLL+CL-4)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
154	1.2(DL+COLL+CL-1)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
155	1.2(DL+COLL+CL-2)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
156	1.2(DL+COLL+CL-3)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
157	1.2(DL+COLL+CL-4)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
158	1.2(DL+COLL+CL-1)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
159	1.2(DL+COLL+CL-2)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
160	1.2(DL+COLL+CL-3)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
161	1.2(DL+COLL+CL-4)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
162	1.2(DL+COLL+CL-1)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
163	1.2(DL+COLL+CL-2)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
164	1.2(DL+COLL+CL-3)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
165	1.2(DL+COLL+CL-4)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
166	1.2(DL+COLL+CL-1)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
167	1.2(DL+COLL+CL-2)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
168	1.2(DL+COLL+CL-3)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
169	1.2(DL+COLL+CL-4)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
170	1.2(DL+COLL+CL-1)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
171	1.2(DL+COLL+CL-2)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
172	1.2(DL+COLL+CL-3)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
173	1.2(DL+COLL+CL-4)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
174	1.2(DL+COLL+CL-1)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
175	1.2(DL+COLL+CL-2)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
176	1.2(DL+COLL+CL-3)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
177	1.2(DL+COLL+CL-4)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
178	1.2(DL+COLL+CL-1)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02

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179	1.2(DL+COLL+CL-2)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
180	1.2(DL+COLL+CL-3)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
181	1.2(DL+COLL+CL-4)+1.6(W	175.25	106.13	205.29	34.27	5.69	54.02
182	1.2(DL+COLL+CL-1)+1.0(S	188.34	83.54	233.90	40.71	-2.61	64.70
183	1.2(DL+COLL+CL-2)+1.0(S	188.34	83.54	233.90	40.71	-2.61	64.70
184	1.2(DL+COLL+CL-3)+1.0(S	188.34	83.54	233.90	40.71	-2.61	64.70
185	1.2(DL+COLL+CL-4)+1.0(S	188.34	83.54	233.90	40.71	-2.61	64.70
186	1.2(DL+COLL+CL-1)+1.0(S	162.16	128.73	176.70	27.82	14.00	43.34
187	1.2(DL+COLL+CL-2)+1.0(S	162.16	128.73	176.70	27.82	14.00	43.34
188	1.2(DL+COLL+CL-3)+1.0(S	162.16	128.73	176.70	27.82	14.00	43.34
189	1.2(DL+COLL+CL-4)+1.0(S	162.16	128.73	176.70	27.82	14.00	43.34
190	0.9(DL+COLL+CL-1)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
191	0.9(DL+COLL+CL-2)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
192	0.9(DL+COLL+CL-3)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
193	0.9(DL+COLL+CL-4)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
194	0.9(DL+COLL+CL-1)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
195	0.9(DL+COLL+CL-2)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
196	0.9(DL+COLL+CL-3)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
197	0.9(DL+COLL+CL-4)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
198	0.9(DL+COLL+CL-1)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
199	0.9(DL+COLL+CL-2)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
200	0.9(DL+COLL+CL-3)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
201	0.9(DL+COLL+CL-4)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
202	0.9(DL+COLL+CL-1)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
203	0.9(DL+COLL+CL-2)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
204	0.9(DL+COLL+CL-3)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
205	0.9(DL+COLL+CL-4)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
206	0.9(DL+COLL+CL-1)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
207	0.9(DL+COLL+CL-2)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
208	0.9(DL+COLL+CL-3)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
209	0.9(DL+COLL+CL-4)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
210	0.9(DL+COLL+CL-1)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
211	0.9(DL+COLL+CL-2)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
212	0.9(DL+COLL+CL-3)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52

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213	0.9(DL+COLL+CL-4)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
214	0.9(DL+COLL+CL-1)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
215	0.9(DL+COLL+CL-2)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
216	0.9(DL+COLL+CL-3)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
217	0.9(DL+COLL+CL-4)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
218	0.9(DL+COLL+CL-1)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
219	0.9(DL+COLL+CL-2)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
220	0.9(DL+COLL+CL-3)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
221	0.9(DL+COLL+CL-4)+1.6(W	86.11	63.67	95.86	11.77	2.49	18.52
222	0.9(DL+COLL+CL-1)+1.0(S	99.20	41.08	124.47	18.21	-5.82	29.20
223	0.9(DL+COLL+CL-2)+1.0(S	99.20	41.08	124.47	18.21	-5.82	29.20
224	0.9(DL+COLL+CL-3)+1.0(S	99.20	41.08	124.47	18.21	-5.82	29.20
225	0.9(DL+COLL+CL-4)+1.0(S	99.20	41.08	124.47	18.21	-5.82	29.20
226	0.9(DL+COLL+CL-1)+1.0(S	86.27	73.02	92.02	10.79	5.32	16.82
227	0.9(DL+COLL+CL-2)+1.0(S	86.27	73.02	92.02	10.79	5.32	16.82
228	0.9(DL+COLL+CL-3)+1.0(S	86.27	73.02	92.02	10.79	5.32	16.82
229	0.9(DL+COLL+CL-4)+1.0(S	86.27	73.02	92.02	10.79	5.32	16.82

Sample calculation of BM & SF

Load case : 101

p_{1z} = 139.65-(139.65-46.56)*0.625/2= 110.56 kN/m^2

p_{2z} = 46.56+(139.65-46.56)*0.625/2= 75.65 kN/m^2

p_{3z} = 139.65-(139.65-46.56)*(0.625-0.326)/2= 125.73 kN/m^2

Overhang = 0.625 m

Effective depth = 0.326 m

BM_{1z} = (110.56-38.4)*0.625*0.625/2+0.5*(139.65-110.56)*0.625*0.625*0.67 = 17.90 kN-m/m

BM_{2z} = (75.65-38.4)*0.625*0.625/2-0.5*(-46.56+75.65)*0.625*0.625*0.67 = 3.47 kN-m/m

SF_{3z} = (125.73-38.4)*(0.625-0.326)+0.5*(139.65-125.73)*(0.625-0.326) = 28.19 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	81.39	0.766	1.021	0.0025	846.27	586.80	16ϕT	237.59	300	200	1005.3
At top	0.00	0.000	0.000	0.0000	0.00	293.40	10ϕT	267.69	250	250	314.2

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 128.74 kN/m

Shear stress , τ_v= (128.74*1000)/(1000*326)

= 0.39 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	**200**	c/c
	At Top	10ϕT	**250**	c/c
Along Z Direction :				
	At Bottom	16ϕT	**200**	c/c
	At Top	10ϕT	**250**	c/c

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5.7 DESIGN OF CALCULATION OF FOOTING F7

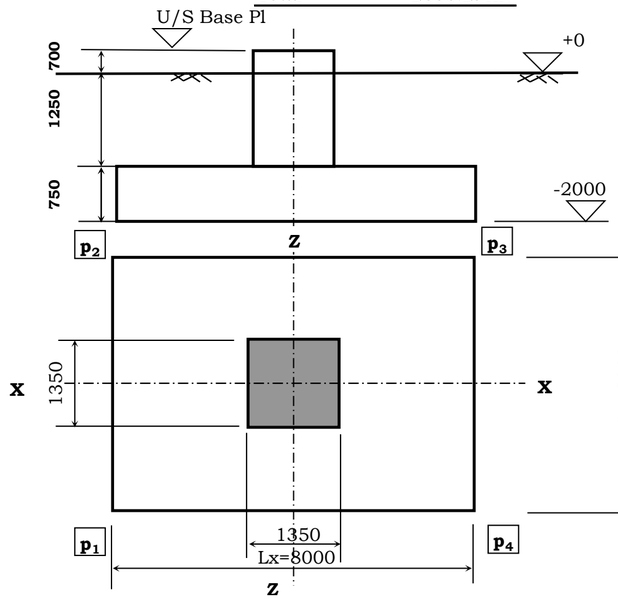
	Along-X	Along-Z
Size of column :	1.35m	1.35m
Size of fdn. raft :	8.00m	8.00m
Thickness of raft :	0.75 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-2.00 m	
U/S level Base Plate :	0.70 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Live Load	LL
Crane Load	CL
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (8*8*0.75*24)+(1.35*1.35*1.95*24) = 1237.29KN
Weight of soil = ((8*8)-(1.35*1.35))*1.25*18 = 1398.99KN
Total = 2636.29KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	25mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	25mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 687.5mm = 0.6875m

Effective depth of slab along Z direction = 662.5mm = 0.6625m
Clear projection along Z direction = 3.325m
Clear projection along X direction = 3.325m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

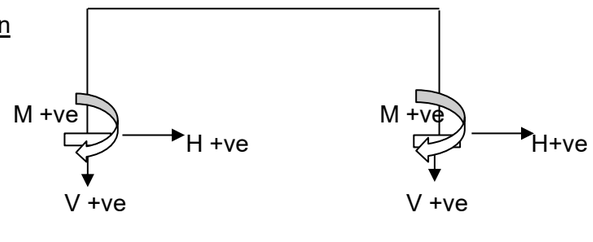
M_X=Moment about X-X direction

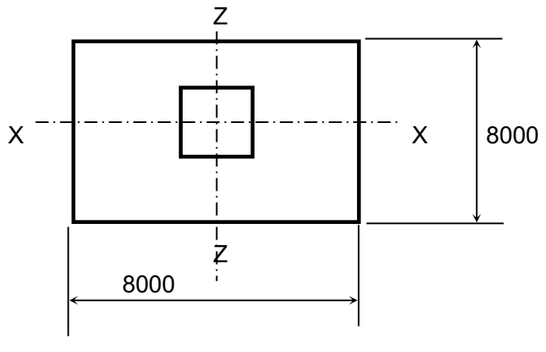
Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F7

Sign convention





Height from BOF to top of column about X (l_{ux}) = 2700mm
Height from BOF to top of column about Z (l_{uz}) = 2700mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	293.2416		-188.9868		-148.536
2	COLLATERAL LOAD (COLL)	0		0		0
3	ROOF LIVE LOAD (RLL)	115.1316		-24.3936		-92.3748
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	769.6536		-584.1576		-373.536
5	CRANE LOAD-1 (CL-1)	0		0		0
6	CRANE LOAD-2 (CL-2)	0		0		0
7	CRANE LOAD-3 (CL-3)	0		0		0
8	CRANE LOAD-4 (CL-4)	0		0		0
9	WIND LEFT-1 (WLL-1)	-40.8672		77.8884		126.1464
10	WIND LEFT-2 (WLL-2)	-111.317		8.9736		58.0164
11	WIND RIGHT-1 (WLR-1)	-119.645		114.8016		249.2124
12	WIND RIGHT-2 (WLR-2)	-108.77		39.0276		121.0284
13	WIND LONG-1 (WLLG-1)	-190.853	72.2196		222.6684	
14	WIND LONG-2 (WLLG-2)	-142.639	31.5156		124.9272	
15	WIND LONG-3 (WLLG-3)	-213.926	75.0516		191.3232	
16	WIND LONG-4 (WLLG-4)	-167.699	36.0168		109.2432	
17	SEISMIC LOAD SL'+'	2.598	-11.4348	-11.4348	-30.3744	-30.3744
18	SEISMIC LOAD SL'-'	-2.598	11.4348	11.4348	30.3744	30.3744
19	TEMPERATURE LOAD (TL1)	-11.5176		-36.564		-166.2624
20	TEMPERATURE LOAD (TL2)	11.5176		36.564		166.2624
21	BRACING REACTION (BR-WLLG1)	51.7	-41		0	
22	BRACING REACTION (BR-WLLG2)	51.7	-41		0	
23	BRACING REACTION (BR-WLLG3)	-51.7	41		0	
24	BRACING REACTION (BR-WLLG4)	-51.7	41		0	
25	BRACING REACTION (BR-SL'+')	91.34	-96.33	-96.33	0	0
26	BRACING REACTION (BR-SL'-')	-91.34	96.33	96.33	0	0

Ultimate Condition

Combined Load Cases:		Load at top of foundation			Load at bottom of foundation		
Load Case	Load Combination	Factored Load/Moment					
		Pu (kN)	Muz (kN-m)	Mux (kN-m)	Pu (kN)	Muz (kN-m)	Mux (kN-m)
101	1.4(DL+COLL)	410.54	0.0	922.3	3046.8	0.0	922.3
102	1.2(DL+COLL+CL-1)+1.6(RLL+PLL / MLL)	1767.5	0.0	4165.0	4403.8	0.0	4165.0
103	1.2(DL+COLL+CL-2)+1.6(RLL+PLL / MLL)	1767.5	0.0	4165.0	4403.8	0.0	4165.0
104	1.2(DL+COLL+CL-3)+1.6(RLL+PLL / MLL)	1767.5	0.0	4165.0	4403.8	0.0	4165.0
105	1.2(DL+COLL+CL-4)+1.6(RLL+PLL / MLL)	1767.5	0.0	4165.0	4403.8	0.0	4165.0
106	1.4(DL+COLL+CL-1)	410.5	0.0	922.3	3046.8	0.0	922.3
107	1.4(DL+COLL+CL-2)	410.5	0.0	922.3	3046.8	0.0	922.3
108	1.4(DL+COLL+CL-3)	410.5	0.0	922.3	3046.8	0.0	922.3
109	1.4(DL+COLL+CL-4)	410.5	0.0	922.3	3046.8	0.0	922.3
110	1.2(DL+COLL+CL-1+TL1)+1.6(RLL+PLL / MLL)	1753.7	0.0	4482.9	4390.0	0.0	4482.9
111	1.2(DL+COLL+CL-2+TL1)+1.6(RLL+PLL / MLL)	1753.7	0.0	4482.9	4390.0	0.0	4482.9
112	1.2(DL+COLL+CL-3+TL1)+1.6(RLL+PLL / MLL)	1753.7	0.0	4482.9	4390.0	0.0	4482.9
113	1.2(DL+COLL+CL-4+TL1)+1.6(RLL+PLL / MLL)	1753.7	0.0	4482.9	4390.0	0.0	4482.9
114	1.2(DL+COLL+CL-1+TL2)+1.6(RLL+PLL / MLL)	1781.4	0.0	3847.0	4417.7	0.0	3847.0
115	1.2(DL+COLL+CL-2+TL2)+1.6(RLL+PLL / MLL)	1781.4	0.0	3847.0	4417.7	0.0	3847.0
116	1.2(DL+COLL+CL-3+TL2)+1.6(RLL+PLL / MLL)	1781.4	0.0	3847.0	4417.7	0.0	3847.0
117	1.2(DL+COLL+CL-4+TL2)+1.6(RLL+PLL / MLL)	1781.4	0.0	3847.0	4417.7	0.0	3847.0
118	1.2(DL+COLL+CL-1)+0.8(WLL-1)	319.2	0.0	521.4	2955.5	0.0	521.4
119	1.2(DL+COLL+CL-2)+0.8(WLL-1)	319.2	0.0	521.4	2955.5	0.0	521.4
120	1.2(DL+COLL+CL-3)+0.8(WLL-1)	319.2	0.0	521.4	2955.5	0.0	521.4
121	1.2(DL+COLL+CL-4)+0.8(WLL-1)	319.2	0.0	521.4	2955.5	0.0	521.4
122	1.2(DL+COLL+CL-1)+0.8(WLL-2)	262.8	0.0	724.8	2899.1	0.0	724.8
123	1.2(DL+COLL+CL-2)+0.8(WLL-2)	262.8	0.0	724.8	2899.1	0.0	724.8
124	1.2(DL+COLL+CL-3)+0.8(WLL-2)	262.8	0.0	724.8	2899.1	0.0	724.8
125	1.2(DL+COLL+CL-4)+0.8(WLL-2)	262.8	0.0	724.8	2899.1	0.0	724.8
126	1.2(DL+COLL+CL-1)+0.8(WLR-1)	256.2	0.0	343.2	2892.5	0.0	343.2
127	1.2(DL+COLL+CL-2)+0.8(WLR-1)	256.2	0.0	343.2	2892.5	0.0	343.2
128	1.2(DL+COLL+CL-3)+0.8(WLR-1)	256.2	0.0	343.2	2892.5	0.0	343.2
129	1.2(DL+COLL+CL-4)+0.8(WLR-1)	256.2	0.0	343.2	2892.5	0.0	343.2
130	1.2(DL+COLL+CL-1)+0.8(WLR-2)	264.9	0.0	609.4	2901.2	0.0	609.4
131	1.2(DL+COLL+CL-2)+0.8(WLR-2)	264.9	0.0	609.4	2901.2	0.0	609.4
132	1.2(DL+COLL+CL-3)+0.8(WLR-2)	264.9	0.0	609.4	2901.2	0.0	609.4
133	1.2(DL+COLL+CL-4)+0.8(WLR-2)	264.9	0.0	609.4	2901.2	0.0	609.4
134	1.2(DL+COLL+CL-1)+0.8(WLLG-1+BR-WLLG-1)	240.6	245.6	790.6	2876.9	245.6	790.6
135	1.2(DL+COLL+CL-2)+0.8(WLLG-1+BR-WLLG-1)	240.6	245.6	790.6	2876.9	245.6	790.6
136	1.2(DL+COLL+CL-3)+0.8(WLLG-1+BR-WLLG-1)	240.6	245.6	790.6	2876.9	245.6	790.6
137	1.2(DL+COLL+CL-4)+0.8(WLLG-1+BR-WLLG-1)	240.6	245.6	790.6	2876.9	245.6	790.6
138	1.2(DL+COLL+CL-1)+0.8(WLLG-2+BR-WLLG-2)	279.1	79.5	790.6	2915.4	79.5	790.6
139	1.2(DL+COLL+CL-2)+0.8(WLLG-2+BR-WLLG-2)	279.1	79.5	790.6	2915.4	79.5	790.6
140	1.2(DL+COLL+CL-3)+0.8(WLLG-2+BR-WLLG-2)	279.1	79.5	790.6	2915.4	79.5	790.6
141	1.2(DL+COLL+CL-4)+0.8(WLLG-2+BR-WLLG-2)	279.1	79.5	790.6	2915.4	79.5	790.6
142	1.2(DL+COLL+CL-1)+0.8(WLLG-3+BR-WLLG-3)	139.4	403.7	790.6	2775.7	403.7	790.6
143	1.2(DL+COLL+CL-2)+0.8(WLLG-3+BR-WLLG-3)	139.4	403.7	790.6	2775.7	403.7	790.6
144	1.2(DL+COLL+CL-3)+0.8(WLLG-3+BR-WLLG-3)	139.4	403.7	790.6	2775.7	403.7	790.6
145	1.2(DL+COLL+CL-4)+0.8(WLLG-3+BR-WLLG-3)	139.4	403.7	790.6	2775.7	403.7	790.6
146	1.2(DL+COLL+CL-1)+0.8(WLLG-4+BR-WLLG-4)	176.4	253.8	790.6	2812.7	253.8	790.6

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147	1.2(DL+COLL+CL-2)+0.8(WLLG-4+BR-WLLG-4)	176.4	253.8	790.6	2812.7	253.8	790.6
148	1.2(DL+COLL+CL-3)+0.8(WLLG-4+BR-WLLG-4)	176.4	253.8	790.6	2812.7	253.8	790.6
149	1.2(DL+COLL+CL-4)+0.8(WLLG-4+BR-WLLG-4)	176.4	253.8	790.6	2812.7	253.8	790.6
150	1.2(DL+COLL+CL-1)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	728.9	0.0	1306.7	3365.2	0.0	1306.7
151	1.2(DL+COLL+CL-2)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	728.9	0.0	1306.7	3365.2	0.0	1306.7
152	1.2(DL+COLL+CL-3)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	728.9	0.0	1306.7	3365.2	0.0	1306.7
153	1.2(DL+COLL+CL-4)+1.6(WLL-1)+0.5(RLL+PLL / MLL)	728.9	0.0	1306.7	3365.2	0.0	1306.7
154	1.2(DL+COLL+CL-1)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	616.2	0.0	1713.5	3252.5	0.0	1713.5
155	1.2(DL+COLL+CL-2)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	616.2	0.0	1713.5	3252.5	0.0	1713.5
156	1.2(DL+COLL+CL-3)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	616.2	0.0	1713.5	3252.5	0.0	1713.5
157	1.2(DL+COLL+CL-4)+1.6(WLL-2)+0.5(RLL+PLL / MLL)	616.2	0.0	1713.5	3252.5	0.0	1713.5
158	1.2(DL+COLL+CL-1)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	602.9	0.0	950.4	3239.1	0.0	950.4
159	1.2(DL+COLL+CL-2)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	602.9	0.0	950.4	3239.1	0.0	950.4
160	1.2(DL+COLL+CL-3)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	602.9	0.0	950.4	3239.1	0.0	950.4
161	1.2(DL+COLL+CL-4)+1.6(WLR-1)+0.5(RLL+PLL / MLL)	602.9	0.0	950.4	3239.1	0.0	950.4
162	1.2(DL+COLL+CL-1)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	620.2	0.0	1482.8	3256.5	0.0	1482.8
163	1.2(DL+COLL+CL-2)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	620.2	0.0	1482.8	3256.5	0.0	1482.8
164	1.2(DL+COLL+CL-3)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	620.2	0.0	1482.8	3256.5	0.0	1482.8
165	1.2(DL+COLL+CL-4)+1.6(WLR-2)+0.5(RLL+PLL / MLL)	620.2	0.0	1482.8	3256.5	0.0	1482.8
166	1.2(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	571.6	491.1	1845.1	3207.9	491.1	1845.1
167	1.2(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	571.6	491.1	1845.1	3207.9	491.1	1845.1
168	1.2(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	571.6	491.1	1845.1	3207.9	491.1	1845.1
169	1.2(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)+0.5(RLL+P	571.6	491.1	1845.1	3207.9	491.1	1845.1
170	1.2(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	648.8	158.9	1845.1	3285.1	158.9	1845.1
171	1.2(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	648.8	158.9	1845.1	3285.1	158.9	1845.1
172	1.2(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	648.8	158.9	1845.1	3285.1	158.9	1845.1
173	1.2(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)+0.5(RLL+P	648.8	158.9	1845.1	3285.1	158.9	1845.1
174	1.2(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	369.3	807.5	1845.1	3005.6	807.5	1845.1
175	1.2(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	369.3	807.5	1845.1	3005.6	807.5	1845.1
176	1.2(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	369.3	807.5	1845.1	3005.6	807.5	1845.1
177	1.2(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)+0.5(RLL+P	369.3	807.5	1845.1	3005.6	807.5	1845.1
178	1.2(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	443.2	507.5	1845.1	3079.5	507.5	1845.1
179	1.2(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	443.2	507.5	1845.1	3079.5	507.5	1845.1
180	1.2(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	443.2	507.5	1845.1	3079.5	507.5	1845.1
181	1.2(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG4)+0.5(RLL+PL	443.2	507.5	1845.1	3079.5	507.5	1845.1
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	888.2	321.3	2166.4	3524.5	321.3	2166.4
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	888.2	321.3	2166.4	3524.5	321.3	2166.4
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	888.2	321.3	2166.4	3524.5	321.3	2166.4
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')+0.5(RLL+PLL / ML	888.2	321.3	2166.4	3524.5	321.3	2166.4
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	700.3	321.3	1523.7	3336.6	321.3	1523.7
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-)+0.5(RLL+PLL / MLL	700.3	321.3	1523.7	3336.6	321.3	1523.7
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	700.3	321.3	1523.7	3336.6	321.3	1523.7
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')+0.5(RLL+PLL / MLL	700.3	321.3	1523.7	3336.6	321.3	1523.7
190	0.9(DL+COLL+CL-1)+1.6(WLL-1)	198.5	0.0	54.6	2834.8	0.0	54.6
191	0.9(DL+COLL+CL-2)+1.6(WLL-1)	198.5	0.0	54.6	2834.8	0.0	54.6
192	0.9(DL+COLL+CL-3)+1.6(WLL-1)	198.5	0.0	54.6	2834.8	0.0	54.6
193	0.9(DL+COLL+CL-4)+1.6(WLL-1)	198.5	0.0	54.6	2834.8	0.0	54.6
194	0.9(DL+COLL+CL-1)+1.6(WLL-2)	85.8	0.0	461.3	2722.1	0.0	461.3
195	0.9(DL+COLL+CL-2)+1.6(WLL-2)	85.8	0.0	461.3	2722.1	0.0	461.3
196	0.9(DL+COLL+CL-3)+1.6(WLL-2)	85.8	0.0	461.3	2722.1	0.0	461.3
197	0.9(DL+COLL+CL-4)+1.6(WLL-2)	85.8	0.0	461.3	2722.1	0.0	461.3
198	0.9(DL+COLL+CL-1)+1.6(WLR-1)	72.5	0.0	301.8	2708.8	0.0	301.8
199	0.9(DL+COLL+CL-2)+1.6(WLR-1)	72.5	0.0	301.8	2708.8	0.0	301.8
200	0.9(DL+COLL+CL-3)+1.6(WLR-1)	72.5	0.0	301.8	2708.8	0.0	301.8
201	0.9(DL+COLL+CL-4)+1.6(WLR-1)	72.5	0.0	301.8	2708.8	0.0	301.8
202	0.9(DL+COLL+CL-1)+1.6(WLR-2)	89.9	0.0	230.7	2726.2	0.0	230.7
203	0.9(DL+COLL+CL-2)+1.6(WLR-2)	89.9	0.0	230.7	2726.2	0.0	230.7
204	0.9(DL+COLL+CL-3)+1.6(WLR-2)	89.9	0.0	230.7	2726.2	0.0	230.7
205	0.9(DL+COLL+CL-4)+1.6(WLR-2)	89.9	0.0	230.7	2726.2	0.0	230.7
206	0.9(DL+COLL+CL-1)+1.6(WLLG-1+BR-WLLG-1)	41.3	491.1	592.9	2677.6	491.1	592.9
207	0.9(DL+COLL+CL-2)+1.6(WLLG-1+BR-WLLG-1)	41.3	491.1	592.9	2677.6	491.1	592.9
208	0.9(DL+COLL+CL-3)+1.6(WLLG-1+BR-WLLG-1)	41.3	491.1	592.9	2677.6	491.1	592.9
209	0.9(DL+COLL+CL-4)+1.6(WLLG-1+BR-WLLG-1)	41.3	491.1	592.9	2677.6	491.1	592.9
210	0.9(DL+COLL+CL-1)+1.6(WLLG-2+BR-WLLG-2)	118.4	158.9	592.9	2754.7	158.9	592.9
211	0.9(DL+COLL+CL-2)+1.6(WLLG-2+BR-WLLG-2)	118.4	158.9	592.9	2754.7	158.9	592.9
212	0.9(DL+COLL+CL-3)+1.6(WLLG-2+BR-WLLG-2)	118.4	158.9	592.9	2754.7	158.9	592.9
213	0.9(DL+COLL+CL-4)+1.6(WLLG-2+BR-WLLG-2)	118.4	158.9	592.9	2754.7	158.9	592.9
214	0.9(DL+COLL+CL-1)+1.6(WLLG-3+BR-WLLG-3)	-161.1	807.5	592.9	2475.2	807.5	592.9
215	0.9(DL+COLL+CL-2)+1.6(WLLG-3+BR-WLLG-3)	-161.1	807.5	592.9	2475.2	807.5	592.9
216	0.9(DL+COLL+CL-3)+1.6(WLLG-3+BR-WLLG-3)	-161.1	807.5	592.9	2475.2	807.5	592.9
217	0.9(DL+COLL+CL-4)+1.6(WLLG-3+BR-WLLG-3)	-161.1	807.5	592.9	2475.2	807.5	592.9
218	0.9(DL+COLL+CL-1)+1.6(WLLG-4+BR-WLLG-4)	-87.1	507.5	592.9	2549.2	507.5	592.9
219	0.9(DL+COLL+CL-2)+1.6(WLLG-4+BR-WLLG-4)	-87.1	507.5	592.9	2549.2	507.5	592.9
220	0.9(DL+COLL+CL-3)+1.6(WLLG-4+BR-WLLG-4)	-87.1	507.5	592.9	2549.2	507.5	592.9
221	0.9(DL+COLL+CL-4)+1.6(WLLG-4+BR-WLLG-4)	-87.1	507.5	592.9	2549.2	507.5	592.9
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+BR-SL'+')	357.9	321.3	914.3	2994.1	321.3	914.3
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+BR-SL'+')	357.9	321.3	914.3	2994.1	321.3	914.3
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+BR-SL'+')	357.9	321.3	914.3	2994.1	321.3	914.3
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+BR-SL'+')	357.9	321.3	914.3	2994.1	321.3	914.3
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+BR-SL'-')	170.0	321.3	271.6	2806.3	321.3	271.6
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+BR-SL'-')	170.0	321.3	271.6	2806.3	321.3	271.6
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+BR-SL'-')	170.0	321.3	271.6	2806.3	321.3	271.6
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+BR-SL'-')	170.0	321.3	271.6	2806.3	321.3	271.6

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Allowable stress design

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
301	1.0(DL+COLL)	293.2416	0	0.0	-189.0	-658.8	2929.5	0.0	0.0	-189.0	-658.8
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL	1166.509	0	0.0	-834.1	-3032.8	3802.8	0.0	0.0	-834.1	-3032.8
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL / MLL	1166.509	0	0.0	-834.1	-3032.8	3802.8	0.0	0.0	-834.1	-3032.8
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL / MLL	1166.509	0	0.0	-834.1	-3032.8	3802.8	0.0	0.0	-834.1	-3032.8
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL / MLL	1166.509	0	0.0	-834.1	-3032.8	3802.8	0.0	0.0	-834.1	-3032.8
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL / MLL	1189.544	0	0.0	-761.0	-2502.8	3825.8	0.0	0.0	-761.0	-2502.8
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL / MLL	1189.544	0	0.0	-761.0	-2502.8	3825.8	0.0	0.0	-761.0	-2502.8
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL / MLL	1189.544	0	0.0	-761.0	-2502.8	3825.8	0.0	0.0	-761.0	-2502.8
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL / MLL	1189.544	0	0.0	-761.0	-2502.8	3825.8	0.0	0.0	-761.0	-2502.8
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL+PLL	948.19	0	0.0	-672.8	-2439.3	3584.5	0.0	0.0	-672.8	-2439.3
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL+PLL	948.19	0	0.0	-672.8	-2439.3	3584.5	0.0	0.0	-672.8	-2439.3
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL+PLL	948.19	0	0.0	-672.8	-2439.3	3584.5	0.0	0.0	-672.8	-2439.3
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL+PLL	948.19	0	0.0	-672.8	-2439.3	3584.5	0.0	0.0	-672.8	-2439.3
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL+PLL	965.47	0	0.0	-618.0	-2041.8	3601.8	0.0	0.0	-618.0	-2041.8
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL+PLL	965.47	0	0.0	-618.0	-2041.8	3601.8	0.0	0.0	-618.0	-2041.8
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL+PLL	965.47	0	0.0	-618.0	-2041.8	3601.8	0.0	0.0	-618.0	-2041.8
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL+PLL	965.47	0	0.0	-618.0	-2041.8	3601.8	0.0	0.0	-618.0	-2041.8
318	1.0(DL+COLL+CL-1)	293.2416	0	0.0	-189.0	-658.8	2929.5	0.0	0.0	-189.0	-658.8
319	1.0(DL+COLL+CL-2)	293.2416	0	0.0	-189.0	-658.8	2929.5	0.0	0.0	-189.0	-658.8
320	1.0(DL+COLL+CL-3)	293.2416	0	0.0	-189.0	-658.8	2929.5	0.0	0.0	-189.0	-658.8
321	1.0(DL+COLL+CL-4)	293.2416	0	0.0	-189.0	-658.8	2929.5	0.0	0.0	-189.0	-658.8
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	252.3744	0	0.0	-111.1	-322.4	2888.7	0.0	0.0	-111.1	-322.4
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	252.3744	0	0.0	-111.1	-322.4	2888.7	0.0	0.0	-111.1	-322.4
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	252.3744	0	0.0	-111.1	-322.4	2888.7	0.0	0.0	-111.1	-322.4
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	252.3744	0	0.0	-111.1	-322.4	2888.7	0.0	0.0	-111.1	-322.4
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	181.9248	0	0.0	-180.0	-576.6	2818.2	0.0	0.0	-180.0	-576.6
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	181.9248	0	0.0	-180.0	-576.6	2818.2	0.0	0.0	-180.0	-576.6
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	181.9248	0	0.0	-180.0	-576.6	2818.2	0.0	0.0	-180.0	-576.6
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	181.9248	0	0.0	-180.0	-576.6	2818.2	0.0	0.0	-180.0	-576.6
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	173.5968	0	0.0	-74.2	-99.6	2809.9	0.0	0.0	-74.2	-99.6
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	173.5968	0	0.0	-74.2	-99.6	2809.9	0.0	0.0	-74.2	-99.6
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	173.5968	0	0.0	-74.2	-99.6	2809.9	0.0	0.0	-74.2	-99.6
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	173.5968	0	0.0	-74.2	-99.6	2809.9	0.0	0.0	-74.2	-99.6
334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	184.4712	0	0.0	-150.0	-432.4	2820.8	0.0	0.0	-150.0	-432.4
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	184.4712	0	0.0	-150.0	-432.4	2820.8	0.0	0.0	-150.0	-432.4
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	184.4712	0	0.0	-150.0	-432.4	2820.8	0.0	0.0	-150.0	-432.4
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	184.4712	0	0.0	-150.0	-432.4	2820.8	0.0	0.0	-150.0	-432.4
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	154.0888	31.2196	632.4	-189.0	-658.8	2790.4	31.2	632.4	-189.0	-658.8
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	154.0888	31.2196	632.4	-189.0	-658.8	2790.4	31.2	632.4	-189.0	-658.8
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	154.0888	31.2196	632.4	-189.0	-658.8	2790.4	31.2	632.4	-189.0	-658.8
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	154.0888	31.2196	632.4	-189.0	-658.8	2790.4	31.2	632.4	-189.0	-658.8
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	202.3024	-9.4844	327.8	-189.0	-658.8	2838.6	-9.5	327.8	-189.0	-658.8
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	202.3024	-9.4844	327.8	-189.0	-658.8	2838.6	-9.5	327.8	-189.0	-658.8
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	202.3024	-9.4844	327.8	-189.0	-658.8	2838.6	-9.5	327.8	-189.0	-658.8
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	202.3024	-9.4844	327.8	-189.0	-658.8	2838.6	-9.5	327.8	-189.0	-658.8
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	27.6152	116.0516	632.6	-189.0	-658.8	2663.9	116.1	632.6	-189.0	-658.8
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	27.6152	116.0516	632.6	-189.0	-658.8	2663.9	116.1	632.6	-189.0	-658.8
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	27.6152	116.0516	632.6	-189.0	-658.8	2663.9	116.1	632.6	-189.0	-658.8
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	27.6152	116.0516	632.6	-189.0	-658.8	2663.9	116.1	632.6	-189.0	-658.8
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	73.8428	77.0168	372.0	-189.0	-658.8	2710.1	77.0	372.0	-189.0	-658.8
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	73.8428	77.0168	372.0	-189.0	-658.8	2710.1	77.0	372.0	-189.0	-658.8
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	73.8428	77.0168	372.0	-189.0	-658.8	2710.1	77.0	372.0	-189.0	-658.8
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	73.8428	77.0168	372.0	-189.0	-658.8	2710.1	77.0	372.0	-189.0	-658.8
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(RLL+	926.18	0	0.0	-587.0	-1988.2	3562.5	0.0	0.0	-587.0	-1988.2
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(RLL+	926.18	0	0.0	-587.0	-1988.2	3562.5	0.0	0.0	-587.0	-1988.2
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(RLL+	926.18	0	0.0	-587.0	-1988.2	3562.5	0.0	0.0	-587.0	-1988.2
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(RLL+	926.18	0	0.0	-587.0	-1988.2	3562.5	0.0	0.0	-587.0	-1988.2
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(RLL+	873.34	0	0.0	-638.7	-2178.9	3509.6	0.0	0.0	-638.7	-2178.9
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(RLL+	873.34	0	0.0	-638.7	-2178.9	3509.6	0.0	0.0	-638.7	-2178.9
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(RLL+	873.34	0	0.0	-638.7	-2178.9	3509.6	0.0	0.0	-638.7	-2178.9
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(RLL+	873.34	0	0.0	-638.7	-2178.9	3509.6	0.0	0.0	-638.7	-2178.9
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(RLL+	867.10	0	0.0	-559.3	-1821.2	3503.4	0.0	0.0	-559.3	-1821.2
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(RLL+	867.10	0	0.0	-559.3	-1821.2	3503.4	0.0	0.0	-559.3	-1821.2
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(RLL+	867.10	0	0.0	-559.3	-1821.2	3503.4	0.0	0.0	-559.3	-1821.2
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(RLL+	867.10	0	0.0	-559.3	-1821.2	3503.4	0.0	0.0	-559.3	-1821.2
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(RLL+	875.25	0	0.0	-616.1	-2070.7	3511.5	0.0	0.0	-616.1	-2070.7
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(RLL+	875.25	0	0.0	-616.1	-2070.7	3511.5	0.0	0.0	-616.1	-2070.7
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(RLL+	875.25	0	0.0	-616.1	-2070.7	3511.5	0.0	0.0	-616.1	-2070.7
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(RLL+	875.25	0	0.0	-616.1	-2070.7	3511.5	0.0	0.0	-616.1	-2070.7
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-WLLG	852.47	23.4147	474.3	-645.4	-2240.5	3488.8	23.4	474.3	-645.4	-2240.5
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-WLLG	852.47	23.4147	474.3	-645.4	-2240.5	3488.8	23.4	474.3	-645.4	-2240.5
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-WLLG	852.47	23.4147	474.3	-645.4	-2240.5	3488.8	23.4	474.3	-645.4	-2240.5
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-WLLG	852.47	23.4147	474.3	-645.4	-2240.5	3488.8	23.4	474.3	-645.4	-2240.5
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-WLLG	888.63	-7.1133	245.9	-645.4	-2240.5	3524.9	-7.1	245.9	-645.4	-2240.5
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-WLLG	888.63	-7.1133	245.9	-645.4	-2240.5	3524.9	-7.1	245.9	-645.4	-2240.5
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-WLLG	888.63	-7.1133	245.9	-645.4	-2240.5	3524.9	-7.1	245.9	-645.4	-2240.5
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-WLLG	888.63	-7.1133	245.9	-645.4	-2240.5	3524.9	-7.1	245.9	-645.4	-2240.5
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-WLLG	757.61	87.0387	474.5	-645.4	-2240.5	3393.9	87.0	474.5	-645.4	-2240.5
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-WLLG	757.61	87.0387	474.5	-645.4	-2240.5	3393.9	87.0	474.5	-645.4	-2240.5
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-WLLG	757.61	87.0387	474.5	-645.4	-2240.5	3393.9	87.0	474.5	-645.4	-2240.5
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-WLLG	757.61	87.0387	474.5	-645.4	-2240.5	3393.9	87.0	474.5	-645.4	-2240.5
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-WLLG	792.28	57.7626	279.0	-645.4	-2240.5	3428.6	57.8	279.0	-645.4	-2240.5
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-WLLG	792.28	57.7626	279.0	-645.4	-2240.5	3428.6	57.8	279.0	-645.4	-2240.5
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-WLLG	792.28	57.7626	279.0	-645.4	-2240.5	3428.6	57.8	279.0	-645.4	-2240.5
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-WLLG	792.28	57.7626	279.0	-645.4	-2240.5	3428.6	57.8	279.0	-645.4	-2240.5
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+')	363.70	-80.8236	-142.3	-269.8	-801.1	3000.0	-80.8	-142.3	-269.8	-801.1
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+")	363.70	-80.8236	-142.3	-269.8	-801.1	3000.0	-80.8	-142.3	-269.8	-801.1
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+')	363.70	-80.8236	-142.3	-269.8	-801.1	3000.0	-80.8	-142.3	-269.8	-801.1
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+')	363.70	-80.8236	-142.3	-269.8	-801.1	3000.0	-80.8	-142.3	-269.8	-801.1

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390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-')	222.79	80.8236	142.3	-108.2	-516.5	2859.1	80.8	142.3	-108.2	-516.5
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-')	222.79	80.8236	142.3	-108.2	-516.5	2859.1	80.8	142.3	-108.2	-516.5
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-')	222.79	80.8236	142.3	-108.2	-516.5	2859.1	80.8	142.3	-108.2	-516.5
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-')	222.79	80.8236	142.3	-108.2	-516.5	2859.1	80.8	142.3	-108.2	-516.5
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'+')+0	1006.15	-56.57652	-99.6	-702.0	-2340.2	3642.4	-56.6	-99.6	-702.0	-2340.2
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'+')+0	1006.15	-56.57652	-99.6	-702.0	-2340.2	3642.4	-56.6	-99.6	-702.0	-2340.2
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'+')+0	1006.15	-56.57652	-99.6	-702.0	-2340.2	3642.4	-56.6	-99.6	-702.0	-2340.2
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'+')+0	1006.15	-56.57652	-99.6	-702.0	-2340.2	3642.4	-56.6	-99.6	-702.0	-2340.2
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-')+0.7	907.51	56.57652	99.6	-588.8	-2140.9	3543.8	56.6	99.6	-588.8	-2140.9
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-')+0.7	907.51	56.57652	99.6	-588.8	-2140.9	3543.8	56.6	99.6	-588.8	-2140.9
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-')+0.7	907.51	56.57652	99.6	-588.8	-2140.9	3543.8	56.6	99.6	-588.8	-2140.9
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-')+0.7	907.51	56.57652	99.6	-588.8	-2140.9	3543.8	56.6	99.6	-588.8	-2140.9
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	135.08	0	0.0	-35.5	-58.8	2771.4	0.0	0.0	-35.5	-58.8
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	135.08	0	0.0	-35.5	-58.8	2771.4	0.0	0.0	-35.5	-58.8
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	135.08	0	0.0	-35.5	-58.8	2771.4	0.0	0.0	-35.5	-58.8
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	135.08	0	0.0	-35.5	-58.8	2771.4	0.0	0.0	-35.5	-58.8
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	64.62816	0	0.0	-104.4	-313.0	2700.9	0.0	0.0	-104.4	-313.0
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	64.62816	0	0.0	-104.4	-313.0	2700.9	0.0	0.0	-104.4	-313.0
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	64.62816	0	0.0	-104.4	-313.0	2700.9	0.0	0.0	-104.4	-313.0
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	64.62816	0	0.0	-104.4	-313.0	2700.9	0.0	0.0	-104.4	-313.0
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	56.30016	0	0.0	1.4	163.9	2692.6	0.0	0.0	1.4	163.9
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	56.30016	0	0.0	1.4	163.9	2692.6	0.0	0.0	1.4	163.9
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	56.30016	0	0.0	1.4	163.9	2692.6	0.0	0.0	1.4	163.9
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	56.30016	0	0.0	1.4	163.9	2692.6	0.0	0.0	1.4	163.9
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	67.17456	0	0.0	-74.4	-168.9	2703.5	0.0	0.0	-74.4	-168.9
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	67.17456	0	0.0	-74.4	-168.9	2703.5	0.0	0.0	-74.4	-168.9
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	67.17456	0	0.0	-74.4	-168.9	2703.5	0.0	0.0	-74.4	-168.9
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	67.17456	0	0.0	-74.4	-168.9	2703.5	0.0	0.0	-74.4	-168.9
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-WLLG-	36.79216	31.2196	632.4	-113.4	-395.3	2673.1	31.2	632.4	-113.4	-395.3
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-WLLG-	36.79216	31.2196	632.4	-113.4	-395.3	2673.1	31.2	632.4	-113.4	-395.3
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-WLLG-	36.79216	31.2196	632.4	-113.4	-395.3	2673.1	31.2	632.4	-113.4	-395.3
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-WLLG-	36.79216	31.2196	632.4	-113.4	-395.3	2673.1	31.2	632.4	-113.4	-395.3
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-WLLG-	85.00576	-9.4844	327.8	-113.4	-395.3	2721.3	-9.5	327.8	-113.4	-395.3
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-WLLG-	85.00576	-9.4844	327.8	-113.4	-395.3	2721.3	-9.5	327.8	-113.4	-395.3
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-WLLG-	85.00576	-9.4844	327.8	-113.4	-395.3	2721.3	-9.5	327.8	-113.4	-395.3
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-WLLG-	85.00576	-9.4844	327.8	-113.4	-395.3	2721.3	-9.5	327.8	-113.4	-395.3
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-WLLG-	-89.68144	116.0516	632.6	-113.4	-395.3	2546.6	116.1	632.6	-113.4	-395.3
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-WLLG-	-89.68144	116.0516	632.6	-113.4	-395.3	2546.6	116.1	632.6	-113.4	-395.3
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-WLLG-	-89.68144	116.0516	632.6	-113.4	-395.3	2546.6	116.1	632.6	-113.4	-395.3
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-WLLG-	-89.68144	116.0516	632.6	-113.4	-395.3	2546.6	116.1	632.6	-113.4	-395.3
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-WLLG-	-43.45384	77.0168	372.0	-113.4	-395.3	2592.8	77.0	372.0	-113.4	-395.3
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-WLLG-	-43.45384	77.0168	372.0	-113.4	-395.3	2592.8	77.0	372.0	-113.4	-395.3
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-WLLG-	-43.45384	77.0168	372.0	-113.4	-395.3	2592.8	77.0	372.0	-113.4	-395.3
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-WLLG-	-43.45384	77.0168	372.0	-113.4	-395.3	2592.8	77.0	372.0	-113.4	-395.3
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	241.70	-75.43536	-132.8	-188.8	-528.1	2878.0	-75.4	-132.8	-188.8	-528.1
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	241.70	-75.43536	-132.8	-188.8	-528.1	2878.0	-75.4	-132.8	-188.8	-528.1
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	241.70	-75.43536	-132.8	-188.8	-528.1	2878.0	-75.4	-132.8	-188.8	-528.1
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	241.70	-75.43536	-132.8	-188.8	-528.1	2878.0	-75.4	-132.8	-188.8	-528.1
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	217.18	-75.43536	-132.8	-142.1	-326.3	2853.5	-75.4	-132.8	-142.1	-326.3
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	174.91	-75.43536	-132.8	-183.4	-478.8	2811.2	-75.4	-132.8	-183.4	-478.8
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	169.91	-75.43536	-132.8	-119.9	-192.6	2806.2	-75.4	-132.8	-119.9	-192.6
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	176.44	-75.43536	-132.8	-165.4	-392.3	2812.7	-75.4	-132.8	-165.4	-392.3

Check for overturning and sliding:

Load Case	Load Combination	Load at bottom of foundation					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
301	1.0(DL+COLL)	2929.53	0.00	0.00	188.99	658.80	NA	16.01	NA	5.58
302	1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL /	3802.80	0.00	0.00	834.10	3032.78	NA	4.51	NA	1.64
303	1.0(DL+COLL+CL-2+TL1)+1.0(RLL+PLL /	3802.80	0.00	0.00	834.10	3032.78	NA	4.51	NA	1.64
304	1.0(DL+COLL+CL-3+TL1)+1.0(RLL+PLL /	3802.80	0.00	0.00	834.10	3032.78	NA	4.51	NA	1.64
305	1.0(DL+COLL+CL-4+TL1)+1.0(RLL+PLL /	3802.80	0.00	0.00	834.10	3032.78	NA	4.51	NA	1.64
306	1.0(DL+COLL+CL-1+TL2)+1.0(RLL+PLL /	3825.83	0.00	0.00	760.97	2502.81	NA	5.50	NA	1.81
307	1.0(DL+COLL+CL-2+TL2)+1.0(RLL+PLL /	3825.83	0.00	0.00	760.97	2502.81	NA	5.50	NA	1.81
308	1.0(DL+COLL+CL-3+TL2)+1.0(RLL+PLL /	3825.83	0.00	0.00	760.97	2502.81	NA	5.50	NA	1.81
309	1.0(DL+COLL+CL-4+TL2)+1.0(RLL+PLL /	3825.83	0.00	0.00	760.97	2502.81	NA	5.50	NA	1.81
310	1.0(DL+COLL+CL-1)+0.75(TL1)+0.75(RLL	3584.48	0.00	0.00	672.82	2439.29	NA	5.29	NA	1.92
311	1.0(DL+COLL+CL-2)+0.75(TL1)+0.75(RLL	3584.48	0.00	0.00	672.82	2439.29	NA	5.29	NA	1.92
312	1.0(DL+COLL+CL-3)+0.75(TL1)+0.75(RLL	3584.48	0.00	0.00	672.82	2439.29	NA	5.29	NA	1.92
313	1.0(DL+COLL+CL-4)+0.75(TL1)+0.75(RLL	3584.48	0.00	0.00	672.82	2439.29	NA	5.29	NA	1.92
314	1.0(DL+COLL+CL-1)+0.75(TL2)+0.75(RLL	3601.76	0.00	0.00	617.98	2041.81	NA	6.35	NA	2.10
315	1.0(DL+COLL+CL-2)+0.75(TL2)+0.75(RLL	3601.76	0.00	0.00	617.98	2041.81	NA	6.35	NA	2.10
316	1.0(DL+COLL+CL-3)+0.75(TL2)+0.75(RLL	3601.76	0.00	0.00	617.98	2041.81	NA	6.35	NA	2.10
317	1.0(DL+COLL+CL-4)+0.75(TL2)+0.75(RLL	3601.76	0.00	0.00	617.98	2041.81	NA	6.35	NA	2.10
318	1.0(DL+COLL+CL-1)	2929.53	0.00	0.00	188.99	658.80	NA	16.01	NA	5.58
319	1.0(DL+COLL+CL-2)	2929.53	0.00	0.00	188.99	658.80	NA	16.01	NA	5.58
320	1.0(DL+COLL+CL-3)	2929.53	0.00	0.00	188.99	658.80	NA	16.01	NA	5.58
321	1.0(DL+COLL+CL-4)	2929.53	0.00	0.00	188.99	658.80	NA	16.01	NA	5.58
322	1.0(DL+COLL+CL-1)+1.0(WLL-1)	2888.66	0.00	0.00	111.10	322.36	NA	32.26	NA	9.36
323	1.0(DL+COLL+CL-2)+1.0(WLL-1)	2888.66	0.00	0.00	111.10	322.36	NA	32.26	NA	9.36
324	1.0(DL+COLL+CL-3)+1.0(WLL-1)	2888.66	0.00	0.00	111.10	322.36	NA	32.26	NA	9.36
325	1.0(DL+COLL+CL-4)+1.0(WLL-1)	2888.66	0.00	0.00	111.10	322.36	NA	32.26	NA	9.36
326	1.0(DL+COLL+CL-1)+1.0(WLL-2)	2818.21	0.00	0.00	180.01	576.56	NA	17.60	NA	5.64
327	1.0(DL+COLL+CL-2)+1.0(WLL-2)	2818.21	0.00	0.00	180.01	576.56	NA	17.60	NA	5.64
328	1.0(DL+COLL+CL-3)+1.0(WLL-2)	2818.21	0.00	0.00	180.01	576.56	NA	17.60	NA	5.64
329	1.0(DL+COLL+CL-4)+1.0(WLL-2)	2818.21	0.00	0.00	180.01	576.56	NA	17.60	NA	5.64
330	1.0(DL+COLL+CL-1)+1.0(WLR-1)	2809.89	0.00	0.00	74.19	99.62	NA	101.54	NA	13.64
331	1.0(DL+COLL+CL-2)+1.0(WLR-1)	2809.89	0.00	0.00	74.19	99.62	NA	101.54	NA	13.64
332	1.0(DL+COLL+CL-3)+1.0(WLR-1)	2809.89	0.00	0.00	74.19	99.62	NA	101.54	NA	13.64
333	1.0(DL+COLL+CL-4)+1.0(WLR-1)	2809.89	0.00	0.00	74.19	99.62	NA	101.54	NA	13.64

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334	1.0(DL+COLL+CL-1)+1.0(WLR-2)	2820.76	0.00	0.00	149.96	432.40	NA	23.48	NA	6.77
335	1.0(DL+COLL+CL-2)+1.0(WLR-2)	2820.76	0.00	0.00	149.96	432.40	NA	23.48	NA	6.77
336	1.0(DL+COLL+CL-3)+1.0(WLR-2)	2820.76	0.00	0.00	149.96	432.40	NA	23.48	NA	6.77
337	1.0(DL+COLL+CL-4)+1.0(WLR-2)	2820.76	0.00	0.00	149.96	432.40	NA	23.48	NA	6.77
338	1.0(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	2790.38	31.22	632.42	188.99	658.80	15.88	15.25	35.75	5.32
339	1.0(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	2790.38	31.22	632.42	188.99	658.80	15.88	15.25	35.75	5.32
340	1.0(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	2790.38	31.22	632.42	188.99	658.80	15.88	15.25	35.75	5.32
341	1.0(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	2790.38	31.22	632.42	188.99	658.80	15.88	15.25	35.75	5.32
342	1.0(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	2838.59	-9.48	327.82	188.99	658.80	31.17	15.51	119.72	5.41
343	1.0(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	2838.59	-9.48	327.82	188.99	658.80	31.17	15.51	119.72	5.41
344	1.0(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	2838.59	-9.48	327.82	188.99	658.80	31.17	15.51	119.72	5.41
345	1.0(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	2838.59	-9.48	327.82	188.99	658.80	31.17	15.51	119.72	5.41
346	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	2663.91	116.05	632.62	188.99	658.80	15.16	14.56	9.18	5.07
347	1.0(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	2663.91	116.05	632.62	188.99	658.80	15.16	14.56	9.18	5.07
348	1.0(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	2663.91	116.05	632.62	188.99	658.80	15.16	14.56	9.18	5.07
349	1.0(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	2663.91	116.05	632.62	188.99	658.80	15.16	14.56	9.18	5.07
350	1.0(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	2710.13	77.02	371.97	188.99	658.80	26.23	14.81	14.08	5.16
351	1.0(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	2710.13	77.02	371.97	188.99	658.80	26.23	14.81	14.08	5.16
352	1.0(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	2710.13	77.02	371.97	188.99	658.80	26.23	14.81	14.08	5.16
353	1.0(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	2710.13	77.02	371.97	188.99	658.80	26.23	14.81	14.08	5.16
354	1.0(DL+COLL+CL-1)+0.75(WLL-1)+0.75(F	3562.47	0.00	0.00	586.98	1988.22	NA	6.45	NA	2.18
355	1.0(DL+COLL+CL-2)+0.75(WLL-1)+0.75(F	3562.47	0.00	0.00	586.98	1988.22	NA	6.45	NA	2.18
356	1.0(DL+COLL+CL-3)+0.75(WLL-1)+0.75(F	3562.47	0.00	0.00	586.98	1988.22	NA	6.45	NA	2.18
357	1.0(DL+COLL+CL-4)+0.75(WLL-1)+0.75(F	3562.47	0.00	0.00	586.98	1988.22	NA	6.45	NA	2.18
358	1.0(DL+COLL+CL-1)+0.75(WLL-2)+0.75(F	3509.63	0.00	0.00	638.67	2178.87	NA	5.80	NA	1.98
359	1.0(DL+COLL+CL-2)+0.75(WLL-2)+0.75(F	3509.63	0.00	0.00	638.67	2178.87	NA	5.80	NA	1.98
360	1.0(DL+COLL+CL-3)+0.75(WLL-2)+0.75(F	3509.63	0.00	0.00	638.67	2178.87	NA	5.80	NA	1.98
361	1.0(DL+COLL+CL-4)+0.75(WLL-2)+0.75(F	3509.63	0.00	0.00	638.67	2178.87	NA	5.80	NA	1.98
362	1.0(DL+COLL+CL-1)+0.75(WLR-1)+0.75(	3503.39	0.00	0.00	559.30	1821.17	NA	6.93	NA	2.26
363	1.0(DL+COLL+CL-2)+0.75(WLR-1)+0.75(	3503.39	0.00	0.00	559.30	1821.17	NA	6.93	NA	2.26
364	1.0(DL+COLL+CL-3)+0.75(WLR-1)+0.75(	3503.39	0.00	0.00	559.30	1821.17	NA	6.93	NA	2.26
365	1.0(DL+COLL+CL-4)+0.75(WLR-1)+0.75(	3503.39	0.00	0.00	559.30	1821.17	NA	6.93	NA	2.26
366	1.0(DL+COLL+CL-1)+0.75(WLR-2)+0.75(	3511.54	0.00	0.00	616.13	2070.75	NA	6.10	NA	2.05
367	1.0(DL+COLL+CL-2)+0.75(WLR-2)+0.75(	3511.54	0.00	0.00	616.13	2070.75	NA	6.10	NA	2.05
368	1.0(DL+COLL+CL-3)+0.75(WLR-2)+0.75(	3511.54	0.00	0.00	616.13	2070.75	NA	6.10	NA	2.05
369	1.0(DL+COLL+CL-4)+0.75(WLR-2)+0.75(	3511.54	0.00	0.00	616.13	2070.75	NA	6.10	NA	2.05
370	1.0(DL+COLL+CL-1)+0.75(WLLG-1+BR-V	3488.76	23.41	474.32	645.40	2240.55	26.48	5.61	59.60	1.95
371	1.0(DL+COLL+CL-2)+0.75(WLLG-1+BR-V	3488.76	23.41	474.32	645.40	2240.55	26.48	5.61	59.60	1.95
372	1.0(DL+COLL+CL-3)+0.75(WLLG-1+BR-V	3488.76	23.41	474.32	645.40	2240.55	26.48	5.61	59.60	1.95
373	1.0(DL+COLL+CL-4)+0.75(WLLG-1+BR-V	3488.76	23.41	474.32	645.40	2240.55	26.48	5.61	59.60	1.95
374	1.0(DL+COLL+CL-1)+0.75(WLLG-2+BR-V	3524.92	-7.11	245.86	645.40	2240.55	51.61	5.66	198.22	1.97
375	1.0(DL+COLL+CL-2)+0.75(WLLG-2+BR-V	3524.92	-7.11	245.86	645.40	2240.55	51.61	5.66	198.22	1.97
376	1.0(DL+COLL+CL-3)+0.75(WLLG-2+BR-V	3524.92	-7.11	245.86	645.40	2240.55	51.61	5.66	198.22	1.97
377	1.0(DL+COLL+CL-4)+0.75(WLLG-2+BR-V	3524.92	-7.11	245.86	645.40	2240.55	51.61	5.66	198.22	1.97
378	1.0(DL+COLL+CL-1)+0.75(WLLG-3+BR-V	3393.90	87.04	474.47	645.40	2240.55	25.75	5.45	15.60	1.89
379	1.0(DL+COLL+CL-2)+0.75(WLLG-3+BR-V	3393.90	87.04	474.47	645.40	2240.55	25.75	5.45	15.60	1.89
380	1.0(DL+COLL+CL-3)+0.75(WLLG-3+BR-V	3393.90	87.04	474.47	645.40	2240.55	25.75	5.45	15.60	1.89
381	1.0(DL+COLL+CL-4)+0.75(WLLG-3+BR-V	3393.90	87.04	474.47	645.40	2240.55	25.75	5.45	15.60	1.89
382	1.0(DL+COLL+CL-1)+0.75(WLLG-4+BR-V	3428.57	57.76	278.98	645.40	2240.55	44.24	5.51	23.74	1.91
383	1.0(DL+COLL+CL-2)+0.75(WLLG-4+BR-V	3428.57	57.76	278.98	645.40	2240.55	44.24	5.51	23.74	1.91
384	1.0(DL+COLL+CL-3)+0.75(WLLG-4+BR-V	3428.57	57.76	278.98	645.40	2240.55	44.24	5.51	23.74	1.91
385	1.0(DL+COLL+CL-4)+0.75(WLLG-4+BR-V	3428.57	57.76	278.98	645.40	2240.55	44.24	5.51	23.74	1.91
386	1.0(DL+COLL+CL-1)+0.75(SL'+'+BR-SL'+	2999.99	-80.82	-142.33	269.81	801.13	75.88	13.48	14.85	4.00
387	1.0(DL+COLL+CL-2)+0.75(SL'+'+BR-SL'+	2999.99	-80.82	-142.33	269.81	801.13	75.88	13.48	14.85	4.00
388	1.0(DL+COLL+CL-3)+0.75(SL'+'+BR-SL'+	2999.99	-80.82	-142.33	269.81	801.13	75.88	13.48	14.85	4.00
389	1.0(DL+COLL+CL-4)+0.75(SL'+'+BR-SL'+	2999.99	-80.82	-142.33	269.81	801.13	75.88	13.48	14.85	4.00
390	1.0(DL+COLL+CL-1)+0.75(SL'-'+BR-SL'-)	2859.08	80.82	142.33	108.16	516.47	72.31	19.93	14.15	9.52
391	1.0(DL+COLL+CL-2)+0.75(SL'-'+BR-SL'-)	2859.08	80.82	142.33	108.16	516.47	72.31	19.93	14.15	9.52
392	1.0(DL+COLL+CL-3)+0.75(SL'-'+BR-SL'-)	2859.08	80.82	142.33	108.16	516.47	72.31	19.93	14.15	9.52
393	1.0(DL+COLL+CL-4)+0.75(SL'-'+BR-SL'-)	2859.08	80.82	142.33	108.16	516.47	72.31	19.93	14.15	9.52
394	1.0(DL+COLL+CL-1)+0.525(SL'+'+BR-SL'	3642.44	-56.58	-99.63	701.98	2340.18	131.61	5.60	25.75	1.87
395	1.0(DL+COLL+CL-2)+0.525(SL'+'+BR-SL'	3642.44	-56.58	-99.63	701.98	2340.18	131.61	5.60	25.75	1.87
396	1.0(DL+COLL+CL-3)+0.525(SL'+'+BR-SL'	3642.44	-56.58	-99.63	701.98	2340.18	131.61	5.60	25.75	1.87
397	1.0(DL+COLL+CL-4)+0.525(SL'+'+BR-SL'	3642.44	-56.58	-99.63	701.98	2340.18	131.61	5.60	25.75	1.87
398	1.0(DL+COLL+CL-1)+0.525(SL'-'+BR-SL'-	3543.80	56.58	99.63	588.82	2140.92	128.05	5.96	25.05	2.17
399	1.0(DL+COLL+CL-2)+0.525(SL'-'+BR-SL'-	3543.80	56.58	99.63	588.82	2140.92	128.05	5.96	25.05	2.17
400	1.0(DL+COLL+CL-3)+0.525(SL'-'+BR-SL'-	3543.80	56.58	99.63	588.82	2140.92	128.05	5.96	25.05	2.17
401	1.0(DL+COLL+CL-4)+0.525SL'-'+BR-SL'-'	3543.80	56.58	99.63	588.82	2140.92	128.05	5.96	25.05	2.17
402	0.6(DL+COLL+CL-1)+1.0(WLL-1)	2771.37	0.00	0.00	35.50	58.84	NA	169.57	NA	28.10
403	0.6(DL+COLL+CL-2)+1.0(WLL-1)	2771.37	0.00	0.00	35.50	58.84	NA	169.57	NA	28.10
404	0.6(DL+COLL+CL-3)+1.0(WLL-1)	2771.37	0.00	0.00	35.50	58.84	NA	169.57	NA	28.10
405	0.6(DL+COLL+CL-4)+1.0(WLL-1)	2771.37	0.00	0.00	35.50	58.84	NA	169.57	NA	28.10
406	0.6(DL+COLL+CL-1)+1.0(WLL-2)	2700.92	0.00	0.00	104.42	313.04	NA	31.06	NA	9.31
407	0.6(DL+COLL+CL-2)+1.0(WLL-2)	2700.92	0.00	0.00	104.42	313.04	NA	31.06	NA	9.31
408	0.6(DL+COLL+CL-3)+1.0(WLL-2)	2700.92	0.00	0.00	104.42	313.04	NA	31.06	NA	9.31
409	0.6(DL+COLL+CL-4)+1.0(WLL-2)	2700.92	0.00	0.00	104.42	313.04	NA	31.06	NA	9.31
410	0.6(DL+COLL+CL-1)+1.0(WLR-1)	2692.59	0.00	0.00	1.41	163.90	NA	59.14	NA	687.70
411	0.6(DL+COLL+CL-2)+1.0(WLR-1)	2692.59	0.00	0.00	1.41	163.90	NA	59.14	NA	687.70
412	0.6(DL+COLL+CL-3)+1.0(WLR-1)	2692.59	0.00	0.00	1.41	163.90	NA	59.14	NA	687.70
413	0.6(DL+COLL+CL-4)+1.0(WLR-1)	2692.59	0.00	0.00	1.41	163.90	NA	59.14	NA	687.70
414	0.6(DL+COLL+CL-1)+1.0(WLR-2)	2703.46	0.00	0.00	74.36	168.88	NA	57.63	NA	13.09
415	0.6(DL+COLL+CL-2)+1.0(WLR-2)	2703.46	0.00	0.00	74.36	168.88	NA	57.63	NA	13.09
416	0.6(DL+COLL+CL-3)+1.0(WLR-2)	2703.46	0.00	0.00	74.36	168.88	NA	57.63	NA	13.09
417	0.6(DL+COLL+CL-4)+1.0(WLR-2)	2703.46	0.00	0.00	74.36	168.88	NA	57.63	NA	13.09
418	0.6(DL+COLL+CL-1)+1.0(WLLG-1+BR-W	2673.08	31.22	632.42	113.39	395.28	15.22	24.34	34.25	8.49
419	0.6(DL+COLL+CL-2)+1.0(WLLG-1+BR-W	2673.08	31.22	632.42	113.39	395.28	15.22	24.34	34.25	8.49
420	0.6(DL+COLL+CL-3)+1.0(WLLG-1+BR-W	2673.08	31.22	632.42	113.39	395.28	15.22	24.34	34.25	8.49
421	0.6(DL+COLL+CL-4)+1.0(WLLG-1+BR-W	2673.08	31.22	632.42	113.39	395.28	15.22	24.34	34.25	8.49
422	0.6(DL+COLL+CL-1)+1.0(WLLG-2+BR-W	2721.30	-9.48	327.82	113.39	395.28	29.88	24.78	114.77	8.64
423	0.6(DL+COLL+CL-2)+1.0(WLLG-2+BR-W	2721.30	-9.48	327.82	113.39	395.28	29.88	24.78	114.77	8.64
424	0.6(DL+COLL+CL-3)+1.0(WLLG-2+BR-W	2721.30	-9.48	327.82	113.39	395.28	29.88	24.78	114.77	8.64
425	0.6(DL+COLL+CL-4)+1.0(WLLG-2+BR-W	2721.30	-9.48	327.82	113.39	395.28	29.88	24.78	114.77	8.64
426	0.6(DL+COLL+CL-1)+1.0(WLLG-3+BR-W	2546.61	116.05	632.62	113.39	395.28	14.49	23.19	8.78	8.09
427	0.6(DL+COLL+CL-2)+1.0(WLLG-3+BR-W	2546.61	116.05	632.62	113.39	395.28	14.49	23.19	8.78	8.09
428	0.6(DL+COLL+CL-3)+1.0(WLLG-3+BR-W	2546.61	116.05	632.62	113.39	395.28	14.49	23.19	8.78	8.09
429	0.6(DL+COLL+CL-4)+1.0(WLLG-3+BR-W	2546.61	116.05	632.62	113.39	395.28	14.49	23.19	8.78	8.09
430	0.6(DL+COLL+CL-1)+1.0(WLLG-4+BR-W	2592.84	77.02	371.97	113.39	395.28	25.09	23.61	13.47	8.23
431	0.6(DL+COLL+CL-2)+1.0(WLLG-4+BR-W	2592.84	77.02	371.97	113.39	395.28	25.09	23.61	13.47	8.23
432	0.6(DL+COLL+CL-3)+1.0(WLLG-4+BR-W	2592.84	77.02	371.97	113.39	395.28	25.09	23.61	13.47	8.23
433	0.6(DL+COLL+CL-4)+1.0(WLLG-4+BR-W	2592.84	77.02	371.97	113.39	395.28	25.09	23.61	13.47	8.23
434	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'+')	2877.99	-75.44	-132.84	188.83	528.12	77.99	19.62	15.26	5.49
435	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'+')	2877.99	-75.44	-132.84	188.83	528.12	77.99	19.62	15.26	5.49
436	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'+')	2877.99	-75.44	-132.84	188.83	528.12	77.99	19.62	15.26	5.49
437	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'+')	2877.99	-75.44	-132.84	188.83	528.12	77.99	19.62	15.26	5.49
438	0.6(DL+COLL+CL-1)+0.7(SL'+'+BR-SL'-')	2853.47	-75.44	-132.84	142.09	326.26	77.33	31.49	15.13	7.23
439	0.6(DL+COLL+CL-2)+0.7(SL'+'+BR-SL'-')	2811.20	-75.44	-132.84	183.44	478.78	76.18	21.14	14.91	5.52
440	0.6(DL+COLL+CL-3)+0.7(SL'+'+BR-SL'-')	2806.20	-75.44	-132.84	119.95	192.62	76.05	52.45	14.88	8.42
441	0.6(DL+COLL+CL-4)+0.7(SL'+'+BR-SL'-')	2812.73	-75.44	-132.84	165.41	392.28	76.22	25.81	14.91	6.12
						Min.=	**14.49**	**4.51**	**8.78**	**1.64**
							OK	OK	OK	OK

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BASE PRESSURE CHECK OF FOOTING F7

Load Combination & Calculation of Base pressure

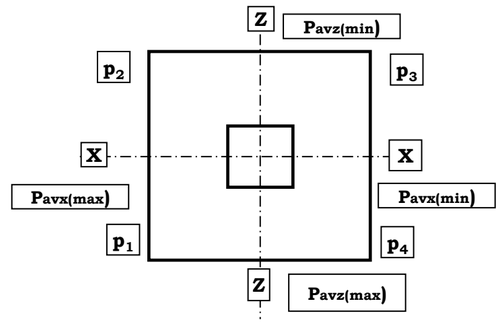
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 436 kN/m^2 For WL/SL case = 536 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
301	1.0(DL+COLL)	2929.53	0.00	-658.80	45.77	0.00	7.72	53.49		38.05	53.49	38.05	17.49	45.77	45.77	53.49	38.05	O.K
302	1.0(DL+COLL+CL-1+TL1)+1.(	3802.80	0.00	-3032.78	59.42	0.00	35.54	94.96		23.88	94.96	23.88	58.96	59.42	59.42	94.96	23.88	O.K
303	1.0(DL+COLL+CL-2+TL1)+1.(	3802.80	0.00	-3032.78	59.42	0.00	35.54	94.96		23.88	94.96	23.88	58.96	59.42	59.42	94.96	23.88	O.K
304	1.0(DL+COLL+CL-3+TL1)+1.(	3802.80	0.00	-3032.78	59.42	0.00	35.54	94.96		23.88	94.96	23.88	58.96	59.42	59.42	94.96	23.88	O.K
305	1.0(DL+COLL+CL-4+TL1)+1.(	3802.80	0.00	-3032.78	59.42	0.00	35.54	94.96		23.88	94.96	23.88	58.96	59.42	59.42	94.96	23.88	O.K
306	1.0(DL+COLL+CL-1+TL2)+1.(	3825.83	0.00	-2502.81	59.78	0.00	29.33	89.11		30.45	89.11	30.45	53.11	59.78	59.78	89.11	30.45	O.K
307	1.0(DL+COLL+CL-2+TL2)+1.(	3825.83	0.00	-2502.81	59.78	0.00	29.33	89.11		30.45	89.11	30.45	53.11	59.78	59.78	89.11	30.45	O.K
308	1.0(DL+COLL+CL-3+TL2)+1.(	3825.83	0.00	-2502.81	59.78	0.00	29.33	89.11		30.45	89.11	30.45	53.11	59.78	59.78	89.11	30.45	O.K
309	1.0(DL+COLL+CL-4+TL2)+1.(	3825.83	0.00	-2502.81	59.78	0.00	29.33	89.11		30.45	89.11	30.45	53.11	59.78	59.78	89.11	30.45	O.K
310	1.0(DL+COLL+CL-1)+0.75(TL	3584.48	0.00	-2439.29	56.01	0.00	28.59	84.59		27.42	84.59	27.42	48.59	56.01	56.01	84.59	27.42	O.K
311	1.0(DL+COLL+CL-2)+0.75(TL	3584.48	0.00	-2439.29	56.01	0.00	28.59	84.59		27.42	84.59	27.42	48.59	56.01	56.01	84.59	27.42	O.K
312	1.0(DL+COLL+CL-3)+0.75(TL	3584.48	0.00	-2439.29	56.01	0.00	28.59	84.59		27.42	84.59	27.42	48.59	56.01	56.01	84.59	27.42	O.K
313	1.0(DL+COLL+CL-4)+0.75(TL	3584.48	0.00	-2439.29	56.01	0.00	28.59	84.59		27.42	84.59	27.42	48.59	56.01	56.01	84.59	27.42	O.K
314	1.0(DL+COLL+CL-1)+0.75(TL	3601.76	0.00	-2041.81	56.28	0.00	23.93	80.20		32.35	80.20	32.35	44.20	56.28	56.28	80.20	32.35	O.K
315	1.0(DL+COLL+CL-2)+0.75(TL	3601.76	0.00	-2041.81	56.28	0.00	23.93	80.20		32.35	80.20	32.35	44.20	56.28	56.28	80.20	32.35	O.K
316	1.0(DL+COLL+CL-3)+0.75(TL	3601.76	0.00	-2041.81	56.28	0.00	23.93	80.20		32.35	80.20	32.35	44.20	56.28	56.28	80.20	32.35	O.K

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317	1.0(DL+COLL+CL-4)+0.75(TL	3601.76	0.00	-2041.81	56.28	0.00	23.93	80.20		32.35	80.20	32.35	44.20	56.28	56.28	80.20	32.35	O.K
318	1.0(DL+COLL+CL-1)	2929.53	0.00	-658.80	45.77	0.00	7.72	53.49		38.05	53.49	38.05	17.49	45.77	45.77	53.49	38.05	O.K
319	1.0(DL+COLL+CL-2)	2929.53	0.00	-658.80	45.77	0.00	7.72	53.49		38.05	53.49	38.05	17.49	45.77	45.77	53.49	38.05	O.K
320	1.0(DL+COLL+CL-3)	2929.53	0.00	-658.80	45.77	0.00	7.72	53.49		38.05	53.49	38.05	17.49	45.77	45.77	53.49	38.05	O.K
321	1.0(DL+COLL+CL-4)	2929.53	0.00	-658.80	45.77	0.00	7.72	53.49		38.05	53.49	38.05	17.49	45.77	45.77	53.49	38.05	O.K
322	1.0(DL+COLL+CL-1)+1.0(WLI	2888.66	0.00	-322.36	45.14	0.00	3.78	48.91		41.36	48.91	41.36	12.91	45.14	45.14	48.91	41.36	O.K
323	1.0(DL+COLL+CL-2)+1.0(WLI	2888.66	0.00	-322.36	45.14	0.00	3.78	48.91		41.36	48.91	41.36	12.91	45.14	45.14	48.91	41.36	O.K
324	1.0(DL+COLL+CL-3)+1.0(WLI	2888.66	0.00	-322.36	45.14	0.00	3.78	48.91		41.36	48.91	41.36	12.91	45.14	45.14	48.91	41.36	O.K
325	1.0(DL+COLL+CL-4)+1.0(WLI	2888.66	0.00	-322.36	45.14	0.00	3.78	48.91		41.36	48.91	41.36	12.91	45.14	45.14	48.91	41.36	O.K
326	1.0(DL+COLL+CL-1)+1.0(WLI	2818.21	0.00	-576.56	44.03	0.00	6.76	50.79		37.28	50.79	37.28	14.79	44.04	44.04	50.79	37.28	O.K
327	1.0(DL+COLL+CL-2)+1.0(WLI	2818.21	0.00	-576.56	44.03	0.00	6.76	50.79		37.28	50.79	37.28	14.79	44.04	44.04	50.79	37.28	O.K
328	1.0(DL+COLL+CL-3)+1.0(WLI	2818.21	0.00	-576.56	44.03	0.00	6.76	50.79		37.28	50.79	37.28	14.79	44.04	44.04	50.79	37.28	O.K
329	1.0(DL+COLL+CL-4)+1.0(WLI	2818.21	0.00	-576.56	44.03	0.00	6.76	50.79		37.28	50.79	37.28	14.79	44.04	44.04	50.79	37.28	O.K
330	1.0(DL+COLL+CL-1)+1.0(WLI	2809.89	0.00	-99.62	43.90	0.00	1.17	45.07		42.74	45.07	42.74	9.07	43.91	43.91	45.07	42.74	O.K
331	1.0(DL+COLL+CL-2)+1.0(WLI	2809.89	0.00	-99.62	43.90	0.00	1.17	45.07		42.74	45.07	42.74	9.07	43.91	43.91	45.07	42.74	O.K
332	1.0(DL+COLL+CL-3)+1.0(WLI	2809.89	0.00	-99.62	43.90	0.00	1.17	45.07		42.74	45.07	42.74	9.07	43.91	43.91	45.07	42.74	O.K
333	1.0(DL+COLL+CL-4)+1.0(WLI	2809.89	0.00	-99.62	43.90	0.00	1.17	45.07		42.74	45.07	42.74	9.07	43.91	43.91	45.07	42.74	O.K
334	1.0(DL+COLL+CL-1)+1.0(WLI	2820.76	0.00	-432.40	44.07	0.00	5.07	49.14		39.01	49.14	39.01	13.14	44.08	44.08	49.14	39.01	O.K
335	1.0(DL+COLL+CL-2)+1.0(WLI	2820.76	0.00	-432.40	44.07	0.00	5.07	49.14		39.01	49.14	39.01	13.14	44.08	44.08	49.14	39.01	O.K
336	1.0(DL+COLL+CL-3)+1.0(WLI	2820.76	0.00	-432.40	44.07	0.00	5.07	49.14		39.01	49.14	39.01	13.14	44.08	44.08	49.14	39.01	O.K
337	1.0(DL+COLL+CL-4)+1.0(WLI	2820.76	0.00	-432.40	44.07	0.00	5.07	49.14		39.01	49.14	39.01	13.14	44.08	44.08	49.14	39.01	O.K
338	1.0(DL+COLL+CL-1)+1.0(WLI	2790.38	632.42	-658.80	43.60	7.41	7.72	58.73		43.29	43.91	28.47	22.73	51.01	36.19	51.32	35.88	O.K
339	1.0(DL+COLL+CL-2)+1.0(WLI	2790.38	632.42	-658.80	43.60	7.41	7.72	58.73		43.29	43.91	28.47	22.73	51.01	36.19	51.32	35.88	O.K
340	1.0(DL+COLL+CL-3)+1.0(WLI	2790.38	632.42	-658.80	43.60	7.41	7.72	58.73		43.29	43.91	28.47	22.73	51.01	36.19	51.32	35.88	O.K
341	1.0(DL+COLL+CL-4)+1.0(WLI	2790.38	632.42	-658.80	43.60	7.41	7.72	58.73		43.29	43.91	28.47	22.73	51.01	36.19	51.32	35.88	O.K
342	1.0(DL+COLL+CL-1)+1.0(WLI	2838.59	327.82	-658.80	44.35	3.84	7.72	55.91		40.47	48.23	32.79	19.91	48.19	40.51	52.07	36.63	O.K
343	1.0(DL+COLL+CL-2)+1.0(WLI	2838.59	327.82	-658.80	44.35	3.84	7.72	55.91		40.47	48.23	32.79	19.91	48.19	40.51	52.07	36.63	O.K
344	1.0(DL+COLL+CL-3)+1.0(WLI	2838.59	327.82	-658.80	44.35	3.84	7.72	55.91		40.47	48.23	32.79	19.91	48.19	40.51	52.07	36.63	O.K
345	1.0(DL+COLL+CL-4)+1.0(WLI	2838.59	327.82	-658.80	44.35	3.84	7.72	55.91		40.47	48.23	32.79	19.91	48.19	40.51	52.07	36.63	O.K
346	1.0(DL+COLL+CL-1)+1.0(WLI	2663.91	632.62	-658.80	41.62	7.41	7.72	56.76		41.32	41.93	26.49	20.76	49.04	34.21	49.35	33.91	O.K
347	1.0(DL+COLL+CL-2)+1.0(WLI	2663.91	632.62	-658.80	41.62	7.41	7.72	56.76		41.32	41.93	26.49	20.76	49.04	34.21	49.35	33.91	O.K

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348	1.0(DL+COLL+CL-3)+1.0(WL	2663.91	632.62	-658.80	41.62	7.41	7.72	56.76		41.32	41.93	26.49	20.76	49.04	34.21	49.35	33.91	O.K
349	1.0(DL+COLL+CL-4)+1.0(WL	2663.91	632.62	-658.80	41.62	7.41	7.72	56.76		41.32	41.93	26.49	20.76	49.04	34.21	49.35	33.91	O.K
350	1.0(DL+COLL+CL-1)+1.0(WL	2710.13	371.97	-658.80	42.35	4.36	7.72	54.43		38.98	45.71	30.27	18.43	46.71	37.99	50.07	34.63	O.K
351	1.0(DL+COLL+CL-2)+1.0(WL	2710.13	371.97	-658.80	42.35	4.36	7.72	54.43		38.98	45.71	30.27	18.43	46.71	37.99	50.07	34.63	O.K
352	1.0(DL+COLL+CL-3)+1.0(WL	2710.13	371.97	-658.80	42.35	4.36	7.72	54.43		38.98	45.71	30.27	18.43	46.71	37.99	50.07	34.63	O.K
353	1.0(DL+COLL+CL-4)+1.0(WL	2710.13	371.97	-658.80	42.35	4.36	7.72	54.43		38.98	45.71	30.27	18.43	46.71	37.99	50.07	34.63	O.K
354	1.0(DL+COLL+CL-1)+0.75(WI	3562.47	0.00	-1988.22	55.66	0.00	23.30	78.96		32.36	78.96	32.36	42.96	55.66	55.66	78.96	32.36	O.K
355	1.0(DL+COLL+CL-2)+0.75(WI	3562.47	0.00	-1988.22	55.66	0.00	23.30	78.96		32.36	78.96	32.36	42.96	55.66	55.66	78.96	32.36	O.K
356	1.0(DL+COLL+CL-3)+0.75(WI	3562.47	0.00	-1988.22	55.66	0.00	23.30	78.96		32.36	78.96	32.36	42.96	55.66	55.66	78.96	32.36	O.K
357	1.0(DL+COLL+CL-4)+0.75(WI	3562.47	0.00	-1988.22	55.66	0.00	23.30	78.96		32.36	78.96	32.36	42.96	55.66	55.66	78.96	32.36	O.K
358	1.0(DL+COLL+CL-1)+0.75(WI	3509.63	0.00	-2178.87	54.84	0.00	25.53	80.37		29.30	80.37	29.30	44.37	54.84	54.84	80.37	29.30	O.K
359	1.0(DL+COLL+CL-2)+0.75(WI	3509.63	0.00	-2178.87	54.84	0.00	25.53	80.37		29.30	80.37	29.30	44.37	54.84	54.84	80.37	29.30	O.K
360	1.0(DL+COLL+CL-3)+0.75(WI	3509.63	0.00	-2178.87	54.84	0.00	25.53	80.37		29.30	80.37	29.30	44.37	54.84	54.84	80.37	29.30	O.K
361	1.0(DL+COLL+CL-4)+0.75(WI	3509.63	0.00	-2178.87	54.84	0.00	25.53	80.37		29.30	80.37	29.30	44.37	54.84	54.84	80.37	29.30	O.K
362	1.0(DL+COLL+CL-1)+0.75(WI	3503.39	0.00	-1821.17	54.74	0.00	21.34	76.08		33.40	76.08	33.40	40.08	54.74	54.74	76.08	33.40	O.K
363	1.0(DL+COLL+CL-2)+0.75(WI	3503.39	0.00	-1821.17	54.74	0.00	21.34	76.08		33.40	76.08	33.40	40.08	54.74	54.74	76.08	33.40	O.K
364	1.0(DL+COLL+CL-3)+0.75(WI	3503.39	0.00	-1821.17	54.74	0.00	21.34	76.08		33.40	76.08	33.40	40.08	54.74	54.74	76.08	33.40	O.K
365	1.0(DL+COLL+CL-4)+0.75(WI	3503.39	0.00	-1821.17	54.74	0.00	21.34	76.08		33.40	76.08	33.40	40.08	54.74	54.74	76.08	33.40	O.K
366	1.0(DL+COLL+CL-1)+0.75(WI	3511.54	0.00	-2070.75	54.87	0.00	24.27	79.13		30.60	79.13	30.60	43.13	54.87	54.87	79.13	30.60	O.K
367	1.0(DL+COLL+CL-2)+0.75(WI	3511.54	0.00	-2070.75	54.87	0.00	24.27	79.13		30.60	79.13	30.60	43.13	54.87	54.87	79.13	30.60	O.K
368	1.0(DL+COLL+CL-3)+0.75(WI	3511.54	0.00	-2070.75	54.87	0.00	24.27	79.13		30.60	79.13	30.60	43.13	54.87	54.87	79.13	30.60	O.K
369	1.0(DL+COLL+CL-4)+0.75(WI	3511.54	0.00	-2070.75	54.87	0.00	24.27	79.13		30.60	79.13	30.60	43.13	54.87	54.87	79.13	30.60	O.K
370	1.0(DL+COLL+CL-1)+0.75(WI	3488.76	474.32	-2240.55	54.51	5.56	26.26	86.33		33.81	75.21	22.70	50.33	60.07	48.96	80.77	28.26	O.K
371	1.0(DL+COLL+CL-2)+0.75(WI	3488.76	474.32	-2240.55	54.51	5.56	26.26	86.33		33.81	75.21	22.70	50.33	60.07	48.96	80.77	28.26	O.K
372	1.0(DL+COLL+CL-3)+0.75(WI	3488.76	474.32	-2240.55	54.51	5.56	26.26	86.33		33.81	75.21	22.70	50.33	60.07	48.96	80.77	28.26	O.K
373	1.0(DL+COLL+CL-4)+0.75(WI	3488.76	474.32	-2240.55	54.51	5.56	26.26	86.33		33.81	75.21	22.70	50.33	60.07	48.96	80.77	28.26	O.K
374	1.0(DL+COLL+CL-1)+0.75(WI	3524.92	245.86	-2240.55	55.08	2.88	26.26	84.21		31.70	78.45	25.94	48.21	57.96	52.20	81.33	28.82	O.K
375	1.0(DL+COLL+CL-2)+0.75(WI	3524.92	245.86	-2240.55	55.08	2.88	26.26	84.21		31.70	78.45	25.94	48.21	57.96	52.20	81.33	28.82	O.K
376	1.0(DL+COLL+CL-3)+0.75(WI	3524.92	245.86	-2240.55	55.08	2.88	26.26	84.21		31.70	78.45	25.94	48.21	57.96	52.20	81.33	28.82	O.K
377	1.0(DL+COLL+CL-4)+0.75(WI	3524.92	245.86	-2240.55	55.08	2.88	26.26	84.21		31.70	78.45	25.94	48.21	57.96	52.20	81.33	28.82	O.K
378	1.0(DL+COLL+CL-1)+0.75(WI	3393.90	474.47	-2240.55	53.03	5.56	26.26	84.85		32.33	73.73	21.21	48.85	58.59	47.47	79.29	26.77	O.K

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379	1.0(DL+COLL+CL-2)+0.75(WI	3393.90	474.47	-2240.55	53.03	5.56	26.26	84.85		32.33	73.73	21.21	48.85	58.59	47.47	79.29	26.77	O.K
380	1.0(DL+COLL+CL-3)+0.75(WI	3393.90	474.47	-2240.55	53.03	5.56	26.26	84.85		32.33	73.73	21.21	48.85	58.59	47.47	79.29	26.77	O.K
381	1.0(DL+COLL+CL-4)+0.75(WI	3393.90	474.47	-2240.55	53.03	5.56	26.26	84.85		32.33	73.73	21.21	48.85	58.59	47.47	79.29	26.77	O.K
382	1.0(DL+COLL+CL-1)+0.75(WI	3428.57	278.98	-2240.55	53.57	3.27	26.26	83.10		30.58	76.56	24.05	47.10	56.84	50.31	79.83	27.32	O.K
383	1.0(DL+COLL+CL-2)+0.75(WI	3428.57	278.98	-2240.55	53.57	3.27	26.26	83.10		30.58	76.56	24.05	47.10	56.84	50.31	79.83	27.32	O.K
384	1.0(DL+COLL+CL-3)+0.75(WI	3428.57	278.98	-2240.55	53.57	3.27	26.26	83.10		30.58	76.56	24.05	47.10	56.84	50.31	79.83	27.32	O.K
385	1.0(DL+COLL+CL-4)+0.75(WI	3428.57	278.98	-2240.55	53.57	3.27	26.26	83.10		30.58	76.56	24.05	47.10	56.84	50.31	79.83	27.32	O.K
386	1.0(DL+COLL+CL-1)+0.75(SL	2999.99	-142.33	-801.13	46.87	1.67	9.39	57.93		39.15	54.60	35.82	21.93	48.54	45.21	56.27	37.49	O.K
387	1.0(DL+COLL+CL-2)+0.75(SL	2999.99	-142.33	-801.13	46.87	1.67	9.39	57.93		39.15	54.60	35.82	21.93	48.54	45.21	56.27	37.49	O.K
388	1.0(DL+COLL+CL-3)+0.75(SL	2999.99	-142.33	-801.13	46.87	1.67	9.39	57.93		39.15	54.60	35.82	21.93	48.54	45.21	56.27	37.49	O.K
389	1.0(DL+COLL+CL-4)+0.75(SL	2999.99	-142.33	-801.13	46.87	1.67	9.39	57.93		39.15	54.60	35.82	21.93	48.54	45.21	56.27	37.49	O.K
390	1.0(DL+COLL+CL-1)+0.75(SL	2859.08	142.33	-516.47	44.67	1.67	6.05	52.39		40.29	49.06	36.95	16.39	46.34	43.01	50.73	38.62	O.K
391	1.0(DL+COLL+CL-2)+0.75(SL	2859.08	142.33	-516.47	44.67	1.67	6.05	52.39		40.29	49.06	36.95	16.39	46.34	43.01	50.73	38.62	O.K
392	1.0(DL+COLL+CL-3)+0.75(SL	2859.08	142.33	-516.47	44.67	1.67	6.05	52.39		40.29	49.06	36.95	16.39	46.34	43.01	50.73	38.62	O.K
393	1.0(DL+COLL+CL-4)+0.75(SL	2859.08	142.33	-516.47	44.67	1.67	6.05	52.39		40.29	49.06	36.95	16.39	46.34	43.01	50.73	38.62	O.K
394	1.0(DL+COLL+CL-1)+0.525(S	3642.44	-99.63	-2340.18	56.91	1.17	27.42	85.50		30.66	83.17	28.32	49.50	58.08	55.75	84.34	29.49	O.K
395	1.0(DL+COLL+CL-2)+0.525(S	3642.44	-99.63	-2340.18	56.91	1.17	27.42	85.50		30.66	83.17	28.32	49.50	58.08	55.75	84.34	29.49	O.K
396	1.0(DL+COLL+CL-3)+0.525(S	3642.44	-99.63	-2340.18	56.91	1.17	27.42	85.50		30.66	83.17	28.32	49.50	58.08	55.75	84.34	29.49	O.K
397	1.0(DL+COLL+CL-4)+0.525(S	3642.44	-99.63	-2340.18	56.91	1.17	27.42	85.50		30.66	83.17	28.32	49.50	58.08	55.75	84.34	29.49	O.K
398	1.0(DL+COLL+CL-1)+0.525(S	3543.80	99.63	-2140.92	55.37	1.17	25.09	81.63		31.45	79.29	29.12	45.63	56.54	54.21	80.46	30.29	O.K
399	1.0(DL+COLL+CL-2)+0.525(S	3543.80	99.63	-2140.92	55.37	1.17	25.09	81.63		31.45	79.29	29.12	45.63	56.54	54.21	80.46	30.29	O.K
400	1.0(DL+COLL+CL-3)+0.525(S	3543.80	99.63	-2140.92	55.37	1.17	25.09	81.63		31.45	79.29	29.12	45.63	56.54	54.21	80.46	30.29	O.K
401	1.0(DL+COLL+CL-4)+0.525SI	3543.80	99.63	-2140.92	55.37	1.17	25.09	81.63		31.45	79.29	29.12	45.63	56.54	54.21	80.46	30.29	O.K
402	0.6(DL+COLL+CL-1)+1.0(WLI	2771.37	0.00	-58.84	43.30	0.00	0.69	43.99		42.61	43.99	42.61	7.99	43.30	43.30	43.99	42.61	O.K
403	0.6(DL+COLL+CL-2)+1.0(WLI	2771.37	0.00	-58.84	43.30	0.00	0.69	43.99		42.61	43.99	42.61	7.99	43.30	43.30	43.99	42.61	O.K
404	0.6(DL+COLL+CL-3)+1.0(WLI	2771.37	0.00	-58.84	43.30	0.00	0.69	43.99		42.61	43.99	42.61	7.99	43.30	43.30	43.99	42.61	O.K
405	0.6(DL+COLL+CL-4)+1.0(WLI	2771.37	0.00	-58.84	43.30	0.00	0.69	43.99		42.61	43.99	42.61	7.99	43.30	43.30	43.99	42.61	O.K
406	0.6(DL+COLL+CL-1)+1.0(WLI	2700.92	0.00	-313.04	42.20	0.00	3.67	45.87		38.53	45.87	38.53	9.87	42.20	42.20	45.87	38.53	O.K
407	0.6(DL+COLL+CL-2)+1.0(WLI	2700.92	0.00	-313.04	42.20	0.00	3.67	45.87		38.53	45.87	38.53	9.87	42.20	42.20	45.87	38.53	O.K
408	0.6(DL+COLL+CL-3)+1.0(WLI	2700.92	0.00	-313.04	42.20	0.00	3.67	45.87		38.53	45.87	38.53	9.87	42.20	42.20	45.87	38.53	O.K
409	0.6(DL+COLL+CL-4)+1.0(WLI	2700.92	0.00	-313.04	42.20	0.00	3.67	45.87		38.53	45.87	38.53	9.87	42.20	42.20	45.87	38.53	O.K

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410	0.6(DL+COLL+CL-1)+1.0(WLI	2692.59	0.00	163.90	42.07	0.00	1.92	43.99		40.15	43.99	40.15	7.99	42.07	42.07	43.99	40.15	O.K
411	0.6(DL+COLL+CL-2)+1.0(WLI	2692.59	0.00	163.90	42.07	0.00	1.92	43.99		40.15	43.99	40.15	7.99	42.07	42.07	43.99	40.15	O.K
412	0.6(DL+COLL+CL-3)+1.0(WLI	2692.59	0.00	163.90	42.07	0.00	1.92	43.99		40.15	43.99	40.15	7.99	42.07	42.07	43.99	40.15	O.K
413	0.6(DL+COLL+CL-4)+1.0(WLI	2692.59	0.00	163.90	42.07	0.00	1.92	43.99		40.15	43.99	40.15	7.99	42.07	42.07	43.99	40.15	O.K
414	0.6(DL+COLL+CL-1)+1.0(WLI	2703.46	0.00	-168.88	42.24	0.00	1.98	44.22		40.26	44.22	40.26	8.22	42.24	42.24	44.22	40.26	O.K
415	0.6(DL+COLL+CL-2)+1.0(WLI	2703.46	0.00	-168.88	42.24	0.00	1.98	44.22		40.26	44.22	40.26	8.22	42.24	42.24	44.22	40.26	O.K
416	0.6(DL+COLL+CL-3)+1.0(WLI	2703.46	0.00	-168.88	42.24	0.00	1.98	44.22		40.26	44.22	40.26	8.22	42.24	42.24	44.22	40.26	O.K
417	0.6(DL+COLL+CL-4)+1.0(WLI	2703.46	0.00	-168.88	42.24	0.00	1.98	44.22		40.26	44.22	40.26	8.22	42.24	42.24	44.22	40.26	O.K
418	0.6(DL+COLL+CL-1)+1.0(WLI	2673.08	632.42	-395.28	41.77	7.41	4.63	53.81		44.55	38.99	29.72	17.81	49.18	34.36	46.40	37.14	O.K
419	0.6(DL+COLL+CL-2)+1.0(WLI	2673.08	632.42	-395.28	41.77	7.41	4.63	53.81		44.55	38.99	29.72	17.81	49.18	34.36	46.40	37.14	O.K
420	0.6(DL+COLL+CL-3)+1.0(WLI	2673.08	632.42	-395.28	41.77	7.41	4.63	53.81		44.55	38.99	29.72	17.81	49.18	34.36	46.40	37.14	O.K
421	0.6(DL+COLL+CL-4)+1.0(WLI	2673.08	632.42	-395.28	41.77	7.41	4.63	53.81		44.55	38.99	29.72	17.81	49.18	34.36	46.40	37.14	O.K
422	0.6(DL+COLL+CL-1)+1.0(WLI	2721.30	327.82	-395.28	42.52	3.84	4.63	50.99		41.73	43.31	34.05	14.99	46.36	38.68	47.15	37.89	O.K
423	0.6(DL+COLL+CL-2)+1.0(WLI	2721.30	327.82	-395.28	42.52	3.84	4.63	50.99		41.73	43.31	34.05	14.99	46.36	38.68	47.15	37.89	O.K
424	0.6(DL+COLL+CL-3)+1.0(WLI	2721.30	327.82	-395.28	42.52	3.84	4.63	50.99		41.73	43.31	34.05	14.99	46.36	38.68	47.15	37.89	O.K
425	0.6(DL+COLL+CL-4)+1.0(WLI	2721.30	327.82	-395.28	42.52	3.84	4.63	50.99		41.73	43.31	34.05	14.99	46.36	38.68	47.15	37.89	O.K
426	0.6(DL+COLL+CL-1)+1.0(WLI	2546.61	632.62	-395.28	39.79	7.41	4.63	51.84		42.57	37.01	27.75	15.84	47.21	32.38	44.43	35.16	O.K
427	0.6(DL+COLL+CL-2)+1.0(WLI	2546.61	632.62	-395.28	39.79	7.41	4.63	51.84		42.57	37.01	27.75	15.84	47.21	32.38	44.43	35.16	O.K
428	0.6(DL+COLL+CL-3)+1.0(WLI	2546.61	632.62	-395.28	39.79	7.41	4.63	51.84		42.57	37.01	27.75	15.84	47.21	32.38	44.43	35.16	O.K
429	0.6(DL+COLL+CL-4)+1.0(WLI	2546.61	632.62	-395.28	39.79	7.41	4.63	51.84		42.57	37.01	27.75	15.84	47.21	32.38	44.43	35.16	O.K
430	0.6(DL+COLL+CL-1)+1.0(WLI	2592.84	371.97	-395.28	40.51	4.36	4.63	49.50		40.24	40.79	31.52	13.50	44.87	36.16	45.15	35.88	O.K
431	0.6(DL+COLL+CL-2)+1.0(WLI	2592.84	371.97	-395.28	40.51	4.36	4.63	49.50		40.24	40.79	31.52	13.50	44.87	36.16	45.15	35.88	O.K
432	0.6(DL+COLL+CL-3)+1.0(WLI	2592.84	371.97	-395.28	40.51	4.36	4.63	49.50		40.24	40.79	31.52	13.50	44.87	36.16	45.15	35.88	O.K
433	0.6(DL+COLL+CL-4)+1.0(WLI	2592.84	371.97	-395.28	40.51	4.36	4.63	49.50		40.24	40.79	31.52	13.50	44.87	36.16	45.15	35.88	O.K
434	0.6(DL+COLL+CL-1)+0.7(SL'+	2877.99	-132.84	-528.12	44.97	1.56	6.19	52.71		40.34	49.60	37.22	16.71	46.53	43.41	51.16	38.78	O.K
435	0.6(DL+COLL+CL-2)+0.7(SL'+	2877.99	-132.84	-528.12	44.97	1.56	6.19	52.71		40.34	49.60	37.22	16.71	46.53	43.41	51.16	38.78	O.K
436	0.6(DL+COLL+CL-3)+0.7(SL'+	2877.99	-132.84	-528.12	44.97	1.56	6.19	52.71		40.34	49.60	37.22	16.71	46.53	43.41	51.16	38.78	O.K
437	0.6(DL+COLL+CL-4)+0.7(SL'+	2877.99	-132.84	-528.12	44.97	1.56	6.19	52.71		40.34	49.60	37.22	16.71	46.53	43.41	51.16	38.78	O.K
438	0.6(DL+COLL+CL-1)+0.7(SL'+	2853.47	-132.84	-326.26	44.59	1.56	3.82	49.97		42.32	46.85	39.21	13.97	46.15	43.03	48.41	40.77	O.K
439	0.6(DL+COLL+CL-2)+0.7(SL'+	2811.20	-132.84	-478.78	43.93	1.56	5.61	51.09		39.87	47.98	36.76	15.09	45.48	42.37	49.54	38.32	O.K
440	0.6(DL+COLL+CL-3)+0.7(SL'+	2806.20	-132.84	-192.62	43.85	1.56	2.26	47.66		43.15	44.55	40.03	11.66	45.41	42.29	46.11	41.59	O.K
441	0.6(DL+COLL+CL-4)+0.7(SL'+	2812.73	-132.84	-392.28	43.95	1.56	4.60	50.10		40.91	46.99	37.80	14.10	45.51	42.40	48.55	39.36	O.K

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Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})		p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
	Ultimate Conditions																	
101	1.4(DL+COLL)	3046.83	0.00	922.32	47.61	0.00	10.81	58.42		36.80	58.42	36.80	22.42	47.61	47.61	58.42	36.80	
102	1.2(DL+COLL+CL-1)+1.6(RLL	4403.84	0.00	4164.96	68.81	0.00	48.81	117.62		20.00	117.62	20.00	81.62	68.81	68.81	117.62	20.00	
103	1.2(DL+COLL+CL-2)+1.6(RLL	4403.84	0.00	4164.96	68.81	0.00	48.81	117.62		20.00	117.62	20.00	81.62	68.81	68.81	117.62	20.00	
104	1.2(DL+COLL+CL-3)+1.6(RLL	4403.84	0.00	4164.96	68.81	0.00	48.81	117.62		20.00	117.62	20.00	81.62	68.81	68.81	117.62	20.00	
105	1.2(DL+COLL+CL-4)+1.6(RLL	4403.84	0.00	4164.96	68.81	0.00	48.81	117.62		20.00	117.62	20.00	81.62	68.81	68.81	117.62	20.00	
106	1.4(DL+COLL+CL-1)	3046.83	0.00	922.32	47.61	0.00	10.81	58.42		36.80	58.42	36.80	22.42	47.61	47.61	58.42	36.80	
107	1.4(DL+COLL+CL-2)	3046.83	0.00	922.32	47.61	0.00	10.81	58.42		36.80	58.42	36.80	22.42	47.61	47.61	58.42	36.80	
108	1.4(DL+COLL+CL-3)	3046.83	0.00	922.32	47.61	0.00	10.81	58.42		36.80	58.42	36.80	22.42	47.61	47.61	58.42	36.80	
109	1.4(DL+COLL+CL-4)	3046.83	0.00	922.32	47.61	0.00	10.81	58.42		36.80	58.42	36.80	22.42	47.61	47.61	58.42	36.80	
110	1.2(DL+COLL+CL-1+TL1)+1.6	4390.02	0.00	4482.94	68.59	0.00	52.53	121.13		16.06	121.13	16.06	85.13	68.60	68.60	121.13	16.06	
111	1.2(DL+COLL+CL-2+TL1)+1.6	4390.02	0.00	4482.94	68.59	0.00	52.53	121.13		16.06	121.13	16.06	85.13	68.60	68.60	121.13	16.06	
112	1.2(DL+COLL+CL-3+TL1)+1.6	4390.02	0.00	4482.94	68.59	0.00	52.53	121.13		16.06	121.13	16.06	85.13	68.60	68.60	121.13	16.06	
113	1.2(DL+COLL+CL-4+TL1)+1.6	4390.02	0.00	4482.94	68.59	0.00	52.53	121.13		16.06	121.13	16.06	85.13	68.60	68.60	121.13	16.06	
114	1.2(DL+COLL+CL-1+TL2)+1.6	4417.66	0.00	3846.98	69.03	0.00	45.08	114.11		23.94	114.11	23.94	78.11	69.03	69.03	114.11	23.94	
115	1.2(DL+COLL+CL-2+TL2)+1.6	4417.66	0.00	3846.98	69.03	0.00	45.08	114.11		23.94	114.11	23.94	78.11	69.03	69.03	114.11	23.94	
116	1.2(DL+COLL+CL-3+TL2)+1.6	4417.66	0.00	3846.98	69.03	0.00	45.08	114.11		23.94	114.11	23.94	78.11	69.03	69.03	114.11	23.94	
117	1.2(DL+COLL+CL-4+TL2)+1.6	4417.66	0.00	3846.98	69.03	0.00	45.08	114.11		23.94	114.11	23.94	78.11	69.03	69.03	114.11	23.94	
118	1.2(DL+COLL+CL-1)+0.8(WLI	2955.49	0.00	521.40	46.18	0.00	6.11	52.29		40.07	52.29	40.07	16.29	46.18	46.18	52.29	40.07	
119	1.2(DL+COLL+CL-2)+0.8(WLI	2955.49	0.00	521.40	46.18	0.00	6.11	52.29		40.07	52.29	40.07	16.29	46.18	46.18	52.29	40.07	
120	1.2(DL+COLL+CL-3)+0.8(WLI	2955.49	0.00	521.40	46.18	0.00	6.11	52.29		40.07	52.29	40.07	16.29	46.18	46.18	52.29	40.07	
121	1.2(DL+COLL+CL-4)+0.8(WLI	2955.49	0.00	521.40	46.18	0.00	6.11	52.29		40.07	52.29	40.07	16.29	46.18	46.18	52.29	40.07	
122	1.2(DL+COLL+CL-1)+0.8(WLI	2899.13	0.00	724.76	45.30	0.00	8.49	53.79		36.81	53.79	36.81	17.79	45.30	45.30	53.79	36.81	
123	1.2(DL+COLL+CL-2)+0.8(WLI	2899.13	0.00	724.76	45.30	0.00	8.49	53.79		36.81	53.79	36.81	17.79	45.30	45.30	53.79	36.81	
124	1.2(DL+COLL+CL-3)+0.8(WLI	2899.13	0.00	724.76	45.30	0.00	8.49	53.79		36.81	53.79	36.81	17.79	45.30	45.30	53.79	36.81	

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125	1.2(DL+COLL+CL-4)+0.8(WLI	2899.13	0.00	724.76	45.30	0.00	8.49	53.79		36.81	53.79	36.81	17.79	45.30	45.30	53.79	36.81	
126	1.2(DL+COLL+CL-1)+0.8(WLI	2892.46	0.00	343.22	45.19	0.00	4.02	49.22		41.17	49.22	41.17	13.22	45.20	45.20	49.22	41.17	
127	1.2(DL+COLL+CL-2)+0.8(WLI	2892.46	0.00	343.22	45.19	0.00	4.02	49.22		41.17	49.22	41.17	13.22	45.20	45.20	49.22	41.17	
128	1.2(DL+COLL+CL-3)+0.8(WLI	2892.46	0.00	343.22	45.19	0.00	4.02	49.22		41.17	49.22	41.17	13.22	45.20	45.20	49.22	41.17	
129	1.2(DL+COLL+CL-4)+0.8(WLI	2892.46	0.00	343.22	45.19	0.00	4.02	49.22		41.17	49.22	41.17	13.22	45.20	45.20	49.22	41.17	
130	1.2(DL+COLL+CL-1)+0.8(WLI	2901.16	0.00	609.44	45.33	0.00	7.14	52.47		38.19	52.47	38.19	16.47	45.33	45.33	52.47	38.19	
131	1.2(DL+COLL+CL-2)+0.8(WLI	2901.16	0.00	609.44	45.33	0.00	7.14	52.47		38.19	52.47	38.19	16.47	45.33	45.33	52.47	38.19	
132	1.2(DL+COLL+CL-3)+0.8(WLI	2901.16	0.00	609.44	45.33	0.00	7.14	52.47		38.19	52.47	38.19	16.47	45.33	45.33	52.47	38.19	
133	1.2(DL+COLL+CL-4)+0.8(WLI	2901.16	0.00	609.44	45.33	0.00	7.14	52.47		38.19	52.47	38.19	16.47	45.33	45.33	52.47	38.19	
134	1.2(DL+COLL+CL-1)+0.8(WLI	2876.86	245.57	790.56	44.95	2.88	9.26	57.09		38.56	51.34	32.81	21.09	47.83	42.08	54.22	35.69	
135	1.2(DL+COLL+CL-2)+0.8(WLI	2876.86	245.57	790.56	44.95	2.88	9.26	57.09		38.56	51.34	32.81	21.09	47.83	42.08	54.22	35.69	
136	1.2(DL+COLL+CL-3)+0.8(WLI	2876.86	245.57	790.56	44.95	2.88	9.26	57.09		38.56	51.34	32.81	21.09	47.83	42.08	54.22	35.69	
137	1.2(DL+COLL+CL-4)+0.8(WLI	2876.86	245.57	790.56	44.95	2.88	9.26	57.09		38.56	51.34	32.81	21.09	47.83	42.08	54.22	35.69	
138	1.2(DL+COLL+CL-1)+0.8(WLI	2915.43	79.46	790.56	45.55	0.93	9.26	55.75		37.22	53.89	35.36	19.75	46.49	44.63	54.82	36.29	
139	1.2(DL+COLL+CL-2)+0.8(WLI	2915.43	79.46	790.56	45.55	0.93	9.26	55.75		37.22	53.89	35.36	19.75	46.49	44.63	54.82	36.29	
140	1.2(DL+COLL+CL-3)+0.8(WLI	2915.43	79.46	790.56	45.55	0.93	9.26	55.75		37.22	53.89	35.36	19.75	46.49	44.63	54.82	36.29	
141	1.2(DL+COLL+CL-4)+0.8(WLI	2915.43	79.46	790.56	45.55	0.93	9.26	55.75		37.22	53.89	35.36	19.75	46.49	44.63	54.82	36.29	
142	1.2(DL+COLL+CL-1)+0.8(WLI	2775.68	403.73	790.56	43.37	4.73	9.26	57.37		38.84	47.90	29.37	21.37	48.11	38.64	52.64	34.11	
143	1.2(DL+COLL+CL-2)+0.8(WLI	2775.68	403.73	790.56	43.37	4.73	9.26	57.37		38.84	47.90	29.37	21.37	48.11	38.64	52.64	34.11	
144	1.2(DL+COLL+CL-3)+0.8(WLI	2775.68	403.73	790.56	43.37	4.73	9.26	57.37		38.84	47.90	29.37	21.37	48.11	38.64	52.64	34.11	
145	1.2(DL+COLL+CL-4)+0.8(WLI	2775.68	403.73	790.56	43.37	4.73	9.26	57.37		38.84	47.90	29.37	21.37	48.11	38.64	52.64	34.11	
146	1.2(DL+COLL+CL-1)+0.8(WLI	2812.66	253.75	790.56	43.95	2.97	9.26	56.19		37.66	50.24	31.71	20.19	46.93	40.98	53.22	34.69	
147	1.2(DL+COLL+CL-2)+0.8(WLI	2812.66	253.75	790.56	43.95	2.97	9.26	56.19		37.66	50.24	31.71	20.19	46.93	40.98	53.22	34.69	
148	1.2(DL+COLL+CL-3)+0.8(WLI	2812.66	253.75	790.56	43.95	2.97	9.26	56.19		37.66	50.24	31.71	20.19	46.93	40.98	53.22	34.69	
149	1.2(DL+COLL+CL-4)+0.8(WLI	2812.66	253.75	790.56	43.95	2.97	9.26	56.19		37.66	50.24	31.71	20.19	46.93	40.98	53.22	34.69	
150	1.2(DL+COLL+CL-1)+1.6(WLI	3365.19	0.00	1306.75	52.58	0.00	15.31	67.89		37.27	67.89	37.27	31.89	52.58	52.58	67.89	37.27	
151	1.2(DL+COLL+CL-2)+1.6(WLI	3365.19	0.00	1306.75	52.58	0.00	15.31	67.89		37.27	67.89	37.27	31.89	52.58	52.58	67.89	37.27	
152	1.2(DL+COLL+CL-3)+1.6(WLI	3365.19	0.00	1306.75	52.58	0.00	15.31	67.89		37.27	67.89	37.27	31.89	52.58	52.58	67.89	37.27	
153	1.2(DL+COLL+CL-4)+1.6(WLI	3365.19	0.00	1306.75	52.58	0.00	15.31	67.89		37.27	67.89	37.27	31.89	52.58	52.58	67.89	37.27	
154	1.2(DL+COLL+CL-1)+1.6(WLI	3252.47	0.00	1713.47	50.82	0.00	20.08	70.90		30.74	70.90	30.74	34.90	50.82	50.82	70.90	30.74	
155	1.2(DL+COLL+CL-2)+1.6(WLI	3252.47	0.00	1713.47	50.82	0.00	20.08	70.90		30.74	70.90	30.74	34.90	50.82	50.82	70.90	30.74	
156	1.2(DL+COLL+CL-3)+1.6(WLI	3252.47	0.00	1713.47	50.82	0.00	20.08	70.90		30.74	70.90	30.74	34.90	50.82	50.82	70.90	30.74	

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157	1.2(DL+COLL+CL-4)+1.6(WLI	3252.47	0.00	1713.47	50.82	0.00	20.08	70.90		30.74	70.90	30.74	34.90	50.82	50.82	70.90	30.74	
158	1.2(DL+COLL+CL-1)+1.6(WLI	3239.14	0.00	950.38	50.61	0.00	11.14	61.75		39.47	61.75	39.47	25.75	50.61	50.61	61.75	39.47	
159	1.2(DL+COLL+CL-2)+1.6(WLI	3239.14	0.00	950.38	50.61	0.00	11.14	61.75		39.47	61.75	39.47	25.75	50.61	50.61	61.75	39.47	
160	1.2(DL+COLL+CL-3)+1.6(WLI	3239.14	0.00	950.38	50.61	0.00	11.14	61.75		39.47	61.75	39.47	25.75	50.61	50.61	61.75	39.47	
161	1.2(DL+COLL+CL-4)+1.6(WLI	3239.14	0.00	950.38	50.61	0.00	11.14	61.75		39.47	61.75	39.47	25.75	50.61	50.61	61.75	39.47	
162	1.2(DL+COLL+CL-1)+1.6(WLI	3256.54	0.00	1482.82	50.88	0.00	17.38	68.26		33.51	68.26	33.51	32.26	50.89	50.89	68.26	33.51	
163	1.2(DL+COLL+CL-2)+1.6(WLI	3256.54	0.00	1482.82	50.88	0.00	17.38	68.26		33.51	68.26	33.51	32.26	50.89	50.89	68.26	33.51	
164	1.2(DL+COLL+CL-3)+1.6(WLI	3256.54	0.00	1482.82	50.88	0.00	17.38	68.26		33.51	68.26	33.51	32.26	50.89	50.89	68.26	33.51	
165	1.2(DL+COLL+CL-4)+1.6(WLI	3256.54	0.00	1482.82	50.88	0.00	17.38	68.26		33.51	68.26	33.51	32.26	50.89	50.89	68.26	33.51	
166	1.2(DL+COLL+CL-1)+1.6(WLI	3207.93	491.14	1845.06	50.12	5.76	21.62	77.50		34.26	65.99	22.75	41.50	55.88	44.37	71.75	28.51	
167	1.2(DL+COLL+CL-2)+1.6(WLI	3207.93	491.14	1845.06	50.12	5.76	21.62	77.50		34.26	65.99	22.75	41.50	55.88	44.37	71.75	28.51	
168	1.2(DL+COLL+CL-3)+1.6(WLI	3207.93	491.14	1845.06	50.12	5.76	21.62	77.50		34.26	65.99	22.75	41.50	55.88	44.37	71.75	28.51	
169	1.2(DL+COLL+CL-4)+1.6(WLI	3207.93	491.14	1845.06	50.12	5.76	21.62	77.50		34.26	65.99	22.75	41.50	55.88	44.37	71.75	28.51	
170	1.2(DL+COLL+CL-1)+1.6(WLI	3285.07	158.91	1845.06	51.33	1.86	21.62	74.81		31.57	71.09	27.85	38.81	53.19	49.47	72.95	29.71	
171	1.2(DL+COLL+CL-2)+1.6(WLI	3285.07	158.91	1845.06	51.33	1.86	21.62	74.81		31.57	71.09	27.85	38.81	53.19	49.47	72.95	29.71	
172	1.2(DL+COLL+CL-3)+1.6(WLI	3285.07	158.91	1845.06	51.33	1.86	21.62	74.81		31.57	71.09	27.85	38.81	53.19	49.47	72.95	29.71	
173	1.2(DL+COLL+CL-4)+1.6(WLI	3285.07	158.91	1845.06	51.33	1.86	21.62	74.81		31.57	71.09	27.85	38.81	53.19	49.47	72.95	29.71	
174	1.2(DL+COLL+CL-1)+1.6(WLI	3005.57	807.46	1845.06	46.96	9.46	21.62	78.05		34.80	59.12	15.88	42.05	56.43	37.50	68.59	25.34	
175	1.2(DL+COLL+CL-2)+1.6(WLI	3005.57	807.46	1845.06	46.96	9.46	21.62	78.05		34.80	59.12	15.88	42.05	56.43	37.50	68.59	25.34	
176	1.2(DL+COLL+CL-3)+1.6(WLI	3005.57	807.46	1845.06	46.96	9.46	21.62	78.05		34.80	59.12	15.88	42.05	56.43	37.50	68.59	25.34	
177	1.2(DL+COLL+CL-4)+1.6(WLI	3005.57	807.46	1845.06	46.96	9.46	21.62	78.05		34.80	59.12	15.88	42.05	56.43	37.50	68.59	25.34	
178	1.2(DL+COLL+CL-1)+1.6(WLI	3079.53	507.50	1845.06	48.12	5.95	21.62	75.69		32.44	63.79	20.55	39.69	54.07	42.17	69.74	26.50	
179	1.2(DL+COLL+CL-2)+1.6(WLI	3079.53	507.50	1845.06	48.12	5.95	21.62	75.69		32.44	63.79	20.55	39.69	54.07	42.17	69.74	26.50	
180	1.2(DL+COLL+CL-3)+1.6(WLI	3079.53	507.50	1845.06	48.12	5.95	21.62	75.69		32.44	63.79	20.55	39.69	54.07	42.17	69.74	26.50	
181	1.2(DL+COLL+CL-4)+1.6(WLI	3079.53	507.50	1845.06	48.12	5.95	21.62	75.69		32.44	63.79	20.55	39.69	54.07	42.17	69.74	26.50	
182	1.2(DL+COLL+CL-1)+1.0(SL'+	3524.51	321.34	2166.40	55.07	3.77	25.39	84.22		33.45	76.69	25.92	48.22	58.84	51.31	80.46	29.69	
183	1.2(DL+COLL+CL-2)+1.0(SL'+	3524.51	321.34	2166.40	55.07	3.77	25.39	84.22		33.45	76.69	25.92	48.22	58.84	51.31	80.46	29.69	
184	1.2(DL+COLL+CL-3)+1.0(SL'+	3524.51	321.34	2166.40	55.07	3.77	25.39	84.22		33.45	76.69	25.92	48.22	58.84	51.31	80.46	29.69	
185	1.2(DL+COLL+CL-4)+1.0(SL'+	3524.51	321.34	2166.40	55.07	3.77	25.39	84.22		33.45	76.69	25.92	48.22	58.84	51.31	80.46	29.69	
186	1.2(DL+COLL+CL-1)+1.0(SL'-	3336.63	321.34	1523.72	52.13	3.77	17.86	73.76		38.04	66.23	30.51	37.76	55.90	48.37	70.00	34.28	
187	1.2(DL+COLL+CL-2)+1.0(SL'-	3336.63	321.34	1523.72	52.13	3.77	17.86	73.76		38.04	66.23	30.51	37.76	55.90	48.37	70.00	34.28	

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188	1.2(DL+COLL+CL-3)+1.0(SL'-	3336.63	321.34	1523.72	52.13	3.77	17.86	73.76		38.04	66.23	30.51	37.76	55.90	48.37	70.00	34.28	
189	1.2(DL+COLL+CL-4)+1.0(SL'-	3336.63	321.34	1523.72	52.13	3.77	17.86	73.76		38.04	66.23	30.51	37.76	55.90	48.37	70.00	34.28	
190	0.9(DL+COLL+CL-1)+1.6(WLI	2834.82	0.00	54.61	44.29	0.00	0.64	44.93		43.65	44.93	43.65	8.93	44.29	44.29	44.93	43.65	
191	0.9(DL+COLL+CL-2)+1.6(WLI	2834.82	0.00	54.61	44.29	0.00	0.64	44.93		43.65	44.93	43.65	8.93	44.29	44.29	44.93	43.65	
192	0.9(DL+COLL+CL-3)+1.6(WLI	2834.82	0.00	54.61	44.29	0.00	0.64	44.93		43.65	44.93	43.65	8.93	44.29	44.29	44.93	43.65	
193	0.9(DL+COLL+CL-4)+1.6(WLI	2834.82	0.00	54.61	44.29	0.00	0.64	44.93		43.65	44.93	43.65	8.93	44.29	44.29	44.93	43.65	
194	0.9(DL+COLL+CL-1)+1.6(WLI	2722.10	0.00	461.33	42.53	0.00	5.41	47.94		37.13	47.94	37.13	11.94	42.54	42.54	47.94	37.13	
195	0.9(DL+COLL+CL-2)+1.6(WLI	2722.10	0.00	461.33	42.53	0.00	5.41	47.94		37.13	47.94	37.13	11.94	42.54	42.54	47.94	37.13	
196	0.9(DL+COLL+CL-3)+1.6(WLI	2722.10	0.00	461.33	42.53	0.00	5.41	47.94		37.13	47.94	37.13	11.94	42.54	42.54	47.94	37.13	
197	0.9(DL+COLL+CL-4)+1.6(WLI	2722.10	0.00	461.33	42.53	0.00	5.41	47.94		37.13	47.94	37.13	11.94	42.54	42.54	47.94	37.13	
198	0.9(DL+COLL+CL-1)+1.6(WLI	2708.78	0.00	301.76	42.32	0.00	3.54	45.86		38.79	45.86	38.79	9.86	42.33	42.33	45.86	38.79	
199	0.9(DL+COLL+CL-2)+1.6(WLI	2708.78	0.00	301.76	42.32	0.00	3.54	45.86		38.79	45.86	38.79	9.86	42.33	42.33	45.86	38.79	
200	0.9(DL+COLL+CL-3)+1.6(WLI	2708.78	0.00	301.76	42.32	0.00	3.54	45.86		38.79	45.86	38.79	9.86	42.33	42.33	45.86	38.79	
201	0.9(DL+COLL+CL-4)+1.6(WLI	2708.78	0.00	301.76	42.32	0.00	3.54	45.86		38.79	45.86	38.79	9.86	42.33	42.33	45.86	38.79	
202	0.9(DL+COLL+CL-1)+1.6(WLI	2726.17	0.00	230.68	42.60	0.00	2.70	45.30		39.89	45.30	39.89	9.30	42.60	42.60	45.30	39.89	
203	0.9(DL+COLL+CL-2)+1.6(WLI	2726.17	0.00	230.68	42.60	0.00	2.70	45.30		39.89	45.30	39.89	9.30	42.60	42.60	45.30	39.89	
204	0.9(DL+COLL+CL-3)+1.6(WLI	2726.17	0.00	230.68	42.60	0.00	2.70	45.30		39.89	45.30	39.89	9.30	42.60	42.60	45.30	39.89	
205	0.9(DL+COLL+CL-4)+1.6(WLI	2726.17	0.00	230.68	42.60	0.00	2.70	45.30		39.89	45.30	39.89	9.30	42.60	42.60	45.30	39.89	
206	0.9(DL+COLL+CL-1)+1.6(WLI	2677.56	491.14	592.92	41.84	5.76	6.95	54.54		40.64	43.03	29.13	18.54	47.59	36.08	48.79	34.89	
207	0.9(DL+COLL+CL-2)+1.6(WLI	2677.56	491.14	592.92	41.84	5.76	6.95	54.54		40.64	43.03	29.13	18.54	47.59	36.08	48.79	34.89	
208	0.9(DL+COLL+CL-3)+1.6(WLI	2677.56	491.14	592.92	41.84	5.76	6.95	54.54		40.64	43.03	29.13	18.54	47.59	36.08	48.79	34.89	
209	0.9(DL+COLL+CL-4)+1.6(WLI	2677.56	491.14	592.92	41.84	5.76	6.95	54.54		40.64	43.03	29.13	18.54	47.59	36.08	48.79	34.89	
210	0.9(DL+COLL+CL-1)+1.6(WLI	2754.70	158.91	592.92	43.04	1.86	6.95	51.85		37.96	48.13	34.23	15.85	44.91	41.18	49.99	36.10	
211	0.9(DL+COLL+CL-2)+1.6(WLI	2754.70	158.91	592.92	43.04	1.86	6.95	51.85		37.96	48.13	34.23	15.85	44.91	41.18	49.99	36.10	
212	0.9(DL+COLL+CL-3)+1.6(WLI	2754.70	158.91	592.92	43.04	1.86	6.95	51.85		37.96	48.13	34.23	15.85	44.91	41.18	49.99	36.10	
213	0.9(DL+COLL+CL-4)+1.6(WLI	2754.70	158.91	592.92	43.04	1.86	6.95	51.85		37.96	48.13	34.23	15.85	44.91	41.18	49.99	36.10	
214	0.9(DL+COLL+CL-1)+1.6(WLI	2475.21	807.46	592.92	38.68	9.46	6.95	55.09		41.19	36.16	22.26	19.09	48.14	29.21	45.63	31.73	
215	0.9(DL+COLL+CL-2)+1.6(WLI	2475.21	807.46	592.92	38.68	9.46	6.95	55.09		41.19	36.16	22.26	19.09	48.14	29.21	45.63	31.73	
216	0.9(DL+COLL+CL-3)+1.6(WLI	2475.21	807.46	592.92	38.68	9.46	6.95	55.09		41.19	36.16	22.26	19.09	48.14	29.21	45.63	31.73	
217	0.9(DL+COLL+CL-4)+1.6(WLI	2475.21	807.46	592.92	38.68	9.46	6.95	55.09		41.19	36.16	22.26	19.09	48.14	29.21	45.63	31.73	
218	0.9(DL+COLL+CL-1)+1.6(WLI	2549.17	507.50	592.92	39.83	5.95	6.95	52.73		38.83	40.83	26.94	16.73	45.78	33.89	46.78	32.89	
219	0.9(DL+COLL+CL-2)+1.6(WLI	2549.17	507.50	592.92	39.83	5.95	6.95	52.73		38.83	40.83	26.94	16.73	45.78	33.89	46.78	32.89	

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220	0.9(DL+COLL+CL-3)+1.6(WL	2549.17	507.50	592.92	39.83	5.95	6.95	52.73		38.83	40.83	26.94	16.73	45.78	33.89	46.78	32.89	
221	0.9(DL+COLL+CL-4)+1.6(WL	2549.17	507.50	592.92	39.83	5.95	6.95	52.73		38.83	40.83	26.94	16.73	45.78	33.89	46.78	32.89	
222	0.9(DL+COLL+CL-1)+1.0(SL'+	2994.15	321.34	914.26	46.78	3.77	10.71	61.26		39.84	53.73	32.30	25.26	50.55	43.02	57.50	36.07	
223	0.9(DL+COLL+CL-2)+1.0(SL'+	2994.15	321.34	914.26	46.78	3.77	10.71	61.26		39.84	53.73	32.30	25.26	50.55	43.02	57.50	36.07	
224	0.9(DL+COLL+CL-3)+1.0(SL'+	2994.15	321.34	914.26	46.78	3.77	10.71	61.26		39.84	53.73	32.30	25.26	50.55	43.02	57.50	36.07	
225	0.9(DL+COLL+CL-4)+1.0(SL'+	2994.15	321.34	914.26	46.78	3.77	10.71	61.26		39.84	53.73	32.30	25.26	50.55	43.02	57.50	36.07	
226	0.9(DL+COLL+CL-1)+1.0(SL'-	2806.27	321.34	271.58	43.85	3.77	3.18	50.80		44.43	43.26	36.90	14.80	47.62	40.08	47.03	40.67	
227	0.9(DL+COLL+CL-2)+1.0(SL'-	2806.27	321.34	271.58	43.85	3.77	3.18	50.80		44.43	43.26	36.90	14.80	47.62	40.08	47.03	40.67	
228	0.9(DL+COLL+CL-3)+1.0(SL'-	2806.27	321.34	271.58	43.85	3.77	3.18	50.80		44.43	43.26	36.90	14.80	47.62	40.08	47.03	40.67	
229	0.9(DL+COLL+CL-4)+1.0(SL'-	2806.27	321.34	271.58	43.85	3.77	3.18	50.80		44.43	43.26	36.90	14.80	47.62	40.08	47.03	40.67	

Max :	3826	633	164
Min :	2547	-142	-3033

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 17.49 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 27.49 kN/m²
Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 50.33 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 60.33 kN/m²

Sample calculation of values of base pressure:

Load case : 302 1.0(DL+COLL+CL-1+TL1)+1.0(RLL+PLL / MLL)

Maximum pressure, p_1 (pmax) = 59.42+0.00+35.54= 94.96 kN/m²

p_2= 59.42+0.00-35.54= 23.88 kN/m²

p_3= 59.42-(0.00+35.54)= 23.88 kN/m²

p_4= 59.42-0.00+35.54= 94.96 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 94.96-2*18= 58.96 kN/m²

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DESIGN OF ISOLATED FOOTING F7

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

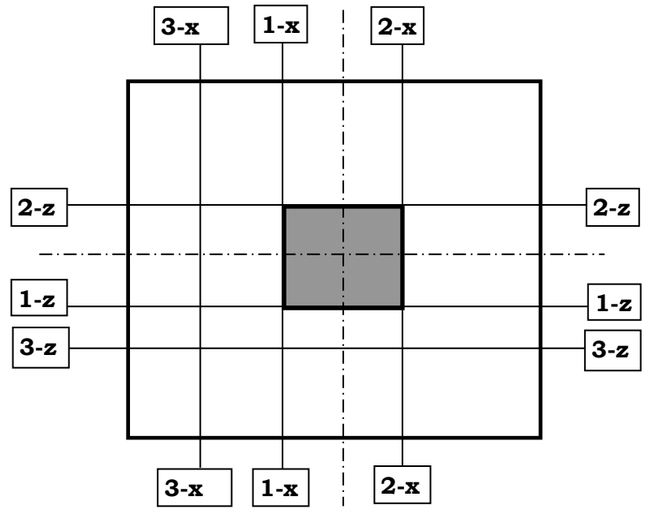
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((1.35+2*0.5*0.6625)+(1.35+2*0.5*0.6625))*2
= 8.05 m

Punching Area= ((1.35+2*0.5*0.6625)*(1.35+2*0.5*0.6625))
= 4.05 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.75*24+1.25*18)
= 40.50 kN/m^2

Area of footing, A_f= 64.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4(DL+COLL)	47.61	47.61	47.61	39.30	39.30	18.75	426.05
102	1.2(DL+COLL+CL-1)+1.6(R	68.81	68.81	68.81	156.49	156.49	74.67	1697.18
103	1.2(DL+COLL+CL-2)+1.6(R	68.81	68.81	68.81	156.49	156.49	74.67	1697.18
104	1.2(DL+COLL+CL-3)+1.6(R	68.81	68.81	68.81	156.49	156.49	74.67	1697.18
105	1.2(DL+COLL+CL-4)+1.6(R	68.81	68.81	68.81	156.49	156.49	74.67	1697.18
106	1.4(DL+COLL+CL-1)	47.61	47.61	47.61	39.30	39.30	18.75	426.05
107	1.4(DL+COLL+CL-2)	47.61	47.61	47.61	39.30	39.30	18.75	426.05
108	1.4(DL+COLL+CL-3)	47.61	47.61	47.61	39.30	39.30	18.75	426.05
109	1.4(DL+COLL+CL-4)	47.61	47.61	47.61	39.30	39.30	18.75	426.05
110	1.2(DL+COLL+CL-1+TL1)+	68.60	68.60	68.60	155.33	155.33	74.11	1684.23
111	1.2(DL+COLL+CL-2+TL1)+	68.60	68.60	68.60	155.33	155.33	74.11	1684.23
112	1.2(DL+COLL+CL-3+TL1)+	68.60	68.60	68.60	155.33	155.33	74.11	1684.23

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113	1.2(DL+COLL+CL-4+TL1)+	68.60	68.60	68.60	155.33	155.33	74.11	1684.23
114	1.2(DL+COLL+CL-1+TL2)+	69.03	69.03	69.03	157.71	157.71	75.25	1710.13
115	1.2(DL+COLL+CL-2+TL2)+	69.03	69.03	69.03	157.71	157.71	75.25	1710.13
116	1.2(DL+COLL+CL-3+TL2)+	69.03	69.03	69.03	157.71	157.71	75.25	1710.13
117	1.2(DL+COLL+CL-4+TL2)+	69.03	69.03	69.03	157.71	157.71	75.25	1710.13
118	1.2(DL+COLL+CL-1)+0.8(W	46.18	46.18	46.18	31.40	31.40	14.98	340.48
119	1.2(DL+COLL+CL-2)+0.8(W	46.18	46.18	46.18	31.40	31.40	14.98	340.48
120	1.2(DL+COLL+CL-3)+0.8(W	46.18	46.18	46.18	31.40	31.40	14.98	340.48
121	1.2(DL+COLL+CL-4)+0.8(W	46.18	46.18	46.18	31.40	31.40	14.98	340.48
122	1.2(DL+COLL+CL-1)+0.8(W	45.30	45.30	45.30	26.53	26.53	12.66	287.69
123	1.2(DL+COLL+CL-2)+0.8(W	45.30	45.30	45.30	26.53	26.53	12.66	287.69
124	1.2(DL+COLL+CL-3)+0.8(W	45.30	45.30	45.30	26.53	26.53	12.66	287.69
125	1.2(DL+COLL+CL-4)+0.8(W	45.30	45.30	45.30	26.53	26.53	12.66	287.69
126	1.2(DL+COLL+CL-1)+0.8(W	45.20	45.20	45.20	25.98	25.98	12.40	281.45
127	1.2(DL+COLL+CL-2)+0.8(W	45.20	45.20	45.20	25.98	25.98	12.40	281.45
128	1.2(DL+COLL+CL-3)+0.8(W	45.20	45.20	45.20	25.98	25.98	12.40	281.45
129	1.2(DL+COLL+CL-4)+0.8(W	45.20	45.20	45.20	25.98	25.98	12.40	281.45
130	1.2(DL+COLL+CL-1)+0.8(W	45.33	45.33	45.33	26.70	26.70	12.74	289.60
131	1.2(DL+COLL+CL-2)+0.8(W	45.33	45.33	45.33	26.70	26.70	12.74	289.60
132	1.2(DL+COLL+CL-3)+0.8(W	45.33	45.33	45.33	26.70	26.70	12.74	289.60
133	1.2(DL+COLL+CL-4)+0.8(W	45.33	45.33	45.33	26.70	26.70	12.74	289.60
134	1.2(DL+COLL+CL-1)+0.8(W	45.44	44.47	45.93	36.12	13.14	16.83	266.83
135	1.2(DL+COLL+CL-2)+0.8(W	45.44	44.47	45.93	36.12	13.14	16.83	266.83
136	1.2(DL+COLL+CL-3)+0.8(W	45.44	44.47	45.93	36.12	13.14	16.83	266.83
137	1.2(DL+COLL+CL-4)+0.8(W	45.44	44.47	45.93	36.12	13.14	16.83	266.83
138	1.2(DL+COLL+CL-1)+0.8(W	45.72	45.40	45.88	31.69	24.25	14.99	302.96
139	1.2(DL+COLL+CL-2)+0.8(W	45.72	45.40	45.88	31.69	24.25	14.99	302.96
140	1.2(DL+COLL+CL-3)+0.8(W	45.72	45.40	45.88	31.69	24.25	14.99	302.96
141	1.2(DL+COLL+CL-4)+0.8(W	45.72	45.40	45.88	31.69	24.25	14.99	302.96
142	1.2(DL+COLL+CL-1)+0.8(W	44.17	42.58	44.99	34.81	-3.03	15.95	172.06
143	1.2(DL+COLL+CL-2)+0.8(W	44.17	42.58	44.99	34.81	-3.03	15.95	172.06

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144	1.2(DL+COLL+CL-3)+0.8(W	44.17	42.58	44.99	34.81	-3.03	15.95	172.06
145	1.2(DL+COLL+CL-4)+0.8(W	44.17	42.58	44.99	34.81	-3.03	15.95	172.06
146	1.2(DL+COLL+CL-1)+0.8(W	44.46	43.45	44.97	30.99	7.21	14.37	206.70
147	1.2(DL+COLL+CL-2)+0.8(W	44.46	43.45	44.97	30.99	7.21	14.37	206.70
148	1.2(DL+COLL+CL-3)+0.8(W	44.46	43.45	44.97	30.99	7.21	14.37	206.70
149	1.2(DL+COLL+CL-4)+0.8(W	44.46	43.45	44.97	30.99	7.21	14.37	206.70
150	1.2(DL+COLL+CL-1)+1.6(W	52.58	52.58	52.58	66.78	66.78	31.86	724.26
151	1.2(DL+COLL+CL-2)+1.6(W	52.58	52.58	52.58	66.78	66.78	31.86	724.26
152	1.2(DL+COLL+CL-3)+1.6(W	52.58	52.58	52.58	66.78	66.78	31.86	724.26
153	1.2(DL+COLL+CL-4)+1.6(W	52.58	52.58	52.58	66.78	66.78	31.86	724.26
154	1.2(DL+COLL+CL-1)+1.6(W	50.82	50.82	50.82	57.05	57.05	27.22	618.67
155	1.2(DL+COLL+CL-2)+1.6(W	50.82	50.82	50.82	57.05	57.05	27.22	618.67
156	1.2(DL+COLL+CL-3)+1.6(W	50.82	50.82	50.82	57.05	57.05	27.22	618.67
157	1.2(DL+COLL+CL-4)+1.6(W	50.82	50.82	50.82	57.05	57.05	27.22	618.67
158	1.2(DL+COLL+CL-1)+1.6(W	50.61	50.61	50.61	55.89	55.89	26.67	606.19
159	1.2(DL+COLL+CL-2)+1.6(W	50.61	50.61	50.61	55.89	55.89	26.67	606.19
160	1.2(DL+COLL+CL-3)+1.6(W	50.61	50.61	50.61	55.89	55.89	26.67	606.19
161	1.2(DL+COLL+CL-4)+1.6(W	50.61	50.61	50.61	55.89	55.89	26.67	606.19
162	1.2(DL+COLL+CL-1)+1.6(W	50.89	50.89	50.89	57.43	57.43	27.40	622.49
163	1.2(DL+COLL+CL-2)+1.6(W	50.89	50.89	50.89	57.43	57.43	27.40	622.49
164	1.2(DL+COLL+CL-3)+1.6(W	50.89	50.89	50.89	57.43	57.43	27.40	622.49
165	1.2(DL+COLL+CL-4)+1.6(W	50.89	50.89	50.89	57.43	57.43	27.40	622.49
166	1.2(DL+COLL+CL-1)+1.6(W	51.10	49.15	52.09	76.20	30.21	35.56	576.95
167	1.2(DL+COLL+CL-2)+1.6(W	51.10	49.15	52.09	76.20	30.21	35.56	576.95
168	1.2(DL+COLL+CL-3)+1.6(W	51.10	49.15	52.09	76.20	30.21	35.56	576.95
169	1.2(DL+COLL+CL-4)+1.6(W	51.10	49.15	52.09	76.20	30.21	35.56	576.95
170	1.2(DL+COLL+CL-1)+1.6(W	51.64	51.02	51.96	67.30	52.43	31.85	649.21
171	1.2(DL+COLL+CL-2)+1.6(W	51.64	51.02	51.96	67.30	52.43	31.85	649.21
172	1.2(DL+COLL+CL-3)+1.6(W	51.64	51.02	51.96	67.30	52.43	31.85	649.21
173	1.2(DL+COLL+CL-4)+1.6(W	51.64	51.02	51.96	67.30	52.43	31.85	649.21
174	1.2(DL+COLL+CL-1)+1.6(W	48.56	45.37	50.19	73.56	-2.09	33.79	387.40

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175	1.2(DL+COLL+CL-2)+1.6(W	48.56	45.37	50.19	73.56	-2.09	33.79	387.40
176	1.2(DL+COLL+CL-3)+1.6(W	48.56	45.37	50.19	73.56	-2.09	33.79	387.40
177	1.2(DL+COLL+CL-4)+1.6(W	48.56	45.37	50.19	73.56	-2.09	33.79	387.40
178	1.2(DL+COLL+CL-1)+1.6(W	49.12	47.12	50.15	65.90	18.34	30.62	456.68
179	1.2(DL+COLL+CL-2)+1.6(W	49.12	47.12	50.15	65.90	18.34	30.62	456.68
180	1.2(DL+COLL+CL-3)+1.6(W	49.12	47.12	50.15	65.90	18.34	30.62	456.68
181	1.2(DL+COLL+CL-4)+1.6(W	49.12	47.12	50.15	65.90	18.34	30.62	456.68
182	1.2(DL+COLL+CL-1)+1.0(S	55.71	54.44	56.36	95.61	65.52	45.10	873.50
183	1.2(DL+COLL+CL-2)+1.0(S	55.71	54.44	56.36	95.61	65.52	45.10	873.50
184	1.2(DL+COLL+CL-3)+1.0(S	55.71	54.44	56.36	95.61	65.52	45.10	873.50
185	1.2(DL+COLL+CL-4)+1.0(S	55.71	54.44	56.36	95.61	65.52	45.10	873.50
186	1.2(DL+COLL+CL-1)+1.0(S	52.77	51.50	53.42	79.36	49.27	37.34	697.51
187	1.2(DL+COLL+CL-2)+1.0(S	52.77	51.50	53.42	79.36	49.27	37.34	697.51
188	1.2(DL+COLL+CL-3)+1.0(S	52.77	51.50	53.42	79.36	49.27	37.34	697.51
189	1.2(DL+COLL+CL-4)+1.0(S	52.77	51.50	53.42	79.36	49.27	37.34	697.51
190	0.9(DL+COLL+CL-1)+1.6(W	44.29	44.29	44.29	20.95	20.95	10.00	227.45
191	0.9(DL+COLL+CL-2)+1.6(W	44.29	44.29	44.29	20.95	20.95	10.00	227.45
192	0.9(DL+COLL+CL-3)+1.6(W	44.29	44.29	44.29	20.95	20.95	10.00	227.45
193	0.9(DL+COLL+CL-4)+1.6(W	44.29	44.29	44.29	20.95	20.95	10.00	227.45
194	0.9(DL+COLL+CL-1)+1.6(W	42.54	42.54	42.54	11.28	11.28	5.38	121.87
195	0.9(DL+COLL+CL-2)+1.6(W	42.54	42.54	42.54	11.28	11.28	5.38	121.87
196	0.9(DL+COLL+CL-3)+1.6(W	42.54	42.54	42.54	11.28	11.28	5.38	121.87
197	0.9(DL+COLL+CL-4)+1.6(W	42.54	42.54	42.54	11.28	11.28	5.38	121.87
198	0.9(DL+COLL+CL-1)+1.6(W	42.33	42.33	42.33	10.12	10.12	4.83	109.39
199	0.9(DL+COLL+CL-2)+1.6(W	42.33	42.33	42.33	10.12	10.12	4.83	109.39
200	0.9(DL+COLL+CL-3)+1.6(W	42.33	42.33	42.33	10.12	10.12	4.83	109.39
201	0.9(DL+COLL+CL-4)+1.6(W	42.33	42.33	42.33	10.12	10.12	4.83	109.39
202	0.9(DL+COLL+CL-1)+1.6(W	42.60	42.60	42.60	11.61	11.61	5.54	125.68
203	0.9(DL+COLL+CL-2)+1.6(W	42.60	42.60	42.60	11.61	11.61	5.54	125.68
204	0.9(DL+COLL+CL-3)+1.6(W	42.60	42.60	42.60	11.61	11.61	5.54	125.68
205	0.9(DL+COLL+CL-4)+1.6(W	42.60	42.60	42.60	11.61	11.61	5.54	125.68

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206	0.9(DL+COLL+CL-1)+1.6(W	42.81	40.86	43.80	30.38	-15.62	13.70	80.15
207	0.9(DL+COLL+CL-2)+1.6(W	42.81	40.86	43.80	30.38	-15.62	13.70	80.15
208	0.9(DL+COLL+CL-3)+1.6(W	42.81	40.86	43.80	30.38	-15.62	13.70	80.15
209	0.9(DL+COLL+CL-4)+1.6(W	42.81	40.86	43.80	30.38	-15.62	13.70	80.15
210	0.9(DL+COLL+CL-1)+1.6(W	43.36	42.73	43.68	21.52	6.62	10.01	152.41
211	0.9(DL+COLL+CL-2)+1.6(W	43.36	42.73	43.68	21.52	6.62	10.01	152.41
212	0.9(DL+COLL+CL-3)+1.6(W	43.36	42.73	43.68	21.52	6.62	10.01	152.41
213	0.9(DL+COLL+CL-4)+1.6(W	43.36	42.73	43.68	21.52	6.62	10.01	152.41
214	0.9(DL+COLL+CL-1)+1.6(W	40.27	37.08	41.90	27.74	-47.91	11.92	-109.40
215	0.9(DL+COLL+CL-2)+1.6(W	40.27	37.08	41.90	27.74	-47.91	11.92	-109.40
216	0.9(DL+COLL+CL-3)+1.6(W	40.27	37.08	41.90	27.74	-47.91	11.92	-109.40
217	0.9(DL+COLL+CL-4)+1.6(W	40.27	37.08	41.90	27.74	-47.91	11.92	-109.40
218	0.9(DL+COLL+CL-1)+1.6(W	40.84	38.83	41.86	20.08	-27.43	8.76	-40.12
219	0.9(DL+COLL+CL-2)+1.6(W	40.84	38.83	41.86	20.08	-27.43	8.76	-40.12
220	0.9(DL+COLL+CL-3)+1.6(W	40.84	38.83	41.86	20.08	-27.43	8.76	-40.12
221	0.9(DL+COLL+CL-4)+1.6(W	40.84	38.83	41.86	20.08	-27.43	8.76	-40.12
222	0.9(DL+COLL+CL-1)+1.0(S	47.42	46.15	48.07	49.79	19.70	23.23	376.70
223	0.9(DL+COLL+CL-2)+1.0(S	47.42	46.15	48.07	49.79	19.70	23.23	376.70
224	0.9(DL+COLL+CL-3)+1.0(S	47.42	46.15	48.07	49.79	19.70	23.23	376.70
225	0.9(DL+COLL+CL-4)+1.0(S	47.42	46.15	48.07	49.79	19.70	23.23	376.70
226	0.9(DL+COLL+CL-1)+1.0(S	44.49	43.21	45.13	33.58	3.45	15.50	200.71
227	0.9(DL+COLL+CL-2)+1.0(S	44.49	43.21	45.13	33.58	3.45	15.50	200.71
228	0.9(DL+COLL+CL-3)+1.0(S	44.49	43.21	45.13	33.58	3.45	15.50	200.71
229	0.9(DL+COLL+CL-4)+1.0(S	44.49	43.21	45.13	33.58	3.45	15.50	200.71

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Sample calculation of BM & SF

Load case : 101 1.4(DL+COLL)

p_{1x} = 47.61-(47.61-47.61)*3.325/8 = 47.61 kN/m^2 Overhang = 3.325 m

p_{2x} = 47.61+(47.61-47.61)*3.325/8 = 47.61 kN/m^2 Effective depth = 0.6875 m

p_{3x} = 47.61-(47.61-47.61)*(3.325-0.6875)/8 = 47.61 kN/m^2

BM_{1x} = (47.61-40.5)*3.325*3.325/2+0.5*(47.61-47.61)*3.325*3.325*0.67 = 39.30 kN-m/m

BM_{2x} = (47.61-40.5)*3.325*3.325/2-0.5*(-47.61+47.61)*3.325*3.325*0.67 = 39.30 kN-m/m

SF_{3x} = (47.61-40.5)*(3.325-0.6875)+0.5*(47.61-47.61)*(3.325-0.6875) = 18.75 kN/m

$S.F_{(punching)}$ = (64-4.05)*(47.61-40.5) = 426.05 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	157.71	0.33	0.44	0.001	733.76	1237.50	25ϕT	396.67	300	200	2454.37
At top	-47.91	-0.10	0.14	0.000	221.74	618.75	12ϕT	182.78	300	150	753.98

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 75.25 kN/m

Shear stress , τ_v= (75.25*1000)/(1000*687.5)

= 0.11 N/mm^2 **O.K.**

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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 1710.13 kN

Strength of Concrete in shear Vc1 = 16370.90 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 10117.42 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 10913.93 kN **Table 22.6.5.2_ (c)**

Design Vc= 10117.42 kN

Allowable shear stress, (ϕ Vc)/bd = 1.37 N/mm^2

Design punching shear stress, τ = (1710.13*1000)/(8050*687.5)

= 0.31 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4(DL+COLL)	49.43	45.79	51.22	82.67	-4.06	38.13
102	1.2(DL+COLL+CL-1)+1.6(R	77.05	60.57	85.13	352.29	-39.31	162.08
103	1.2(DL+COLL+CL-2)+1.6(R	77.05	60.57	85.13	352.29	-39.31	162.08
104	1.2(DL+COLL+CL-3)+1.6(R	77.05	60.57	85.13	352.29	-39.31	162.08
105	1.2(DL+COLL+CL-4)+1.6(R	77.05	60.57	85.13	352.29	-39.31	162.08
106	1.4(DL+COLL+CL-1)	49.43	45.79	51.22	82.67	-4.06	38.13
107	1.4(DL+COLL+CL-2)	49.43	45.79	51.22	82.67	-4.06	38.13
108	1.4(DL+COLL+CL-3)	49.43	45.79	51.22	82.67	-4.06	38.13

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109	1.4(DL+COLL+CL-4)	49.43	45.79	51.22	82.67	-4.06	38.13
110	1.2(DL+COLL+CL-1+TL1)+	77.46	59.73	86.16	366.05	-55.44	168.13
111	1.2(DL+COLL+CL-2+TL1)+	77.46	59.73	86.16	366.05	-55.44	168.13
112	1.2(DL+COLL+CL-3+TL1)+	77.46	59.73	86.16	366.05	-55.44	168.13
113	1.2(DL+COLL+CL-4+TL1)+	77.46	59.73	86.16	366.05	-55.44	168.13
114	1.2(DL+COLL+CL-1+TL2)+	76.63	61.42	84.10	338.54	-23.18	156.04
115	1.2(DL+COLL+CL-2+TL2)+	76.63	61.42	84.10	338.54	-23.18	156.04
116	1.2(DL+COLL+CL-3+TL2)+	76.63	61.42	84.10	338.54	-23.18	156.04
117	1.2(DL+COLL+CL-4+TL2)+	76.63	61.42	84.10	338.54	-23.18	156.04
118	1.2(DL+COLL+CL-1)+0.8(W	47.21	45.15	48.22	55.91	6.89	25.98
119	1.2(DL+COLL+CL-2)+0.8(W	47.21	45.15	48.22	55.91	6.89	25.98
120	1.2(DL+COLL+CL-3)+0.8(W	47.21	45.15	48.22	55.91	6.89	25.98
121	1.2(DL+COLL+CL-4)+0.8(W	47.21	45.15	48.22	55.91	6.89	25.98
122	1.2(DL+COLL+CL-1)+0.8(W	46.73	43.87	48.14	60.59	-7.52	27.86
123	1.2(DL+COLL+CL-2)+0.8(W	46.73	43.87	48.14	60.59	-7.52	27.86
124	1.2(DL+COLL+CL-3)+0.8(W	46.73	43.87	48.14	60.59	-7.52	27.86
125	1.2(DL+COLL+CL-4)+0.8(W	46.73	43.87	48.14	60.59	-7.52	27.86
126	1.2(DL+COLL+CL-1)+0.8(W	45.87	44.52	46.54	42.10	9.81	19.65
127	1.2(DL+COLL+CL-2)+0.8(W	45.87	44.52	46.54	42.10	9.81	19.65
128	1.2(DL+COLL+CL-3)+0.8(W	45.87	44.52	46.54	42.10	9.81	19.65
129	1.2(DL+COLL+CL-4)+0.8(W	45.87	44.52	46.54	42.10	9.81	19.65
130	1.2(DL+COLL+CL-1)+0.8(W	46.53	44.13	47.72	55.34	-1.94	25.54
131	1.2(DL+COLL+CL-2)+0.8(W	46.53	44.13	47.72	55.34	-1.94	25.54
132	1.2(DL+COLL+CL-3)+0.8(W	46.53	44.13	47.72	55.34	-1.94	25.54
133	1.2(DL+COLL+CL-4)+0.8(W	46.53	44.13	47.72	55.34	-1.94	25.54
134	1.2(DL+COLL+CL-1)+0.8(W	46.52	43.39	48.05	61.79	-12.54	28.32
135	1.2(DL+COLL+CL-2)+0.8(W	46.52	43.39	48.05	61.79	-12.54	28.32
136	1.2(DL+COLL+CL-3)+0.8(W	46.52	43.39	48.05	61.79	-12.54	28.32
137	1.2(DL+COLL+CL-4)+0.8(W	46.52	43.39	48.05	61.79	-12.54	28.32
138	1.2(DL+COLL+CL-1)+0.8(W	47.12	43.99	48.65	65.11	-9.22	29.92
139	1.2(DL+COLL+CL-2)+0.8(W	47.12	43.99	48.65	65.11	-9.22	29.92
140	1.2(DL+COLL+CL-3)+0.8(W	47.12	43.99	48.65	65.11	-9.22	29.92
141	1.2(DL+COLL+CL-4)+0.8(W	47.12	43.99	48.65	65.11	-9.22	29.92

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142	1.2(DL+COLL+CL-1)+0.8(W	44.94	41.81	46.47	53.06	-21.27	24.11
143	1.2(DL+COLL+CL-2)+0.8(W	44.94	41.81	46.47	53.06	-21.27	24.11
144	1.2(DL+COLL+CL-3)+0.8(W	44.94	41.81	46.47	53.06	-21.27	24.11
145	1.2(DL+COLL+CL-4)+0.8(W	44.94	41.81	46.47	53.06	-21.27	24.11
146	1.2(DL+COLL+CL-1)+0.8(W	45.52	42.39	47.05	56.26	-18.07	25.66
147	1.2(DL+COLL+CL-2)+0.8(W	45.52	42.39	47.05	56.26	-18.07	25.66
148	1.2(DL+COLL+CL-3)+0.8(W	45.52	42.39	47.05	56.26	-18.07	25.66
149	1.2(DL+COLL+CL-4)+0.8(W	45.52	42.39	47.05	56.26	-18.07	25.66
150	1.2(DL+COLL+CL-1)+1.6(W	55.16	50.00	57.70	128.19	5.36	59.36
151	1.2(DL+COLL+CL-2)+1.6(W	55.16	50.00	57.70	128.19	5.36	59.36
152	1.2(DL+COLL+CL-3)+1.6(W	55.16	50.00	57.70	128.19	5.36	59.36
153	1.2(DL+COLL+CL-4)+1.6(W	55.16	50.00	57.70	128.19	5.36	59.36
154	1.2(DL+COLL+CL-1)+1.6(W	54.21	47.43	57.53	137.60	-23.50	63.15
155	1.2(DL+COLL+CL-2)+1.6(W	54.21	47.43	57.53	137.60	-23.50	63.15
156	1.2(DL+COLL+CL-3)+1.6(W	54.21	47.43	57.53	137.60	-23.50	63.15
157	1.2(DL+COLL+CL-4)+1.6(W	54.21	47.43	57.53	137.60	-23.50	63.15
158	1.2(DL+COLL+CL-1)+1.6(W	52.49	48.73	54.33	100.57	11.20	46.71
159	1.2(DL+COLL+CL-2)+1.6(W	52.49	48.73	54.33	100.57	11.20	46.71
160	1.2(DL+COLL+CL-3)+1.6(W	52.49	48.73	54.33	100.57	11.20	46.71
161	1.2(DL+COLL+CL-4)+1.6(W	52.49	48.73	54.33	100.57	11.20	46.71
162	1.2(DL+COLL+CL-1)+1.6(W	53.82	47.95	56.69	127.11	-12.29	58.51
163	1.2(DL+COLL+CL-2)+1.6(W	53.82	47.95	56.69	127.11	-12.29	58.51
164	1.2(DL+COLL+CL-3)+1.6(W	53.82	47.95	56.69	127.11	-12.29	58.51
165	1.2(DL+COLL+CL-4)+1.6(W	53.82	47.95	56.69	127.11	-12.29	58.51
166	1.2(DL+COLL+CL-1)+1.6(W	53.78	46.48	57.36	139.96	-33.50	64.05
167	1.2(DL+COLL+CL-2)+1.6(W	53.78	46.48	57.36	139.96	-33.50	64.05
168	1.2(DL+COLL+CL-3)+1.6(W	53.78	46.48	57.36	139.96	-33.50	64.05
169	1.2(DL+COLL+CL-4)+1.6(W	53.78	46.48	57.36	139.96	-33.50	64.05
170	1.2(DL+COLL+CL-1)+1.6(W	54.98	47.68	58.56	146.59	-26.86	67.24
171	1.2(DL+COLL+CL-2)+1.6(W	54.98	47.68	58.56	146.59	-26.86	67.24
172	1.2(DL+COLL+CL-3)+1.6(W	54.98	47.68	58.56	146.59	-26.86	67.24
173	1.2(DL+COLL+CL-4)+1.6(W	54.98	47.68	58.56	146.59	-26.86	67.24
174	1.2(DL+COLL+CL-1)+1.6(W	50.61	43.32	54.20	122.49	-51.01	55.63

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175	1.2(DL+COLL+CL-2)+1.6(W	50.61	43.32	54.20	122.49	-51.01	55.63
176	1.2(DL+COLL+CL-3)+1.6(W	50.61	43.32	54.20	122.49	-51.01	55.63
177	1.2(DL+COLL+CL-4)+1.6(W	50.61	43.32	54.20	122.49	-51.01	55.63
178	1.2(DL+COLL+CL-1)+1.6(W	51.77	44.47	55.35	128.85	-44.61	58.69
179	1.2(DL+COLL+CL-2)+1.6(W	51.77	44.47	55.35	128.85	-44.61	58.69
180	1.2(DL+COLL+CL-3)+1.6(W	51.77	44.47	55.35	128.85	-44.61	58.69
181	1.2(DL+COLL+CL-4)+1.6(W	51.77	44.47	55.35	128.85	-44.61	58.69
182	1.2(DL+COLL+CL-1)+1.0(S	59.36	50.79	63.56	182.40	-21.26	83.90
183	1.2(DL+COLL+CL-2)+1.0(S	59.36	50.79	63.56	182.40	-21.26	83.90
184	1.2(DL+COLL+CL-3)+1.0(S	59.36	50.79	63.56	182.40	-21.26	83.90
185	1.2(DL+COLL+CL-4)+1.0(S	59.36	50.79	63.56	182.40	-21.26	83.90
186	1.2(DL+COLL+CL-1)+1.0(S	55.15	49.13	58.11	135.99	-7.30	62.72
187	1.2(DL+COLL+CL-2)+1.0(S	55.15	49.13	58.11	135.99	-7.30	62.72
188	1.2(DL+COLL+CL-3)+1.0(S	55.15	49.13	58.11	135.99	-7.30	62.72
189	1.2(DL+COLL+CL-4)+1.0(S	55.15	49.13	58.11	135.99	-7.30	62.72
190	0.9(DL+COLL+CL-1)+1.6(W	44.40	44.18	44.50	23.52	18.38	11.23
191	0.9(DL+COLL+CL-2)+1.6(W	44.40	44.18	44.50	23.52	18.38	11.23
192	0.9(DL+COLL+CL-3)+1.6(W	44.40	44.18	44.50	23.52	18.38	11.23
193	0.9(DL+COLL+CL-4)+1.6(W	44.40	44.18	44.50	23.52	18.38	11.23
194	0.9(DL+COLL+CL-1)+1.6(W	43.45	41.62	44.34	32.93	-10.43	15.02
195	0.9(DL+COLL+CL-2)+1.6(W	43.45	41.62	44.34	32.93	-10.43	15.02
196	0.9(DL+COLL+CL-3)+1.6(W	43.45	41.62	44.34	32.93	-10.43	15.02
197	0.9(DL+COLL+CL-4)+1.6(W	43.45	41.62	44.34	32.93	-10.43	15.02
198	0.9(DL+COLL+CL-1)+1.6(W	42.92	41.73	43.51	24.27	-4.09	11.14
199	0.9(DL+COLL+CL-2)+1.6(W	42.92	41.73	43.51	24.27	-4.09	11.14
200	0.9(DL+COLL+CL-3)+1.6(W	42.92	41.73	43.51	24.27	-4.09	11.14
201	0.9(DL+COLL+CL-4)+1.6(W	42.92	41.73	43.51	24.27	-4.09	11.14
202	0.9(DL+COLL+CL-1)+1.6(W	43.05	42.14	43.50	22.43	0.73	10.38
203	0.9(DL+COLL+CL-2)+1.6(W	43.05	42.14	43.50	22.43	0.73	10.38
204	0.9(DL+COLL+CL-3)+1.6(W	43.05	42.14	43.50	22.43	0.73	10.38
205	0.9(DL+COLL+CL-4)+1.6(W	43.05	42.14	43.50	22.43	0.73	10.38
206	0.9(DL+COLL+CL-1)+1.6(W	43.01	40.67	44.16	35.29	-20.47	15.91
207	0.9(DL+COLL+CL-2)+1.6(W	43.01	40.67	44.16	35.29	-20.47	15.91
208	0.9(DL+COLL+CL-3)+1.6(W	43.01	40.67	44.16	35.29	-20.47	15.91

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209	0.9(DL+COLL+CL-4)+1.6(W	43.01	40.67	44.16	35.29	-20.47	15.91
210	0.9(DL+COLL+CL-1)+1.6(W	44.22	41.87	45.37	41.93	-13.79	19.11
211	0.9(DL+COLL+CL-2)+1.6(W	44.22	41.87	45.37	41.93	-13.79	19.11
212	0.9(DL+COLL+CL-3)+1.6(W	44.22	41.87	45.37	41.93	-13.79	19.11
213	0.9(DL+COLL+CL-4)+1.6(W	44.22	41.87	45.37	41.93	-13.79	19.11
214	0.9(DL+COLL+CL-1)+1.6(W	39.85	37.51	41.00	17.82	-37.94	7.50
215	0.9(DL+COLL+CL-2)+1.6(W	39.85	37.51	41.00	17.82	-37.94	7.50
216	0.9(DL+COLL+CL-3)+1.6(W	39.85	37.51	41.00	17.82	-37.94	7.50
217	0.9(DL+COLL+CL-4)+1.6(W	39.85	37.51	41.00	17.82	-37.94	7.50
218	0.9(DL+COLL+CL-1)+1.6(W	41.01	38.66	42.16	24.18	-31.54	10.57
219	0.9(DL+COLL+CL-2)+1.6(W	41.01	38.66	42.16	24.18	-31.54	10.57
220	0.9(DL+COLL+CL-3)+1.6(W	41.01	38.66	42.16	24.18	-31.54	10.57
221	0.9(DL+COLL+CL-4)+1.6(W	41.01	38.66	42.16	24.18	-31.54	10.57
222	0.9(DL+COLL+CL-1)+1.0(S	48.59	44.98	50.37	77.73	-8.24	35.77
223	0.9(DL+COLL+CL-2)+1.0(S	48.59	44.98	50.37	77.73	-8.24	35.77
224	0.9(DL+COLL+CL-3)+1.0(S	48.59	44.98	50.37	77.73	-8.24	35.77
225	0.9(DL+COLL+CL-4)+1.0(S	48.59	44.98	50.37	77.73	-8.24	35.77
226	0.9(DL+COLL+CL-1)+1.0(S	44.39	43.31	44.91	31.27	5.76	14.57
227	0.9(DL+COLL+CL-2)+1.0(S	44.39	43.31	44.91	31.27	5.76	14.57
228	0.9(DL+COLL+CL-3)+1.0(S	44.39	43.31	44.91	31.27	5.76	14.57
229	0.9(DL+COLL+CL-4)+1.0(S	44.39	43.31	44.91	31.27	5.76	14.57

Sample calculation of BM & SF

Load case : 101

p_{1z} = 58.42-(58.42-36.8)*3.325/8= 49.43 kN/m^2

p_{2z} = 36.8+(58.42-36.8)*3.325/8= 45.79 kN/m^2

p_{3z} = 58.42-(58.42-36.8)*(3.325-0.6625)/8= 51.22 kN/m^2

Overhang = 3.325 m

Effective depth = 0.6625 m

BM_{1z} = (49.43-40.5)*3.325*3.325/2+0.5*(58.42-49.43)*3.325*3.325*0.67 = 82.67 kN-m/m

BM_{2z} = (45.79-40.5)*3.325*3.325/2-0.5*(-36.8+45.79)*3.325*3.325*0.67 = -4.06 kN-m/m

SF_{3z} = (51.22-40.5)*(3.325-0.6625)+0.5*(58.42-51.22)*(3.325-0.6625) = 38.13 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	366.05	0.834	1.112	0.0027	1855.58	1192.50	25ϕT	264.54	300	200	2454.4
At top	-55.44	-0.126	-0.168	-0.0004	-2.65	596.25	12ϕT	189.68	250	150	754.0

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 168.13 kN/m

Shear stress , τ_v= (168.13*1000)/(1000*662.5)

= 0.25 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	25ϕT	**200**	c/c
	At Top	12ϕT	**150**	c/c
Along Z Direction :				
	At Bottom	25ϕT	**200**	c/c
	At Top	12ϕT	**150**	c/c

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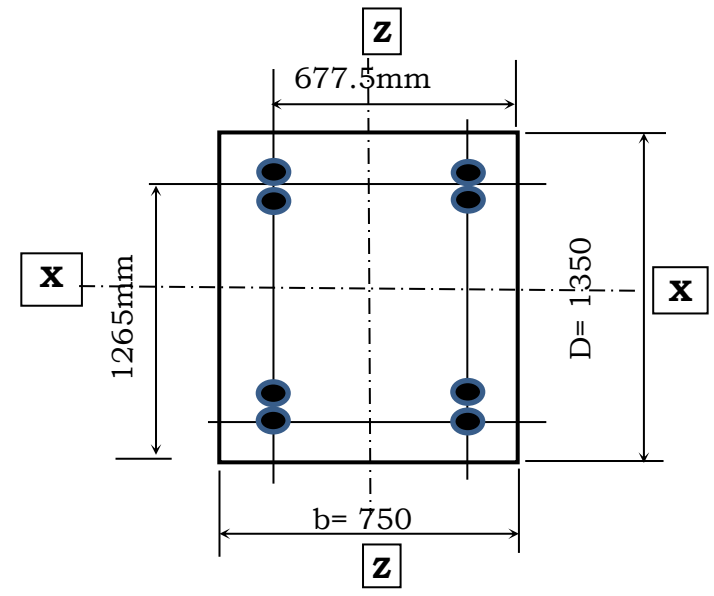
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5.8 DESIGN OF COLUMN MKD C1

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	85mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	1.00%
Max. Steel Ratio	4.00%
Dimension parallel to X-X axis	750mm
Dimension parallel to Z-Z axis	1350mm

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	2100mm	γ_x =	0.89
Unsupproted length of column about Z (l_{uz}) =	2100mm	γ_z =	0.83

Where, Effective length factor K= 2

Slenderness ratio, $K l_{ux} / r_x$ = 10.37 < 22 SECTION 6.2.5a Radius of gyration r_x= 405 mm

$K l_{uz} / r_z$ = 18.67 < 22 SECTION 6.2.5a Radius of gyration r_z= 225 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	30.0672		21.8664		85.5336
2	COLLATERAL LOAD (COLL)	-0.0048		-0.0048		0.0048
3	ROOF LIVE LOAD (RLL)	68.052		55.8516		201.347
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	3.5328		10.9272		64.5576
5	CRANE LOAD-1 (CL-1)	0		0		0
6	CRANE LOAD-2 (CL-2)	0		0		0
7	CRANE LOAD-3 (CL-3)	0		0		0
8	CRANE LOAD-4 (CL-4)	0		0		0
9	WIND LEFT-1 (WLL-1)	-139.5168		-127.1076		-420.995
10	WIND LEFT-2 (WLL-2)	-77.3916		-106.3572		-291.737
11	WIND RIGHT-1 (WLR-1)	-105.1044		-52.0608		-263.651
12	WIND RIGHT-2 (WLR-2)	-43.1652		-32.3664		-141.863
13	WIND LONG-1 (WLLG-1)	-179.3952	-78.9444		-363.3672	
14	WIND LONG-2 (WLLG-2)	-117.4728	-59.1216		-240.3684	
15	WIND LONG-3 (WLLG-3)	-150.9552	-71.3664		-351.4164	
16	WIND LONG-4 (WLLG-4)	-89.0328	-51.5436		-228.4164	
17	SEISMIC LOAD SL'+'	-0.1668	-1.1688	-1.1688	-7.488	-7.488
18	SEISMIC LOAD SL'-'	0.1668	1.1688	1.1688	7.488	7.488
19	TEMPERATURE LOAD (TL1)	1.662		9.8352		69.972
20	TEMPERATURE LOAD (TL2)	-1.662		-9.8352		-69.972
21	BRACING REACTION (BR-WLLG1)	51.7	-41		0	
22	BRACING REACTION (BR-WLLG2)	51.7	-41		0	
23	BRACING REACTION (BR-WLLG3)	-51.7	41		0	
24	BRACING REACTION (BR-WLLG4)	-51.7	41		0	
25	BRACING REACTION (BR-SL'+')	-11	-10	-10	0	0
26	BRACING REACTION (BR-SL'-')	11	10	10	0	0

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4(DL+COLL)	4.21	0.0	18.4	6.5	0.0	28.3	0.00
102	1.2(DL+COLL+CL-1)+1.6(RLL+P	15.1	0.0	80.8	23.2	0.0	124.2	0.01
103	1.2(DL+COLL+CL-2)+1.6(RLL+P	15.1	0.0	80.8	23.2	0.0	124.2	0.01
104	1.2(DL+COLL+CL-3)+1.6(RLL+P	15.1	0.0	80.8	23.2	0.0	124.2	0.01
105	1.2(DL+COLL+CL-4)+1.6(RLL+P	15.1	0.0	80.8	23.2	0.0	124.2	0.01
106	1.4(DL+COLL+CL-1)	4.2	0.0	18.4	6.5	0.0	28.3	0.00
107	1.4(DL+COLL+CL-2)	4.2	0.0	18.4	6.5	0.0	28.3	0.00
108	1.4(DL+COLL+CL-3)	4.2	0.0	18.4	6.5	0.0	28.3	0.00
109	1.4(DL+COLL+CL-4)	4.2	0.0	18.4	6.5	0.0	28.3	0.00
110	1.2(DL+COLL+CL-1+TL1)+1.6(R	15.3	0.0	91.6	23.5	0.0	141.0	0.01
111	1.2(DL+COLL+CL-2+TL1)+1.6(R	15.3	0.0	91.6	23.5	0.0	141.0	0.01
112	1.2(DL+COLL+CL-3+TL1)+1.6(R	15.3	0.0	91.6	23.5	0.0	141.0	0.01
113	1.2(DL+COLL+CL-4+TL1)+1.6(R	15.3	0.0	91.6	23.5	0.0	141.0	0.01
114	1.2(DL+COLL+CL-1+TL2)+1.6(R	14.9	0.0	69.9	22.9	0.0	107.5	0.01
115	1.2(DL+COLL+CL-2+TL2)+1.6(R	14.9	0.0	69.9	22.9	0.0	107.5	0.01
116	1.2(DL+COLL+CL-3+TL2)+1.6(R	14.9	0.0	69.9	22.9	0.0	107.5	0.01
117	1.2(DL+COLL+CL-4+TL2)+1.6(R	14.9	0.0	69.9	22.9	0.0	107.5	0.01
118	1.2(DL+COLL+CL-1)+0.8(WLL-1	-7.6	0.0	39.3	-11.6	0.0	60.4	0.00
119	1.2(DL+COLL+CL-2)+0.8(WLL-1	-7.6	0.0	39.3	-11.6	0.0	60.4	0.00
120	1.2(DL+COLL+CL-3)+0.8(WLL-1	-7.6	0.0	39.3	-11.6	0.0	60.4	0.00
121	1.2(DL+COLL+CL-4)+0.8(WLL-1	-7.6	0.0	39.3	-11.6	0.0	60.4	0.00
122	1.2(DL+COLL+CL-1)+0.8(WLL-2	-2.6	0.0	25.4	-4.0	0.0	39.1	0.00
123	1.2(DL+COLL+CL-2)+0.8(WLL-2	-2.6	0.0	25.4	-4.0	0.0	39.1	0.00
124	1.2(DL+COLL+CL-3)+0.8(WLL-2	-2.6	0.0	25.4	-4.0	0.0	39.1	0.00
125	1.2(DL+COLL+CL-4)+0.8(WLL-2	-2.6	0.0	25.4	-4.0	0.0	39.1	0.00
126	1.2(DL+COLL+CL-1)+0.8(WLR-1	-4.8	0.0	14.1	-7.4	0.0	21.6	0.00
127	1.2(DL+COLL+CL-2)+0.8(WLR-1	-4.8	0.0	14.1	-7.4	0.0	21.6	0.00
128	1.2(DL+COLL+CL-3)+0.8(WLR-1	-4.8	0.0	14.1	-7.4	0.0	21.6	0.00
129	1.2(DL+COLL+CL-4)+0.8(WLR-1	-4.8	0.0	14.1	-7.4	0.0	21.6	0.00
130	1.2(DL+COLL+CL-1)+0.8(WLR-2	0.2	0.0	1.0	0.2	0.0	1.6	0.00
131	1.2(DL+COLL+CL-2)+0.8(WLR-2	0.2	0.0	1.0	0.2	0.0	1.6	0.00
132	1.2(DL+COLL+CL-3)+0.8(WLR-2	0.2	0.0	1.0	0.2	0.0	1.6	0.00
133	1.2(DL+COLL+CL-4)+0.8(WLR-2	0.2	0.0	1.0	0.2	0.0	1.6	0.00
134	1.2(DL+COLL+CL-1)+0.8(WLLG	-6.6	49.2	15.8	-10.2	75.7	24.3	0.00
135	1.2(DL+COLL+CL-2)+0.8(WLLG	-6.6	49.2	15.8	-10.2	75.7	24.3	0.00
136	1.2(DL+COLL+CL-3)+0.8(WLLG	-6.6	49.2	15.8	-10.2	75.7	24.3	0.00
137	1.2(DL+COLL+CL-4)+0.8(WLLG	-6.6	49.2	15.8	-10.2	75.7	24.3	0.00
138	1.2(DL+COLL+CL-1)+0.8(WLLG	-1.7	36.0	15.8	-2.5	55.5	24.3	0.00
139	1.2(DL+COLL+CL-2)+0.8(WLLG	-1.7	36.0	15.8	-2.5	55.5	24.3	0.00
140	1.2(DL+COLL+CL-3)+0.8(WLLG	-1.7	36.0	15.8	-2.5	55.5	24.3	0.00
141	1.2(DL+COLL+CL-4)+0.8(WLLG	-1.7	36.0	15.8	-2.5	55.5	24.3	0.00
142	1.2(DL+COLL+CL-1)+0.8(WLLG	-12.6	33.2	15.8	-19.4	51.1	24.3	-0.01
143	1.2(DL+COLL+CL-2)+0.8(WLLG	-12.6	33.2	15.8	-19.4	51.1	24.3	-0.01
144	1.2(DL+COLL+CL-3)+0.8(WLLG	-12.6	33.2	15.8	-19.4	51.1	24.3	-0.01
145	1.2(DL+COLL+CL-4)+0.8(WLLG	-12.6	33.2	15.8	-19.4	51.1	24.3	-0.01
146	1.2(DL+COLL+CL-1)+0.8(WLLG	-7.7	20.0	15.8	-11.8	30.8	24.3	0.00
147	1.2(DL+COLL+CL-2)+0.8(WLLG	-7.7	20.0	15.8	-11.8	30.8	24.3	0.00
148	1.2(DL+COLL+CL-3)+0.8(WLLG	-7.7	20.0	15.8	-11.8	30.8	24.3	0.00
149	1.2(DL+COLL+CL-4)+0.8(WLLG	-7.7	20.0	15.8	-11.8	30.8	24.3	0.00
150	1.2(DL+COLL+CL-1)+1.6(WLL-1	-15.1	0.0	74.0	-23.3	0.0	113.8	-0.01
151	1.2(DL+COLL+CL-2)+1.6(WLL-1	-15.1	0.0	74.0	-23.3	0.0	113.8	-0.01
152	1.2(DL+COLL+CL-3)+1.6(WLL-1	-15.1	0.0	74.0	-23.3	0.0	113.8	-0.01

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153	1.2(DL+COLL+CL-4)+1.6(WLL-1	-15.1	0.0	74.0	-23.3	0.0	113.8	-0.01
154	1.2(DL+COLL+CL-1)+1.6(WLL-2	-5.2	0.0	46.3	-8.0	0.0	71.3	0.00
155	1.2(DL+COLL+CL-2)+1.6(WLL-2	-5.2	0.0	46.3	-8.0	0.0	71.3	0.00
156	1.2(DL+COLL+CL-3)+1.6(WLL-2	-5.2	0.0	46.3	-8.0	0.0	71.3	0.00
157	1.2(DL+COLL+CL-4)+1.6(WLL-2	-5.2	0.0	46.3	-8.0	0.0	71.3	0.00
158	1.2(DL+COLL+CL-1)+1.6(WLR-1	-9.6	0.0	23.6	-14.8	0.0	36.3	0.00
159	1.2(DL+COLL+CL-2)+1.6(WLR-1	-9.6	0.0	23.6	-14.8	0.0	36.3	0.00
160	1.2(DL+COLL+CL-3)+1.6(WLR-1	-9.6	0.0	23.6	-14.8	0.0	36.3	0.00
161	1.2(DL+COLL+CL-4)+1.6(WLR-1	-9.6	0.0	23.6	-14.8	0.0	36.3	0.00
162	1.2(DL+COLL+CL-1)+1.6(WLR-2	0.3	0.0	2.5	0.4	0.0	3.9	0.00
163	1.2(DL+COLL+CL-2)+1.6(WLR-2	0.3	0.0	2.5	0.4	0.0	3.9	0.00
164	1.2(DL+COLL+CL-3)+1.6(WLR-2	0.3	0.0	2.5	0.4	0.0	3.9	0.00
165	1.2(DL+COLL+CL-4)+1.6(WLR-2	0.3	0.0	2.5	0.4	0.0	3.9	0.00
166	1.2(DL+COLL+CL-1)+1.6(WLLG-	-13.2	98.4	36.1	-20.4	151.4	55.5	-0.01
167	1.2(DL+COLL+CL-2)+1.6(WLLG-	-13.2	98.4	36.1	-20.4	151.4	55.5	-0.01
168	1.2(DL+COLL+CL-3)+1.6(WLLG-	-13.2	98.4	36.1	-20.4	151.4	55.5	-0.01
169	1.2(DL+COLL+CL-4)+1.6(WLLG-	-13.2	98.4	36.1	-20.4	151.4	55.5	-0.01
170	1.2(DL+COLL+CL-1)+1.6(WLLG-	-3.3	72.1	36.1	-5.1	110.9	55.5	0.00
171	1.2(DL+COLL+CL-2)+1.6(WLLG-	-3.3	72.1	36.1	-5.1	110.9	55.5	0.00
172	1.2(DL+COLL+CL-3)+1.6(WLLG-	-3.3	72.1	36.1	-5.1	110.9	55.5	0.00
173	1.2(DL+COLL+CL-4)+1.6(WLLG-	-3.3	72.1	36.1	-5.1	110.9	55.5	0.00
174	1.2(DL+COLL+CL-1)+1.6(WLLG-	-25.2	66.4	36.1	-38.8	102.2	55.5	-0.01
175	1.2(DL+COLL+CL-2)+1.6(WLLG-	-25.2	66.4	36.1	-38.8	102.2	55.5	-0.01
176	1.2(DL+COLL+CL-3)+1.6(WLLG-	-25.2	66.4	36.1	-38.8	102.2	55.5	-0.01
177	1.2(DL+COLL+CL-4)+1.6(WLLG-	-25.2	66.4	36.1	-38.8	102.2	55.5	-0.01
178	1.2(DL+COLL+CL-1)+1.6(WLLG-	-15.3	40.1	36.1	-23.6	61.7	55.5	-0.01
179	1.2(DL+COLL+CL-2)+1.6(WLLG-	-15.3	40.1	36.1	-23.6	61.7	55.5	-0.01
180	1.2(DL+COLL+CL-3)+1.6(WLLG-	-15.3	40.1	36.1	-23.6	61.7	55.5	-0.01
181	1.2(DL+COLL+CL-4)+1.6(WLLG-	-15.3	40.1	36.1	-23.6	61.7	55.5	-0.01
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+I	6.1	3.1	33.0	9.3	4.8	50.7	0.00
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+I	6.1	3.1	33.0	9.3	4.8	50.7	0.00
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+I	6.1	3.1	33.0	9.3	4.8	50.7	0.00
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+I	6.1	3.1	33.0	9.3	4.8	50.7	0.00
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+E	8.3	3.1	39.2	12.8	4.8	60.3	0.00
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+E	8.3	3.1	39.2	12.8	4.8	60.3	0.00
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+E	8.3	3.1	39.2	12.8	4.8	60.3	0.00
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+E	8.3	3.1	39.2	12.8	4.8	60.3	0.00
190	0.9(DL+COLL+CL-1)+1.6(WLL-1	-19.6	0.0	98.2	-30.2	0.0	151.1	-0.01
191	0.9(DL+COLL+CL-2)+1.6(WLL-1	-19.6	0.0	98.2	-30.2	0.0	151.1	-0.01
192	0.9(DL+COLL+CL-3)+1.6(WLL-1	-19.6	0.0	98.2	-30.2	0.0	151.1	-0.01
193	0.9(DL+COLL+CL-4)+1.6(WLL-1	-19.6	0.0	98.2	-30.2	0.0	151.1	-0.01
194	0.9(DL+COLL+CL-1)+1.6(WLL-2	-9.7	0.0	70.6	-14.9	0.0	108.6	0.00
195	0.9(DL+COLL+CL-2)+1.6(WLL-2	-9.7	0.0	70.6	-14.9	0.0	108.6	0.00
196	0.9(DL+COLL+CL-3)+1.6(WLL-2	-9.7	0.0	70.6	-14.9	0.0	108.6	0.00
197	0.9(DL+COLL+CL-4)+1.6(WLL-2	-9.7	0.0	70.6	-14.9	0.0	108.6	0.00
198	0.9(DL+COLL+CL-1)+1.6(WLR-1	-14.1	0.0	47.8	-21.7	0.0	73.6	-0.01
199	0.9(DL+COLL+CL-2)+1.6(WLR-1	-14.1	0.0	47.8	-21.7	0.0	73.6	-0.01
200	0.9(DL+COLL+CL-3)+1.6(WLR-1	-14.1	0.0	47.8	-21.7	0.0	73.6	-0.01
201	0.9(DL+COLL+CL-4)+1.6(WLR-1	-14.1	0.0	47.8	-21.7	0.0	73.6	-0.01
202	0.9(DL+COLL+CL-1)+1.6(WLR-2	-4.2	0.0	21.7	-6.5	0.0	33.5	0.00
203	0.9(DL+COLL+CL-2)+1.6(WLR-2	-4.2	0.0	21.7	-6.5	0.0	33.5	0.00
204	0.9(DL+COLL+CL-3)+1.6(WLR-2	-4.2	0.0	21.7	-6.5	0.0	33.5	0.00
205	0.9(DL+COLL+CL-4)+1.6(WLR-2	-4.2	0.0	21.7	-6.5	0.0	33.5	0.00
206	0.9(DL+COLL+CL-1)+1.6(WLLG-	-17.7	98.4	11.8	-27.3	151.4	18.2	-0.01
207	0.9(DL+COLL+CL-2)+1.6(WLLG-	-17.7	98.4	11.8	-27.3	151.4	18.2	-0.01
208	0.9(DL+COLL+CL-3)+1.6(WLLG-	-17.7	98.4	11.8	-27.3	151.4	18.2	-0.01
209	0.9(DL+COLL+CL-4)+1.6(WLLG-	-17.7	98.4	11.8	-27.3	151.4	18.2	-0.01

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210	0.9(DL+COLL+CL-1)+1.6(WLLG	-7.8	72.1	11.8	-12.0	110.9	18.2	0.00
211	0.9(DL+COLL+CL-2)+1.6(WLLG	-7.8	72.1	11.8	-12.0	110.9	18.2	0.00
212	0.9(DL+COLL+CL-3)+1.6(WLLG	-7.8	72.1	11.8	-12.0	110.9	18.2	0.00
213	0.9(DL+COLL+CL-4)+1.6(WLLG	-7.8	72.1	11.8	-12.0	110.9	18.2	0.00
214	0.9(DL+COLL+CL-1)+1.6(WLLG	-29.7	66.4	11.8	-45.7	102.2	18.2	-0.01
215	0.9(DL+COLL+CL-2)+1.6(WLLG	-29.7	66.4	11.8	-45.7	102.2	18.2	-0.01
216	0.9(DL+COLL+CL-3)+1.6(WLLG	-29.7	66.4	11.8	-45.7	102.2	18.2	-0.01
217	0.9(DL+COLL+CL-4)+1.6(WLLG	-29.7	66.4	11.8	-45.7	102.2	18.2	-0.01
218	0.9(DL+COLL+CL-1)+1.6(WLLG	-19.8	40.1	11.8	-30.5	61.7	18.2	-0.01
219	0.9(DL+COLL+CL-2)+1.6(WLLG	-19.8	40.1	11.8	-30.5	61.7	18.2	-0.01
220	0.9(DL+COLL+CL-3)+1.6(WLLG	-19.8	40.1	11.8	-30.5	61.7	18.2	-0.01
221	0.9(DL+COLL+CL-4)+1.6(WLLG	-19.8	40.1	11.8	-30.5	61.7	18.2	-0.01
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+I	1.6	3.1	8.7	2.4	4.8	13.4	0.00
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+I	1.6	3.1	8.7	2.4	4.8	13.4	0.00
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+I	1.6	3.1	8.7	2.4	4.8	13.4	0.00
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+I	1.6	3.1	8.7	2.4	4.8	13.4	0.00
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+E	3.8	3.1	14.9	5.9	4.8	23.0	0.00
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+E	3.8	3.1	14.9	5.9	4.8	23.0	0.00
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+E	3.8	3.1	14.9	5.9	4.8	23.0	0.00
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+E	3.8	3.1	14.9	5.9	4.8	23.0	0.00

COLUMN DESIGN AND INTERACTION RATIO CHECK

Reinforcement: Provide,

4-Nos	25 mm Dia
16-Nos	25 mm Dia
16-Nos	25 mm Dia

1964.29 Sq mm (At Corner) Asc pvd. = 17678.57 Sq mm
7857.14 Sq mm (Along Z axis)
7857.14 Sq mm (Along X axis) Total Asc = 1.75 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 3702.0938 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.9 (COLUMNS 3.3.4)		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
101	0.085	0.085	225.9	406.6	0.55	1.16	0.05	O.K
102	0.085	0.085	225.9	406.6	0.55	1.16	0.25	O.K
103	0.085	0.085	225.9	406.6	0.55	1.16	0.25	O.K
104	0.085	0.085	225.9	406.6	0.55	1.16	0.25	O.K
105	0.085	0.085	225.9	406.6	0.55	1.16	0.25	O.K
106	0.085	0.085	225.9	406.6	0.55	1.16	0.05	O.K
107	0.085	0.085	225.9	406.6	0.55	1.16	0.05	O.K
108	0.085	0.085	225.9	406.6	0.55	1.16	0.05	O.K
109	0.085	0.085	225.9	406.6	0.55	1.16	0.05	O.K
110	0.085	0.085	225.9	406.6	0.55	1.16	0.29	O.K
111	0.085	0.085	225.9	406.6	0.55	1.16	0.29	O.K
112	0.085	0.085	225.9	406.6	0.55	1.16	0.29	O.K
113	0.085	0.085	225.9	406.6	0.55	1.16	0.29	O.K
114	0.085	0.085	225.9	406.6	0.55	1.16	0.21	O.K
115	0.085	0.085	225.9	406.6	0.55	1.16	0.21	O.K
116	0.085	0.085	225.9	406.6	0.55	1.16	0.21	O.K
117	0.085	0.085	225.9	406.6	0.55	1.16	0.21	O.K
118	0.085	0.085	225.9	406.6	0.55	1.16	0.11	O.K
119	0.085	0.085	225.9	406.6	0.55	1.16	0.11	O.K
120	0.085	0.085	225.9	406.6	0.55	1.16	0.11	O.K
121	0.085	0.085	225.9	406.6	0.55	1.16	0.11	O.K
122	0.085	0.085	225.9	406.6	0.55	1.16	0.07	O.K
123	0.085	0.085	225.9	406.6	0.55	1.16	0.07	O.K
124	0.085	0.085	225.9	406.6	0.55	1.16	0.07	O.K
125	0.085	0.085	225.9	406.6	0.55	1.16	0.07	O.K
126	0.085	0.085	225.9	406.6	0.55	1.16	0.03	O.K
127	0.085	0.085	225.9	406.6	0.55	1.16	0.03	O.K

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128	0.085	0.085	225.9	406.6	0.55	1.16	0.03	O.K
129	0.085	0.085	225.9	406.6	0.55	1.16	0.03	O.K
130	0.085	0.085	225.9	406.6	0.55	1.16	0.00	O.K
131	0.085	0.085	225.9	406.6	0.55	1.16	0.00	O.K
132	0.085	0.085	225.9	406.6	0.55	1.16	0.00	O.K
133	0.085	0.085	225.9	406.6	0.55	1.16	0.00	O.K
134	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
135	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
136	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
137	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
138	0.085	0.085	225.9	406.6	0.55	1.16	0.23	O.K
139	0.085	0.085	225.9	406.6	0.55	1.16	0.23	O.K
140	0.085	0.085	225.9	406.6	0.55	1.16	0.23	O.K
141	0.085	0.085	225.9	406.6	0.55	1.16	0.23	O.K
142	0.085	0.085	225.9	406.6	0.55	1.16	0.22	O.K
143	0.085	0.085	225.9	406.6	0.55	1.16	0.22	O.K
144	0.085	0.085	225.9	406.6	0.55	1.16	0.22	O.K
145	0.085	0.085	225.9	406.6	0.55	1.16	0.22	O.K
146	0.085	0.085	225.9	406.6	0.55	1.16	0.14	O.K
147	0.085	0.085	225.9	406.6	0.55	1.16	0.14	O.K
148	0.085	0.085	225.9	406.6	0.55	1.16	0.14	O.K
149	0.085	0.085	225.9	406.6	0.55	1.16	0.14	O.K
150	0.085	0.085	225.9	406.6	0.55	1.16	0.23	O.K
151	0.085	0.085	225.9	406.6	0.55	1.16	0.23	O.K
152	0.085	0.085	225.9	406.6	0.55	1.16	0.23	O.K
153	0.085	0.085	225.9	406.6	0.55	1.16	0.23	O.K
154	0.085	0.085	225.9	406.6	0.55	1.16	0.13	O.K
155	0.085	0.085	225.9	406.6	0.55	1.16	0.13	O.K
156	0.085	0.085	225.9	406.6	0.55	1.16	0.13	O.K
157	0.085	0.085	225.9	406.6	0.55	1.16	0.13	O.K
158	0.085	0.085	225.9	406.6	0.55	1.16	0.06	O.K
159	0.085	0.085	225.9	406.6	0.55	1.16	0.06	O.K
160	0.085	0.085	225.9	406.6	0.55	1.16	0.06	O.K
161	0.085	0.085	225.9	406.6	0.55	1.16	0.06	O.K
162	0.085	0.085	225.9	406.6	0.55	1.16	0.00	O.K
163	0.085	0.085	225.9	406.6	0.55	1.16	0.00	O.K
164	0.085	0.085	225.9	406.6	0.55	1.16	0.00	O.K
165	0.085	0.085	225.9	406.6	0.55	1.16	0.00	O.K
166	0.085	0.085	225.9	406.6	0.55	1.16	0.73	O.K
167	0.085	0.085	225.9	406.6	0.55	1.16	0.73	O.K
168	0.085	0.085	225.9	406.6	0.55	1.16	0.73	O.K
169	0.085	0.085	225.9	406.6	0.55	1.16	0.73	O.K
170	0.085	0.085	225.9	406.6	0.55	1.16	0.54	O.K
171	0.085	0.085	225.9	406.6	0.55	1.16	0.54	O.K
172	0.085	0.085	225.9	406.6	0.55	1.16	0.54	O.K
173	0.085	0.085	225.9	406.6	0.55	1.16	0.54	O.K
174	0.085	0.085	225.9	406.6	0.55	1.16	0.50	O.K
175	0.085	0.085	225.9	406.6	0.55	1.16	0.50	O.K
176	0.085	0.085	225.9	406.6	0.55	1.16	0.50	O.K
177	0.085	0.085	225.9	406.6	0.55	1.16	0.50	O.K
178	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
179	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
180	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
181	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
182	0.085	0.085	225.9	406.6	0.55	1.16	0.10	O.K
183	0.085	0.085	225.9	406.6	0.55	1.16	0.10	O.K
184	0.085	0.085	225.9	406.6	0.55	1.16	0.10	O.K
185	0.085	0.085	225.9	406.6	0.55	1.16	0.10	O.K
186	0.085	0.085	225.9	406.6	0.55	1.16	0.12	O.K
187	0.085	0.085	225.9	406.6	0.55	1.16	0.12	O.K
188	0.085	0.085	225.9	406.6	0.55	1.16	0.12	O.K
189	0.085	0.085	225.9	406.6	0.55	1.16	0.12	O.K
190	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K

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191	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
192	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
193	0.085	0.085	225.9	406.6	0.55	1.16	0.32	O.K
194	0.085	0.085	225.9	406.6	0.55	1.16	0.22	O.K
195	0.085	0.085	225.9	406.6	0.55	1.16	0.22	O.K
196	0.085	0.085	225.9	406.6	0.55	1.16	0.22	O.K
197	0.085	0.085	225.9	406.6	0.55	1.16	0.22	O.K
198	0.085	0.085	225.9	406.6	0.55	1.16	0.14	O.K
199	0.085	0.085	225.9	406.6	0.55	1.16	0.14	O.K
200	0.085	0.085	225.9	406.6	0.55	1.16	0.14	O.K
201	0.085	0.085	225.9	406.6	0.55	1.16	0.14	O.K
202	0.085	0.085	225.9	406.6	0.55	1.16	0.06	O.K
203	0.085	0.085	225.9	406.6	0.55	1.16	0.06	O.K
204	0.085	0.085	225.9	406.6	0.55	1.16	0.06	O.K
205	0.085	0.085	225.9	406.6	0.55	1.16	0.06	O.K
206	0.085	0.085	225.9	406.6	0.55	1.16	0.66	O.K
207	0.085	0.085	225.9	406.6	0.55	1.16	0.66	O.K
208	0.085	0.085	225.9	406.6	0.55	1.16	0.66	O.K
209	0.085	0.085	225.9	406.6	0.55	1.16	0.66	O.K
210	0.085	0.085	225.9	406.6	0.55	1.16	0.47	O.K
211	0.085	0.085	225.9	406.6	0.55	1.16	0.47	O.K
212	0.085	0.085	225.9	406.6	0.55	1.16	0.47	O.K
213	0.085	0.085	225.9	406.6	0.55	1.16	0.47	O.K
214	0.085	0.085	225.9	406.6	0.55	1.16	0.43	O.K
215	0.085	0.085	225.9	406.6	0.55	1.16	0.43	O.K
216	0.085	0.085	225.9	406.6	0.55	1.16	0.43	O.K
217	0.085	0.085	225.9	406.6	0.55	1.16	0.43	O.K
218	0.085	0.085	225.9	406.6	0.55	1.16	0.25	O.K
219	0.085	0.085	225.9	406.6	0.55	1.16	0.25	O.K
220	0.085	0.085	225.9	406.6	0.55	1.16	0.25	O.K
221	0.085	0.085	225.9	406.6	0.55	1.16	0.25	O.K
222	0.085	0.085	225.9	406.6	0.55	1.16	0.03	O.K
223	0.085	0.085	225.9	406.6	0.55	1.16	0.03	O.K
224	0.085	0.085	225.9	406.6	0.55	1.16	0.03	O.K
225	0.085	0.085	225.9	406.6	0.55	1.16	0.03	O.K
226	0.085	0.085	225.9	406.6	0.55	1.16	0.05	O.K
227	0.085	0.085	225.9	406.6	0.55	1.16	0.05	O.K
228	0.085	0.085	225.9	406.6	0.55	1.16	0.05	O.K
229	0.085	0.085	225.9	406.6	0.55	1.16	0.05	O.K
							Max = **0.73**	**O.K**

Links, 10mm dia @ 300 C/C

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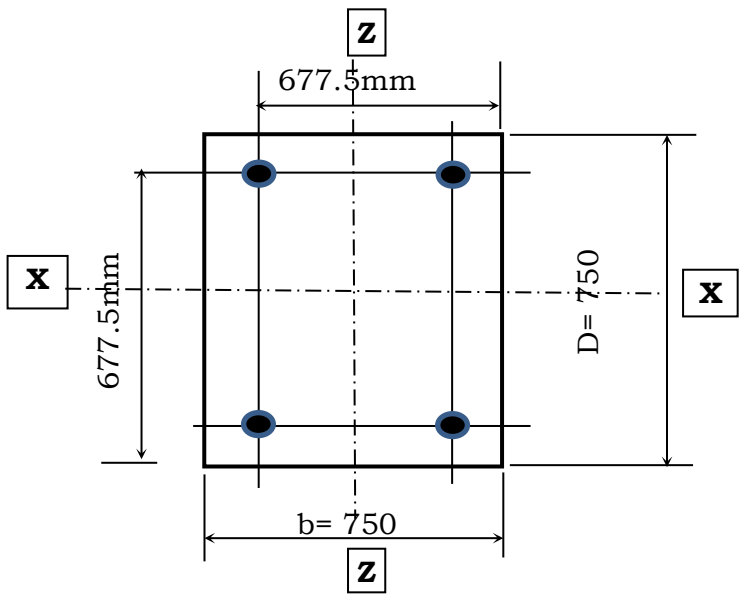
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5.9 DESIGN OF COLUMN MKD C2

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	72.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	1.00%
Max. Steel Ratio	4.00%
Dimension parallel to X-X axis	750mm
Dimension parallel to Z-Z axis	750mm

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	2300mm	γ_x =	0.83
Unsupproted length of column about Z (l_{uz}) =	2300mm	γ_z =	0.83

Where, Effective length factor K= 2

Slenderness ratio,	K l_{ux} / r_x =	20.44	< 22 SECTION 6.2.5a	Radius of gyration r_x=	225	mm
	K l_{uz} / r_z =	20.44	< 22 SECTION 6.2.5a	Radius of gyration r_z=	225	mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	13.1292		-6.702		0
2	COLLATERAL LOAD (COLL)	0.0012		-0.0024		0
3	ROOF LIVE LOAD (RLL)	36.4752		-20.7252		0
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	0.0036		-0.006		0
5	CRANE LOAD-1 (CL-1)	3.6888		-5.1264		0
6	CRANE LOAD-2 (CL-2)	-0.3096		0.4176		0
7	CRANE LOAD-3 (CL-3)	0.5304		-0.8856		0
8	CRANE LOAD-4 (CL-4)	-3.48		4.6848		0
9	WIND LEFT-1 (WLL-1)	-51.7152		18.342		0
10	WIND LEFT-2 (WLL-2)	-18.3552		7.35		0
11	WIND RIGHT-1 (WLR-1)	-82.6872		50.4228		0
12	WIND RIGHT-2 (WLR-2)	-16.1352		25.7268		0
13	WIND LONG-1 (WLLG-1)	-85.0944	29.0928		0	
14	WIND LONG-2 (WLLG-2)	-55.7412	21.4044		0	
15	WIND LONG-3 (WLLG-3)	-83.8356	27.3528		0	
16	WIND LONG-4 (WLLG-4)	-54.5208	19.8132		0	
17	SEISMIC LOAD SL'+'	0.1116	-0.306	-0.306	0	0
18	SEISMIC LOAD SL'-'	-0.1248	0.3252	0.3252	0	0
19	TEMPERATURE LOAD (TL1)	1.0284		-1.644		0
20	TEMPERATURE LOAD (TL2)	-1.0296		1.6488		0
21	BRACING REACTION (BR-WLLG1)	0	0		0	
22	BRACING REACTION (BR-WLLG2)	0	0		0	
23	BRACING REACTION (BR-WLLG3)	0	0		0	
24	BRACING REACTION (BR-WLLG4)	0	0		0	
25	BRACING REACTION (BR-SL'+')	0	0	0	0	0
26	BRACING REACTION (BR-SL'-')	0	0	0	0	0

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4(DL+COLL)	1.84	0.0	2.2	2.8	0.0	3.3	0.00
102	1.2(DL+COLL+CL-1)+1.6(RLL+P	7.9	0.0	10.9	12.1	0.0	16.8	0.01
103	1.2(DL+COLL+CL-2)+1.6(RLL+P	7.4	0.0	9.4	11.3	0.0	14.4	0.01
104	1.2(DL+COLL+CL-3)+1.6(RLL+P	7.5	0.0	9.7	11.5	0.0	15.0	0.01
105	1.2(DL+COLL+CL-4)+1.6(RLL+P	7.0	0.0	8.2	10.8	0.0	12.6	0.01
106	1.4(DL+COLL+CL-1)	2.4	0.0	3.8	3.6	0.0	5.9	0.00
107	1.4(DL+COLL+CL-2)	1.8	0.0	2.0	2.8	0.0	3.1	0.00
108	1.4(DL+COLL+CL-3)	1.9	0.0	2.4	2.9	0.0	3.8	0.00
109	1.4(DL+COLL+CL-4)	1.4	0.0	0.7	2.1	0.0	1.0	0.00
110	1.2(DL+COLL+CL-1+TL1)+1.6(R	8.0	0.0	11.3	12.3	0.0	17.5	0.01
111	1.2(DL+COLL+CL-2+TL1)+1.6(R	7.5	0.0	9.8	11.5	0.0	15.1	0.01
112	1.2(DL+COLL+CL-3+TL1)+1.6(R	7.6	0.0	10.2	11.7	0.0	15.7	0.01
113	1.2(DL+COLL+CL-4+TL1)+1.6(R	7.1	0.0	8.6	11.0	0.0	13.3	0.01
114	1.2(DL+COLL+CL-1+TL2)+1.6(R	7.7	0.0	10.4	11.9	0.0	16.1	0.01
115	1.2(DL+COLL+CL-2+TL2)+1.6(R	7.3	0.0	8.9	11.2	0.0	13.7	0.01
116	1.2(DL+COLL+CL-3+TL2)+1.6(R	7.4	0.0	9.3	11.3	0.0	14.3	0.01
117	1.2(DL+COLL+CL-4+TL2)+1.6(R	6.9	0.0	7.7	10.6	0.0	11.9	0.01
118	1.2(DL+COLL+CL-1)+0.8(WLL-1	-2.1	0.0	0.1	-3.3	0.0	0.2	0.00
119	1.2(DL+COLL+CL-2)+0.8(WLL-1	-2.6	0.0	1.6	-4.0	0.0	2.5	0.00
120	1.2(DL+COLL+CL-3)+0.8(WLL-1	-2.5	0.0	1.3	-3.8	0.0	2.0	0.00
121	1.2(DL+COLL+CL-4)+0.8(WLL-1	-3.0	0.0	2.8	-4.6	0.0	4.3	0.00
122	1.2(DL+COLL+CL-1)+0.8(WLL-2	0.5	0.0	1.9	0.8	0.0	2.9	0.00
123	1.2(DL+COLL+CL-2)+0.8(WLL-2	0.1	0.0	0.4	0.1	0.0	0.6	0.00
124	1.2(DL+COLL+CL-3)+0.8(WLL-2	0.2	0.0	0.7	0.3	0.0	1.1	0.00
125	1.2(DL+COLL+CL-4)+0.8(WLL-2	-0.3	0.0	0.8	-0.5	0.0	1.2	0.00
126	1.2(DL+COLL+CL-1)+0.8(WLR-1	-4.6	0.0	6.0	-7.1	0.0	9.2	0.00
127	1.2(DL+COLL+CL-2)+0.8(WLR-1	-5.1	0.0	7.5	-7.8	0.0	11.6	0.00
128	1.2(DL+COLL+CL-3)+0.8(WLR-1	-5.0	0.0	7.2	-7.7	0.0	11.1	0.00
129	1.2(DL+COLL+CL-4)+0.8(WLR-1	-5.5	0.0	8.7	-8.4	0.0	13.4	0.00
130	1.2(DL+COLL+CL-1)+0.8(WLR-2	0.7	0.0	1.5	1.1	0.0	2.3	0.00
131	1.2(DL+COLL+CL-2)+0.8(WLR-2	0.2	0.0	3.0	0.4	0.0	4.6	0.00
132	1.2(DL+COLL+CL-3)+0.8(WLR-2	0.3	0.0	2.6	0.5	0.0	4.1	0.00
133	1.2(DL+COLL+CL-4)+0.8(WLR-2	-0.1	0.0	4.2	-0.2	0.0	6.4	0.00
134	1.2(DL+COLL+CL-1)+0.8(WLLG-	-4.8	5.4	3.3	-7.4	8.2	5.0	0.00
135	1.2(DL+COLL+CL-2)+0.8(WLLG-	-5.3	5.4	1.7	-8.1	8.2	2.7	0.00
136	1.2(DL+COLL+CL-3)+0.8(WLLG-	-5.2	5.4	2.1	-8.0	8.2	3.2	0.00
137	1.2(DL+COLL+CL-4)+0.8(WLLG-	-5.6	5.4	0.6	-8.7	8.2	0.9	0.00
138	1.2(DL+COLL+CL-1)+0.8(WLLG-	-2.4	3.9	3.3	-3.8	6.1	5.0	0.00
139	1.2(DL+COLL+CL-2)+0.8(WLLG-	-2.9	3.9	1.7	-4.5	6.1	2.7	0.00
140	1.2(DL+COLL+CL-3)+0.8(WLLG-	-2.8	3.9	2.1	-4.3	6.1	3.2	0.00
141	1.2(DL+COLL+CL-4)+0.8(WLLG-	-3.3	3.9	0.6	-5.1	6.1	0.9	0.00
142	1.2(DL+COLL+CL-1)+0.8(WLLG-	-4.7	5.0	3.3	-7.2	7.7	5.0	0.00
143	1.2(DL+COLL+CL-2)+0.8(WLLG-	-5.2	5.0	1.7	-8.0	7.7	2.7	0.00
144	1.2(DL+COLL+CL-3)+0.8(WLLG-	-5.1	5.0	2.1	-7.8	7.7	3.2	0.00
145	1.2(DL+COLL+CL-4)+0.8(WLLG-	-5.5	5.0	0.6	-8.5	7.7	0.9	0.00
146	1.2(DL+COLL+CL-1)+0.8(WLLG-	-2.3	3.6	3.3	-3.6	5.6	5.0	0.00
147	1.2(DL+COLL+CL-2)+0.8(WLLG-	-2.8	3.6	1.7	-4.3	5.6	2.7	0.00
148	1.2(DL+COLL+CL-3)+0.8(WLLG-	-2.7	3.6	2.1	-4.2	5.6	3.2	0.00
149	1.2(DL+COLL+CL-4)+0.8(WLLG-	-3.2	3.6	0.6	-4.9	5.6	0.9	0.00
150	1.2(DL+COLL+CL-1)+1.6(WLL-1	-4.4	0.0	1.1	-6.8	0.0	1.7	0.00
151	1.2(DL+COLL+CL-2)+1.6(WLL-1	-4.9	0.0	2.6	-7.6	0.0	4.0	0.00
152	1.2(DL+COLL+CL-3)+1.6(WLL-1	-4.8	0.0	2.3	-7.4	0.0	3.5	0.00

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153	1.2(DL+COLL+CL-4)+1.6(WLL-1	-5.3	0.0	3.8	-8.1	0.0	5.9	0.00
154	1.2(DL+COLL+CL-1)+1.6(WLL-2	0.9	0.0	2.9	1.4	0.0	4.5	0.00
155	1.2(DL+COLL+CL-2)+1.6(WLL-2	0.4	0.0	1.4	0.7	0.0	2.2	0.00
156	1.2(DL+COLL+CL-3)+1.6(WLL-2	0.5	0.0	1.8	0.8	0.0	2.7	0.00
157	1.2(DL+COLL+CL-4)+1.6(WLL-2	0.0	0.0	0.2	0.1	0.0	0.4	0.00
158	1.2(DL+COLL+CL-1)+1.6(WLR-1	-9.4	0.0	12.9	-14.4	0.0	19.9	-0.01
159	1.2(DL+COLL+CL-2)+1.6(WLR-1	-9.9	0.0	14.4	-15.2	0.0	22.2	-0.01
160	1.2(DL+COLL+CL-3)+1.6(WLR-1	-9.8	0.0	14.1	-15.0	0.0	21.7	-0.01
161	1.2(DL+COLL+CL-4)+1.6(WLR-1	-10.2	0.0	15.6	-15.8	0.0	24.0	-0.01
162	1.2(DL+COLL+CL-1)+1.6(WLR-2	1.3	0.0	3.8	1.9	0.0	5.9	0.00
163	1.2(DL+COLL+CL-2)+1.6(WLR-2	0.8	0.0	5.3	1.2	0.0	8.2	0.00
164	1.2(DL+COLL+CL-3)+1.6(WLR-2	0.9	0.0	5.0	1.4	0.0	7.7	0.00
165	1.2(DL+COLL+CL-4)+1.6(WLR-2	0.4	0.0	6.5	0.6	0.0	10.0	0.00
166	1.2(DL+COLL+CL-1)+1.6(WLLG-	-9.8	10.7	5.6	-15.0	16.5	8.7	-0.01
167	1.2(DL+COLL+CL-2)+1.6(WLLG-	-10.3	10.7	4.1	-15.8	16.5	6.3	-0.01
168	1.2(DL+COLL+CL-3)+1.6(WLLG-	-10.2	10.7	4.5	-15.6	16.5	6.9	-0.01
169	1.2(DL+COLL+CL-4)+1.6(WLLG-	-10.6	10.7	2.9	-16.4	16.5	4.5	-0.01
170	1.2(DL+COLL+CL-1)+1.6(WLLG-	-5.1	7.9	5.6	-7.8	12.1	8.7	0.00
171	1.2(DL+COLL+CL-2)+1.6(WLLG-	-5.6	7.9	4.1	-8.5	12.1	6.3	0.00
172	1.2(DL+COLL+CL-3)+1.6(WLLG-	-5.5	7.9	4.5	-8.4	12.1	6.9	0.00
173	1.2(DL+COLL+CL-4)+1.6(WLLG-	-5.9	7.9	2.9	-9.1	12.1	4.5	0.00
174	1.2(DL+COLL+CL-1)+1.6(WLLG-	-9.6	10.1	5.6	-14.7	15.5	8.7	-0.01
175	1.2(DL+COLL+CL-2)+1.6(WLLG-	-10.1	10.1	4.1	-15.5	15.5	6.3	-0.01
176	1.2(DL+COLL+CL-3)+1.6(WLLG-	-10.0	10.1	4.5	-15.3	15.5	6.9	-0.01
177	1.2(DL+COLL+CL-4)+1.6(WLLG-	-10.4	10.1	2.9	-16.0	15.5	4.5	-0.01
178	1.2(DL+COLL+CL-1)+1.6(WLLG-	-4.9	7.3	5.6	-7.5	11.2	8.7	0.00
179	1.2(DL+COLL+CL-2)+1.6(WLLG-	-5.4	7.3	4.1	-8.2	11.2	6.3	0.00
180	1.2(DL+COLL+CL-3)+1.6(WLLG-	-5.3	7.3	4.5	-8.1	11.2	6.9	0.00
181	1.2(DL+COLL+CL-4)+1.6(WLLG-	-5.7	7.3	2.9	-8.8	11.2	4.5	0.00
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+I	3.9	0.1	5.7	5.9	0.1	8.8	0.00
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+I	3.4	0.1	4.2	5.2	0.1	6.4	0.00
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+I	3.5	0.1	4.5	5.3	0.1	7.0	0.00
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+I	3.0	0.1	3.0	4.6	0.1	4.6	0.00
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+E	3.8	0.1	5.6	5.9	0.1	8.6	0.00
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+E	3.3	0.1	4.0	5.2	0.1	6.2	0.00
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+E	3.5	0.1	4.4	5.3	0.1	6.8	0.00
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+E	3.0	0.1	2.9	4.6	0.1	4.4	0.00
190	0.9(DL+COLL+CL-1)+1.6(WLL-1	-6.8	0.0	4.3	-10.4	0.0	6.6	-0.01
191	0.9(DL+COLL+CL-2)+1.6(WLL-1	-7.1	0.0	5.4	-11.0	0.0	8.4	-0.01
192	0.9(DL+COLL+CL-3)+1.6(WLL-1	-7.0	0.0	5.2	-10.8	0.0	8.0	-0.01
193	0.9(DL+COLL+CL-4)+1.6(WLL-1	-7.4	0.0	6.3	-11.4	0.0	9.7	-0.01
194	0.9(DL+COLL+CL-1)+1.6(WLL-2	-1.4	0.0	0.3	-2.2	0.0	0.4	0.00
195	0.9(DL+COLL+CL-2)+1.6(WLL-2	-1.8	0.0	1.4	-2.7	0.0	2.2	0.00
196	0.9(DL+COLL+CL-3)+1.6(WLL-2	-1.7	0.0	1.1	-2.6	0.0	1.7	0.00
197	0.9(DL+COLL+CL-4)+1.6(WLL-2	-2.1	0.0	2.3	-3.2	0.0	3.5	0.00
198	0.9(DL+COLL+CL-1)+1.6(WLR-1	-11.7	0.0	16.1	-18.0	0.0	24.8	-0.01
199	0.9(DL+COLL+CL-2)+1.6(WLR-1	-12.1	0.0	17.3	-18.6	0.0	26.5	-0.01
200	0.9(DL+COLL+CL-3)+1.6(WLR-1	-12.0	0.0	17.0	-18.5	0.0	26.1	-0.01
201	0.9(DL+COLL+CL-4)+1.6(WLR-1	-12.4	0.0	18.1	-19.0	0.0	27.9	-0.01
202	0.9(DL+COLL+CL-1)+1.6(WLR-2	-1.1	0.0	7.0	-1.6	0.0	10.8	0.00
203	0.9(DL+COLL+CL-2)+1.6(WLR-2	-1.4	0.0	8.2	-2.2	0.0	12.6	0.00
204	0.9(DL+COLL+CL-3)+1.6(WLR-2	-1.4	0.0	7.9	-2.1	0.0	12.1	0.00
205	0.9(DL+COLL+CL-4)+1.6(WLR-2	-1.7	0.0	9.0	-2.6	0.0	13.9	0.00
206	0.9(DL+COLL+CL-1)+1.6(WLLG-	-12.1	10.7	2.4	-18.6	16.5	3.8	-0.01
207	0.9(DL+COLL+CL-2)+1.6(WLLG-	-12.5	10.7	1.3	-19.2	16.5	2.0	-0.01
208	0.9(DL+COLL+CL-3)+1.6(WLLG-	-12.4	10.7	1.6	-19.1	16.5	2.4	-0.01
209	0.9(DL+COLL+CL-4)+1.6(WLLG-	-12.7	10.7	0.4	-19.6	16.5	0.6	-0.01

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210	0.9(DL+COLL+CL-1)+1.6(WLLG	-7.4	7.9	2.4	-11.4	12.1	3.8	-0.01
211	0.9(DL+COLL+CL-2)+1.6(WLLG	-7.8	7.9	1.3	-11.9	12.1	2.0	-0.01
212	0.9(DL+COLL+CL-3)+1.6(WLLG	-7.7	7.9	1.6	-11.8	12.1	2.4	-0.01
213	0.9(DL+COLL+CL-4)+1.6(WLLG	-8.1	7.9	0.4	-12.4	12.1	0.6	-0.01
214	0.9(DL+COLL+CL-1)+1.6(WLLG	-11.9	10.1	2.4	-18.3	15.5	3.8	-0.01
215	0.9(DL+COLL+CL-2)+1.6(WLLG	-12.3	10.1	1.3	-18.9	15.5	2.0	-0.01
216	0.9(DL+COLL+CL-3)+1.6(WLLG	-12.2	10.1	1.6	-18.7	15.5	2.4	-0.01
217	0.9(DL+COLL+CL-4)+1.6(WLLG	-12.5	10.1	0.4	-19.3	15.5	0.6	-0.01
218	0.9(DL+COLL+CL-1)+1.6(WLLG	-7.2	7.3	2.4	-11.1	11.2	3.8	-0.01
219	0.9(DL+COLL+CL-2)+1.6(WLLG	-7.6	7.3	1.3	-11.6	11.2	2.0	-0.01
220	0.9(DL+COLL+CL-3)+1.6(WLLG	-7.5	7.3	1.6	-11.5	11.2	2.4	-0.01
221	0.9(DL+COLL+CL-4)+1.6(WLLG	-7.9	7.3	0.4	-12.1	11.2	0.6	-0.01
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+E	1.5	0.1	2.5	2.3	0.1	3.9	0.00
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+E	1.2	0.1	1.4	1.8	0.1	2.1	0.00
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+E	1.2	0.1	1.6	1.9	0.1	2.5	0.00
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+E	0.9	0.1	0.5	1.4	0.1	0.8	0.00
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+E	1.5	0.1	2.4	2.3	0.1	3.7	0.00
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+E	1.1	0.1	1.2	1.8	0.1	1.9	0.00
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+E	1.2	0.1	1.5	1.9	0.1	2.3	0.00
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+E	0.9	0.1	0.3	1.3	0.1	0.5	0.00

COLUMN DESIGN AND INTERACTION RATIO CHECK

Reinforcement: Provide,

4-Nos	25 mm Dia
6-Nos	25 mm Dia
6-Nos	25 mm Dia

1964.29 Sq mm (At Corner) Asc pvd. = 7857.14 Sq mm
2946.43 Sq mm (Along Z axis)
2946.43 Sq mm (Along X axis) Total Asc = 1.40 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1980.0625 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8 (COLUMNS 3.3.3) $R_{nz}=M_{noz}/f'_c b^2 D$	From Chart of SP17(M)-R35-420.8 (COLUMNS 3.3.3) $R_{nx}=M_{nox}/f'_c b^2 D$	M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
101	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
102	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
103	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
104	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
105	0.05	0.05	73.8	73.8	0.55	1.16	0.13	O.K
106	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
107	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
108	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
109	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
110	0.05	0.05	73.8	73.8	0.55	1.16	0.19	O.K
111	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
112	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
113	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
114	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
115	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
116	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
117	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
118	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
119	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
120	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
121	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
122	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
123	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
124	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
125	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
126	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
127	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K

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128	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
129	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
130	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
131	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
132	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
133	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
134	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
135	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
136	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
137	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
138	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
139	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
140	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
141	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
142	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
143	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
144	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
145	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
146	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
147	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
148	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
149	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
150	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
151	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
152	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
153	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
154	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
155	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
156	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
157	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
158	0.05	0.05	73.8	73.8	0.55	1.16	0.22	O.K
159	0.05	0.05	73.8	73.8	0.55	1.16	0.25	O.K
160	0.05	0.05	73.8	73.8	0.55	1.16	0.24	O.K
161	0.05	0.05	73.8	73.8	0.55	1.16	0.27	O.K
162	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
163	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
164	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
165	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
166	0.05	0.05	73.8	73.8	0.55	1.16	0.26	O.K
167	0.05	0.05	73.8	73.8	0.55	1.16	0.23	O.K
168	0.05	0.05	73.8	73.8	0.55	1.16	0.24	O.K
169	0.05	0.05	73.8	73.8	0.55	1.16	0.21	O.K
170	0.05	0.05	73.8	73.8	0.55	1.16	0.21	O.K
171	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
172	0.05	0.05	73.8	73.8	0.55	1.16	0.19	O.K
173	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
174	0.05	0.05	73.8	73.8	0.55	1.16	0.25	O.K
175	0.05	0.05	73.8	73.8	0.55	1.16	0.22	O.K
176	0.05	0.05	73.8	73.8	0.55	1.16	0.23	O.K
177	0.05	0.05	73.8	73.8	0.55	1.16	0.20	O.K
178	0.05	0.05	73.8	73.8	0.55	1.16	0.20	O.K
179	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
180	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
181	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
182	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
183	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
184	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
185	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
186	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
187	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
188	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
189	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
190	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K

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191	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
192	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
193	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
194	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
195	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
196	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
197	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
198	0.05	0.05	73.8	73.8	0.55	1.16	0.28	O.K
199	0.05	0.05	78.3	78.3	0.55	1.16	0.29	O.K
200	0.05	0.05	73.8	73.8	0.55	1.16	0.30	O.K
201	0.05	0.05	78.3	78.3	0.55	1.16	0.30	O.K
202	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
203	0.05	0.05	73.8	73.8	0.55	1.16	0.13	O.K
204	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
205	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
206	0.05	0.05	73.8	73.8	0.55	1.16	0.21	O.K
207	0.05	0.05	73.8	73.8	0.55	1.16	0.19	O.K
208	0.05	0.05	73.8	73.8	0.55	1.16	0.19	O.K
209	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
210	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
211	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
212	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
213	0.05	0.05	73.8	73.8	0.55	1.16	0.13	O.K
214	0.05	0.05	73.8	73.8	0.55	1.16	0.20	O.K
215	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
216	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
217	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
218	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
219	0.05	0.05	73.8	73.8	0.55	1.16	0.13	O.K
220	0.05	0.05	73.8	73.8	0.55	1.16	0.13	O.K
221	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
222	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
223	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
224	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
225	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
226	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
227	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
228	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
229	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
							Max = **0.30**	**O.K**

Links, 10mm dia @ 300 C/C

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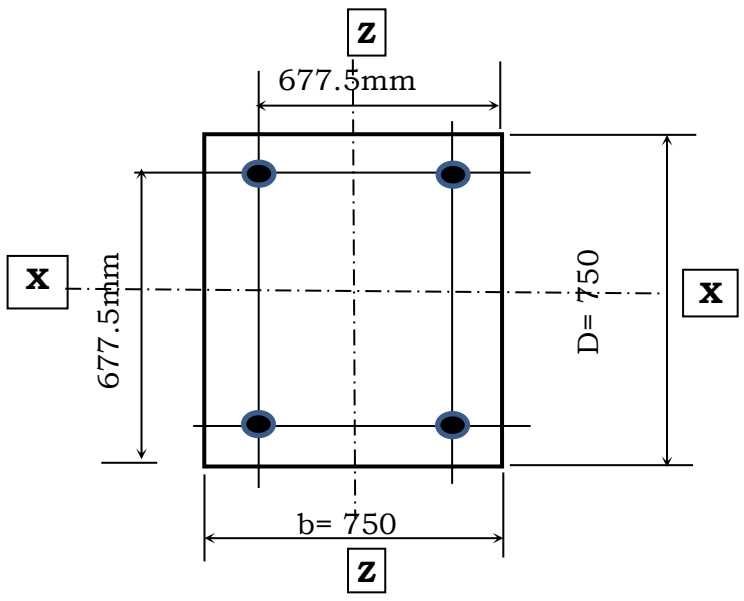
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5.10 DESIGN OF COLUMN MKD C3

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	72.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	1.00%
Max. Steel Ratio	4.00%
Dimension parallel to X-X axis	750mm
Dimension parallel to Z-Z axis	750mm

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	2300mm	γ_x =	0.83
Unsupproted length of column about Z (l_{uz}) =	2300mm	γ_z =	0.83

Where, Effective length factor K= 2

Slenderness ratio,	K l_{ux} / r_x =	20.44	< 22 SECTION 6.2.5a	Radius of gyration r_x=	225	mm
	K l_{uz} / r_z =	20.44	< 22 SECTION 6.2.5a	Radius of gyration r_z=	225	mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	23.0856	0.00	0		
2	COLLATERAL LOAD (COLL)	0	0.00	0		
3	ROOF LIVE LOAD (RLL)	25.7592	0.00	0		
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	0	0.00	0		
5	CRANE LOAD-1 (CL-1)	0	0.00	0		
6	CRANE LOAD-2 (CL-2)	0	0.00	0		
7	CRANE LOAD-3 (CL-3)	0	0.00	0		
8	CRANE LOAD-4 (CL-4)	0	0.00	0		
9	WIND LEFT-1 (WLL-1)	-66.6984	-47.28	-1.4136		
10	WIND LEFT-2 (WLL-2)	-44.7144	-19.05	-0.0024		
11	WIND RIGHT-1 (WLR-1)	-61.0212	-47.47	-12.4428		
12	WIND RIGHT-2 (WLR-2)	-36.3216	19.12	-9.9036		
13	WIND LONG-1 (WLLG-1)	-61.3284	-47.34	-4.6668		
14	WIND LONG-2 (WLLG-2)	-38.0136	19.08	-2.1324		
15	WIND LONG-3 (WLLG-3)	-59.6376	-47.41	-8.4576		
16	WIND LONG-4 (WLLG-4)	-36.1956	47.41	-5.9148		
17	SEISMIC LOAD SL'+'	0	0.00	0		
18	SEISMIC LOAD SL'-'	0	0.00	0		
19	TEMPERATURE LOAD (TL1)	-27.1104	0.00	-2.94		
20	TEMPERATURE LOAD (TL2)	27.1344	0.00	0		
21	BRACING REACTION (BR-WLLG1)	0	0.00	0		
22	BRACING REACTION (BR-WLLG2)	0	0.00	0		
23	BRACING REACTION (BR-WLLG3)	0	0.00	0		
24	BRACING REACTION (BR-WLLG4)	0	0.00	0		
25	BRACING REACTION (BR-SL'+')	0	0.00	0		
26	BRACING REACTION (BR-SL'-')	0	0.00	0		

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4(DL+COLL)	3.23	0.0	0.0	5.0	0.0	0.0	0.00
102	1.2(DL+COLL+CL-1)+1.6(RLL+P	6.9	0.0	0.0	10.6	0.0	0.0	0.01
103	1.2(DL+COLL+CL-2)+1.6(RLL+P	6.9	0.0	0.0	10.6	0.0	0.0	0.01
104	1.2(DL+COLL+CL-3)+1.6(RLL+P	6.9	0.0	0.0	10.6	0.0	0.0	0.01
105	1.2(DL+COLL+CL-4)+1.6(RLL+P	6.9	0.0	0.0	10.6	0.0	0.0	0.01
106	1.4(DL+COLL+CL-1)	3.2	0.0	0.0	5.0	0.0	0.0	0.00
107	1.4(DL+COLL+CL-2)	3.2	0.0	0.0	5.0	0.0	0.0	0.00
108	1.4(DL+COLL+CL-3)	3.2	0.0	0.0	5.0	0.0	0.0	0.00
109	1.4(DL+COLL+CL-4)	3.2	0.0	0.0	5.0	0.0	0.0	0.00
110	1.2(DL+COLL+CL-1+TL1)+1.6(R	3.6	0.0	0.8	5.6	0.0	1.2	0.00
111	1.2(DL+COLL+CL-2+TL1)+1.6(R	3.6	0.0	0.8	5.6	0.0	1.2	0.00
112	1.2(DL+COLL+CL-3+TL1)+1.6(R	3.6	0.0	0.8	5.6	0.0	1.2	0.00
113	1.2(DL+COLL+CL-4+TL1)+1.6(R	3.6	0.0	0.8	5.6	0.0	1.2	0.00
114	1.2(DL+COLL+CL-1+TL2)+1.6(R	10.1	0.0	0.0	15.6	0.0	0.0	0.01
115	1.2(DL+COLL+CL-2+TL2)+1.6(R	10.1	0.0	0.0	15.6	0.0	0.0	0.01
116	1.2(DL+COLL+CL-3+TL2)+1.6(R	10.1	0.0	0.0	15.6	0.0	0.0	0.01
117	1.2(DL+COLL+CL-4+TL2)+1.6(R	10.1	0.0	0.0	15.6	0.0	0.0	0.01
118	1.2(DL+COLL+CL-1)+0.8(WLL-1	-2.6	8.7	0.3	-3.9	13.4	0.4	0.00
119	1.2(DL+COLL+CL-2)+0.8(WLL-1	-2.6	8.7	0.3	-3.9	13.4	0.4	0.00
120	1.2(DL+COLL+CL-3)+0.8(WLL-1	-2.6	8.7	0.3	-3.9	13.4	0.4	0.00
121	1.2(DL+COLL+CL-4)+0.8(WLL-1	-2.6	8.7	0.3	-3.9	13.4	0.4	0.00
122	1.2(DL+COLL+CL-1)+0.8(WLL-2	-0.8	3.5	0.0	-1.2	5.4	0.0	0.00
123	1.2(DL+COLL+CL-2)+0.8(WLL-2	-0.8	3.5	0.0	-1.2	5.4	0.0	0.00
124	1.2(DL+COLL+CL-3)+0.8(WLL-2	-0.8	3.5	0.0	-1.2	5.4	0.0	0.00
125	1.2(DL+COLL+CL-4)+0.8(WLL-2	-0.8	3.5	0.0	-1.2	5.4	0.0	0.00
126	1.2(DL+COLL+CL-1)+0.8(WLR-1	-2.1	8.7	2.3	-3.2	13.4	3.5	0.00
127	1.2(DL+COLL+CL-2)+0.8(WLR-1	-2.1	8.7	2.3	-3.2	13.4	3.5	0.00
128	1.2(DL+COLL+CL-3)+0.8(WLR-1	-2.1	8.7	2.3	-3.2	13.4	3.5	0.00
129	1.2(DL+COLL+CL-4)+0.8(WLR-1	-2.1	8.7	2.3	-3.2	13.4	3.5	0.00
130	1.2(DL+COLL+CL-1)+0.8(WLR-2	-0.1	3.5	1.8	-0.2	5.4	2.8	0.00
131	1.2(DL+COLL+CL-2)+0.8(WLR-2	-0.1	3.5	1.8	-0.2	5.4	2.8	0.00
132	1.2(DL+COLL+CL-3)+0.8(WLR-2	-0.1	3.5	1.8	-0.2	5.4	2.8	0.00
133	1.2(DL+COLL+CL-4)+0.8(WLR-2	-0.1	3.5	1.8	-0.2	5.4	2.8	0.00
134	1.2(DL+COLL+CL-1)+0.8(WLLG-	-2.1	8.7	0.9	-3.3	13.4	1.3	0.00
135	1.2(DL+COLL+CL-2)+0.8(WLLG-	-2.1	8.7	0.9	-3.3	13.4	1.3	0.00
136	1.2(DL+COLL+CL-3)+0.8(WLLG-	-2.1	8.7	0.9	-3.3	13.4	1.3	0.00
137	1.2(DL+COLL+CL-4)+0.8(WLLG-	-2.1	8.7	0.9	-3.3	13.4	1.3	0.00
138	1.2(DL+COLL+CL-1)+0.8(WLLG-	-0.3	3.5	0.4	-0.4	5.4	0.6	0.00
139	1.2(DL+COLL+CL-2)+0.8(WLLG-	-0.3	3.5	0.4	-0.4	5.4	0.6	0.00
140	1.2(DL+COLL+CL-3)+0.8(WLLG-	-0.3	3.5	0.4	-0.4	5.4	0.6	0.00
141	1.2(DL+COLL+CL-4)+0.8(WLLG-	-0.3	3.5	0.4	-0.4	5.4	0.6	0.00
142	1.2(DL+COLL+CL-1)+0.8(WLLG-	-2.0	8.7	1.6	-3.1	13.4	2.4	0.00
143	1.2(DL+COLL+CL-2)+0.8(WLLG-	-2.0	8.7	1.6	-3.1	13.4	2.4	0.00
144	1.2(DL+COLL+CL-3)+0.8(WLLG-	-2.0	8.7	1.6	-3.1	13.4	2.4	0.00
145	1.2(DL+COLL+CL-4)+0.8(WLLG-	-2.0	8.7	1.6	-3.1	13.4	2.4	0.00
146	1.2(DL+COLL+CL-1)+0.8(WLLG-	-0.1	8.7	1.1	-0.2	13.4	1.7	0.00
147	1.2(DL+COLL+CL-2)+0.8(WLLG-	-0.1	8.7	1.1	-0.2	13.4	1.7	0.00
148	1.2(DL+COLL+CL-3)+0.8(WLLG-	-0.1	8.7	1.1	-0.2	13.4	1.7	0.00
149	1.2(DL+COLL+CL-4)+0.8(WLLG-	-0.1	8.7	1.1	-0.2	13.4	1.7	0.00
150	1.2(DL+COLL+CL-1)+1.6(WLL-1	-6.6	17.4	0.5	-10.2	26.8	0.8	-0.01
151	1.2(DL+COLL+CL-2)+1.6(WLL-1	-6.6	17.4	0.5	-10.2	26.8	0.8	-0.01
152	1.2(DL+COLL+CL-3)+1.6(WLL-1	-6.6	17.4	0.5	-10.2	26.8	0.8	-0.01

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153	1.2(DL+COLL+CL-4)+1.6(WLL-1	-6.6	17.4	0.5	-10.2	26.8	0.8	-0.01
154	1.2(DL+COLL+CL-1)+1.6(WLL-2	-3.1	7.0	0.0	-4.8	10.8	0.0	0.00
155	1.2(DL+COLL+CL-2)+1.6(WLL-2	-3.1	7.0	0.0	-4.8	10.8	0.0	0.00
156	1.2(DL+COLL+CL-3)+1.6(WLL-2	-3.1	7.0	0.0	-4.8	10.8	0.0	0.00
157	1.2(DL+COLL+CL-4)+1.6(WLL-2	-3.1	7.0	0.0	-4.8	10.8	0.0	0.00
158	1.2(DL+COLL+CL-1)+1.6(WLR-1	-5.7	17.5	4.6	-8.8	26.9	7.0	0.00
159	1.2(DL+COLL+CL-2)+1.6(WLR-1	-5.7	17.5	4.6	-8.8	26.9	7.0	0.00
160	1.2(DL+COLL+CL-3)+1.6(WLR-1	-5.7	17.5	4.6	-8.8	26.9	7.0	0.00
161	1.2(DL+COLL+CL-4)+1.6(WLR-1	-5.7	17.5	4.6	-8.8	26.9	7.0	0.00
162	1.2(DL+COLL+CL-1)+1.6(WLR-2	-1.8	17.5	3.6	-2.7	26.9	5.6	0.00
163	1.2(DL+COLL+CL-2)+1.6(WLR-2	-1.8	17.5	3.6	-2.7	26.9	5.6	0.00
164	1.2(DL+COLL+CL-3)+1.6(WLR-2	-1.8	17.5	3.6	-2.7	26.9	5.6	0.00
165	1.2(DL+COLL+CL-4)+1.6(WLR-2	-1.8	7.0	3.6	-2.7	10.8	5.6	0.00
166	1.2(DL+COLL+CL-1)+1.6(WLLG-	-5.8	17.4	1.7	-8.9	26.8	2.6	0.00
167	1.2(DL+COLL+CL-2)+1.6(WLLG-	-5.8	17.4	1.7	-8.9	26.8	2.6	0.00
168	1.2(DL+COLL+CL-3)+1.6(WLLG-	-5.8	17.4	1.7	-8.9	26.8	2.6	0.00
169	1.2(DL+COLL+CL-4)+1.6(WLLG-	-5.8	17.4	1.7	-8.9	26.8	2.6	0.00
170	1.2(DL+COLL+CL-1)+1.6(WLLG-	-2.0	7.0	0.8	-3.1	10.8	1.2	0.00
171	1.2(DL+COLL+CL-2)+1.6(WLLG-	-2.0	7.0	0.8	-3.1	10.8	1.2	0.00
172	1.2(DL+COLL+CL-3)+1.6(WLLG-	-2.0	7.0	0.8	-3.1	10.8	1.2	0.00
173	1.2(DL+COLL+CL-4)+1.6(WLLG-	-2.0	7.0	0.8	-3.1	10.8	1.2	0.00
174	1.2(DL+COLL+CL-1)+1.6(WLLG-	-5.5	17.4	3.1	-8.4	26.8	4.8	0.00
175	1.2(DL+COLL+CL-2)+1.6(WLLG-	-5.5	17.4	3.1	-8.4	26.8	4.8	0.00
176	1.2(DL+COLL+CL-3)+1.6(WLLG-	-5.5	17.4	3.1	-8.4	26.8	4.8	0.00
177	1.2(DL+COLL+CL-4)+1.6(WLLG-	-5.5	17.4	3.1	-8.4	26.8	4.8	0.00
178	1.2(DL+COLL+CL-1)+1.6(WLLG-	-1.7	17.4	2.2	-2.7	26.8	3.3	0.00
179	1.2(DL+COLL+CL-2)+1.6(WLLG-	-1.7	17.4	2.2	-2.7	26.8	3.3	0.00
180	1.2(DL+COLL+CL-3)+1.6(WLLG-	-1.7	17.4	2.2	-2.7	26.8	3.3	0.00
181	1.2(DL+COLL+CL-4)+1.6(WLLG-	-1.7	17.4	2.2	-2.7	26.8	3.3	0.00
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+I	4.1	0.0	0.0	6.2	0.0	0.0	0.00
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+I	4.1	0.0	0.0	6.2	0.0	0.0	0.00
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+I	4.1	0.0	0.0	6.2	0.0	0.0	0.00
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+I	4.1	0.0	0.0	6.2	0.0	0.0	0.00
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+E	4.1	0.0	0.0	6.2	0.0	0.0	0.00
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+E	4.1	0.0	0.0	6.2	0.0	0.0	0.00
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+E	4.1	0.0	0.0	6.2	0.0	0.0	0.00
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+E	4.1	0.0	0.0	6.2	0.0	0.0	0.00
190	0.9(DL+COLL+CL-1)+1.6(WLL-1	-8.6	17.4	0.5	-13.2	26.8	0.8	-0.01
191	0.9(DL+COLL+CL-2)+1.6(WLL-1	-8.6	17.4	0.5	-13.2	26.8	0.8	-0.01
192	0.9(DL+COLL+CL-3)+1.6(WLL-1	-8.6	17.4	0.5	-13.2	26.8	0.8	-0.01
193	0.9(DL+COLL+CL-4)+1.6(WLL-1	-8.6	17.4	0.5	-13.2	26.8	0.8	-0.01
194	0.9(DL+COLL+CL-1)+1.6(WLL-2	-5.1	7.0	0.0	-7.8	10.8	0.0	0.00
195	0.9(DL+COLL+CL-2)+1.6(WLL-2	-5.1	7.0	0.0	-7.8	10.8	0.0	0.00
196	0.9(DL+COLL+CL-3)+1.6(WLL-2	-5.1	7.0	0.0	-7.8	10.8	0.0	0.00
197	0.9(DL+COLL+CL-4)+1.6(WLL-2	-5.1	7.0	0.0	-7.8	10.8	0.0	0.00
198	0.9(DL+COLL+CL-1)+1.6(WLR-1	-7.7	17.5	4.6	-11.8	26.9	7.0	-0.01
199	0.9(DL+COLL+CL-2)+1.6(WLR-1	-7.7	17.5	4.6	-11.8	26.9	7.0	-0.01
200	0.9(DL+COLL+CL-3)+1.6(WLR-1	-7.7	17.5	4.6	-11.8	26.9	7.0	-0.01
201	0.9(DL+COLL+CL-4)+1.6(WLR-1	-7.7	17.5	4.6	-11.8	26.9	7.0	-0.01
202	0.9(DL+COLL+CL-1)+1.6(WLR-2	-3.7	7.0	3.6	-5.7	10.8	5.6	0.00
203	0.9(DL+COLL+CL-2)+1.6(WLR-2	-3.7	7.0	3.6	-5.7	10.8	5.6	0.00
204	0.9(DL+COLL+CL-3)+1.6(WLR-2	-3.7	7.0	3.6	-5.7	10.8	5.6	0.00
205	0.9(DL+COLL+CL-4)+1.6(WLR-2	-3.7	7.0	3.6	-5.7	10.8	5.6	0.00
206	0.9(DL+COLL+CL-1)+1.6(WLLG-	-7.7	17.4	1.7	-11.9	26.8	2.6	-0.01
207	0.9(DL+COLL+CL-2)+1.6(WLLG-	-7.7	17.4	1.7	-11.9	26.8	2.6	-0.01
208	0.9(DL+COLL+CL-3)+1.6(WLLG-	-7.7	17.4	1.7	-11.9	26.8	2.6	-0.01
209	0.9(DL+COLL+CL-4)+1.6(WLLG-	-7.7	17.4	1.7	-11.9	26.8	2.6	-0.01

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210	0.9(DL+COLL+CL-1)+1.6(WLLG	-4.0	7.0	0.8	-6.2	10.8	1.2	0.00
211	0.9(DL+COLL+CL-2)+1.6(WLLG	-4.0	7.0	0.8	-6.2	10.8	1.2	0.00
212	0.9(DL+COLL+CL-3)+1.6(WLLG	-4.0	7.0	0.8	-6.2	10.8	1.2	0.00
213	0.9(DL+COLL+CL-4)+1.6(WLLG	-4.0	7.0	0.8	-6.2	10.8	1.2	0.00
214	0.9(DL+COLL+CL-1)+1.6(WLLG	-7.5	17.4	3.1	-11.5	26.8	4.8	-0.01
215	0.9(DL+COLL+CL-2)+1.6(WLLG	-7.5	17.4	3.1	-11.5	26.8	4.8	-0.01
216	0.9(DL+COLL+CL-3)+1.6(WLLG	-7.5	17.4	3.1	-11.5	26.8	4.8	-0.01
217	0.9(DL+COLL+CL-4)+1.6(WLLG	-7.5	17.4	3.1	-11.5	26.8	4.8	-0.01
218	0.9(DL+COLL+CL-1)+1.6(WLLG	-3.7	17.4	2.2	-5.7	26.8	3.3	0.00
219	0.9(DL+COLL+CL-2)+1.6(WLLG	-3.7	17.4	2.2	-5.7	26.8	3.3	0.00
220	0.9(DL+COLL+CL-3)+1.6(WLLG	-3.7	17.4	2.2	-5.7	26.8	3.3	0.00
221	0.9(DL+COLL+CL-4)+1.6(WLLG	-3.7	17.4	2.2	-5.7	26.8	3.3	0.00
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+E	2.1	0.0	0.0	3.2	0.0	0.0	0.00
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+E	2.1	0.0	0.0	3.2	0.0	0.0	0.00
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+E	2.1	0.0	0.0	3.2	0.0	0.0	0.00
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+E	2.1	0.0	0.0	3.2	0.0	0.0	0.00
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+E	2.1	0.0	0.0	3.2	0.0	0.0	0.00
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+E	2.1	0.0	0.0	3.2	0.0	0.0	0.00
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+E	2.1	0.0	0.0	3.2	0.0	0.0	0.00
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+E	2.1	0.0	0.0	3.2	0.0	0.0	0.00

COLUMN DESIGN AND INTERACTION RATIO CHECK

Reinforcement: Provide,

4-Nos	25 mm Dia
6-Nos	20 mm Dia
6-Nos	20 mm Dia

1964.29 Sq mm (At Corner) Asc pvd. = 5735.71 Sq mm
1885.71 Sq mm (Along Z axis)
1885.71 Sq mm (Along X axis) Total Asc = 1.02 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1897.2738 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8 (COLUMNS 3.3.3)		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
101	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
102	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
103	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
104	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
105	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
106	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
107	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
108	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
109	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
110	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
111	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
112	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
113	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
114	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
115	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
116	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
117	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
118	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
119	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
120	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
121	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
122	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
123	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
124	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
125	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
126	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
127	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K

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128	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
129	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
130	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
131	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
132	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
133	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
134	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
135	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
136	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
137	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
138	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
139	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
140	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
141	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
142	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
143	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
144	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
145	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
146	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
147	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
148	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
149	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
150	0.05	0.05	73.8	73.8	0.55	1.16	0.31	O.K
151	0.05	0.05	73.8	73.8	0.55	1.16	0.31	O.K
152	0.05	0.05	73.8	73.8	0.55	1.16	0.31	O.K
153	0.05	0.05	73.8	73.8	0.55	1.16	0.31	O.K
154	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
155	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
156	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
157	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
158	0.05	0.05	73.8	73.8	0.55	1.16	0.38	O.K
159	0.05	0.05	73.8	73.8	0.55	1.16	0.38	O.K
160	0.05	0.05	73.8	73.8	0.55	1.16	0.38	O.K
161	0.05	0.05	73.8	73.8	0.55	1.16	0.38	O.K
162	0.05	0.05	73.8	73.8	0.55	1.16	0.36	O.K
163	0.05	0.05	73.8	73.8	0.55	1.16	0.36	O.K
164	0.05	0.05	73.8	73.8	0.55	1.16	0.36	O.K
165	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
166	0.05	0.05	73.8	73.8	0.55	1.16	0.33	O.K
167	0.05	0.05	73.8	73.8	0.55	1.16	0.33	O.K
168	0.05	0.05	73.8	73.8	0.55	1.16	0.33	O.K
169	0.05	0.05	73.8	73.8	0.55	1.16	0.33	O.K
170	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
171	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
172	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
173	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
174	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
175	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
176	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
177	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
178	0.05	0.05	73.8	73.8	0.55	1.16	0.34	O.K
179	0.05	0.05	73.8	73.8	0.55	1.16	0.34	O.K
180	0.05	0.05	73.8	73.8	0.55	1.16	0.34	O.K
181	0.05	0.05	73.8	73.8	0.55	1.16	0.34	O.K
182	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
183	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
184	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
185	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
186	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
187	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
188	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
189	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
190	0.05	0.05	73.8	73.8	0.55	1.16	0.31	O.K

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191	0.05	0.05	73.8	73.8	0.55	1.16	0.31	O.K
192	0.05	0.05	73.8	73.8	0.55	1.16	0.31	O.K
193	0.05	0.05	73.8	73.8	0.55	1.16	0.31	O.K
194	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
195	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
196	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
197	0.05	0.05	73.8	73.8	0.55	1.16	0.11	O.K
198	0.05	0.05	73.8	73.8	0.55	1.16	0.38	O.K
199	0.05	0.05	73.8	73.8	0.55	1.16	0.38	O.K
200	0.05	0.05	73.8	73.8	0.55	1.16	0.38	O.K
201	0.05	0.05	73.8	73.8	0.55	1.16	0.38	O.K
202	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
203	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
204	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
205	0.05	0.05	73.8	73.8	0.55	1.16	0.16	O.K
206	0.05	0.05	73.8	73.8	0.55	1.16	0.33	O.K
207	0.05	0.05	73.8	73.8	0.55	1.16	0.33	O.K
208	0.05	0.05	73.8	73.8	0.55	1.16	0.33	O.K
209	0.05	0.05	73.8	73.8	0.55	1.16	0.33	O.K
210	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
211	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
212	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
213	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
214	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
215	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
216	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
217	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
218	0.05	0.05	73.8	73.8	0.55	1.16	0.34	O.K
219	0.05	0.05	73.8	73.8	0.55	1.16	0.34	O.K
220	0.05	0.05	73.8	73.8	0.55	1.16	0.34	O.K
221	0.05	0.05	73.8	73.8	0.55	1.16	0.34	O.K
222	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
223	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
224	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
225	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
226	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
227	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
228	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
229	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
						Max =	**0.38**	**O.K**

Links, 10mm dia @ 300 C/C

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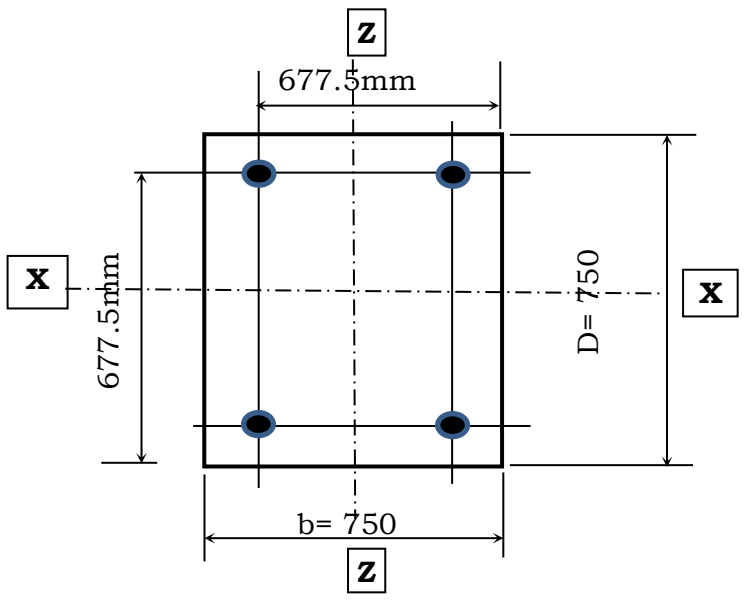
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5.11 DESIGN OF COLUMN MKD C4

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	72.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	1.00%
Max. Steel Ratio	4.00%
Dimension parallel to X-X axis	750mm
Dimension parallel to Z-Z axis	750mm

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	2300mm	γ_x =	0.83
Unsupproted length of column about Z (l_{uz}) =	2300mm	γ_z =	0.83

Where, Effective length factor K= 2

Slenderness ratio,	K l_{ux} / r_x =	20.44	< 22 SECTION 6.2.5a	Radius of gyration r_x=	225	mm
	K l_{uz} / r_z =	20.44	< 22 SECTION 6.2.5a	Radius of gyration r_z=	225	mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	11.4432		-4.4832		0
2	COLLATERAL LOAD (COLL)	-0.0048		-0.0048		0
3	ROOF LIVE LOAD (RLL)	31.74		-14.0124		0
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	-0.3456		0.1128		0
5	CRANE LOAD-1 (CL-1)	0		0		0
6	CRANE LOAD-2 (CL-2)	0		0		0
7	CRANE LOAD-3 (CL-3)	0		0		0
8	CRANE LOAD-4 (CL-4)	0		0		0
9	WIND LEFT-1 (WLL-1)	24.1668		-12.5448		0
10	WIND LEFT-2 (WLL-2)	-14.4024		9.132		0
11	WIND RIGHT-1 (WLR-1)	59.2452		-22.59		0
12	WIND RIGHT-2 (WLR-2)	-54.1632		20.9604		0
13	WIND LONG-1 (WLLG-1)	-17.5764	8.22		0	
14	WIND LONG-2 (WLLG-2)	-34.266	15.2148		0	
15	WIND LONG-3 (WLLG-3)	-17.394	8.0052		0	
16	WIND LONG-4 (WLLG-4)	-34.0836	15		0	
17	SEISMIC LOAD SL'+'	0.174	-0.2232	-0.2232	0	0
18	SEISMIC LOAD SL'-'	-0.174	0.2232	0.2232	0	0
19	TEMPERATURE LOAD (TL1)	0.408		-0.9384		0
20	TEMPERATURE LOAD (TL2)	-0.408		0.9384		0
21	BRACING REACTION (BR-WLLG1)	0	0		0	
22	BRACING REACTION (BR-WLLG2)	0	0		0	
23	BRACING REACTION (BR-WLLG3)	0	0		0	
24	BRACING REACTION (BR-WLLG4)	0	0		0	
25	BRACING REACTION (BR-SL'+')	0	0	0	0	0
26	BRACING REACTION (BR-SL'-')	0	0	0	0	0

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4(DL+COLL)	1.60	0.0	1.4	2.5	0.0	2.2	0.00
102	1.2(DL+COLL+CL-1)+1.6(RLL+P	6.4	0.0	6.4	9.8	0.0	9.8	0.00
103	1.2(DL+COLL+CL-2)+1.6(RLL+P	6.4	0.0	6.4	9.8	0.0	9.8	0.00
104	1.2(DL+COLL+CL-3)+1.6(RLL+P	6.4	0.0	6.4	9.8	0.0	9.8	0.00
105	1.2(DL+COLL+CL-4)+1.6(RLL+P	6.4	0.0	6.4	9.8	0.0	9.8	0.00
106	1.4(DL+COLL+CL-1)	1.6	0.0	1.4	2.5	0.0	2.2	0.00
107	1.4(DL+COLL+CL-2)	1.6	0.0	1.4	2.5	0.0	2.2	0.00
108	1.4(DL+COLL+CL-3)	1.6	0.0	1.4	2.5	0.0	2.2	0.00
109	1.4(DL+COLL+CL-4)	1.6	0.0	1.4	2.5	0.0	2.2	0.00
110	1.2(DL+COLL+CL-1+TL1)+1.6(R	6.4	0.0	6.6	9.9	0.0	10.2	0.01
111	1.2(DL+COLL+CL-2+TL1)+1.6(R	6.4	0.0	6.6	9.9	0.0	10.2	0.01
112	1.2(DL+COLL+CL-3+TL1)+1.6(R	6.4	0.0	6.6	9.9	0.0	10.2	0.01
113	1.2(DL+COLL+CL-4+TL1)+1.6(R	6.4	0.0	6.6	9.9	0.0	10.2	0.01
114	1.2(DL+COLL+CL-1+TL2)+1.6(R	6.3	0.0	6.1	9.8	0.0	9.4	0.00
115	1.2(DL+COLL+CL-2+TL2)+1.6(R	6.3	0.0	6.1	9.8	0.0	9.4	0.00
116	1.2(DL+COLL+CL-3+TL2)+1.6(R	6.3	0.0	6.1	9.8	0.0	9.4	0.00
117	1.2(DL+COLL+CL-4+TL2)+1.6(R	6.3	0.0	6.1	9.8	0.0	9.4	0.00
118	1.2(DL+COLL+CL-1)+0.8(WLL-1	3.3	0.0	3.5	5.1	0.0	5.5	0.00
119	1.2(DL+COLL+CL-2)+0.8(WLL-1	3.3	0.0	3.5	5.1	0.0	5.5	0.00
120	1.2(DL+COLL+CL-3)+0.8(WLL-1	3.3	0.0	3.5	5.1	0.0	5.5	0.00
121	1.2(DL+COLL+CL-4)+0.8(WLL-1	3.3	0.0	3.5	5.1	0.0	5.5	0.00
122	1.2(DL+COLL+CL-1)+0.8(WLL-2	0.2	0.0	0.4	0.3	0.0	0.7	0.00
123	1.2(DL+COLL+CL-2)+0.8(WLL-2	0.2	0.0	0.4	0.3	0.0	0.7	0.00
124	1.2(DL+COLL+CL-3)+0.8(WLL-2	0.2	0.0	0.4	0.3	0.0	0.7	0.00
125	1.2(DL+COLL+CL-4)+0.8(WLL-2	0.2	0.0	0.4	0.3	0.0	0.7	0.00
126	1.2(DL+COLL+CL-1)+0.8(WLR-1	6.1	0.0	5.4	9.4	0.0	8.3	0.00
127	1.2(DL+COLL+CL-2)+0.8(WLR-1	6.1	0.0	5.4	9.4	0.0	8.3	0.00
128	1.2(DL+COLL+CL-3)+0.8(WLR-1	6.1	0.0	5.4	9.4	0.0	8.3	0.00
129	1.2(DL+COLL+CL-4)+0.8(WLR-1	6.1	0.0	5.4	9.4	0.0	8.3	0.00
130	1.2(DL+COLL+CL-1)+0.8(WLR-2	-3.0	0.0	2.6	-4.6	0.0	4.0	0.00
131	1.2(DL+COLL+CL-2)+0.8(WLR-2	-3.0	0.0	2.6	-4.6	0.0	4.0	0.00
132	1.2(DL+COLL+CL-3)+0.8(WLR-2	-3.0	0.0	2.6	-4.6	0.0	4.0	0.00
133	1.2(DL+COLL+CL-4)+0.8(WLR-2	-3.0	0.0	2.6	-4.6	0.0	4.0	0.00
134	1.2(DL+COLL+CL-1)+0.8(WLLG-	0.0	1.5	1.2	-0.1	2.3	1.9	0.00
135	1.2(DL+COLL+CL-2)+0.8(WLLG-	0.0	1.5	1.2	-0.1	2.3	1.9	0.00
136	1.2(DL+COLL+CL-3)+0.8(WLLG-	0.0	1.5	1.2	-0.1	2.3	1.9	0.00
137	1.2(DL+COLL+CL-4)+0.8(WLLG-	0.0	1.5	1.2	-0.1	2.3	1.9	0.00
138	1.2(DL+COLL+CL-1)+0.8(WLLG-	-1.4	2.8	1.2	-2.1	4.3	1.9	0.00
139	1.2(DL+COLL+CL-2)+0.8(WLLG-	-1.4	2.8	1.2	-2.1	4.3	1.9	0.00
140	1.2(DL+COLL+CL-3)+0.8(WLLG-	-1.4	2.8	1.2	-2.1	4.3	1.9	0.00
141	1.2(DL+COLL+CL-4)+0.8(WLLG-	-1.4	2.8	1.2	-2.1	4.3	1.9	0.00
142	1.2(DL+COLL+CL-1)+0.8(WLLG-	0.0	1.5	1.2	0.0	2.3	1.9	0.00
143	1.2(DL+COLL+CL-2)+0.8(WLLG-	0.0	1.5	1.2	0.0	2.3	1.9	0.00
144	1.2(DL+COLL+CL-3)+0.8(WLLG-	0.0	1.5	1.2	0.0	2.3	1.9	0.00
145	1.2(DL+COLL+CL-4)+0.8(WLLG-	0.0	1.5	1.2	0.0	2.3	1.9	0.00
146	1.2(DL+COLL+CL-1)+0.8(WLLG-	-1.4	2.8	1.2	-2.1	4.2	1.9	0.00
147	1.2(DL+COLL+CL-2)+0.8(WLLG-	-1.4	2.8	1.2	-2.1	4.2	1.9	0.00
148	1.2(DL+COLL+CL-3)+0.8(WLLG-	-1.4	2.8	1.2	-2.1	4.2	1.9	0.00
149	1.2(DL+COLL+CL-4)+0.8(WLLG-	-1.4	2.8	1.2	-2.1	4.2	1.9	0.00
150	1.2(DL+COLL+CL-1)+1.6(WLL-1	6.8	0.0	7.5	10.5	0.0	11.5	0.01
151	1.2(DL+COLL+CL-2)+1.6(WLL-1	6.8	0.0	7.5	10.5	0.0	11.5	0.01
152	1.2(DL+COLL+CL-3)+1.6(WLL-1	6.8	0.0	7.5	10.5	0.0	11.5	0.01

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153	1.2(DL+COLL+CL-4)+1.6(WLL-1	6.8	0.0	7.5	10.5	0.0	11.5	0.01
154	1.2(DL+COLL+CL-1)+1.6(WLL-2	0.6	0.0	0.5	1.0	0.0	0.8	0.00
155	1.2(DL+COLL+CL-2)+1.6(WLL-2	0.6	0.0	0.5	1.0	0.0	0.8	0.00
156	1.2(DL+COLL+CL-3)+1.6(WLL-2	0.6	0.0	0.5	1.0	0.0	0.8	0.00
157	1.2(DL+COLL+CL-4)+1.6(WLL-2	0.6	0.0	0.5	1.0	0.0	0.8	0.00
158	1.2(DL+COLL+CL-1)+1.6(WLR-1	12.4	0.0	11.2	19.1	0.0	17.2	0.01
159	1.2(DL+COLL+CL-2)+1.6(WLR-1	12.4	0.0	11.2	19.1	0.0	17.2	0.01
160	1.2(DL+COLL+CL-3)+1.6(WLR-1	12.4	0.0	11.2	19.1	0.0	17.2	0.01
161	1.2(DL+COLL+CL-4)+1.6(WLR-1	12.4	0.0	11.2	19.1	0.0	17.2	0.01
162	1.2(DL+COLL+CL-1)+1.6(WLR-2	-5.7	0.0	4.9	-8.8	0.0	7.5	0.00
163	1.2(DL+COLL+CL-2)+1.6(WLR-2	-5.7	0.0	4.9	-8.8	0.0	7.5	0.00
164	1.2(DL+COLL+CL-3)+1.6(WLR-2	-5.7	0.0	4.9	-8.8	0.0	7.5	0.00
165	1.2(DL+COLL+CL-4)+1.6(WLR-2	-5.7	0.0	4.9	-8.8	0.0	7.5	0.00
166	1.2(DL+COLL+CL-1)+1.6(WLLG-	0.1	3.0	2.8	0.2	4.7	4.4	0.00
167	1.2(DL+COLL+CL-2)+1.6(WLLG-	0.1	3.0	2.8	0.2	4.7	4.4	0.00
168	1.2(DL+COLL+CL-3)+1.6(WLLG-	0.1	3.0	2.8	0.2	4.7	4.4	0.00
169	1.2(DL+COLL+CL-4)+1.6(WLLG-	0.1	3.0	2.8	0.2	4.7	4.4	0.00
170	1.2(DL+COLL+CL-1)+1.6(WLLG-	-2.5	5.6	2.8	-3.9	8.6	4.4	0.00
171	1.2(DL+COLL+CL-2)+1.6(WLLG-	-2.5	5.6	2.8	-3.9	8.6	4.4	0.00
172	1.2(DL+COLL+CL-3)+1.6(WLLG-	-2.5	5.6	2.8	-3.9	8.6	4.4	0.00
173	1.2(DL+COLL+CL-4)+1.6(WLLG-	-2.5	5.6	2.8	-3.9	8.6	4.4	0.00
174	1.2(DL+COLL+CL-1)+1.6(WLLG-	0.2	2.9	2.8	0.2	4.5	4.4	0.00
175	1.2(DL+COLL+CL-2)+1.6(WLLG-	0.2	2.9	2.8	0.2	4.5	4.4	0.00
176	1.2(DL+COLL+CL-3)+1.6(WLLG-	0.2	2.9	2.8	0.2	4.5	4.4	0.00
177	1.2(DL+COLL+CL-4)+1.6(WLLG-	0.2	2.9	2.8	0.2	4.5	4.4	0.00
178	1.2(DL+COLL+CL-1)+1.6(WLLG-	-2.5	5.5	2.8	-3.9	8.5	4.4	0.00
179	1.2(DL+COLL+CL-2)+1.6(WLLG-	-2.5	5.5	2.8	-3.9	8.5	4.4	0.00
180	1.2(DL+COLL+CL-3)+1.6(WLLG-	-2.5	5.5	2.8	-3.9	8.5	4.4	0.00
181	1.2(DL+COLL+CL-4)+1.6(WLLG-	-2.5	5.5	2.8	-3.9	8.5	4.4	0.00
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+I	3.0	0.1	2.9	4.6	0.1	4.4	0.00
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+I	3.0	0.1	2.9	4.6	0.1	4.4	0.00
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+I	3.0	0.1	2.9	4.6	0.1	4.4	0.00
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+I	3.0	0.1	2.9	4.6	0.1	4.4	0.00
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+E	2.9	0.1	2.8	4.5	0.1	4.3	0.00
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+E	2.9	0.1	2.8	4.5	0.1	4.3	0.00
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+E	2.9	0.1	2.8	4.5	0.1	4.3	0.00
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+E	2.9	0.1	2.8	4.5	0.1	4.3	0.00
190	0.9(DL+COLL+CL-1)+1.6(WLL-1	4.9	0.0	5.5	7.5	0.0	8.5	0.00
191	0.9(DL+COLL+CL-2)+1.6(WLL-1	4.9	0.0	5.5	7.5	0.0	8.5	0.00
192	0.9(DL+COLL+CL-3)+1.6(WLL-1	4.9	0.0	5.5	7.5	0.0	8.5	0.00
193	0.9(DL+COLL+CL-4)+1.6(WLL-1	4.9	0.0	5.5	7.5	0.0	8.5	0.00
194	0.9(DL+COLL+CL-1)+1.6(WLL-2	-1.3	0.0	2.4	-2.0	0.0	3.7	0.00
195	0.9(DL+COLL+CL-2)+1.6(WLL-2	-1.3	0.0	2.4	-2.0	0.0	3.7	0.00
196	0.9(DL+COLL+CL-3)+1.6(WLL-2	-1.3	0.0	2.4	-2.0	0.0	3.7	0.00
197	0.9(DL+COLL+CL-4)+1.6(WLL-2	-1.3	0.0	2.4	-2.0	0.0	3.7	0.00
198	0.9(DL+COLL+CL-1)+1.6(WLR-1	10.5	0.0	9.2	16.2	0.0	14.2	0.01
199	0.9(DL+COLL+CL-2)+1.6(WLR-1	10.5	0.0	9.2	16.2	0.0	14.2	0.01
200	0.9(DL+COLL+CL-3)+1.6(WLR-1	10.5	0.0	9.2	16.2	0.0	14.2	0.01
201	0.9(DL+COLL+CL-4)+1.6(WLR-1	10.5	0.0	9.2	16.2	0.0	14.2	0.01
202	0.9(DL+COLL+CL-1)+1.6(WLR-2	-7.6	0.0	6.8	-11.7	0.0	10.4	-0.01
203	0.9(DL+COLL+CL-2)+1.6(WLR-2	-7.6	0.0	6.8	-11.7	0.0	10.4	-0.01
204	0.9(DL+COLL+CL-3)+1.6(WLR-2	-7.6	0.0	6.8	-11.7	0.0	10.4	-0.01
205	0.9(DL+COLL+CL-4)+1.6(WLR-2	-7.6	0.0	6.8	-11.7	0.0	10.4	-0.01
206	0.9(DL+COLL+CL-1)+1.6(WLLG-	-1.8	3.0	0.9	-2.7	4.7	1.4	0.00
207	0.9(DL+COLL+CL-2)+1.6(WLLG-	-1.8	3.0	0.9	-2.7	4.7	1.4	0.00
208	0.9(DL+COLL+CL-3)+1.6(WLLG-	-1.8	3.0	0.9	-2.7	4.7	1.4	0.00
209	0.9(DL+COLL+CL-4)+1.6(WLLG-	-1.8	3.0	0.9	-2.7	4.7	1.4	0.00

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210	0.9(DL+COLL+CL-1)+1.6(WLLG	-4.5	5.6	0.9	-6.9	8.6	1.4	0.00
211	0.9(DL+COLL+CL-2)+1.6(WLLG	-4.5	5.6	0.9	-6.9	8.6	1.4	0.00
212	0.9(DL+COLL+CL-3)+1.6(WLLG	-4.5	5.6	0.9	-6.9	8.6	1.4	0.00
213	0.9(DL+COLL+CL-4)+1.6(WLLG	-4.5	5.6	0.9	-6.9	8.6	1.4	0.00
214	0.9(DL+COLL+CL-1)+1.6(WLLG	-1.8	2.9	0.9	-2.7	4.5	1.4	0.00
215	0.9(DL+COLL+CL-2)+1.6(WLLG	-1.8	2.9	0.9	-2.7	4.5	1.4	0.00
216	0.9(DL+COLL+CL-3)+1.6(WLLG	-1.8	2.9	0.9	-2.7	4.5	1.4	0.00
217	0.9(DL+COLL+CL-4)+1.6(WLLG	-1.8	2.9	0.9	-2.7	4.5	1.4	0.00
218	0.9(DL+COLL+CL-1)+1.6(WLLG	-4.4	5.5	0.9	-6.8	8.5	1.4	0.00
219	0.9(DL+COLL+CL-2)+1.6(WLLG	-4.4	5.5	0.9	-6.8	8.5	1.4	0.00
220	0.9(DL+COLL+CL-3)+1.6(WLLG	-4.4	5.5	0.9	-6.8	8.5	1.4	0.00
221	0.9(DL+COLL+CL-4)+1.6(WLLG	-4.4	5.5	0.9	-6.8	8.5	1.4	0.00
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+E	1.0	0.1	1.0	1.6	0.1	1.5	0.00
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+E	1.0	0.1	1.0	1.6	0.1	1.5	0.00
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+E	1.0	0.1	1.0	1.6	0.1	1.5	0.00
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+E	1.0	0.1	1.0	1.6	0.1	1.5	0.00
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+E	1.0	0.1	0.9	1.6	0.1	1.4	0.00
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+E	1.0	0.1	0.9	1.6	0.1	1.4	0.00
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+E	1.0	0.1	0.9	1.6	0.1	1.4	0.00
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+E	1.0	0.1	0.9	1.6	0.1	1.4	0.00

COLUMN DESIGN AND INTERACTION RATIO CHECK

Reinforcement: Provide,

4-Nos	25 mm Dia
6-Nos	20 mm Dia
6-Nos	20 mm Dia

1964.29 Sq mm (At Corner) Asc pvd. = 5735.71 Sq mm
1885.71 Sq mm (Along Z axis)
1885.71 Sq mm (Along X axis) Total Asc = 1.02 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1897.2738 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8 (COLUMNS 3.3.3)		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
101	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
102	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
103	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
104	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
105	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
106	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
107	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
108	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
109	0.05	0.05	73.8	73.8	0.55	1.16	0.02	O.K
110	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
111	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
112	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
113	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
114	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
115	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
116	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
117	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
118	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
119	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
120	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
121	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
122	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
123	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
124	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
125	0.05	0.05	73.8	73.8	0.55	1.16	0.00	O.K
126	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
127	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K

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128	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
129	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
130	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
131	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
132	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
133	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
134	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
135	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
136	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
137	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
138	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
139	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
140	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
141	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
142	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
143	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
144	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
145	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
146	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
147	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
148	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
149	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
150	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
151	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
152	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
153	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
154	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
155	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
156	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
157	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
158	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
159	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
160	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
161	0.05	0.05	73.8	73.8	0.55	1.16	0.18	O.K
162	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
163	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
164	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
165	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
166	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
167	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
168	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
169	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
170	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
171	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
172	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
173	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
174	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
175	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
176	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
177	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
178	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
179	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
180	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
181	0.05	0.05	73.8	73.8	0.55	1.16	0.12	O.K
182	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
183	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
184	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
185	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
186	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
187	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
188	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
189	0.05	0.05	73.8	73.8	0.55	1.16	0.04	O.K
190	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K

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191	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
192	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
193	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
194	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
195	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
196	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
197	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
198	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
199	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
200	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
201	0.05	0.05	73.8	73.8	0.55	1.16	0.15	O.K
202	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
203	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
204	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
205	0.05	0.05	73.8	73.8	0.55	1.16	0.10	O.K
206	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
207	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
208	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
209	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
210	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
211	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
212	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
213	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
214	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
215	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
216	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
217	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
218	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
219	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
220	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
221	0.05	0.05	73.8	73.8	0.55	1.16	0.09	O.K
222	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
223	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
224	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
225	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
226	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
227	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
228	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
229	0.05	0.05	73.8	73.8	0.55	1.16	0.01	O.K
							Max = **0.18**	**O.K**

Links, 10mm dia @ 300 C/C

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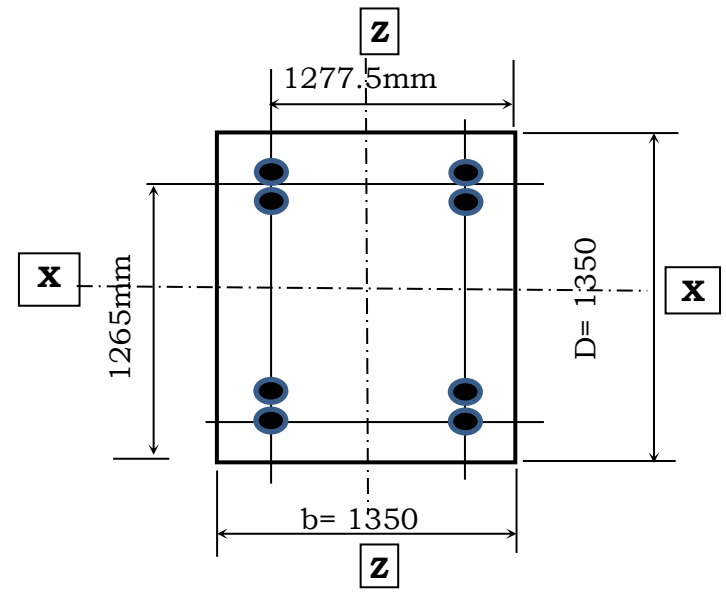
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5.12 DESIGN OF COLUMN MKD C5

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	85mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	1.00%
Max. Steel Ratio	4.00%
Dimension parallel to X-X axis	1350mm
Dimension parallel to Z-Z axis	1350mm

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) = 2000mm γ_x = 0.89

Unsupproted length of column about Z (l_{uz}) = 2000mm γ_z = 0.91

Where, Effective length factor K= 2

Slenderness ratio, K l_{ux} / r_x = 9.88 < 22 SECTION 6.2.5a Radius of gyration r_x= 405 mm

K l_{uz} / r_z = 9.88 < 22 SECTION 6.2.5a Radius of gyration r_z= 405 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	533.598		-154.95		-207.016
2	COLLATERAL LOAD (COLL)	-0.0048		-0.0048		0.0048
3	ROOF LIVE LOAD (RLL)	114.132		-31.7076		-114.187
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	1539.994		-445.7676		-479.687
5	CRANE LOAD-1 (CL-1)	0		0		0
6	CRANE LOAD-2 (CL-2)	0		0		0
7	CRANE LOAD-3 (CL-3)	0		0		0
8	CRANE LOAD-4 (CL-4)	0		0		0
9	WIND LEFT-1 (WLL-1)	-40.7604		76.8		190.832
10	WIND LEFT-2 (WLL-2)	-109.5012		12.198		63.2544
11	WIND RIGHT-1 (WLR-1)	-117.4008		115.3872		311.694
12	WIND RIGHT-2 (WLR-2)	-104.0304		34.7412		116.862
13	WIND LONG-1 (WLLG-1)	-189.3672	86.5608		272.4492	
14	WIND LONG-2 (WLLG-2)	-143.3808	44.7336		159.9156	
15	WIND LONG-3 (WLLG-3)	-208.0128	73.3788		224.2368	
16	WIND LONG-4 (WLLG-4)	-162.0264	31.5516		111.702	
17	SEISMIC LOAD SL'+'	6.606	-23.5272	-23.5272	-69.4092	-69.4092
18	SEISMIC LOAD SL'-'	-6.606	23.5272	23.5272	69.4092	69.4092
19	TEMPERATURE LOAD (TL1)	-10.194		-125.8584		-222.756
20	TEMPERATURE LOAD (TL2)	10.194		125.8584		222.756
21	BRACING REACTION (BR-WLLG1)	51.7	-41		0	
22	BRACING REACTION (BR-WLLG2)	51.7	-41		0	
23	BRACING REACTION (BR-WLLG3)	-51.7	41		0	
24	BRACING REACTION (BR-WLLG4)	-51.7	41		0	
25	BRACING REACTION (BR-SL'+')	91.34	-96.33	-96.33	0	0
26	BRACING REACTION (BR-SL'-')	-91.34	96.33	96.33	0	0

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4(DL+COLL)	74.70	0.0	72.4	114.9	0.0	111.3	0.02
102	1.2(DL+COLL+CL-1)+1.6(RLL+P	328.7	0.0	309.8	505.7	0.0	476.7	0.08
103	1.2(DL+COLL+CL-2)+1.6(RLL+P	328.7	0.0	309.8	505.7	0.0	476.7	0.08
104	1.2(DL+COLL+CL-3)+1.6(RLL+P	328.7	0.0	309.8	505.7	0.0	476.7	0.08
105	1.2(DL+COLL+CL-4)+1.6(RLL+P	328.7	0.0	309.8	505.7	0.0	476.7	0.08
106	1.4(DL+COLL+CL-1)	74.7	0.0	72.4	114.9	0.0	111.3	0.02
107	1.4(DL+COLL+CL-2)	74.7	0.0	72.4	114.9	0.0	111.3	0.02
108	1.4(DL+COLL+CL-3)	74.7	0.0	72.4	114.9	0.0	111.3	0.02
109	1.4(DL+COLL+CL-4)	74.7	0.0	72.4	114.9	0.0	111.3	0.02
110	1.2(DL+COLL+CL-1+TL1)+1.6(R	327.5	0.0	366.8	503.8	0.0	564.3	0.08
111	1.2(DL+COLL+CL-2+TL1)+1.6(R	327.5	0.0	366.8	503.8	0.0	564.3	0.08
112	1.2(DL+COLL+CL-3+TL1)+1.6(R	327.5	0.0	366.8	503.8	0.0	564.3	0.08
113	1.2(DL+COLL+CL-4+TL1)+1.6(R	327.5	0.0	366.8	503.8	0.0	564.3	0.08
114	1.2(DL+COLL+CL-1+TL2)+1.6(R	329.9	0.0	252.9	507.6	0.0	389.1	0.08
115	1.2(DL+COLL+CL-2+TL2)+1.6(R	329.9	0.0	252.9	507.6	0.0	389.1	0.08
116	1.2(DL+COLL+CL-3+TL2)+1.6(R	329.9	0.0	252.9	507.6	0.0	389.1	0.08
117	1.2(DL+COLL+CL-4+TL2)+1.6(R	329.9	0.0	252.9	507.6	0.0	389.1	0.08
118	1.2(DL+COLL+CL-1)+0.8(WLL-1	60.8	0.0	34.5	93.5	0.0	53.0	0.01
119	1.2(DL+COLL+CL-2)+0.8(WLL-1	60.8	0.0	34.5	93.5	0.0	53.0	0.01
120	1.2(DL+COLL+CL-3)+0.8(WLL-1	60.8	0.0	34.5	93.5	0.0	53.0	0.01
121	1.2(DL+COLL+CL-4)+0.8(WLL-1	60.8	0.0	34.5	93.5	0.0	53.0	0.01
122	1.2(DL+COLL+CL-1)+0.8(WLL-2	55.3	0.0	55.0	85.0	0.0	84.6	0.01
123	1.2(DL+COLL+CL-2)+0.8(WLL-2	55.3	0.0	55.0	85.0	0.0	84.6	0.01
124	1.2(DL+COLL+CL-3)+0.8(WLL-2	55.3	0.0	55.0	85.0	0.0	84.6	0.01
125	1.2(DL+COLL+CL-4)+0.8(WLL-2	55.3	0.0	55.0	85.0	0.0	84.6	0.01
126	1.2(DL+COLL+CL-1)+0.8(WLR-1	54.6	0.0	18.6	84.1	0.0	28.7	0.01
127	1.2(DL+COLL+CL-2)+0.8(WLR-1	54.6	0.0	18.6	84.1	0.0	28.7	0.01
128	1.2(DL+COLL+CL-3)+0.8(WLR-1	54.6	0.0	18.6	84.1	0.0	28.7	0.01
129	1.2(DL+COLL+CL-4)+0.8(WLR-1	54.6	0.0	18.6	84.1	0.0	28.7	0.01
130	1.2(DL+COLL+CL-1)+0.8(WLR-2	55.7	0.0	47.1	85.7	0.0	72.5	0.01
131	1.2(DL+COLL+CL-2)+0.8(WLR-2	55.7	0.0	47.1	85.7	0.0	72.5	0.01
132	1.2(DL+COLL+CL-3)+0.8(WLR-2	55.7	0.0	47.1	85.7	0.0	72.5	0.01
133	1.2(DL+COLL+CL-4)+0.8(WLR-2	55.7	0.0	47.1	85.7	0.0	72.5	0.01
134	1.2(DL+COLL+CL-1)+0.8(WLLG	53.0	29.1	62.0	81.6	44.7	95.4	0.01
135	1.2(DL+COLL+CL-2)+0.8(WLLG	53.0	29.1	62.0	81.6	44.7	95.4	0.01
136	1.2(DL+COLL+CL-3)+0.8(WLLG	53.0	29.1	62.0	81.6	44.7	95.4	0.01
137	1.2(DL+COLL+CL-4)+0.8(WLLG	53.0	29.1	62.0	81.6	44.7	95.4	0.01
138	1.2(DL+COLL+CL-1)+0.8(WLLG	56.7	13.4	62.0	87.2	20.6	95.4	0.01
139	1.2(DL+COLL+CL-2)+0.8(WLLG	56.7	13.4	62.0	87.2	20.6	95.4	0.01
140	1.2(DL+COLL+CL-3)+0.8(WLLG	56.7	13.4	62.0	87.2	20.6	95.4	0.01
141	1.2(DL+COLL+CL-4)+0.8(WLLG	56.7	13.4	62.0	87.2	20.6	95.4	0.01
142	1.2(DL+COLL+CL-1)+0.8(WLLG	43.3	36.2	62.0	66.5	55.8	95.4	0.01
143	1.2(DL+COLL+CL-2)+0.8(WLLG	43.3	36.2	62.0	66.5	55.8	95.4	0.01
144	1.2(DL+COLL+CL-3)+0.8(WLLG	43.3	36.2	62.0	66.5	55.8	95.4	0.01
145	1.2(DL+COLL+CL-4)+0.8(WLLG	43.3	36.2	62.0	66.5	55.8	95.4	0.01
146	1.2(DL+COLL+CL-1)+0.8(WLLG	46.9	20.5	62.0	72.2	31.6	95.4	0.01
147	1.2(DL+COLL+CL-2)+0.8(WLLG	46.9	20.5	62.0	72.2	31.6	95.4	0.01
148	1.2(DL+COLL+CL-3)+0.8(WLLG	46.9	20.5	62.0	72.2	31.6	95.4	0.01
149	1.2(DL+COLL+CL-4)+0.8(WLLG	46.9	20.5	62.0	72.2	31.6	95.4	0.01
150	1.2(DL+COLL+CL-1)+1.6(WLL-1	140.2	0.0	84.4	215.7	0.0	129.8	0.03
151	1.2(DL+COLL+CL-2)+1.6(WLL-1	140.2	0.0	84.4	215.7	0.0	129.8	0.03
152	1.2(DL+COLL+CL-3)+1.6(WLL-1	140.2	0.0	84.4	215.7	0.0	129.8	0.03

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153	1.2(DL+COLL+CL-4)+1.6(WLL-1	140.2	0.0	84.4	215.7	0.0	129.8	0.03
154	1.2(DL+COLL+CL-1)+1.6(WLL-2	129.2	0.0	125.4	198.8	0.0	193.0	0.03
155	1.2(DL+COLL+CL-2)+1.6(WLL-2	129.2	0.0	125.4	198.8	0.0	193.0	0.03
156	1.2(DL+COLL+CL-3)+1.6(WLL-2	129.2	0.0	125.4	198.8	0.0	193.0	0.03
157	1.2(DL+COLL+CL-4)+1.6(WLL-2	129.2	0.0	125.4	198.8	0.0	193.0	0.03
158	1.2(DL+COLL+CL-1)+1.6(WLR-1	128.0	0.0	52.7	196.9	0.0	81.0	0.03
159	1.2(DL+COLL+CL-2)+1.6(WLR-1	128.0	0.0	52.7	196.9	0.0	81.0	0.03
160	1.2(DL+COLL+CL-3)+1.6(WLR-1	128.0	0.0	52.7	196.9	0.0	81.0	0.03
161	1.2(DL+COLL+CL-4)+1.6(WLR-1	128.0	0.0	52.7	196.9	0.0	81.0	0.03
162	1.2(DL+COLL+CL-1)+1.6(WLR-2	130.1	0.0	109.7	200.1	0.0	168.7	0.03
163	1.2(DL+COLL+CL-2)+1.6(WLR-2	130.1	0.0	109.7	200.1	0.0	168.7	0.03
164	1.2(DL+COLL+CL-3)+1.6(WLR-2	130.1	0.0	109.7	200.1	0.0	168.7	0.03
165	1.2(DL+COLL+CL-4)+1.6(WLR-2	130.1	0.0	109.7	200.1	0.0	168.7	0.03
166	1.2(DL+COLL+CL-1)+1.6(WLLG-	124.7	58.2	139.5	191.9	89.5	214.6	0.03
167	1.2(DL+COLL+CL-2)+1.6(WLLG-	124.7	58.2	139.5	191.9	89.5	214.6	0.03
168	1.2(DL+COLL+CL-3)+1.6(WLLG-	124.7	58.2	139.5	191.9	89.5	214.6	0.03
169	1.2(DL+COLL+CL-4)+1.6(WLLG-	124.7	58.2	139.5	191.9	89.5	214.6	0.03
170	1.2(DL+COLL+CL-1)+1.6(WLLG-	132.1	26.8	139.5	203.2	41.2	214.6	0.03
171	1.2(DL+COLL+CL-2)+1.6(WLLG-	132.1	26.8	139.5	203.2	41.2	214.6	0.03
172	1.2(DL+COLL+CL-3)+1.6(WLLG-	132.1	26.8	139.5	203.2	41.2	214.6	0.03
173	1.2(DL+COLL+CL-4)+1.6(WLLG-	132.1	26.8	139.5	203.2	41.2	214.6	0.03
174	1.2(DL+COLL+CL-1)+1.6(WLLG-	105.2	72.5	139.5	161.8	111.5	214.6	0.03
175	1.2(DL+COLL+CL-2)+1.6(WLLG-	105.2	72.5	139.5	161.8	111.5	214.6	0.03
176	1.2(DL+COLL+CL-3)+1.6(WLLG-	105.2	72.5	139.5	161.8	111.5	214.6	0.03
177	1.2(DL+COLL+CL-4)+1.6(WLLG-	105.2	72.5	139.5	161.8	111.5	214.6	0.03
178	1.2(DL+COLL+CL-1)+1.6(WLLG-	112.5	41.1	139.5	173.1	63.2	214.6	0.03
179	1.2(DL+COLL+CL-2)+1.6(WLLG-	112.5	41.1	139.5	173.1	63.2	214.6	0.03
180	1.2(DL+COLL+CL-3)+1.6(WLLG-	112.5	41.1	139.5	173.1	63.2	214.6	0.03
181	1.2(DL+COLL+CL-4)+1.6(WLLG-	112.5	41.1	139.5	173.1	63.2	214.6	0.03
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+E	156.5	30.9	170.4	240.8	47.6	262.1	0.04
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+E	156.5	30.9	170.4	240.8	47.6	262.1	0.04
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+E	156.5	30.9	170.4	240.8	47.6	262.1	0.04
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+E	156.5	30.9	170.4	240.8	47.6	262.1	0.04
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+E	136.9	30.9	108.6	210.7	47.6	167.0	0.03
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+E	136.9	30.9	108.6	210.7	47.6	167.0	0.03
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+E	136.9	30.9	108.6	210.7	47.6	167.0	0.03
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+E	136.9	30.9	108.6	210.7	47.6	167.0	0.03
190	0.9(DL+COLL+CL-1)+1.6(WLL-1	41.5	0.0	8.6	63.8	0.0	13.2	0.01
191	0.9(DL+COLL+CL-2)+1.6(WLL-1	41.5	0.0	8.6	63.8	0.0	13.2	0.01
192	0.9(DL+COLL+CL-3)+1.6(WLL-1	41.5	0.0	8.6	63.8	0.0	13.2	0.01
193	0.9(DL+COLL+CL-4)+1.6(WLL-1	41.5	0.0	8.6	63.8	0.0	13.2	0.01
194	0.9(DL+COLL+CL-1)+1.6(WLL-2	30.5	0.0	32.5	46.9	0.0	50.0	0.01
195	0.9(DL+COLL+CL-2)+1.6(WLL-2	30.5	0.0	32.5	46.9	0.0	50.0	0.01
196	0.9(DL+COLL+CL-3)+1.6(WLL-2	30.5	0.0	32.5	46.9	0.0	50.0	0.01
197	0.9(DL+COLL+CL-4)+1.6(WLL-2	30.5	0.0	32.5	46.9	0.0	50.0	0.01
198	0.9(DL+COLL+CL-1)+1.6(WLR-1	29.2	0.0	40.3	45.0	0.0	62.0	0.01
199	0.9(DL+COLL+CL-2)+1.6(WLR-1	29.2	0.0	40.3	45.0	0.0	62.0	0.01
200	0.9(DL+COLL+CL-3)+1.6(WLR-1	29.2	0.0	40.3	45.0	0.0	62.0	0.01
201	0.9(DL+COLL+CL-4)+1.6(WLR-1	29.2	0.0	40.3	45.0	0.0	62.0	0.01
202	0.9(DL+COLL+CL-1)+1.6(WLR-2	31.4	0.0	16.7	48.3	0.0	25.7	0.01
203	0.9(DL+COLL+CL-2)+1.6(WLR-2	31.4	0.0	16.7	48.3	0.0	25.7	0.01
204	0.9(DL+COLL+CL-3)+1.6(WLR-2	31.4	0.0	16.7	48.3	0.0	25.7	0.01
205	0.9(DL+COLL+CL-4)+1.6(WLR-2	31.4	0.0	16.7	48.3	0.0	25.7	0.01
206	0.9(DL+COLL+CL-1)+1.6(WLLG-	26.0	58.2	46.5	40.0	89.5	71.6	0.01
207	0.9(DL+COLL+CL-2)+1.6(WLLG-	26.0	58.2	46.5	40.0	89.5	71.6	0.01
208	0.9(DL+COLL+CL-3)+1.6(WLLG-	26.0	58.2	46.5	40.0	89.5	71.6	0.01
209	0.9(DL+COLL+CL-4)+1.6(WLLG-	26.0	58.2	46.5	40.0	89.5	71.6	0.01

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210	0.9(DL+COLL+CL-1)+1.6(WLLG	33.4	26.8	46.5	51.3	41.2	71.6	0.01
211	0.9(DL+COLL+CL-2)+1.6(WLLG	33.4	26.8	46.5	51.3	41.2	71.6	0.01
212	0.9(DL+COLL+CL-3)+1.6(WLLG	33.4	26.8	46.5	51.3	41.2	71.6	0.01
213	0.9(DL+COLL+CL-4)+1.6(WLLG	33.4	26.8	46.5	51.3	41.2	71.6	0.01
214	0.9(DL+COLL+CL-1)+1.6(WLLG	6.5	72.5	46.5	10.0	111.5	71.6	0.00
215	0.9(DL+COLL+CL-2)+1.6(WLLG	6.5	72.5	46.5	10.0	111.5	71.6	0.00
216	0.9(DL+COLL+CL-3)+1.6(WLLG	6.5	72.5	46.5	10.0	111.5	71.6	0.00
217	0.9(DL+COLL+CL-4)+1.6(WLLG	6.5	72.5	46.5	10.0	111.5	71.6	0.00
218	0.9(DL+COLL+CL-1)+1.6(WLLG	13.8	41.1	46.5	21.3	63.2	71.6	0.00
219	0.9(DL+COLL+CL-2)+1.6(WLLG	13.8	41.1	46.5	21.3	63.2	71.6	0.00
220	0.9(DL+COLL+CL-3)+1.6(WLLG	13.8	41.1	46.5	21.3	63.2	71.6	0.00
221	0.9(DL+COLL+CL-4)+1.6(WLLG	13.8	41.1	46.5	21.3	63.2	71.6	0.00
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+E	57.8	30.9	77.4	89.0	47.6	119.1	0.01
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+E	57.8	30.9	77.4	89.0	47.6	119.1	0.01
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+E	57.8	30.9	77.4	89.0	47.6	119.1	0.01
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+E	57.8	30.9	77.4	89.0	47.6	119.1	0.01
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+E	38.2	30.9	15.6	58.8	47.6	24.0	0.01
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+E	38.2	30.9	15.6	58.8	47.6	24.0	0.01
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+E	38.2	30.9	15.6	58.8	47.6	24.0	0.01
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+E	38.2	30.9	15.6	58.8	47.6	24.0	0.01

COLUMN INTERACTION RATIO CHECK

Reinforcement: Provide,

4-Nos	25 mm Dia
20-Nos	25 mm Dia
20-Nos	25 mm Dia

1964.29 Sq mm (At Corner) Asc pvd. = 21607.14 Sq mm
9821.43 Sq mm (Along Z axis)
9821.43 Sq mm (Along X axis) Total Asc = 1.2 %

$\beta = P_n / P_0$ (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 6265.1563 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.9 (COLUMNS 3.3.4)		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
101	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
102	0.07	0.07	637.2	637.2	0.55	1.16	0.71	O.K
103	0.07	0.07	637.2	637.2	0.55	1.16	0.71	O.K
104	0.07	0.07	637.2	637.2	0.55	1.16	0.71	O.K
105	0.07	0.07	637.2	637.2	0.55	1.16	0.71	O.K
106	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
107	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
108	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
109	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
110	0.07	0.07	637.2	637.2	0.55	1.16	0.87	O.K
111	0.07	0.07	637.2	637.2	0.55	1.16	0.87	O.K
112	0.07	0.07	637.2	637.2	0.55	1.16	0.87	O.K
113	0.07	0.07	637.2	637.2	0.55	1.16	0.87	O.K
114	0.07	0.07	637.2	637.2	0.55	1.16	0.56	O.K
115	0.07	0.07	637.2	637.2	0.55	1.16	0.56	O.K
116	0.07	0.07	637.2	637.2	0.55	1.16	0.56	O.K
117	0.07	0.07	637.2	637.2	0.55	1.16	0.56	O.K
118	0.07	0.07	637.2	637.2	0.55	1.16	0.06	O.K
119	0.07	0.07	637.2	637.2	0.55	1.16	0.06	O.K
120	0.07	0.07	637.2	637.2	0.55	1.16	0.06	O.K
121	0.07	0.07	637.2	637.2	0.55	1.16	0.06	O.K
122	0.07	0.07	637.2	637.2	0.55	1.16	0.10	O.K
123	0.07	0.07	637.2	637.2	0.55	1.16	0.10	O.K
124	0.07	0.07	637.2	637.2	0.55	1.16	0.10	O.K
125	0.07	0.07	637.2	637.2	0.55	1.16	0.10	O.K
126	0.07	0.07	637.2	637.2	0.55	1.16	0.03	O.K
127	0.07	0.07	637.2	637.2	0.55	1.16	0.03	O.K

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128	0.07	0.07	637.2	637.2	0.55	1.16	0.03	O.K
129	0.07	0.07	637.2	637.2	0.55	1.16	0.03	O.K
130	0.07	0.07	637.2	637.2	0.55	1.16	0.08	O.K
131	0.07	0.07	637.2	637.2	0.55	1.16	0.08	O.K
132	0.07	0.07	637.2	637.2	0.55	1.16	0.08	O.K
133	0.07	0.07	637.2	637.2	0.55	1.16	0.08	O.K
134	0.07	0.07	637.2	637.2	0.55	1.16	0.16	O.K
135	0.07	0.07	637.2	637.2	0.55	1.16	0.16	O.K
136	0.07	0.07	637.2	637.2	0.55	1.16	0.16	O.K
137	0.07	0.07	637.2	637.2	0.55	1.16	0.16	O.K
138	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
139	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
140	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
141	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
142	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
143	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
144	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
145	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
146	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
147	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
148	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
149	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
150	0.07	0.07	637.2	637.2	0.55	1.16	0.16	O.K
151	0.07	0.07	637.2	637.2	0.55	1.16	0.16	O.K
152	0.07	0.07	637.2	637.2	0.55	1.16	0.16	O.K
153	0.07	0.07	637.2	637.2	0.55	1.16	0.16	O.K
154	0.07	0.07	637.2	637.2	0.55	1.16	0.25	O.K
155	0.07	0.07	637.2	637.2	0.55	1.16	0.25	O.K
156	0.07	0.07	637.2	637.2	0.55	1.16	0.25	O.K
157	0.07	0.07	637.2	637.2	0.55	1.16	0.25	O.K
158	0.07	0.07	637.2	637.2	0.55	1.16	0.09	O.K
159	0.07	0.07	637.2	637.2	0.55	1.16	0.09	O.K
160	0.07	0.07	637.2	637.2	0.55	1.16	0.09	O.K
161	0.07	0.07	637.2	637.2	0.55	1.16	0.09	O.K
162	0.07	0.07	637.2	637.2	0.55	1.16	0.21	O.K
163	0.07	0.07	637.2	637.2	0.55	1.16	0.21	O.K
164	0.07	0.07	637.2	637.2	0.55	1.16	0.21	O.K
165	0.07	0.07	637.2	637.2	0.55	1.16	0.21	O.K
166	0.07	0.07	637.2	637.2	0.55	1.16	0.39	O.K
167	0.07	0.07	637.2	637.2	0.55	1.16	0.39	O.K
168	0.07	0.07	637.2	637.2	0.55	1.16	0.39	O.K
169	0.07	0.07	637.2	637.2	0.55	1.16	0.39	O.K
170	0.07	0.07	637.2	637.2	0.55	1.16	0.32	O.K
171	0.07	0.07	637.2	637.2	0.55	1.16	0.32	O.K
172	0.07	0.07	637.2	637.2	0.55	1.16	0.32	O.K
173	0.07	0.07	637.2	637.2	0.55	1.16	0.32	O.K
174	0.07	0.07	637.2	637.2	0.55	1.16	0.42	O.K
175	0.07	0.07	637.2	637.2	0.55	1.16	0.42	O.K
176	0.07	0.07	637.2	637.2	0.55	1.16	0.42	O.K
177	0.07	0.07	637.2	637.2	0.55	1.16	0.42	O.K
178	0.07	0.07	637.2	637.2	0.55	1.16	0.35	O.K
179	0.07	0.07	637.2	637.2	0.55	1.16	0.35	O.K
180	0.07	0.07	637.2	637.2	0.55	1.16	0.35	O.K
181	0.07	0.07	637.2	637.2	0.55	1.16	0.35	O.K
182	0.07	0.07	637.2	637.2	0.55	1.16	0.41	O.K
183	0.07	0.07	637.2	637.2	0.55	1.16	0.41	O.K
184	0.07	0.07	637.2	637.2	0.55	1.16	0.41	O.K
185	0.07	0.07	637.2	637.2	0.55	1.16	0.41	O.K
186	0.07	0.07	637.2	637.2	0.55	1.16	0.26	O.K
187	0.07	0.07	637.2	637.2	0.55	1.16	0.26	O.K
188	0.07	0.07	637.2	637.2	0.55	1.16	0.26	O.K
189	0.07	0.07	637.2	637.2	0.55	1.16	0.26	O.K
190	0.07	0.07	637.2	637.2	0.55	1.16	0.01	O.K

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191	0.07	0.07	637.2	637.2	0.55	1.16	0.01	O.K
192	0.07	0.07	637.2	637.2	0.55	1.16	0.01	O.K
193	0.07	0.07	637.2	637.2	0.55	1.16	0.01	O.K
194	0.07	0.07	637.2	637.2	0.55	1.16	0.05	O.K
195	0.07	0.07	637.2	637.2	0.55	1.16	0.05	O.K
196	0.07	0.07	637.2	637.2	0.55	1.16	0.05	O.K
197	0.07	0.07	637.2	637.2	0.55	1.16	0.05	O.K
198	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
199	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
200	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
201	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
202	0.07	0.07	637.2	637.2	0.55	1.16	0.02	O.K
203	0.07	0.07	637.2	637.2	0.55	1.16	0.02	O.K
204	0.07	0.07	637.2	637.2	0.55	1.16	0.02	O.K
205	0.07	0.07	637.2	637.2	0.55	1.16	0.02	O.K
206	0.07	0.07	637.2	637.2	0.55	1.16	0.18	O.K
207	0.07	0.07	637.2	637.2	0.55	1.16	0.18	O.K
208	0.07	0.07	637.2	637.2	0.55	1.16	0.18	O.K
209	0.07	0.07	637.2	637.2	0.55	1.16	0.18	O.K
210	0.07	0.07	637.2	637.2	0.55	1.16	0.12	O.K
211	0.07	0.07	637.2	637.2	0.55	1.16	0.12	O.K
212	0.07	0.07	637.2	637.2	0.55	1.16	0.12	O.K
213	0.07	0.07	637.2	637.2	0.55	1.16	0.12	O.K
214	0.07	0.07	637.2	637.2	0.55	1.16	0.21	O.K
215	0.07	0.07	637.2	637.2	0.55	1.16	0.21	O.K
216	0.07	0.07	637.2	637.2	0.55	1.16	0.21	O.K
217	0.07	0.07	637.2	637.2	0.55	1.16	0.21	O.K
218	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
219	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
220	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
221	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
222	0.07	0.07	637.2	637.2	0.55	1.16	0.19	O.K
223	0.07	0.07	637.2	637.2	0.55	1.16	0.19	O.K
224	0.07	0.07	637.2	637.2	0.55	1.16	0.19	O.K
225	0.07	0.07	637.2	637.2	0.55	1.16	0.19	O.K
226	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
227	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
228	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
229	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
							Max = **0.87**	**O.K**

Links, 10mm dia @ 300 C/C

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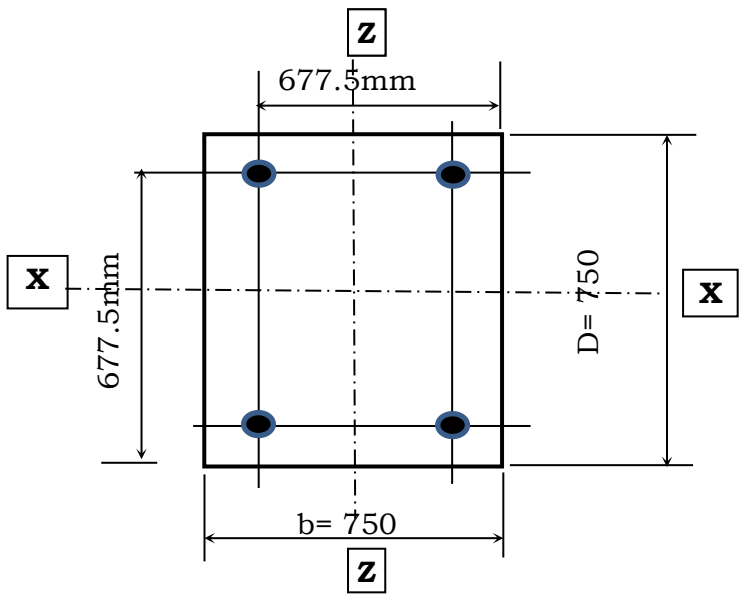
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5.13 DESIGN OF COLUMN MKD C6

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	72.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	1.00%
Max. Steel Ratio	4.00%
Dimension parallel to X-X axis	750mm
Dimension parallel to Z-Z axis	750mm

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	2300mm	γ_x =	0.83
Unsupproted length of column about Z (l_{uz}) =	2300mm	γ_z =	0.83

Where, Effective length factor K= 2

Slenderness ratio,	K l_{ux} / r_x =	20.44	< 22 SECTION 6.2.5a	Radius of gyration r_x=	225	mm
	K l_{uz} / r_z =	20.44	< 22 SECTION 6.2.5a	Radius of gyration r_z=	225	mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	145.698	0.00	-16.4172		
2	COLLATERAL LOAD (COLL)	0	0.00	0		
3	ROOF LIVE LOAD (RLL)	0	0.00	0		
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	438.948	0.00	-51.6228		
5	CRANE LOAD-1 (CL-1)	0	0.00	0		
6	CRANE LOAD-2 (CL-2)	0	0.00	0		
7	CRANE LOAD-3 (CL-3)	0	0.00	0		
8	CRANE LOAD-4 (CL-4)	0	0.00	0		
9	WIND LEFT-1 (WLL-1)	0	0.00	0		
10	WIND LEFT-2 (WLL-2)	0	0.00	0		
11	WIND RIGHT-1 (WLR-1)	0	0.00	0		
12	WIND RIGHT-2 (WLR-2)	0	0.00	0		
13	WIND LONG-1 (WLLG-1)	0	0.00	0		
14	WIND LONG-2 (WLLG-2)	0	0.00	0		
15	WIND LONG-3 (WLLG-3)	0	0.00	0		
16	WIND LONG-4 (WLLG-4)	0	0.00	0		
17	SEISMIC LOAD SL'+'	-19.001	0.00	-23.494		
18	SEISMIC LOAD SL'-'	19.001	0.00	23.494		
19	TEMPERATURE LOAD (TL1)	0	0.00	-15.1896		
20	TEMPERATURE LOAD (TL2)	0	0.00	15.1896		
21	BRACING REACTION (BR-WLLG1)	0	0.00	0		
22	BRACING REACTION (BR-WLLG2)	0	0.00	0		
23	BRACING REACTION (BR-WLLG3)	0	0.00	0		
24	BRACING REACTION (BR-WLLG4)	0	0.00	0		
25	BRACING REACTION (BR-SL'+')	0	0.00	0		
26	BRACING REACTION (BR-SL'-')	0	0.00	0		

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4(DL+COLL)	20.40	0.0	5.3	31.4	0.0	8.1	0.02
102	1.2(DL+COLL+CL-1)+1.6(RLL+P	87.7	0.0	23.5	134.9	0.0	36.2	0.07
103	1.2(DL+COLL+CL-2)+1.6(RLL+P	87.7	0.0	23.5	134.9	0.0	36.2	0.07
104	1.2(DL+COLL+CL-3)+1.6(RLL+P	87.7	0.0	23.5	134.9	0.0	36.2	0.07
105	1.2(DL+COLL+CL-4)+1.6(RLL+P	87.7	0.0	23.5	134.9	0.0	36.2	0.07
106	1.4(DL+COLL+CL-1)	20.4	0.0	5.3	31.4	0.0	8.1	0.02
107	1.4(DL+COLL+CL-2)	20.4	0.0	5.3	31.4	0.0	8.1	0.02
108	1.4(DL+COLL+CL-3)	20.4	0.0	5.3	31.4	0.0	8.1	0.02
109	1.4(DL+COLL+CL-4)	20.4	0.0	5.3	31.4	0.0	8.1	0.02
110	1.2(DL+COLL+CL-1+TL1)+1.6(R	87.7	0.0	27.7	134.9	0.0	42.6	0.07
111	1.2(DL+COLL+CL-2+TL1)+1.6(R	87.7	0.0	27.7	134.9	0.0	42.6	0.07
112	1.2(DL+COLL+CL-3+TL1)+1.6(R	87.7	0.0	27.7	134.9	0.0	42.6	0.07
113	1.2(DL+COLL+CL-4+TL1)+1.6(R	87.7	0.0	27.7	134.9	0.0	42.6	0.07
114	1.2(DL+COLL+CL-1+TL2)+1.6(R	87.7	0.0	19.3	134.9	0.0	29.7	0.07
115	1.2(DL+COLL+CL-2+TL2)+1.6(R	87.7	0.0	19.3	134.9	0.0	29.7	0.07
116	1.2(DL+COLL+CL-3+TL2)+1.6(R	87.7	0.0	19.3	134.9	0.0	29.7	0.07
117	1.2(DL+COLL+CL-4+TL2)+1.6(R	87.7	0.0	19.3	134.9	0.0	29.7	0.07
118	1.2(DL+COLL+CL-1)+0.8(WLL-1	17.5	0.0	4.5	26.9	0.0	7.0	0.01
119	1.2(DL+COLL+CL-2)+0.8(WLL-1	17.5	0.0	4.5	26.9	0.0	7.0	0.01
120	1.2(DL+COLL+CL-3)+0.8(WLL-1	17.5	0.0	4.5	26.9	0.0	7.0	0.01
121	1.2(DL+COLL+CL-4)+0.8(WLL-1	17.5	0.0	4.5	26.9	0.0	7.0	0.01
122	1.2(DL+COLL+CL-1)+0.8(WLL-2	17.5	0.0	4.5	26.9	0.0	7.0	0.01
123	1.2(DL+COLL+CL-2)+0.8(WLL-2	17.5	0.0	4.5	26.9	0.0	7.0	0.01
124	1.2(DL+COLL+CL-3)+0.8(WLL-2	17.5	0.0	4.5	26.9	0.0	7.0	0.01
125	1.2(DL+COLL+CL-4)+0.8(WLL-2	17.5	0.0	4.5	26.9	0.0	7.0	0.01
126	1.2(DL+COLL+CL-1)+0.8(WLR-1	17.5	0.0	4.5	26.9	0.0	7.0	0.01
127	1.2(DL+COLL+CL-2)+0.8(WLR-1	17.5	0.0	4.5	26.9	0.0	7.0	0.01
128	1.2(DL+COLL+CL-3)+0.8(WLR-1	17.5	0.0	4.5	26.9	0.0	7.0	0.01
129	1.2(DL+COLL+CL-4)+0.8(WLR-1	17.5	0.0	4.5	26.9	0.0	7.0	0.01
130	1.2(DL+COLL+CL-1)+0.8(WLR-2	17.5	0.0	4.5	26.9	0.0	7.0	0.01
131	1.2(DL+COLL+CL-2)+0.8(WLR-2	17.5	0.0	4.5	26.9	0.0	7.0	0.01
132	1.2(DL+COLL+CL-3)+0.8(WLR-2	17.5	0.0	4.5	26.9	0.0	7.0	0.01
133	1.2(DL+COLL+CL-4)+0.8(WLR-2	17.5	0.0	4.5	26.9	0.0	7.0	0.01
134	1.2(DL+COLL+CL-1)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
135	1.2(DL+COLL+CL-2)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
136	1.2(DL+COLL+CL-3)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
137	1.2(DL+COLL+CL-4)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
138	1.2(DL+COLL+CL-1)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
139	1.2(DL+COLL+CL-2)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
140	1.2(DL+COLL+CL-3)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
141	1.2(DL+COLL+CL-4)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
142	1.2(DL+COLL+CL-1)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
143	1.2(DL+COLL+CL-2)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
144	1.2(DL+COLL+CL-3)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
145	1.2(DL+COLL+CL-4)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
146	1.2(DL+COLL+CL-1)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
147	1.2(DL+COLL+CL-2)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
148	1.2(DL+COLL+CL-3)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
149	1.2(DL+COLL+CL-4)+0.8(WLLG-	17.5	0.0	4.5	26.9	0.0	7.0	0.01
150	1.2(DL+COLL+CL-1)+1.6(WLL-1	39.4	0.0	10.5	60.7	0.0	16.1	0.03
151	1.2(DL+COLL+CL-2)+1.6(WLL-1	39.4	0.0	10.5	60.7	0.0	16.1	0.03
152	1.2(DL+COLL+CL-3)+1.6(WLL-1	39.4	0.0	10.5	60.7	0.0	16.1	0.03

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153	1.2(DL+COLL+CL-4)+1.6(WLL-1	39.4	0.0	10.5	60.7	0.0	16.1	0.03
154	1.2(DL+COLL+CL-1)+1.6(WLL-2	39.4	0.0	10.5	60.7	0.0	16.1	0.03
155	1.2(DL+COLL+CL-2)+1.6(WLL-2	39.4	0.0	10.5	60.7	0.0	16.1	0.03
156	1.2(DL+COLL+CL-3)+1.6(WLL-2	39.4	0.0	10.5	60.7	0.0	16.1	0.03
157	1.2(DL+COLL+CL-4)+1.6(WLL-2	39.4	0.0	10.5	60.7	0.0	16.1	0.03
158	1.2(DL+COLL+CL-1)+1.6(WLR-1	39.4	0.0	10.5	60.7	0.0	16.1	0.03
159	1.2(DL+COLL+CL-2)+1.6(WLR-1	39.4	0.0	10.5	60.7	0.0	16.1	0.03
160	1.2(DL+COLL+CL-3)+1.6(WLR-1	39.4	0.0	10.5	60.7	0.0	16.1	0.03
161	1.2(DL+COLL+CL-4)+1.6(WLR-1	39.4	0.0	10.5	60.7	0.0	16.1	0.03
162	1.2(DL+COLL+CL-1)+1.6(WLR-2	39.4	0.0	10.5	60.7	0.0	16.1	0.03
163	1.2(DL+COLL+CL-2)+1.6(WLR-2	39.4	0.0	10.5	60.7	0.0	16.1	0.03
164	1.2(DL+COLL+CL-3)+1.6(WLR-2	39.4	0.0	10.5	60.7	0.0	16.1	0.03
165	1.2(DL+COLL+CL-4)+1.6(WLR-2	39.4	0.0	10.5	60.7	0.0	16.1	0.03
166	1.2(DL+COLL+CL-1)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
167	1.2(DL+COLL+CL-2)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
168	1.2(DL+COLL+CL-3)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
169	1.2(DL+COLL+CL-4)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
170	1.2(DL+COLL+CL-1)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
171	1.2(DL+COLL+CL-2)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
172	1.2(DL+COLL+CL-3)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
173	1.2(DL+COLL+CL-4)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
174	1.2(DL+COLL+CL-1)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
175	1.2(DL+COLL+CL-2)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
176	1.2(DL+COLL+CL-3)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
177	1.2(DL+COLL+CL-4)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
178	1.2(DL+COLL+CL-1)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
179	1.2(DL+COLL+CL-2)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
180	1.2(DL+COLL+CL-3)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
181	1.2(DL+COLL+CL-4)+1.6(WLLG-	39.4	0.0	10.5	60.7	0.0	16.1	0.03
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+I	37.5	0.0	15.9	57.7	0.0	24.4	0.03
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+I	37.5	0.0	15.9	57.7	0.0	24.4	0.03
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+I	37.5	0.0	15.9	57.7	0.0	24.4	0.03
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+I	37.5	0.0	15.9	57.7	0.0	24.4	0.03
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+E	41.3	0.0	5.1	63.6	0.0	7.8	0.03
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+E	41.3	0.0	5.1	63.6	0.0	7.8	0.03
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+E	41.3	0.0	5.1	63.6	0.0	7.8	0.03
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+E	41.3	0.0	5.1	63.6	0.0	7.8	0.03
190	0.9(DL+COLL+CL-1)+1.6(WLL-1	13.1	0.0	3.4	20.2	0.0	5.2	0.01
191	0.9(DL+COLL+CL-2)+1.6(WLL-1	13.1	0.0	3.4	20.2	0.0	5.2	0.01
192	0.9(DL+COLL+CL-3)+1.6(WLL-1	13.1	0.0	3.4	20.2	0.0	5.2	0.01
193	0.9(DL+COLL+CL-4)+1.6(WLL-1	13.1	0.0	3.4	20.2	0.0	5.2	0.01
194	0.9(DL+COLL+CL-1)+1.6(WLL-2	13.1	0.0	3.4	20.2	0.0	5.2	0.01
195	0.9(DL+COLL+CL-2)+1.6(WLL-2	13.1	0.0	3.4	20.2	0.0	5.2	0.01
196	0.9(DL+COLL+CL-3)+1.6(WLL-2	13.1	0.0	3.4	20.2	0.0	5.2	0.01
197	0.9(DL+COLL+CL-4)+1.6(WLL-2	13.1	0.0	3.4	20.2	0.0	5.2	0.01
198	0.9(DL+COLL+CL-1)+1.6(WLR-1	13.1	0.0	3.4	20.2	0.0	5.2	0.01
199	0.9(DL+COLL+CL-2)+1.6(WLR-1	13.1	0.0	3.4	20.2	0.0	5.2	0.01
200	0.9(DL+COLL+CL-3)+1.6(WLR-1	13.1	0.0	3.4	20.2	0.0	5.2	0.01
201	0.9(DL+COLL+CL-4)+1.6(WLR-1	13.1	0.0	3.4	20.2	0.0	5.2	0.01
202	0.9(DL+COLL+CL-1)+1.6(WLR-2	13.1	0.0	3.4	20.2	0.0	5.2	0.01
203	0.9(DL+COLL+CL-2)+1.6(WLR-2	13.1	0.0	3.4	20.2	0.0	5.2	0.01
204	0.9(DL+COLL+CL-3)+1.6(WLR-2	13.1	0.0	3.4	20.2	0.0	5.2	0.01
205	0.9(DL+COLL+CL-4)+1.6(WLR-2	13.1	0.0	3.4	20.2	0.0	5.2	0.01
206	0.9(DL+COLL+CL-1)+1.6(WLLG-	13.1	0.0	3.4	20.2	0.0	5.2	0.01
207	0.9(DL+COLL+CL-2)+1.6(WLLG-	13.1	0.0	3.4	20.2	0.0	5.2	0.01
208	0.9(DL+COLL+CL-3)+1.6(WLLG-	13.1	0.0	3.4	20.2	0.0	5.2	0.01
209	0.9(DL+COLL+CL-4)+1.6(WLLG-	13.1	0.0	3.4	20.2	0.0	5.2	0.01

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210	0.9(DL+COLL+CL-1)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
211	0.9(DL+COLL+CL-2)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
212	0.9(DL+COLL+CL-3)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
213	0.9(DL+COLL+CL-4)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
214	0.9(DL+COLL+CL-1)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
215	0.9(DL+COLL+CL-2)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
216	0.9(DL+COLL+CL-3)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
217	0.9(DL+COLL+CL-4)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
218	0.9(DL+COLL+CL-1)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
219	0.9(DL+COLL+CL-2)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
220	0.9(DL+COLL+CL-3)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
221	0.9(DL+COLL+CL-4)+1.6(WLLG	13.1	0.0	3.4	20.2	0.0	5.2	0.01
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+E	11.2	0.0	8.8	17.3	0.0	13.5	0.01
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+E	11.2	0.0	8.8	17.3	0.0	13.5	0.01
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+E	11.2	0.0	8.8	17.3	0.0	13.5	0.01
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+E	11.2	0.0	8.8	17.3	0.0	13.5	0.01
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+E	15.0	0.0	2.0	23.1	0.0	3.1	0.01
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+E	15.0	0.0	2.0	23.1	0.0	3.1	0.01
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+E	15.0	0.0	2.0	23.1	0.0	3.1	0.01
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+E	15.0	0.0	2.0	23.1	0.0	3.1	0.01

COLUMN DESIGN AND INTERACTION RATIO CHECK

Reinforcement: Provide,

4-Nos	25 mm Dia
6-Nos	20 mm Dia
6-Nos	20 mm Dia

1964.29 Sq mm (At Corner) Asc pvd. = 5735.71 Sq mm
1885.71 Sq mm (Along Z axis)
1885.71 Sq mm (Along X axis) Total Asc = 1.02 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1897.2738 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8 (COLUMNS 3.3.3) $R_{nz}=M_{noz}/f'_c b^2 D$	From Chart of SP17(M)-R35-420.8 (COLUMNS 3.3.3) $R_{nx}=M_{nox}/f'_c b^2 D$	M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
101	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
102	0.05	0.05	73.8	73.8	0.55	1.16	0.44	O.K
103	0.05	0.05	73.8	73.8	0.55	1.16	0.44	O.K
104	0.05	0.05	73.8	73.8	0.55	1.16	0.44	O.K
105	0.05	0.05	73.8	73.8	0.55	1.16	0.44	O.K
106	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
107	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
108	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
109	0.05	0.05	73.8	73.8	0.55	1.16	0.08	O.K
110	0.05	0.05	73.8	73.8	0.55	1.16	0.53	O.K
111	0.05	0.05	73.8	73.8	0.55	1.16	0.53	O.K
112	0.05	0.05	73.8	73.8	0.55	1.16	0.53	O.K
113	0.05	0.05	73.8	73.8	0.55	1.16	0.53	O.K
114	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
115	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
116	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
117	0.05	0.05	73.8	73.8	0.55	1.16	0.35	O.K
118	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
119	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
120	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
121	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
122	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
123	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
124	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
125	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
126	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
127	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K

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128	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
129	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
130	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
131	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
132	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
133	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
134	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
135	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
136	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
137	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
138	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
139	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
140	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
141	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
142	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
143	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
144	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
145	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
146	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
147	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
148	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
149	0.05	0.05	73.8	73.8	0.55	1.16	0.06	O.K
150	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
151	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
152	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
153	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
154	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
155	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
156	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
157	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
158	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
159	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
160	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
161	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
162	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
163	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
164	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
165	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
166	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
167	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
168	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
169	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
170	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
171	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
172	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
173	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
174	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
175	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
176	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
177	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
178	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
179	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
180	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
181	0.05	0.05	73.8	73.8	0.55	1.16	0.17	O.K
182	0.05	0.05	73.8	73.8	0.55	1.16	0.28	O.K
183	0.05	0.05	73.8	73.8	0.55	1.16	0.28	O.K
184	0.05	0.05	73.8	73.8	0.55	1.16	0.28	O.K
185	0.05	0.05	73.8	73.8	0.55	1.16	0.28	O.K
186	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
187	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
188	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
189	0.05	0.05	73.8	73.8	0.55	1.16	0.07	O.K
190	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K

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191	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
192	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
193	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
194	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
195	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
196	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
197	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
198	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
199	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
200	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
201	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
202	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
203	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
204	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
205	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
206	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
207	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
208	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
209	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
210	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
211	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
212	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
213	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
214	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
215	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
216	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
217	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
218	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
219	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
220	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
221	0.05	0.05	73.8	73.8	0.55	1.16	0.05	O.K
222	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
223	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
224	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
225	0.05	0.05	73.8	73.8	0.55	1.16	0.14	O.K
226	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
227	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
228	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
229	0.05	0.05	73.8	73.8	0.55	1.16	0.03	O.K
							Max = **0.53**	**O.K**

Links, 10mm dia @ 300 C/C

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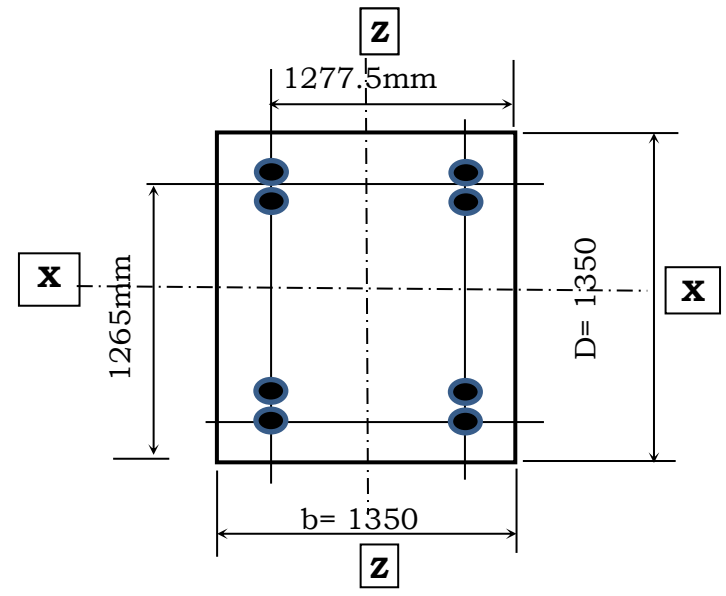
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5.14 DESIGN OF COLUMN MKD C7

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	85mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	1.00%
Max. Steel Ratio	4.00%
Dimension parallel to X-X axis	1350mm
Dimension parallel to Z-Z axis	1350mm

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	2000mm	γ_x =	0.89
Unsupproted length of column about Z (l_{uz}) =	2000mm	γ_z =	0.91

Where, Effective length factor K= 2

Slenderness ratio,	K l_{ux} / r_x =	9.88	< 22	SECTION 6.2.5a	Radius of gyration r_x=	405 mm
	K l_{uz} / r_z =	9.88	< 22	SECTION 6.2.5a	Radius of gyration r_z=	405 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	DEAD LOAD (DL)	293.2416		-188.9868		-148.536
2	COLLATERAL LOAD (COLL)	0		0		0
3	ROOF LIVE LOAD (RLL)	115.1316		-24.3936		-92.3748
4	PLATFORM OR MEZZ. LIVE LOAD (PLL / MLL)	769.6536		-584.1576		-373.536
5	CRANE LOAD-1 (CL-1)	0		0		0
6	CRANE LOAD-2 (CL-2)	0		0		0
7	CRANE LOAD-3 (CL-3)	0		0		0
8	CRANE LOAD-4 (CL-4)	0		0		0
9	WIND LEFT-1 (WLL-1)	-40.8672		77.8884		126.146
10	WIND LEFT-2 (WLL-2)	-111.3168		8.9736		58.0164
11	WIND RIGHT-1 (WLR-1)	-119.6448		114.8016		249.212
12	WIND RIGHT-2 (WLR-2)	-108.7704		39.0276		121.028
13	WIND LONG-1 (WLLG-1)	-190.8528	72.2196		222.6684	
14	WIND LONG-2 (WLLG-2)	-142.6392	31.5156		124.9272	
15	WIND LONG-3 (WLLG-3)	-213.9264	75.0516		191.3232	
16	WIND LONG-4 (WLLG-4)	-167.6988	36.0168		109.2432	
17	SEISMIC LOAD SL'+'	2.598	-11.4348	-11.4348	-30.3744	-30.3744
18	SEISMIC LOAD SL'-'	-2.598	11.4348	11.4348	30.3744	30.3744
19	TEMPERATURE LOAD (TL1)	-11.5176		-36.564		-166.262
20	TEMPERATURE LOAD (TL2)	11.5176		36.564		166.262
21	BRACING REACTION (BR-WLLG1)	51.7	-41		0	
22	BRACING REACTION (BR-WLLG2)	51.7	-41		0	
23	BRACING REACTION (BR-WLLG3)	-51.7	41		0	
24	BRACING REACTION (BR-WLLG4)	-51.7	41		0	
25	BRACING REACTION (BR-SL'+')	91.34	-96.33	-96.33	0	0
26	BRACING REACTION (BR-SL'-')	-91.34	96.33	96.33	0	0

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4(DL+COLL)	41.05	0.0	73.7	63.2	0.0	113.4	0.01
102	1.2(DL+COLL+CL-1)+1.6(RLL+P	176.8	0.0	332.5	271.9	0.0	511.5	0.04
103	1.2(DL+COLL+CL-2)+1.6(RLL+P	176.8	0.0	332.5	271.9	0.0	511.5	0.04
104	1.2(DL+COLL+CL-3)+1.6(RLL+P	176.8	0.0	332.5	271.9	0.0	511.5	0.04
105	1.2(DL+COLL+CL-4)+1.6(RLL+P	176.8	0.0	332.5	271.9	0.0	511.5	0.04
106	1.4(DL+COLL+CL-1)	41.1	0.0	73.7	63.2	0.0	113.4	0.01
107	1.4(DL+COLL+CL-2)	41.1	0.0	73.7	63.2	0.0	113.4	0.01
108	1.4(DL+COLL+CL-3)	41.1	0.0	73.7	63.2	0.0	113.4	0.01
109	1.4(DL+COLL+CL-4)	41.1	0.0	73.7	63.2	0.0	113.4	0.01
110	1.2(DL+COLL+CL-1+TL1)+1.6(R	175.4	0.0	361.2	269.8	0.0	555.7	0.04
111	1.2(DL+COLL+CL-2+TL1)+1.6(R	175.4	0.0	361.2	269.8	0.0	555.7	0.04
112	1.2(DL+COLL+CL-3+TL1)+1.6(R	175.4	0.0	361.2	269.8	0.0	555.7	0.04
113	1.2(DL+COLL+CL-4+TL1)+1.6(R	175.4	0.0	361.2	269.8	0.0	555.7	0.04
114	1.2(DL+COLL+CL-1+TL2)+1.6(R	178.1	0.0	303.7	274.1	0.0	467.3	0.04
115	1.2(DL+COLL+CL-2+TL2)+1.6(R	178.1	0.0	303.7	274.1	0.0	467.3	0.04
116	1.2(DL+COLL+CL-3+TL2)+1.6(R	178.1	0.0	303.7	274.1	0.0	467.3	0.04
117	1.2(DL+COLL+CL-4+TL2)+1.6(R	178.1	0.0	303.7	274.1	0.0	467.3	0.04
118	1.2(DL+COLL+CL-1)+0.8(WLL-1	31.9	0.0	40.6	49.1	0.0	62.5	0.01
119	1.2(DL+COLL+CL-2)+0.8(WLL-1	31.9	0.0	40.6	49.1	0.0	62.5	0.01
120	1.2(DL+COLL+CL-3)+0.8(WLL-1	31.9	0.0	40.6	49.1	0.0	62.5	0.01
121	1.2(DL+COLL+CL-4)+0.8(WLL-1	31.9	0.0	40.6	49.1	0.0	62.5	0.01
122	1.2(DL+COLL+CL-1)+0.8(WLL-2	26.3	0.0	57.1	40.4	0.0	87.9	0.01
123	1.2(DL+COLL+CL-2)+0.8(WLL-2	26.3	0.0	57.1	40.4	0.0	87.9	0.01
124	1.2(DL+COLL+CL-3)+0.8(WLL-2	26.3	0.0	57.1	40.4	0.0	87.9	0.01
125	1.2(DL+COLL+CL-4)+0.8(WLL-2	26.3	0.0	57.1	40.4	0.0	87.9	0.01
126	1.2(DL+COLL+CL-1)+0.8(WLR-1	25.6	0.0	24.9	39.4	0.0	38.3	0.01
127	1.2(DL+COLL+CL-2)+0.8(WLR-1	25.6	0.0	24.9	39.4	0.0	38.3	0.01
128	1.2(DL+COLL+CL-3)+0.8(WLR-1	25.6	0.0	24.9	39.4	0.0	38.3	0.01
129	1.2(DL+COLL+CL-4)+0.8(WLR-1	25.6	0.0	24.9	39.4	0.0	38.3	0.01
130	1.2(DL+COLL+CL-1)+0.8(WLR-2	26.5	0.0	47.3	40.7	0.0	72.7	0.01
131	1.2(DL+COLL+CL-2)+0.8(WLR-2	26.5	0.0	47.3	40.7	0.0	72.7	0.01
132	1.2(DL+COLL+CL-3)+0.8(WLR-2	26.5	0.0	47.3	40.7	0.0	72.7	0.01
133	1.2(DL+COLL+CL-4)+0.8(WLR-2	26.5	0.0	47.3	40.7	0.0	72.7	0.01
134	1.2(DL+COLL+CL-1)+0.8(WLLG	24.1	22.8	63.2	37.0	35.1	97.2	0.01
135	1.2(DL+COLL+CL-2)+0.8(WLLG	24.1	22.8	63.2	37.0	35.1	97.2	0.01
136	1.2(DL+COLL+CL-3)+0.8(WLLG	24.1	22.8	63.2	37.0	35.1	97.2	0.01
137	1.2(DL+COLL+CL-4)+0.8(WLLG	24.1	22.8	63.2	37.0	35.1	97.2	0.01
138	1.2(DL+COLL+CL-1)+0.8(WLLG	27.9	8.5	63.2	42.9	13.0	97.2	0.01
139	1.2(DL+COLL+CL-2)+0.8(WLLG	27.9	8.5	63.2	42.9	13.0	97.2	0.01
140	1.2(DL+COLL+CL-3)+0.8(WLLG	27.9	8.5	63.2	42.9	13.0	97.2	0.01
141	1.2(DL+COLL+CL-4)+0.8(WLLG	27.9	8.5	63.2	42.9	13.0	97.2	0.01
142	1.2(DL+COLL+CL-1)+0.8(WLLG	13.9	33.9	63.2	21.4	52.1	97.2	0.00
143	1.2(DL+COLL+CL-2)+0.8(WLLG	13.9	33.9	63.2	21.4	52.1	97.2	0.00
144	1.2(DL+COLL+CL-3)+0.8(WLLG	13.9	33.9	63.2	21.4	52.1	97.2	0.00
145	1.2(DL+COLL+CL-4)+0.8(WLLG	13.9	33.9	63.2	21.4	52.1	97.2	0.00
146	1.2(DL+COLL+CL-1)+0.8(WLLG	17.6	21.1	63.2	27.1	32.4	97.2	0.00
147	1.2(DL+COLL+CL-2)+0.8(WLLG	17.6	21.1	63.2	27.1	32.4	97.2	0.00
148	1.2(DL+COLL+CL-3)+0.8(WLLG	17.6	21.1	63.2	27.1	32.4	97.2	0.00
149	1.2(DL+COLL+CL-4)+0.8(WLLG	17.6	21.1	63.2	27.1	32.4	97.2	0.00
150	1.2(DL+COLL+CL-1)+1.6(WLL-1	72.9	0.0	102.2	112.1	0.0	157.3	0.02
151	1.2(DL+COLL+CL-2)+1.6(WLL-1	72.9	0.0	102.2	112.1	0.0	157.3	0.02
152	1.2(DL+COLL+CL-3)+1.6(WLL-1	72.9	0.0	102.2	112.1	0.0	157.3	0.02

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153	1.2(DL+COLL+CL-4)+1.6(WLL-1	72.9	0.0	102.2	112.1	0.0	157.3	0.02
154	1.2(DL+COLL+CL-1)+1.6(WLL-2	61.6	0.0	135.2	94.8	0.0	208.0	0.01
155	1.2(DL+COLL+CL-2)+1.6(WLL-2	61.6	0.0	135.2	94.8	0.0	208.0	0.01
156	1.2(DL+COLL+CL-3)+1.6(WLL-2	61.6	0.0	135.2	94.8	0.0	208.0	0.01
157	1.2(DL+COLL+CL-4)+1.6(WLL-2	61.6	0.0	135.2	94.8	0.0	208.0	0.01
158	1.2(DL+COLL+CL-1)+1.6(WLR-1	60.3	0.0	70.7	92.7	0.0	108.8	0.01
159	1.2(DL+COLL+CL-2)+1.6(WLR-1	60.3	0.0	70.7	92.7	0.0	108.8	0.01
160	1.2(DL+COLL+CL-3)+1.6(WLR-1	60.3	0.0	70.7	92.7	0.0	108.8	0.01
161	1.2(DL+COLL+CL-4)+1.6(WLR-1	60.3	0.0	70.7	92.7	0.0	108.8	0.01
162	1.2(DL+COLL+CL-1)+1.6(WLR-2	62.0	0.0	115.5	95.4	0.0	177.7	0.01
163	1.2(DL+COLL+CL-2)+1.6(WLR-2	62.0	0.0	115.5	95.4	0.0	177.7	0.01
164	1.2(DL+COLL+CL-3)+1.6(WLR-2	62.0	0.0	115.5	95.4	0.0	177.7	0.01
165	1.2(DL+COLL+CL-4)+1.6(WLR-2	62.0	0.0	115.5	95.4	0.0	177.7	0.01
166	1.2(DL+COLL+CL-1)+1.6(WLLG-	57.2	45.6	147.3	87.9	70.2	226.7	0.01
167	1.2(DL+COLL+CL-2)+1.6(WLLG-	57.2	45.6	147.3	87.9	70.2	226.7	0.01
168	1.2(DL+COLL+CL-3)+1.6(WLLG-	57.2	45.6	147.3	87.9	70.2	226.7	0.01
169	1.2(DL+COLL+CL-4)+1.6(WLLG-	57.2	45.6	147.3	87.9	70.2	226.7	0.01
170	1.2(DL+COLL+CL-1)+1.6(WLLG-	64.9	17.0	147.3	99.8	26.1	226.7	0.02
171	1.2(DL+COLL+CL-2)+1.6(WLLG-	64.9	17.0	147.3	99.8	26.1	226.7	0.02
172	1.2(DL+COLL+CL-3)+1.6(WLLG-	64.9	17.0	147.3	99.8	26.1	226.7	0.02
173	1.2(DL+COLL+CL-4)+1.6(WLLG-	64.9	17.0	147.3	99.8	26.1	226.7	0.02
174	1.2(DL+COLL+CL-1)+1.6(WLLG-	36.9	67.7	147.3	56.8	104.2	226.7	0.01
175	1.2(DL+COLL+CL-2)+1.6(WLLG-	36.9	67.7	147.3	56.8	104.2	226.7	0.01
176	1.2(DL+COLL+CL-3)+1.6(WLLG-	36.9	67.7	147.3	56.8	104.2	226.7	0.01
177	1.2(DL+COLL+CL-4)+1.6(WLLG-	36.9	67.7	147.3	56.8	104.2	226.7	0.01
178	1.2(DL+COLL+CL-1)+1.6(WLLG-	44.3	42.1	147.3	68.2	64.8	226.7	0.01
179	1.2(DL+COLL+CL-2)+1.6(WLLG-	44.3	42.1	147.3	68.2	64.8	226.7	0.01
180	1.2(DL+COLL+CL-3)+1.6(WLLG-	44.3	42.1	147.3	68.2	64.8	226.7	0.01
181	1.2(DL+COLL+CL-4)+1.6(WLLG-	44.3	42.1	147.3	68.2	64.8	226.7	0.01
182	1.2(DL+COLL+CL-1)+1.0(SL'+'+E	88.8	24.6	171.9	136.6	37.8	264.5	0.02
183	1.2(DL+COLL+CL-2)+1.0(SL'+'+E	88.8	24.6	171.9	136.6	37.8	264.5	0.02
184	1.2(DL+COLL+CL-3)+1.0(SL'+'+E	88.8	24.6	171.9	136.6	37.8	264.5	0.02
185	1.2(DL+COLL+CL-4)+1.0(SL'+'+E	88.8	24.6	171.9	136.6	37.8	264.5	0.02
186	1.2(DL+COLL+CL-1)+1.0(SL'-'+E	70.0	24.6	122.7	107.7	37.8	188.8	0.02
187	1.2(DL+COLL+CL-2)+1.0(SL'-'+E	70.0	24.6	122.7	107.7	37.8	188.8	0.02
188	1.2(DL+COLL+CL-3)+1.0(SL'-'+E	70.0	24.6	122.7	107.7	37.8	188.8	0.02
189	1.2(DL+COLL+CL-4)+1.0(SL'-'+E	70.0	24.6	122.7	107.7	37.8	188.8	0.02
190	0.9(DL+COLL+CL-1)+1.6(WLL-1	19.9	0.0	2.3	30.5	0.0	3.5	0.00
191	0.9(DL+COLL+CL-2)+1.6(WLL-1	19.9	0.0	2.3	30.5	0.0	3.5	0.00
192	0.9(DL+COLL+CL-3)+1.6(WLL-1	19.9	0.0	2.3	30.5	0.0	3.5	0.00
193	0.9(DL+COLL+CL-4)+1.6(WLL-1	19.9	0.0	2.3	30.5	0.0	3.5	0.00
194	0.9(DL+COLL+CL-1)+1.6(WLL-2	8.6	0.0	35.2	13.2	0.0	54.2	0.00
195	0.9(DL+COLL+CL-2)+1.6(WLL-2	8.6	0.0	35.2	13.2	0.0	54.2	0.00
196	0.9(DL+COLL+CL-3)+1.6(WLL-2	8.6	0.0	35.2	13.2	0.0	54.2	0.00
197	0.9(DL+COLL+CL-4)+1.6(WLL-2	8.6	0.0	35.2	13.2	0.0	54.2	0.00
198	0.9(DL+COLL+CL-1)+1.6(WLR-1	7.2	0.0	29.2	11.2	0.0	45.0	0.00
199	0.9(DL+COLL+CL-2)+1.6(WLR-1	7.2	0.0	29.2	11.2	0.0	45.0	0.00
200	0.9(DL+COLL+CL-3)+1.6(WLR-1	7.2	0.0	29.2	11.2	0.0	45.0	0.00
201	0.9(DL+COLL+CL-4)+1.6(WLR-1	7.2	0.0	29.2	11.2	0.0	45.0	0.00
202	0.9(DL+COLL+CL-1)+1.6(WLR-2	9.0	0.0	15.5	13.8	0.0	23.9	0.00
203	0.9(DL+COLL+CL-2)+1.6(WLR-2	9.0	0.0	15.5	13.8	0.0	23.9	0.00
204	0.9(DL+COLL+CL-3)+1.6(WLR-2	9.0	0.0	15.5	13.8	0.0	23.9	0.00
205	0.9(DL+COLL+CL-4)+1.6(WLR-2	9.0	0.0	15.5	13.8	0.0	23.9	0.00
206	0.9(DL+COLL+CL-1)+1.6(WLLG-	4.1	45.6	47.4	6.3	70.2	72.9	0.00
207	0.9(DL+COLL+CL-2)+1.6(WLLG-	4.1	45.6	47.4	6.3	70.2	72.9	0.00
208	0.9(DL+COLL+CL-3)+1.6(WLLG-	4.1	45.6	47.4	6.3	70.2	72.9	0.00
209	0.9(DL+COLL+CL-4)+1.6(WLLG-	4.1	45.6	47.4	6.3	70.2	72.9	0.00

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210	0.9(DL+COLL+CL-1)+1.6(WLLG	11.8	17.0	47.4	18.2	26.1	72.9	0.00
211	0.9(DL+COLL+CL-2)+1.6(WLLG	11.8	17.0	47.4	18.2	26.1	72.9	0.00
212	0.9(DL+COLL+CL-3)+1.6(WLLG	11.8	17.0	47.4	18.2	26.1	72.9	0.00
213	0.9(DL+COLL+CL-4)+1.6(WLLG	11.8	17.0	47.4	18.2	26.1	72.9	0.00
214	0.9(DL+COLL+CL-1)+1.6(WLLG	-16.1	67.7	47.4	-24.8	104.2	72.9	0.00
215	0.9(DL+COLL+CL-2)+1.6(WLLG	-16.1	67.7	47.4	-24.8	104.2	72.9	0.00
216	0.9(DL+COLL+CL-3)+1.6(WLLG	-16.1	67.7	47.4	-24.8	104.2	72.9	0.00
217	0.9(DL+COLL+CL-4)+1.6(WLLG	-16.1	67.7	47.4	-24.8	104.2	72.9	0.00
218	0.9(DL+COLL+CL-1)+1.6(WLLG	-8.7	42.1	47.4	-13.4	64.8	72.9	0.00
219	0.9(DL+COLL+CL-2)+1.6(WLLG	-8.7	42.1	47.4	-13.4	64.8	72.9	0.00
220	0.9(DL+COLL+CL-3)+1.6(WLLG	-8.7	42.1	47.4	-13.4	64.8	72.9	0.00
221	0.9(DL+COLL+CL-4)+1.6(WLLG	-8.7	42.1	47.4	-13.4	64.8	72.9	0.00
222	0.9(DL+COLL+CL-1)+1.0(SL'+'+E	35.8	24.6	72.0	55.1	37.8	110.7	0.01
223	0.9(DL+COLL+CL-2)+1.0(SL'+'+E	35.8	24.6	72.0	55.1	37.8	110.7	0.01
224	0.9(DL+COLL+CL-3)+1.0(SL'+'+E	35.8	24.6	72.0	55.1	37.8	110.7	0.01
225	0.9(DL+COLL+CL-4)+1.0(SL'+'+E	35.8	24.6	72.0	55.1	37.8	110.7	0.01
226	0.9(DL+COLL+CL-1)+1.0(SL'-'+E	17.0	24.6	22.8	26.2	37.8	35.1	0.00
227	0.9(DL+COLL+CL-2)+1.0(SL'-'+E	17.0	24.6	22.8	26.2	37.8	35.1	0.00
228	0.9(DL+COLL+CL-3)+1.0(SL'-'+E	17.0	24.6	22.8	26.2	37.8	35.1	0.00
229	0.9(DL+COLL+CL-4)+1.0(SL'-'+E	17.0	24.6	22.8	26.2	37.8	35.1	0.00

COLUMN INTERACTION RATIO CHECK

Reinforcement: Provide,

4-Nos	25 mm Dia
20-Nos	25 mm Dia
20-Nos	25 mm Dia

1964.29 Sq mm (At Corner) Asc pvd. = 21607.14 Sq mm
9821.43 Sq mm (Along Z axis)
9821.43 Sq mm (Along X axis) Total Asc = 1.19 %

$\beta = P_n / P_0$ (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 6265.1563 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.9 (COLUMNS 3.3.4)		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
101	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
102	0.07	0.07	637.2	637.2	0.55	1.16	0.78	O.K
103	0.07	0.07	637.2	637.2	0.55	1.16	0.78	O.K
104	0.07	0.07	637.2	637.2	0.55	1.16	0.78	O.K
105	0.07	0.07	637.2	637.2	0.55	1.16	0.78	O.K
106	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
107	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
108	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
109	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
110	0.07	0.07	637.2	637.2	0.55	1.16	0.85	O.K
111	0.07	0.07	637.2	637.2	0.55	1.16	0.85	O.K
112	0.07	0.07	637.2	637.2	0.55	1.16	0.85	O.K
113	0.07	0.07	637.2	637.2	0.55	1.16	0.85	O.K
114	0.07	0.07	637.2	637.2	0.55	1.16	0.70	O.K
115	0.07	0.07	637.2	637.2	0.55	1.16	0.70	O.K
116	0.07	0.07	637.2	637.2	0.55	1.16	0.70	O.K
117	0.07	0.07	637.2	637.2	0.55	1.16	0.70	O.K
118	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
119	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
120	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
121	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
122	0.07	0.07	637.2	637.2	0.55	1.16	0.10	O.K
123	0.07	0.07	637.2	637.2	0.55	1.16	0.10	O.K
124	0.07	0.07	637.2	637.2	0.55	1.16	0.10	O.K
125	0.07	0.07	637.2	637.2	0.55	1.16	0.10	O.K
126	0.07	0.07	637.2	637.2	0.55	1.16	0.04	O.K
127	0.07	0.07	637.2	637.2	0.55	1.16	0.04	O.K

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128	0.07	0.07	637.2	637.2	0.55	1.16	0.04	O.K
129	0.07	0.07	637.2	637.2	0.55	1.16	0.04	O.K
130	0.07	0.07	637.2	637.2	0.55	1.16	0.08	O.K
131	0.07	0.07	637.2	637.2	0.55	1.16	0.08	O.K
132	0.07	0.07	637.2	637.2	0.55	1.16	0.08	O.K
133	0.07	0.07	637.2	637.2	0.55	1.16	0.08	O.K
134	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
135	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
136	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
137	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
138	0.07	0.07	637.2	637.2	0.55	1.16	0.12	O.K
139	0.07	0.07	637.2	637.2	0.55	1.16	0.12	O.K
140	0.07	0.07	637.2	637.2	0.55	1.16	0.12	O.K
141	0.07	0.07	637.2	637.2	0.55	1.16	0.12	O.K
142	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
143	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
144	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
145	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
146	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
147	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
148	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
149	0.07	0.07	637.2	637.2	0.55	1.16	0.14	O.K
150	0.07	0.07	637.2	637.2	0.55	1.16	0.20	O.K
151	0.07	0.07	637.2	637.2	0.55	1.16	0.20	O.K
152	0.07	0.07	637.2	637.2	0.55	1.16	0.20	O.K
153	0.07	0.07	637.2	637.2	0.55	1.16	0.20	O.K
154	0.07	0.07	637.2	637.2	0.55	1.16	0.27	O.K
155	0.07	0.07	637.2	637.2	0.55	1.16	0.27	O.K
156	0.07	0.07	637.2	637.2	0.55	1.16	0.27	O.K
157	0.07	0.07	637.2	637.2	0.55	1.16	0.27	O.K
158	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
159	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
160	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
161	0.07	0.07	637.2	637.2	0.55	1.16	0.13	O.K
162	0.07	0.07	637.2	637.2	0.55	1.16	0.23	O.K
163	0.07	0.07	637.2	637.2	0.55	1.16	0.23	O.K
164	0.07	0.07	637.2	637.2	0.55	1.16	0.23	O.K
165	0.07	0.07	637.2	637.2	0.55	1.16	0.23	O.K
166	0.07	0.07	637.2	637.2	0.55	1.16	0.38	O.K
167	0.07	0.07	637.2	637.2	0.55	1.16	0.38	O.K
168	0.07	0.07	637.2	637.2	0.55	1.16	0.38	O.K
169	0.07	0.07	637.2	637.2	0.55	1.16	0.38	O.K
170	0.07	0.07	637.2	637.2	0.55	1.16	0.33	O.K
171	0.07	0.07	637.2	637.2	0.55	1.16	0.33	O.K
172	0.07	0.07	637.2	637.2	0.55	1.16	0.33	O.K
173	0.07	0.07	637.2	637.2	0.55	1.16	0.33	O.K
174	0.07	0.07	637.2	637.2	0.55	1.16	0.42	O.K
175	0.07	0.07	637.2	637.2	0.55	1.16	0.42	O.K
176	0.07	0.07	637.2	637.2	0.55	1.16	0.42	O.K
177	0.07	0.07	637.2	637.2	0.55	1.16	0.42	O.K
178	0.07	0.07	637.2	637.2	0.55	1.16	0.37	O.K
179	0.07	0.07	637.2	637.2	0.55	1.16	0.37	O.K
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187	0.07	0.07	637.2	637.2	0.55	1.16	0.28	O.K
188	0.07	0.07	637.2	637.2	0.55	1.16	0.28	O.K
189	0.07	0.07	637.2	637.2	0.55	1.16	0.28	O.K
190	0.07	0.07	637.2	637.2	0.55	1.16	0.00	O.K

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191	0.07	0.07	637.2	637.2	0.55	1.16	0.00	O.K
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213	0.07	0.07	637.2	637.2	0.55	1.16	0.11	O.K
214	0.07	0.07	637.2	637.2	0.55	1.16	0.20	O.K
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217	0.07	0.07	637.2	637.2	0.55	1.16	0.20	O.K
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219	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
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221	0.07	0.07	637.2	637.2	0.55	1.16	0.15	O.K
222	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
223	0.07	0.07	637.2	637.2	0.55	1.16	0.17	O.K
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226	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
227	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
228	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
229	0.07	0.07	637.2	637.2	0.55	1.16	0.07	O.K
							Max = **0.85**	**O.K**

Links, 10mm dia @ 300 C/C

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1

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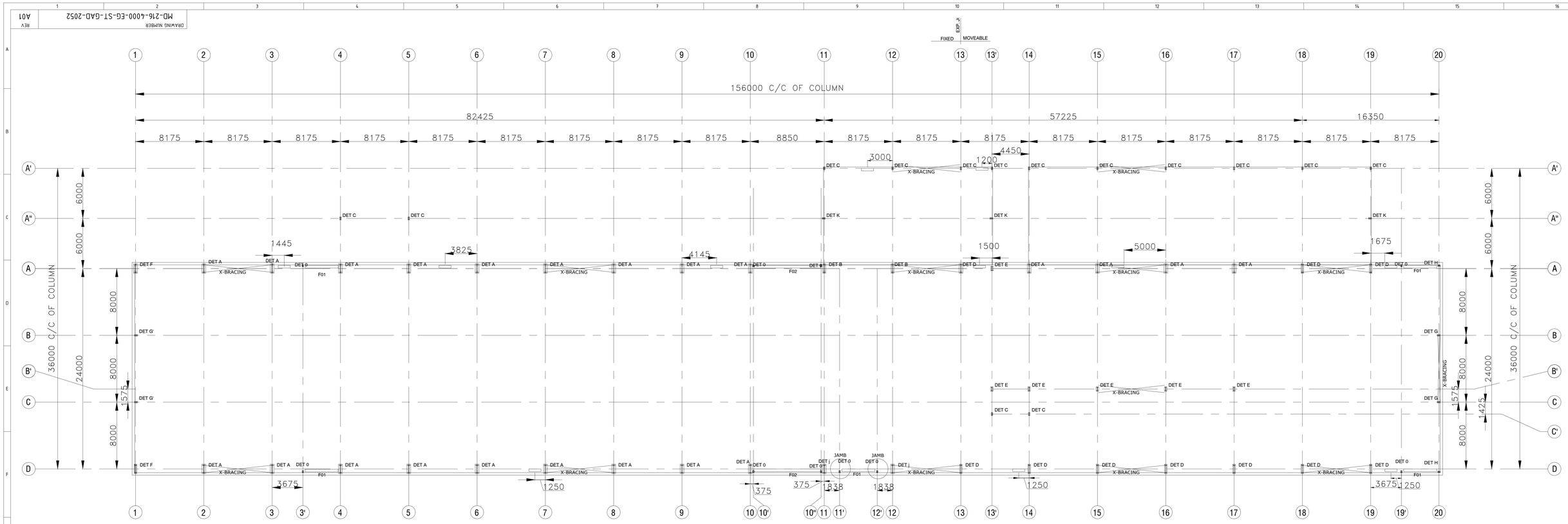
ATTACHMENT 1 LOAD DATA DRAWINGS

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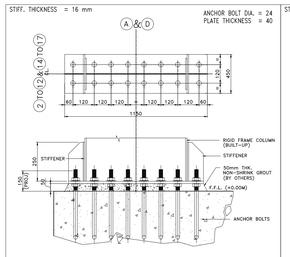
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DRAWING NUMBER: MD-216-4000-EG-ST-GAD-2052
REV: A01

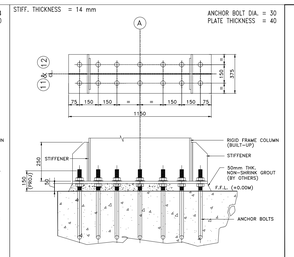


ANCHOR BOLT PLAN

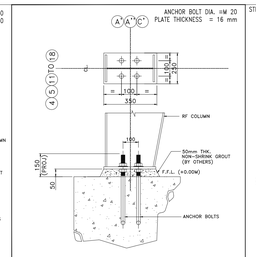
XB– DENOTES TO BRACING FULL HEIGHT.



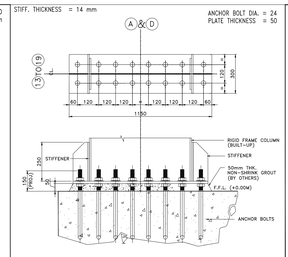
DETAIL–(A)



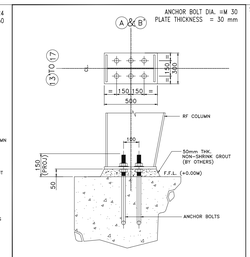
DETAIL–(B)



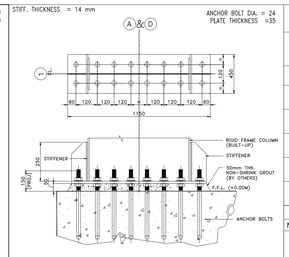
DETAIL–(C)



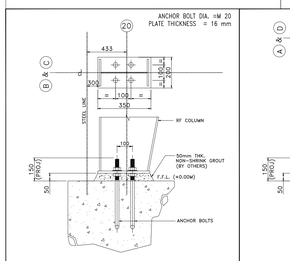
DETAIL–(D)



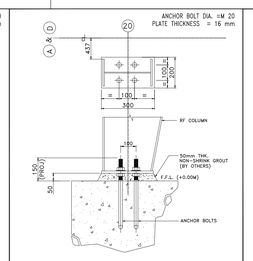
DETAIL–(E)



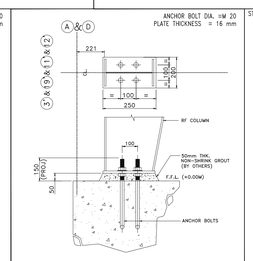
DETAIL–(F)



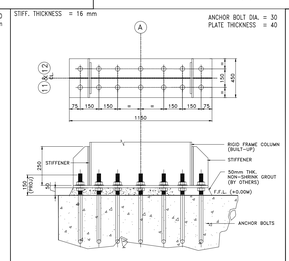
DETAIL–(G)



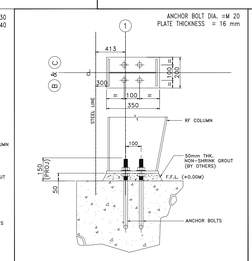
DETAIL–(H)



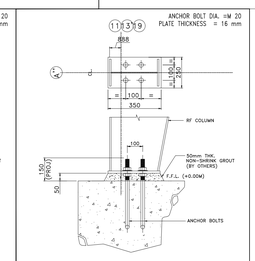
DETAIL–(O)



DETAIL–(J)



DETAIL–(G')



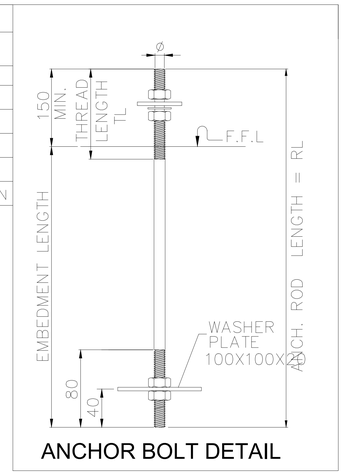
DETAIL–(K)

NOTE: ANCHOR BOLT SPECIFICATION F1554(GRADE36,55or105)

QTY.	DIA. OF BOLT	DIA. OF HOLE	THREAD LENGTH	ANCHOR ROD LENGTH
	ø	D	TL	RL
112	M20	26	150	450
550	M24	30	150	500
96	M30	36	200	700
	M36	42	200	1000
	>M36	ø + 8	200	AS PER DESIGN

MATERIAL SPECIFICATIONS

NO.	MATERIALS		GRADE & SPECIFICATION	STRENGTH
01	ANCHOR BOLTS	J-type	ASTM F1554 Grade A36M, Hot dip galvanized to ASTM A153M, Heavy Hex Nuts A563M-105 Washers-ASTMF436	Fu = 400 N/mm²
		STRAIGHT -type	BS EN ISO 898-1 GRDE 8.8, Hot dip galvanized to BS EN ISO1461 BS EN ISO 898-2 10 STYLE2 WASHER-BS4320	Fu = 800 N/mm²
02	BUILT-UP SECTIONS		ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm²
03	HOT ROLLED SECTIONS	Tubes	E10025 S355HR or equivalent	Fy = 345 N/mm²
		I-Sections	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm²
		Channels	E10025 S355JR or equivalent	Fy = 355 N/mm²
		Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm²
		Pipes	ASTM A53 Grade B or equivalent	Fy = 240 N/mm²
04	Primary Connection bolts		ASTM A325M TYPE-1; Fully threaded hot dip galvanized	Ft = 827 N/mm²
05	DIAGONAL "X" BRACING	Cables	ASTM A475-Extra High Strength Class A, HDG	BREAKING LOAD pu= 119.7 kN
		HR Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm²
06	Cold formed secondary members		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm²
07	Secondary Connection bolts (for purlins,girts and flange braces)		Grade 4.6 Fully threaded hot dip galvanized	Ft = 400 N/mm²
08	Sag Rods		ASTM Grade A36M Electro galvanized, Fully threaded with 4 nuts Coating Z275 (Grade 90)	Fy = 245 N/mm²
09	Mezzanine Deck Panels		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm²



ANCHOR BOLT DETAIL

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-4000-EG-ME-GAD-1001-A03	MECH. DWG								
MD-216-4000-EG-ME-GAD-1002-A03	MECH. DWG								
MD-216-4000-EG-AR-GAD-3005-A01 R1	ELEC DWG								
MD-216-4000-EG-AR-GAD-3003-A01 R1	ELEC DWG								
MD-216-4000-EG-AR-GAD-3005-A01	ARCH DWG								
MD-216-4000-EG-AR-GAD-3004-A01	ARCH DWG								
MD-216-4000-EG-AR-GAD-3003-A01	ARCH DWG	00							
DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	xx/xx/xxxx	ISSUED FOR REFERENCE	--	MNP	--	QRS	

aic arabian international company
P.O.Box : 41017 Jeddah 21521,K.S.A
Tel. : +966 2 591 0999
Fax : +966 2 591 0099
E-mail : aic@aicsteel.com
Site : www.aicsteel.com

DRAWING NUMBER: MD-P20131-B01-A001 — REVISION: 00
DATE 18/05/2020 | DATE 18/05/2020 | DATE 18/05/2020
DRAWN 7017-7019 | CHECKED 7013 | APPROVED 7001

LARSEN & TOUBRO DRAWING NUMBER: 019045_501_S_G_26005_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 __ GAU01 __ AN03 __ MTL01__ 40041 — DOCUMENT ID: OU10067696 — REVISION: 0

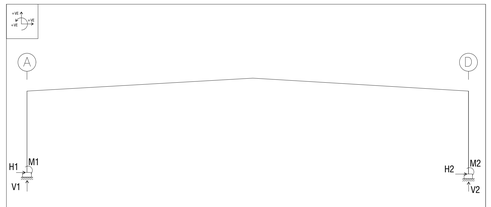
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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	--	XX/XX/XXXX	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	WAREHOUSE & WORKSHOP
APPROVED	--	XX/XX/XXXX	ANCHOR BOLT PLAN
EMPLOYER			
SCALE	A0		DRAWING NUMBER: MD-216-4000-EG-ST-GAD-2052 — REV: A01

DRAWING NUMBER: MD-216-4000-EG-ST-GAD-2053 | REV: A01



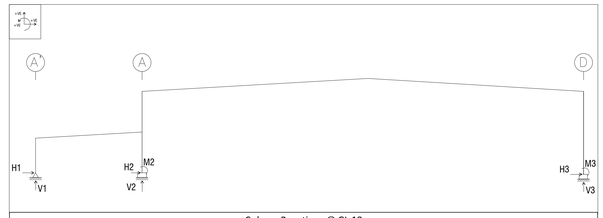
Column Reactions @ GL 2 TO 10

CATEGORY	LOAD CASE	COL.@GL.A			COL.@GL. D		
		H1 (KN)	V1 (KN)	M1 (KN)	H2 (KN)	V2 (KN)	M2 (KN)
Seismic	Seismic	-1.25	0.37	11.48	-1.47	-0.37	12.57
	-Seismic	1.25	-0.37	-11.48	1.47	0.37	-12.57
Dead	Dead	16.09	46.23	-71.19	-16.09	45.48	71.46
Collateral	Collateral	0.24	5.10	-0.56	-0.24	5.10	0.56
Live	Roof Live	36.92	71.48	-164.28	-36.92	71.48	164.28
Live	PLATEFORM.LIVE	0.00	15.81	0.00	0.00	15.81	0.00
CRANE	CRANE 1	-6.19	301.36	214.33	-43.94	83.68	273.42
	CRANE 2	43.84	306.73	-100.72	6.29	78.32	-40.84
	CRANE 3	-6.24	78.33	40.36	-43.89	307.77	100.40
	CRANE 4	43.99	83.70	-273.91	6.14	302.40	-214.65
Wind	Wind_Left1	-118.67	-159.09	537.93	-11.66	-115.41	3.30
	Wind_Left2	-130.23	-89.80	501.63	0.03	-46.06	41.13
	Wind_Right1	11.66	-115.41	-3.30	118.67	-159.09	-537.93
	Wind_Right2	-0.03	-46.06	-41.13	130.23	-89.80	-501.63
	Wind_Long1	-6.77	-199.09	157.60	10.53	-171.26	-226.35
	Wind_Long2	-23.26	-131.49	161.03	18.21	-100.69	-153.55
	Wind_Long3	-9.76	-171.62	215.51	9.76	-200.73	-179.37
	Wind_Long4	-21.12	-101.93	177.54	21.12	-131.04	-140.89
Temperture	T1	1.86	0.00	-20.69	-1.86	0.00	20.69
	T2	-1.86	0.00	20.69	1.86	0.00	-20.69



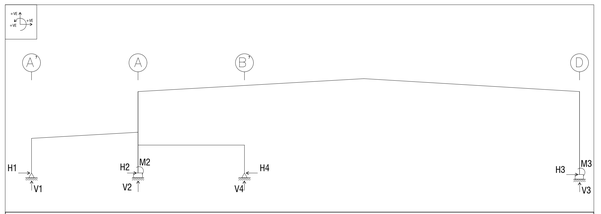
Column Reactions @ GL 11&12

CATEGORY	LOAD CASE	COL.@GL.A'		COL.@GL.A			COL.@GL. D		
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	M2 (KN)	H3 (KN)	V3 (KN)	M3 (KN)
Seismic	Seismic	-0.21	0.03	-2.31	0.30	15.62	-1.56	-0.32	14.21
	-Seismic	0.23	-0.04	2.53	-0.28	-16.80	1.32	0.32	-12.92
Dead	Dead	-7.52	15.31	-8.55	61.33	67.05	16.06	47.37	-69.57
Collateral	Collateral	0.00	0.00	-0.24	5.10	0.53	0.24	5.10	-0.56
Live	Roof Live	-22.05	38.31	-14.03	103.97	124.75	36.08	71.24	-157.75
Live	PLATEFORM.LIVE	-0.01	0.00	0.00	15.81	-0.03	0.00	15.81	-0.01
CRANE	CRANE 1	-4.22	3.04	-40.74	80.29	253.63	-5.18	301.72	205.86
	CRANE 2	0.60	-0.44	5.83	78.78	-37.72	43.70	306.70	-99.16
	CRANE 3	-0.55	0.30	-43.39	306.42	98.32	-6.20	78.33	39.88
	CRANE 4	4.31	-3.19	2.97	304.92	-193.61	42.86	83.31	-264.31
Wind	Wind_Left1	-20.24	30.36	57.80	-53.20	-109.92	-119.28	-159.34	545.88
	Wind_Left2	32.96	-69.48	-34.99	-84.38	121.05	-129.86	-89.54	500.94
	Wind_Right1	-25.53	62.89	136.99	-126.08	-570.81	10.33	-115.95	9.90
	Wind_Right2	30.90	-45.35	41.61	-149.41	-325.52	-1.13	-46.26	-29.49
	Wind_Long1	15.27	-23.07	41.40	-185.93	-273.27	-5.56	-199.05	148.15
	Wind_Long2	24.35	-41.83	11.44	-135.47	-159.99	-17.91	-129.76	116.31
	Wind_Long3	13.90	-22.07	38.66	-215.00	-210.54	-8.99	-170.98	214.54
	Wind_Long4	23.19	-40.83	15.63	-165.11	-121.82	-20.60	-101.53	176.31
Temperture	T1	-1.43	0.88	-0.73	-0.94	14.55	2.15	0.06	-23.56
	T2	1.43	-0.88	0.73	0.94	-14.55	-2.15	-0.06	23.56



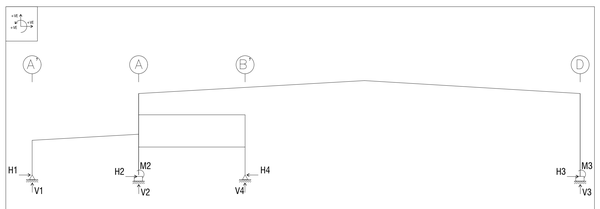
Column Reactions @ GL 18

CATEGORY	LOAD CASE	COL.@GL.A'		COL.@GL.A			COL.@GL. D		
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	M2 (KN)	H3 (KN)	V3 (KN)	M3 (KN)
Seismic	Seismic	-0.20	0.02	-2.19	0.22	11.60	-1.69	-0.24	10.93
	-Seismic	0.20	-0.02	2.19	-0.22	-11.60	1.69	0.24	-10.93
Dead	Dead	-7.30	15.27	-14.56	49.79	66.22	21.86	35.12	-69.92
Live	Roof Live	-8.81	17.29	-40.67	86.08	151.18	49.48	68.49	-154.82
Wind	Wind_Left1	-11.74	23.64	91.02	-35.68	-213.66	-124.74	-141.93	400.51
	Wind_Left2	10.70	-15.67	15.56	-103.55	-100.15	-98.00	-76.87	236.51
	Wind_Right1	-19.95	57.09	128.61	-103.22	-365.35	-44.91	-107.32	203.25
	Wind_Right2	22.84	-55.49	41.04	-97.28	-164.78	-26.41	-43.96	94.27
	Wind_Long1	12.23	-20.73	96.82	-170.05	-364.49	-59.65	-178.76	229.08
	Wind_Long2	17.79	-36.26	44.92	-129.05	-204.72	-47.60	-117.33	158.41
	Wind_Long3	11.20	-19.83	96.74	-197.88	-326.30	-61.96	-153.62	273.39
	Wind_Long4	17.18	-35.72	47.84	-156.97	-187.50	-45.32	-90.65	176.74
Temperture	T1	-1.68	1.17	-2.94	-1.33	26.20	4.61	0.17	-36.20
	T2	1.68	-1.17	2.94	1.33	-26.20	-4.61	-0.17	36.20



Column Reactions @ GL 16

CATEGORY	LOAD CASE	COL.@GL.A'		COL.@GL.A			COL @ GL.B'		COL.@GL. D		
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	M2 (KN)	H4 (KN)	V4 (KN)	H3 (KN)	V3 (KN)	M3 (KN)
Seismic	Seismic	-0.11	-0.05	-10.61	2.80	30.16	-4.77	-2.61	-1.09	-0.13	6.59
	-Seismic	0.11	0.05	10.61	-2.80	-30.16	4.77	2.61	1.09	0.13	-6.59
Dead	Dead	-6.87	15.41	-162.24	264.55	139.20	139.17	208.82	29.95	38.22	-122.53
Collateral	Collateral	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Live	Roof Live	-13.97	31.71	-24.39	115.13	92.37	-15.15	-8.78	53.51	68.46	-189.27
Live	MEZZ.	2.29	-2.15	-584.16	769.65	373.54	567.50	794.62	14.37	3.55	-91.67
Wind	Wind_Left1	-12.88	24.37	77.89	-40.87	-126.15	7.19	2.97	-124.77	-140.44	414.25
	Wind_Left2	9.09	-14.33	8.97	-111.32	-58.02	11.36	6.76	-102.91	-77.21	275.92
	Wind_Right1	-22.79	59.18	114.80	-119.64	-249.21	26.24	13.89	-51.13	-106.88	259.11
	Wind_Right2	20.91	-53.93	39.03	-108.77	-121.03	17.79	10.12	-31.52	-44.15	135.39
	Wind_Long1	8.10	-17.44	72.22	-190.85	-222.67	32.62	18.55	-72.93	-179.80	333.00
	Wind_Long2	15.09	-34.06	31.52	-142.64	-124.93	20.99	12.21	-56.50	-118.15	227.16
	Wind_Long3	7.80	-17.17	75.05	-213.93	-191.32	23.81	13.15	-70.07	-153.38	344.06
	Wind_Long4	14.90	-33.88	36.02	-167.70	-109.24	15.57	8.84	-50.81	-90.60	223.72
Temperture	T1	-1.89	1.32	-36.56	-11.52	166.26	34.27	10.04	4.18	0.15	-32.73
	T2	1.89	-1.32	36.56	11.52	-166.26	-34.27	-10.04	-4.18	-0.15	32.73



Column Reactions @ GL 14&15

CATEGORY	LOAD CASE	COL.@GL.A'		COL.@GL.A			COL @ GL.B'		COL.@GL. D		
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	M2 (KN)	H4 (KN)	V4 (KN)	H3 (KN)	V3 (KN)	M3 (KN)
Seismic	Seismic	-0.43	0.21	-41.44	10.86	123.48	-6.80	-10.72	-2.20	-0.34	14.30
	-Seismic	0.43	-0.21	41.44	-10.86	-123.48	6.80	10.72	2.20	0.34	-14.30
Dead	Dead	-5.68	14.18	-103.30	387.55	130.61	74.73	329.94	25.31	35.17	-94.72
Collateral	Collateral	0.00	0.00	-0.04	1.35	0.08	-0.01	0.00	0.05	1.35	-0.07
Live	Roof Live	-14.01	31.74	-31.71	114.13	114.19	-10.13	-7.41	55.85	68.05	-201.35
Live	MEZZ.	0.11	-0.35	-445.77	1539.99	479.69	434.73	1588.17	10.93	3.53	-64.56
Wind	Wind_Left1	-12.54	24.17	76.80	-40.76	-190.83	22.54	1.66	-127.11	-139.52	420.99
	Wind_Left2	9.13	-14.40	12.20	-109.50	-63.25	4.66	5.59	-106.36	-77.39	291.74
	Wind_Right1	-22.59	59.25	115.39	-117.40	-311.69	34.50	9.33	-52.06	-105.10	263.65
	Wind_Right2	20.96	-54.16	34.74	-104.03	-116.86	11.73	5.02	-32.37	-43.17	141.86
	Wind_Long1	8.22	-17.58	86.56	-189.37	-272.45	24.16	16.99	-78.94	-179.40	363.37
	Wind_Long2	15.21	-34.27	44.73	-143.38	-159.92	12.31	13.29	-59.12	-117.47	240.37
	Wind_Long3	8.01	-17.39	73.38	-208.01	-224.24	23.15	7.02	-71.37	-150.96	351.42
	Wind_Long4	15.00	-34.08	31.55	-162.03	-111.70	11.30	3.32	-51.54	-89.03	228.42
Temperture	T1	-0.94	0.41	-125.86	-10.19	222.76	116.96	8.12	9.84	1.66	-69.97
	T2	0.94	-0.41	125.86	10.19	-222.76	-116.96	-8.12	-9.84	-1.66	69.97



Column Reactions @ GL 13

CATEGORY	LOAD CASE	COL.@GL.A'		COL.@GL.A			COL.@GL. D		
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	M2 (KN)	H3 (KN)	V3 (KN)	M3 (KN)
Seismic	Seismic	-0.35	0.14	-3.30	0.42	22.53	-3.15	-0.56	24.31
	-Seismic	0.38	-0.16	3.79	-0.40	-24.72	2.64	0.56	-21.91
Dead	Dead	-7.40	15.22	-5.92	64.68	50.41	13.32	50.24	-53.33
Collateral	Collateral	0.00	0.00	-0.20	5.10	0.41	0.21	5.10	-0.43
Live	Roof Live	-20.73	36.48	-3.89	100.06	76.01	24.62	68.38	-103.24
Live	PLATEFORM.LIVE	-0.01	0.00	0.00	15.81	-0.03	0.00	15.81	-0.02
CRANE	CRANE 1	-5.13	3.69	-48.24	79.83	287.56	3.23	301.53	160.18
	CRANE 2	0.42	-0.31	0.91	78.71	-19.17	48.81	306.65	-120.85
	CRANE 3	-0.89	0.53	-48.89	306.16	120.48	-0.36	78.36	15.77
	CRANE 4	4.68	-3.48	-0.17	305.05	-186.16	45.62	83.48	-265.03
Wind	Wind_Left1	18.34	-51.72	-50.10	-186.27	126.89	-94.46	-158.98	407.68
	Wind_Left2	7.35	-18.36	-32.65	-78.91	140.60	-106.17	-89.90	382.20
	Wind_Right1	50.42	-82.69	33.72	-201.18	-303.22	25.12	-115.35	-45.81
	Wind_Right2	25.73	-16.14	65.41	-96.85	-337.04	15.71	-45.98	-88.37
	Wind_Long1	29.09	-85.09	-15.23	-228.32	-116.42	2.41	-199.41	103.78
	Wind_Long2	21.40	-55.74	-7.62	-135.21	-85.92	-9.70	-130.24	78.09
	Wind_Long3	27.35	-83.84	-18.61	-258.08	-44.80	-0.01	-170.90	167.00
	Wind_Long4	19.81	-54.52	-6.36	-165.38	-30.09	-11.03	-101.61	135.27
Temperture	T1	-1.64	1.03	-5.79	-1.13	31.32	7.44	0.11	-41.14
	T2	1.65	-1.03	5.79	1.14	-31.34	-7.44	-0.11	41.17

NOTES FOR REACTIONS

Building reactions are based on the following building data:

ROOF LOADS		
DEAD LOAD (kN/m2)	=	0.16
ROOF LIVE LOAD (kN/m2)	=	1.00 REDUCIBLE AS PER SBC
BASIC LOADS		
WIND LOAD	Wind Speed (3 sec)=155 (Km/Hr) WORKING	
	Wind Code	= SBC-301
	EXPOSURE	= C
	CLASSIFICATION	= ENCLOSED
SEIS LOAD IBC 2006	Ss	= 0.1g
	S1	= 0.02g
	SITE CLASS	= C
TEMPERATURE LOAD	TEMPERATURE LOAD	= ± 25 C
MEZZANINE LOAD		
DEAD LOAD (kN/m2)	=	3.5
LIVE LOAD (kN/m2)	=	12
CRANE LOAD		
CRANE CAPACITY	=	25 TON
JIB CRANE LOAD		
CRANE CAPACITY	=	1 TON
PLATFORM LOADS		
DEAD LOAD (kN/m2)	=	1.00
LIVE LOAD (kN/m2)	=	4.00

LOADS LEGEND	
Wind_Left1	Denotes wind load from left to right with internal pressure
Wind_Right1	Denotes wind load from right to left with internal pressure
Wind_Left2	Denotes wind load from left to right with internal suction
Wind_Right2	Denotes wind load from right to left with internal suction
Wind_Left2	Denotes wind load from left to right with internal pressure
Wind_Right2	Denotes wind load from right to left with internal pressure
Wind_Left3	Denotes wind load from left to right with internal suction
Wind_Right4	Denotes wind load from right to left with internal suction
Wind_Long1	Denotes wind load parallel to ridge from end wall with internal pressure
Wind_Long2	Denotes wind load parallel to ridge from end wall with internal suction
Seismic_L	Denotes seismic load from left
Seismic_R	Denotes seismic load from right

IMPORTANT NOTES :
(1) BEARING CAPACITY OF CONCRETE UNDER BASE PLATE SHOULD NOT BE LESS THAN 1.05 KN/CM2
(2) GIVEN REACTIONS ARE FOR UNFACTORED LOADS.
(3) BRACING REACTIONS TO BE ADDED WITH MAIN FRAME COLUMN REACTIONS FOR THE DESIGN OF FOUNDATION.
(4) LOADS IN THE SAME CATEGORY , NOT TO BE COMBINED IN THE SAME LOAD COMBINATION .

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-4000-EG-ME-GAD-3001-A03	MECH. DWG								
MD-216-4000-EG-ME-GAD-3002-A03	MECH. DWG								
MD-216-4000-EG-AR-GAD-3005-A01 R1	ELEC DWG								
MD-216-4000-EG-AR-GAD-3003-A01 R1	ELEC DWG								
MD-216-4000-EG-AR-GAD-3005-A01	ARCH DWG								
MD-216-4000-EG-AR-GAD-3004-A01	ARCH DWG								
MD-216-4000-EG-AR-GAD-3003-A01	ARCH DWG	00							
		A01	xx/xx/xxxx	ISSUED FOR REFERENCE	--	MNP	--	QRS	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

P.O.Box : 41017 Jeddah 21521,K.S.A
Tel: +966 2 591 0999
Fax +966 2 591 0099
E-mail aic@aicsteel.com
Site www.aicsteel.com

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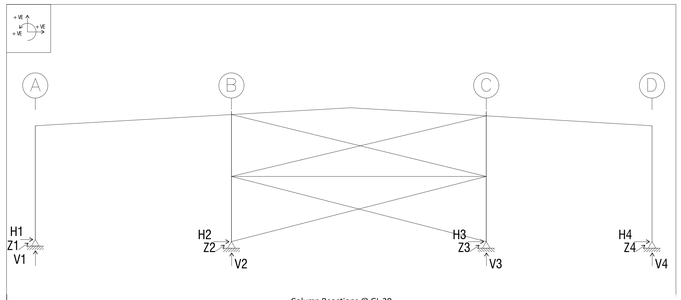
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DRAWING NUMBER	MD-P20131-B01-A002		REVISION 00
DRAWN	DATE 18/05/2020	CHECKED DATE 18/05/2020	APPROVED DATE 18/05/2020
	7017-7019	7013	7001

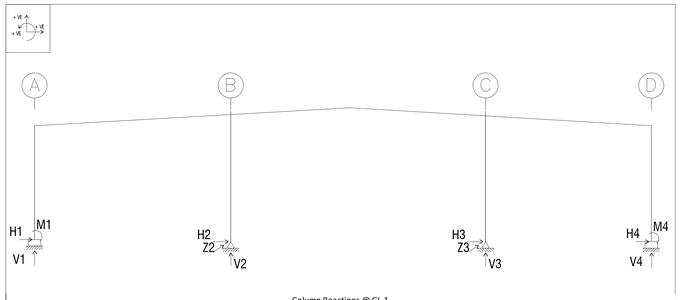
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OUTOTEC DRAWING NUMBER	900707 _ GAU01 _ AN03 _ MTL01 _ 40042	DOCUMENT ID 0U1006764J	REVISION 0
DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	--	XX/XX/XXXX	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	WAREHOUSE & WORKSHOP
APPROVED	--	XX/XX/XXXX	COLUMN REACTIONS(1/2)
EMPLOYER			
SCALE	A0	DRAWING NUMBER	MD-216-4000-EG-ST-GAD-2053 REV A01

DRAWING NUMBER MD-216-4000-EG-ST-GAD-2055 REV A01



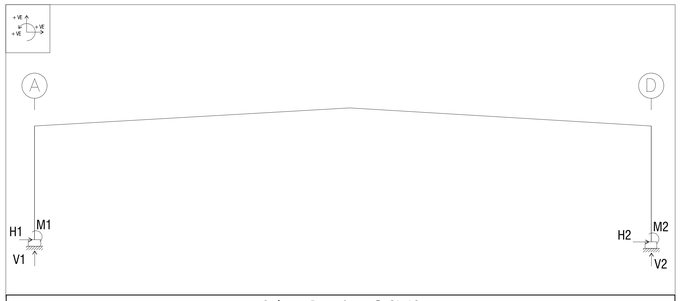
Column Reactions @ GL 20

CATEGORY	LOAD CASE	COL.@GL.A			COL.@GL. B			COL.@GL. C			COL.@GL. D		
		H1 (KN)	V1 (KN)	Z1 (KN)	H2 (KN)	V2 (KN)	Z2 (KN)	H3 (KN)	V3 (KN)	Z3 (KN)	H4 (KN)	V4 (KN)	Z4 (KN)
Seismic	Seismic	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	-Seismic	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Dead	Dead	0.00	6.29	0.00	0.00	23.09	0.00	0.00	23.09	0.00	0.00	6.29	0.00
Live	Roof Live	0.00	9.72	0.00	0.00	25.76	0.00	0.00	25.76	0.00	0.00	9.72	0.00
Wind	Wind_Left1	0.00	-30.46	-21.73	-1.41	-66.70	-47.28	12.44	-61.02	-47.47	0.00	-17.50	-21.73
	Wind_Left2	0.00	-21.70	-8.76	0.00	-44.71	-19.05	9.90	-36.32	-19.12	0.00	-8.72	-8.76
	Wind_Right1	0.00	-17.50	21.73	-12.44	-61.02	-47.47	1.41	-66.70	-47.28	0.00	-30.46	21.73
	Wind_Right2	0.00	-8.72	8.76	-9.90	-36.32	19.12	0.00	-44.71	19.05	0.00	-21.70	8.76
	Wind_Long1	0.00	-30.33	-17.52	-4.67	-61.33	-47.34	8.43	-59.39	-47.41	0.00	-21.26	-17.52
	Wind_Long2	0.00	-21.70	-8.76	-2.13	-38.01	19.08	5.90	-36.07	19.10	0.00	-12.49	-8.76
	Wind_Long3	0.00	-21.41	-21.73	-8.46	-59.64	-47.41	4.69	-61.58	-47.34	0.00	-30.54	-21.73
	Wind_Long4	0.00	-12.58	8.76	-5.91	-36.20	47.41	2.15	-38.13	47.34	0.00	-21.71	8.76
Temperture	T1	0.00	27.11	0.00	-2.94	-27.11	0.00	2.94	-27.11	0.00	0.00	27.11	0.00
	T2	0.00	-27.13	0.00	0.00	27.13	0.00	0.00	27.13	0.00	0.00	-27.13	0.00



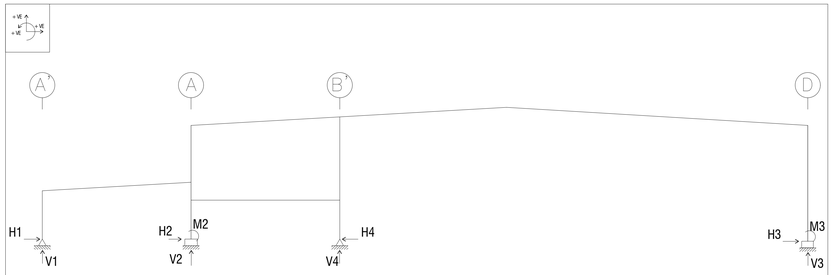
Column Reactions @ GL 1

CATEGORY	LOAD CASE	COL.@GL.A			COL.@GL. B			COL.@GL. C			COL.@GL. D		
		H1 (KN)	V1 (KN)	M1 (KN)	H2 (KN)	V2 (KN)	Z2 (KN)	H3 (KN)	V3 (KN)	Z3 (KN)	H4 (KN)	V4 (KN)	M4 (KN)
Seismic	Seismic	-1.36	0.95	11.38	0.00	-1.17	0.00	0.00	-1.17	0.00	-1.36	-0.62	11.56
	-Seismic	1.36	-0.95	-11.38	0.00	1.17	0.00	0.00	1.17	0.00	1.36	0.62	-11.56
Dead	Dead	-1.31	19.30	4.99	0.00	15.93	0.00	0.00	15.93	0.00	1.31	19.30	-4.99
Collateral	Collateral	-0.25	5.11	0.57	0.00	-0.01	0.00	0.00	-0.01	0.00	0.25	5.11	-0.57
Live	Roof Live	-2.38	11.46	9.75	0.00	22.83	0.00	0.00	22.83	0.00	2.38	11.46	-9.75
Live	PLATEFORM.LIVE	0.03	15.78	-0.12	0.00	0.03	0.00	0.00	0.03	0.00	-0.03	15.78	0.12
CRANE	CRANE 1	-6.41	89.56	56.41	0.00	1.40	0.00	0.00	-14.14	0.00	6.42	260.14	72.78
	CRANE 2	-30.54	82.90	143.27	0.00	8.04	0.00	0.00	-7.47	0.00	30.54	253.49	-14.65
	CRANE 3	-6.42	260.14	-72.78	0.00	-14.14	0.00	0.00	1.40	0.00	6.41	89.56	-56.41
	CRANE 4	-30.54	253.49	14.65	0.00	-7.47	0.00	0.00	8.04	0.00	30.54	82.90	-143.27
Wind	Wind_Left1	-45.35	-16.78	186.07	0.00	-63.74	-63.55	0.00	-54.61	-63.55	-47.94	-46.53	213.81
	Wind_Left2	-27.29	-2.44	148.32	0.00	-44.73	-27.24	0.00	-35.69	-27.24	-66.06	-32.26	252.45
	Wind_Right1	47.94	-46.53	-213.81	0.00	-54.61	63.55	0.00	-63.74	63.55	45.35	-16.78	-186.07
	Wind_Right2	66.06	-32.26	-252.45	0.00	-35.69	27.24	0.00	-44.73	27.24	27.29	-2.44	-148.32
	Wind_Long1	-27.83	-34.34	42.30	0.00	-54.70	63.55	0.00	-44.70	63.55	29.61	-43.74	-68.52
	Wind_Long2	-9.74	-20.04	4.06	0.00	-35.74	27.24	0.00	-25.72	27.24	11.53	-29.43	-30.42
	Wind_Long3	-52.41	-27.88	258.56	0.00	-34.81	0.00	0.00	-75.02	0.00	-66.48	-39.77	314.37
	Wind_Long4	-11.53	-29.43	30.42	0.00	-25.72	0.00	0.00	-35.74	0.00	9.74	-20.04	-4.06
Temperture	T1	-4.12	3.86	28.44	0.00	-3.86	0.00	0.00	-3.86	0.00	4.12	3.86	-28.44
	T2	4.13	-3.87	-28.45	0.00	3.87	0.00	0.00	3.87	0.00	-4.13	-3.87	28.45



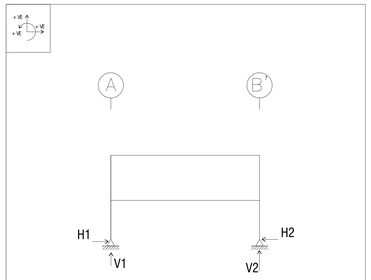
Column Reactions @ GL 19

CATEGORY	LOAD CASE	COL.@GL.A			COL.@GL. D		
		H1 (KN)	V1 (KN)	M1 (KN)	H2 (KN)	V2 (KN)	M2 (KN)
Seismic	Seismic	-1.19	0.24	8.16	-1.53	-0.24	9.46
	-Seismic	1.19	-0.24	-8.16	1.53	0.24	-9.46
Dead	Dead	-22.14	34.97	72.98	22.14	34.97	-72.98
Live	Roof Live	50.18	68.64	-161.06	50.18	68.64	-161.06
Wind	Wind_Left1	-119.05	-140.34	368.67	-119.05	-140.34	368.67
	Wind_Left2	-104.13	-78.12	279.46	-104.13	-78.12	279.46
	Wind_Right1	-38.78	-105.91	165.78	-38.78	-105.91	165.78
	Wind_Right2	-23.85	-43.68	76.57	-23.85	-43.68	76.57
	Wind_Long1	-61.47	-178.96	245.96	-61.47	-178.96	245.96
	Wind_Long2	-50.62	-117.96	180.86	-50.62	-117.96	180.86
	Wind_Long3	-64.32	-153.92	293.68	-64.32	-153.92	293.68
	Wind_Long4	-49.16	-91.49	204.14	-49.16	-91.49	204.14
Temperture	T1	3.79	0.00	-30.66	3.79	0.00	-30.66
	T2	-3.79	0.00	30.66	-3.79	0.00	30.66



Column Reactions @ GL 17

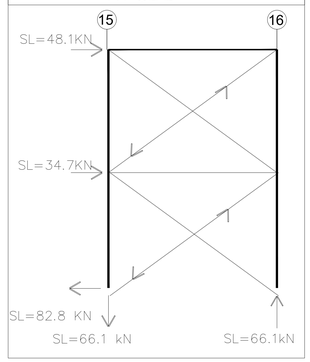
CATEGORY	LOAD CASE	COL.@GL.A'		COL.@GL.A			COL @ GL.B'			COL.@GL. D		
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	MZ2 (KN)	H4 (KN)	V4 (KN)	Z4 (KN)	H3 (KN)	V3 (KN)	M3 (KN)
Seismic	Seismic	-0.21	0.02	-12.42	1.73	40.03	-2.44	-1.54	0.00	-1.52	-0.20	9.69
	-Seismic	0.21	-0.02	12.42	-1.73	-40.03	2.44	1.54	0.00	1.52	0.20	-9.69
Dead	Dead	-6.51	14.56	-109.28	174.49	101.51	37.81	115.49	0.00	27.53	36.00	-111.32
Collateral	Collateral	0.00	0.00	0.00	0.13	-0.02	0.00	0.00	0.00	0.00	0.00	-0.01
Live	Roof Live	-14.28	31.97	-28.11	111.65	115.23	-8.75	-4.91	-4.91	51.15	67.81	-174.63
Live	MEZZ.	1.93	-1.77	-317.22	389.06	192.54	303.18	393.37	393.37	12.12	2.97	-77.52
CRANE	CRANE 1	0.26	-0.20	2.03	-2.43	-27.67	-5.02	61.86	9.74	2.73	0.77	-17.31
Wind	Wind_Left1	-12.87	24.40	80.87	-39.81	-136.10	3.52	1.58	1.58	-124.08	-140.15	410.73
	Wind_Left2	9.34	-14.54	11.58	-108.62	-74.42	6.56	3.74	3.74	-100.96	-76.67	263.85
	Wind_Right1	-22.36	58.87	122.58	-114.24	-291.37	14.86	7.92	7.92	-47.96	-106.00	239.47
	Wind_Right2	21.26	-54.21	43.58	-104.77	-148.23	10.23	5.68	5.68	-28.87	-43.43	118.92
	Wind_Long1	8.77	-17.97	80.64	-183.54	-274.58	18.98	10.54	10.54	-68.38	-178.57	304.62
	Wind_Long2	15.55	-34.43	36.65	-137.80	-157.59	12.25	6.89	6.89	-53.35	-117.31	207.51
	Wind_Long3	8.22	-17.49	81.56	-208.78	-226.23	13.35	7.28	7.28	-66.53	-152.34	322.64
	Wind_Long4	15.19	-34.11	40.02	-164.21	-131.16	8.74	4.84	4.84	-48.27	-89.85	208.29
Temperture	T1	-1.78	1.24	-14.62	-5.64	87.99	11.66	4.13	4.13	4.74	0.27	-36.49
	T2	1.78	-1.24	14.62	5.64	-87.99	-11.66	-4.13	-4.13	-4.74	-0.27	36.49



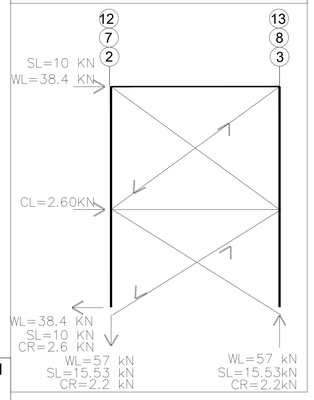
Column Reactions @ GL 13'

CATEGORY	LOAD CASE	COL.@GL.A		COL.@GL. D	
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)
Seismic	Seismic	-23.49	-19.00	-23.49	-19.00
	-Seismic	23.49	19.00	23.49	19.00
Dead	Dead	-16.42	145.70	16.42	145.86
Live	MEZZ. Live	-51.62	438.95	51.62	438.95
Temperture	T1	-15.19	0.00	15.19	0.00
	T2	15.19	0.00	-15.19	0.00

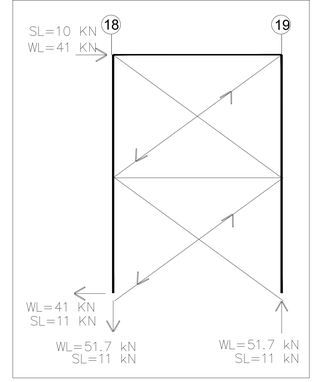
SIDEWALL (XB) BRACING REACTION AT GL B`



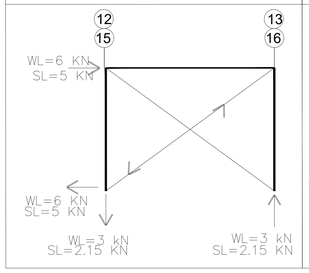
SIDEWALL (XB) BRACING REACTION AT GL A &D



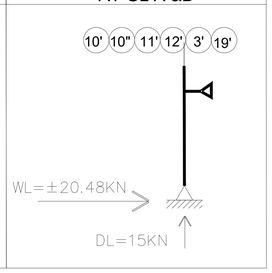
SIDEWALL (XB) BRACING REACTION AT GL A &D



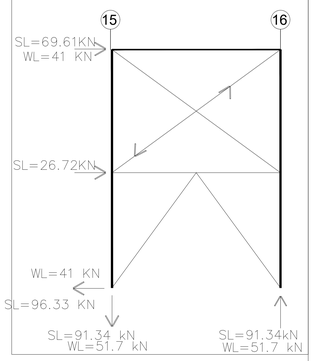
SIDEWALL (XB) BRACING REACTION AT GL A'



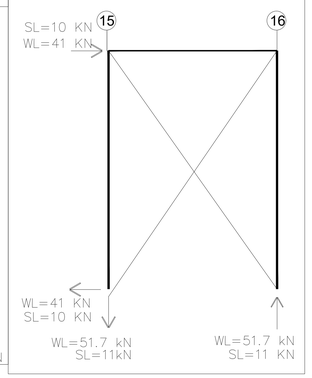
JAMB COLUMN REACTION AT GL A &D



SIDEWALL (XB) BRACING REACTION AT GL A



SIDEWALL (XB) BRACING REACTION AT GL D



LOADS LEGEND	
Wind_Left1	Denotes wind load from left to right with internal pressure
Wind_Right1	Denotes wind load from right to left with internal pressure
Wind_Left2	Denotes wind load from left to right with internal suction
Wind_Right2	Denotes wind load from right to left with internal suction
Wind_Left2	Denotes wind load from left to right with internal pressure
Wind_Right2	Denotes wind load from right to left with internal pressure
Wind_Left3	Denotes wind load from left to right with internal suction
Wind_Right4	Denotes wind load from right to left with internal suction
Wind_Long1	Denotes wind load parallel to ridge from end wall with internal pressure
Wind_Long2	Denotes wind load parallel to ridge from end wall with internal suction
Seismic_L	Denotes seismic load from left
Seismic_R	Denotes seismic load from right

IMPORTANT NOTES :
(1) BEARING CAPACITY OF CONCRETE UNDER BASE PLATE SHOULD NOT BE LESS THAN 1.05 KN/CM2
(2) GIVEN REACTIONS ARE FOR UNFACTORED LOADS.
(3) BRACING REACTIONS TO BE ADDED WITH MAIN FRAME COLUMN REACTIONS FOR THE DESIGN OF FOUNDATION.
(4) LOADS IN THE SAME CATEGORY , NOT TO BE COMBINED IN THE SAME LOAD COMBINATION .

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-4000-EG-ME-GAD-1001-A03	MECH. DWG								
MD-216-4000-EG-ME-GAD-1002-A03	MECH. DWG								
MD-216-4000-EG-AR-GAD-3005-A01 R1	ELEC DWG								
MD-216-4000-EG-AR-GAD-3003-A01 R1	ELEC DWG								
MD-216-4000-EG-AR-GAD-3005-A01	ARCH DWG								
MD-216-4000-EG-AR-GAD-3004-A01	ARCH DWG								
MD-216-4000-EG-AR-GAD-3003-A01	ARCH DWG	00							

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	xx/xx/xxxx	ISSUED FOR REFERENCE	--	MNP	--	QRS	

AIC Arabian International Company
P.O.Box : 41017 Jeddah 21521,K.S.A
Tel. : +966 2 591 0999
Fax : +966 2 591 0099
E-mail : aic@aicsteel.com
Site : www.aicsteel.com

DRAWING NUMBER		REVISION
MD-P20131-B01-A003		00
DATE 18/05/2020	DATE 18/05/2020	DATE 18/05/2020
DRAWN 7017-7019	CHECKED 7013	APPROVED 7001

LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_501_S_G_26005_003		0
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707 __ GAU01 __ AN03 __ MTL01__ 40043	OU10067498	0

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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	--	XX/XX/XXXX	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	WAREHOUSE & WORKSHOP
APPROVED	--	XX/XX/XXXX	COLUMN REACTIONS (2/2)
EMPLOYER			
SCALE	A0		DRAWING NUMBER: MD-216-4000-EG-ST-GAD-2054 REV: A01